建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

I N S T R U C T I O N M A N U A L

（Vol.1/2）

CUSTOMER ： FHI/LMYP-1

JOB NO. ： XZ-EPNE26，XZ-EPNE27

EQUIP. NAME ： S.A.FAN、P.A.FAN、I.D.FAN

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建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

I N S T R U C T I O N M A N U A L

（Vol.2/2）

CUSTOMER ： FHI/LMYP-1

JOB NO. ： XZ-EPNE26，XZ-EPNE27

EQUIP. NAME ： S.A.FAN、P.A.FAN、I.D.FAN

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建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

LMYP-1 Limay, Philippine
S.A.Fan Specification Sheet
送風機規範書資料

本文編號 RF2-0580-DOC-6-0

						CERTIFIED	
						PROJ.	
						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2013/09/16	PROJECT NO.	PAGE
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建成機械股份有限公司
CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

☐PROPOSAL ■FOR CONSTRUCTION ☐FINAL

CUSTOMER ： FHI

INQUIRY NO. ：

PROJECT NO. ： LMYP-1

DATE ： Jan. / 14 / 2015

S. A. FAN

FOR SCPC REFERENCE ONLY

1. DESIGN CONDITION：

CAPACITY： 4222 ACMM
STATIC PRESS. (SUC. / DISC. / DIFF.)： -153 / 1922 / 2075 mmAQ
OPERATION TEMP.： 40 ℃
FAN OPER. SPEED： 1780 RPM
TE. / SE.： 85 / 84 %
MAX. BREAK POWER： 1698 BKW @ DES. CONDITION
DRIVER： ■ELECTRIC MOTOR ☐TURBINE
1850 KW × 4 P × 6600 V × 3 ∅ × 60 Hz × GD^2 3000 KG-M^2
TEFC TYPE F CLASS SUPPLY BY CCMC

FAN 1st CRITICAL SPEED： 2440 RPM
VIBRATION (DIS.)： ISO 10816-3
SOUND PRESSURE LEVEL： 90 dBA AT 1 M (AFTER FAN IS COMPLETELY INSULATED)

2. FAN CONSTRUCTION：

FAN TYPE： ☐AIR FOIL ☐RADIAL ■BACKWARD ☐RADIAL TIPS CENTRIFUGAL HEAVY DUTY
FAN SIZE： BCT 2118A / 935 DWDI A3D2
FAN Q'TY： 2 SET(S)
ARRANGEMENT： A.M.C.A. ■A-3 ☐A-8 ☐A-1 ☐A-7 ☐A-4
TYPE OF DRIVE： ■DIRECT BY MOTOR ☐BY BELT
☐BY V.S. HYDRAULIC COUPLING

2.1 HOUSING：

CONSTRUCTION：	SPLIT HOUSING ALL WELDED TYPE			
SIDE PLATE：	■ASTM A-36	☐ASTM A-588	☐304SS	☐
SCROLL：	■ASTM A-36	☐ASTM A-588	☐304SS	☐
STIFF.：	■ASTM A-36	☐ASTM A-588	☐304SS	☐
INLET BOX：	■ASTM A-36	☐ASTM A-588	☐304SS	☐

2.2 WHEEL：

CONSTRUCTION： ■DOUBLE SUCTION ALL WELDED TYPE
☐SINGLE SUCTION ALL WELDED TYPE
EFFECTIVE DIA.： D 1900 mm

SIDE PLATE：	■ASTM A-514	☐ASTM A-588	☐304SS	☐HITEN
CENTRAL PLATE：	■ASTM A-514	☐ASTM A-588	☐304SS	☐HITEN
BLADE：	■ASTM A-514	☐ASTM A-588	☐304SS	☐HITEN
LINER：	☐FIRMAX	☐TOP B	■NONE	
HUB：	☐ASTM A-27	☐JIS SCS13	■NONE	

GD^2： 3000 KG-M^2
TIP SPEED： 177 M/S

2.3 SHAFT, BEARING & BEARING HOUSING：

SHAFT MATERIAL： ☐AISI 4130 ■ASTM A-668 ☐304SS ☐
DIA AT HUB / BEARING： ∅ 480 mm / ∅ 160 mm
SHAFT SEAL： ■WITH ☐WITHOUT
TYPE ☐LABYRITHIAN ■GLAND
BEARING TYPE： ■SLEEVE ☐ANTI-FRICTION
BEARING LIFE： ■NO LIMITED ☐L10H OVER 60,000
BEARING HOUSING： ASTM A-536
LUBRICATION METHOD： ■OIL BATH ☐GREASING ☐FORCED LUB. (AS. 3.6)
LUBRICATE： ■VG 32 (CPC R32) ☐CPC R3
HOUSING TYPE： ■ZOLLERN STAND. WATER COOLING HORIZONTAL SPLIT TYPE OR EQ'L
☐SKF. STAND. AIR COOLING HORIZONTAL SPLIT TYPE OR EQ'L
COOLING WATER RATE / PRESS.： 3 T/H / 2 KG / CM^2 (PRESS. DIFF.)

FAN SPECIFICATION SHEET

2.4 AUXILIARY ITEMS：

CONN. COUPLING：	■WITH	□WITHOUT
TYPE	■DISC □STEELFLEX □_____	
COUPLING GUARD：	■WITH	□WITHOUT
	■NON-SPARKING TYPE	
BELT：	_____ (BANDO)	
PULLEY：	□WITH	■WITHOUT
ANCHOR BOLT W/FLAME：	■WITH	□WITHOUT
ACCESS DOOR：	■WITH	□WITHOUT
	■FAN HOUSING	
	■INLET BOX × 2 SET	
ISOLATOR：	□WITH	■WITHOUT
SAFELY COVER：	■WITH	□WITHOUT
COMMON BASE：	□WITH	■WITHOUT
LIFTING LUG：	■WITH	□WITHOUT
DRAIN W/VALVE：	■WITH	□WITHOUT
MOTOR BASE W/ADJ. BOLT：	■WITH	□WITHOUT
OIL LEVEL GAUGE：	■WITH	□WITHOUT
GREASE NIPPLE：	□WITH	■WITHOUT
THERMO COUPLE：	■WITH	□WITHOUT
COOLING WHEEL：	□WITH	■WITHOUT
VIBRATION DETECTOR：	■WITH	□WITHOUT

3. ACCESSORIES：

3.1 INLET

3.1.1	SILENCER：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.2	RAIN HOOD：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.3	SCREEN：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.4	FILTER：	□WITH		■WITHOUT
3.1.5	FLOW MEASUREMENT：	■VENTURI	□AIR FOIL	□NONE
	MAT'L：	■ASTM A-588 +	■304SS	□
3.1.6	INLET VOLUME CONTROL：	■WITH		□WITHOUT
		■VANE	□DAMPER	
3.1.7	DAMPER：	□WITH		■WITHOUT
	CONTROL METHOD：	□ON-OFF	□SERVO	□CHECK（AUTO.）
3.1.8	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	■304SS	□
3.1.9	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	■ASTM A-36

3.2 OUTLET

3.2.1	SILENCER：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.2.2	DAMPER：	□WITH		■WITHOUT
	CONTROL METHOD：	□ON-OFF	□SERVO	□CHECK（AUTO.）
3.2.3	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	□
3.2.4	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	□

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CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

3.3	ACTUATOR：	■WITH (MARKER： ROTORK)	□WITHOUT
3.3.1	FOR INLET：	■VANE □DAMPER	□WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	■ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	■ELECTRIC
		□IA. PRESSURE	□STEP MOTOR
		□ _____ Kg/cm²	■ 3 ∅ 460 V
	TORQUE：	■ 320 KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	■WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	■WITH	□WITHOUT
		□ _____ ∅ _____ V	
	SIGNAL INPUT：	■4～20 mA.	□ _____
	SENSITIVITY：	■0.5 %	□0.3 %
	RESPONSE TIME：	■ 1800 CYCLE/H	■ 0.5 Hz
	FAILURE：	■LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	■WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	■WITH	□WITHOUT
3.3.2	FOR OUTLET DAMPER：	□WITH	■WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	□ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	□ _____
	TORQUE：	□ _____ KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	□WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	□WITH	□WITHOUT
	SIGNAL INPUT：	□4～20 mA.	□ _____
	SENSITIVITY：	□0.5 %	□0.3 %
	RESPONSE TIME：	□ _____ CYCLE/H	□ _____ Hz
	FAILURE：	□LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	□WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	□WITH	□WITHOUT
3.4	V.S. HYDRAULIC COUPLING：	□WITH (MARKER： _____)	■WITHOUT
	MODEL：	□ _____	□ _____
	SPEED RANGE：	□ _____ ~ _____ RPM	□ _____
	WORK OIL：	□SUPPLY BY CCMC	□OUT SCOPE
	HEAT EXCHANGER & PIPING：	□WITH	□WITHOUT
	WORK OIL COOLER METHOD：	□WATER	□ _____
	COOLING WATER FLOW RATE：	□ _____ TONS/H	□ _____
3.4.1	ACTUATOR：	□WITH (MARKER： _____)	□WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	□ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	□ELECTRIC
		□IA. PRESSURE	□STEP MOTOR
		□ _____ Kg/cm²	□ _____ ∅ _____ V
	TORQUE：	□ _____ KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	□WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	□WITH	□WITHOUT
		□ _____ ∅ _____ V	
	SIGNAL INPUT：	□4～20 mA.	□ _____
	SENSITIVITY：	□0.5 %	□0.3 %
	RESPONSE TIME：	□ _____ CYCLE/H	□ _____ Hz
	FAILURE：	□LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	□WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	□WITH	□WITHOUT
3.5	TURNING MOTOR & OTHERS EQUIPMENT：		
	OVERRUNNING CLUTCH：	□WITH	■WITHOUT
	STANDBY MOTOR：	_____ KW × _____ P	
	REVOLUTION SPEED：	_____ RPM	
	FLUID TEMP. LIMIT：	_____ ℃	
	CONN. COUPLING：	□DISC	□STEELFLEX
	REQ'D：	_____ SET(S)	

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FAN SPECIFICATION SHEET

3.6	FORCED LUB. SYSTEM FOR BEARING：	☐WITH	■WITHOUT
	OIL PUMP W/MOTOR：	☐DUAL	☐SINGLE
	OPERATION：	☐STAND BY	☐ALL TIME
	HEAT EXCHANGER：	☐DUAL	☐SINGLE
	FILTER：	☐DUAL	☐SINGLE
	LOW OIL PRESS. SWITCH：		
	LOW PRESS.：	☐ALARM	☐
	LOW LOW OIL PRESS.：	☐SHUT DOWN	☐
	HIGH OIL TEMP. SWITCH：		
	HIGH TEMP.	☐ALARM	☐
	HIGH HIGH TEMP.：	☐SHUT DOWN	☐

4. TEST & INSPECTION：

(1)	DYNAMIC BALANCING：	ISO. 1940 G2.5		
(2)	MECH. RUNNING TEST：	☐AT SHOP	■AT FIELD	
(3)	PERFORMANCE TEST：	☐AT SHOP	■AT FIELD	（FOLLOW CNS）
(4)	BEARING HOUSING HYDRAULIC TEST：	■AT SHOP	☐AT FIELD	
(5)	DIMENSION CHECK：	■AT SHOP	☐AT FIELD	（CCMC STANDARD）

5. MISCELLANEOUS：

(1)	FAN NET WEIGHT：	13000 KGS (W/MOTOR)	
(2)	MAINTENANCE WEIGHT：	4000 KGS	
(3)	PAINTING：	☐CCMC STD.	■CUSTOMER STD.
	SANDBLAST：	CLASS 2.5	
	PRIMER：	YES	
	FINISH：	YES	

6. GUARANTEE：

(1) PERIOD：1 YEAR FOR OPERATION OR 18 MOTHS FOR DELIVERY
(2) OPERATION CONDITION：IN NORMAL DESIGN CONDITION AND MAINTENANCE

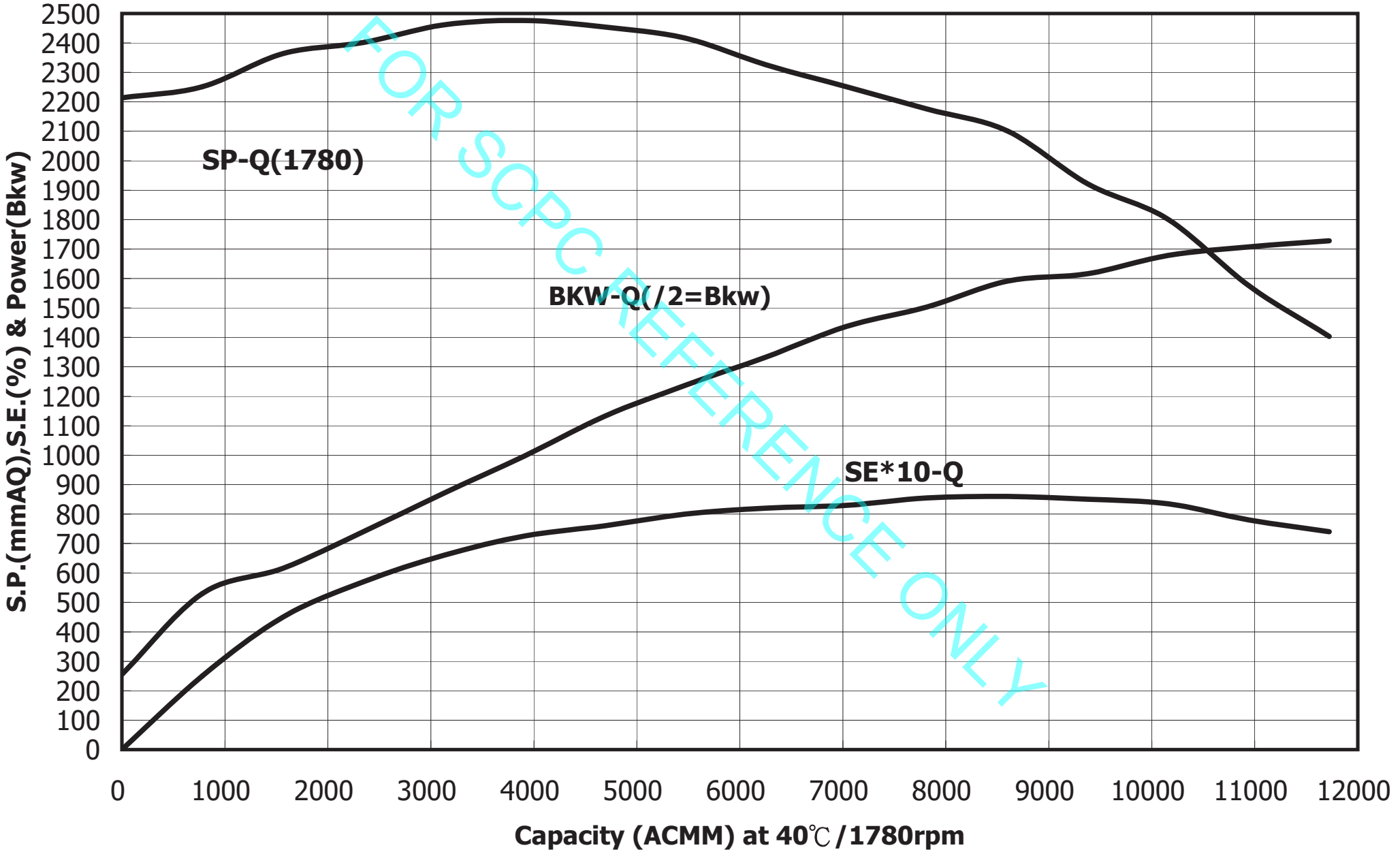
7. SPARE PART：

NO NECESSARY

8. MAKER：

FOR SCPC REFERENCE ONLY

D2118A TLT/935 DWDI Curve
For:FHI(股)公司 Proj.:LMYP-1 SAF



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CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

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LMYP-1 Limay, Philippine
P.A.Fan Specification Sheet
送風機規範書資料

本文編號 RF2-0581-DOC-6-0

						CERTIFIED	
						PROJ.	
						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2013/09/16	PROJECT NO.	PAGE
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	**LMYP-1**	1

建成機械股份有限公司
CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

☐PROPOSAL ■FOR CONSTRUCTION ☐FINAL

CUSTOMER ： FHI

INQUIRY NO. ：

PROJECT NO. ： LMYP-1

DATE ： Jan. / 14 / 2015

P. A. FAN

FOR SCPC REFERENCE ONLY

1. DESIGN CONDITION：

CAPACITY： 6907 ACMM
STATIC PRESS. (SUC. / DISC. / DIFF.)： -153 / 2328 / _____ mmAQ
OPERATION TEMP.： 40 ℃
FAN OPER. SPEED： 1780 RPM
TE. / SE.： 84.5 / 83.7 %
MAX. BREAK POWER： 3341 BKW @ DES. CONDITION
DRIVER： ■ELECTRIC MOTOR ☐TURBINE
3700 KW × 4 P × 6600 V × 3 ∅ × 60 Hz × GD^2 5000 KG-M^2
TEFC TYPE F CLASS SUPPLY BY CCMC

FAN 1st CRITICAL SPEED： 2400 RPM
VIBRATION (DIS.)： ISO 10816-3
SOUND PRESSURE LEVEL： 90 dBA AT 1 M (AFTER FAN IS COMPLETELY INSULATED)

2. FAN CONSTRUCTION：

FAN TYPE： ☐AIR FOIL ☐RADIAL
■BACKWARD ☐RADIAL TIPS CENTRIFUGAL HEAVY DUTY

FAN SIZE： BCT 2118A / 1027 DWDI A3D2
FAN Q'TY： 2 SET(S)
ARRANGEMENT： A.M.C.A. ■A-3 ☐A-8 ☐A-1 ☐A-7 ☐A-4
TYPE OF DRIVE： ■DIRECT BY MOTOR ☐BY BELT
☐BY V.S. HYDRAULIC COUPLING

2.1 HOUSING：
CONSTRUCTION： SPLIT HOUSING ALL WELDED TYPE
SIDE PLATE： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
SCROLL： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
STIFF.： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
INLET BOX： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐

2.2 WHEEL：
CONSTRUCTION： ■DOUBLE SUCTION ALL WELDED TYPE
☐SINGLE SUCTION ALL WELDED TYPE
EFFECTIVE DIA.： D 2166 mm
SIDE PLATE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
CENTRAL PLATE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
BLADE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
LINER： ☐FIRMAX ☐TOP B ■NONE
HUB： ☐ASTM A-27 ☐JIS SCS13 ■NONE
GD^2： 5000 KG-M^2
TIP SPEED： 202 M/S

2.3 SHAFT, BEARING & BEARING HOUSING：
SHAFT MATERIAL： ☐AISI 4130 ■ASTM A-668 ☐304SS ☐
DIA AT HUB / BEARING： ∅ 560 mm / ∅ 180 mm
SHAFT SEAL： ■WITH ☐WITHOUT
TYPE ☐LABYRITHIAN ■GLAND
BEARING TYPE： ■SLEEVE ☐ANTI-FRICTION
BEARING LIFE： ■NO LIMITED ☐L10H OVER 60,000
BEARING HOUSING： ASTM A-536
LUBRICATION METHOD： ☐OIL BATH ☐GREASING ■FORCED LUB. (AS. 3.6)
LUBRICATE： ■VG 46 (CPC R46) ☐CPC R3
HOUSING TYPE： ■ZOLLERN STAND. WATER COOLING HORIZONTAL SPLIT TYPE OR EQ'L
☐SKF. STAND. AIR COOLING HORIZONTAL SPLIT TYPE OR EQ'L
COOLING WATER RATE / PRESS.： 2 T/H / 2 KG / CM^2 (PRESS. DIFF.)

FAN SPECIFICATION SHEET

2.4 AUXILIARY ITEMS：

CONN. COUPLING：	■WITH	□WITHOUT
TYPE	■DISC □STEELFLEX □_____	
COUPLING GUARD：	■WITH	□WITHOUT
	■NON-SPARKING TYPE	
BELT：	_____ (BANDO)	
PULLEY：	□WITH	■WITHOUT
ANCHOR BOLT W/FLAME：	■WITH	□WITHOUT
ACCESS DOOR：	■WITH	□WITHOUT
	■FAN HOUSING	
	■INLET BOX × 2 SET	
ISOLATOR：	□WITH	■WITHOUT
SAFELY COVER：	■WITH	□WITHOUT
COMMON BASE：	□WITH	■WITHOUT
LIFTING LUG：	■WITH	□WITHOUT
DRAIN W/VALVE：	■WITH	□WITHOUT
MOTOR BASE W/ADJ. BOLT：	■WITH	□WITHOUT
OIL LEVEL GAUGE：	■WITH	□WITHOUT
GREASE NIPPLE：	□WITH	■WITHOUT
THERMO COUPLE：	■WITH	□WITHOUT
COOLING WHEEL：	□WITH	■WITHOUT
VIBRATION DETECTOR：	■WITH	□WITHOUT

3. ACCESSORIES：

3.1 INLET

3.1.1	SILENCER：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.2	RAIN HOOD：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.3	SCREEN：	■WITH		□WITHOUT
	MAT'L：	□ASTM A-588	■304SS	□
3.1.4	FILTER：	□WITH		■WITHOUT
3.1.5	FLOW MEASUREMENT：	■VENTURI	□AIR FOIL	□NONE
	MAT'L：	■ASTM A-588 +	■304SS	□
3.1.6	INLET VOLUME CONTROL：	■WITH		□WITHOUT
		■VANE	□DAMPER	
3.1.7	DAMPER：	□WITH		■WITHOUT
	CONTROL METHOD：	□ON-OFF	□SERVO	□CHECK（AUTO.）
3.1.8	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	■304SS	□
3.1.9	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	■ASTM A-36

3.2 OUTLET

3.2.1	SILENCER：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.2.2	DAMPER：	□WITH		■WITHOUT
	CONTROL METHOD：	□ON-OFF	□SERVO	□CHECK（AUTO.）
3.2.3	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	□
3.2.4	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	□

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CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

3.3	ACTUATOR：	■WITH (MARKER： ROTORK)	□WITHOUT
3.3.1	FOR INLET：	■VANE □DAMPER	□WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	■ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	■ELECTRIC
		□IA. PRESSURE	□STEP MOTOR
		□ _____ Kg/cm²	■ 3 ∅ 460 V
	TORQUE：	■ 540 KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	■WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	■WITH	□WITHOUT
		□ _____ ∅ _____ V	
	SIGNAL INPUT：	■4～20 mA.	□ _____
	SENSITIVITY：	■0.5 %	□0.3 %
	RESPONSE TIME：	■ 1800 CYCLE/H	■ 0.5 Hz
	FAILURE：	■LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	■WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	■WITH	□WITHOUT
3.3.2	FOR OUTLET DAMPER：	□WITH	■WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	□ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	□ _____
	TORQUE：	□ _____ KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	□WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	□WITH	□WITHOUT
	SIGNAL INPUT：	□4～20 mA.	□ _____
	SENSITIVITY：	□0.5 %	□0.3 %
	RESPONSE TIME：	□ _____ CYCLE/H	□ _____ Hz
	FAILURE：	□LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	□WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	□WITH	□WITHOUT
3.4	V.S. HYDRAULIC COUPLING：	□WITH (MARKER： _____)	■WITHOUT
	MODEL：	□ _____	□ _____
	SPEED RANGE：	□ _____ ~ _____ RPM	□ _____
	WORK OIL：	□SUPPLY BY CCMC	□OUT SCOPE
	HEAT EXCHANGER & PIPING：	□WITH	□WITHOUT
	WORK OIL COOLER METHOD：	□WATER	□ _____
	COOLING WATER FLOW RATE：	□ _____ TONS/H	□ _____
3.4.1	ACTUATOR：	□WITH (MARKER： _____)	□WITHOUT
	DRIVEN METHOD：	□PNEUMATIC	□ELECTRIC
	POWER SOURCE：	□INSTRUMENT AIR	□ELECTRIC
		□IA. PRESSURE	□STEP MOTOR
		□ _____ Kg/cm²	□ _____ ∅ _____ V
	TORQUE：	□ _____ KG-M	□ _____ Lb-ft
	I/P CONVERTER W/POSITIONED：	□WITH	□WITHOUT
	MANUAL (HAND WHEEL)：	□WITH	□WITHOUT
		□ _____ ∅ _____ V	
	SIGNAL INPUT：	□4～20 mA.	□ _____
	SENSITIVITY：	□0.5 %	□0.3 %
	RESPONSE TIME：	□ _____ CYCLE/H	□ _____ Hz
	FAILURE：	□LOCK POSITION □OPEN	□CLOSED
	LIMIT SWITCH：	□WITH (CLOSE, OPEN)	□WITHOUT
	FEED BACK SIGNAL：	□WITH	□WITHOUT
3.5	TURNING MOTOR & OTHERS EQUIPMENT：		
	OVERRUNNING CLUTCH：	□WITH	■WITHOUT
	STANDBY MOTOR：	_____ KW × _____ P	
	REVOLUTION SPEED：	_____ RPM	
	FLUID TEMP. LIMIT：	_____ ℃	
	CONN. COUPLING：	□DISC	□STEELFLEX
	REQ'D：	_____ SET(S)	

FOR SCPC REFERENCE ONLY

FAN SPECIFICATION SHEET

3.6	FORCED LUB. SYSTEM FOR BEARING：	■WITH	□WITHOUT
	OIL PUMP W/MOTOR：	■DUAL	□SINGLE
	OPERATION：	□STAND BY	■ALL TIME
	HEAT EXCHANGER：	■DUAL	□SINGLE
	FILTER：	■DUAL	□SINGLE
	LOW OIL PRESS. SWITCH：		
	LOW PRESS.：	■ALARM	□
	LOW LOW OIL PRESS.：	■SHUT DOWN	□
	HIGH OIL TEMP. SWITCH：		
	HIGH TEMP.	■ALARM	□
	HIGH HIGH TEMP.：	■SHUT DOWN	□

4. TEST & INSPECTION：

(1)	DYNAMIC BALANCING：	ISO. 1940 G2.5		
(2)	MECH. RUNNING TEST：	□AT SHOP	■AT FIELD	
(3)	PERFORMANCE TEST：	□AT SHOP	■AT FIELD	（FOLLOW CNS）
(4)	BEARING HOUSING HYDRAULIC TEST：	■AT SHOP	□AT FIELD	
(5)	DIMENSION CHECK：	■AT SHOP	□AT FIELD	（CCMC STANDARD）

5. MISCELLANEOUS：

(1)	FAN NET WEIGHT：	40000 KGS (W/MOTOR)	
(2)	MAINTENANCE WEIGHT：	12000 KGS	
(3)	PAINTING：	□CCMC STD.	■CUSTOMER STD.
	SANDBLAST：	CLASS 2.5	
	PRIMER：	YES	
	FINISH：	YES	

6. GUARANTEE：

(1) PERIOD：1 YEAR FOR OPERATION OR 18 MOTHS FOR DELIVERY
(2) OPERATION CONDITION：IN NORMAL DESIGN CONDITION AND MAINTENANCE

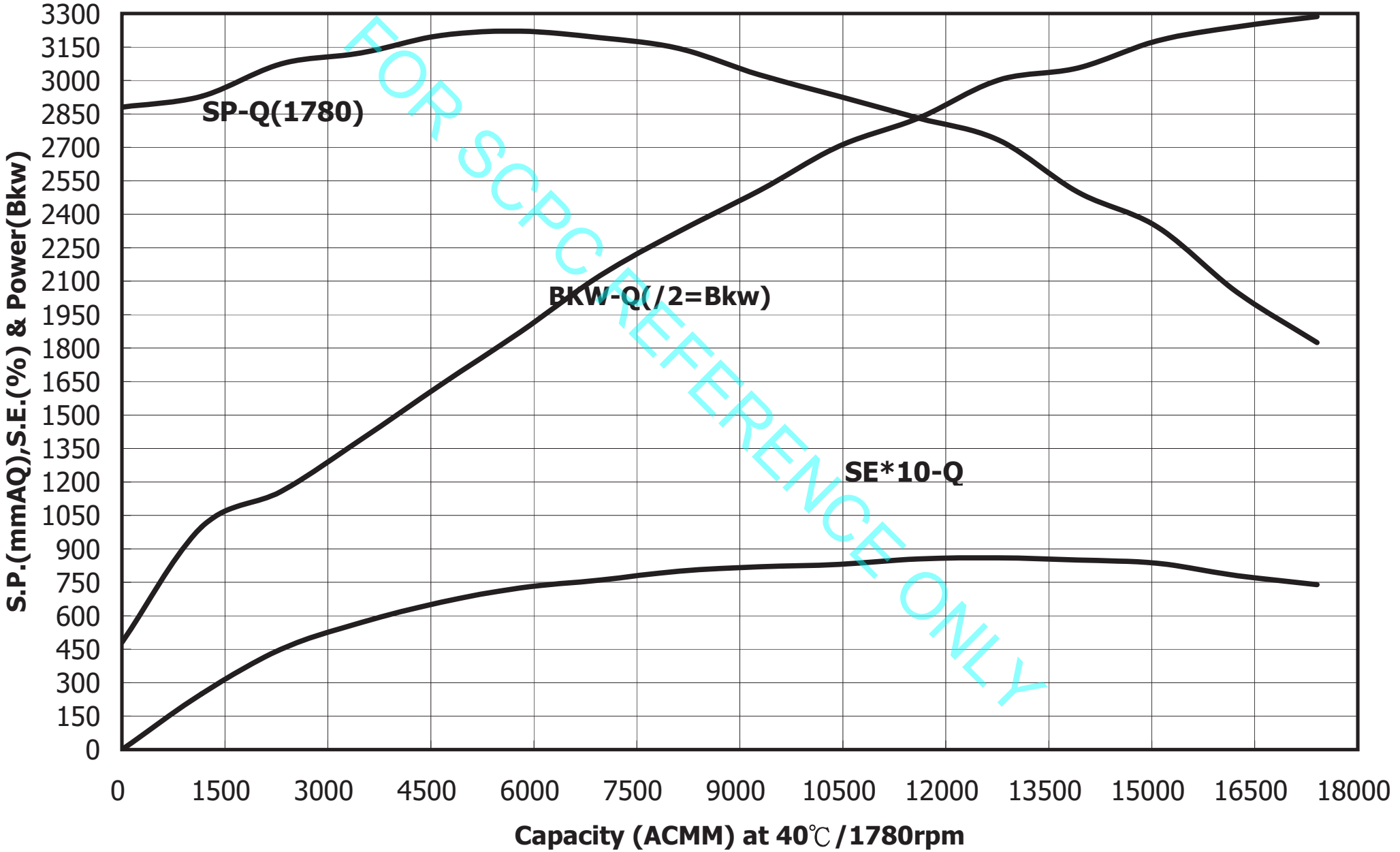
7. SPARE PART：

NO NECESSARY

8. MAKER：

FOR SCPC REFERENCE ONLY

D2118A TLT/1027 DWDI Curve
For:FHI(股)公司 Proj.:LMYP-1 PAF



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建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

LMYP-1 Limay, Philippine
I.D.Fan Specification Sheet
引風機規範書資料

本文編號 RF2-0582-DOC-6-0

						CERTIFIED	
						PROJ.	
						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2013/09/16	PROJECT NO.	PAGE
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	**LMYP-1**	1

建成機械股份有限公司
CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

☐PROPOSAL ■FOR CONSTRUCTION ☐FINAL

CUSTOMER ： FHI

INQUIRY NO. ：

PROJECT NO. ： LMYP-1

DATE ： Jan. / 14 / 2015

I. D. .FAN

FOR SCPC REFERENCE ONLY

1. DESIGN CONDITION：

CAPACITY： 17621 ACMM
STATIC PRESS. (SUC. / DISC. / DIFF.)： -825 / 33 / 858 mmAQ
OPERATION TEMP.： 150 ℃
FAN OPER. SPEED： 1140 RPM
TE. / SE.： 87 / 84 %
MAX. BREAK POWER： 2850 BKW @ DES. CONDITION
DRIVER： ■ELECTRIC MOTOR ☐TURBINE
3300 KW × 6 P × 6600 V × 3 ∅ × 60 Hz × GD^2 15000 KG-M^2
TEFC TYPE F CLASS SUPPLY BY CCMC

FAN 1st CRITICAL SPEED： 1600 RPM
VIBRATION (DIS.)： ISO 10816-3
SOUND PRESSURE LEVEL： 90 dBA AT 1 M (AFTER FAN IS COMPLETELY INSULATED)

2. FAN CONSTRUCTION：

FAN TYPE： ■AIR FOIL ☐RADIAL ☐BACKWARD ☐RADIAL TIPS CENTRIFUGAL HEAVY DUTY
FAN SIZE： D1902 / 10600 DWDI A3D2
FAN Q'TY： 2 SET(S)
ARRANGEMENT： A.M.C.A. ■A-3 ☐A-8 ☐A-1 ☐A-7 ☐A-4
TYPE OF DRIVE： ☐DIRECT BY MOTOR ☐BY BELT
■BY V.S. HYDRAULIC COUPLING

2.1 HOUSING：
CONSTRUCTION： SPLIT HOUSING ALL WELDED TYPE
SIDE PLATE： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
SCROLL： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
STIFF.： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐
INLET BOX： ■ASTM A-36 ☐ASTM A-588 ☐304SS ☐

2.2 WHEEL：
CONSTRUCTION： ■DOUBLE SUCTION ALL WELDED TYPE
☐SINGLE SUCTION ALL WELDED TYPE
EFFECTIVE DIA.： D 2695 mm
SIDE PLATE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
CENTRAL PLATE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
BLADE： ■ASTM A-514 ☐ASTM A-588 ☐304SS ☐HITEN
LINER： ■FIRMAX ☐TOP B ☐NONE
HUB： ☐ASTM A-27 ☐JIS SCS13 ■NONE
GD^2： 15000 KG-M^2
TIP SPEED： 161 M/S

2.3 SHAFT, BEARING & BEARING HOUSING：
SHAFT MATERIAL： ☐AISI 4130 ■ASTM A-668 ☐304SS ☐
DIA AT HUB / BEARING： ∅ 790 mm / ∅ 200 mm
SHAFT SEAL： ■WITH ☐WITHOUT
TYPE ☐LABYRITHIAN ■GLAND
BEARING TYPE： ■SLEEVE ☐ANTI-FRICTION
BEARING LIFE： ■NO LIMITED ☐L10H OVER 60,000
BEARING HOUSING： ASTM A-536
LUBRICATION METHOD： ■OIL BATH ☐GREASING ☐FORCED LUB. (AS. 3.6)
LUBRICATE： ■VG 46 (CPC R46) ☐CPC R3
HOUSING TYPE： ■DODGE STAND. WATER COOLING HORIZONTAL SPLIT TYPE OR EQ'L
☐SKF. STAND. AIR COOLING HORIZONTAL SPLIT TYPE OR EQ'L
COOLING WATER RATE / PRESS.： 4 T/H / 2 KG / CM^2 (PRESS. DIFF.)

FAN SPECIFICATION SHEET

2.4 AUXILIARY ITEMS：

CONN. COUPLING：	■WITH	□WITHOUT
TYPE	■DISC × 2 □STEELFLEX	□_____
COUPLING GUARD：	■WITH	□WITHOUT
	■NON-SPARKING TYPE	
BELT：	_____ (BANDO)	
PULLEY：	□WITH	■WITHOUT
ANCHOR BOLT W/FLAME：	■WITH	□WITHOUT
ACCESS DOOR：	■WITH	□WITHOUT
	■FAN HOUSING	
	■INLET BOX × ___2___ SET	
ISOLATOR：	□WITH	■WITHOUT
SAFELY COVER：	■WITH	□WITHOUT
COMMON BASE：	□WITH	■WITHOUT
LIFTING LUG：	■WITH	□WITHOUT
DRAIN W/VALVE：	■WITH	□WITHOUT
MOTOR BASE W/ADJ. BOLT：	■WITH	□WITHOUT
OIL LEVEL GAUGE：	■WITH	□WITHOUT
GREASE NIPPLE：	□WITH	■WITHOUT
THERMO COUPLE：	■WITH	□WITHOUT
COOLING WHEEL：	■WITH	□WITHOUT
VIBRATION DETECTOR：	■WITH	□WITHOUT

3. ACCESSORIES：

3.1 INLET

3.1.1	SILENCER：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.1.2	RAIN HOOD：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.1.3	SCREEN：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.1.4	FILTER：	□WITH		■WITHOUT
3.1.5	FLOW MEASUREMENT：	□VENTURI	■AIR FOIL	□NONE
	MAT'L：	□ASTM A-588	□304SS	□
3.1.6	INLET VOLUME CONTROL：	■WITH		□WITHOUT
		□VANE	■DAMPER	
3.1.7	DAMPER：	■WITH		□WITHOUT
	CONTROL METHOD：	□ON-OFF	■SERVO	□CHECK（AUTO.）
3.1.8	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	■304SS	□
3.1.9	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	■ASTM A-36

3.2 OUTLET

3.2.1	SILENCER：	□WITH		■WITHOUT
	MAT'L：	□ASTM A-588	□304SS	□
3.2.2	DAMPER：	□WITH		■WITHOUT
	CONTROL METHOD：	□ON-OFF	□SERVO	□CHECK（AUTO.）
3.2.3	EXPANSION JOINT：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	□
3.2.4	CONN. FLANGE：	■WITH		□WITHOUT
	MAT'L：	□CORTEN	□304SS	■ASTM A-36

FOR SCPC REFERENCE ONLY

建成機械股份有限公司
CHENG CHEN MACHINERY CO., LTD.

FAN SPECIFICATION SHEET

3.3 ACTUATOR：	■WITH (MARKER： STI)	□WITHOUT
3.3.1 FOR INLET：	□VANE ■DAMPER	□WITHOUT
DRIVEN METHOD：	□PNEUMATIC	■ELECTRIC
POWER SOURCE：	□INSTRUMENT AIR	■ELECTRIC
	□IA. PRESSURE	□STEP MOTOR
	□ _____ Kg/cm^2	□ _____ ∅ _____ V
TORQUE：	■ 420 KG-M	□ _____ Lb-ft
I/P CONVERTER W/POSITIONED：	■WITH	□WITHOUT
MANUAL (HAND WHEEL)：	■WITH	□WITHOUT
	□ _____ ∅ _____ V	
SIGNAL INPUT：	■4～20 mA.	□ _____
SENSITIVITY：	■0.5 %	□0.3 %
RESPONSE TIME：	■ Continuous ~~CYCLE/H~~	□ _____ Hz
FAILURE：	■LOCK POSITION □OPEN	□CLOSED
LIMIT SWITCH：	■WITH (CLOSE, OPEN)	□WITHOUT
FEED BACK SIGNAL：	■WITH	□WITHOUT
3.3.2 FOR OUTLET DAMPER：	□WITH	■WITHOUT
DRIVEN METHOD：	□PNEUMATIC	□ELECTRIC
POWER SOURCE：	□INSTRUMENT AIR	□ _____
TORQUE：	□ _____ KG-M	□ _____ Lb-ft
I/P CONVERTER W/POSITIONED：	□WITH	□WITHOUT
MANUAL (HAND WHEEL)：	□WITH	□WITHOUT
SIGNAL INPUT：	□4～20 mA.	□ _____
SENSITIVITY：	□0.5 %	□0.3 %
RESPONSE TIME：	□ _____ CYCLE/H	□ _____ Hz
FAILURE：	□LOCK POSITION □OPEN	□CLOSED
LIMIT SWITCH：	□WITH (CLOSE, OPEN)	□WITHOUT
FEED BACK SIGNAL：	□WITH	□WITHOUT
3.4 V.S. HYDRAULIC COUPLING：	■WITH (MARKER： VOITH)	□WITHOUT
MODEL：	■ 1150 SVNL22.32	□ _____
SPEED RANGE：	■ 350 ~ 1140 RPM	□ _____
WORK OIL：	■SUPPLY BY CCMC	□OUT SCOPE
HEAT EXCHANGER & PIPING：	■WITH	□WITHOUT
WORK OIL COOLER METHOD：	■WATER	□ _____
COOLING WATER FLOW RATE：	■ 53.9 TONS/H	□ _____
3.4.1 ACTUATOR：	■WITH (MARKER： ABB or EQ'L)	□WITHOUT
DRIVEN METHOD：	□PNEUMATIC	■ELECTRIC
POWER SOURCE：	□INSTRUMENT AIR	■ELECTRIC
	□IA. PRESSURE	■STEP MOTOR
	□ _____ Kg/cm^2	■ 1 ∅ 220 V
TORQUE：	■ **HOLD** KG-M	□ _____ Lb-ft
I/P CONVERTER W/POSITIONED：	■WITH	□WITHOUT
MANUAL (HAND WHEEL)：	■WITH	□WITHOUT
	□ _____ ∅ _____ V	
SIGNAL INPUT：	■4～20 mA.	□ _____
SENSITIVITY：	■0.5 %	□0.3 %
RESPONSE TIME：	■ Continuous CYCLE/H	□ _____ Hz
FAILURE：	■LOCK POSITION □OPEN	□CLOSED
LIMIT SWITCH：	■WITH (CLOSE, OPEN)	□WITHOUT
FEED BACK SIGNAL：	■WITH	□WITHOUT
3.5 TURNING MOTOR & OTHERS EQUIPMENT：		
OVERRUNNING CLUTCH：	□WITH	■WITHOUT
STANDBY MOTOR：	_____ KW × _____ P	
REVOLUTION SPEED：	_____ RPM	
FLUID TEMP. LIMIT：	_____ ℃	
CONN. COUPLING：	□DISC	□STEELFLEX
REQ'D：	_____ SET(S)	

FOR SCPC REFERENCE ONLY

FAN SPECIFICATION SHEET

3.6	FORCED LUB. SYSTEM FOR BEARING：	☐WITH	■WITHOUT
	OIL PUMP W/MOTOR：	☐DUAL	☐SINGLE
	OPERATION：	☐STAND BY	☐ALL TIME
	HEAT EXCHANGER：	☐DUAL	☐SINGLE
	FILTER：	☐DUAL	☐SINGLE
	LOW OIL PRESS. SWITCH：		
	LOW PRESS.：	☐ALARM	☐
	LOW LOW OIL PRESS.：	☐SHUT DOWN	☐
	HIGH OIL TEMP. SWITCH：		
	HIGH TEMP.	☐ALARM	☐
	HIGH HIGH TEMP.：	☐SHUT DOWN	☐

4. TEST & INSPECTION：

(1)	DYNAMIC BALANCING：	ISO. 1940 G2.5		
(2)	MECH. RUNNING TEST：	☐AT SHOP	■AT FIELD	
(3)	PERFORMANCE TEST：	☐AT SHOP	■AT FIELD	（FOLLOW CNS）
(4)	BEARING HOUSING HYDRAULIC TEST：	■AT SHOP	☐AT FIELD	
(5)	DIMENSION CHECK：	■AT SHOP	☐AT FIELD	（CCMC STANDARD）

5. MISCELLANEOUS：

(1)	FAN NET WEIGHT：	60000 KGS (W/MOTOR)	
(2)	MAINTENANCE WEIGHT：	25000 KGS	
(3)	PAINTING：	☐CCMC STD.	■CUSTOMER STD.
	SANDBLAST：	CLASS 2.5	
	PRIMER：	YES	
	FINISH：	YES	

6. GUARANTEE：

(1) PERIOD：1 YEAR FOR OPERATION OR 18 MOTHS FOR DELIVERY
(2) OPERATION CONDITION：IN NORMAL DESIGN CONDITION AND MAINTENANCE

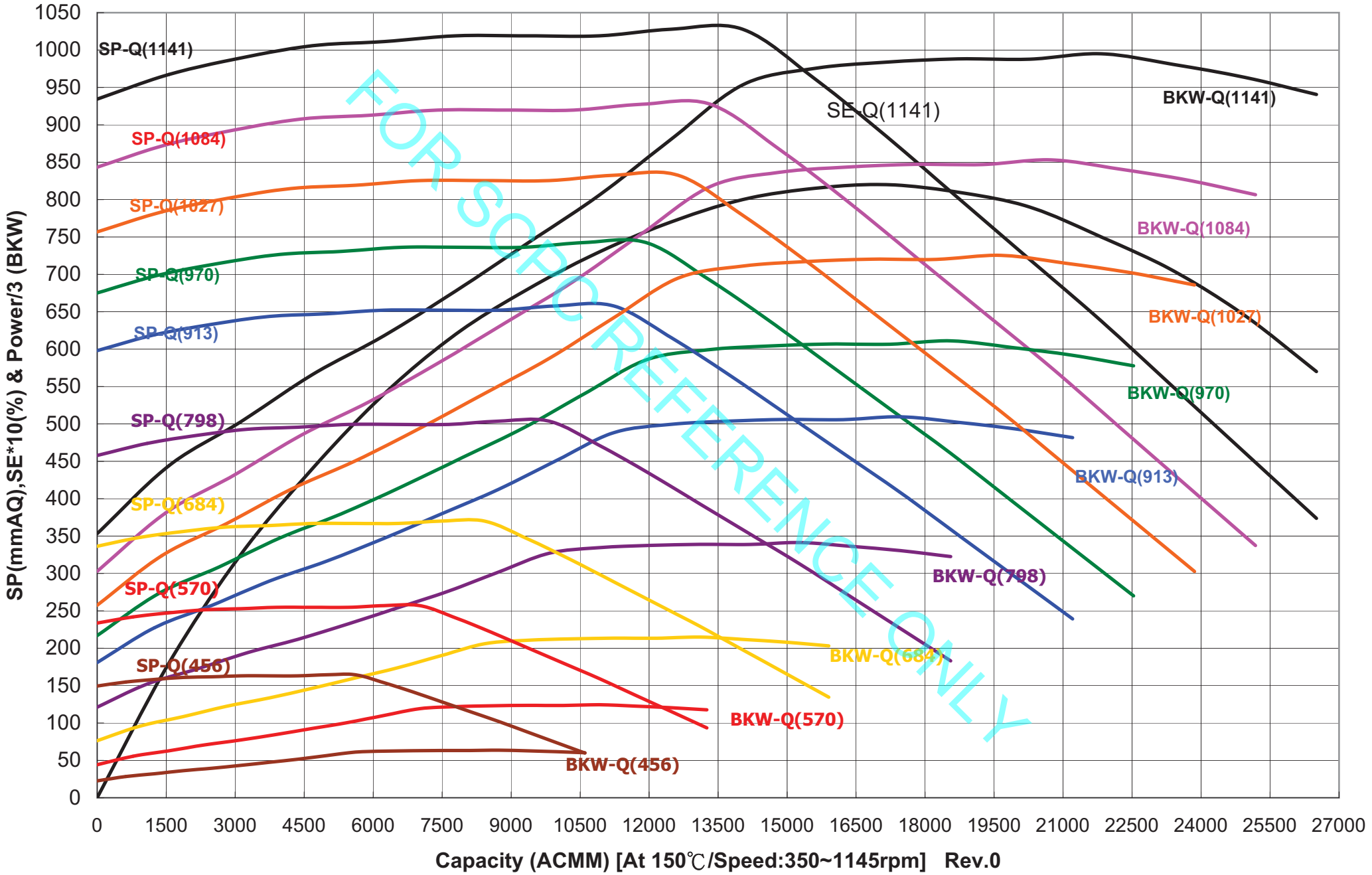
7. SPARE PART：

NO NECESSARY

8. MAKER：

FOR SCPC REFERENCE ONLY

D1902/10600 DWDI VS Performance Curve(W/B)
For:FHI Proj:LMYP-1 C.F.B. I.D.Fan



1. DRAWING

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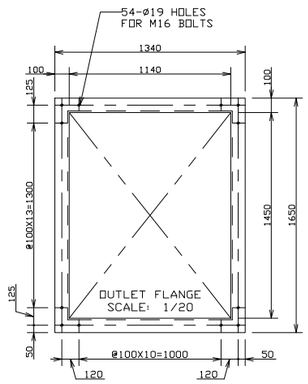
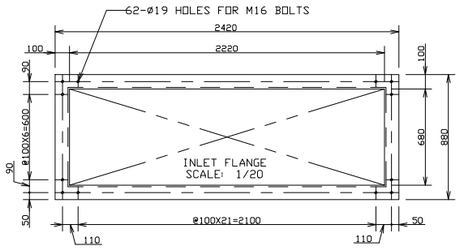
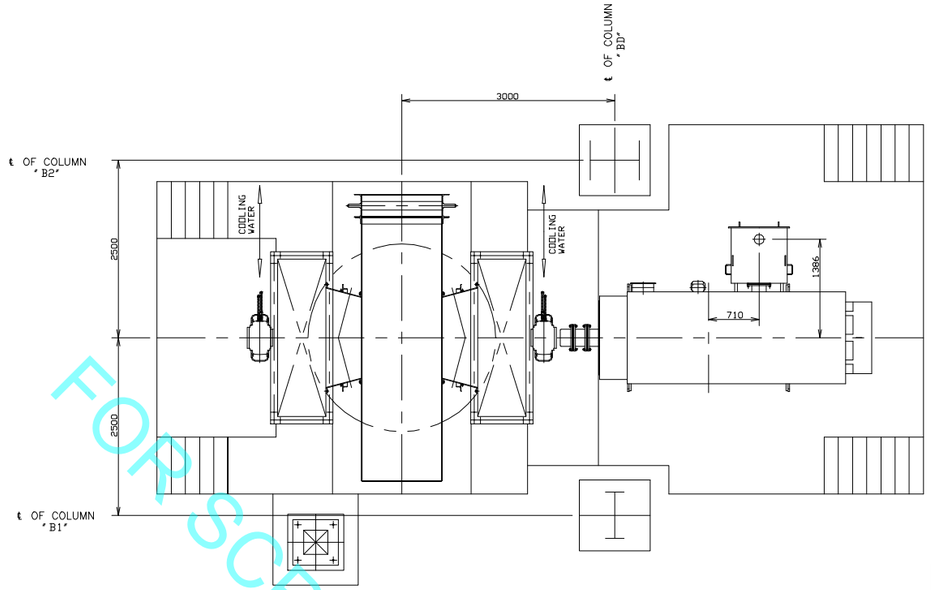
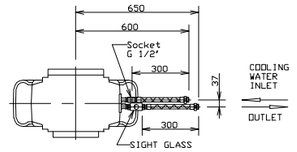
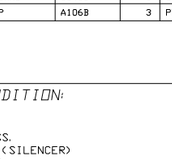


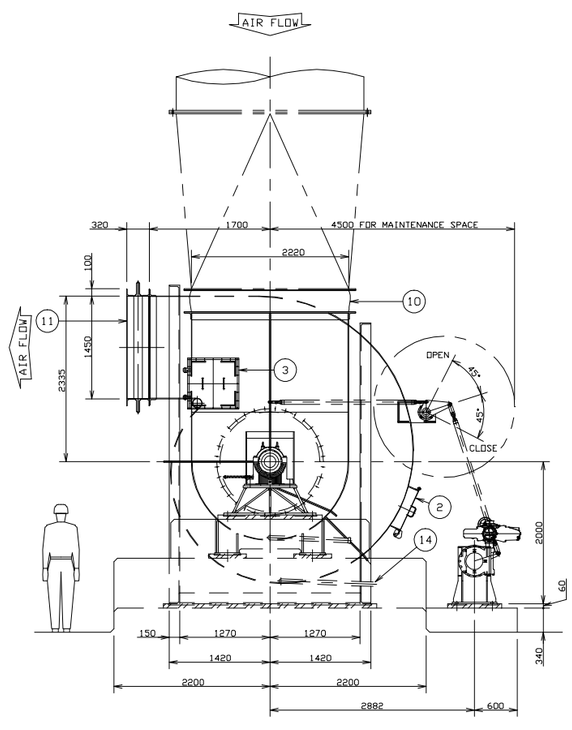


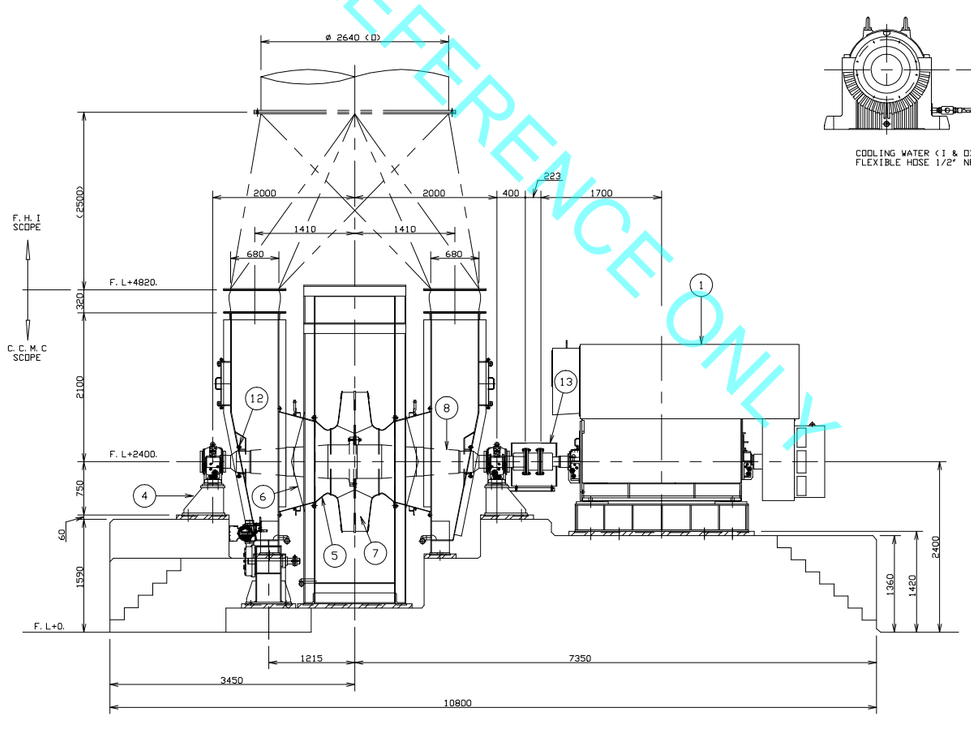






WHEEL & CONE GAP
SCALE: 1/2





ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	MOTOR	TEFC	1	SEE NOTE
2	CLEAN-OUT DOOR	SS400	1	B-RF2-0555-12
3	CLEAN-OUT DOOR	SS400	2	B-RF2-0555-13
4	BEARING PEDESTAL	SS400	2	C-RF2-0555-15
5	INLET CONE	SS400	2	C-RF2-0555-41
6	I.V.C ASS'Y	SS400	2	A-RF2-0555-42
7	WHEEL ASS'Y	ASTM-A514	1	B-RF2-0555-51
8	SHAFT	ASTM-A668	1	A-RF2-0552-61
9	MOTOR BASE	SS400	1	B-RF2-0555-71
10	EXPANSION JOINTER (I)	CANVAS & SUS304	2	B-RF2-0555-73
11	EXPANSION JOINTER (O)	SUS304	1	B-RF2-0555-74
12	SHAFT SEAL	#V6500	2	C-RF2-0555-91
13	COUPLING GUARD	SS400	1	C-RF2-0555-92
14	DRAIN & CAP	A1068	3	PT 2"

BASIC CONDITION:

ITEM/	DESIGN
CAPACITY	4222 ACMM
STATIC PRESS.	2075 mmAQ
SUCTION (SILENCER)	-153 mmAQ
DISCHARGE	1922 mmAQ
OPERATION TEMP.	40 deg C
STATIC EFF.	84 %
REVOLUTION SPEED	1780 RPM
BRAKE POWER	1698 BKW
NOISE(AT 1 M)	85 DBA

NOTE:

1). MOTOR WILL BE TEFC, OUTDOOR SERVIC, 1850KWX4PX60HZX6600V GD^2=3000KG-M^2 SUPPLY BY C.C.M.C.
2). COUPLING WILL BE DISC COUPLING TYPE : TIEN YI T61-165PF06.
3). BEARING WILL BE ZOLLERN SLEEVE BEARING. ZRWLK 18-160 FOR FIXED SIDE BEARING ZRWLQ 18-160 FOR FREE SIDE BEARING.
4). BEARING HOUSING IS ZOLLERN STANDARD PARTS WITH WATER COOLING, HORIZONTAL SPLIT, OIL LUB. TYPE.
5). LUBRICATION OIL WILL BE ISO VG32 FOR BEARING.
6). COOLING WATER RATE FOR EACH BEARING HOUSING IS 3 T/H. TEMP. RISE 2°C. PRESSURE DIFF. : 2Kg/CM^2. INLET PRESSURE : 4Kg/CM^2.
7). I.V.C TORQUE WILL BE 320 KG-M. ACTUATOR SUPPY BY C.C.M.C.
8). INSULATION WORK FOR FAN HOUSING WILL BE 300m/m. THICKNESS BY CUSTOMER. BUT NUT, STUD BOLT ARE C.C.M.C SCOPE.
9). NOISE LEVEL IS 785 DBA AT 1M AFTER FAN IS COMPLETE INSULATION.
10). BEARING HOUS'G WILL INSTALLATION TEMP. RTD.
11). VIBRATION GAUGE. METRIX ST5491E-022-020-00.

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
D	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
C	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
B	CHANGE MOTOR DIMENSION	陳玉樹	陳玉樹	傅泰錫

SCALE: 1/40

APPD	CKD
傅泰錫 08/19/'13	陳玉樹 08/16/'13
DESN	DWN
陳玉樹 08/15/'13	陳玉樹 08/15/'13
PROJECT NO.	SHEET NO.
RF2-0580	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX(035)981149

NAME: BCT2118/925 DWDI CWTH 107% A/3
S.A. FAN ASS'Y

DWG NO. A-RF2-0580-01 | UNIT | MM | REV | 0

REG. NO.	:
P.T.R. NO.	:
DATE	:
PLACE	:
T.I.N.	:

ENGINEER

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

COPY

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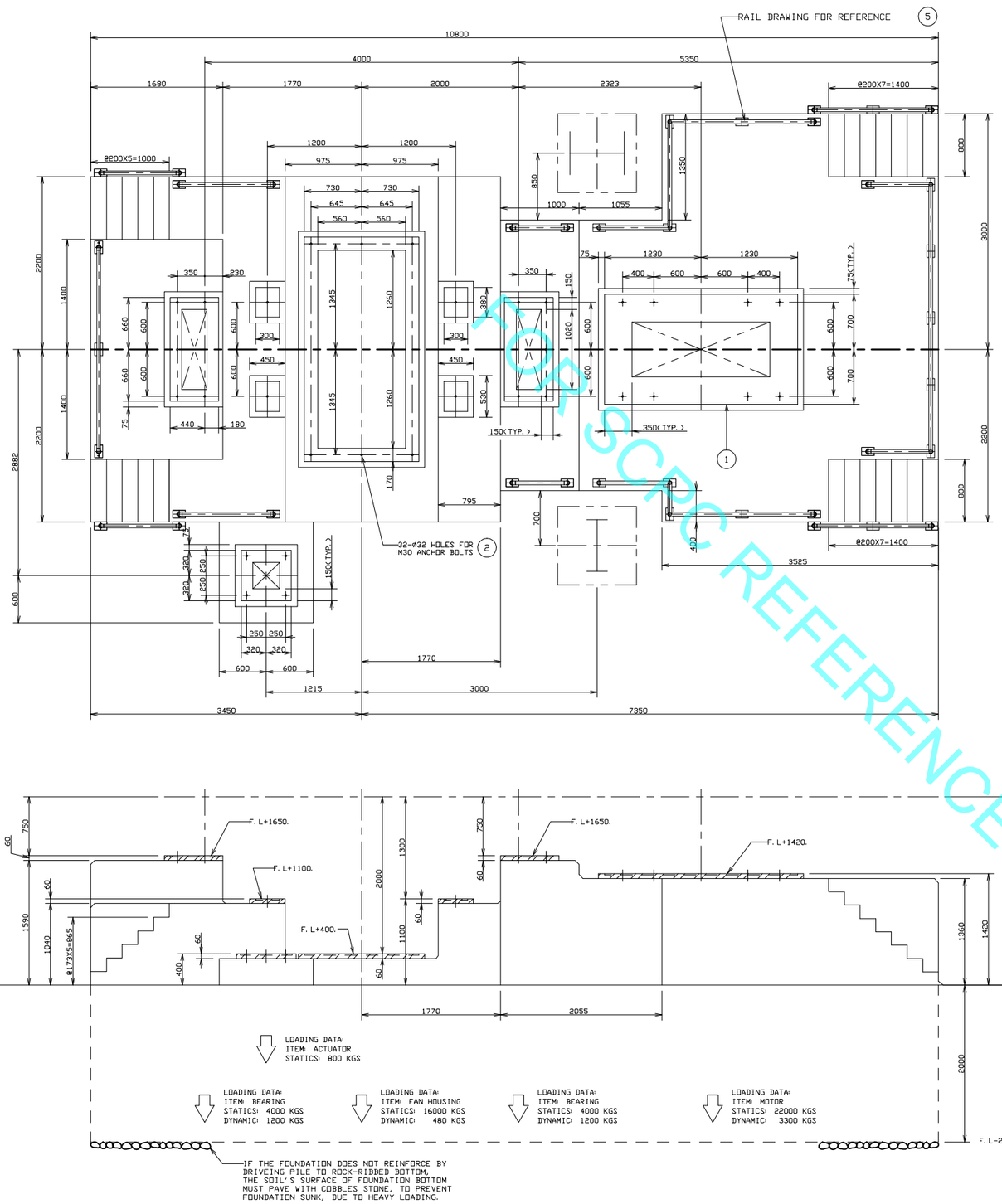
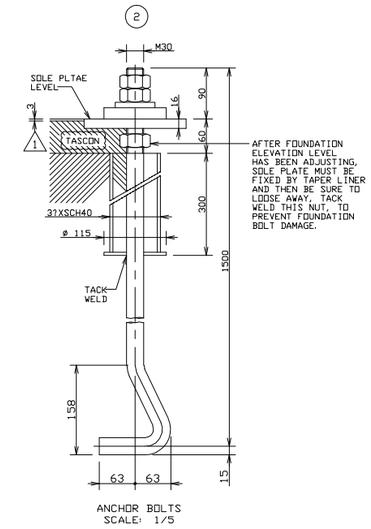
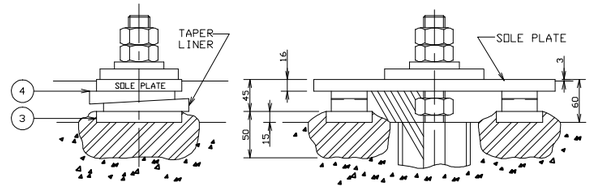
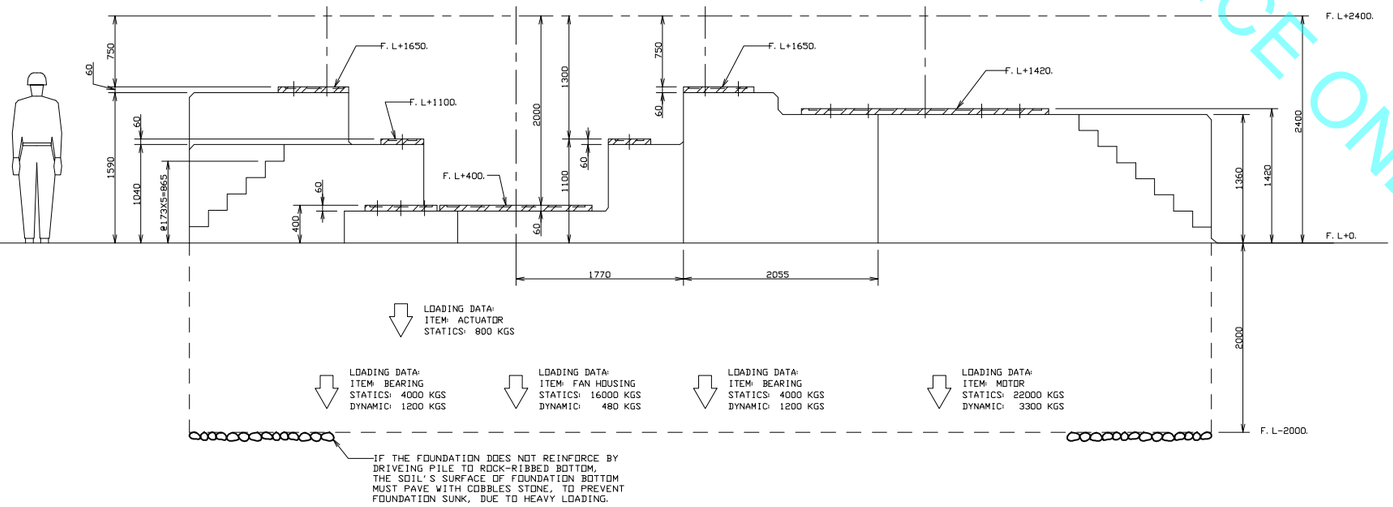
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	SOLE PLATE	SS400	9	16t SEE DWG
2	ANCHOR BOLTS	SS400	32	M30 SEE DWG
3	PLATE	SS400	80	10X75X150
4	TAPER LINER (1/50)	SS400	160	19X65X150
5	RAIL	SUS304	1	BY FHI SCOPE





ANCHOR BOLTS
SCALE: 1/5





FOR SCPC REFERENCE ONLY

COPY

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
1	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
B	ADD DIMENSION	陳玉樹	陳玉樹	傅泰錫

SCALE: 1/30

APPD	CKD
傅泰錫	陳玉樹
08/19/'13	08/16/'13
DESN	DWN
陳玉樹	陳玉樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0580	NO. OF

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:(035)981149

NAME: BCT2118/925 DWDI CWTH 107% A/3
S. A. F FOUNDATION DETAIL

DWG NO. A-RF2-0580-02 UNIT MM REV 1

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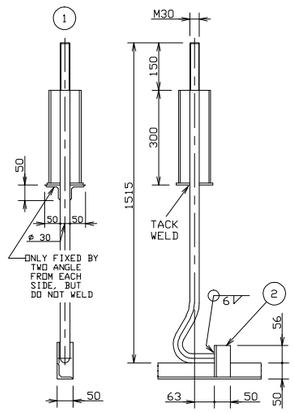
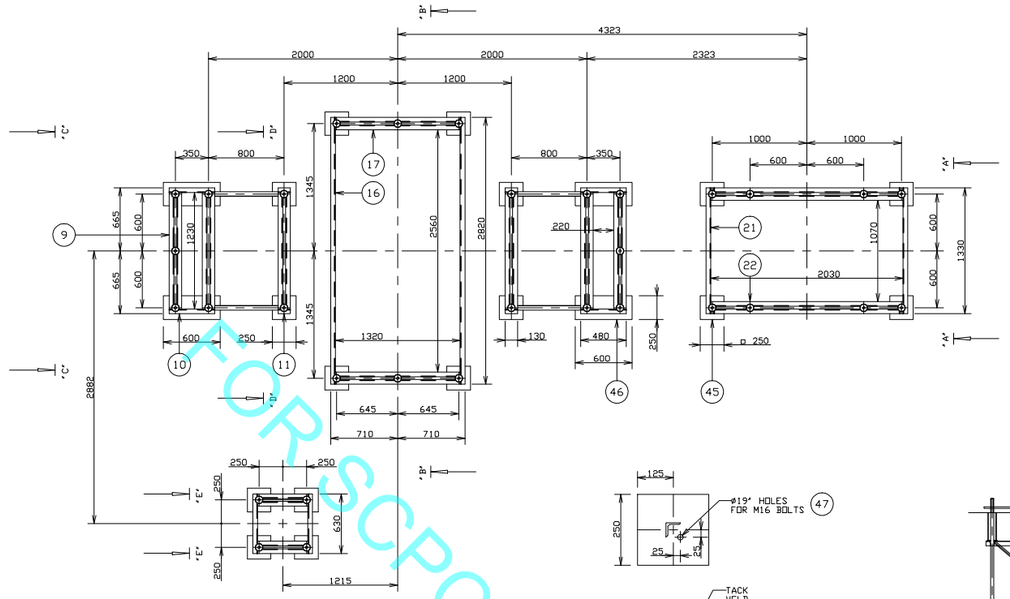
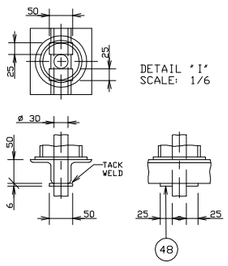
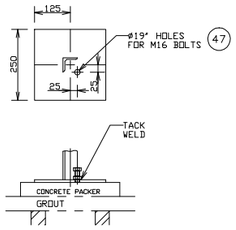


SCALE: 1/9

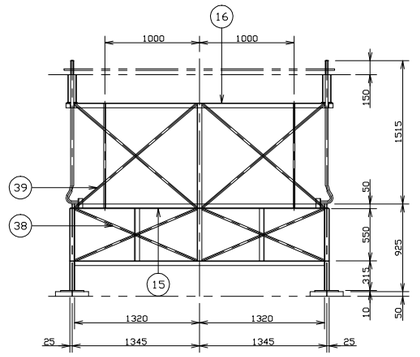




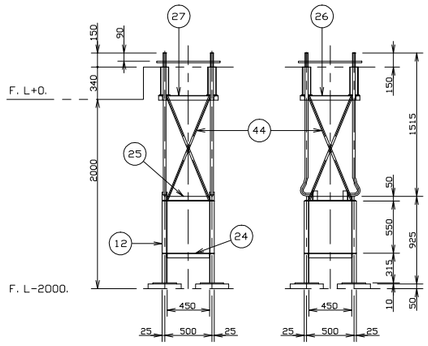


CUT OFF ALL PILE END TO SAME ELEVATION LEVEL, THEN PAVE WITH THIN GROUT

DETAIL 'H'
SCALE: 1/10



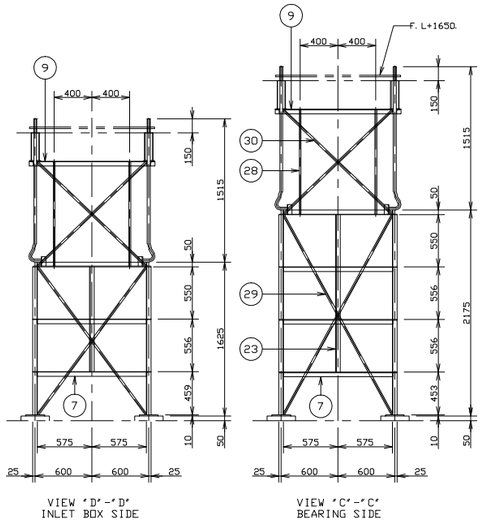
VIEW 'B'-'B'
FAN HOUSING SIDE



VIEW 'E'-'E'
FOR ACTUATOR

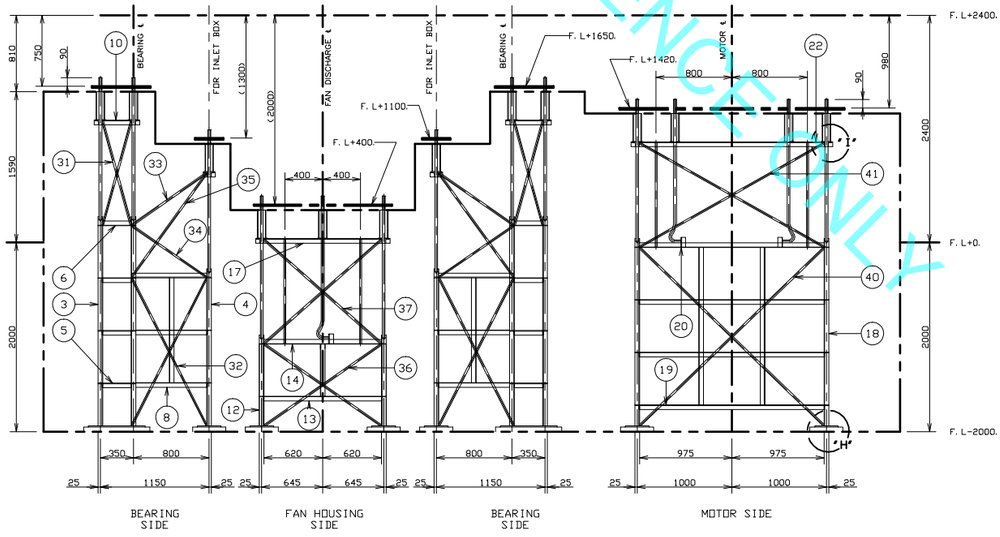
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	ANCHOR BOLTS	SS400	32	M30 SEE DWG
2	ANGLE	SS400	32	L 50X50X6X100L
3	ANGLE	SS400	8	L 50X50X6X2115L
4	ANGLE	SS400	4	L 50X50X6X1565L
5	ANGLE	SS400	12	L 50X50X6X300L
6	ANGLE	SS400	4	L 50X50X6X400L
7	ANGLE	SS400	30	L 50X50X6X1150L
8	ANGLE	SS400	12	L 50X50X6X794L
9	ANGLE	SS400	12	L 50X50X6X1230L
10	ANGLE	SS400	4	L 50X50X6X480L
11	ANGLE	SS400	4	L 50X50X6X130L
12	ANGLE	SS400	8	L 50X50X6X865L
13	ANGLE	SS400	2	L 50X50X6X1240L
14	ANGLE	SS400	2	L 50X50X6X1340L
15	ANGLE	SS400	4	L 50X50X6X2640L
16	ANGLE	SS400	2	L 50X50X6X2820L
17	ANGLE	SS400	4	L 50X50X6X1320L
18	ANGLE	SS400	4	L 50X50X6X1885L
19	ANGLE	SS400	6	L 50X50X6X1950L
20	ANGLE	SS400	2	L 50X50X6X2150L
21	ANGLE	SS400	2	L 50X50X6X1330L
22	ANGLE	SS400	4	L 50X50X6X2030L
23	ANGLE	SS400	50	L 50X50X6X550L
24	ANGLE	SS400	6	L 50X50X6X450L
25	ANGLE	SS400	2	L 50X50X6X550L
26	ANGLE	SS400	2	L 50X50X6X630L
27	ANGLE	SS400	4	L 50X50X6X530L
28	REINFORCING STEEL	SS400	26	D 3/4"X1103L
29	REINFORCING STEEL	SS400	8	D 3/4"X2410L
30	REINFORCING STEEL	SS400	12	D 3/4"X1600L
31	REINFORCING STEEL	SS400	8	D 3/4"X1145L
32	REINFORCING STEEL	SS400	8	D 3/4"X1750L
33	REINFORCING STEEL	SS400	4	D 3/4"X970L
34	REINFORCING STEEL	SS400	4	D 3/4"X965L
35	REINFORCING STEEL	SS400	4	D 3/4"X1360L
36	REINFORCING STEEL	SS400	4	D 3/4"X1515L
37	REINFORCING STEEL	SS400	4	D 3/4"X1660L
38	REINFORCING STEEL	SS400	8	D 3/4"X1410L
39	REINFORCING STEEL	SS400	8	D 3/4"X1705L
40	REINFORCING STEEL	SS400	4	D 3/4"X2715L
41	REINFORCING STEEL	SS400	4	D 3/4"X2245L
42	REINFORCING STEEL	SS400	4	D 3/4"X2210L
43	REINFORCING STEEL	SS400	4	D 3/4"X1600L
44	REINFORCING STEEL	SS400	8	D 3/4"X1190L
45	PLATE	SS400	12	10tX250X250
46	PLATE	SS400	4	10tX250X600
47	HEX-BOLT,NUT	SS400	24	M16X35L
48	F. B	SS400	64	6tX25X50
49	SOLE PLATE	SS400	8	A-RF2-0555-02
50				

Total Weight : 2400 Kgs

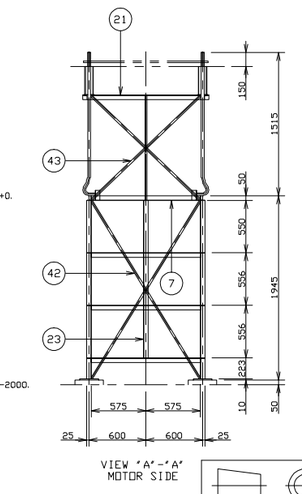


VIEW 'D'-'D'
INLET BOX SIDE

VIEW 'C'-'C'
BEARING SIDE



BEARING SIDE | FAN HOUSING SIDE | BEARING SIDE | MOTOR SIDE



VIEW 'A'-'A'
MOTOR SIDE

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳孟樹	陳孟樹	傅泰錫

SCALE: 1/30

APPD	CKD
傅泰錫 08/19/'13	陳孟樹 08/16/'13
DESN	DWN
陳孟樹 08/15/'13	陳孟樹 08/15/'13
PROJECT NO. RF2-0580	SHEET NO. NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX(035)981149

NAME: BCT2118/925 DWDI CWTH 107% A/3
S. A. F ANCHOR FRAME DETAIL

DWG NO. A-RF2-0580-03 | UNIT MM | REV 0

ENGINEER	
REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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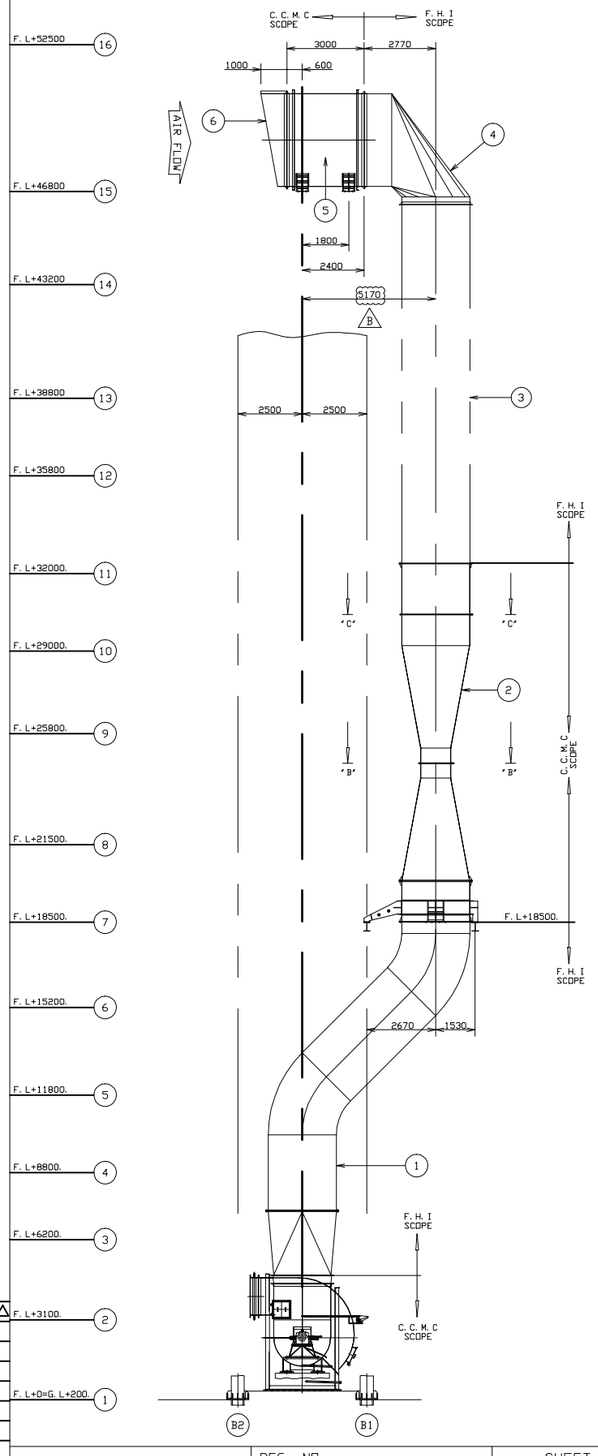
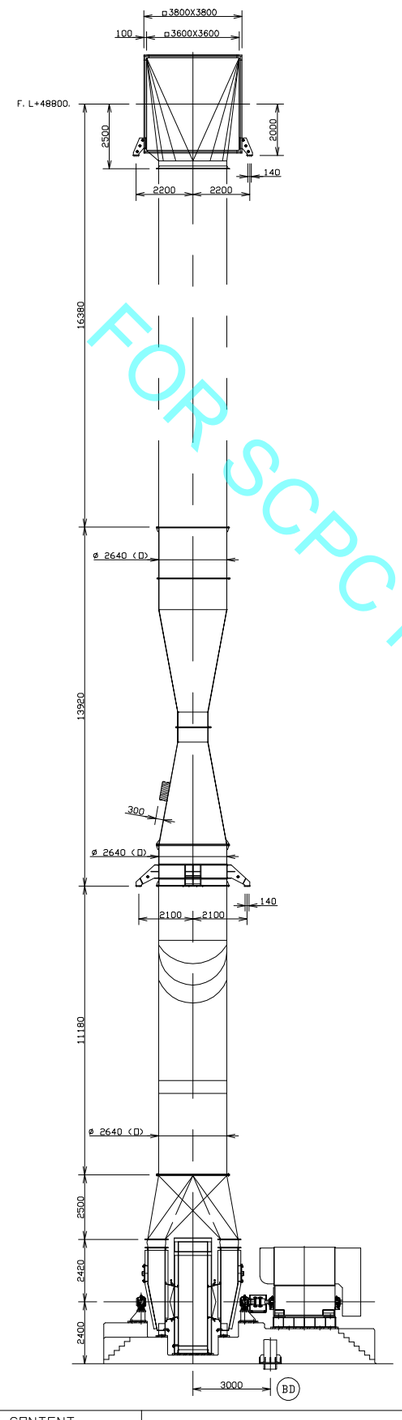
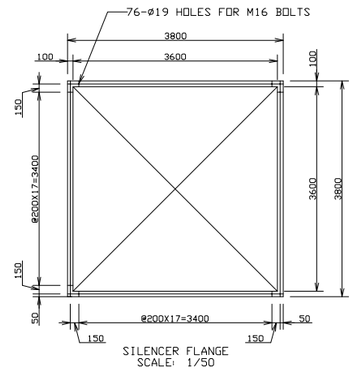
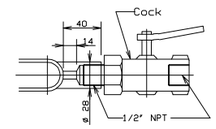






SILENCER FLANGE
SCALE: 1/50



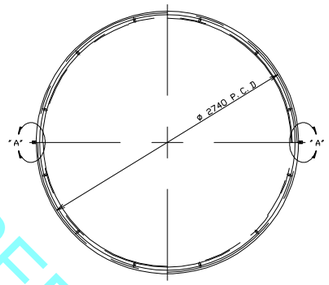
DETAIL 'A'
SCALE: 1/3

ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	DUCT	SS400	1	BY FHI SCOPE
2	VENTURI	SS400	1	
3	DUCT	SS400	1	BY FHI SCOPE
4	ELBOW	SS400	1	BY FHI SCOPE
5	SILENCER	SS400	1	
6	INLET SCREEN	SS400	1	

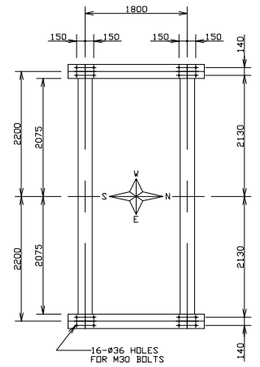
NOTE:

1). INSULATION WORK FOR VENTURI WILL BE 300m/m THICKNESS BY CUSTOMER. BUT NUT, STUD BOLT ARE C. C. M. C SCOPE.

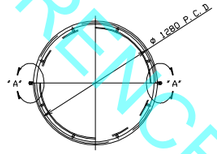
2). ESTIMATED WEIGHT
a). VENTURI : 13000 Kg
b). SILENCER : 13000 Kg



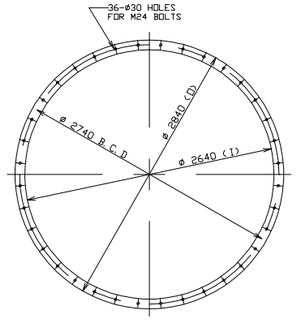
SECTION 'C'-'C'
SCALE: 1/30



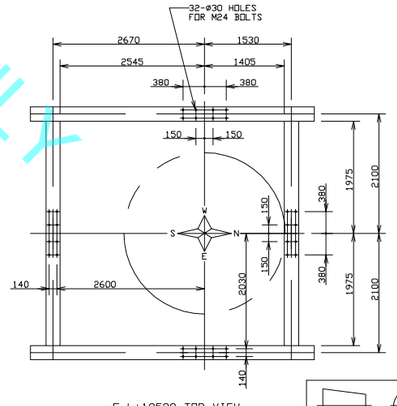
F. L+46800 TOP VIEW
SILENCER SUPPORT
SCALE: 1/50



SECTION 'B'-'B'
SCALE: 1/30



VENTURI & ITEM 4
FLANGE DETAIL
SCALE : 1/30



F. L+18500 TOP VIEW
VENTURI SUPPORT
SCALE: 1/50

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COPY

ENGINEER	REG. NO. :	SHEET CONTENT	
	P. T. R. NO. :		
	DATE :		
	PLACE :		
	T. I. N. :		

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳孟樹	陳孟樹	傅泰錫
B	TO COMPLY WITH CUSTOMER'S REQUIRED	陳孟樹	陳孟樹	傅泰錫

SCALE: 1/110

APPD	CKD
傅泰錫	陳孟樹
08/19/'13	08/16/'13
DESN	DWN
陳孟樹	陳孟樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0580	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:(035)981149

NAME: BCT2118/925 DWDI CWTH 107% A/3
S. A. FAN CONNECTING ASS'Y

DWG NO. A-RF2-0580-05 UNIT MM REV 0

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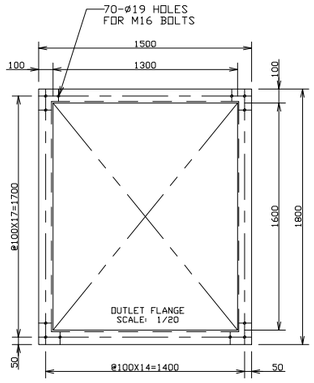
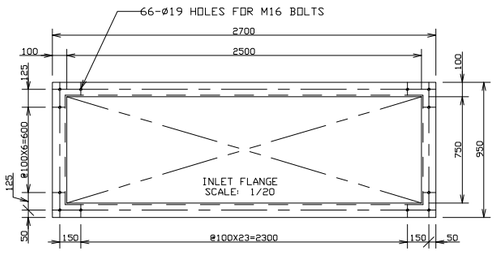
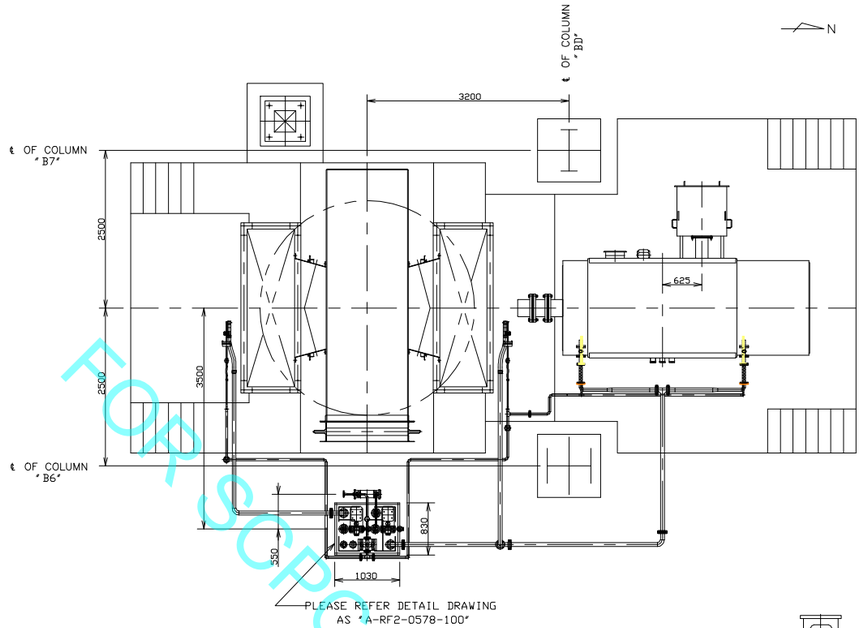






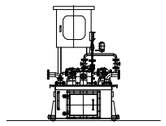
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	MOTOR	TEFC	1	SEE NOTE
2	CLEAN-OUT DOOR	SS400	1	B-RF2-0556-12
3	CLEAN-OUT DOOR	SS400	2	B-RF2-0556-13
4	BEARING PEDESTAL	SS400	2	C-RF2-0556-15
5	INLET CONE	SS400	2	C-RF2-0556-41
6	I.V.C ASS'Y	SS400	2	A-RF2-0556-42
7	WHEEL ASS'Y	ASTM-A514	1	B-RF2-0556-51
8	SHAFT	ASTM-A668	1	A-RF2-0556-61
9	MOTOR BASE	SS400	1	B-RF2-0556-71
10	EXPANSION JOINTER (I)	CANVAS & SUS304	2	B-RF2-0556-73
11	EXPANSION JOINTER (D)	SUS304	1	B-RF2-0556-74
12	SHAFT SEAL	#V6500	2	C-RF2-0556-91
13	COUPLING GUARD	SS400	1	C-RF2-0556-92
14	DRAIN & CAP	A106B	3	PT 2"

BASIC CONDITION:

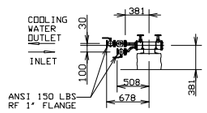
ITEM/	DESIGN
CAPACITY	6907 ACMM
STATIC PRESS.	2481 mmAQ
SUCTION (SILENCER)	-153 mmAQ
DISCHARGE	2328 mmAQ
OPERATION TEMP.	40 deg C
STATIC EFF.	83.7 %
REVOLUTION SPEED	1780 RPM
BRAKE POWER	3341 BKW
NOISE(AT 1 M)	85 DBA

NOTE:

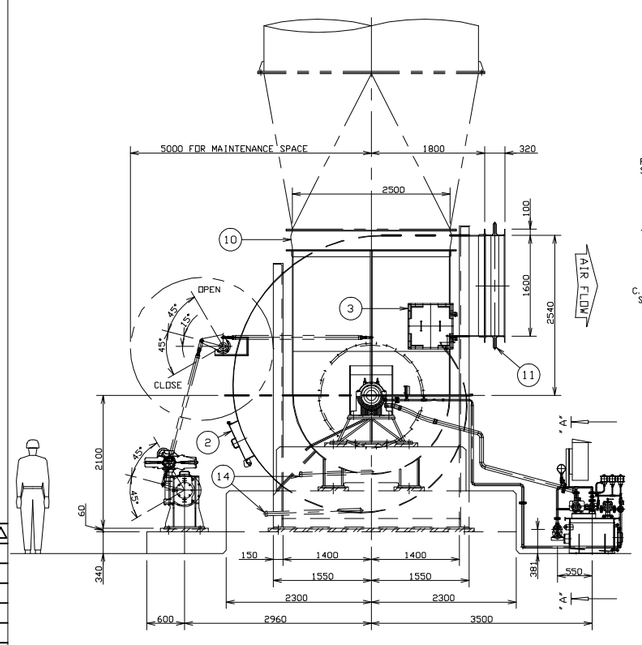
1). MOTOR WILL BE TEFC, OUTDOOR SERVIC.
   370OKWX4PX60HZX6600V GD^2=5000KG-M^2
   SUPPLY BY C.C.M.C.
2). COUPLING WILL BE DISC COUPLING
   TYPE : TIEN YI T81-187PF08.
3). BEARING WILL BE ZOLLERN SLEEVE BEARING.
   ZRZLK 18-180 FOR FIXED SIDE BEARING
   ZRZLQ 18-180 FOR FREE SIDE BEARING.
4). LUBRICATION BY OIL CIRCULATION
   WITH EXTERNAL OIL COOLING. LING.
5). LUBRICATION OIL WILL BE ISO VG46 FOR BEARING.
6). COOLING WATER RATE FOR OIL CIRCULATION SYSTEM
   IS 2 T/H. TEMP. RISE 2°C.
   PRESSURE DIFF. : 2Kg/CM^2.
   INLET PRESSURE : 4Kg/CM^2.
7). I.V.C TORQUE WILL BE 540 KG-M.
   ACTUATOR SUPPY BY C.C.M.C.
8). INSULATION WORK FOR FAN HOUSING WILL
   BE 300m/m, THICKNESS BY CUSTOMER. BUT
   NUT, STUD BOLT ARE C.C.M.C SCOPE.
9). NOISE LEVEL IS ≤85 DBA AT 1M AFTER FAN
   IS COMPLETE INSULATION.
10). BEARING HOUS'G WILL INSTALLATION TEMP. RTD.
11). VIBRATION GAUGE. METRIX ST5491E-022-020-00.

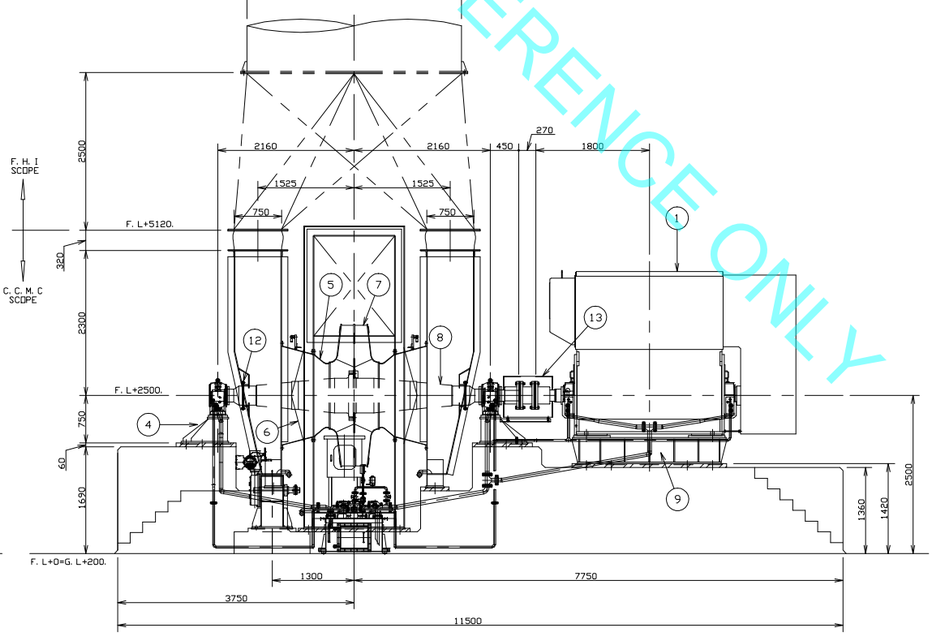


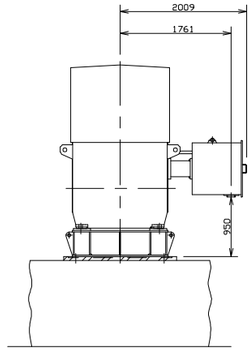
CIRCULATING OIL SYSTEMS

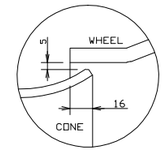


VIEW "A"-"A"









WHEEL & CONE GAP
SCALE: 1/2

COPY

O	FOR CONSTRUCTION	陳孟樹	陳孟樹	傅泰錫
E	ADD OIL PIPING	陳孟樹	陳孟樹	傅泰錫
D	TO COMPLY WITH CUSTOMER'S REQUIRED	陳孟樹	陳孟樹	傅泰錫
C	CHANGE MOTOR DIMENSION	陳孟樹	陳孟樹	傅泰錫
B	ADD DIMENSION	陳孟樹	陳孟樹	傅泰錫
REV.	DESCRIPTION	REVISED	CHECKED	APPROVED

SCALE: 1/40

APPD	CKD
傅泰錫	陳孟樹
08/19/'13	08/16/'13
DESN	DWN
陳孟樹	陳孟樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0581	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:(035)981149

NAME: BCT2118/1027 DWDI CCWTH 110% A/3
P.A. FAN ASS'Y

DWG NO. A-RF2-0581-01 | UNIT | MM | REV | 0

ENGINEER

REG. NO.	:
P.T.R. NO.	:
DATE	:
PLACE	:
T.I.N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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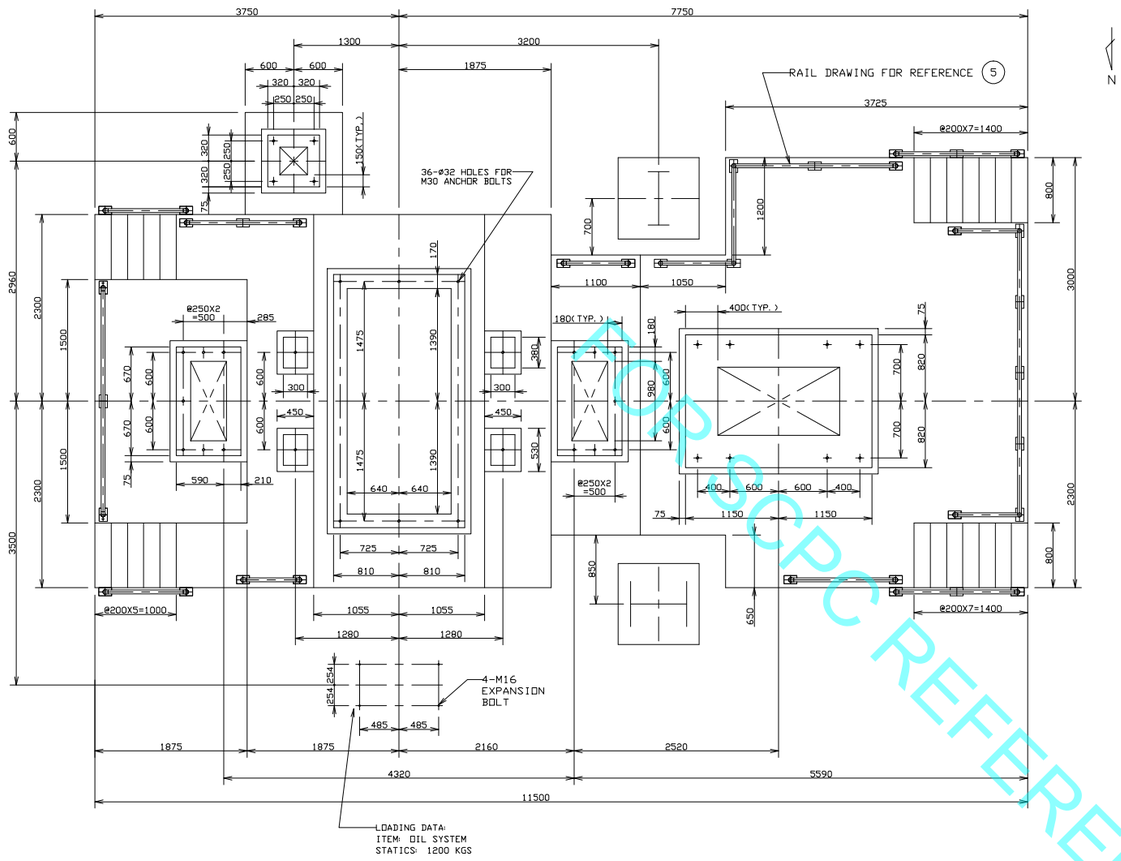
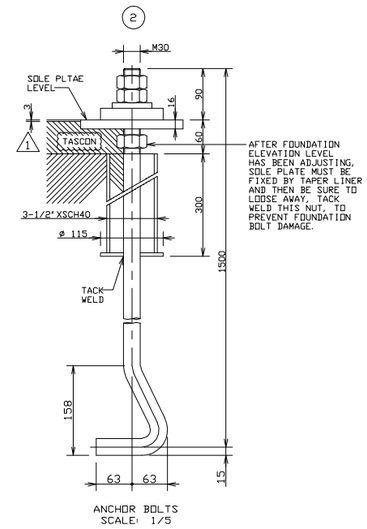
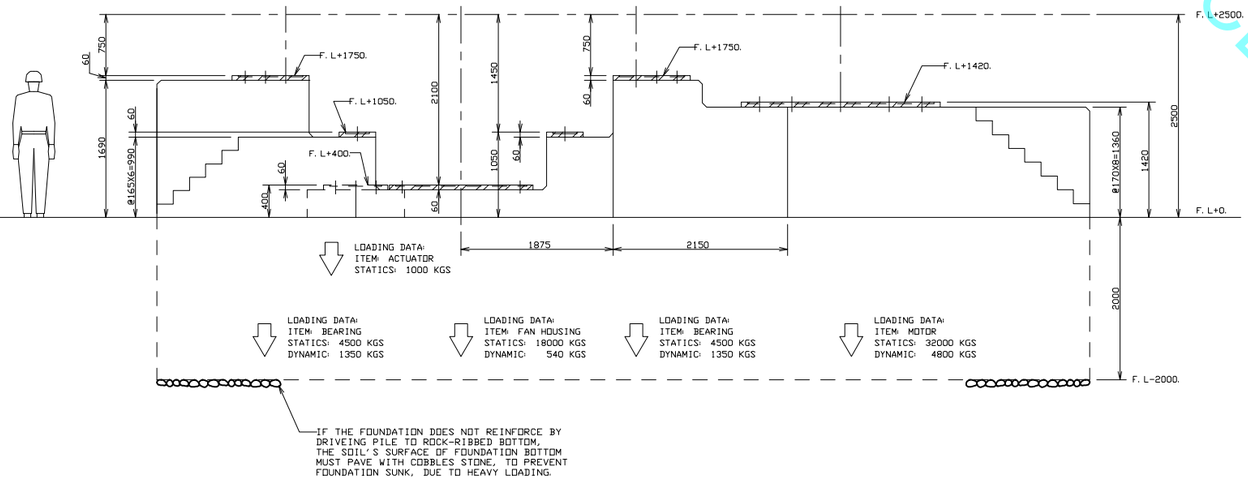
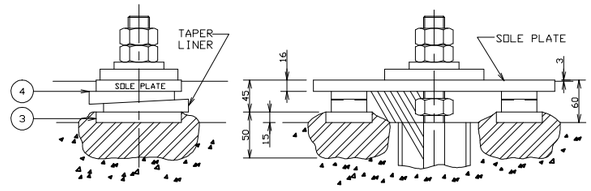
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	SOLE PLATE	SS400	9	16t SEE DWG
2	ANCHOR BOLTS	SS400	36	M30 SEE DWG
3	PLATE	SS400	80	10X75X150
4	TAPER LINER (1/50)	SS400	160	19X65X150
5	RAIL	SUS304	1	BY FHI SCOPE









FOR SCPC REFERENCE ONLY

COPY

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
1	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
B	ADD DIMENSION	陳玉樹	陳玉樹	傅泰錫

SCALE: 1/30

APPD	CKD
傅泰錫 08/19/'13	陳玉樹 08/16/'13
DESN	DWN
陳玉樹 08/15/'13	陳玉樹 08/15/'13
PROJECT NO.	SHEET NO.
RF2-0581	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX(035)981149

NAME: BCT2118/1027 DWDI CCWTH 109% A/3
P. A. F FOUNDATION DETAIL
DWG NO. A-RF2-0581-02 UNIT MM REV 1

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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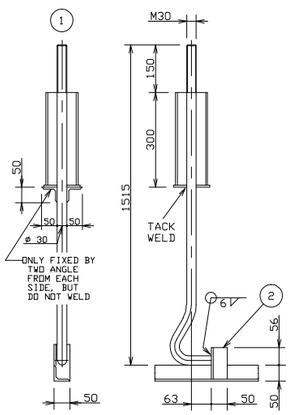
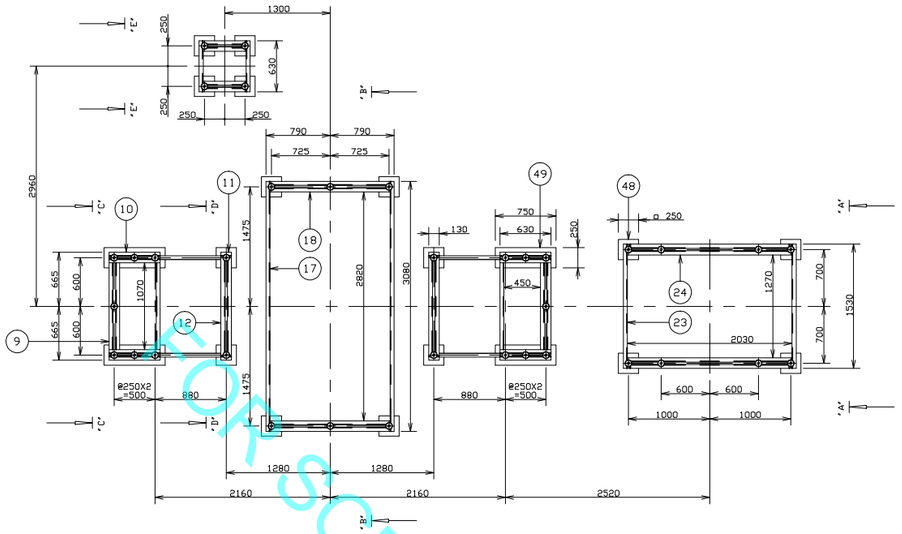
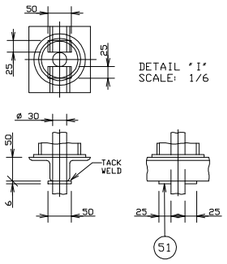
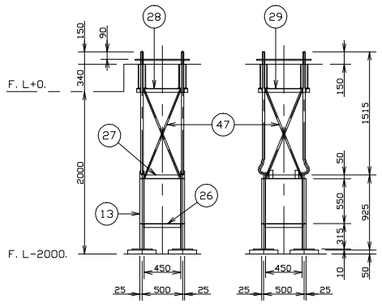


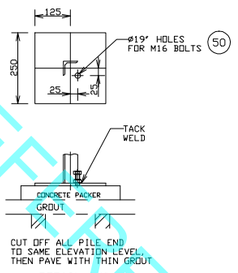
SCALE: 1/9



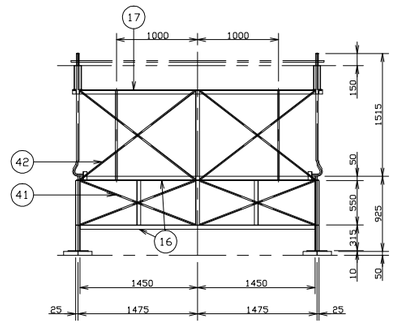




VIEW 'E'-'E'
FOR ACTUATOR



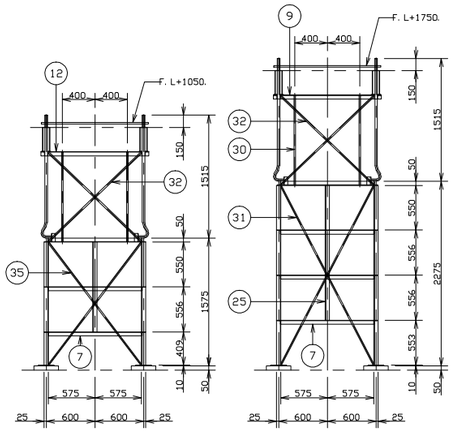
DETAIL 'H'
SCALE: 1/10



VIEW 'B'-'B'
FAN HOUSING SIDE

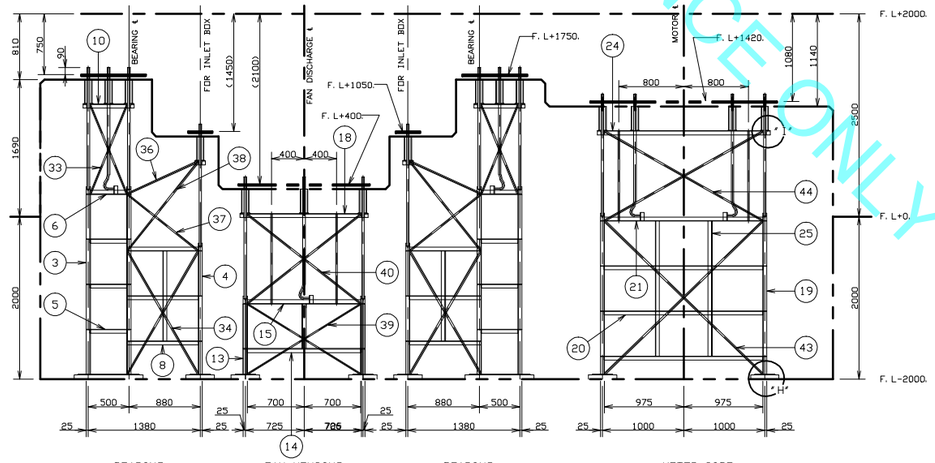
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	ANCHOR BOLTS	SS400	36	M30 SEE DWG
2	ANGLE	SS400	36	L 50X50X6X100L
3	ANGLE	SS400	8	L 50X50X6X2215L
4	ANGLE	SS400	4	L 50X50X6X1514L
5	ANGLE	SS400	12	L 50X50X6X450L
6	ANGLE	SS400	4	L 50X50X6X550L
7	ANGLE	SS400	22	L 50X50X6X1150L
8	ANGLE	SS400	12	L 50X50X6X874L
9	ANGLE	SS400	6	L 50X50X6X1070L
10	ANGLE	SS400	8	L 50X50X6X630L
11	ANGLE	SS400	4	L 50X50X6X130L
12	ANGLE	SS400	4	L 50X50X6X1230L
13	ANGLE	SS400	8	L 50X50X6X865L
14	ANGLE	SS400	2	L 50X50X6X1400L
15	ANGLE	SS400	2	L 50X50X6X1500L
16	ANGLE	SS400	4	L 50X50X6X2900L
17	ANGLE	SS400	2	L 50X50X6X3080L
18	ANGLE	SS400	4	L 50X50X6X1480L
19	ANGLE	SS400	4	L 50X50X6X1885L
20	ANGLE	SS400	6	L 50X50X6X1950L
21	ANGLE	SS400	2	L 50X50X6X2050L
22	ANGLE	SS400	8	L 50X50X6X1350L
23	ANGLE	SS400	2	L 50X50X6X1530L
24	ANGLE	SS400	4	L 50X50X6X2030L
25	ANGLE	SS400	46	L 50X50X6X550L
26	ANGLE	SS400	6	L 50X50X6X450L
27	ANGLE	SS400	2	L 50X50X6X550L
28	ANGLE	SS400	2	L 50X50X6X630L
29	ANGLE	SS400	4	L 50X50X6X530L
30	REINFORCING STEEL	SS400	28	D 3/4'X1103L
31	REINFORCING STEEL	SS400	8	D 3/4'X2500L
32	REINFORCING STEEL	SS400	12	D 3/4'X1600L
33	REINFORCING STEEL	SS400	8	D 3/4'X1190L
34	REINFORCING STEEL	SS400	8	D 3/4'X1750L
35	REINFORCING STEEL	SS400	2	D 3/4'X1905L
36	REINFORCING STEEL	SS400	4	D 3/4'X960L
37	REINFORCING STEEL	SS400	4	D 3/4'X1120L
38	REINFORCING STEEL	SS400	4	D 3/4'X1410L
39	REINFORCING STEEL	SS400	4	D 3/4'X1650L
40	REINFORCING STEEL	SS400	4	D 3/4'X1785L
41	REINFORCING STEEL	SS400	8	D 3/4'X1530L
42	REINFORCING STEEL	SS400	8	D 3/4'X1805L
43	REINFORCING STEEL	SS400	4	D 3/4'X2715L
44	REINFORCING STEEL	SS400	4	D 3/4'X2245L
45	REINFORCING STEEL	SS400	4	D 3/4'X2320L
46	REINFORCING STEEL	SS400	4	D 3/4'X1745L
47	REINFORCING STEEL	SS400	8	D 3/4'X1190L
48	PLATE	SS400	12	10tX250X250
49	PLATE	SS400	4	10tX250X750
50	HEX-BOLT,NUT	SS400	24	M16X35L
51	F. B	SS400	72	6tX25X50
52	SOLE PLATE	SS400	8	A-RF2-0556-02

Total Weight : 2720 Kgs

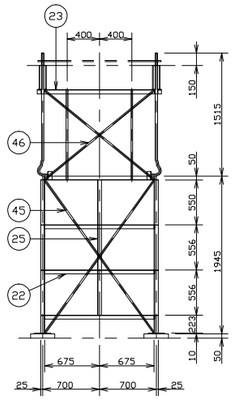


VIEW 'D'-'D'
INLET BOX SIDE

VIEW 'C'-'C'
BEARING SIDE



BEARING SIDE | FAN HOUSING SIDE | BEARING SIDE | MOTOR SIDE



VIEW 'A'-'A'
MOTOR SIDE

FOR SCPC REFERENCE ONLY

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳孟樹	陳孟樹	傅泰錫

SCALE: 1/40

APPD	CKD
傅泰錫 08/19/'13	陳孟樹 08/16/'13
DESN	DWN
陳孟樹 08/15/'13	陳孟樹 08/15/'13
PROJECT NO.	SHEET NO.
RF2-0581	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:(035)981149

NAME: BCT2118/1027 DWDI CCWTH
P. A. F ANCHOR FRAME DETAIL

DWG NO. A-RF2-0581-03 | UNIT MM | REV 0

COPY

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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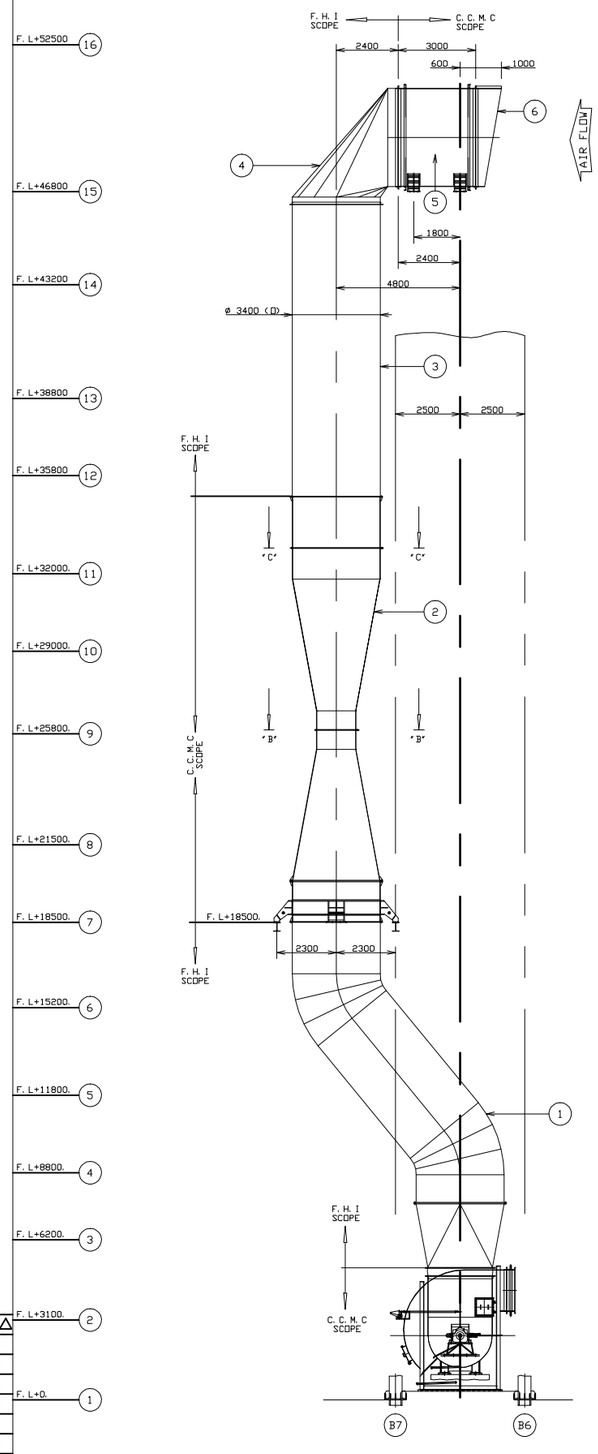
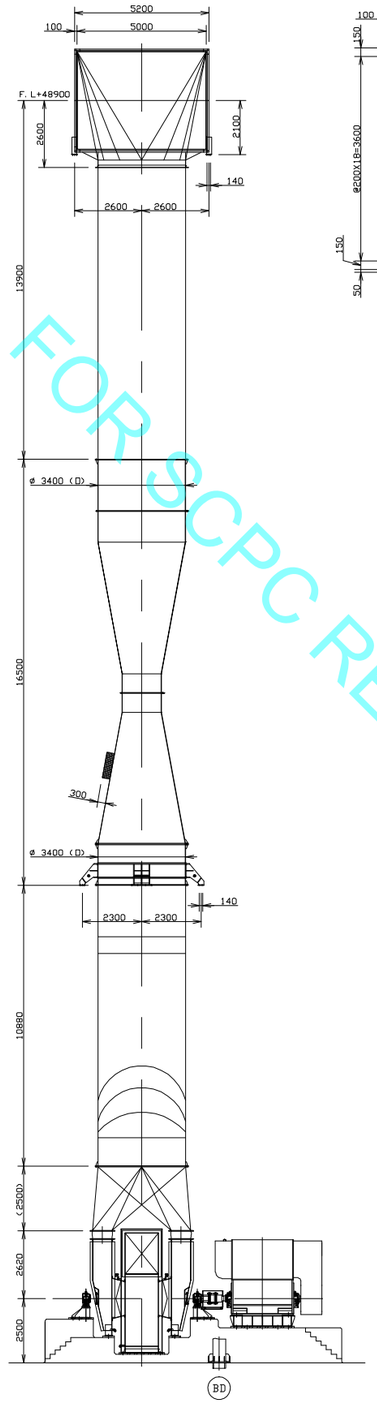
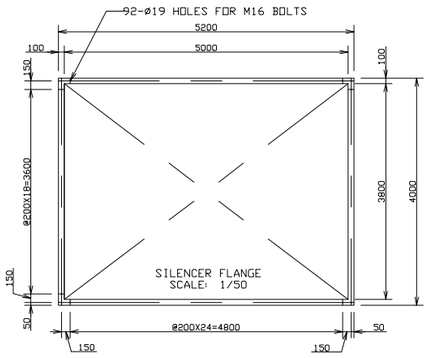
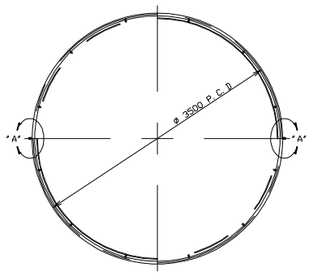
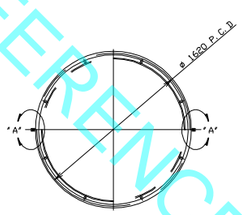




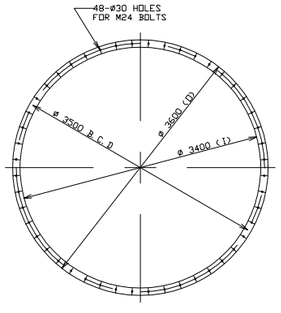




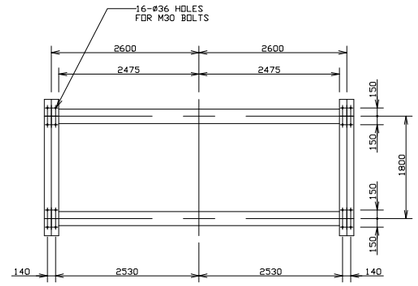
SECTION 'C'-'C'
SCALE: 1/40



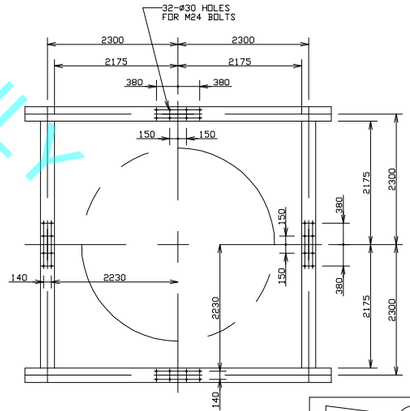
SECTION 'B'-'B'
SCALE: 1/30



VENTURI & ITEM 4
FLANGE DETAIL
SCALE : 1/40



F. L+46800 TOP VIEW
SILENCER SUPPORT
SCALE: 1/50



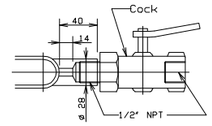
F. L+18500 TOP VIEW
VENTURI SUPPORT
SCALE: 1/50

ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	DUCT	SS400	1	BY FHI SCOPE
2	VENTURI	SS400	1	
3	DUCT	SS400	1	BY FHI SCOPE
4	ELBOW	SS400	1	BY FHI SCOPE
5	SILENCER	SS400	1	
6	INLET SCREEN	SS400	1	

NOTE:

1). INSULATION WORK FOR VENTURI WILL BE 300m/m THICKNESS BY CUSTOMER. BUT NUT, STUD BOLT ARE C. C. M. C SCOPE.

2). ESTIMATED WEIGHT
   a). VENTURI : 17000 Kg
   b). SILENCER : 16000 Kg



DETAIL 'A'
SCALE: 1/3

FOR SCPC REFERENCE ONLY

COPY

			0	FOR CONSTRUCTION	陳孟樹 陳孟樹 傅泰錫
			REV.	DESCRIPTION	REVISED CHECKED APPROVED

SCALE: 1/110

APPD	CKD
傅泰錫	陳孟樹
08/19/'13	08/16/'13
DESN	DWN
陳孟樹	陳孟樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0581	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX(035)981149

NAME: BCT2118/1027 DWDI CCWTH P. A. FAN CONNECTING ASS'Y

DWG NO. A-RF2-0581-05 | UNIT MM | REV 0

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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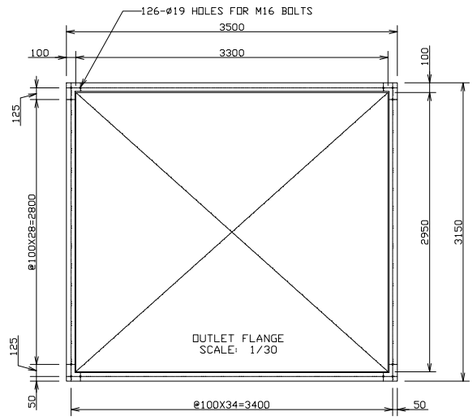
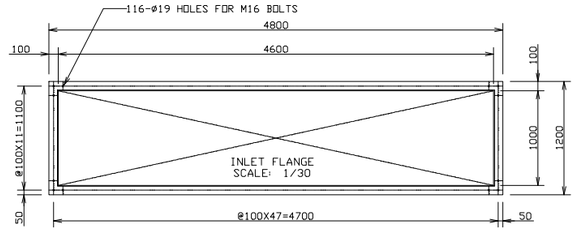
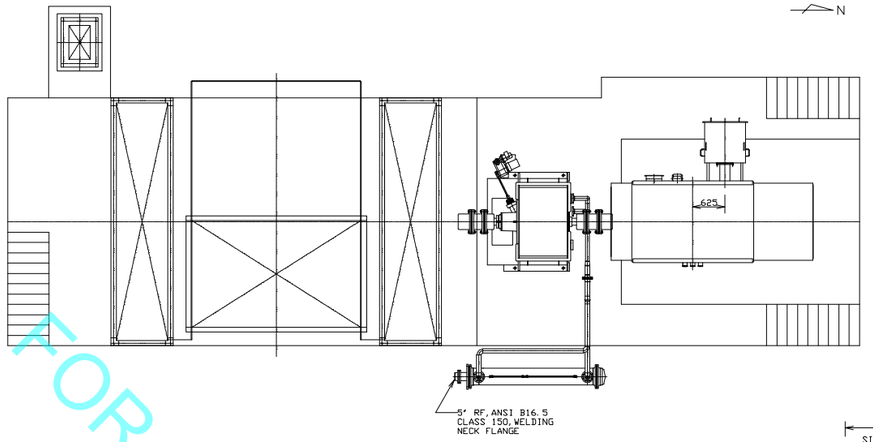
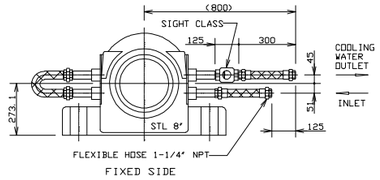
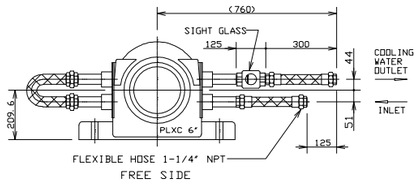
FOR SCPC REFERENCE ONLY

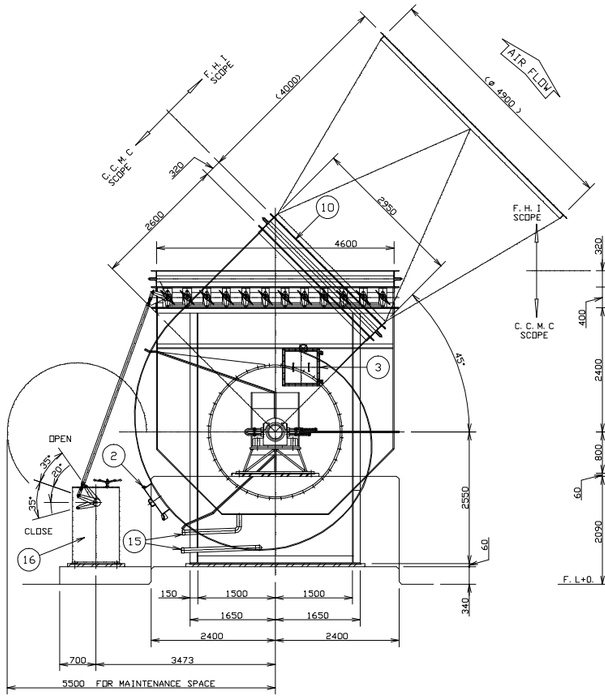


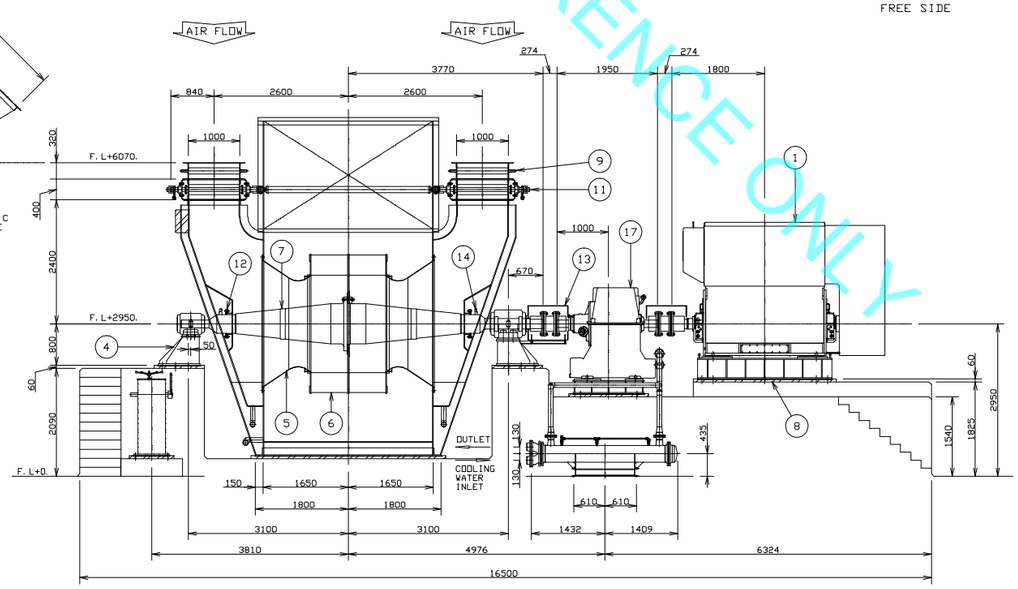


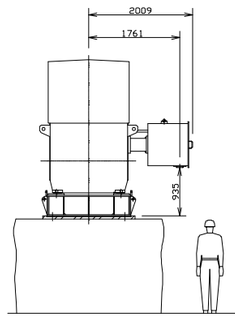


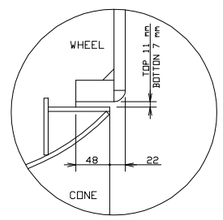












WHEEL & CONE GAP
SCALE: 1/4

ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	MOTOR	TEFC	1	SEE NOTE
2	CLEAN-OUT DOOR	SS400	1	B-RF2-0557-12
3	CLEAN-OUT DOOR	SS400	2	B-RF2-0557-13
4	BEARING PEDESTAL	SS400	2	C-RF2-0557-15
5	INLET CONE	SS400	2	B-RF2-0557-41
6	WHEEL ASS'Y	ASTM-A514	1	B-RF2-0557-51
7	SHAFT	ASTM-A668	1	A-RF2-0557-61
8	MOTOR BASE	SS400	1	B-RF2-0557-71
9	EXPANSION JOINTER (I)	SUS304	2	B-RF2-0557-73
10	EXPANSION JOINTER (D)	SUS304	1	B-RF2-0557-74
11	INLET DAMPER	SS400	1	A-RF2-0557-74
12	SHAFT SEAL	CARBON	2	B-RF1-0258-91
13	COUPLING GUARD	SS400	2	C-RF2-0557-92
14	COOLING WHEEL	AC2B	1	C-RF2-0557-96
15	DRAIN & CAP	A106B	3	PT 3"
16	ACTUATOR	STI	1	CMV3
17	HYDRALIC COUPLING	VOITH	1	
18				

BASIC CONDITION:

ITEM/	DESIGN
CAPACITY	17621 ACMM
STATIC PRESS.	858 mmAQ
SUCTION	-825 mmAQ
DISCHARGE	33 mmAQ
OPERATION TEMP.	150 deg C
STATIC EFF.	84 %
REVOLUTION SPEED	1140 RPM
BRAKE POWER	2850 BKW
NOISE(AT 1 M)	85 DBA

NOTE:

1). MOTOR WILL BE TEFC, OUTDOOR SERVIC, 3300KWX6PX60HZX6600V GD^2=15000KG-M^2 SUPPLY BY C.C.M.C.
2). COUPLING WILL BE DISC COUPLING TYPE : TIEN YI T81-205PF08.
3). BEARING WILL BE DODGE SLEEVE BEARING. 8" STL FOR FIXED SIDE BEARING AND 6" STL FOR FREE SIDE BEARING.
4). BEARING HOUSING IS DODGE STANDARD PARTS WITH WATER COOLING, HORIZONTAL SPLIT, OIL LUB. TYPE.
5). LUBRICATION OIL WILL BE ISO VG46 FOR BEARING.
6). COOLING WATER RATE FOR EACH BEARING HOUSING IS 4 T/H. TEMP. RISE 2°C. PRESSURE DIFF. : 2Kg/CM^2. INLET PRESSURE : 4Kg/CM^2.
7). INLET DAMPER TORQUE WILL BE 420 KG-M. ACTUATOR SUPPY BY C.C.M.C.
8). INSULATION WORK FOR FAN HOUSING WILL BE 250m/m. THICKNESS BY CUSTOMER. BUT NUT, STUD BOLT ARE C.C.M.C SCOPE.
9). NOISE LEVEL IS 85 DBA AT 1M AFTER FAN IS COMPLETE INSULATION.
10). BEARING HOUS'G WILL INSTALLATION TEMP. RTD.
11). VIBRATION GAUGE, METRIX ST5491E-022-020-00.
12). HYDRALIC COUPLING SUPPLY BY C.C.M.C. RECOMMEND MODEL: VOITH 1150 SVNL 22-32
13). COOLING WATER FOR HYDRAULIC COUPLING IS 53.9 T/H. TEMP. RISE 8°C. WORKING PRESSURE : 4Kg/CM^2. PRESSURE DIFF. : 0.12Kg/CM^2.

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
O	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
H	CHANGE COOLING WATER FLOW RATE	陳玉樹	陳玉樹	傅泰錫
G	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
F	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
E	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
D	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
C	CHANGE MOTOR DIMENSION	陳玉樹	陳玉樹	傅泰錫
B	ADD NOTE	陳玉樹	陳玉樹	傅泰錫

SCALE: 1/50

APPD	CKD
08/19/'13	08/16/'13
DESN	DWN
陳玉樹	陳玉樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0582	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:035)981149

NAME: D1902/10600 DWDI CCWTAU 109% A/3
I. D. FAN ASS'Y

DWG NO. A-RF2-0582-01 UNIT MM REV 0

COPY

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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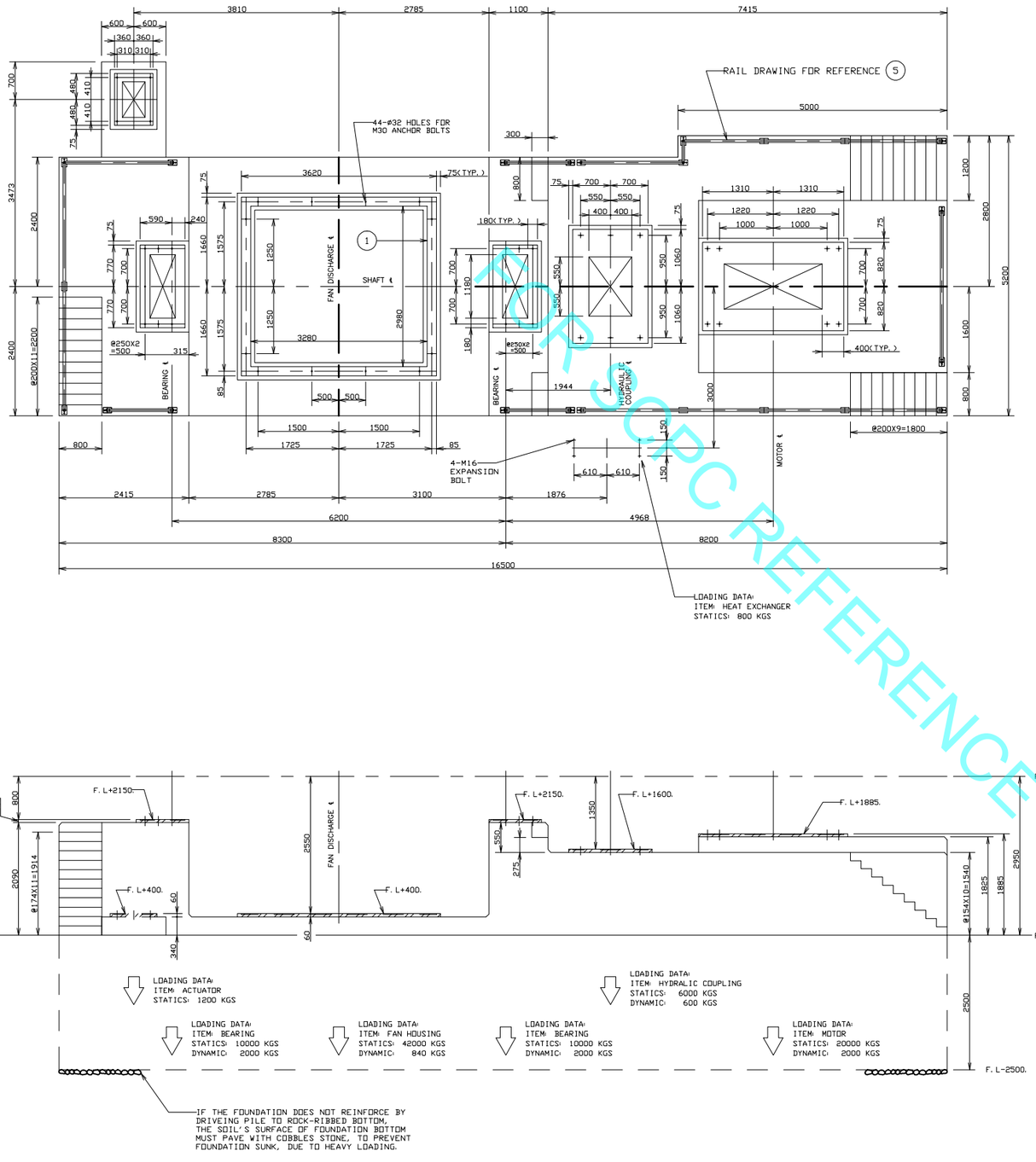
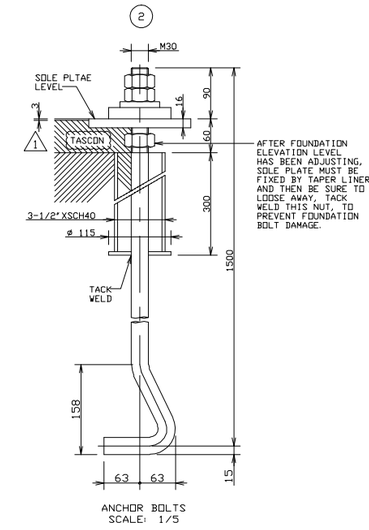
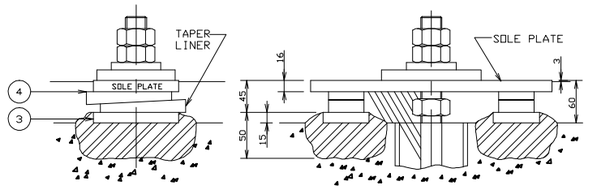
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	SOLE PLATE	SS400	6	16t SEE DWG
2	ANCHOR BOLTS	SS400	42	M30 SEE DWG
3	PLATE	SS400	90	10X75X150
4	TAPER LINER (1/50)	SS400	180	19X65X150
5	RAIL	SUS304	1	BY FHI SCOPE







FOR SOPC REFERENCE ONLY

COPY

1	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
B	ADD DIMENSION	陳玉樹	陳玉樹	傅泰錫
REV.	DESCRIPTION	REVISED	CHECKED	APPROVED

SCALE: 1/50

APPD	CKD
傅泰錫	陳玉樹
08/19/'13	08/16/'13
DESN	DWN
陳玉樹	陳玉樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0582	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX(035)981149

NAME: D1902/10500 DWDI CCWTAU 110% A/3
I. D. F FOUNDATION DETAIL

DWG NO. A-RF2-0582-02 UNIT MM REV 1

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

ENGINEER

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

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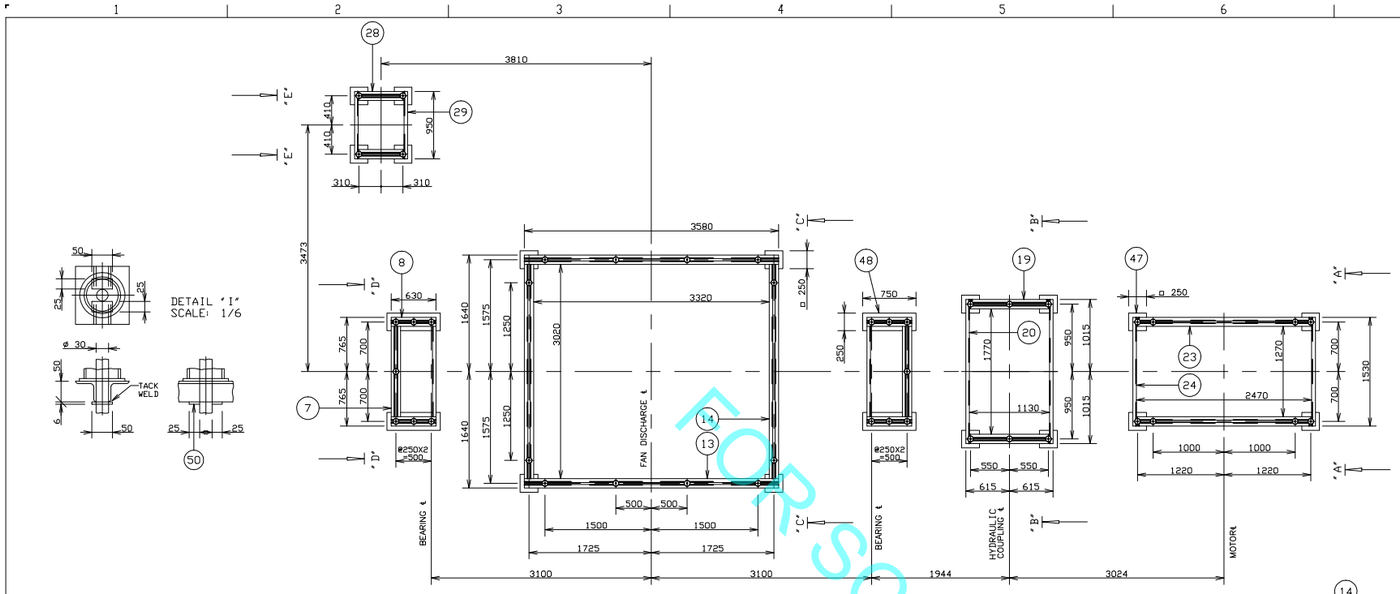
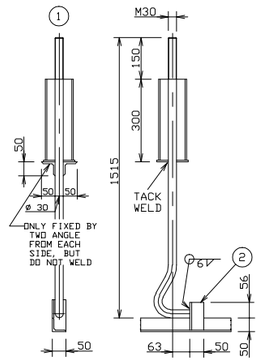
FOR SCPC REFERENCE ONLY

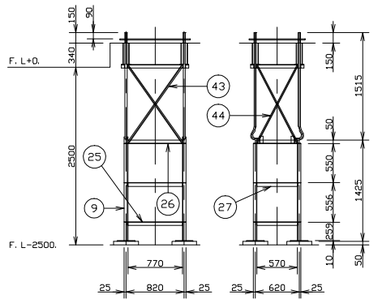




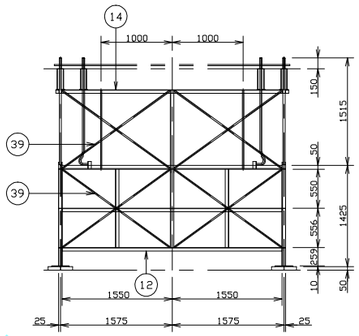
SCALE: 1/9

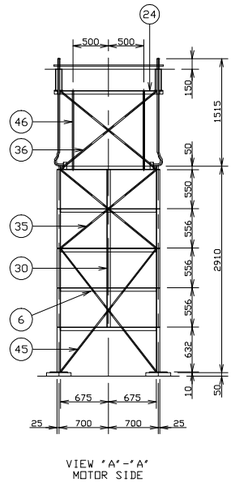
ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	ANCHOR BOLTS	SS400	44	M30 SEE DWG
2	ANGLE	SS400	44	L 50X50X6X100L
3	ANGLE	SS400	8	L 50X50X6X3115L
4	ANGLE	SS400	20	L 50X50X6X450L
5	ANGLE	SS400	4	L 50X50X6X550L
6	ANGLE	SS400	34	L 50X50X6X1350L
7	ANGLE	SS400	6	L 50X50X6X1270L
8	ANGLE	SS400	4	L 50X50X6X630L
9	ANGLE	SS400	8	L 50X50X6X1365L
10	ANGLE	SS400	4	L 50X50X6X3400L
11	ANGLE	SS400	2	L 50X50X6X3500L
12	ANGLE	SS400	6	L 50X50X6X3100L
13	ANGLE	SS400	4	L 50X50X6X3580L
14	ANGLE	SS400	4	L 50X50X6X3020L
15	ANGLE	SS400	4	L 50X50X6X2565L
16	ANGLE	SS400	8	L 50X50X6X1050L
17	ANGLE	SS400	2	L 50X50X6X1150L
18	ANGLE	SS400	10	L 50X50X6X1850L
19	ANGLE	SS400	4	L 50X50X6X1130L
20	ANGLE	SS400	2	L 50X50X6X2030L
21	ANGLE	SS400	4	L 50X50X6X2850L
22	ANGLE	SS400	8	L 50X50X6X2390L
23	ANGLE	SS400	4	L 50X50X6X2470L
24	ANGLE	SS400	2	L 50X50X6X1530L
25	ANGLE	SS400	4	L 50X50X6X770L
26	ANGLE	SS400	2	L 50X50X6X780L
27	ANGLE	SS400	6	L 50X50X6X570L
28	ANGLE	SS400	4	L 50X50X6X650L
29	ANGLE	SS400	2	L 50X50X6X950L
30	ANGLE	SS400	84	L 50X50X6X550L
31	ANGLE	SS400	6	L 50X50X6X1103L
32	ANGLE	SS400	2	L 50X50X6X2490L
33	REINFORCING STEEL	SS400	8	D 3/4"X1190L
34	REINFORCING STEEL	SS400	4	D 3/4"X1620L
35	REINFORCING STEEL	SS400	20	D 3/4"X1750L
36	REINFORCING STEEL	SS400	12	D 3/4"X1745L
37	REINFORCING STEEL	SS400	8	D 3/4"X2010L
38	REINFORCING STEEL	SS400	8	D 3/4"X2005L
39	REINFORCING STEEL	SS400	16	D 3/4"X1885L
40	REINFORCING STEEL	SS400	12	D 3/4"X1525L
41	REINFORCING STEEL	SS400	24	D 3/4"X1620L
42	REINFORCING STEEL	SS400	12	D 3/4"X2155L
43	REINFORCING STEEL	SS400	4	D 3/4"X1345L
44	REINFORCING STEEL	SS400	4	D 3/4"X1245L
45	REINFORCING STEEL	SS400	4	D 3/4"X2200L
46	REINFORCING STEEL	SS400	32	D 3/4"X1103L
47	PLATE	SS400	16	10tX250X250
48	PLATE	SS400	4	10tX250X750
49	HEX-BOLT,NUT	SS400	24	M16X35L
50	F.B	SS400	92	6tX25X50
51	SOLE PLATE	SS400	6	A-RF2-0557-02
52				

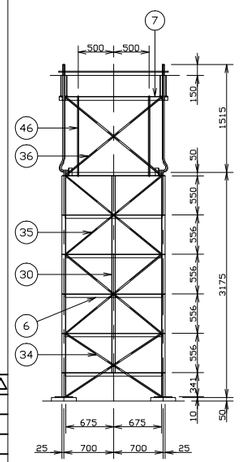
Total Weight : 4110 Kgs

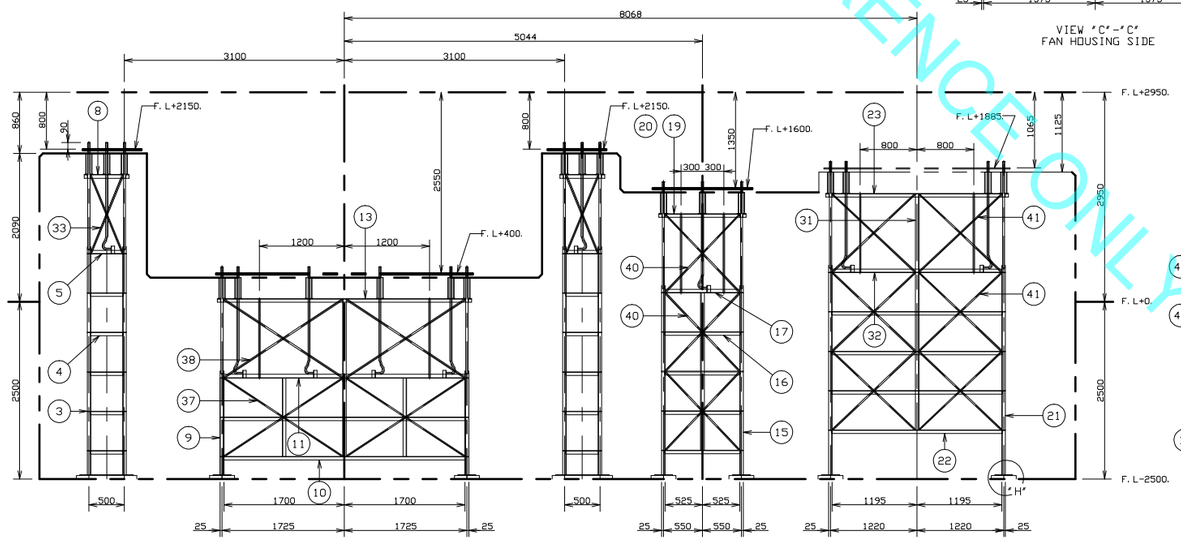


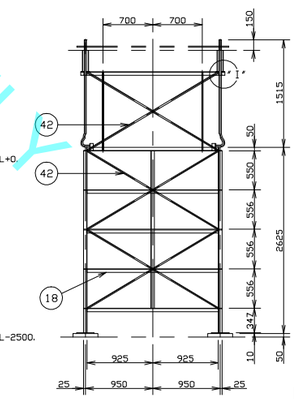
VIEW 'E'-'E'
FOR ACTUATOR

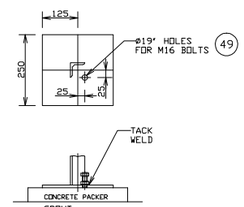

VIEW 'C'-'C'
FAN HOUSING SIDE


VIEW 'A'-'A'
MOTOR SIDE


VIEW 'D'-'D'
BEARING SIDE




VIEW 'B'-'B'
HYDRAULIC COUPLING SIDE



CUT OFF ALL PILE END TO SAME ELEVATION LEVEL, THEN PAVE WITH THIN GROUT

DETAIL 'H'
SCALE: 1/10

COPY

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION

SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS

SMC Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

SCALE: 1/50

APPD	CKD
傅泰錫	陳孟樹
08/19/'13	08/16/'13
DES'N	DWN
陳孟樹	陳孟樹
08/15/'13	08/15/'13
PROJECT NO.	SHEET NO.
RF2-0582	NO. OF

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳孟樹	陳孟樹	傅泰錫

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:(035)981149

NAME: D1902/10600 DWDI CCWTAU 109% A/3
I. D. F ANCHOR FRAME DETAIL

DWG NO. A-RF2-0582-03 UNIT MM REV 0

建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

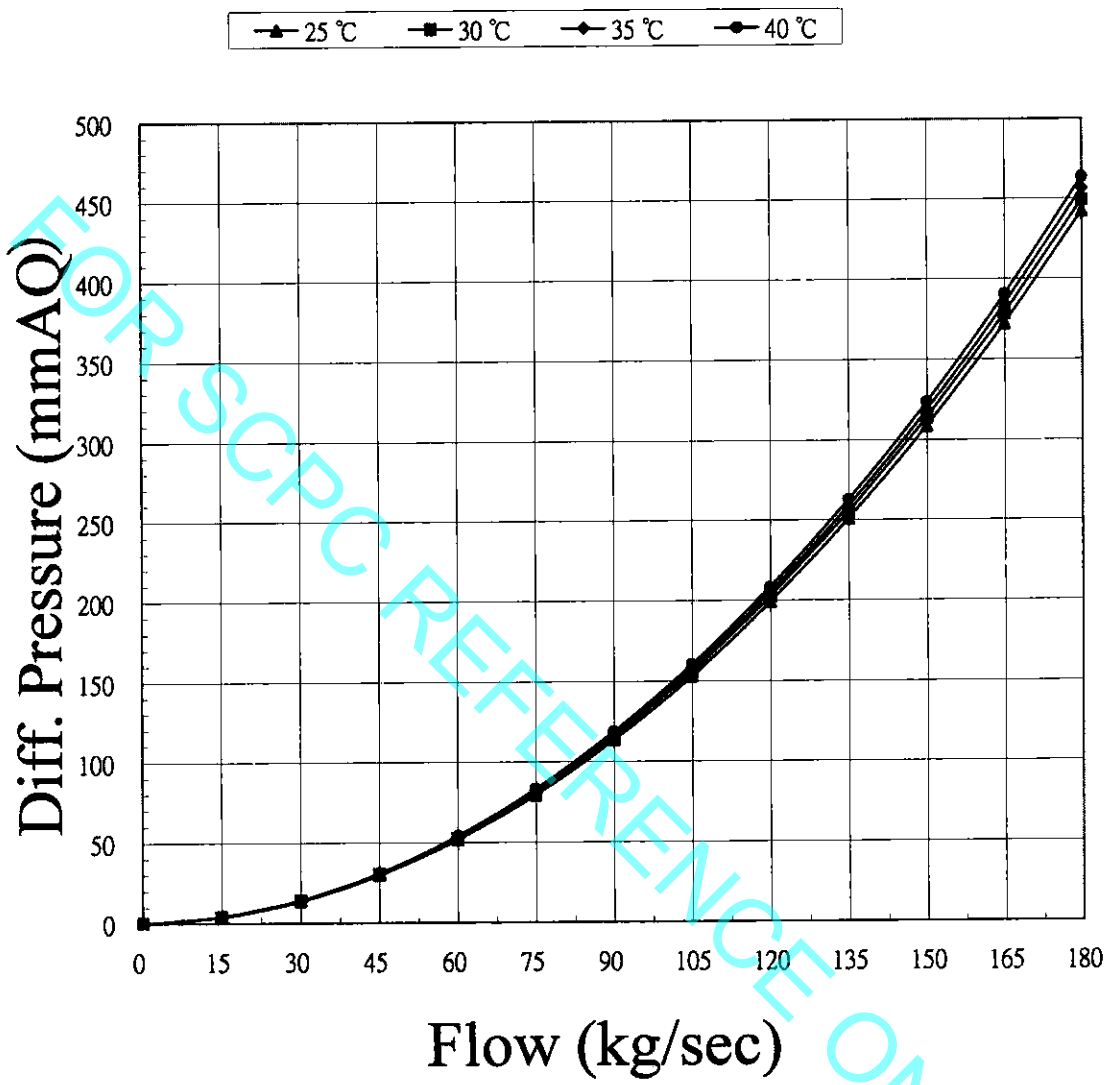
LMYP-1 Limay,Philippine
Flow Curve For Fan
送風機流量曲線資料

本文編號 RF2-0580-DOC-5-0

						CERTIFIED	
						PROJ.	
						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	林良彥	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	林良彥	陳玉樹	傅泰錫	2013/09/11	PROJECT NO.	PAGE
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	**LMYP-1**	1/7

LMYP-1 P.A.FAN REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (Kg/sec)

D.P. (mmAQ)	0	15	30	45	60	75	90	105	120	135	150	165	180
25℃	0	4	14	30	52	79	113	153	199	251	308	372	442
30℃	0	4	14	30	52	81	115	155	202	255	313	378	449
35℃	0	4	14	31	53	82	117	158	205	259	318	384	456
40℃	0	4	14	31	54	83	119	160	208	263	323	390	463

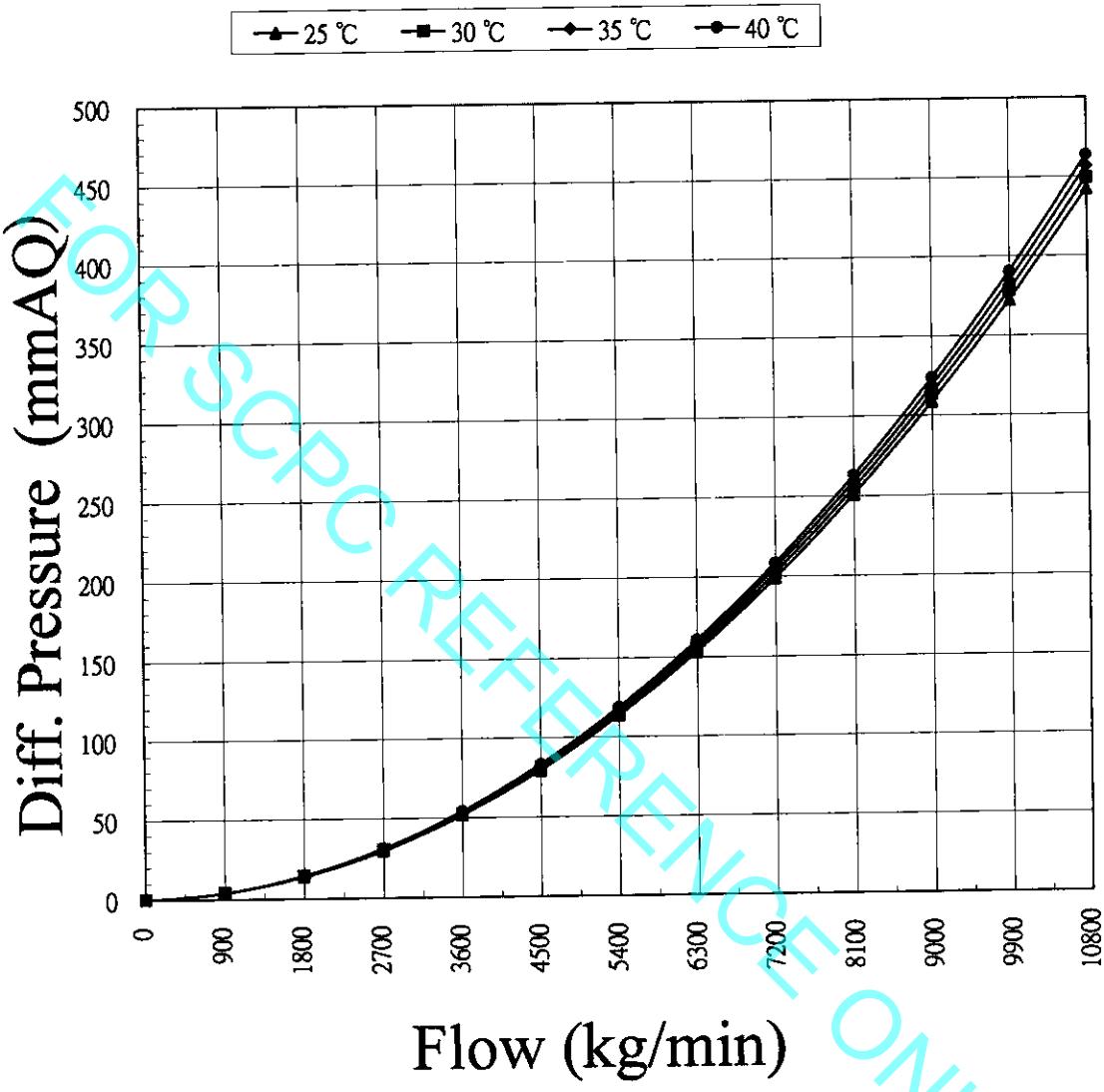
Test Block: 40℃ , 129.2 kg/sec

核准： 傅 92. 9. 16 泰錫　審核： 陳 92. 9. 11 玉樹　經辦： 林 92. 9. 11 良彥

LMYP-1 P.A.FAN

REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (Kg/min)

D.P. (mmAQ)	0	900	1800	2700	3600	4500	5400	6300	7200	8100	9000	9900	10800
25℃	0	4	14	30	52	79	113	153	199	251	308	372	442
30℃	0	4	14	30	52	81	115	155	202	255	313	378	449
35℃	0	4	14	31	53	82	117	158	205	259	318	384	456
40℃	0	4	14	31	54	83	119	160	208	263	323	390	463

Test Block: 40℃ , 7753.3 kg/min

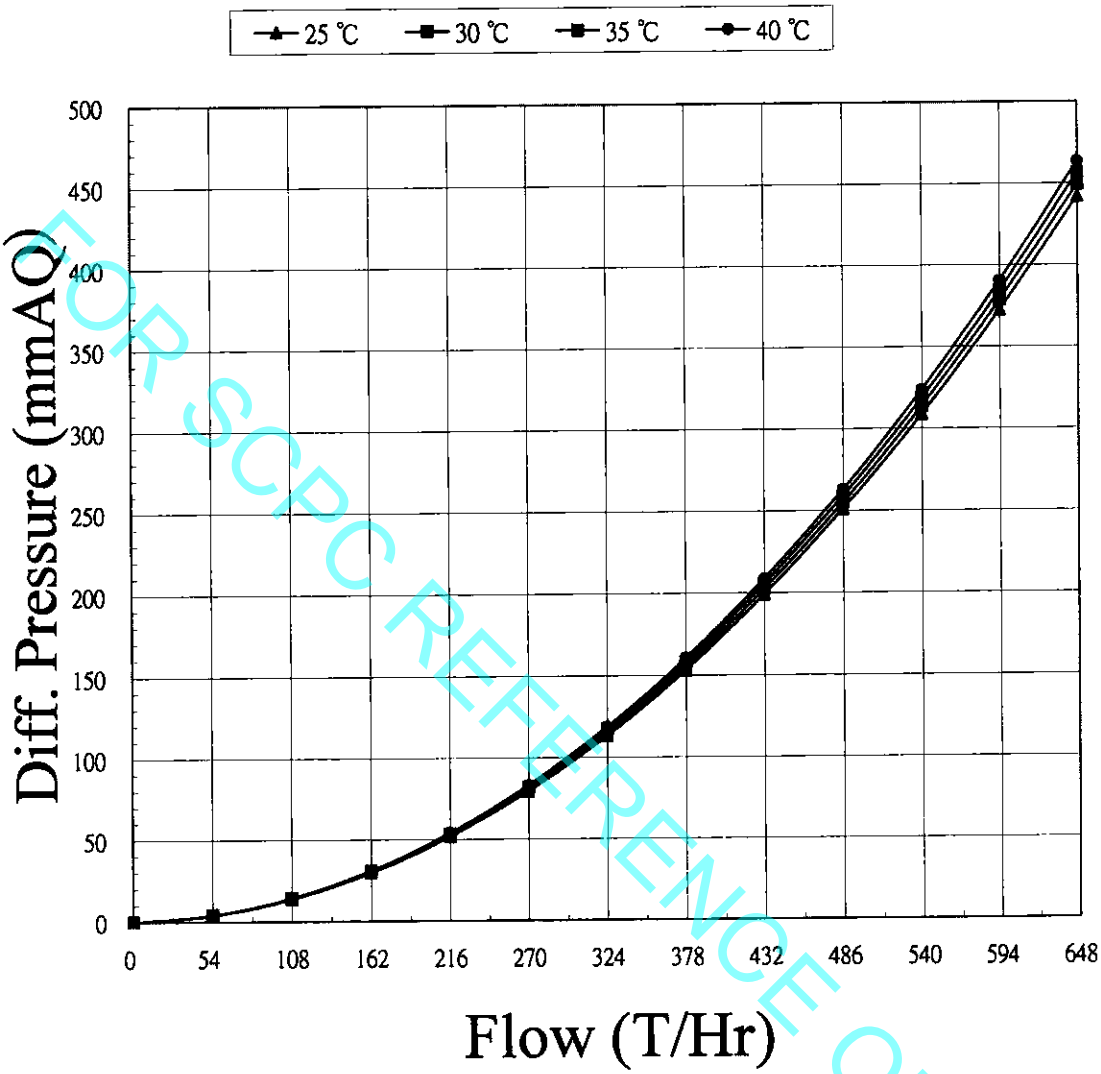
核准： 傅 92. 9. 16 泰鈺

審核： 陳 92. 9. 11 玉樹

經辦： 林 92. 9. 11 良彥

LMYP-1 P.A.FAN REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (T/Hr)

D.P. (mmAQ)	0	54	108	162	216	270	324	378	432	486	540	594	648
25℃	0	4	14	30	52	79	113	153	199	251	308	372	442
30℃	0	4	14	30	52	81	115	155	202	255	313	378	449
35℃	0	4	14	31	53	82	117	158	205	259	318	384	456
40℃	0	4	14	31	54	83	119	160	208	263	323	390	463

Test Block: 43℃ , 465.2 T/hr

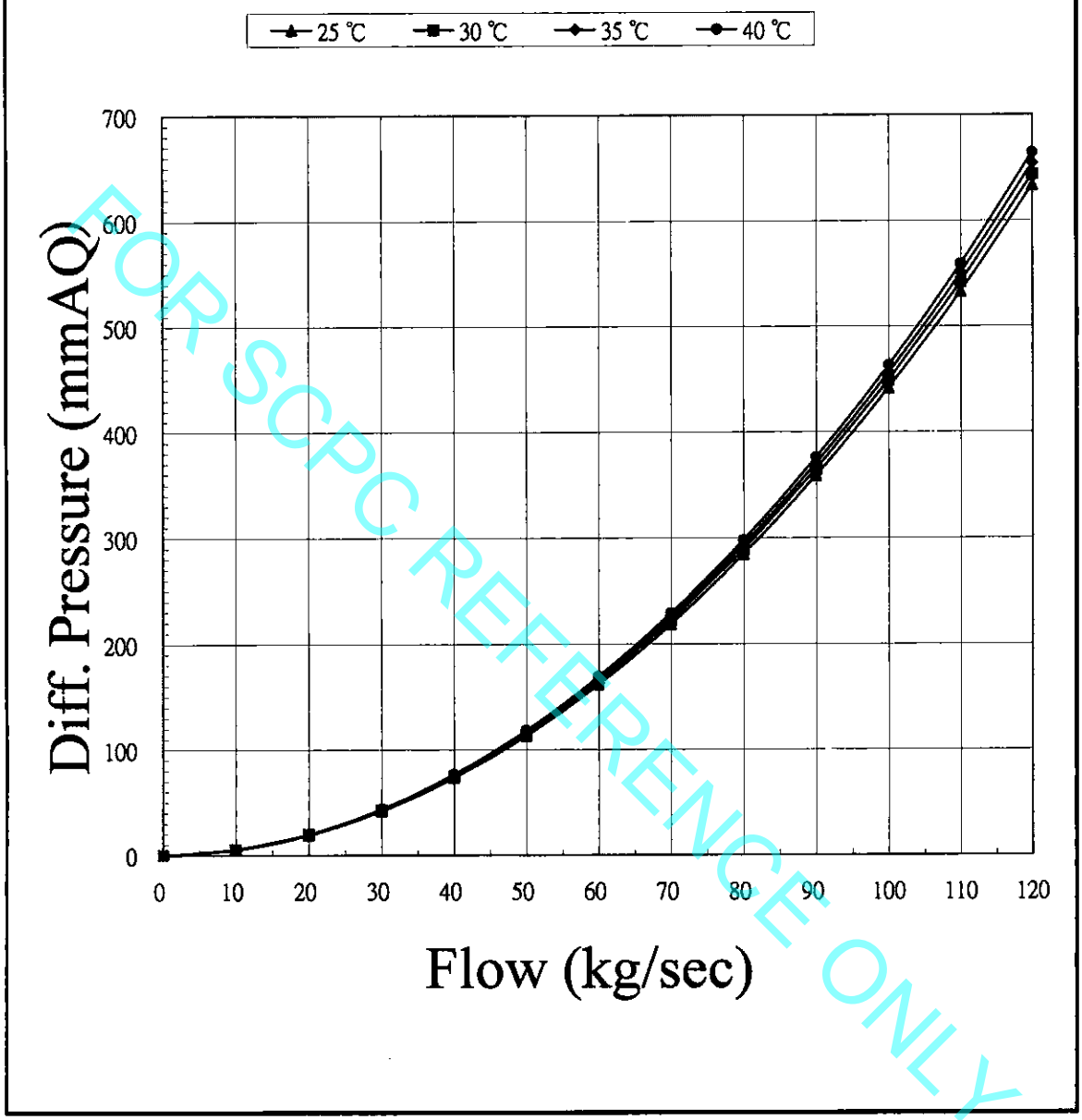
核准： 傅 92. 9. 16

審核： 陳 92. 9. 11 玉樹

經辦： 林 92. 9. 11 良彥

LMYP-1 S.A.FAN REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (Kg/sec)

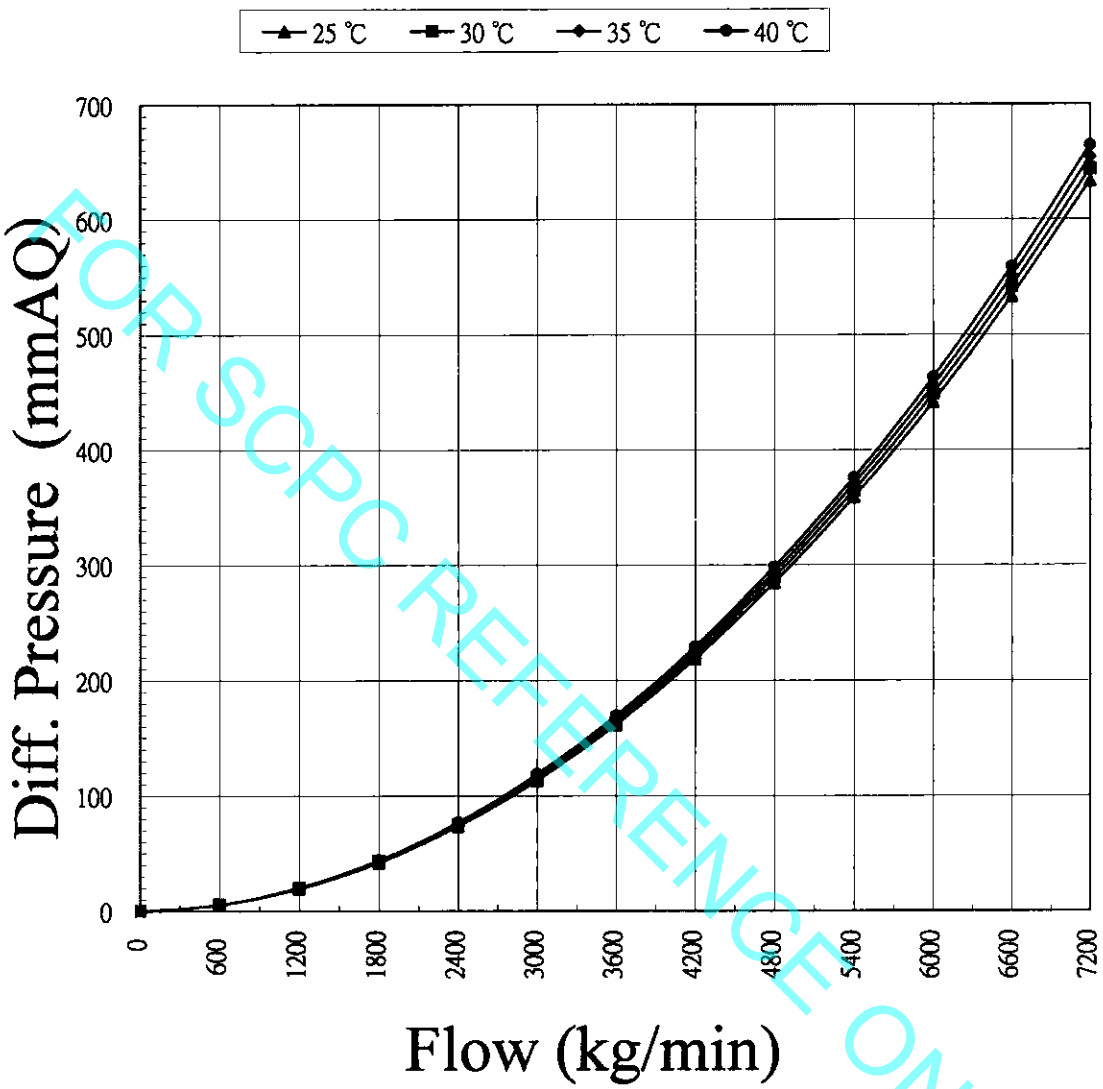
D.P. (mmAQ)	0	10	20	30	40	50	60	70	80	90	100	110	120
25℃	0	5	19	42	73	113	161	219	284	358	441	533	633
30℃	0	5	20	43	74	115	164	222	289	364	449	542	643
35℃	0	6	20	43	75	117	167	225	293	370	456	550	654
40℃	0	6	20	44	77	118	169	229	298	376	463	559	664

Test Block: 40℃ , 79 kg/sec

核准： 傅 92. 9. 16 泰錫　　審核： 陳 92. 9. 11 玉樹　　經辦： 林 92. 9. 11 良彥

LMYP-1 S.A.FAN REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (Kg/min)

D.P. (mmAQ)	0	600	1200	1800	2400	3000	3600	4200	4800	5400	6000	6600	7200
25℃	0	5	19	42	73	113	161	219	284	358	441	533	633
30℃	0	5	20	43	74	115	164	222	289	364	449	542	643
35℃	0	6	20	43	75	117	167	225	293	370	456	550	654
40℃	0	6	20	44	77	118	169	229	298	376	463	559	664

Test Block: 40℃ , 4739 kg/min

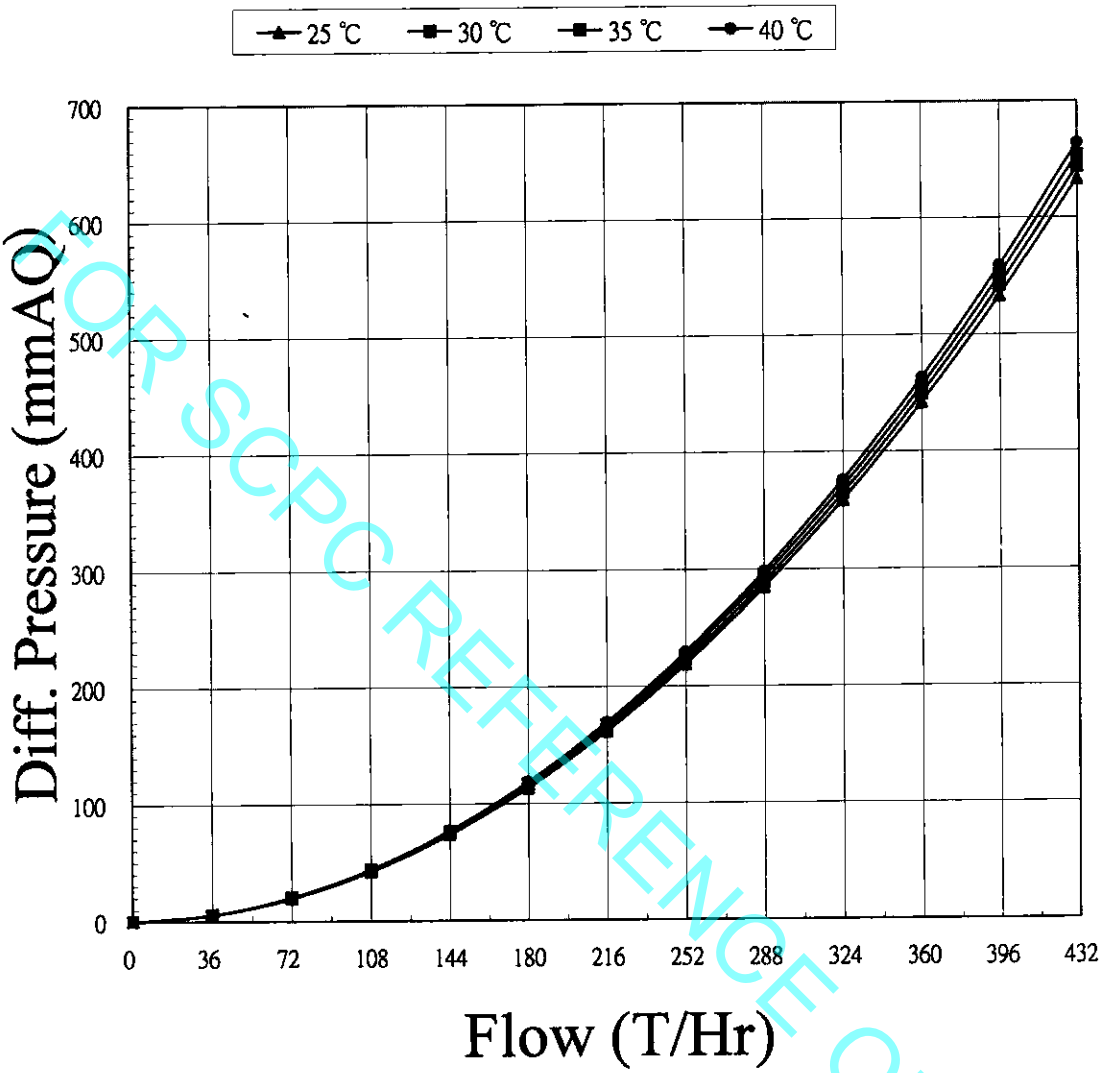
核准： 傅 92. 9. 16 泰

審核： 陳 92. 9. 11 玉樹

經辦： 林 92. 9. 11 良彥

LMYP-1 S.A.FAN REV.0

Flow-Diff. Pressure Curve



FOR SCPC REFERENCE ONLY

Flow (T/Hr)

D.P. (mmAQ)	0	36	72	108	144	180	216	252	288	324	360	396	432
25℃	0	5	19	42	73	113	161	219	284	358	441	533	633
30℃	0	5	20	43	74	115	164	222	289	364	449	542	643
35℃	0	6	20	43	75	117	167	225	293	370	456	550	654
40℃	0	6	20	44	77	118	169	229	298	376	463	559	664

Test Block: 43℃ , 284.3 T/hr

核准： 傅 102.9.16 泰

審核： 陳 102.9.11 玉樹

經辦： 林 102.9.11 良彥

2. INSPALLATTION RECORE

FOR SCPC REFERENCE ONLY

中國鋼鐵 CHINA STEEL

品質證明書
TEST CERTIFICATE

中國鋼鐵股份有限公司
CHINA STEEL CORPORATION
中華民國高雄市小港區中鋼路1號
1 CHUNG KANG ROAD HSIAO KANG, KAOHSIUNG(812)
TAIWAN, REPUBLIC OF CHINA
TEL:(07)802-1111 FAX:(07)802-2511, (07)801-9427

H0356-01 COMPANY REGISTRATION NUMBER: 30414175 0172

客戶名稱 SOLD TO	尤氏股份有限公司 YOUS TRADING CO., LTD.			產品名稱 PRODUCT	HR PLATE(CUT EDGE)
				發票號碼 INVOICE NO.	UF10945698
規格名稱 SPEC.	ASTM A36, GENERAL PURPOSES (GP).			證明書編號 CERTIFICATE NO.	000621H0356
				客戶編號 CUSTOMER NO.	11091908
				中鋼訂單編號 CSC ORDER NO.	JP4434E
				交運日期 SHIPPING DATE	JUN. 20, 2011
				證明書日期 T/C ISSUE DATE	JUN. 21, 2011
檢驗 INSP.	CSC MILL INSPECTION.	T/C	02	客戶訂單編號 CUST ORDER NO.	

項目 ITEM NO.	產品序號 LABEL NO.	尺寸 MATERIAL DESCRIPTION 厚度 THICK. mm	寬度 WIDTH mm	長度 LENGTH mm	數量 QTY	質量 MASS kg	爐號 HEAT NO.	鋼捲號碼 COIL NO.	試片編號 SAMPLE ID.	拉伸試驗 *1 G.L.=200 mm 降伏 Y.S. MPa, N/mm2	抗拉 T.S. N/mm2	伸長 EL %
001	S089560	10.00	1219	2438	20	4,596	5LK61	1222359	1220667	293	454	26.9
									1220670	330	472	25.6
	S089561	"	"	"	20	4,596	5LK61	1222359	1220667	293	454	26.9
									1220670	330	472	25.6
	S089562	"	"	"	20	4,596	5LK61	1222359	1220667	293	454	26.9
									1220670	330	472	25.6
	S089563	"	"	"	16	3,677	5LK61	1222359	1220667	293	454	26.9
									1220670	330	472	25.6
			TOTAL:		76	17,465						

化學成分 CHEMICAL ANALYSIS % C	Mn	P	S	Si	Cu	Ni	Cr	Mo	T. Al	備註 REMARK
	2 X10	2 X10	3 X10	2 X10					3 X10	
16	75	16	5	8	1				A8	
"	"	"	"	"	"				"	
"	"	"	"	"	"				"	
"	"	"	"	"	"				"	

FOR SCRC REFERENCE ONLY

註釋 NOTES

*01 拉伸試驗 TENSILE TEST

1MPa = 1N/mm2

Witnessed
Reviewed
CCMC/Q.C.Dept.
HONDAR TSENG

SURVEYOR TO

茲證明本表所列產品，均依材料規格製造及試驗，並符合規格之要求。
WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.

Chih-Chung Wu
冶金技術處處長
GENERAL MANAGER, METALLURGICAL DEPARTMENT

中國鋼鐵 CHINASTEEL

品質證明書
TEST CERTIFICATE

中國鋼鐵股份有限公司
CHINA STEEL CORPORATION
中華民國高雄市小港區中鋼路1號
1 CHUNG KANG ROAD HSIAO KANG, KAOHSIUNG(812)
TAIWAN, REPUBLIC OF CHINA
TEL:(07)802-1111 FAX:(07)802-2511, (07)801-9427

A0082-01 COMPANY REGISTRATION NUMBER: 30414175 0070

客戶名稱 SOLD TO	尤氏股份有限公司 YOUS TRADING CO., LTD.			產品名稱 PRODUCT	PLATE-CARBON STEEL. (MILL EDGE)
				發票號碼 INVOICE NO.	BY12216236
規格名稱 SPEC.	JIS G3101-10 SS400			證明書編號 CERTIFICATE NO.	010531A0082
				客戶編號 CUSTOMER NO.	11091908
				中鋼訂單編號 CSC ORDER NO.	JP6512V
				交運日期 SHIPPING DATE	MAY. 30, 2012
				證明書日期 T/C ISSUE DATE	MAY. 31, 2012
檢驗 INSP.	CSC MILL INSPECTION.	T/C	02	客戶訂單編號 CUST ORDER NO.	

項目 ITEM NO.	產品序號 SERIAL NO. (PLATE NO)	尺寸 MATERIAL DESCRIPTION 厚度 THICK. mm	寬度 WIDTH mm	長度 LENGTH mm	數量 QTY	質量 MASS kg	爐號 HEAT NO.	胚號 SLAB NO.	試片編號 SAMPLE ID.	拉伸試驗 *1 G.L. = 200 mm 降伏 Y.S. MPa, N/mm2	抗拉 T.S.	伸長 EL %
001	14579	10.00	2438	7315	4	5,600	1YR35	404-4	14579	295	464	28.2
	14582	"	"	"	4	5,600	1YR35	403-2	14579	295	464	28.2
	14584	"	"	"	4	5,600	1YR35	403-1	14579	295	464	28.2
	14586	"	"	"	2	2,800	3NZ87	306-2	14586	296	441	28.3
			TOTAL:		14	19,600						

化學成份 CHEMICAL ANALYSIS %	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Al	REMARK
	2 X10	2 X10	3 X10		2 X10					X10	
	13	69	16	6	30						
	"	"	"	"	"						
	"	"	"	"	"						
	13	65	19	10	28						

註釋 NOTES

*01 拉伸試驗 TENSILE TEST

1MPa = 1N/mm2

☐ Witnessed
☑ Reviewed
CCMC/Q.C.Dept.
HONDAR TSENG

SURVEYOR TO

茲證明本表所列產品，均依材料規格製造及試驗，並符合規格之要求，且經偵測無輻射污染。
WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN MANUFACTURED AND TESTED WITH SATISFACTORY RESULTS IN ACCORDANCE WITH THE REQUIREMENT OF THE ABOVE MATERIAL SPECIFICATION.

Chih-Chung Wu
冶金技術處處長
GENERAL MANAGER, METALLURGICAL DEPARTMENT

FOR SCC REFERENCE ONLY

YYS
有益鋼鐵股份有限公司
YEOU YIH STEEL CO., LTD.

品質證明書
INSPECTION CERTIFICATE

高雄市永安區永工十路2號
NO.2 YEONG GONG 10TH ROAD, YEONG AN DISTRICT,
KAOHSIUNG CITY, TAIWAN, R.O.C.
TEL:886-7-6225616 FAX:886-7-6226808

FOR SCPC REFERENCE ONLY

No.(品證號碼): MT1202102

Contract No.(訂單號碼): S1201018 Customer(客戶): [G024] 力春實業股份有限公司 PAGE: 1/1 **Date: 2012/02/15**

Commodity(品名): HOT ROLLED STAINLESS STEEL PLATE Surface Finish(表面等級): NO. 1 **Solution Treatment 1080 ℃ A.T.**

Specification(規範): ASTM A240-07 Steel Grade(鋼種): 304/304L

Item No.	Product No.	Heat No.	Size	Pcs	Kgs	EN 10028-7 Impact Test(衝擊試驗) (20℃) / Min. 60 J 1	2	3	AVE.	(-196℃) / Min. 60 J 1	2	3	AVE.
1	BU20092	122387	12mm X 1524mm X 6096mm	20	18,015								
			Total	20	18,015								

Item No.	Chemical Composition (化學成份) C	Si	Mn	P	S	Ni	Cr	Mo	N	Cu	Ti	降伏強度 Y.S(MPa) 1%	0.2%	抗拉強度 T.S(MPa)	伸長率 % Elongation	硬度試驗 Hardness HRB/AVE.
Max.	0.030	0.75	2.00	0.045	0.030	12.00	19.50		0.100							92
Min.						8.00	17.50						170	485	40	
1	0.019	0.50	1.05	0.030	0.003	8.05	18.24	0.01	0.050	0.02			290	576	50	77

Remarks:
1.YYS has established a QMS according to ISO 9001:2008, which is certified by TUV.

☐ Witnessed
☑ Reviewed
CCMC/Q.C.Dept.
MONDAR TSENG

新威不銹鋼有限公司
TEL:3665461-3
FAX:3665035
桃園市五山街186號

Manager of Quality Assurance Department

王我源

We hereby certify that material described herein has been manufactured and tested with satisfactory results in accordance with the requirement of the above material specification

We hereby certify that the material described above has been detected and has no radiation contaimination.(鋼輻偵字第206號)

CERTIFICATE OF MATERIAL

HYUNJIN MATERIALS

☒3.1 ACCORDING TO ISO 10474/EN10204 : 2004
☐ 3.2 ACCORDING TO ISO 10474/EN10204 : 2004

QUALITY CONTROL OFFICE ★ LABORATORY SECTION ★ HYUNJIN MATERIALS CO.,LTD.

#1601-1, SongJung-Dong, GangSeo-Ku
Busan, Korea,
Tel. : (051) 602-7752
Fax.: (051) 602-7779

Test Machine	Model no.	Recal. Date
UTM	10-2482	2014.05.07(HPI)
HB Tester	10-K1473	
Impact Tester	10-2742	

Certificate No. : HJ-13-ME-0843
Project No. : -
Product Name : BIG FORGING SHAFT
Condition : As Rough Machined

Customer : GS GLOBAL CORPORATION
Customer P/O No. : GS-HJM(Q)-2013-4012-03
Mat'l Specification : ASTM A668 Class G
Process : E. A. F. + FORGED

Page : 1 of 1
Work No. : 13HJM209577
DWG. No. : RF2-0577/0580-M1 Rev.0
Date : 2013. 12. 04.

No.	Size (Unit : mm, Q'ty)	Serial No.	Mark of Specimen	*1	Tensile Test *3 Size of Test Specimen D(mm)	G.L(mm)	Y.S N/㎟	T.S N/㎟	E.L %	R.A %	Impact Test Temp : - Notch : -				Hardness HB	Remark
					12.5	50.0	min. 345	min. 550	min. 22	min. 38	min. : -			Ave. -	163 -207	
1	OD710 X 4510L=1EA	BFS12	01A	L	12.5	50.0	380	641	24	46		-			183	* Marking Stamped on Side Body
			01B	L	12.5	50.0	375	628	23	45		-			183	
								BLANK								

Chemical Composition (%)

Ingot No.		Ingot Quantity (EA)	Mark of Specimen	C	Si	Mn	P	S	Cr	Mo	-	-
				max. 0.45	max. 0.35	max. 1.10	max. 0.025	max. 0.025	max. 1.10	max. 0.25		
HSC	134501	1	01/A,B	0.43	0.21	1.00	0.014	0.001	0.15	0.02		
							BLANK					

Customer Requirements

Heat treatment *2
N : 870 ℃ × 19.8 HRS
T : 620 ℃ × 18.8 HRS
*Report No. : HJ-13-HT-0781
SR : 580 ℃ × 20 HRS
*Report No. : HJ-13-HT-0788

NDE *4
UT (R.M) : ACCEPTABLE
*Report No.: HJ-13-UT-0843

Dimension
*Report No.
HJ-13-DM-0843

*1: L=Longitudinal, T=Tangential, Tr=Transverse
*2: N=Normalized, A=Annealed, Q=Quenched, T=Tempered, S.R=Stress Relieved, A.C=Air Cooled, F.C=Furnace Cooled, W.C=Water Cooled, O.C=Oil Cooled,
*3: Y.S=Yield Strength, T.S=Tensile Strength, E.L=Elongation, R.A=Reduction of Area, D=Diameter, G.L=Gauge Length,
*4: Surface Condition : A.F=As Forged, R.M=Rough Machined, F.M=Finish Machined,

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

T. H. YOON
Checked by

D. G. KIM
Approved by

FOR SDPC REFERENCE ONLY

CERTIFICATE OF MATERIAL

HYUNJIN MATERIALS

☒3.1 ACCORDING TO ISO 10474/EN10204 : 2004
☐ 3.2 ACCORDING TO ISO 10474/EN10204 : 2004

QUALITY CONTROL OFFICE ★ LABORATORY SECTION ★ HYUNJIN MATERIALS CO.,LTD.

#1601-1, SongJung-Dong, GangSeo-Ku
Busan, Korea,
Tel. : (051) 602-7752
Fax.: (051) 602-7779

Certificate No. : HJ-13-ME-0842
Project No. : -
Product Name : BIG FORGING SHAFT
Condition : As Rough Machined

Customer : GS GLOBAL CORPORATION
Customer P/O No. : GS-HJM(Q)-2013-4012-03
Mat'l Specification : ASTM A668 Class G
Process : E. A. F. + FORGED

Page : 1 of 1
Work No. : 13HJM209577
DWG. No. : RF2-0578/0581-M1 Rev.0
Date : 2013. 12. 04.

Test Machine	Model no.	Recal. Date
UTM	10-2482	2014.05.07(HPI)
HB Tester	10-K1473	
Impact Tester	10-2742	

No.	Size (Unit : ㎜ Q'ty)	Serial No.	Mark of Specimen	*1	Tensile Test *3 Size of Test Specimen D(mm)	G.L(mm)	Y.S N/㎟	T.S N/㎟	E.L %	R.A %	Impact Test Temp : - Notch : -				Hard-ness HB	Remark
					12.5	50.0	min. 345	min. 550	min. 22	min. 38	min. : -			Ave. -	163 -207	
1	OD800 X 4880L=1EA	BFS7	57A	L	12.5	50.0	410	671	23	42		-			187	* Marking Stamped on Side Body
			57B	L	12.5	50.0	409	675	23	41		-			187	
							BLANK									

Chemical Composition (%)

Ingot No.		Ingot Quantity (EA)	Mark of Specimen	C	Si	Mn	P	S	Cr	Mo	-	-
				max. 0.45	max. 0.35	max. 1.10	max. 0.025	max. 0.025	max. 1.10	max. 0.25		
HSC	134557	1	57/A,B	0.43	0.24	1.06	0.015	0.002	0.15	0.02		
							BLANK					

Customer Requirements

Heat treatment *2
N : 870 ℃ × 19.8 HRS
T : 620 ℃ × 18.8 HRS
*Report No. : HJ-13-HT-0781
SR : 580 ℃ × 20 HRS
*Report No. : HJ-13-HT-0788

NDE *4
UT (R.M) : ACCEPTABLE
*Report No.: HJ-13-UT-0842

Dimension
*Report No.
HJ-13-DM-0842

*1: L=Longitudinal, T=Tangential, Tr=Transverse
*2: N=Normalized, A=Annealed, Q=Quenched, T=Tempered, S.R=Stress Relieved, A.C=Air Cooled, F.C=Furnace Cooled, W.C=Water Cooled, O.C=Oil Cooled,
*3: Y.S=Yield Strength, T.S=Tensile Strength, E.L=Elongation, R.A=Reduction of Area, D=Diameter, G.L=Gauge Length,
*4: Surface Condition : A.F=As Forged, R.M=Rough Machined, F.M=Finish Machined,

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

T. H. YOON
Checked by

D. G. KIM
Approved by

FOR PCPC REFERENCE ONLY

CERTIFICATE OF MATERIAL

HYUNJIN MATERIALS

☒3.1 ACCORDING TO ISO 10474/EN10204 : 2004
☐ 3.2 ACCORDING TO ISO 10474/EN10204 : 2004

QUALITY CONTROL OFFICE ★ LABORATORY SECTION ★ HYUNJIN MATERIALS CO.,LTD.

#1601-1, SongJung-Dong, GangSeo-Ku
Busan, Korea,
Tel. : (051) 602-7752
Fax.: (051) 602-7779

Certificate No. : HJ-13-ME-0845
Project No. : -
Product Name : BIG FORGING SHAFT
Condition : As Rough Machined

Customer : GS GLOBAL CORPORATION
Customer P/O No. : GS-HJM(Q)-2013-4012-03
Mat'l Specification : ASTM A668 Class G
Process : E. A. F. + FORGED

Page : 1 of 1
Work No. : 13HJM209577
DWG. No. : RF2-0579/0582-M1 Rev.0
Date : 2013. 12. 04.

Test Machine	Model no.	Recal. Date
UTM	10-2482	2014.05.07(HPI)
HB Tester	10-K1473	
Impact Tester	10-2742	

No.	Size (Unit : mm, Q'ty)	Serial No.	Mark of Specimen	*1	Tensile Test *3: Size of Test Specimen D(mm)	Size of Test Specimen G.L(mm)	Y.S N/㎟	T.S N/㎟	E.L %	R.A %	Impact Test: Temp : - / Notch : -	Impact Test: -	Impact Test: min. -	Impact Test: Ave. -	Hard-ness HB	Remark
					12.5	50.0	min. 345	min. 550	min. 22	min. 38	min. :		-		163 -207	
1	OD1060X7025L=1EA	BFS4	36A	L	12.5	50.0	429	651	23	46		-			187	* Marking Stamped on Side Body
			36B	L	12.5	50.0	425	643	23	45		-			187	
								BLANK								

Chemical Composition (%)

Ingot No.		Ingot Quantity (EA)	Mark of Specimen	C max. 0.45	Si max. 0.35	Mn max. 1.10	P max. 0.025	S max. 0.025	Cr max. 1.10	Mo max. 0.25	-	-
HSC	014036	1	36/A,B	0.43	0.26	1.00	0.015	0.004	0.14	0.03		
							BLANK					

Customer Requirements

Heat treatment *2
N : 870 ℃ × 19.8 HRS
T : 620 ℃ × 18.8 HRS
*Report No. : HJ-13-HT-0781
SR : 580 ℃ × 20 HRS
*Report No. : HJ-13-HT-0788

NDE *4
UT (R.M) : ACCEPTABLE
*Report No.: HJ-13-UT-0845

Dimension
*Report No.
HJ-13-DM-0845

*1: L=Longitudinal, T=Tangential, Tr=Transverse
*2: N=Normalized, A=Annealed, Q=Quenched, T=Tempered, S.R=Stress Relieved, A.C=Air Cooled, F.C=Furnace Cooled, W.C=Water Cooled, O.C=Oil Cooled,
*3: Y.S=Yield Strength, T.S=Tensile Strength, E.L=Elongation, R.A=Reduction of Area, D=Diameter, G.L=Gauge Length,
*4: Surface Condition : A.F=As Forged, R.M=Rough Machined, F.M=Finish Machined,

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

T. H. YOON
Checked by

D. G. KIM
Approved by

FOR SPEC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0580-001
		DATE: 06-18 -2014

JOB NO.	RF2-0580	ITEM NO. / ITEM NAME	VENTURI TUBE(A)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	A-RF2-0555-76	MATERIAL / THICKNESS	SS400/10T

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT		
	PENETRANT / DEVELOPER APPLICATION	☐ BRUSHING	■ SPRAYING	☐	
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE	☐ POST EMULSIFIABLE		
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S			
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2			
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET	☐ DRY	
EXAMINATION CONDITION	TEMPERATURE	27 ℃	PENETRATION TIME	10 min	
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min	
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED:. ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0580(A)

Q'TY:1 PCS

內側焊道(100%)

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	06.18.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	6-18-2014	QC MANAGER REVIEWED	6/18/14	AI REVIEWED	

FOR SCPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0581-001
		DATE: 06-18 -2014

JOB NO.	RF2-0581	ITEM NO. / ITEM NAME	VENTURI TUBE(A)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	A-RF2-0556-76	MATERIAL / THICKNESS	SS400/10T

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST ☐ FLUORESCENT
	PENETRANT / DEVELOPER APPLICATION	☐ BRUSHING ■ SPRAYING ☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ☐ POST EMULSIFIABLE ■ SOLVENT REMOVABLE
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2 ■ WET ☐ DRY

EXAMINATION CONDITION	TEMPERATURE	27	℃	PENETRATION TIME	10 min
	DRYING TIME	※ (1) 3 (2) 5	min	DEVELOPING TIME	10 min
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED:. ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0581(A)

Q'TY:1 PCS

內側焊道(100%)

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	06.18.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	6-18-2014	QC MANAGER REVIEWED	6/18/14	AI REVIEWED	

FOR CPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0580-002
		DATE: 08-05 -2014

JOB NO.	RF2-0580	ITEM NO. / ITEM NAME	WHEEL ASS'Y (A)
CUSTOMER	**台朔重工/LMYP-1**	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0555-51 B-RF2-0555-52	MATERIAL / THICKNESS	A514-B / (8T/19T)
TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐
TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING ☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ☐ POST EMULSIFIABLE ■ SOLVENT REMOVABLE	
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S	
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2	
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET ☐ DRY

EXAMINATION CONDITION	TEMPERATURE	26~28 ℃	PENETRATION TIME	10	min
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10	min
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0580

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	08.05.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	8-5-2014	QC MANAGER REVIEWED	8/5/14	AI REVIEWED	

FOR QPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0581-002
		DATE: 08-25 -2014

JOB NO.	RF2-0581	ITEM NO. / ITEM NAME	WHEEL ASS'Y (A)
CUSTOMER	**台朔重工/LMYP-1**	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0556-51 B-RF2-0556-52	MATERIAL / THICKNESS	A514-B / (8T/10T/22T)

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT		
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING	☐	
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE	☐ POST EMULSIFIABLE		
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S			
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2			
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET	☐ DRY	
EXAMINATION CONDITION	TEMPERATURE	26~28 ℃	PENETRATION TIME	10 min	
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min	
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0581

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	08.25.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	8-25-2014	QC MANAGER REVIEWED	8/25/14	AI REVIEWED	

FOR SCPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0582-002
		DATE: 09-09 -2014

JOB NO.	RF2-0582	ITEM NO. / ITEM NAME	WHEEL ASS'Y (B)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0579-51 B-RF2-0579-52	MATERIAL / THICKNESS	A514-B / (8T/16T/30T/45T)

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT		☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT		☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING		☐ MACHINING		☐
TECHNIQUE	METHOD	■ COLOR CONTRAST		☐ FLUORESCENT	
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING		☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE		☐ POST EMULSIFIABLE	
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S			
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2			
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2		■ WET	☐ DRY
EXAMINATION CONDITION	TEMPERATURE	26~29 ℃	PENETRATION TIME	10 min	
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min	
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0582

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

FOR CPC REFERENCE ONLY

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	09.09.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	9-9-2014	QC MANAGER REVIEWED	9/9/14	AI REVIEWED	

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0580-004
		DATE: 08-26 -2014

JOB NO.	RF2-0580	ITEM NO. / ITEM NAME	VENTURI TUBE(B)
CUSTOMER	**台朔重工/LMYP-1**	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	A-RF2-0555-76	MATERIAL / THICKNESS	SS400/10T

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT	
	PENETRANT / DEVELOPER APPLICATION	☐ BRUSHING	■ SPRAYING	☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE	☐ POST EMULSIFIABLE	
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S		
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2		
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET	☐ DRY

EXAMINATION CONDITION	TEMPERATURE	28 ℃	PENETRATION TIME	10 min
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000LX.			

PART EXAMINED:. ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0580(B)

Q'TY:1 PCS

內側焊道(100%)

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	08.26.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	8.26-2014	QC MANAGER REVIEWED	8/26/14	AI REVIEWED	

FOR QC PC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0581-003
		DATE: 09-01 -2014

JOB NO.	RF2-0581	ITEM NO. / ITEM NAME	VENTURI TUBE(B)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	A-RF2-0556-76	MATERIAL / THICKNESS	SS400/10T
TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT ☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT ☐ AFTER FORMING ☐ ☐		
SURFACE PREPARATION	■ GRINDING ☐ MACHINING ☐		
TECHNIQUE	METHOD: ■ COLOR CONTRAST ☐ FLUORESCENT		
	PENETRANT / DEVELOPER APPLICATION: ☐ BRUSHING ■ SPRAYING ☐		
PENETRANT MATERIALS	TYPE: ☐ WATER WASHABLE ☐ POST EMULSIFIABLE ■ SOLVENT REMOVABLE		
	CLEANER / REMOVER: MAGNAFLUX, Spotcheck: SKC-S		
	PENETRANT: MAGNAFLUX, Spotcheck: SKL-SP2		
	DEVELOPER: MAGNAFLUX, Spotcheck: SKD-S2 ■ WET ☐ DRY		
EXAMINATION CONDITION	TEMPERATURE: 28 ℃	PENETRATION TIME: 10 min	
	DRYING TIME: ※ (1) 3 (2) 5 min	DEVELOPING TIME: 10 min	
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.		

PART EXAMINED:. ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0581(B)

Q'TY:1 PCS

內側焊道(100%)

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	09.01.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	9-1-2014	QC MANAGER REVIEWED	9/1/14	AI REVIEWED	

FOR SCPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0580-003
		DATE: 08-19 -2014

JOB NO.	RF2-0580	ITEM NO. / ITEM NAME	WHEEL ASS'Y (B)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0555-51 B-RF2-0555-52	MATERIAL / THICKNESS	A514-B / (8T/19T)
TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐
TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING ☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ☐ POST EMULSIFIABLE ■ SOLVENT REMOVABLE	
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S	
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2	
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET ☐ DRY

EXAMINATION CONDITION	TEMPERATURE	28 °C	PENETRATION TIME	10	min
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10	min
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0580

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	08.19.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	8/19-2014	QC MANAGER REVIEWED	8/19/14	AI REVIEWED	

FOR SCPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0581-004
		DATE: 09-29 -2014

JOB NO.	RF2-0581	ITEM NO. / ITEM NAME	WHEEL ASS'Y (B)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0556-51 B-RF2-0556-52	MATERIAL / THICKNESS	A514-B / (8T/10T/22T)

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT	
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING	☐
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE	☐ POST EMULSIFIABLE	
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S		
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2		
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET	☐ DRY

EXAMINATION CONDITION	TEMPERATURE	26~28 ℃	PENETRATION TIME	10 min
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000LX.			

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0581

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	09.29.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	9-29-2014	QC MANAGER REVIEWED	9/29/14	AI REVIEWED	

FOR SCPC REFERENCE ONLY

CHENG CHEN MACHINERY CO., LTD.	**LIQUID PENETRANT EXAMINATION REPORT**	NO. PTR-RF2-0582-001
		DATE: 08-25 -2014

JOB NO.	RF2-0582	ITEM NO. / ITEM NAME	WHEEL ASS'Y(A)
CUSTOMER	台朔重工/LMYP-1	CHECKLIST NO.	-
APPLICABLE CODE	ASME SEC.V, SEC.Ⅷ DIV.1 (2010 ED+2011AD)	PROCEDURE NO. / REV. NO.	NDE-PT-01 Rev.10
DWG NO. / REV. NO.	B-RF2-0579-51 B-RF2-0579-52	MATERIAL / THICKNESS	A514-B / (8T/16T/30T/45T)

TEST STAGE	☐ PREPARED EDGE ■ AS WELDED ☐ AFTER PWHT	☐ AFTER HYDRO. TEST ☐ AFTER EX PANDED ☐ BEFORE PWHT	☐ AFTER FORMING ☐ ☐
SURFACE PREPARATION	■ GRINDING	☐ MACHINING	☐

TECHNIQUE	METHOD	■ COLOR CONTRAST	☐ FLUORESCENT		
	PENETRANT / DEVELOPER APPLICATION	■ BRUSHING	☐ SPRAYING	☐	
PENETRANT MATERIALS	TYPE	☐ WATER WASHABLE ■ SOLVENT REMOVABLE	☐ POST EMULSIFIABLE		
	CLEANER / REMOVER	MAGNAFLUX, Spotcheck: SKC-S			
	PENETRANT	MAGNAFLUX, Spotcheck: SKL-SP2			
	DEVELOPER	MAGNAFLUX, Spotcheck: SKD-S2	■ WET	☐ DRY	
EXAMINATION CONDITION	TEMPERATURE	26~28 ℃	PENETRATION TIME	10 min	
	DRYING TIME	※ (1) 3 (2) 5 min	DEVELOPING TIME	10 min	
LIGHTING EQUIPMENT	FLASHLIGHT WAS USED DURING EXAMINATION AND EVALUATION OF INDICATIONS, RESULTED IN LIGHT INTENSITY OVER 1000Lx.				

PART EXAMINED: . ※ (1)AFTER PRE-CLEANING (2)AFTER REMOVAL PENETRANT

RF2-0582

Q'TY:1 PCS

FILLET WELD OF BLADE TO WHEEL PLATE EACH SIDE (100%)-FULL

	LEVEL	SIGN	DATE	JUDGEMENT	ATTACHED REPORT
EXAMINER	II	陳朝升 LEVEL-II	08.25.2014	■ ACCEPTABLE	
APPROVED	III			☐ REJECT	

INSPECTION DIV. HEAD REVIEWED	8-25-2014	QC MANAGER REVIEWED	8/25/14	AI REVIEWED	

FOR SQPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0580	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: S.A.FAN WHEEL(A)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd/3rd N/A μm INTER. COAT	4th/5th N/A μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底漆 PRIMER.COAT : RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT :	面漆 TOP.COAT :

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
WHEEL	Sa 2 1/2	89	N/A	N/A	N/A	OK	ACCEPTABLE	

FOR SCPC REFERENCE ONLY

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y. K. Chen	審查 CHECKED BY	承認 APPROVED BY M.D. Hong

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0580	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: S.A.FAN HOUSING&ACCESSORY(A)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T.	:	1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT	
油漆廠牌/種類 BRAND/KIND	:	底 漆 PRIMER.COAT	: RAINBOW EPOXY ALLOY PRIMER NO.1020	中 塗 漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050
面 漆 TOP.COAT	:	1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2			

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
FAN HOUSING	Sa 2 1 / 2	91	127	210	295	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1 / 2	85	126	222	308	OK	ACCEPTABLE	
DAMPER	Sa 2 1 / 2	85	119	206	292	OK	ACCEPTABLE	
INLET CONE	Sa 2 1 / 2	81	120	208	283	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1 / 2	83	124	213	295	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查	審 查	承 認
INSPECTION BY Y.K. Chen	CHECKED BY	APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0580	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: S.A.FAN SILENCER&RAIN HOOD (A)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底漆 PRIMER.COAT : RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT : 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT : 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
SILENCER	Sa 2 1 / 2	79	115	201	276	OK	ACCEPTABLE	
RAIN HOOD	Sa 2 1 / 2	83	116	199	282	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chew	審查 CHECKED BY	承認 APPROVED BY H.D. Hoang

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO :	RF2-0580	工程名稱 JOB NAME :	LMYP-1 PROJECT
品名 NAME OF PARTS :	S.A.FAN VENTURI (A)	客戶名稱 CUSTOMER :	FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底漆 PRIMER.COAT : RAINBOW EPOXY ALLOY PRIMER NO.1020	中塗漆 INTER.COAT : 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT : 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
VENTURI	Sa 2 1 / 2	83	125	215	288	OK	ACCEPTABLE	
DUCT	Sa 2 1 / 2	84	115	198	269	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0581	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: P.A.FAN WHEEL(A)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T.	: 1st 75 μm PRIMER COAT		2nd/3rd N/A μm INTER. COAT		4th/5th N/A μm TOP COAT
油漆廠牌/種類 BRAND/KIND	:	底漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT	:
面漆 TOP.COAT	:				

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm				外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
		1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT			
WHEEL	Sa 2 1/2	93	N/A	N/A	N/A	OK	ACCEPTABLE	

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會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查	審查	承認
INSPECTION BY Y.K. Chen	CHECKED BY	APPROVED BY H.D. Hong

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO :	RF2-0581	工 程 名 稱 JOB NAME :	LMYP-1 PROJECT
品 名 NAME OF PARTS :	P.A.FAN HOUSING&ACCESSORY(A)	客 戶 名 稱 CUSTOMER :	FHI/LMYP-1

標 準 膜 厚 D.F.T. :	1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT		4th+5th 25 / 35 μm TOP COAT	
油漆廠牌/種類 BRAND/KIND :	底 漆 PRIMER.COAT :	RAINBOW EPOXY ALLOY PRIMER NO.1020	中 塗 漆 INTER.COAT :	1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面 漆 TOP.COAT : 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
FAN HOUSING	Sa 2 1 / 2	84	117	206	280	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1 / 2	85	119	204	281	OK	ACCEPTABLE	
DAMPER	Sa 2 1 / 2	91	127	211	290	OK	ACCEPTABLE	
INLET CONE	Sa 2 1 / 2	94	129	215	289	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1 / 2	85	118	207	286	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0581	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: P.A.FAN SILENCER&RAIN HOOD (A)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND	:	底漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT	: 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
SILENCER	Sa 2 1 / 2	83	118	201	274	OK	ACCEPTABLE	
RAIN HOOD	Sa 2 1 / 2	86	122	208	284	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.P. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO :	RF2-0581	工程名稱 JOB NAME :	LMYP-1 PROJECT
品名 NAME OF PARTS :	P.A.FAN VENTURI (A)	客戶名稱 CUSTOMER :	FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底漆 PRIMER.COAT :	RAINBOW EPOXY ALLOY PRIMER NO.1020	中塗漆 INTER.COAT :	1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT :	1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2	

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
VENTURI	Sa 2 1 / 2	79	120	204	277	OK	ACCEPTABLE	
DUCT	Sa 2 1 / 2	86	129	220	297	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.D. Hong

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檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO :	RF2-0582	工 程 名 稱 JOB NAME :	LMYP-1 PROJECT
品 名 NAME OF PARTS :	I.D.FAN WHEEL(A)	客 戶 名 稱 CUSTOMER :	FHI/LMYP-1

標 準 膜 厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd/3rd N/A μm INTER. COAT	4th/5th N/A μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底 漆 PRIMER.COAT : RAINBOW Inorganic zine primer NO.1011	中 塗 漆 INTER.COAT :	面 漆 TOP.COAT :

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
WHEEL	Sa 2 1/2	91	N/A	N/A	N/A	OK	ACCEPTABLE	

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會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0582	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: I.D.FAN HOUSING&ACCESSORY(A)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T.	: 1st 75 μm PRIMER COAT	N/A μm INTER. COAT	2nd 25 μm TOP COAT

油漆廠牌/種類 BRAND/KIND	:	底 漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中 塗 漆 INTER.COAT	:	面 漆 TOP.COAT	: RAINBOW HEAT-RESISTING PAINT,400℃ NO.1513

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm				外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
		1 PRIMER	INTER. COAT	INTER. COAT	1+2 TOP COAT			
FAN HOUSING	Sa 2 1/2	92	N/A	N/A	135	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1/2	92	N/A	N/A	133	OK	ACCEPTABLE	
DAMPER	Sa 2 1/2	89	N/A	N/A	129	OK	ACCEPTABLE	
INLET CONE	Sa 2 1/2	94	N/A	N/A	131	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1/2	95	N/A	N/A	135	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K.Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D.Hong

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檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0580	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: S.A.FAN WHEEL(B)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T.	: 1st 75 μm PRIMER COAT	2nd/3rd N/A μm INTER. COAT		4th/5th N/A μm TOP COAT	
油漆廠牌/種類 BRAND/KIND	: 底漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT	:	面漆 TOP.COAT :

構件名稱 ITEM NAME	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm				外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
		1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT			
WHEEL	Sa 2 1/2	93	N/A	N/A	N/A	OK	ACCEPTABLE	

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會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.D. Hong

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0580	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: S.A.FAN HOUSING&ACCESSORY(B)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T.	: 1st 75 μm PRIMER COAT		2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT	
油漆廠牌/種類 BRAND/KIND	: 底 漆 PRIMER.COAT	: RAINBOW EPOXY ALLOY PRIMER NO.1020	中 塗 漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面 漆 TOP.COAT : 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
FAN HOUSING	Sa 2 1 / 2	91	130	217	294	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1 / 2	86	123	211	285	OK	ACCEPTABLE	
DAMPER	Sa 2 1 / 2	86	130	219	290	OK	ACCEPTABLE	
INLET CONE	Sa 2 1 / 2	84	120	204	281	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1 / 2	85	212	205	283	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0580	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: S.A.FAN SILENCER&RAIN HOOD (B)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T.	:	1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT	

油漆廠牌/種類 BRAND/KIND	:	底 漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中 塗 漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面 漆 TOP.COAT	: 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
SILENCER	Sa 2 1 / 2	91	122	211	283	OK	ACCEPTABLE	
RAIN HOOD	Sa 2 1 / 2	84	117	203	284	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

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檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0580	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: S.A.FAN VENTURI (B)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底 漆 PRIMER.COAT : RAINBOW EPOXY ALLOY PRIMER NO.1020	中 塗 漆 INTER.COAT : 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面 漆 TOP.COAT : 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
VENTURI	Sa 2 1 / 2	79	114	203	275	OK	ACCEPTABLE	
DUCT	Sa 2 1 / 2	86	122	210	278	OK	ACCEPTABLE	

會 驗 人 員 WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0581	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: P.A.FAN WHEEL(B)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd/3rd N/A μm INTER. COAT	4th/5th N/A μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底 漆 PRIMER.COAT : RAINBOW Inorganic zine primer NO.1011	中 塗 漆 INTER.COAT :	面 漆 TOP.COAT :

構件名稱 ITEM NAME	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm				外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
		1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT			
WHEEL	Sa 2 1/2	90	N/A	N/A	N/A	OK	ACCEPTABLE	

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WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO	: RF2-0581	工 程 名 稱 JOB NAME	: LMYP-1 PROJECT
品 名 NAME OF PARTS	: P.A.FAN HOUSING&ACCESSORY(B)	客 戶 名 稱 CUSTOMER	: FHI/LMYP-1

標 準 膜 厚 D.F.T.	: 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT		4th+5th 25 / 35 μm TOP COAT	
油漆廠牌/種類 BRAND/KIND	:	底 漆 PRIMER.COAT	: RAINBOW EPOXY ALLOY PRIMER NO.1020	中 塗 漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050
面 漆 TOP.COAT	: 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2				

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
FAN HOUSING	Sa 2 1 / 2	86	122	212	289	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1 / 2	91	127	222	303	OK	ACCEPTABLE	
DAMPER	Sa 2 1 / 2	92	133	227	296	OK	ACCEPTABLE	
INLET CONE	Sa 2 1 / 2	92	126	213	287	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1 / 2	86	122	211	288	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0581	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: P.A.FAN SILENCER&RAIN HOOD (B)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND	:	底漆 PRIMER.COAT	: RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT	: 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75 μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
SILENCER	Sa 2 1 / 2	89	121	214	286	OK	ACCEPTABLE	
RAIN HOOD	Sa 2 1 / 2	87	123	212	296	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.P. Hou

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO	: RF2-0581	工程名稱 JOB NAME	: LMYP-1 PROJECT
品名 NAME OF PARTS	: P.A.FAN VENTURI (B)	客戶名稱 CUSTOMER	: FHI/LMYP-1

標準膜厚 D.F.T. : 1st 75 μm PRIMER COAT	2nd+3rd 25 / 75 μm INTER. COAT	4th+5th 25 / 35 μm TOP COAT

油漆廠牌/種類 BRAND/KIND	:	底漆 PRIMER.COAT	: RAINBOW EPOXY ALLOY PRIMER NO.1020	中塗漆 INTER.COAT	: 1.RAINBOW EPOXY ALUMINUM TRIPOLYPHOSPHATE PRIMER NO.1075 2.RAINBOW EPOXY M.I.O. PRIMER NO.1050	面漆 TOP.COAT	: 1.RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2 2. RAINBOWPU finish NO.745(UP-04)MUNSELL NO.#2

構件名稱 ITEM NAME	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1(75μm) PRIMER	1~2(100μm) INTER. COAT	1~3(175μm) INTER. COAT	1~5(235)μm TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
VENTURI	Sa 2 1 / 2	89	122	213	289	OK	ACCEPTABLE	
DUCT	Sa 2 1 / 2	83	119	211	300	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工 號 JOB NO ：	RF2-0582	工 程 名 稱 JOB NAME ：	LMYP-1 PROJECT
品 名 NAME OF PARTS ：	I.D.FAN WHEEL(B)	客 戶 名 稱 CUSTOMER ：	FHI/LMYP-1

標 準 膜 厚 D.F.T. ： 1st 75 μm PRIMER COAT	2nd/3rd N/A μm INTER. COAT	4th/5th N/A μm TOP COAT

油漆廠牌/種類 BRAND/KIND ：	底 漆 PRIMER.COAT ： RAINBOW Inorganic zine primer NO.1011	中 塗 漆 INTER.COAT ：	面 漆 TOP.COAT ：

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1 PRIMER	1+2 INTER. COAT	1+2+3 INTER. COAT	1+2+3+4+5 TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
WHEEL	Sa 2 1/2	93	N/A	N/A	N/A	OK	ACCEPTABLE	

會 驗 人 員
WITNESS BY

NOTE : Radom check for each coat

檢 查 INSPECTION BY Y.K. Chen	審 查 CHECKED BY	承 認 APPROVED BY H.D. Hong

FOR SCPC REFERENCE ONLY

檢 查 記 錄 表

PAINTING INSPECTION REPORT

工號 JOB NO :	RF2-0582	工程名稱 JOB NAME :	LMYP-1 PROJECT
品名 NAME OF PARTS :	I.D.FAN HOUSING&ACCESSORY(B)	客戶名稱 CUSTOMER :	FHI/LMYP-1

標準膜厚 D.F.T. :	1st 75 μm PRIMER COAT	N/A μm INTER. COAT	2nd 25 μm TOP COAT

油漆廠牌/種類 BRAND/KIND :	底漆 PRIMER.COAT : RAINBOW Inorganic zine primer NO.1011	中塗漆 INTER.COAT :	面漆 TOP.COAT : RAINBOW HEAT-RESISTING PAINT,400℃ NO.1513

構件名稱 ITEM NAME.	噴砂程度 SWRFACE PREPARATION	膜厚（D.F.T）μm 1 PRIMER	INTER. COAT	INTER. COAT	1+2 TOP COAT	外觀 APPERANCE	判定 JUDGEMENT	備註 REMARKED
FAN HOUSING	Sa 2 1 / 2	94	N/A	N/A	139	OK	ACCEPTABLE	
BEARING PEDESTAL	Sa 2 1 / 2	95	N/A	N/A	133	OK	ACCEPTABLE	
DAMPER	Sa 2 1 / 2	91	N/A	N/A	132	OK	ACCEPTABLE	
INLET CONE	Sa 2 1 / 2	92	N/A	N/A	133	OK	ACCEPTABLE	
MOTOR BASE	Sa 2 1 / 2	92	N/A	N/A	135	OK	ACCEPTABLE	

會驗人員 WITNESS BY

NOTE : Radom check for each coat

檢查 INSPECTION BY Y.K. Chen	審查 CHECKED BY	承認 APPROVED BY M.D. Hong

FOR SCPC REFERENCE ONLY

RF20580

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機(Induction Motor)

申請號碼 Application No.	FD147202B1
申請日期 Application Date	10/09/2014

製造工號 (Serial No.)	FD147202B1-1	額定輸出(Rated Output)	1850 kW
型　式 (Type)	AECK-S2	額定電壓(Rated Voltage)	6600 V
極　數 (Pole)	4	時間額定(Time Rating)	CONT.
週　率 (HZ)	60	絕緣等級(Insulation Class)	F
相　數 (Phase)	3	框　號(Frame No.)	560C
同步轉速 (Syn. R.P.M)	1800	試驗電源(Power Supply)	6600 V 60 HZ

特性 (Performance)

1.絕緣電阻(Insulation Resistance) MΩ/VDC	定子(Stator)	轉子(Rotor)
	19.2K/5000	------
2.耐壓試驗(Dielectric Test) 1分鐘(For 1 Minute Without Injury)	定子(Stator)	轉子(Rotor)
	14200 VA.C	------ VA.C
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)	電壓(Voltage)	電流(Current)
	------ V	------ A

4.無載電流(No Load Current)	40.9 A	5.滿載電流(Full Load Current)	189.0 A
6.滿載效率(Full Load Efficiency)	95.7 %	7.滿載功因(Full Load Power Factor)	89.5 %
8.滿載轉速(Full Load Speed)	1782 R.P.M.	9.滿載轉差率(Full Load Slip)	0.99 %
10.起動電流(Locked Rotor Current)	1229 A	11.起動轉矩(Locked Rotor Torque)	151 %
12.最大輸出(Maximun Output)	223 %	13.停頓轉矩(Break Down Torque)	253 %
14.構造檢查(Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理之。 2)保固期爲壹年。	

附註(Remarks)：

謝謝惠顧

承　認 Authorized by 品管經理 詹孟擎

審　核 Checked by 品管主任 王世益

試驗者 Tested by 品管班長 陳柏翰

FOR SCPC REFERENCE ONLY

RT2-0581

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機(Induction Motor)

申請號碼 Application No.	FE147703B1
申請日期 Application Date	10/09/2014

製造工號 (Serial No.)	FE147703B1-1	額定輸出(Rated Output)	3700 kW
型　式 (Type)	AEZK-S2	額定電壓(Rated Voltage)	6600 V
極　數 (Pole)	4	時間額定(Time Rating)	CONT.
週　率 (HZ)	60	絕緣等級(Insulation Class)	F
相　數 (Phase)	3	框　號(Frame No.)	630D
同步轉速 (Syn. R.P.M)	1800	試驗電源(Power Supply)	6600 V 60 HZ

特性 (Performance)

1.絕緣電阻(Insulation Resistance) MΩ/VDC		定子(Stator)	轉子(Rotor)	
		10K/5000	------	
2.耐壓試驗(Dielectric Test) 1分鐘(For 1 Minute Without Injury)		定子(Stator)	轉子(Rotor)	
		14200 VA.C	------ VA.C	
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)		電壓(Voltage)	電流(Current)	
		----- V	------ A	
4.無載電流(No Load Current)	79.3 A	5.滿載電流(Full Load Current)	371.0 A	
6.滿載效率(Full Load Efficiency)	96.0 %	7.滿載功因(Full Load Power Factor)	90.8 %	
8.滿載轉速(Full Load Speed)	1788 R.P.M.	9.滿載轉差率(Full Load Slip)	0.67 %	
10.起動電流(Locked Rotor Current)	2404 A	11.起動轉矩(Locked Rotor Torque)	153 %	
12.最大輸出(Maximun Output)	244 %	13.停頓轉矩(Break Down Torque)	255 %	
14.構造檢查(Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理之。 2)保固期爲壹年。		
附註(Remarks)：		謝謝惠顧		

FOR SCPC REFERENCE ONLY

承　認 Authorized by	審　核 Checked by	試驗者 Tested by
品管經理 詹孟學	品管主任 王世益	品管班長 陳柏翰

RT2-0582

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機(Induction Motor)

申請號碼 Application No.	FE147704B1
申請日期 Application Date	10/09/2014

製造工號(Serial No.)	FE147704B1-1	額定輸出(Rated Output)	3300 kW
型　式(Type)	AEZK-S2	額定電壓(Rated Voltage)	6600 V
極　數(Pole)	6	時間額定(Time Rating)	CONT.
週　率(HZ)	60	絕緣等級(Insulation Class)	F
相　數(Phase)	3	框　號(Frame No.)	630D
同步轉速(Syn. R.P.M)	1200	試驗電源(Power Supply)	6600 V 60 HZ

特性(Performance)

1.絕緣電阻(Insulation Resistance) MΩ/VDC		定子(Stator)	轉子(Rotor)	
		7.32K/5000	------	
2.耐壓試驗(Dielectric Test) 1分鐘(For 1 Minute Without Injury)		定子(Stator)	轉子(Rotor)	
		14200 VA.C	------ VA.C	
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)		電壓(Voltage)	電流(Current)	
		----- V	------ A	
4.無載電流(No Load Current)	97.5 A	5.滿載電流(Full Load Current)		343.0 A
6.滿載效率(Full Load Efficiency)	96.2 %	7.滿載功因(Full Load Power Factor)		87.5 %
8.滿載轉速(Full Load Speed)	1193 R.P.M.	9.滿載轉差率(Full Load Slip)		0.58 %
10.起動電流(Locked Rotor Current)	2224 A	11.起動轉矩(Locked Rotor Torque)		171 %
12.最大輸出(Maximun Output)	248 %	13.停頓轉矩(Break Down Torque)		256 %
14.構造檢查(Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理之 2)保固期爲壹年		

附註(Remarks)：

謝謝惠顧

FOR SCPC REFERENCE ONLY

品管經理 詹孟學　品管主任 王世益　品管班長 陳柏翰

承　認 Authorized by _____　審　核 Checked by _____　試驗者 Tested by _____

RF2-0580

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機(Induction Motor)

申請號碼 Application No.	FD145202B1
申請日期 Application Date	07/04/2014

製造工號 (Serial No.)	FD145202B1-1	額定輸出(Rated Output)	1850 kW
型　　式 (Type)	AECK-S2	額定電壓(Rated Voltage)	6600 V
極　　數 (Pole)	4	時間額定(Time Rating)	CONT.
週　　率 (HZ)	60	絕緣等級(Insulation Class)	F
相　　數 (Phase)	3	框　　號(Frame No.)	560C
同步轉速 (Syn. R.P.M)	1800	試驗電源(Power Supply)	6600 V 60 HZ

特性 (Performance)

1.絕緣電阻(Insulation Resistance) MΩ/VDC		定子(Stator)	轉子(Rotor)	
		8.42K/5000	------	
2.耐壓試驗(Dielectric Test) 1分鐘(For 1 Minute Without Injury)		定子(Stator)	轉子(Rotor)	
		14200 VA.C	------ VA.C	
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)		電壓(Voltage)	電流(Current)	
		------ V	------ A	
4.無載電流(No Load Current)	39.7 A	5.滿載電流(Full Load Current)	189.0 A	
6.滿載效率(Full Load Efficiency)	95.6 %	7.滿載功因(Full Load Power Factor)	89.6 %	
8.滿載轉速(Full Load Speed)	1782 R.P.M.	9.滿載轉差率(Full Load Slip)	1.00 %	
10.起動電流(Locked Rotor Current)	1228 A	11.起動轉矩(Locked Rotor Torque)	151 %	
12.最大輸出(Maximun Output)	213 %	13.停頓轉矩(Break Down Torque)	253 %	
14.構造檢查(Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理。 2)保固期為壹年。 謝謝惠顧		
附註(Remarks)：				

FOR SCPC REFERENCE ONLY

承認 Authorized by	審核 Checked by	試驗者 Tested by
品管經理 詹孟學	品管主任 王世益	品管班長 陳柏翰

RF2-0581

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機(Induction Motor)

申請號碼 Application No.	FE145702B1
申請日期 Application Date	07/04/2014

製造工號 (Serial No.)	FE145702B1-1	額定輸出(Rated Output)	3700 kW
型　　式 (Type)	AEZK-S2	額定電壓(Rated Voltage)	6600 V
極　　數 (Pole)	4	時間額定(Time Rating)	CONT.
週　　率 (HZ)	60	絕緣等級(Insulation Class)	F
相　　數 (Phase)	3	框　　號(Frame No.)	630D
同步轉速 (Syn. R.P.M)	1800	試驗電源(Power Supply)	6600 V 60 HZ

特性 (Performance)

1.絕緣電阻(Insulation Resistance) MΩ/VDC		定子(Stator)	轉子(Rotor)	
		14.1K/5000	------	
2.耐壓試驗(Dielectric Test) 1分鐘(For 1 Minute Without Injury)		定子(Stator)	轉子(Rotor)	
		14200 VA.C	------ VA.C	
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)		電壓(Voltage)	電流(Current)	
		------ V	------ A	
4.無載電流(No Load Current)	79.8 A	5.滿載電流(Full Load Current)	371.0 A	
6.滿載效率(Full Load Efficiency)	96.0 %	7.滿載功因(Full Load Power Factor)	90.8 %	
8.滿載轉速(Full Load Speed)	1788 R.P.M.	9.滿載轉差率(Full Load Slip)	0.67 %	
10.起動電流(Locked Rotor Current)	2408 A	11.起動轉矩(Locked Rotor Torque)	151 %	
12.最大輸出(Maximun Output)	245 %	13.停頓轉矩(Break Down Torque)	256 %	
14.構造檢查(Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理之。 2)保固期爲壹年。		

附註(Remarks)： 定子溫昇：87.9℃，周溫：21.9℃
L-Brg溫昇：36.7℃，F-Brg溫昇：35.7℃ (S. F.15)

謝謝惠顧

承認 Authorized by	審核 Checked by	試驗者 Tested by
品管經理 詹孟學	品管主任 王世益	品管班長 陳柏翰

FOR SCPC REFERENCE ONLY

RT2-0582

TECO 東元電機股份有限公司
TECO Electric & Machinery Co., Ltd.

馬達出廠證明暨保證卡
MOTOR TEST CERTIFICATE & WARRANTY

感應電動機 (Induction Motor)

申請號碼 Application No.	FE145703B1
申請日期 Application Date	07/04/2014

製造工號 (Serial No.)	FE145703B1-1	額定輸出 (Rated Output)	3300 kW
型　式 (Type)	AEZK-S2	額定電壓 (Rated Voltage)	6600 V
極　數 (Pole)	6	時間額定 (Time Rating)	CONT.
週　率 (HZ)	60	絕緣等級 (Insulation Class)	F
相　數 (Phase)	3	框　號 (Frame No.)	630D
同步轉速 (Syn. R.P.M)	1200	試驗電源 (Power Supply)	6600 V 60 HZ

特性 (Performance)

1.絕緣電阻 (Insulation Resistance) MΩ/VDC		定子 (Stator)	轉子 (Rotor)	
		9.53K/5000	------	
2.耐壓試驗 (Dielectric Test) 1分鐘(For 1 Minute Without Injury)		定子 (Stator)	轉子 (Rotor)	
		14200 VA.C	------ VA.C	
3.轉子二次電壓二次電流 (Rotor Secondary Volts & Current)		電壓 (Voltage)	電流 (Current)	
		------ V	------ A	
4.無載電流 (No Load Current)	94.6 A	5.滿載電流 (Full Load Current)		344.0 A
6.滿載效率 (Full Load Efficiency)	96.2 %	7.滿載功因 (Full Load Power Factor)		87.3 %
8.滿載轉速 (Full Load Speed)	1193 R.P.M.	9.滿載轉差率 (Full Load Slip)		0.58 %
10.起動電流 (Locked Rotor Current)	2230 A	11.起動轉矩 (Locked Rotor Torque)		172 %
12.最大輸出 (Maximun Output)	237 %	13.停頓轉矩 (Break Down Torque)		2230 %
14.構造檢查 (Formation Check)	符合 Good	15.保固內容： 1)保固期間內發生故障，其責任屬於敝公司時，免費修理之。 2)保固期爲壹年。		

附註 (Remarks)： 定子溫昇：87.4℃，周溫：25.6℃
L-Brg溫昇：53.1℃，F-Brg溫昇：42.1℃ (S. F.15)

謝謝惠顧

承認 Authorized by	審核 Checked by	試驗者 Tested by
品管經理 詹孟學	品管主任 王世益	品管班長 陳柏翰

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No :
工 號 RF2-0580A

Date :
日 期 2014.10.28

Customer :
客 戶 台朔重工/LMYP-1

Type Rotor To Be Balanced :
校 正 物 名 稱 葉輪與軸 S.A.FAN

Rotational Speed In Actual Operation :
實 際 迴 轉 速 1780 r.p.m

STD. Applied
應 用 標 準
- ☐ JIS BO905-1978
- ■ ISO 1940/1-1986(E)
  G: 2.5

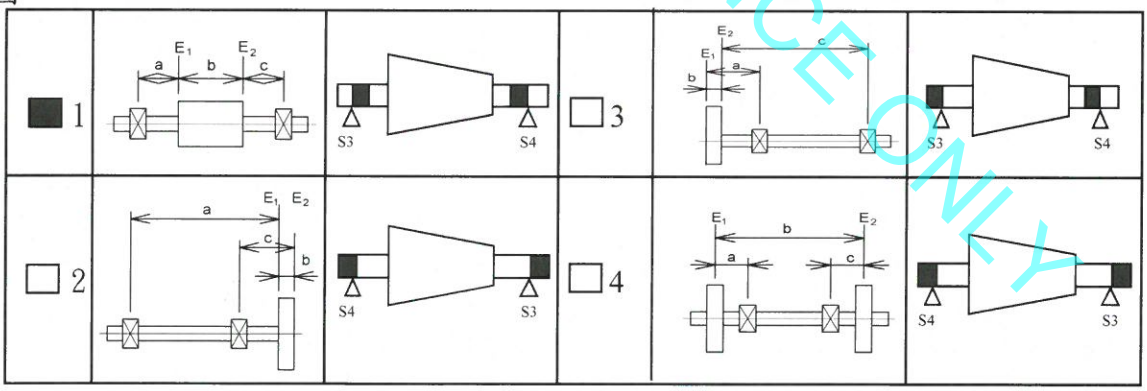
Machine Type
機 種
- ☐ SCHENCK H30N
- ■ SCHENCK H6V
- ☐ IRD 290

Weight Of Rotor :
校 正 物 重 量 4940 Kg

Admissible Unitaty Eccentricity :
容 許 偏 心 量 13.41 μm

Admissible Unbalance For Each Plane :
容 許 單 面 不 平 衡 量 33129.92 g.mm

Sketch
草 圖



a= 1585 mm
b= 420 mm
c= 1580 mm

Radius To Correction
校 正 半 徑

r1（左邊）= 815 mm
r2（右邊）= 815 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 40.65 g

Balancing Record
平衡過程紀錄

RF2-0580A

	RPM 轉速	Lefthand 左側					Righthand 右側				
		Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	209	270			841	822W	295			551	456W
2	343	0			33.5	G	41			18	G
3	343	347			23.8	G	17			13.8	G
4	343	356			16.6	G	23			10.2	G
5	343	90			3.16		29			3.91	
6											
7											
8											

Reading Of Residual Unbalance 殘留不平衡重量	L	3.16	g	R	3.91 g
Reading Unbalance 殘留不平衡量	L	2575.4	g.mm	R	3186.65 g.mm

Result 結 果 (BY Q.C.)

■ Passed 通過
□ Re-balance 重新校正

Date
校正日期 2014/10/18

Operater
校正者 游其川

Superviso
主管 張銘治

Welder
焊工 黃盧朝

Q.C. Witness
品管監校 陳彥氣

Superviso
主管 黃建禪

Customer Witness
客戶監校 (如需會同校正)

MFG DEP'T (製造部) 校正記錄 → Q.C. DEP'T (品管部) 1-COPY → CUSTOMER

1-COPY

1-COPY → ENG DEP'T (技術部)

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No :
工 號 RF2-0581A

Date :
日 期 2014.10.28

Customer :
客 戶 台朔重工/LMYP-1

Type Rotor To Be Balanced :
校 正 物 名 稱 葉輪與軸 P.A.FAN

Rotational Speed In Actual Operation :
實 際 迴 轉 速 1780 r.p.m

STD. Applied
應 用 標 準
- ☐ JIS BO905-1978
- ■ ISO 1940/1-1986(E)
  G: 2.5

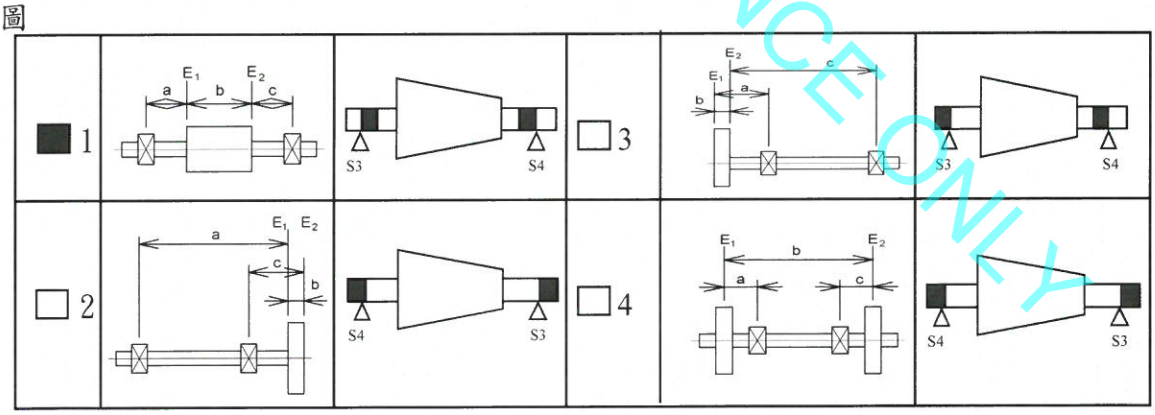
Machine Type
機 種
- ☐ SCHENCK H30N
- ■ SCHENCK H6V
- ☐ IRD 290

Weight Of Rotor :
校 正 物 重 量 6720 Kg

Admissible Unitaty Eccentricity :
容 許 偏 心 量 13.41 μm

Admissible Unbalance For Each Plane :
容 許 單 面 不 平 衡 量 45067.42 g.mm

Sketch
草 圖



a= 1710 mm
b= 480 mm
c= 1710 mm

Radius To Correction
校 正 半 徑

r1（左邊）= 920 mm
r2（右邊）= 920 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 48.99 g

Balancing Record
平衡過程紀錄 RF2-0581A

	RPM	Lefthand 左側					Righthand 右側				
	轉速	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	209	212			176	155W	91			834	835W
2	337	350			16.5	G	97			42.9	G
3	337	347			12	G	104			24.3	G
4	337	352			1.11		138			2.93	
5											
6											
7											
8											

Reading Of Residual Unbalance 殘留不平衡重量	L	1.11 g	R	2.93 g
Reading Unbalance 殘留不平衡量	L	1021.2 g.mm	R	2695.6 g.mm

Result 結果 (BY Q.C.)

■ Passed 通過
□ Re-balance 重新校正

Date
校正日期 2014/10/28

Operater
校正者 游其川

Superviso
主管 張銘治

Welder
焊工 黃盧朝

Q.C. Witness
品管監校 陳彥氫

Superviso
主管 [illegible]

Customer Witness
客戶監校 (如需會同校正)

FOR SCPC REFERENCE ONLY

MFG DEP'T (製造部) 校正記錄 → Q.C. DEP'T (品管部) 1-COPY → CUSTOMER
Q.C. DEP'T → 1-COPY → MFG DEP'T
Q.C. DEP'T --1-COPY--> ENG DEP'T (技術部)

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No :
工 號 RF2-0582A

Date :
日 期 2014, 10, 30

Customer :
客 戶 台朔重工/LMYP-1

Type Rotor To Be Balanced :
校 正 物 名 稱 葉輪與軸 I.D.FAN

Rotational Speed In Actual Operation :
實 際 迴 轉 速 1140 r.p.m

STD. Applied
應 用 標 準
☐ JIS BO905-1978
■ ISO 1940/1-1986(E)
G: 2.5

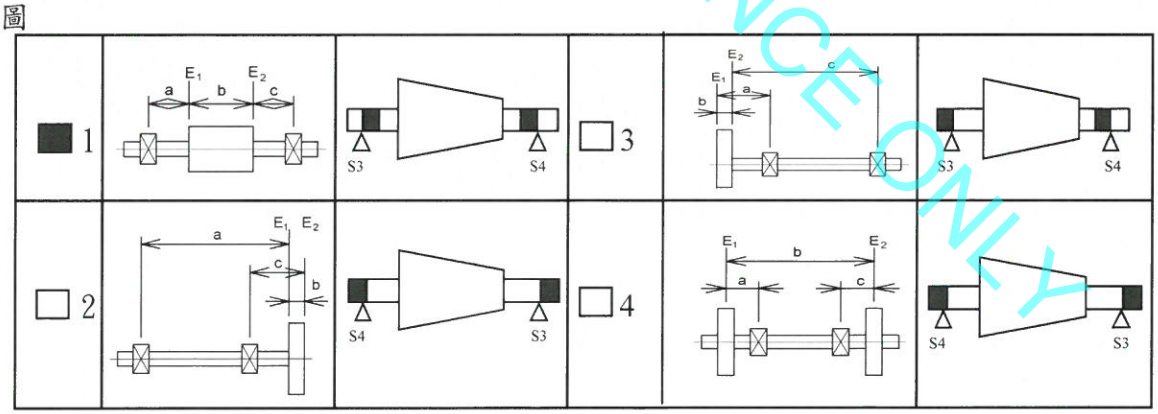
Machine Type
機 種
☐ SCHENCK H30N
☐ SCHENCK H6V
■ IRD 290

Weight Of Rotor :
校 正 物 重 量 15871 Kg

Admissible Unitaty Eccentricity :
容 許 偏 心 量 20.94 μm

Admissible Unbalance For Each Plane :
容 許 單 面 不 平 衡 量 166193.04 g.mm

Sketch
草 圖



a= 2170 mm
b= 1498 mm
c= 2335 mm

Radius To Correction
校 正 半 徑

r1（左邊）= 1111 mm
r2（右邊）= 1111 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 149.57 g

Balancing Record
平衡過程紀錄 RF2-0582A

	RPM 轉速	Lefthand 左側					Righthand 右側				
		Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	230	356			478	626+W	61			453	865+W
2	230	330			36.1		180			45.2	
3											
4											
5											
6											
7											
8											
Reading Of Residual Unbalance 殘留不平衡重量					L 36.1 g					R 45.2 g	
Reading Unbalance 殘留不平衡量					L 40107.1 g.mm					R 50217.2 g.mm	

Result 結 果 (BY Q.C.)

☑ Passed 通過
☐ Re-balance 重新校正

Date
校正日期 _____

Operater
校正者 沈明毅

Superviso
主管 [illegible]

Welder
焊工 黃盧新

Q.C. Witness
品管監校 [illegible]

Superviso
主管 N/A

Customer Witness
客戶監校 李幸勇 10/30

(如需會同校正)

FOR SCPC REFERENCE ONLY

MFG DEP'T (製造部) —校正記錄→ Q.C. DEP'T (品管部) —1-COPY→ CUSTOMER

Q.C. DEP'T —1-COPY→ MFG DEP'T

Q.C. DEP'T —1-COPY→ ENG DEP'T (技術部)

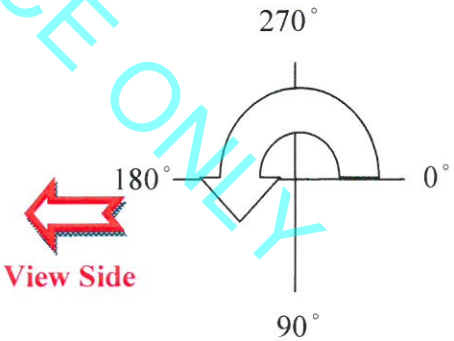
台電電力修護處機械工場
動平衡試驗記錄表

工　　號：IE53312033188　委託單位：建成機械　名　　稱：葉輪與軸(RF2-0582A)
試驗總重量　：15871kg　　試驗轉數：230rpm　傳動方式：□皮帶　■聯軸器
軸 頸 直 徑：左：ϕ152.4mm　右：ϕ203.2mm　　動平衡儀：IRD 295
加 重 半 徑：左：43.74″　　右：43.74″　　試驗日期：103年10月30日

	左側	右側
原有狀況	2.47mil / 332°	45.8mil / 66°
殘留不平衡量	+20.9kg-in / 356°	+198 kg-in / 61°
完成狀況	0.158mil / 330°	0.524mil / 172°
殘留不平衡量	+1.58 kg-in / 330°	+1.98 kg-in / 180°
平衡等級及接受標準	G2.5　7.4868 kg-in	5.5984 kg-in
平　衡　校　正　處　理		
左平衡面	加626g/357°。	
右平衡面	加4365g/66°。	

附註：1. Coupling鍵槽位置為133°，由右側逆時針方向旋轉。
　　2. 實際運轉速1140RPM。

左側　　右側



試驗　洪胡誌　103.10.30　　複核　蕭善敬　103.10.30　　部門主管　蕭善敬　103.10.30

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No : 工 號 RF2-0580B

Date : 日 期 2015.06.08

Customer : 客 戶 台朔重工/LMYP-1

Type Rotor To Be Balanced : 校正物名稱 葉輪與軸 S.A.FAN

Rotational Speed In Actual Operation : 實際迴轉速 1780 r.p.m

STD. Applied 應用標準
- ☐ JIS BO905-1978
- ■ ISO 1940/1-1986(E)
- ■ G: 2.5

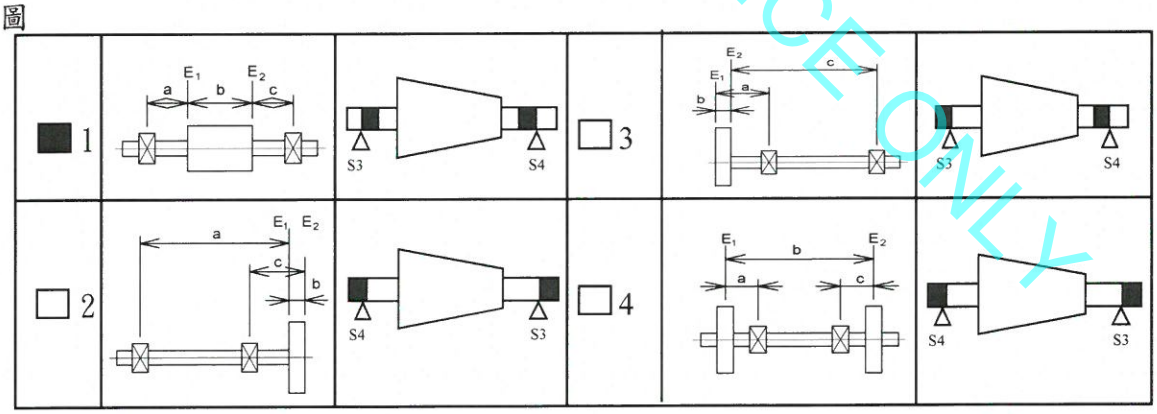
Machine Type 機 種
- ☐ SCHENCK H30N
- ■ SCHENCK H6V
- ☐ IRD 290

Weight Of Rotor : 校正物重量 4940 Kg

Admissible Unitaty Eccentricity : 容許偏心量 13.41 μm

Admissible Unbalance For Each Plane : 容許單面不平衡量 33129.92 g.mm

Sketch 草 圖



a= 1585 mm

b= 420 mm

c= 1580 mm

Radius To Correction 校正半徑

r1（左邊）= 815 mm

r2（右邊）= 815 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 40.65 g

Balancing Record
平衡過程紀錄 RF2-0580B

	RPM	Lefthand 左側					Righthand 右側				
	轉速	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	210	339			321	311+W	186			979	967+W
2	343	136			22.5	G	238			34.3	G
3	342	123			16.5	G	234			21.6	G
4	42	126			14.2	G	236			16.2	G
5	343	122			10.9	G	248			13.4	G
6	343	140			7.76		288			5.74	
7											
8											
Reading Of Residual Unbalance 殘留不平衡重量					L 7.76 g			R 5.74 g			
Reading Unbalance 殘留不平衡量					L 6324.4 g.mm			R 4678.1 g.mm			

Result 結 果 (BY Q.C.)

■ Passed 通過
□ Re-balance 重新校正

Date
校正日期 2015/6/8

Operater
校正者 游其川

Supervisor
主管 張新治

Welder
焊工 劉建池

Q.C. Witness
品管監校 陳彥凱

Supervisor
主管 [illegible]

Customer Witness
客戶監校 (如需會同校正)

FOR SCPC REFERENCE ONLY

MFG DEP'T (製造部) —校正記錄→ Q.C. DEP'T (品管部) —1-COPY→ CUSTOMER
MFG DEP'T ←1-COPY— Q.C. DEP'T
Q.C. DEP'T —1-COPY→ ENG DEP'T (技術部)

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No : 工 號 RF2-0581B

Date : 日 期 2015.06.07

Customer : 客 戶 台朔重工/LMYP-1

Type Rotor To Be Balanced : 校 正 物 名 稱 葉輪與軸 P.A.FAN

Rotational Speed In Actual Operation : 實 際 迴 轉 速 1780 r.p.m

STD. Applied 應 用 標 準
- ☐ JIS BO905-1978
- ■ ISO 1940/1-1986(E) G: 2.5

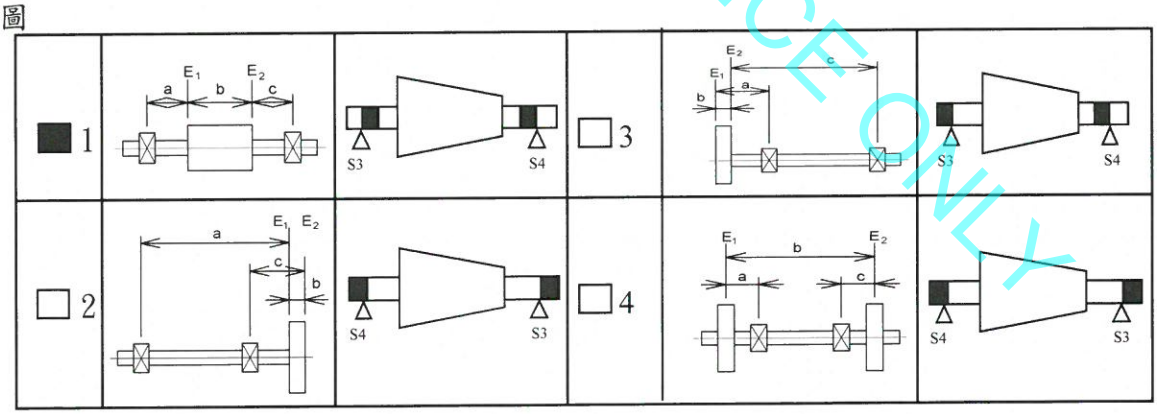
Machine Type 機 種
- ☐ SCHENCK H30N
- ■ SCHENCK H6V
- ☐ IRD 290

Weight Of Rotor : 校 正 物 重 量 6720 Kg

Admissible Unitaty Eccentricity : 容 許 偏 心 量 13.41 μm

Admissible Unbalance For Each Plane : 容 許 單 面 不 平 衡 量 45067.42 g.mm

Sketch 草 圖



a= 1710 mm

b= 480 mm

c= 1710 mm

Radius To Correction 校 正 半 徑

r1（左邊）= 920 mm

r2（右邊）= 920 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 48.99 g

Balancing Record
平衡過程紀錄 RF2-0581B

	RPM 轉速	Lefthand 左側					Righthand 右側				
		Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	209	94			993	961+W	189			65.2	48+W
2	339	73			22.7	G	82			41.7	G
3	338	75			13.6	G	78			31.6	G
4	338	33			10.2	G	75			25.4	G
5	337	355			8.61	G	69			19.8	G
6	337	38			1.36		32			4.44	
7											
8											
Reading Of Residual Unbalance 殘留不平衡重量					L 1.36 g			R 4.44 g			
Reading Unbalance 殘留不平衡量					L 1251.2 g.mm			R 4084.8 g.mm			

Result 結 果 (BY Q.C.)

■ Passed 通過
☐ Re-balance 重新校正

Date 校正日期 2015/6/7

Operater 校正者 游其川

Supervisor 主管 張勝治

Welder 焊工 劉建池

Q.C. Witness 品管監校 陳彩氣

Supervisor 主管 黃世裕

Customer Witness 客戶監校 (如需會同校正)

FOR SCPC REFERENCE ONLY

MFG DEP'T (製造部) 校正記錄 → Q.C. DEP'T (品管部) 1-COPY → CUSTOMER

Q.C. DEP'T (品管部) 1-COPY → MFG DEP'T (製造部)

Q.C. DEP'T (品管部) 1-COPY → ENG DEP'T (技術部)

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

REPORT OF BALANCING
平 衡 校 正 報 告

Order No : 工 號 RF2-0582-B

Date : 日 期 104.05.05

Customer : 客 戶 台朔重工 LMYP-1

Type Rotor To Be Balanced : 校 正 物 名 稱 葉輪與軸

Rotational Speed In Actual Operation : 實 際 迴 轉 速 1140 r.p.m

STD. Applied 應 用 標 準
- ☐ JIS BO905-1978
- ■ ISO 1940/1-1986(E)
- ■ G: 2.5

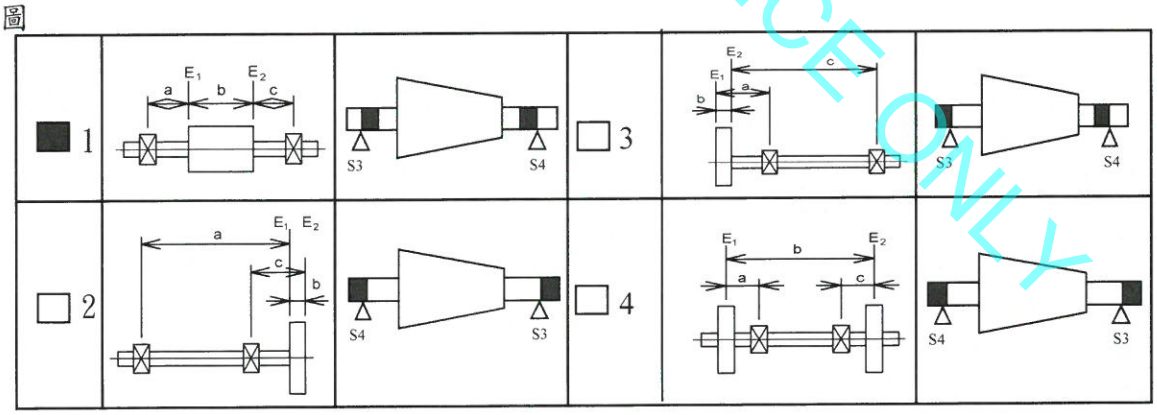
Machine Type 機 種
- ☐ SCHENCK H30N
- ☐ SCHENCK H6V
- ■ IRD 290

Weight Of Rotor : 校 正 物 重 量 15871 Kg

Admissible Unitaty Eccentricity : 容 許 偏 心 量 20.94 μm

Admissible Unbalance For Each Plane : 容許單面不平衡量 166169.37 g.mm

Sketch 草 圖



a= 2170 mm

b= 1498 mm

c= 2335 mm

Radius To Correction 校 正 半 徑

r1（左邊）= 1364 mm

r2（右邊）= 1364 mm

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

Max. Admissible Residual Reading For Either Plane :
一邊容許最大不平衡重量 121.83 g

Balancing Record
平衡過程紀錄 RF2-0582-B

	RPM	Lefthand 左側					Righthand 右側				
	轉速	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量	Ecc.Deg 偏心角度	Read'g 讀數	Sen/div. 感度	Total Ecc 總偏心量	Comp'nWt. 修正量
1	234	308			1903	2270W	327			1772	2270W
2	232	98			49.7		260			52.4	

Reading Of Residual Unbalance 殘留不平衡重量	L	49.7	g	R	52.4 g
Reading Unbalance 殘留不平衡量	L	67791	g.mm	R	71473.6 g.mm

Result 結 果 (BY Q.C.)

■ Passed 通過
□ Re-balance 重新校正

Date
校正日期 104.05.05

Operater
校正者 陳登文

Supervisor
主管 黃文科

Welder
焊工 謝鎮懋

Q.C. Witness
品管監校 [signature]

Supervisor
主管 N/A

Customer Witness
客戶監校 (如需會同校正)

FOR SCPC REFERENCE ONLY

MFG DEP'T (製造部) 校正記錄 → Q.C. DEP'T (品管部) 1-COPY → CUSTOMER
Q.C. DEP'T (品管部) → 1-COPY → MFG DEP'T (製造部)
Q.C. DEP'T (品管部) → 1-COPY → ENG DEP'T (技術部)

3. INSTALLATTION MANUAL

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司

CHENG CHEN MACHINERY CO., LTD.

PURCHASER： FORMOSA HEARY INDUSTRIES CORP.

CUSTOMER： THERMA SOUTH ENERGY COMPANY

INSTRUCTION MANUAL

FOR SCPC REFERENCE ONLY

REF. NO.： XZ-EPNE26-00

FAN NAME： S.A.FAN

MFG. NO.： RF2-0580

FAN TYPE： BCT2118/925 DWDI CWTH A/3

DATE： 07.09.2015

建 成 機 械 股 份 有 限 公 司

CHENG CHEN MACHINERY CO., LTD.

SPECIFICATION

1)	Mfg. No.	:	RF2-0580	
2)	Type	:	BCT2118/925 DWDI CWTH A/3	
3)	Volume	:	4222	ACMM
4)	Pressure	:	2075	mmAQ
5)	Temperature	:	40	℃
6)	Handling gas		AIR	
7)	Fan speed	:	1780	RPM
8)	Driver	:	1850KW×4P×6600V×60HZ	
9)	Volume controlling	:	I.V.C（INLET VALVE CONTROL）	
10)	Transmission method	:	COUPLING	
11)	Coupling type	:	T61-165PF06（PRO-FLEX DISC COUPLING）	
12)	Bearing	:	ZOLLERN ZRWLK 18-160 ZOLLERN ZRWLQ 18-160	
13)	Lubricant	:	ISO VG 32	
14)	No. of fans	:	2 SETS	

FOR SCPC REFERENCE ONLY

建 成 機 械 股 份 有 限 公 司

CHENG CHEN MACHINERY CO., LTD.

PURCHASER： FORMOSA HEARY INDUSTRIES CORP.

CUSTOMER： THERMA SOUTH ENERGY COMPANY

FOR SCPC REFERENCE ONLY

INSTRUCTION MANUAL

REF. NO.： XZ-EPNE27-00

FAN NAME： P.A.FAN

MFG. NO.： RF2-0581

FAN TYPE： BCT2118/1027 DWDI CCWTH A/3

DATE： 07.09.2015

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

SPECIFICATION

1)	Mfg. No.	:	RF2-0581	
2)	Type	:	BCT2118/1027 DWDI CCWTH A/3	
3)	Volume	:	6907	ACMM
4)	Pressure	:	2481	mmAQ
5)	Temperature	:	40	℃
6)	Handling gas		AIR	
7)	Fan speed	:	1780	RPM
8)	Driver	:	3700KW×4P×6600V×60HZ	
9)	Volume controlling	:	I.V.C（INLET VALVE CONTROL）	
10)	Transmission method	:	COUPLING	
11)	Coupling type	:	T81-187PF08（PRO-FLEX DISC COUPLING）	
12)	Bearing	:	ZOLLERN ZRZLK 18-180 ZOLLERN ZRZLQ 18-180	
13)	Lubricant	:	ISO VG 46	
14)	No. of fans	:	2 SETS	

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建 成 機 械 股 份 有 限 公 司

CHENG CHEN MACHINERY CO., LTD.

PURCHASER： FORMOSA HEARY INDUSTRIES CORP.

CUSTOMER： THERMA SOUTH ENERGY COMPANY

FOR SCPC REFERENCE ONLY

INSTRUCTION MANUAL

REF. NO.： XZ-EPNE27-00

FAN NAME： I.D.FAN

MFG. NO.： RF2-0582

FAN TYPE： D1904/10600 DWDI CCWTAU A/3

DATE： 07.09.2015

建 成 機 械 股 份 有 限 公 司
CHENG CHEN MACHINERY CO., LTD.

SPECIFICATION

1)	Mfg. No.	:	RF2-0582	
2)	Type	:	D1904/10600 DWDI CCWTAU A/3	
3)	Volume	:	17621	ACMM
4)	Pressure	:	858	mmAQ
5)	Temperature	:	150	℃
6)	Handling gas		FLUE GAS	
7)	Fan speed	:	1140	RPM
8)	Driver	:	3300KW×6P×6600V×60HZ	
9)	Volume controlling	:	INLET DAMPER	
10)	Transmission method	:	COUPLING	
11)	Coupling type	:	T81-205PF08（PRO-FLEX DISC COUPLING）	
12)	Bearing	:	DODGE 8” RTL PCXC DODGE 6” RTL PCXC	
13)	Lubricant	:	ISO VG 46	
14)	No. of fans	:	2 SETS	

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1. PREFACE

In this INSTRUCTION MANUAL , the essentials and cautions for handling , storage , installation , operation , maintenance and inspection , of the centrifugal fan delivered to customer , is described。

It is required to read this INSTRUCTION MANUAL carefully prior to the operation of the fan at customer ' s site , and then the fan should be handled properly。

Moreover , if any problem or questions occure in the case of operating the fan , the related information should be notified to our head office without hesition。

Items to be noted in the supplied information are as foliows；

1) Manufacturing mumber （stamped on the nameplate）。
2) Fan type（stamped on the nameplate）。
3) Details of problem。

When any defect or trouble occurs becouse of failure to keep the working procedures or precaution described in this INSTRUCTION MANUAL , the cost for repairing , preplacement and the work related to the problem will be charged to the customer。

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2. CAUTIONS FOR HANDLING AND STORAGE OF FAN

2.1. Cautions for handling

2.1.1. In the case of unloading , moving or installing the fan delivered , you should pay attention not to inflict any deformation or damage to the fan by dropping it on the ground or striking it against an object。

You should , in particular , pay special attention when handing the fan so as not to inflict pittings on the rolling surface of bearings by an inflicted impact。

2.1.2. In case electric welding is performed at the site of fan installation , you should provide the welding earth connection to a suitable place so that the bearing of the fan will not be the circuit for the electric welding。

2.1.3. Matchmarks are put or stamped on the fan which is transported after disassembling in order to avoid misassembling at installation site , therefore much care should be taken of these matchmarks where the fan is assembled again。

There are the following kinds of a matchmark ;

1) In case same type fans are more than two , matchmark which means numerical order。（example）

   1 、 2 、 3 、——

2) Matchmark which shows the position and/or direction to be fitted。（example）

   H or H-SIDE ：showing a component fitted on held bearing side。

   H or F-SIDE ：showing a component fitted on held bearing side。

   UP ：meaning to fit a component so as to this mark on the upper part of it in upright direction。

   → ： showing rotational direction arrow of a impeller。

   A — A 、 B — B 、—— ：showing the mutual matchmark adjoining component。

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2.2. Cautions for storage

2.2.1. Storage of fan before installation

1) Delivered fan should be stored indoors。
The article , especially , includes any part which needs rust Prevention , such as bearings , should be stored indoors in a place which has good air-permeability and dry atmosphere
In case components or parts to be unavoidably stored outdoors Such as casing , etc 。should be maintained by use of sleepers And water-proof canvas sheets。
2) Inspection should be carried out periodically to detect rust , dust adherence , partial deformation or breakage , and not to have parts lost during storage。
3) In case the storage goes on for a long term the bearing part should be carried out the visual inspection around onece every 6 months。（In a rainy season or hot and humid environment, especial caution must be paid to rust prevention。）
After the inspection , rust prevention should be carried out as follows;
Rolling bearings lubrcated with oil。
After the residual rust preventives have been wiped off well , rust-proof lubricating oil equivalent to C.P.C. R68 should be applied。

2.2.2. Maintaining the fan under good condition with commisioning。

1) In the case of a short term , at the time of installation , the rust preventives in the bearings must be removed well , and the specified lubricatants should be applied。
2) In the case of a long term , the inspection of bearings and the treatment of rust prevention should be carried out in the same way as described in the aforesaid article 2.2.1 3。
3) The impeller should be turned about 10 times by hand once every month。

FOR SCEC REFERENCE ONLY

2.2.2.

4) In the case of the installed fan used outdoors the bearing part and the motor should be covered with a water-proof canvas sheet to maintain the fan under good condition。
5) In winter , it is necessary to drain water from the cooling water piping in order to avoid the cracking of pipes。
6) Where the oil-lubricated rolling bearing , the bearing shims of 0.1 mm in thickness are put on the split surfaces of horizontal split type bearing case。（Refer to Fig 。1）
This purpose is keeping some clearance in radial direction between the outer ring of bearing and the bearing case for the sliding of bearing caused by the expansion of shaft。
As these shims have very impotant function as mentioned above , they must be put on the split surfaces at the time of reassembling the bearing case after checking it。

FOR SORC REFERENCE ONLY

3. INSTALLATION PROCEDURE

Spquence	Item of execution	Practical procedure
3.1.	Installation of equipment	
3.1.1.	Unpacking	1) Checking of parts。 The articles should be checked with the packing list and it should be confirmed that there is no excess or lack of quantities , deformation or damage to the parts。
3.1.2.	Preparation of foundation	1) Checking the dimension of foundation。 Afterb confirming the center line marked on the concrete foundation on which the fan will be installed , the concrete foundation should be checked in accordance with the layout drawing of foundation bolts which is shown in an attached general view drawing , and it is to be confirmed , whether the relative position of foundation boxes are acceptable or not。 tolerance of reference dimensions are ±10 mm。

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Spquence	Item of execution	Practical procedure
3.1.2.		2) Fixation of packer plate。 The packer plate should be fixed with nonshrinking mortar pad （Nonshrink etc。）at the position where a foundation bolt is situated between two packer plates。 At this time , the height from upper Surface of packer plate to the center Line of fan's shaft should be Adjusted so as to be equal to the Designed height （center height of Fan＋thickness of taper liners）。 The levelness on the upper surface of Each packer plate must be maintained 0.5 mm/m in tolerance , and the difference of height with other plate should be max 。1mm。

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Spquence	Item of execution	Practical procedure
3.1.3.	Installation of the lower casing	1) In case the lower casing is split Vertically , the gaskets（non asbestos yarn）are adhered with adherence by same means as shown on Fig ◦13 to the split flange surface of lower casing , then the split lower casings are assembled to one unit◦ 2) The lower casing is to be lifted up horizontally and shifted to the position upon the location to be installed◦ Then it shall be lowered as both centers of fan shaft and outlet are met each center of foundation and positioned on the soleplates or the taper liners ◦ Foundation bolts for the casing are fitted with the baseplate of it and kept in free condition in the foundation boxes of concrete foundation◦ 3) The gaskets（non asbestos yarn）are adhered with adherence by same means as shown on Fig◦13 to the split flange surface of lower casing◦

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Spquence	Item of execution	Practical procedure
3.1.4.	Installation of	1) Both shaft center lines of held and free side pedestals should be put upon the center line of foundation。 Furthermore they should be positioned as both distance（M1）and （M2）from the center line of outlet of fan to the center line of outlet pedestal are met to the measurement shown on the general view to the measurement shown on the general view drawing。 2) After removal of the rust preventives coated on the machined top surfaces of pedestals , both levelness in axial direction and direction of right angle to the shaft should be measured with a level gauge and adjusted by taper liners under the pedestal so as to be less than 0.5mm/m in the deviation。 At the same time , the difference of mutual height of both bearing pedestals should be measured with a transit , and be adjusted by taper liners so as to be less than 1 mm in tolerance。

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Spquence	Item of execution	Practical procedure
3.1.4.		3) Foundation bolts and washers of pedestals are assembled on the base plates of pedestals and the foundation bolts are kept with hanging condition in the foundation boxes of foundation。

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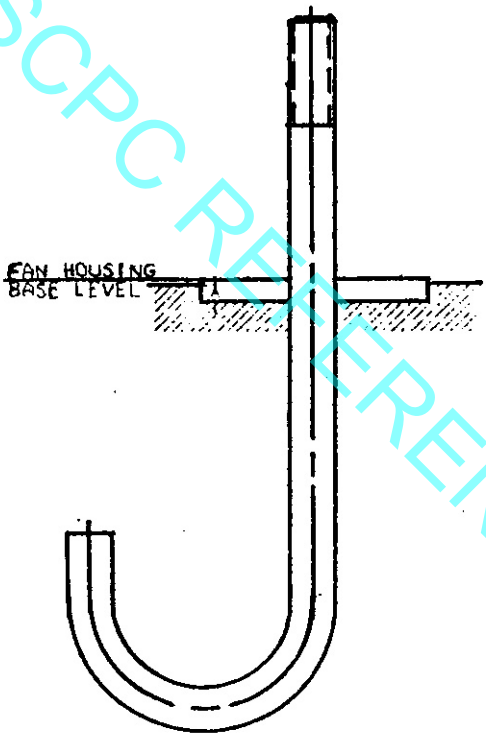
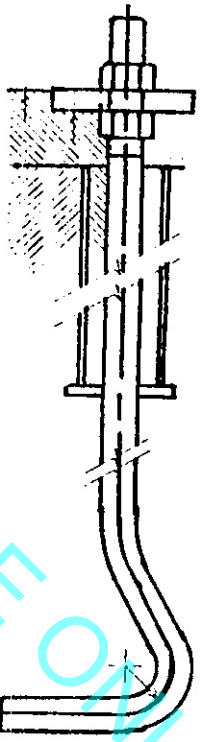


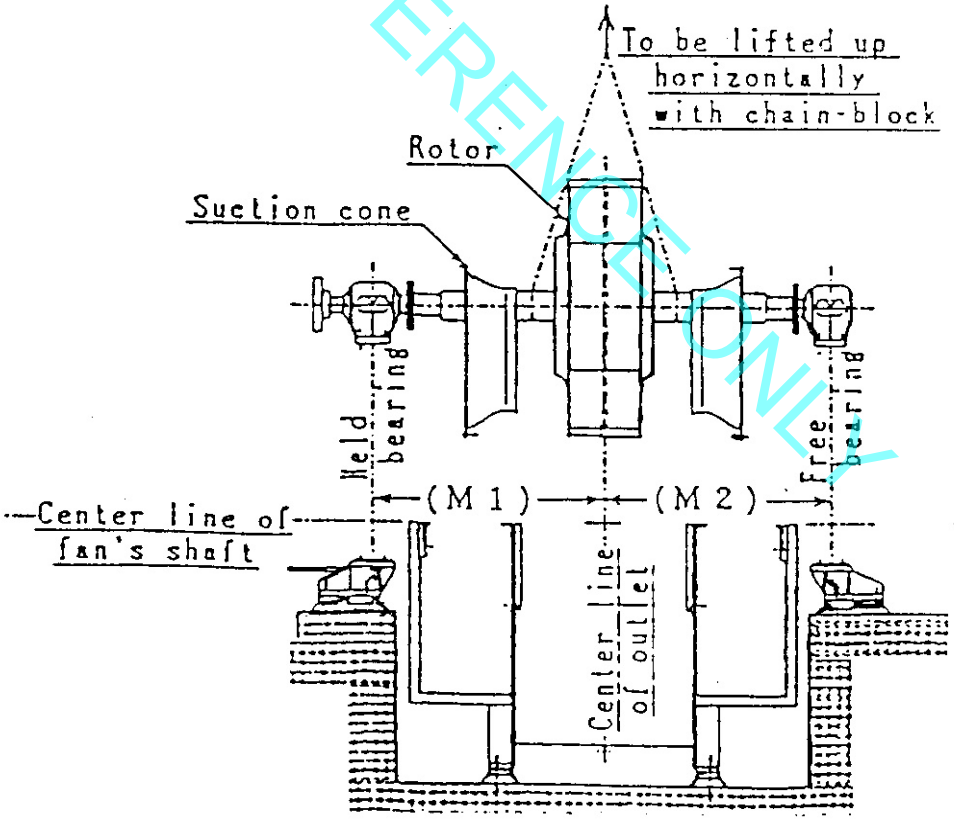
CONTROL DRIVE
ANCHOR BOLT DETAIL



ANCHOR BOLT DETAIL

Spquence	Item of execution	Practical procedure
3.1.5.	Installation of roter	

1) The shaft of roter assembled in auunit with a impeller , shaft and bearings is inserted into suction cone and shifted to the place installed after lifted up horizontally by chain-block。
The roter is lowered slowly , then held and free bearing cases are put on each pedestal。
After knock pins are drived into the knocking holes of both bearing cases and pedestal for positioning , the bearing cases should be fixed on the pedestals with bolts。



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Spquence	Item of execution	Practical procedure
3.1.6.	Mounting of upper casing	1) In case the upper casing is splint vertically , the split upper casings are assembled to one unit by same means as 3.1.4 1。 2) After the opper casing is lifted up and shifted to above the lower casing , the direction of it is confirmed and then it is-lowered slowly as special care should be taken so as not to touch it to the impeller。 The upper casing is placed on the lower casing according to the insertion of the guide rods such as taper pin into the few bolt holes of their split flange during lowering。 3) Both upper and lower casings are tightened uniformly with bolts and nuts。

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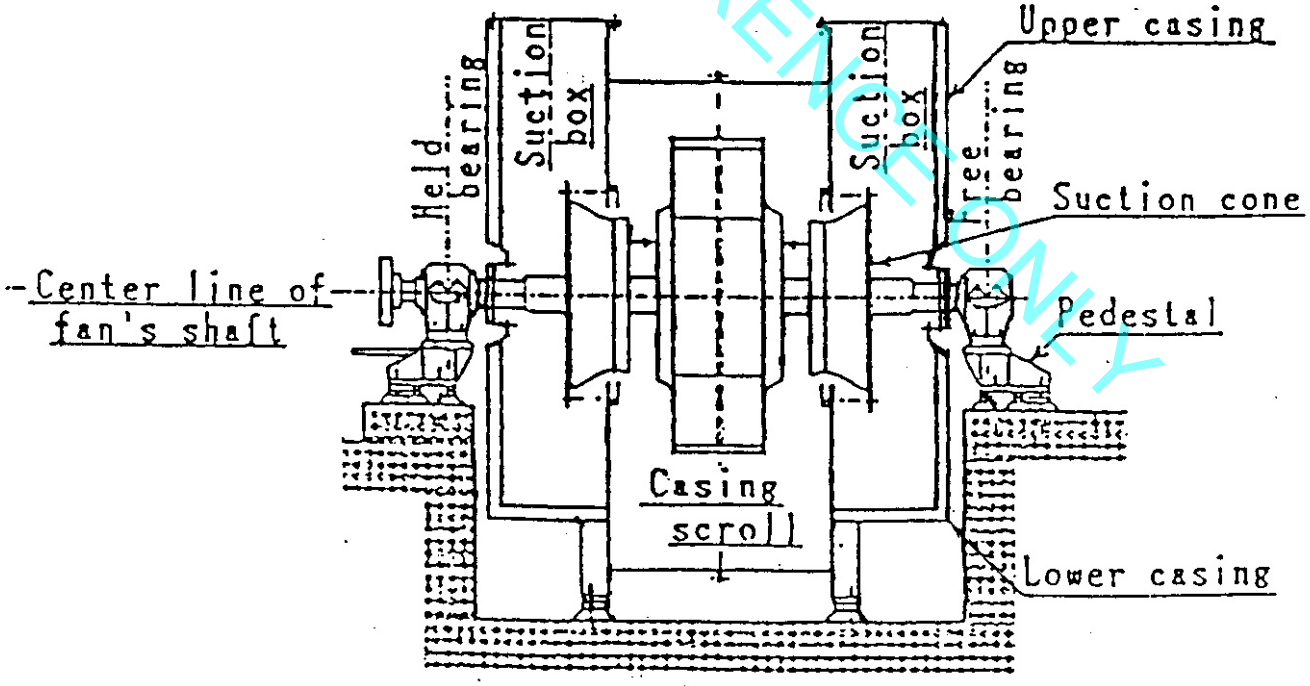
Spquence	Item of execution	Practical procedure
3.1.7.	Assembling of suction cones	1) The matchmarks〝UP〞on suction cones are put so as to the upper part of them , and then they are moved to the position placed according to handling from suction box which is able to enter pass through a inspection holes or inlet。Special care should be taken so as not to touch to the impeller at the time fitting seats of casing。Then the suction cones are positioned so that both holes of suction cones and casing are meet uniformly , and finally they are fixed to the casing with bolts。



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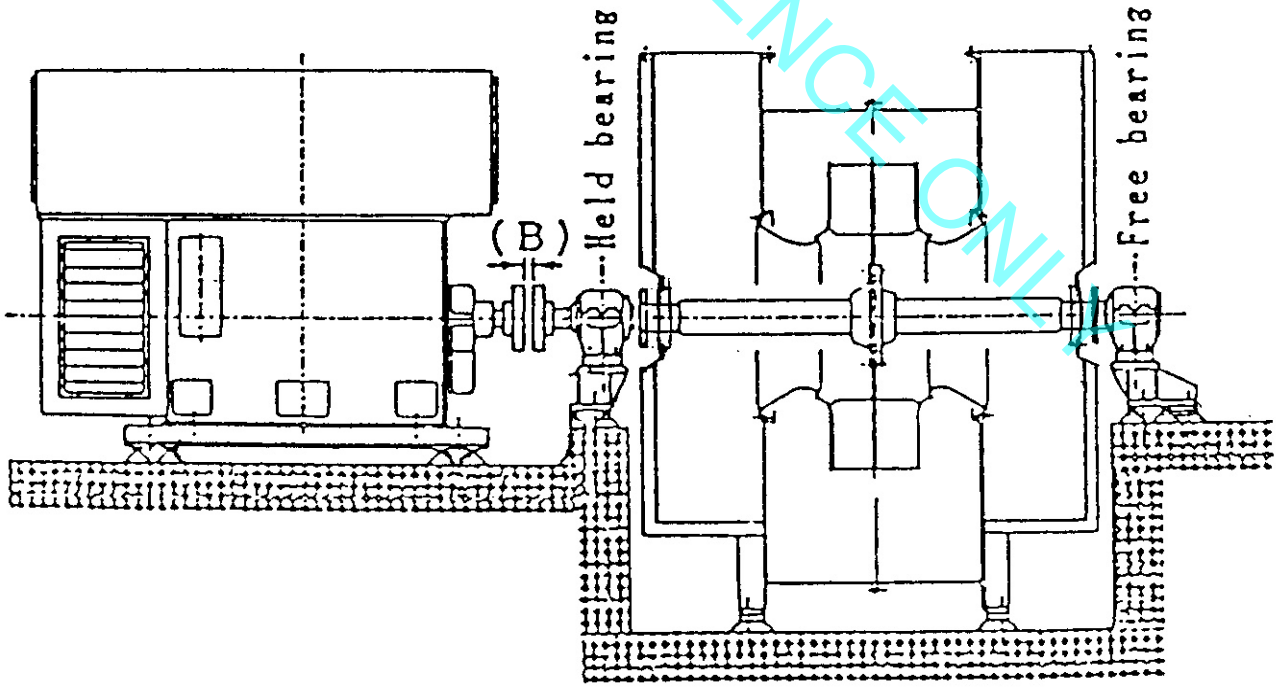
Spquence	Item of execution	Practical procedure
3.1.8.	Installation of motor	

1) The motor is shifted to the position upon the location to be installed after lifting it up horizontally , then it shall be lowered slowly as the center of motor shaft is met to the center of foundation and positioned on the place that the space （B）between the coupling surfaces of fan side and motor side meets a figure shown on the general view drawing。

2) Foundation bolts of motor are fitted with the baseplate of it and kept in free condition in the foundation boxes of concrete foundation。



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Spquence	Item of execution	Practical procedure
3.1.9.	Fixing and tightening of foundation bolts	1) When foundation bolts are fixed with concrete foundation , these bolts should be settled vertically through the center of foundation holes of baseplate。 then concrete is poured into the foundation boxes of concrete foundation and tamped sufficiently with an iron rod。 2) The curing time of the concrete needs more than 150 hours。 3) The foundation bolts should be tightened up uniformly。 In case where the soleplates are put under the baseplate of casing paragraph 3.1.3 2） is refered。

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Spquence	Item of execution	Practical procedure
3.2.	Adjustment of equipment	
3.2.1.	Adjustment of bearing case	

1) The suitable gap in axial direction between the free bearing and free side bearing case against expansion and contraction of the shaft should be provided。 Therefore the end cover is taken off and the unit of bearing case with pedestal should be moved and adjusted so that dimension （F）of Fig .9 is in the allowance of the symbol（F）on the drawing of 〝Allowable dimension on installation〞 。

2) In order to adjust the degree of right angle between shaft and the end surface of outer ring of bearing, the dial gauge point is touched to the end surface of bearing case , and the gradient of bearing case should be adjusted so as the value（C1）and （C2）measured by the turning of shaft at four points in equal top , bottom , right and left on the circumferance are in allowance of the symbol （C1）and（C2）on the drawing of 〝Allowable dimension on installation〞 。

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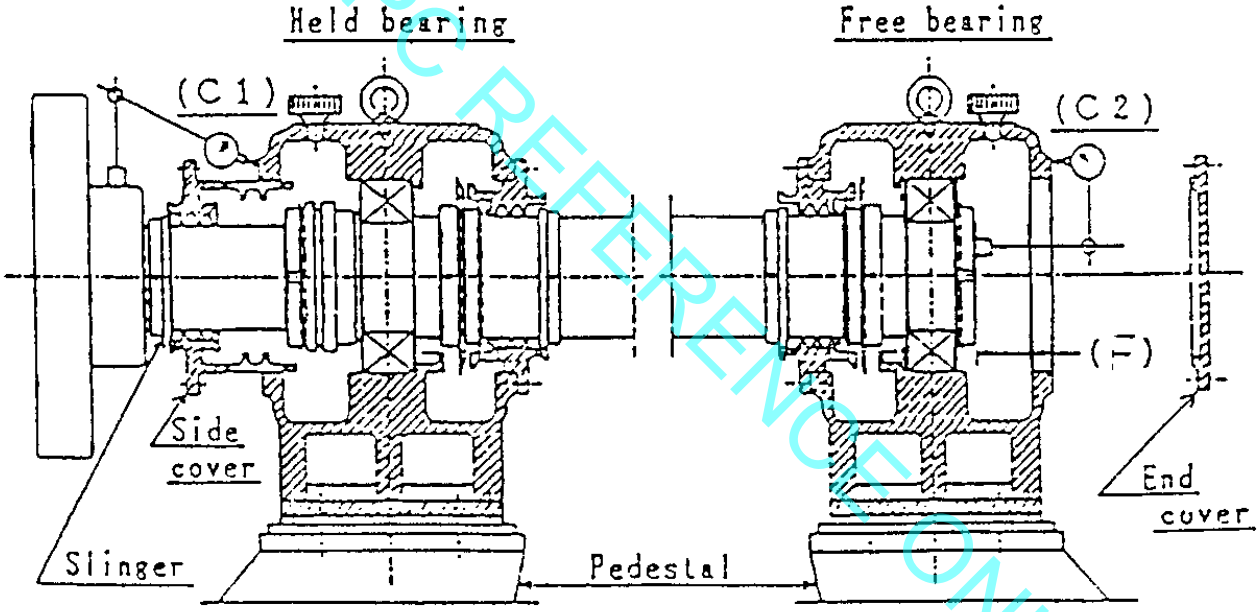
Spquence	Item of execution	Practical procedure
3.2.1.		3) The gap between bearing case in side dimension and bearing out side dimension are measured by plastigage or lead at one point in equal of top and after confirming wether they are in allowance of the symbol（G）（H）show in the drawing of 〝Allowable dimension on in stallation〞。



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Spquence	Item of execution	Practical procedure
3.2.2.	Adjustment of casing	1) The gap between impeller and suction cone are measured by a taper gauge or thickness gauge at four points in equal of top , bottom, right and left on the circumference and after confirming whether they are in allowance of the symbol（F1）, （F2）shown in the drawing of 〝Allowable dimension on installation〞, the record should be kept。 This measurement is performed in the casing scroll according to entering from the inspection hole or outlet。 in case some corrections are required, they should be corrected to the impeller by horizontal movement of the casing unit with suction cone and vertial adjustment according to use of taper liners under the casing in order to be within the allowance。

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Spquence	Item of execution	Practical procedure
3.2.2.		2) The overlaps between impeller and Suction cone are measured by a scale With hook at four points in equal of top, bottom, right and left on the circumference, and after confirming Whether they are in allowance of the symbol（E1）, （E2）shown in the drawing 〝Allowable dimension on installation〞, the record should be kept。 In case some corrections are required, they should be corrected to the impeller by horizontal movement of the casing unit with suction cone in order to be within the allowance。 3) The gaskets（asbestos yarn）are adhered with adherence by same means as shown on fig .13 to the fixing surface of shaft seal cover , and it is fitted to the shaft seal parts of the held and free side of casing so that bolt holes should be meet uniformly。 The gaps of shaft seal part between shaft and casing are measured by a scale and the like at four points in equal of top , bottom , right and left and after confirming whether they are in allowance of the symbol（D1）, （D2）shown in the drawing 〝Allowable dimension on installation〞, the record should be kept。

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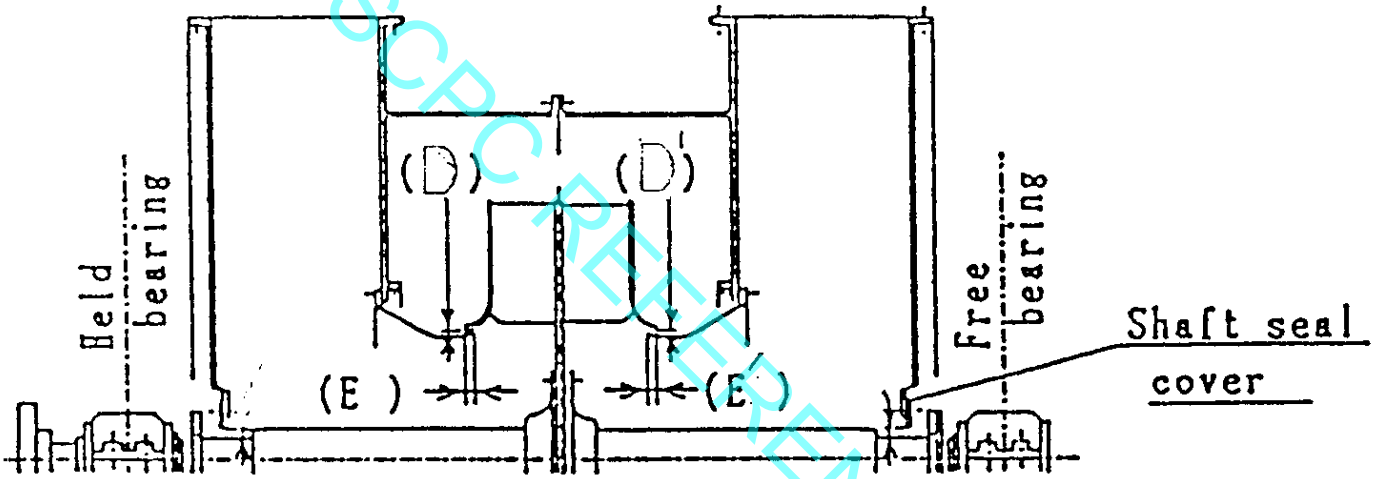
Spquence	Item of execution	Practical procedure
3.2.2.		In case some corrections are required , they should be corrected to the shaft by the same means as described on paragraph 3.2.2 1）in order to be within the allowance。 After the finish of all correction, the shaft seal packings should be fitted with bolts on the casing wall。



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Spquence	Item of execution	Practical procedure

3.2.3. Adjustment of coupling alingnment

1) In order to confirm the alignment of the coupling , first all preventives coated on it should be removed with kerosene to getting clean machined surface。

2) The space between both faces of coupling should be measured with a thickness gauge or calipers in order to confirm whether it is in allowance of symbol（B）shown in the drawing of 〝Allowable dimension on installation〞 。 As the adjusting method , the coupling for variable speed controller side should be adjusted to the fan side coupling by means of horizontally moving the variable speed controller position so as to be within the allowance。

3) The inclination（parallelism）of coupling face should be measured with a dial gauge at four points in equal of top, bottom ,right and left in order to confirm wheter it is in allowance of symbol（B'）SHOWN IN THE DRAWING OF 〝Allowable dimension on installation〞 。
It is adjusted by means of horizontally moving the position of variable speed controller in the direction of right and left ,and adjusting with taper liners in direction of up and down。

4) Misalignment of coupling （A）should be measured with a dial gauge at four points on the coupling periphery in equal top , bottom , right and left in order to confirm whether it is in allowance of symbol（A）shown on the drawing of 〝Allowable dimension on installation〞 。
It is adjusted by the same meaning as descrived on paragraph 2.2.3 3）。

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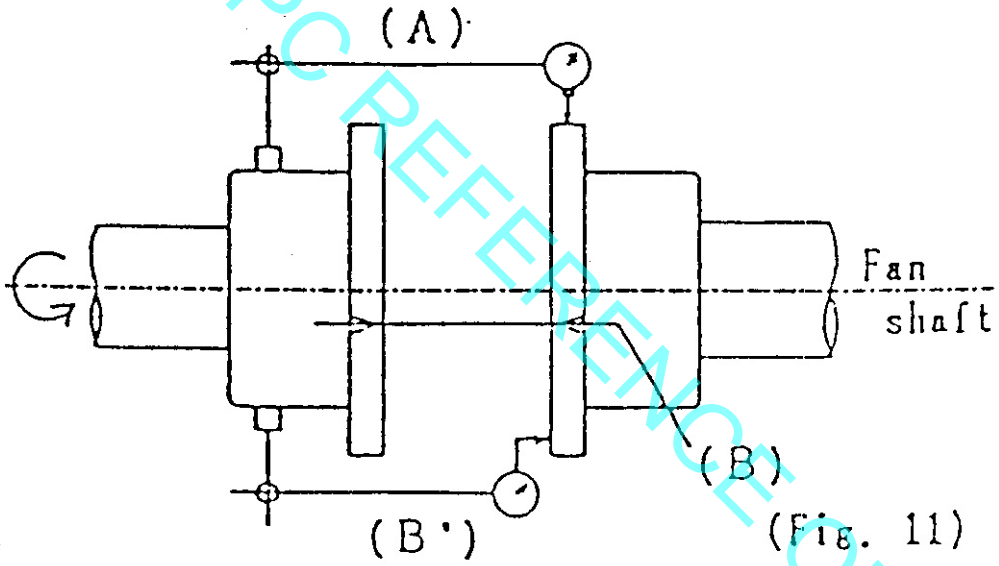
Spquence	Item of execution	Practical procedure

3.2.3.

5) Motor side coupling should be adjusted to the coupling for variable speed controller as same meaning as above。
After finishing the above adjustments , spacers and flex-elements are fitted and coupling bolts are tightened up uniformly。
However ,when the individual operation of motor and variable speed controller might be carried out in the time of trial run ,the spacers and flex-elements should be left dismantled and stored until the individual operation will finish。



(Fig. 11)

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Spquence	Item of execution	Practical procedure
3.2.4.	Spot welding of packer plate with taper liner	1) Snugness of all taper liners under the fan base should be ascertained by tapping with a hammer。 And it should be confirmed that the tapper liners are approximately parallel to the packer plate。 2) The taper liners and packer plate should be fixed by spot welding in order to prevent the assembled taper liners becoming loose or shifting during operation of fan。 But the taper liners should never be welded with the fan base。
3.2.5.	Grouting	1) Mortar is poured into the gap between fan base and concrete foundation , tamped sufficiently with a round rod to the inner part so as to prevent empty spots。 then the spare space having suitable width around the out side of fan base should be surrounded with wooden board and finished with mortar so that the surface become a tier higher than the surface of concrete foundation。

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Spquence	Item of execution	Practical procedure
3.3.	Assembling of accessories and piping work	

1) Assembling of damper and actuator。
After adhereing a gasket（asbestos yarn）on the face of flange , the damper is set up to the fan body, asceryaining the situation and direction of the operating lecer according to the genral view drawing。where the damper actuator is attached , the damper is connected to the actuator with the connecting rod and spindle。

2) Assembling of suction silencer。
The silencers are set up after the gasket has been adhered on the face of flange。

3) Jointing of the duct。
The duct should be jointed to the fan After providing with its own Supports so as not to load the Weight of duct on the fan body。

4) Jointing of cooling water pipes。
The cooling water pipes should be Jointed to the bearing cases after Confirming their inlet and outlet in Accordance with drawings。

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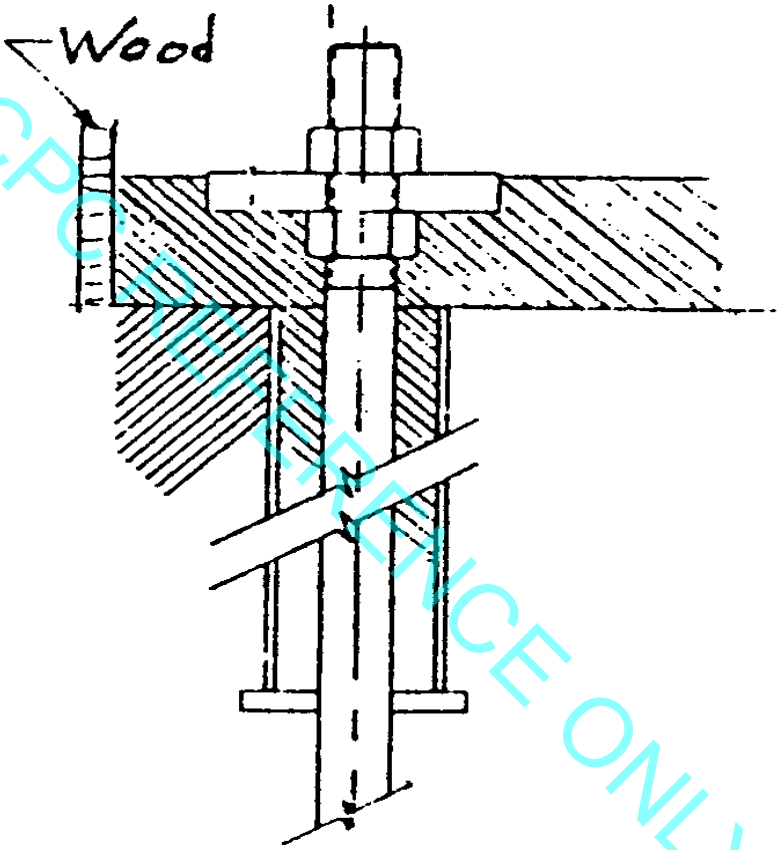
Spquence	Item of execution	Practical procedure
3.3.		5) insulation work。 the insulation work for casing should be carried out in accordance with the 〝insulation drawing〞。



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4. OPERATION PROCEDURE

Spquence	Item of execution	Practical procedure
4.1.	Inspection of each part	1) Refering to the general view drawing , it should be confirmed that the arrangement of the equipment and the fitting of attached equipment are corrected。 2) Tightness of foundation bolts And the fastening bolts should be confirmed by tapping。 3) It must be confirmed that the contact between the rotational and stationary part is not found。 4) After confirming that no bolt nut tool , wooden piece , etc。are remained inside the casing and the connecting duct , their inside should be cleaned up。 5) According to carring out opening and closing operations , smooth operation of the attached damper should be confirmed。

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Spquence	Item of execution	Practical procedure
4.2.	lubricants（Refer to the 〝Specification〞 or the 〝lubricants list〞）	
		1) Grease lubricating part。 although the adequate volume of grease is filled previously in the working part to be lubricated with grease necessary volume of grease should be supplied before test operation。
		2) Oil lubricating part In case of oil lubricating bearing, The lubricating oil is supplied level oiler。 The oil is filled up to the highest oil level line of oil gauge during stoppage , and the oil level has to kept somewhat upper to the middle line of oil gauge during operation。
4.3.	Bearing thermometer	
		1) where an attached thermometer is fitted in a blind hole , lubricating oil should be filled in the hole so as to expedite heat transmission to the remperature sensing element。

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Spquence	Item of execution	Practical procedure
4.4.	Cooling water	1) no leakage in the cooling water piping system should be confirmed。 2) The pressure of cooling water for a bearing should be 0.5～2 Kg/cm^2 in use。 The standard volume of cooling water are as follows , however , in case The bearing temperature tend to be slightly higher or in summer season, The volume should be increased。 nominal diameter of cooling water pipe 20 A（3/4 B）。 volume of cooling water（1/min/Brg.）
4.5.	Saftey measures	1) prior to the operation , a sufficient communication and a previous arrangement with he persons trelated to the operation and the workers of the related equipment should be performed。 2) prior to the operation of fan , it should be confirmed that the preparation of duct connection and related equipment have finished perfectly , according no danger and no injury with the grnerated current of air and pressure occur。

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Spquence	Item of execution	Practical procedure
4.6.	Damper	1) The suction vane damper should be closed entirely。
4.7.	Inching	1) The motor should be inched at first。 2) It should be confirmed that the driving drection agrees with the rotaional direction shown on the fan。if the rotaional direction is reverse , it should be made regular by changing connection with the motor。
4.8.	Operation	1) running of the motor is started at first。 2) After the motor speed reaches specified speed then the suction vane damper should be increased gradually attending to the current of motor, and settled on proper opening condition for working。
4.9.	Bearing temperature	1) The variation of bearing temperature which rises as the operating time goes by should be measured and recorded every 10~15 minutes until the temperature comes to balance。at the same time , the ambient。temperature should also be recorded 2) The bearing temperature must not be over 40℃higher than the ambient temperature。

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Spquence	Item of execution	Practical procedure
4.10.	Vibration	1) Vibration should be measured and recorded on the three directions i.e. horizontal , vertial and axial direction at the center of the fan and the motor。 2) When vibration measured , the fan Should be operated on near the Specific condition so as to keep out Unstable range。 3) The vibration on the bearing case of The fan is desirous to be less than 6 mm/s of the velocity of vibration shown in the figuer； 〝Allowable value of vibration〞 。
4.11.	Abnormal sound	1) It should be confirmed whether any abnormal sound is generating from the fan casing or not , during the operation。 2) The continuity and silentness of revolution sound of bearing should be confirmed with a hearing rod。

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Spquence	Item of execution	Practical procedure
4.12.	Continuous operation	1) The operation of fan can be continued ,when any unusual matter has not be found in the afore-said procedure。 2) The continuous operation in the range of extreme low volume under full closing situation of the damper should be kept out。
4.13.	Stoppage	1) The suction vane damper is entirely Closed at first。 2) Then the motor is stopped。 3) The cooling water for bearing is turned off after the motor has stopped perfectly。

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CHENG-CHEN MACHINERY CO., LTD.

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Allowable value of vibration

(on the bearing case)

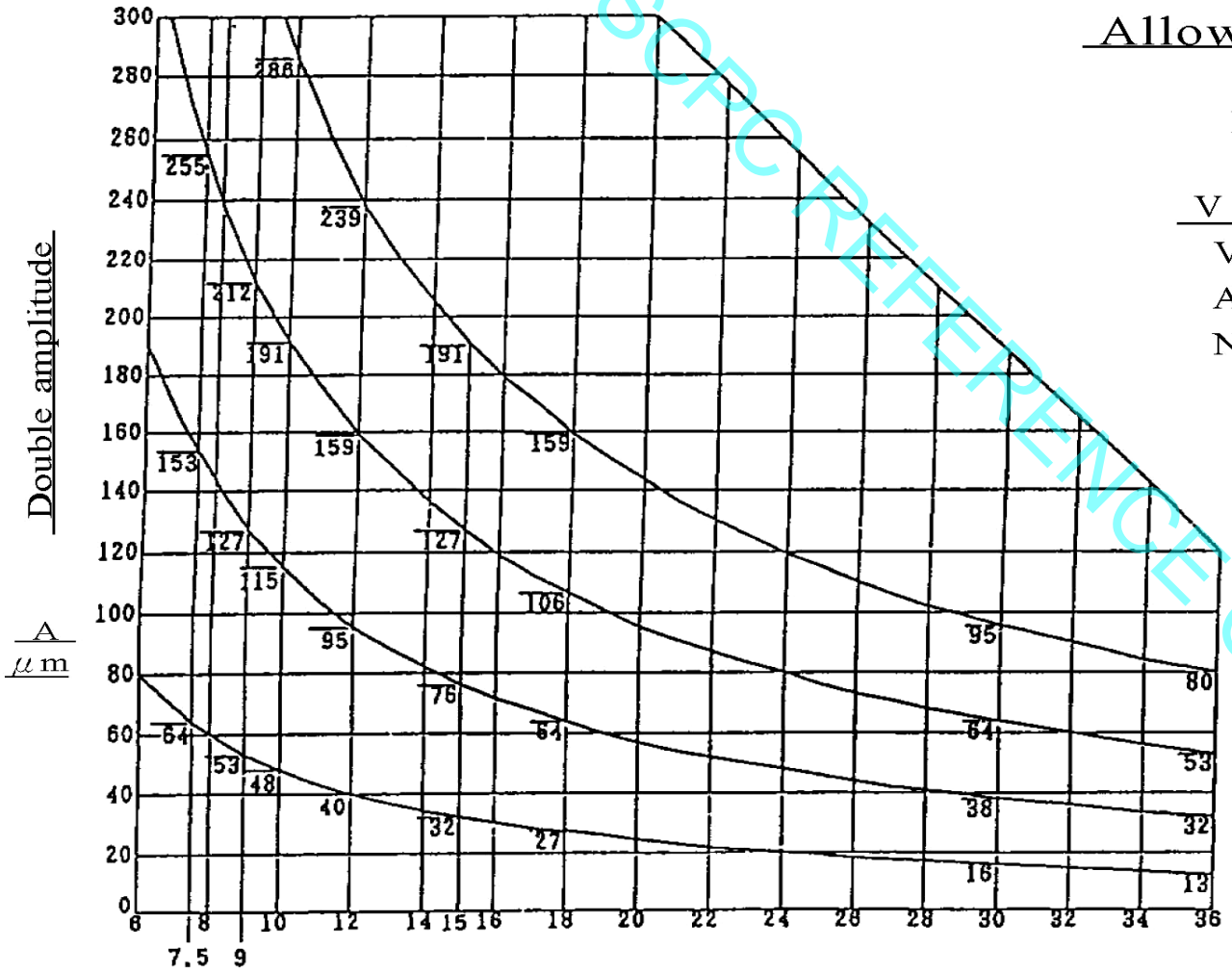
Refered to JIS B8330

$V = A \pi N / 60000$

V : Velocity of vibration mm/s

A : Double amplitude μm

N : Fan speed rpm



V = 15 mm/s (Value of trip)

V = 10 mm/s (Value of alarm)

V = 6 mm/s (Allowable value)

V = 2.5 mm/s (Desired value)

5. MAINTENANCE AND INSPECTION

5.1 Necessity of maintenance and inspection

The periodical maintenance and inspection should be carried out in order to keep the stable operation of fan for a long term。

5.2. daily inspection

it is important that the operating conditions of fan are inspedted at a definit time in every day and recorded。

if any unusual problem should happen , after it is commpared with former records and its cause is made clear , the measures to meet the situation should be taken quickly。

these steps bring the prevention of trouble previous to its occurrence。

5.3 Periodical inspection during stoppage

It is necessary to prolong the fan' s life that the fan is periodically stopped and overhauled and its wrong part is corrected accordingly。

Therefore the fan should be stopped and inspected all over once a year at least by all means。

And also , on the fan which is applied to important service or operated under rigourous condition , the periodical and partial inspections limited to the important part should be carried out at the shortened intervals。

On the other hand , even though the fan is under storage for a long term , the periodical inspection should be carried out and the fan should be kept in such condition as it can be operated according to preparation carried out in a short time。（Refer to paragraph 2.2）

5.4. According to the aforesaid instruction , the inspection items and maintenance procedures required at a minimum for inspection and maintenance of the fan , are shown in the following tables by the inspection period。

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5.4.1 Items to be inspected once every day during operation

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Bearing	Temparature	Tactility or thermometer	To be recorded。In case of something unusual , to supply lubricant or investigate afer stoppage of the fan。	7.
	Vibration	Tactility or vibrometer	Ditto。	7.
	Sound of revolution	Hearing rod	Ditto。	7.
	Oil level	Visual Inspection on Oil gauge	In case of low level Of oil , to sopply lubricant。	4.2 2）
	Existence of Oil leakage	Visual inspection	Oil supply or Investigation after Stoppage of the fan。	4.2 2）
	Volume of cooling water	Visual survey On sight flow	To adjust the volume depending on the bearing temperature。	4.2 2）
casing	vibration	Tactility or vibrometer	In case of something unusual , to investigate after stoppage of the fan。	7.
	Existence of abnormal sound	Sence of hearing	To investigate after Stoppage of the fan。	7.
coupling	Existence of abnormal sound	Sence of hearing	To investigate after Stoppage of the fan。	7.
Shaft Sealing part	Existence of Gas leaking	Visual survey Or smell	Investigation of sealing materials and to replace , if necessary after stoppage of the fan。	7.

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5.4.1

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Motor	Bearing Temperature	Tactility or thermometer	To be recorded In case of something unusual to supply lubricant or investigate after stoppage of the fan	7.
	Vibration	Tactility or vibrometer	Ditto	7.
	Current	Ammeter	Ditto	7.

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5.4.2. Items to be inspected once a year

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Rolling bearing	Deterioration of lubricant	Visual survey	Renewal of lubricant	
Bearing case	Confirmation of alignment	Dial gauge	Adjustment and record	3.2.1 2）
	Existence of contamination of oil gauge	Visual survey	Cleaning or renewal	
	Damage of packings	Visual inspection	Renewal of packing	
	Confirmation of clogging in cooling water pipe	Visual inspection	cleaning	

5.4.2

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Impeller	Existence of something sticked	Visual survey	Removal	
	Existence of corrosion	Visual survey	Repair or replacement	
	Existence of abrasion	Visual inspection	Repair or replacement	
	Confirmation of the gap between impeller and suction cone	Measurement with taper gauge	To be recorded after adjustment	3.2.2
shaft	Existence of something sticked	Visual survey	Removal	
	Existence of corrosion	Visual survey	Repair or replacement	
Shaft seal packings	Confirmation of the degree of abrasion	Visual inspection or measurement with scale	To renew , if neceseary	
coupling	Confirmation of alignment	Dial gauge	To be recorded after adjustment	3.2.3
	Looseness of bolts	Tapping	Aditional tightening of the bolts	
	Confirmation of abrasion degree of rubbre bushing	Visual servey	To renew all bushings , if necessary	

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5.4.2

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Casing and suction cone	Existence of something sticked	Visual servey	removal	
	Existence of corrosion	Visual servey	Repair or replaccment	
	Existence of abrasion	Visual inspection	Repair or replacement	
Damper	Existence of something sticked	Visual servey	Removal	
	Existence of corrosion	Visual servey	Repair or replacement	
	Existence of abrasion	Visual inspection	Repair or replacement	
	Smoothness of opening and closing operation	Operating manually or automatically	To repair and adjust the wrong part	
Silencer	Existence of something sticked	Visual survey	Removal	
	Existence of corrosion	Visual survey	Removal or replacement	
Thermometer	Existence of false indication	Comparing with others	Replacement	
	Existence of damage	Visual survey	Renewal	
Foundation bolts and Fastening bolts	Existence of looseness	Tapping	Aditional tightening of the bolts	
	damage	tapping	Renewal of bolts	

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5.4.2

Part to be inspected	Inspection items	Inspection methods	Maintenance procedures	Reference articles
Painting Surface	Existence of Rust formation	Visual survey	Repainting	
motor	manufacturer' s instruction items		Conforming to manufacturer' s instruction	

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5.4.3 Allowable value and critical value for the maintenance of fan

1) Allowable value of bearing temperature;
   Tmax。= （Ambient temperature）+ 40℃
2) Allowable value of the velocity of vibration on the bearing case:
   Vmax。= 6 mm/s
3) Allowable value related to installation;
   Refer to the drawing 〝Allowable dimension on installation〞 。
4) The designed life of rolling bearing is 5～10 years under the Condition of continuous operation。

6. CAUTION FOR DISASSEMBLING AND REASSEMBLING

6.1 Caution for disassembling

1) The construction of the part to be disassembled should be well understood previously so that the procedure of disassembling must not be made a mistake。
2) Match-marks should be marked at the relative position which must not get out of order mutually。
3) The part that disassembling is difficult because of the close adhesion by long term use should not be damaged by forced disassembling。
4) As the rotational part is the most important part , the handling of this part should be carried out with sufficient care so as not to make deformation or damage。
5) After sufficient sweeping and cleaning the dismantled parts are put on the clean paper or cloth and covered with vinyl plastic sheet。
6) Small parts , bolts and nuts are classified and stored so as to identify the applied part distinctly and not to be lost。
7) After disassembling of bearing , it should be wrapped up in the clean vinyl plastic sheet so that the foreign matters never enter into the bearing。
8) The machining surface of dismsntled parts should be coated With oil so as to prevent rust。
9) It should be investigated whether the parts van be used again or not , and if necessary , the situation of rearrangement of parts and stock for spare parts should be confirmed so as not to cause a waiting in reassembling。

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6.2 Caution for reassembling

1) Reassembling should be carried out in the reverse order of disassembling , and the parts marked with match-marks should be assembled properly conforming each other。

2) The parts that ought to be assembled prior to the others, should be assembled in regular without fail , and the assembling should be carried out after confirmation that there are no foreign matters in the interior。

3) In the case there are very similar parts to be used , the assembling should be carried out carefully so as not to mistake。

4) When a impeller is set in the casing the setting must be carried out slowly by the use of chain block so as not to make deformation by striking in error。

5) It should be confirmed that there are no residues or tools in the casing。

6) After the assembly alingment and adjustment for every part should be carried out and recorded。（Refer to article 3.2 〝adjustment of equipment〞）

7) The instrument or accessories dismantled at the time of disassemling are set up as they ware before

8) After suppling lubricants or cooling water test operation Should be carried out in order to prepare for continuous operation。（Refer to article 4 〝operation procedure〞）

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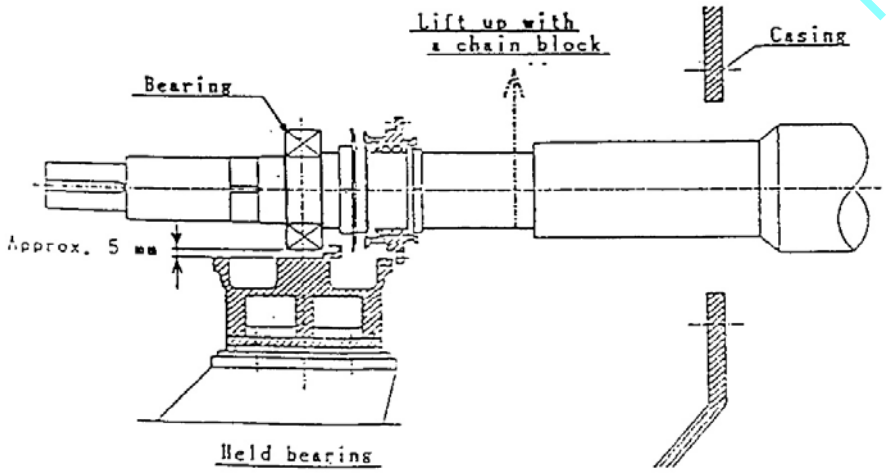
6.3 Replacement of bearing

6.3.1 Dismounting of bearing

1) Remove the thermometer and cooling water pipes for bearing case。
2) Draw out the coupling hub for fan shaft with a puller。
3) Remove the shaft seal packing and packing holder from the casing wall。
4) On held bearing , unbolt the bolt for side cover（2）,（3）then remove the upper bearing case。
5) Draw out the slinger（9）and the side cover（2） from the shaft。
6) Rise up a tab of bearing washer（6）in the gloove of bearing nut（7）, then loosen the bearing nut for rotational direction of impeller with a hoock spanner , then draw out the bearing washer。
7) On free bearing , remove the side cover , end cover , upper bearing case , bearing nut（7）and bearing washer（6）as same as above。

Note：Procedures between next item to 6.3.2 2 ）should be performed on held bearing at first then next on free bearing。

8) Hang the shaft with a wire rope , then lift it up approximate 5 mm with a chain block。
In this time do not touch the shaft to the casing wall（shaft seal part）。
9) Heat up the inner race of bearing uniformity with a gas torch , then draw out the bearing from the shaft with a hammer。



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6.3.2 Mounting of bearing

1) Heat up the new bearing in turbine oil baths ariund 100℃ equally , then draw it in position on the shaft quickly , and tighten it enough with the bearing washer and nut
Bend a tab for bearing washer into the gloove of bearing nut for locking。
2) Then lowered the shaft slowly , and confirm that the bearing locates in correct position at the lower bearing case。
3) The parts disassembled should be reassembled in the reverse order。
4) The coupling hub for fan shaft should be drawn in the shaft, after heating it up around 150℃ with a gas torch。

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7. TROUBLESHOOTING

Defect	Presumptive Causes	Measures
Overheating of bearing	Deficiency of lubricant	Supply of proper volume
	Deterioration of lubricants	Renewal of all of lubricants with proper one
	Invasion of dust into lubricants	Cleaning and renewal of lubricants
	The fault and abrasion of bearing	Replacement
	Defective alignment of bearing cases	Re-aligning and adjusting
	Defect in cooling system	Repair or replacement
	Excessive vibration	To refer to the article 〝excessive vibration of bearing〞
Excessive vibration of bearing	Unbalance of impeller due to corrosion or abrasin	Request of balance adjustment to an expert or replacement of impeller
	Abrasion of bearing	Replacement
	Adherence of dust on impeller	Sufficient cleaning
	Looseness of foundation bolts	Aditional tightening of the bolts
	Operation in unstable bounds	Improvement of the operating condition
	Abrasion of coupling	Replacement
	Poor alignment of coupling	Readjustment of alignment
	Decline of rigidity of foundation	Reinforcement or modification
	Contact between rotational and stationary part	Repair or replacement
	Deformation and damage of impeller	Replacement
	Insufficient warminr up	Improvement of the operation procedure

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7.

Defect	Presumptive causes	Measures
Ceneration of abnormal sound	Defect and abrasion of bearing	Replacement
	Lack of lubricants	Supply of proper volume
	Surging or operation in unstable bounds	Improvement of the operating condition
	Contact between rotational and stationary part	Repair or replacement
	Abrasion or damage of coupling	Replacement
	Invasion of foreign matter	Removal
	Damage of impeller	Supply of lubricants or repair
Poor working of damper	Heavy rusting in link lever and pin	Supply of lubricants or repair
Cas leaking	Defference of pressure balance between suction and discharge side from designed	Re-examination of schematic design
	Abrasion of shaft seal packing	Replacement
	Defect of packing in flange part	Filling of sealing compound or replacement of packings
Lowering of per-formance of fan	Change of the system loss	Inspection of dust and equipment relating to the system
	Change of specific weight of gas	Inprovement of working condition
	Adherence of dust on impeller or casing	Sufficient cleaning
	Deformation corrosion and abrasion of impeller casing and duct	Repair or replacement
	Decreasing of speed	Investigation of power source and motor
	Decreasing of instruments	Renewal

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7.

Defect	Presumptive causes	Measures
Overload of motor	Change of system loss	Inspection of duct and equipment relating to the system
	Contact between rotational and stationary part	Repair or replacement
	Defect of power source	Improvement
	Change of gas temperature	Improvement of working condition
	Poor operation of damper	Correction of operation

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4. ACTUATOR MANUAL

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建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

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LMYP-1 Limay,Philippine

Pneumatic Actuator Data Sheet For I.D.Fan

引風機氣動作動器資料

本文編號 RF2-0582-DOC-3-0

REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	CERTIFIED / PROJECT NO.	PAGE
1						PROJ.	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	MGR. _____ DATE _____	
B	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/03/17	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/01/27	**LMYP-1**	1

CMV PNEUMATIC ACTUATOR DATA SHEET - 133486SC01

Datasheet rev. 2.0

Customer name	FORTUNE TAIWAN	Progressive datasheet	01 of 01 — Revision 1
Project name	CCMC	Quantity	4 — Customer country TAIWAN
End user			Your reference QT-13T253
Project no - Cust. refer.		Schematic	Q070053PD01 — Dimensional dwg DD37625
PO	142001 dated 6 Dec 2012	Wiring diagram	Assembly dwg AD37625
STI job	133486	Plant location	
STI code / serial number	115602 — A1404282 - A1404285		STI PM B. Carrara
STI Quotation	Q132296	Tag	

VALVE	1		Valve Model		
	2		Valve Application / Service		
	3		Ambient condition	STANDARD	
	4		Thrust to open	420 kgm	N
	5		to close	420 kgm	N
	6		to seat		N
	7		Valve weight		kg
	8		Valve plug weight		kg
ACTUATOR - GENERAL FEATURES	9				
	10		Actuator thrust N	747 kgm	
	11		Design temperature range	STANDARD -20 / 70	°C
	12		Accessories electrical certification	Ordinary Location	
	13				
	14		Enclosure rating / Protect class	IP 65 (NEMA4X)	
	15		Electrical certs required	NO	
	16		ATEX actuator	NO	
	17		PED actuator / class	NO	
	18		ACTUATOR type	DOUBLE ACTING	
	19		Type of service	MODULATING	
	20		Size (mm) / Modulating stroke	CMV3 330/400 mm	
	21		Mechanical and modulating stroke mm	402 (Modulating 400 + 2 mm extrastroke)	
	22				
	23		Lever at low signal	Lower Position/Damper	
	24				
	25		End caps mat.	Carbon steel	
	26		Cylinder mat.	Carbon steel	
	27		Tie rod	CF35SMnPb10	
	28		Stem material — Bushing material	C 45 RECT	Bronze
FUNCTION	29		Air failure mode	In place	
	30		Stroke time to Upper position		s
	31		to Lower position		s
	32		Trip to Lower position		s
	33		Trip to Upper position		s
UTILITIES	34		Feed fluid type	AIR	
	35	1	Min / Max supply pressure	4 / 6	bar / Kg/cm2
	36		Air connection	1/2"	NPT
	37	1	Electrical connection	N.3 3/4 G	
HOUSING	38		Housing material - Outside protection	Carbon steel	
	39		Lever operating angle	70°	
	40		Lever dimension and drilling (drawing)	DD33838	
	41		Manual operation	Clutahable handwheel	
	42		Lever accessories	Not included	
POSITIONER	43		Manufacturer and model	ABB/TZID-C	
	44		Electrical certification	IP65	
	45		Inlet signal	4-20 mA Hart	
	46		Valve characteristic	EQUAL PERCENTAGE	
	47				
	48				
	49		Separated IP transducer	Integral	
	50		Gauges Material	Scale	Nr.
	51		Snap acting function	Set	
	52		Motion converter Material		
	53				
	54				
	55				
	56				

ACCESSORIES	57		Air lock (lock up + 2 x 3 way valves)	Size	AL102 1/2"
	58			Set	2.5 bar / Kg/cm2
	59			Material	Aluminum
	60		other 3 way valves	Qty	Size
	61			Material	
	62		Volume booster	Qty 2	Size 1/4"
	63			Material Aluminum	Man. / Model AV-1
	64		Derivative booster	Qty	Size
	65			Material	
	66		Derivative unit	Qty	Size
	67			Material	
	68		Snubber	Yes / No	Material
	69		CS device	Yes / No	Material
FR	70		Filter regulator (with gauge)	Manuf. NORGREN	Model B73 Aluminum
	71	1		Size 1/2"	Set 4 bar / Kg/cm2
TR/LS	72		Position transmitter	Model	Integral into positioner TZID-C
	73			El. certific.	IP66
	74		Limit switches	Type Mechanical	Contact type DPDT
	75			Model	Honeyweel V15T16 into 2130-4M-2S)
	76			Position	1 IN OPEN + 1 IN CLOSE
	77			El. certific.	IP66
SOLENOID	78		Solenoid valves	Manuf.	Qty
	79			Model	Type
	80			Size	Material body
	81			Voltage	Way
	82			El. certific.	
	83			Enclosure	
OTHER	84		Junction box	Manuf.	
	85			Model	
	86			Material	
	87			El. certific.	
	88			Enclosure	
	89		Heater	Manuf.	Model
	90		Pressure switch	Manuf.	Model
	91		Bracket material	Material	Carbon steel
	92		Screw material	Material	Carbon steel
	93		Stroke limitator	Open / Close	at %
TUBING	94		Tubing & Fitting	Tubing size Metric	Material Aisi 316
	95				
	96			Fitting mat. Aisi 316	Manufact. HY-LOK
PAINT	97		Painting specif.	STANDARD GREY RAL 7030 CV31 FOR CMV - CML	
	98		Notes		
	99				
TANK	100		Volume tank	Size liters	Mounting
	101			Certification	Port connection
	102			Material	Supplied by
	103		Check valve	Size	Material
REMARKS	104		(LINE 10) MIN.AVAILABLE TORQUE REFERRED AT 4,5 kg/cm^2 AIR SUPPLY PRESSURE		
	105		AND 90° OPERATING ANGLE OF DAMPER (MAX IN SAME CONDITIONS:1057 kgm)		
	106		(LINE 39) 70° PRIMARY ANGLE OF LEVER ON CONTROL DRIVE;FINAL 90° OPERATING		
	107		ANGLE OF DAMPER OBTAINED THROUGH CORRECT SIZING OF EXT. LEVERS		
	108		STAINLESS STEEL AISI 316 FLEXIBLE HOUSES WHERE NEEDED		
	109				
	110		4 mA EXTERNAL LEVER LOWER AND 20 mA EXTERNAL LEVER UPPER		
	111				
	112	1	(LINE 37) ADDED ADAPTER 3/4" NPT		

VALVE OPERATION TABLE								Rev.	Date	Issued	Checked
AIR SUPPLY	OK	FAIL/LOSS	OK					1	6-Mar-14	M. TAMI	B.CARRARA
SOLENOID VALVE 1								0	25-Nov-13	M. TAMI	B.CARRARA
SOLENOID VALVE 2											
SOLENOID VALVE 3											
SIGNAL	OK	ANY	FAIL/LOSS								
POWER FAIL											
LEVER POSITION	Modulating	In place	In place								

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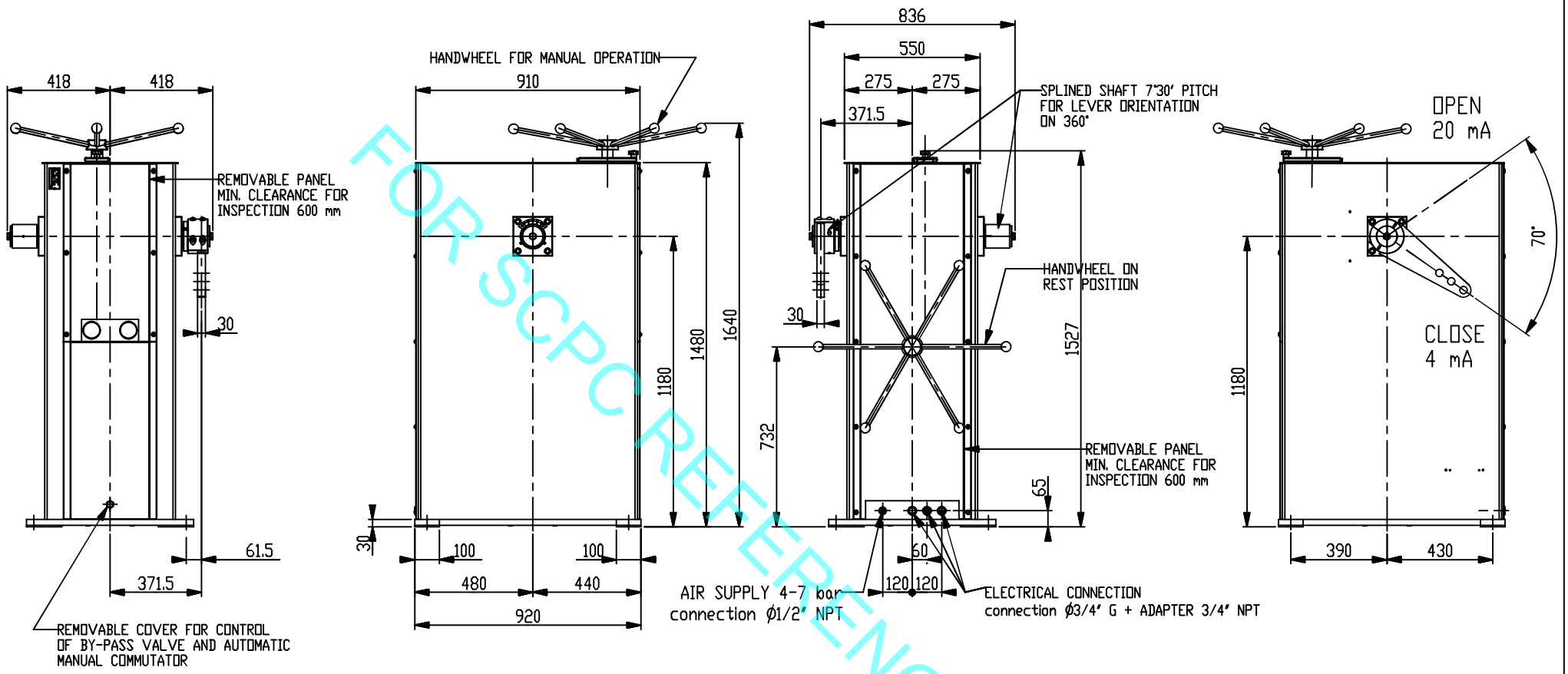

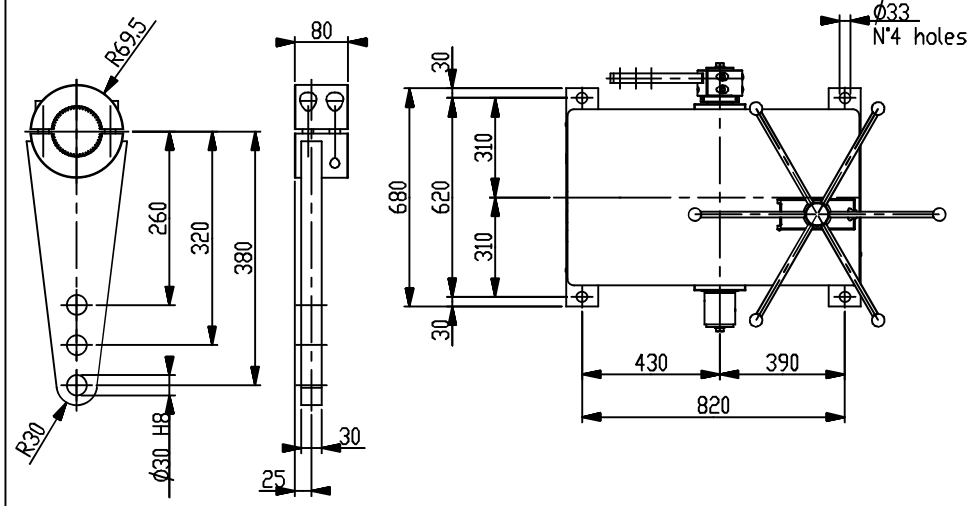

WEIGHT = 725 Kg.

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REV.	DATE	REVISION DESCRIPTION	ISSUED BY	CHECKED/APPR. BY
A	Nov-29-13	FIRST ISSUE	P. PERSICO	I. MOSTOSI
			REPLACE	REPLACED BY

STI

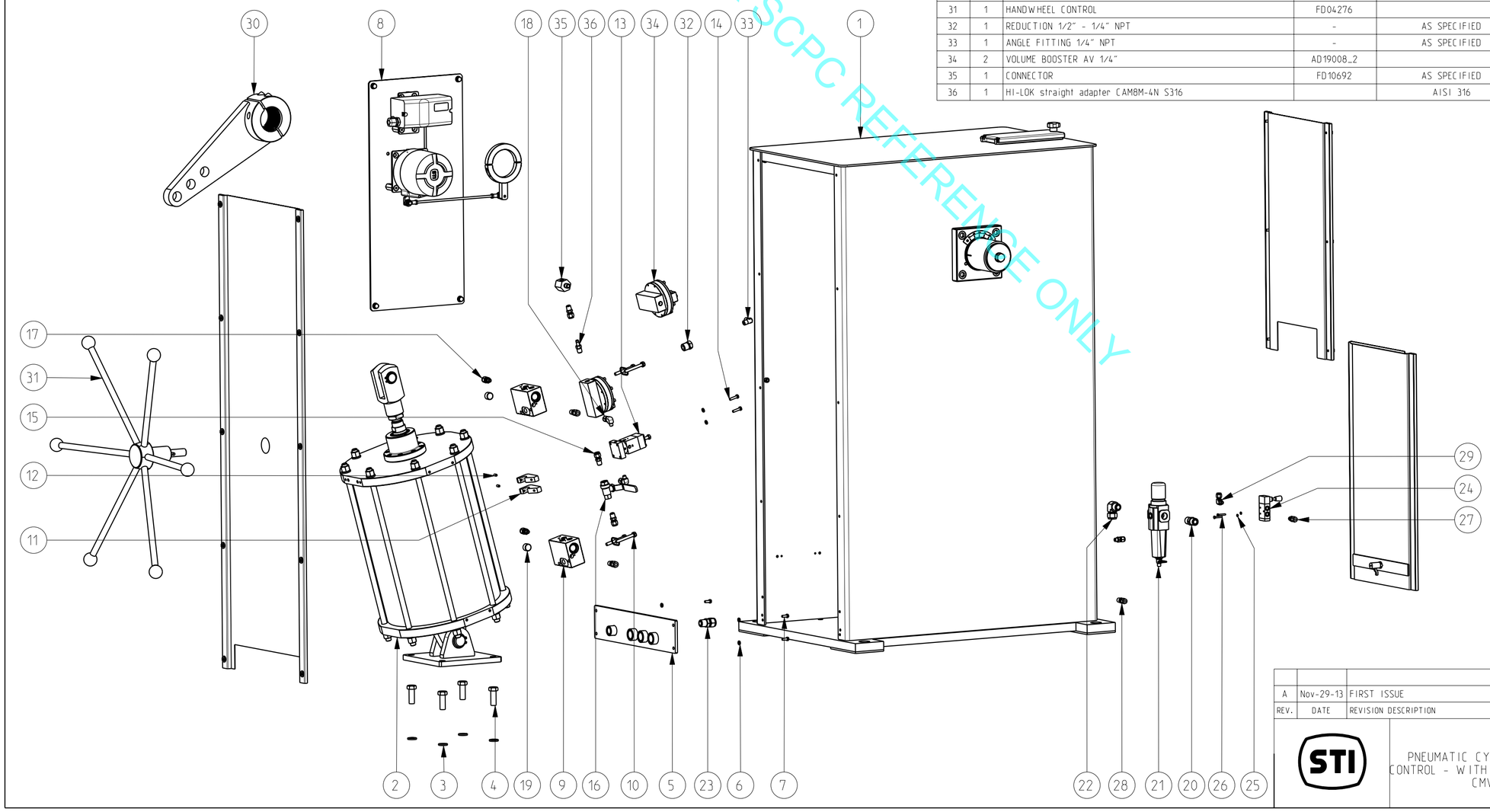
DIMENSIONAL DRAWING
PNEUMATIC CYLINDER ACTUATOR WITH HANDWHEEL
CONTROL - WITH ELECTROPNEUMATIC POSITIONER TZID-C
CMV-3 SERIES TYPE 330/400

DRAWN: DD37625
FILE: DD37625 .DRW
PAG.N° 1 DI N° 1

A037625
Generici

DRAWING SIZE : A3 HORIZONTAL

POS.	Q.TY'	DESCRIPTION	DRAWING N°	MATERIAL	DIMENSIONS	SURFACE TREAT.	NOTE
1	1	PNEUMATIC CYLINDER ACTUATOR CMV-3	ZD35715	S275JR UNI EN 10025	1480 x 920 x 680	VERNICIATO	LEVA A DESTRA
2	1	DOUBLE ACTING PNEUMATIC CYLINDER ACTUATOR SC-2 SERIES	AD29003		SIZE 330/400		
3	4	LOCK WASHER	-	cl. A4	Di6.2xDe27.4sp.3.5 UNI 1751	-	
4	4	SCREW TE	-	AS SPECIFIED	M16x45 UNI 5739		
5	1	EXTERNAL PLATE CONNECTION	ZD04282_8	S235JR UNI EN 10025	100 x 470 x 8	VERNICIATO	
6	6	LOCK WASHER	-	cl.A4	Di6.1xDe11.8sp.1.6 UNI 1751	-	
7	4	SCREW TE	-	cl.A4-70	M6x16 UNI 5739		
8	1	KIT PLATE WITH ACCESSOIRES	ZD17124_17				
9	2	THREE WAY VALVE ND 1/2"	AD12053				
10	2	COUPLING ACCESSORIES KIT	ZD24303_1				
11	2	POSITIONER SUPPORT	FD12434	AS SPECIFIED	60x40x15	ANODIZZ.NEUTRO	
12	2	SCREW STEI	-	cl.A4-70	M6x12 UNI 5927		
13	1	AIR LOCK DEVICE for AIR FAILURE with N°2 CONNEC.	AD10418_2				C.P. FD10377
14	2	SCREW TE	-	AS SPECIFIED	M6x25 UNI 5739		
15	5	HY-LOK straight fitting CMC 8M-4N S316	-	AISI 316	Tube Ø8mm - Pipe 1/4"NPT		
16	1	NEEDLE VALVE M.B. VALVE mod. IVR 660		AISI 316	DN 1/4" NPT		
17	2	HY-LOK straight fitting CMC 8M-2N S316	-	AISI 316	Tube Ø8mm - Pipe 1/8"NPT		
18	2	HY-LOK Elbow fitting CLMA 8M-2N S316	-	AISI 316	Tube Ø8mm - Pipe 1/8"NPT		
19	2	PLUG	-	AISI 316	1/2" NPT		
20	1	NIPPLE 1/2" - 1/2" NPT	-	AISI 316			
21	1	AIR FILTER REGULATOR B73G	-		DN 1/2"		
22	1	HY-LOK elbow fitting CLMA16M-8N S316	-	AISI 316	Tube Ø16mm - Pipe 1/2"NPT		
23	1	HY-LOK elbow fitting CMC 16M-8N S316	-	AISI 316	Tube Ø16 mm - Pipe 1/2" NPT		
24	1	VALVE 3/2 PNEUMAX 228.32.27 DN 1/8"	228.32.27	ALLUMINIO	32 x 18 x74		
25	2	LOCK WASHER	-	A4	Di4.1xDe7.6sp.0.9 UNI 1751	-	
26	2	SCREW TE	-	AS SPECIFIED	M4x25 UNI 5739		
27	2	HY-LOK straight fitting CMC 6M-2N S316		AISI 316	Tube Ø6mm - Pipe 1/8"NPT		
28	1	HY-LOK straight fitting CMC 6M-4N S316		AISI 316	Tube Ø6mm - Pipe 1/4"NPT		
29	1	HY-LOK elbow fitting CLMA 6M-2N S316		AISI 316	Tube Ø6mm - Pipe 1/8"NPT		
30	1	EXTERNAL LEVER	ZD17593	S355JR UNI EN 10025	80 x 479.5	VERNICIATO	
31	1	HANDWHEEL CONTROL	FD04276				
32	1	REDUCTION 1/2" - 1/4" NPT	-	AS SPECIFIED		-	
33	1	ANGLE FITTING 1/4" NPT	-	AS SPECIFIED			
34	2	VOLUME BOOSTER AV 1/4"	AD19008_2				
35	1	CONNECTOR	FD10692	AS SPECIFIED	CH.32 x 49	AS SPECIFIED	
36	1	HI-LOK straight adapter CAM8M-4N S316		AISI 316	Tube Ø8mm - Pipe 1/4" NPT		



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			REPLACE	REPLACED BY
A	Nov-29-13	FIRST ISSUE	P. PERSICO	I. MOSTOSI

STI

ASSEMBLY DRAWING
PNEUMATIC CYLINDER ACTUATOR WITH HANDWHEEL CONTROL - WITH ELECTROPNEUMATIC POSITIONER TZID-C
CMV-3 SERIES TYPE 330/400

DRAW. N° AD37625
FILE AD37625 .DRW
PAG.N° 1 DI N° 1
Generic: AD37625

DRAWING SIZE : A2 HORIZONTAL

ACTUATOR PAINTING SPECIFICATION

CLIENTE: Customer	FORTUNE TAIWAN	DOCUMENTO N° Document no	
RIF. CLIENTE: Your ref.		IMPEGNO N° Job no	133486
IMPIANTO: Project	CCMC	FOGLIO N° Sheet no	1/1
ITEM: Item		CV31 Standard for drive actuator	

1 ACTUATOR HOUSING (Steel frame)

1.1 SURFACE PREPARATION Brushing and cleaning with solvent	.
1.2 PRIMER Epoxy with zinc phosphate primer	thk. 25÷50 micron RAL 7030 Gray
1.3 FINAL COAT Epoxy polyamide enamel	thk. 25÷50 micron

2 ACTUATOR HOUSING (Steel sheet panel)

2.1 SURFACE PREPARATION Brushing and cleaning with solvent	.
2.2 PRIMER Epoxy with zinc phosphate primer	thk. 25÷50 micron RAL 7030 Gray
2.3 FINAL COAT Epoxy polyamide enamel	thk. 25÷50 micron

3 ACTUATORS

3.1 SURFACE PREPARATION Brushing and cleaning with solvent	.
3.2 PRIMER Epoxy	thk. 25÷50 micron RAL 7030 Gray
3.3 FINAL COAT Epoxy polyamide enamel	thk. 25÷50 micron

4 INSIDE ACCESSORIES PANEL

4.1 SURFACE PREPARATION Brushing and cleaning with solvent	.
4.2 PRIMER Epoxy	RAL 7030 Gray
4.3 FINAL COAT Epoxy polyamide enamel	thk. 25÷50 micron

5 ALUMINIUM COMPONENTS FROM SEMIFINISHED MATERIAL

5.1 Anodising	.

6 STAINLESS STEEL AND BRASS COMPONENTS/ACCESSORIES

6.1 Not painted	.

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REV. rev.	DESCRIZIONE Description	EMESSO Issued	CONTROLLATO Checked	DATA Date
1	ISSUE	A. NEGRI	S. ROXAS	14-Feb-06

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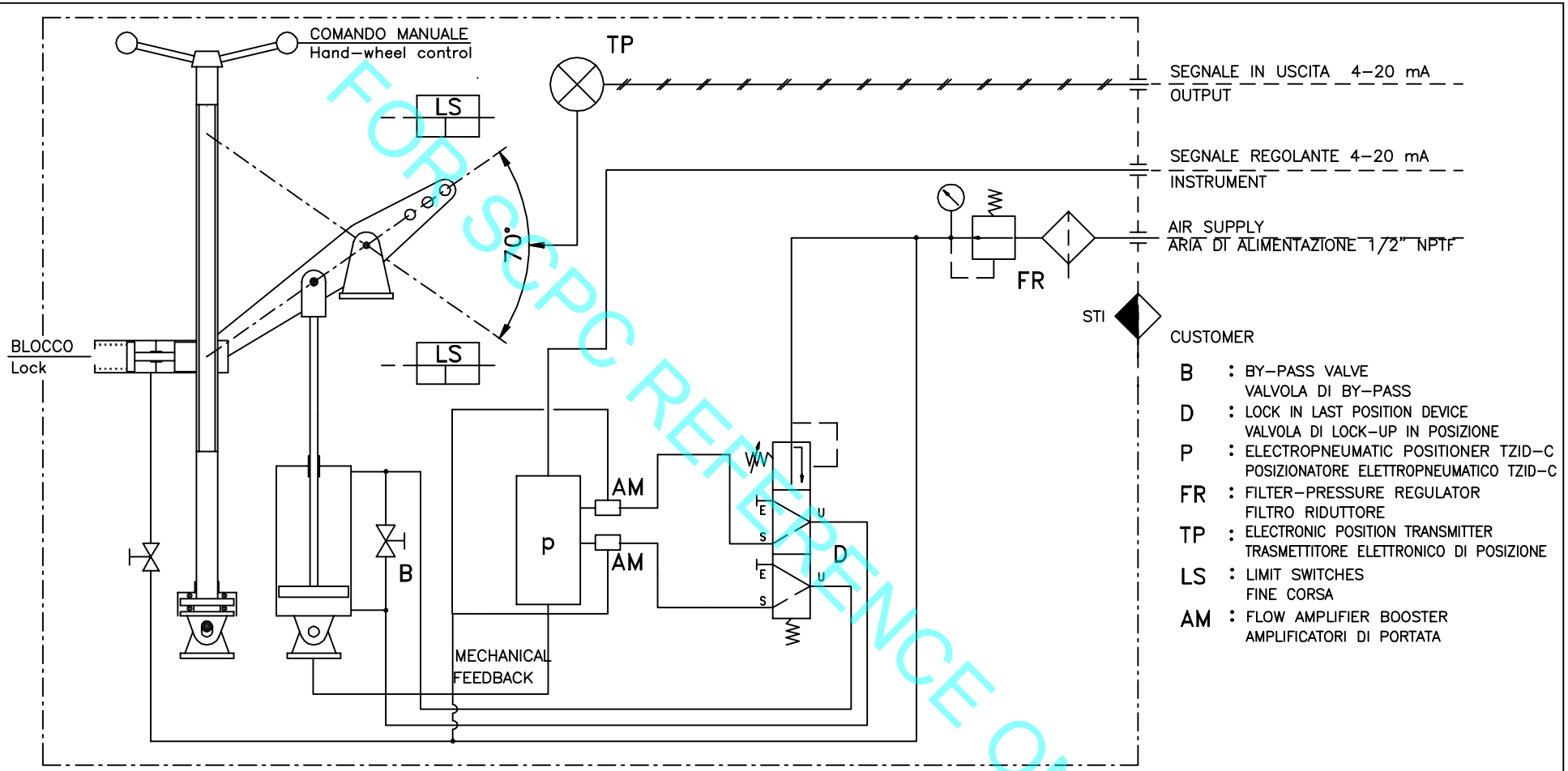

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STRUMENTAZIONE INDUSTRIALE S.p.A.

SCHEMA PNEUMATICO
AZIONATORE PNEUMATICO SERIE CMV2, CMV3 E CMV4 CON COMANDO MANUALE E BLOCCO IN POSIZIONE

PNEUMATIC DIAGRAM
PNEUMATIC ACTUATOR CMV2, CMV3, CMV4 SERIE WITH HANDWHEEL CONTROL AND LOCK IN LAST POSITION

SOSTITUISCE IL	SOSTITUITO DAL
DIS.N°	Q070053PD01
FILE	
PAG. N° 01	DI N° 01

STI S.r.L.

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INSTRUCTION MANUAL
INSTALLATION – SERVICING
PNEUMATIC ACTUATORS CMV SERIES

GENERAL FEATURES

In the CMV actuators series, the housing provides a double function of protection and bearing frame.
This system assures maximum stiffness without making use of additional supporting structures; consequently, the whole volume inside the housing is available for the most rational installation of the actuator and the various accessories.
The possibility of inspection is assured by means of large removable panels.
The actuator used is the cylinder type SC which is equipped with the life time lubrication system patented by STI and the cam positioner type SA/CL-1 available in sizes from ¼" to ½" according to the requested response speed.
The specific features of this actuator are shown on brochure SSP1002.
Of course, on-off actuators are also available on request.
The CMV actuator may be supplied in eight different sizes in order to comply with a large range of torques (see table 1 and diagram 3 and 4).
The shaft equipped with an external lever has an operating angle of 70° and is fitted, at its two terminals, with a splined joint which allows the most convenient orientation of the control lever.
The manual control is provided with a handwheel (separable for CMV2, CMV3 and CMV4) and is mainly composed (see pneumatic diagram fig. 5 and 6) of a threaded shaft with hinge and thrust bearing at the end.
The engaging or disengaging of the manual control is made up by a half-nut (C), controlled by a single acting pneumatic cylinder actuator with spring that causes (when air supply is missing) the coupling between half-nut and threaded shaft.

Table 1

Actuator type	Min. torque (1)	Max. torque (1)	Operating time (2) with positioner		Weight	Dim. dwg
	70° angle 90°	70° angle 90°	SA/CL1 ¼"	SA/CL1 ½"		
CMV1 125/250	140 daNm 99 1033 lb.ft. 727	171 daNm 139 1265 lb.ft. 1030	2,5 sec.	-	kg. 153 lb. 337	DD17834_1
CMV1 160/200	185 daNm 130 1363 lb.ft. 961	226 daNm 184 1668 lb.ft. 1360	3 sec.	-	kg. 163 lb. 359	DD17834_1
CMV1* 200/200	188 daNm 132 1386 lb.ft. 976	230 daNm 187 1699 lb.ft. 1380	5,5 sec.	-	kg. 183 lb. 403	DD17834_1
CMV2 160/300	277 daNm 195 2045 lb.ft. 1443	340 daNm 276 2510 lb.ft. 2040	4,5 sec.	2 sec-	kg. 260 lb. 574	DD17830_2
CMV2 200/300	438 daNm 309 3234 lb.ft. 2278	537 daNm 436 3962 lb.ft. 3221	7 sec.	3,5 sec.	kg. 280 lb. 618	DD17830_2
CMV2 260/300	745 daNm 525 5496 lb.ft. 3873	913 daNm 742 6736 lb.ft. 5476	12 sec.	6 sec.	kg. 305 lb. 673	DD17830_2
CMV3 260/400	993 daNm 700 7330 lb.ft. 5163	1217 daNm 989 8983 lb.ft. 7302	16 sec.	8 sec.	kg. 520 lb. 1148	DD17769_1_2
CMV3 330/400	1627 daNm 1146 12000 lb.ft. 8452	1993 daNm 1620 14700 lb.ft. 11950	27 sec.	13 sec.	kg. 620 lb. 1368	DD17769_2_6
CMV4 390/450	2634 daNm 1816 19427 lb.ft. 13394	3225 daNm 2596 23786 lb.ft. 19147	-	30 sec.	kg. 1550 lb. 3417	DD17768
CMV4 420/450	3026 daNm 2113 22319 lb.ft.15585	3696 daNm 2996 27253 lb.ft. 22097	-	36 sec.	kg. 1600 lb. 3527	DD17768
CMV4 520/450	4638 daNm 3202 34208 lb.ft.22879	5684 daNm 4573 41923 lb.ft. 33729	-	45 sec.	kg. 1700 lb. 3748	DD17768

(1) Torque referred to 100 Psig (690 kPa) air supply pressure.
(2) Operating time without load.
* Max. pressure 65 Psig (448,5 kPa)

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ACTUATOR SELECTION

The selection of the actuator is possible by referring to table 1 which shows the torque values referred to the supply pressure of 100 Psig and the operating times (without load) relating to the positioner SA/CL-1 ND ¼" or ND ½".
Diagrams 3 and 4 allow a quick selection of the actuator according to the available air supply pressure and the requested torque.
The torque values are referred to an operating angle of 70° which is the standard operating angle of the lever in the CMV series. This angle of 70° is considered as the most suitable for an efficient control of dampers, butterfly valves and other final control elements. It is therefore recommendable, when performing the coupling to the final control device, not to amplify this angle.

Only for ON-OFF actuators, it is suggested to use for the final control device a larger operating angle corresponding to about 80/90°.
Generally, in relation to the requested reliability, it is advisable to select the actuator with a max. torque of 50-100% higher than the really requested torque in order to allow the operability of the actuator also in case of decrease of the air supply pressure.
Furthermore, if the actuator is provided with an air-lock system (see diagram fig. 5 and 6) which locks the actuator in the last position, then it is also necessary to make sure that the Set Point pressure of the air-lock is 10-20% higher than the limit pressure necessary to assure the requested torque.

OPERATING PRINCIPLE (Fig. 5 and 6)

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The diagram is shown in the condition of air supply failure. The configuration of the various components corresponds therefore to the function of lock-up in position.

When air supply is missing, it occurs that:
- the pneumatic pressure switch (PS), connected to the air-lock device (D), operates the two 3-way valves and intercepts the two connections of the pneumatic actuator which is locked in the last position.
  At the same time, the air-lock device issues a “status signal” which may be used as simple monitoring or fed into a logic system to allow a possible reset function.
  As the 3-way valve (E) is normally in position H, the air supply failure causes, after the intervention of the air-lock, also the automatic coupling of the half-nut (C) with the manual control. Consequently, the whole system is mechanically locked in the last position.
  In order to operate by means of the handwheel manual control with, it is necessary to open the by-pass (B) and exclude the pneumatic actuator (A).
  Furthermore, if the system is not fitted with a reset device, it is suggested, after the intervention of the air-lock (D), to change over the commutator (E) onto position (I). In such a way the stable coupling of the half-nut is guaranteed even in case of reset of the air supply pressure.

TRANSFER FROM MANUAL CONTROL TO PNEUMATIC CONTROL (Fig. 4 and 5)

When performing the transfer from manual to automatic control, it is important to avoid any deviation in the setting of the final control device; it is furthermore advisable that the half-nut (C) is disengaged only if the torque, generated by the final control device, is previously transferred to the pneumatic actuator.
STI has designed and patented a system enabling to operate under conditions of maximum reliability.
Diagram of fig. 5, shows a semiautomatic configuration for which the presence of an operator is locally required:
When the air supply pressure is reset, the air-lock device is automatically re-activated and therefore, by closing the by-pass (B), the actuator (A) is free to operate.

By slowly closing the control signal in correspondence to the position assumed by the final control device, it is possible to identify the condition in which the actuator can fully support the torque generated by the final control device.
This condition in the actuators series CMV2 to CMV4 is identified by the vertical displacement of the manual control shaft which is provided with a special axial clearance in correspondence of the thrust bearing.
In such balancing conditions, no load at all lies on the manual control and therefore, by commutating the 3-way valve (E) onto position (H), the haft-nut (C) is disengaged without any deviation in the setting of the final control device and the actuator assumes its control function.

TECHNICAL DATA

Air supply:	dry instrument air 35/100 Psig (241/689 kPa)
Operating temperature (without heater):	-20 ÷ +80°C (-4 ÷ +176 °F)
Lever operating angle:	70°

Mechanical position indicator
Lever orientation variable on splined shaft
Pneumatic positioner with characterizable cam type SA/CL-1 ND ¼" or ND ½" for short operating times

Control signal:	standard 3 ÷ 15 Psig (special cams and/or special control signal available on request)

Air-lock device for lock-up in last position on air supply failure included in the supply

MATERIALS OF CONSTRUCTION

Bearing frame:	carbon steel
Housing:	carbon steel
Shaft:	alloy steel 38 NiCrMo4 heat treated
Lever:	carbon steel
Shaft support:	bronze for CMV2 to CMV4; spherical joints for CMV1
Manual control screw:	alloy steel 38 NiCrMo4 heat treated
Standard connections:	brass fittings and copper tubing

Painting: epoxydic

MAINTENANCE

For the various components maintenance and part-list it is recommandable to refer to the specific instruction manuals (i.e. SC pneumatic actuator, SA/CL1 linear positioner, AL/.. air-lock device, 2030 electronic position transmitter).

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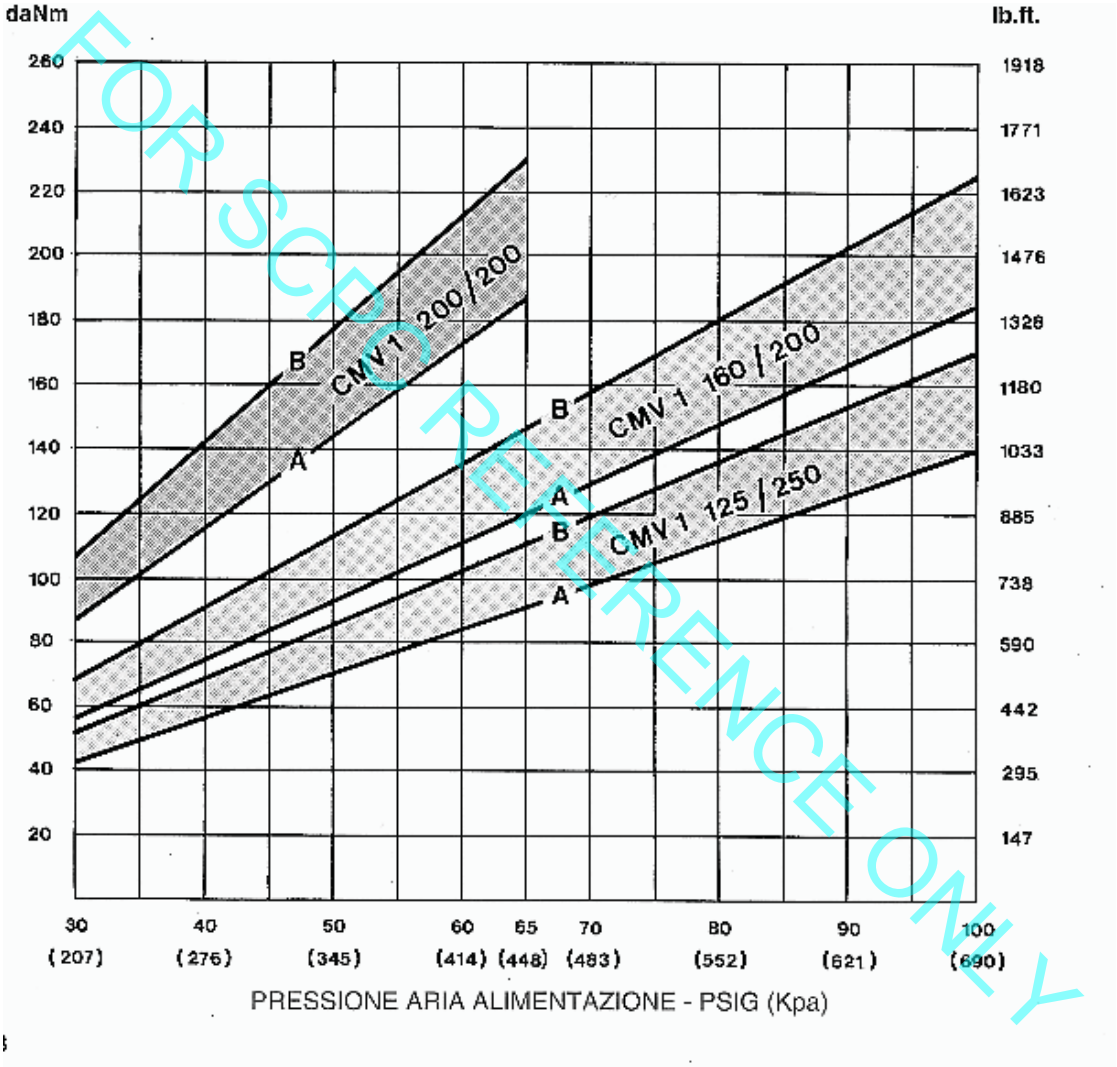
Diagram 3

TORQUE daNm (operating angle 70°)



TORQUE ft.lb. (operating angle 70°)

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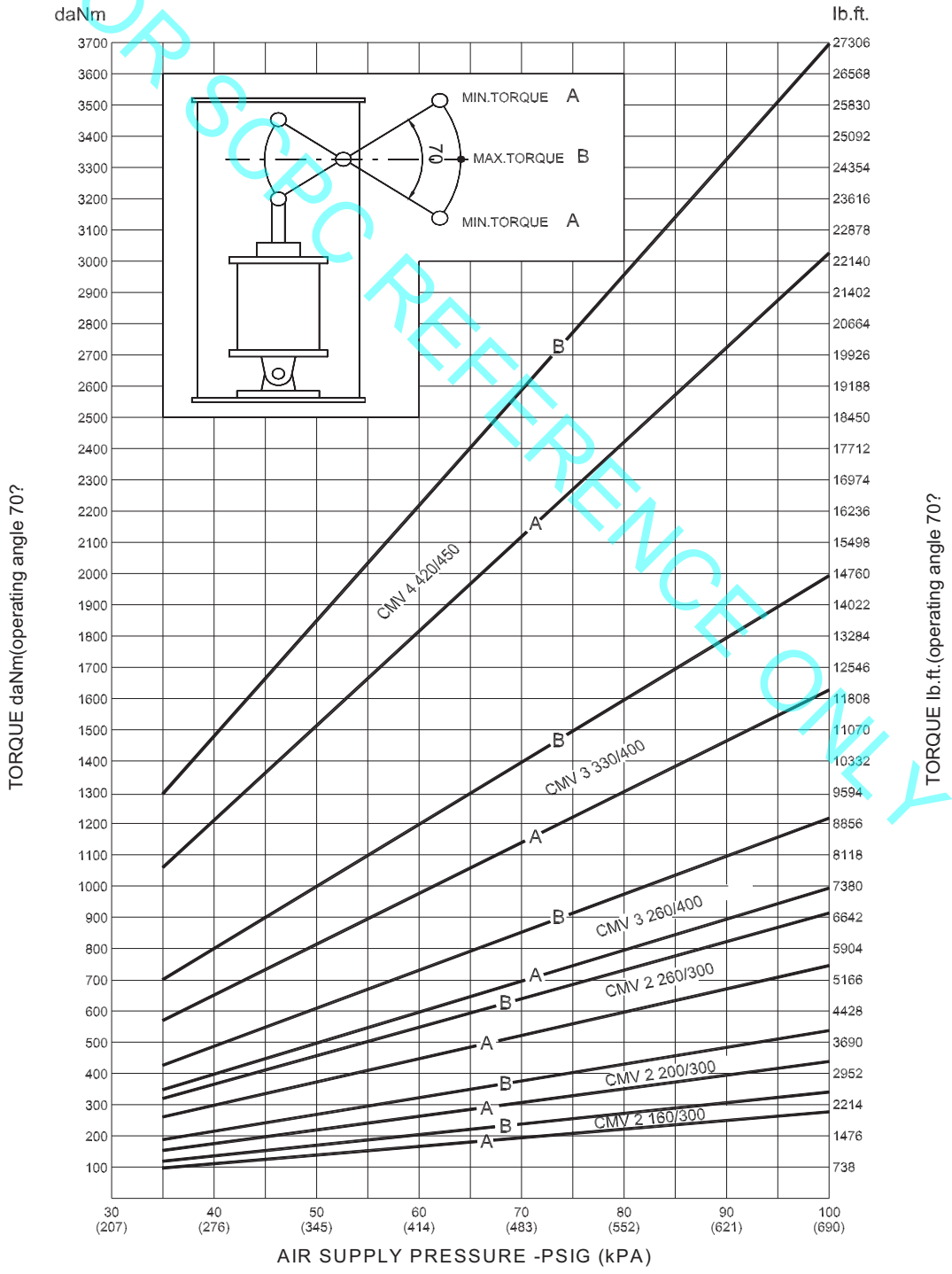
AIR SUPPLY PRESSURE – PSIG (kPa)

Fig. 3

Diagram 4



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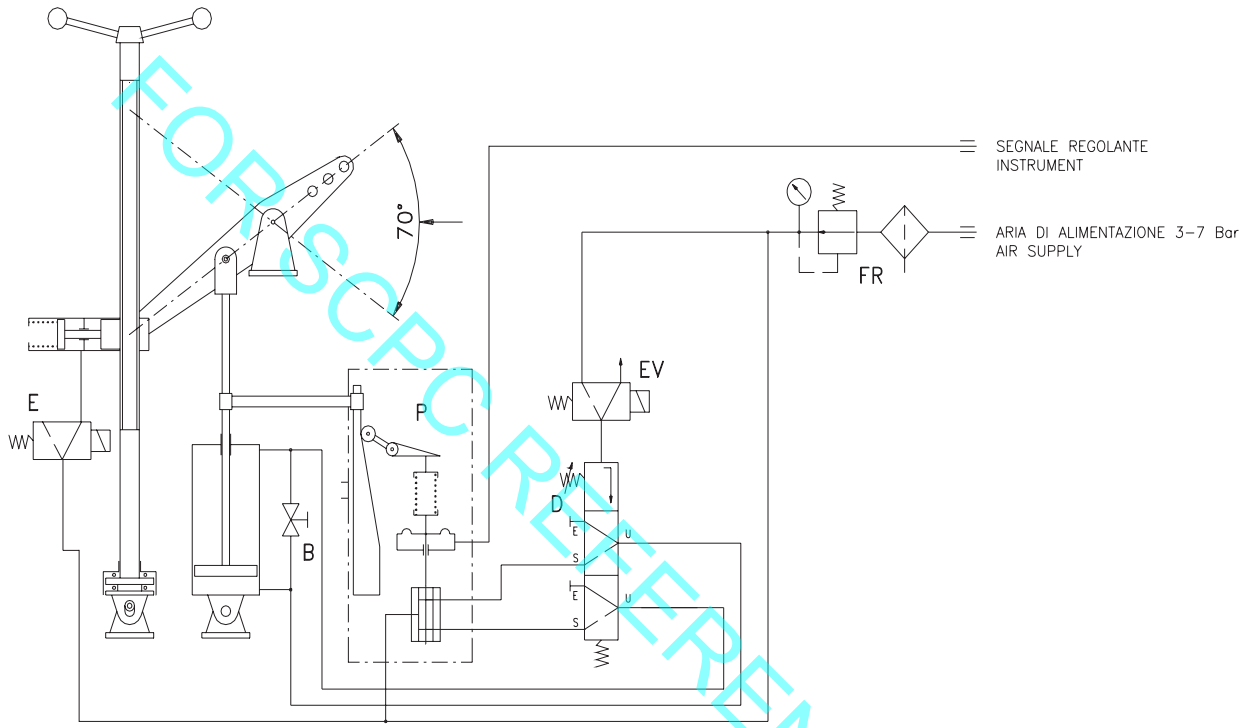
AIR SUPPLY PRESSURE – PSIG (kPa)

Fig. 4

Pneumatic diagram of control actuators series CMV2 to CMV4 with air-lock device for lock-up in last position on air supply failure



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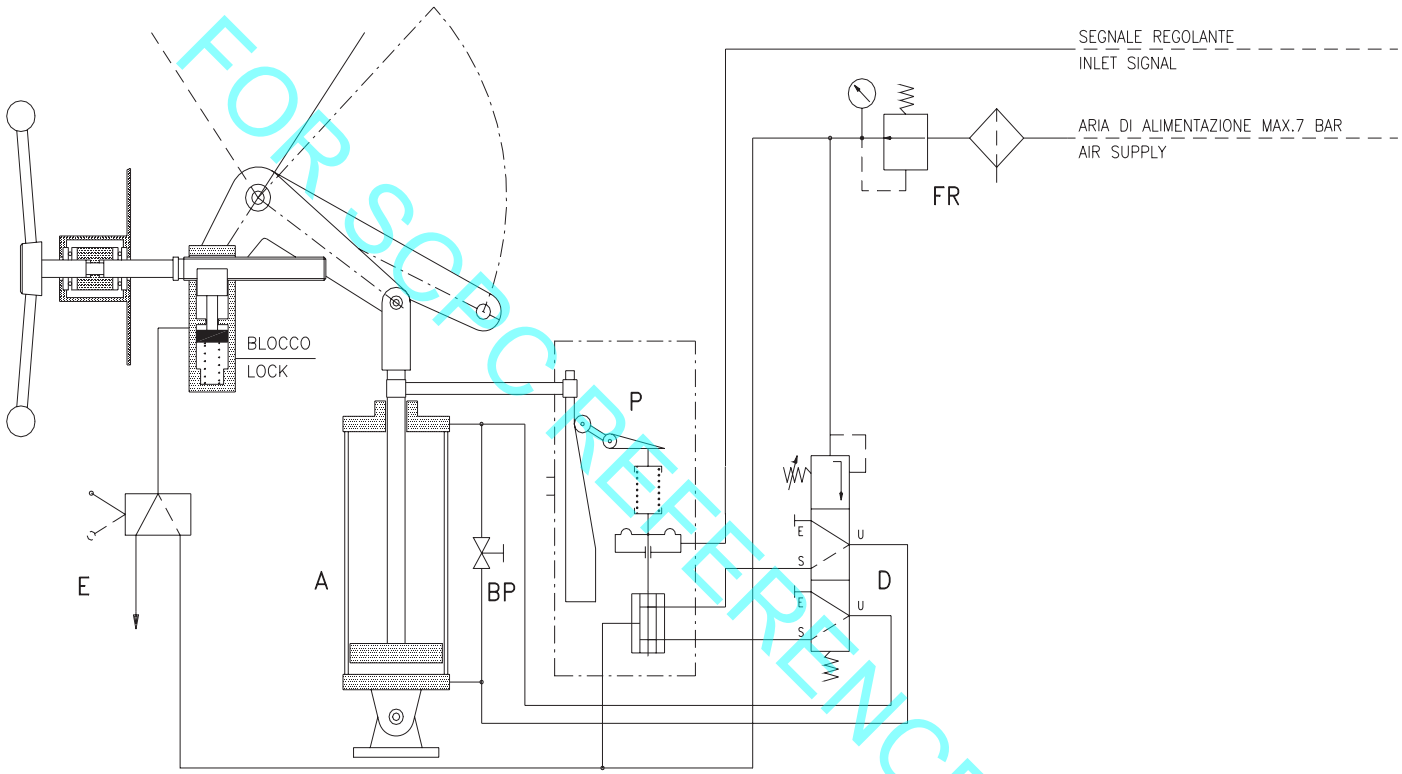
A	Actuator
B	By-pass valve
C	Half-nut
D	Pneumatic commutator
E	3-way valve
F	Filter
P	Positioner
Ps	Pneumatic pressure switch
Bss	Balancing sensor
EV_1	3-way solenoid valve

Fig. 5

Pneumatic diagram of control actuators series CMV1 with air-lock device for lock-up in last position on air supply failure



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A	Actuator
B	By-pass valve
C	Half-nut
D	Pneumatic commutator
E	3-way valve
F	Filter
P	Positioner
Ps	Pneumatic pressure switch
Bss	Balancing sensor
EV_1	3-way solenoid valve

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Instruction, Installation and Maintenance Manual

Electronic position transmitter 2030 series – Limit switches 2130 series

INSTRUCTION MANUAL
Position transmitter – Limit switches

TR

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ANNEX

		/
I	Product description	/
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1. GENERALITA'

1.1 DATI DI IDENTIFICAZIONE DEL FABBRICANTE

Ogni trasmettitore di posizione mod.2030 o fine corsa mod.2130 è provvisto di un'apposita targa che riporta in modo indelebile il numero di matricola per la sua corretta identificazione.

E' vietato asportare la targa e/o sostituirla con altre targhe.

1. GENERAL PRESCRIPTIONS

1.1 MANUFACTURER'S IDENTIFICATION DATA

Every position transmitter mod.2030 or limit switches mod.2130 is provided with a name plate with its serial number.

Do not remove it and/or replace it with other name plates

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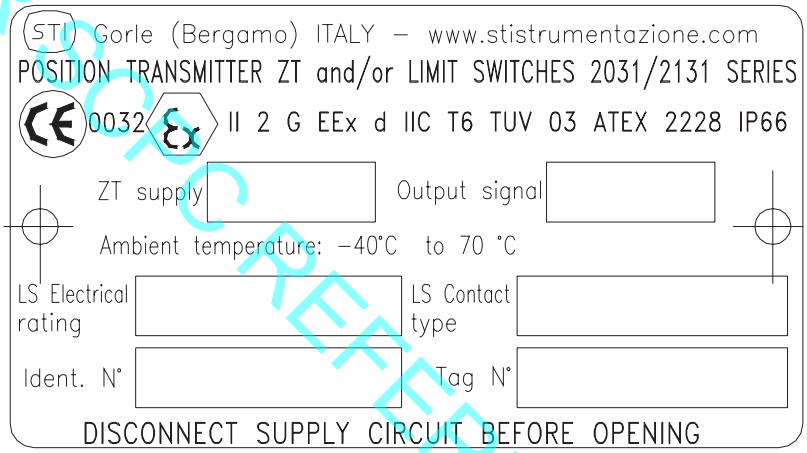

Nella versione antideflagrante il dispositivo è progettato per operare in area potenzialmente esplosiva classificata:

II 2 G EEx d IIC T6

Apparecchio gruppo II per industrie di superficie, categoria 2 per utilizzo in zona 1 con atmosfere esplosive causate da gas, vapori o nebbie. In conformità ai tipi di protezione standardizzati secondo EN 50014 ed EN50018 per idrogeno ed acetilene (classe massima di temperatura 85°C)

The explosionproof version the equipment is designed to operate in potentially explosive atmosphere classified as:

II 2 G EEx d IIC T6

Equipment group II surface industry, category 2 for use in zone 1 with explosive atmospheres caused by gases, vapors or mists. In conformity to types of protection standardized EN 50014 with flameproof enclosure EN 50018 for hydrogen and acetylene (temperature class maximum 85°C).

1.2 AVVERTENZE GENERALI DI SICUREZZA

1.2.1 Prima di effettuare qualsiasi intervento, gli operatori hanno il dovere di leggere il presente manuale, adottando le prescrizioni specifiche di sicurezza previste dalla legislazione della nazione di destinazione.

1.2.2 Gli interventi e le manutenzioni devono essere effettuate da operatori qualificati.

1.2.3 E' severamente vietato al cliente o a terzi (escluso personale della Società STI debitamente autorizzato) apportare modifiche di qualunque genere al trasmettitore di

1.2 GENERAL SAFETY INSTRUCTIONS

1.2.1 Before any operation is made, operators shall follow instructions of this manual and adopt the safety precautions by the country where the product is installed.

1.2.2 Operation and maintenance shall be carried out only by skilled staff.

1.2.3 It is strictly forbidden to customers (except for STI duly authorized staff) to modify the characteristics of position transmitter mod.2030 or limit switches mod.2130.

1.2.4 The apparatus has to be operated according

posizione mod.2030 o fine corsa mod.2130.

1.2.4 L'apparecchio deve essere utilizzato secondo quanto riportato in questo manuale e relativa specifica tecnica.

1.2.5 Quando utilizzato in zone potenzialmente esplosive la custodia deve essere cablata mediante cavi, pressacavi certificati secondo la direttiva 94/9/CE, e le norme EN 50014 ed EN 50018 per temperature tra -25°C e 80°C.

ATTENZIONE: Le connessioni elettriche devono essere eseguite in accordo ai dati di targa, schema e direttiva 94/9/EC (anche per i raccordi, cavi e pressacavi utilizzati)

1.3 CONDIZIONI PER L'IMMAGAZZINAMENTO

1.3.1 Conservare il trasmettitore di posizione mod.2030 o fine corsa mod.2130 in luoghi asciutti e puliti, avendo cura di prendere quelle precauzioni necessarie per impedirne il contatto con polveri, sporcizia ed umidità durante le fasi di immagazzinamento.

1.3.2 Non dovranno essere rimosse le protezioni delle varie connessioni nonché eventuali fermi meccanici.

1.4 MANUTENZIONE

1.4.1 Nessun cambiamento può essere fatto su apparecchi che funzionano in atmosfera potenzialmente esplosiva.

1.4.2 Togliere tensione prima di aprire la custodia

1.4.3 Deve essere eseguita da personale istruito per operare in zone potenzialmente esplosive

1.4.4 Equipaggiamenti ed utensili devono essere adatti all'utilizzo in zone potenzialmente esplosive

1.4.5 Pulire solo con uno straccio umido

1.4.6 Il grasso utilizzato deve essere secondo le prescrizioni del costruttore, e comunque avere un punto di infiammabilità > 140°C

2. TRASMETTITORE DI POSIZIONE

2.1 GENERALITA'

Il trasmettitore elettronico di posizione STI mod.2030 è adatto per installazioni su organi rotanti per angoli fino a 90°. E' stato progettato per soddisfare richieste su applicazioni industriali quali ad esempio attuatori pneumatici.
Il trasmettitore elettronico di posizione STI mod.2030 è anche disponibile con certificazione EExd IIC T6 in accordo alla norme CENELEC EN 50014/50018 per utilizzo in atmosfere potenzialmente esplosive.
Inoltre, la sua custodia può contenere una vasta gamma di fine corsa sia meccanici che induttivi fino a 4. I fine corsa, collegati ad una

to the appropriate data in the data sheet and in this manual.

1.2.5 When mounting in the hazardous area, the explosion proof housing must be provided with cables, cable entries and filler plugs which are at least certified according to directive 94/9/CE, EN 50014 ,EN 50018 and suitable for -45°C up to 80°C.

CAUTION: The electrical connections must be done according to the diagram, label and 94/9/EC directive (also for used fitting, cable and cable gland)

1.3 STORAGE

1.3.1 Store the position transmitter mod.2030 or limit switches mod.2130 in dry and clean places and take all necessary measures to avoid its contact with dust, dirt and humidity during storage.

1.3.2 Do not remove connections, protections and/or mechanical locks.

1.4 MAINTENANCE

1.4.1 No changes can be made to the apparatus , witch are operated in hazardous area.

1.4.2 Disconnect supply circuit before enclosure opening

1.4.3 Must be done by qualified person who is trained to operate into potentially explosive atmosphere

1.4.4 Tools and equipment must be suitable for potentially explosive atmosphere where the actuator is installed

1.4.5 Cleaning must be done with a humid cloth

1.4.6 Greases must be used according to the manufacturer's instructions, however their flammability point must be > 140°C

2. POSITION TRANSMITTER

2.1 GENERAL FEATURES

The electronic position transmitter STI mod. 2030 is suitable for installation on rotary motion devices with angles up to 90°. It has been designed to meet the most various industrial application requirements, i.e. on board of pneumatic actuators.
For use in potentially explosive atmospheres, an explosion proof certification EExd IIC T6 according to CENELEC EN 50014/50018 is also available with the position transmitter.
Besides, the enclosure has the capability to house a wide range of either mechanical or inductive limit switches (up to 4). A shaft with manually-adjustable cam is used for the accurate and easy setting of the limit

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morsettiera, sono impostabili attraverso delle camme regolabili manualmente.

2.1.1 Principio di funzionamento
Il trasmettitore elettronico di posizione STI mod.2030 è basato sull'effetto EDDY (Fig.1).
La rotazione eccentrica di un tamburo viene rilevata da un gruppo di sensori e convertita in un segnale 4÷20 mA.

switches. Limit switches are prewired to a terminal strip.

2.1.1 Operating principle
The operating principle of the electronic position transmitter STI mod. 2030 is based on the EDDY effect (see Fig.1). The eccentric rotation of a cylinder is detected by two sensors and converted into a 4 to 20 mA signal

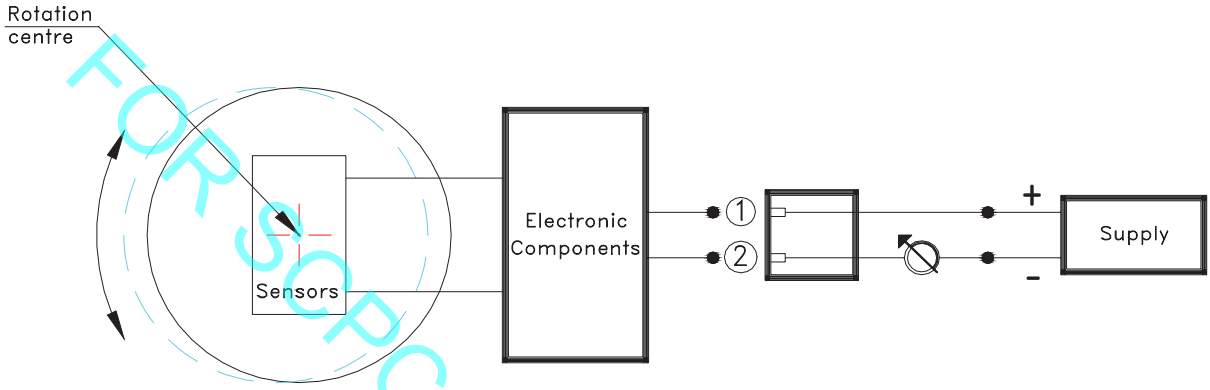

Fig.1

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2.1.2 Gruppo sensori
Il trasmettitore di posizione STI mod.2030 utilizza complessivamente 2 sensori.
La rotazione dell'albero (6) genera una rotazione eccentrica del tamburo(7) (Fig.2-3/allegato 1). Pertanto questo determina una variazione della distanza relativa tra i due sensori (10) ed il tamburo (7).

2.1.2 Sensors
Two sensors are included in the position transmitter.
The rotation of the shaft (6) causes the eccentric cylinder (7) to rotate (see Fig.2-3/enclosure 1). This means while one of the two sensors (10) detects an increasing distance from the cylinder, the other one feels the opposite motion.

2.1.3 Circuito elettronico
Il circuito elettronico del trasmettitore di posizione mod.2030 comprende un generatore ad onda quadra atto ad alimentare i due sensori.
Il segnale, proveniente dai sensori, viene elaborato da un demodulatore e successivamente condizionato ed amplificato, per produrre la corrente di uscita proporzionale alla rotazione in ingresso.
Il trasmettitore elettronico di posizione STI mod.2030 dispone della regolazione indipendente di zero, campo e linearità.

2.1.3 Electronic circuit
The electronic circuit of the position transmitter includes a square wave generator necessary for the sensors feeding. The signal from the sensors is demodulated, subject to conditioning and amplified by the electronic circuit; as a result, a 4 to 20 mA signal proportional to the input rotation is generated.
The electronic position transmitter is provided with independent setting of the zero, range and linearity.

2.2 CARATTERISTICHE TECNICHE

2.2 TECHNICAL FEATURES

MATERIALI / *MATERIALS*	
Contenitore, coperchio / *Housing, cover*	Aluminum G-Al Si 9 UNI 3051
	Stainless Steel A351-CF8M (AISI 316)
Albero / *Shaft*	Stainless Steel AISI 303
	Stainless Steel AISI 316

CARATTERISTICHE TECNICHE / TECHNICAL FEATURES

- Segnale in uscita / *Output signal* :
  4 to 20 mA
- Alimentazione / Supply :
  12 to 32 Vdc
- Azione / Calibration angle :
  Diretta o inversa, reversibile in campo (Rotazione antioraria o oraria)
  Direct or reverse, field reversible (Clock or Counter-clock-wise rotation)
- Grado di Protezione / Protection degree:
  IP66 (explosionproof EExd IIC T6)
- Temperatura di esercizio / *Ambient temperature*:
  -25° to +70°C (-40° to +70°C if specified)
- Angolo operativo / *Operating angle*:
  (α) 30° to 90°
- Tempo di risposta / *Response time* :
  ≤ 0.2 s
- Entrata cavi / *Conduit entries*:
  1 or 2 - ½" NPT F (M20x1.5 option)

Tutti i dati che seguono si riferiscono ad angolo operativo / *Following data are referred to operating angle* α=90°

- Influenza della temperatura / *Temperature influence*:
  < 0.3% every 10°C

- Influenza dell'alimentazione / *Supply influence*:
  < 0.2% for 12 to 32 Vdc

- Accuratezza (linearità, ripetibilità ed isteresi) / *Accuracy including linearity, repeatability,hysteresis* :
  < 0.5%

Questo strumento soddisfa i requisiti per la certificazione di conformità al marchio CE. *This device meets requirements for CE conformity certification*.

Norme di riferimento / *Basic technical specifications*:
CEI EN 55022; CEI EN 50082-2

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2.3 INSTALLAZIONE

Installare il trasmettitore di posizione sull'attuatore usando la connessione VDI/VDE 3845 o , se è montato un albero con attacco std, utilizzando un kit di montaggio adeguato (non fornito con il trasmettitore), generalmente costituito da una leva di accoppiamento.
Nel caso di albero con attacco std, effettuare una corsa completa del servomotore per assicurarsi che la leva si muova liberamente.

Assialità d'accoppiamento steli è necessaria per evitare usura e surriscaldamento

2.4 SETTAGGIO DEL TRASMETTITORE DI POSIZIONE

Il trasmettitore elettronico STI mod.2030 viene settato in fabbrica con un angolo operativo di 90° ed azione diretta. Per azioni ed angoli operativi diversi si dovrà procedere al suo risettaggio come segue (si faccia riferimento alla Fig2 tranne dove diversamente specificato):

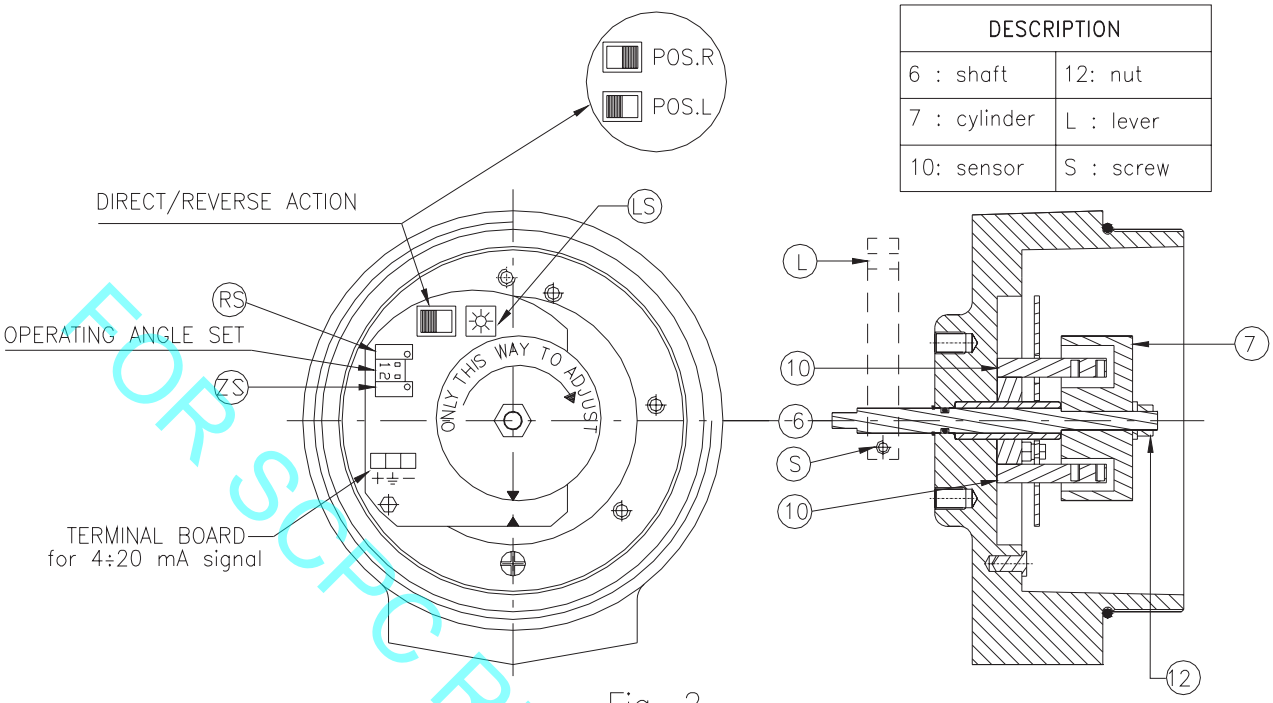
2.3 INSTALLATION

Install the position transmitter on the actuator by using a VDI/VDE 3845 connection. If the actuator is provided with a standard shaft, a suitable mounting kit, not supplied with the transmitter and generally made by a coupling lever, shall be used.
In case of standard shaft, it is also necessary to check the transmitter lever is free to move by operating the actuator throughout a complete stroke.

Axial stems coupling is necessary to prevent wear and overheating

2.4 POSITION TRANSMITTER SETTING

The electronic transmitter STI mod. 2030 is factory set for 90° operating angle and direct action. For different angles or action, it is necessary to readjust it as follows (refer to Fig.2 unless otherwise specified):



DESCRIPTION	
6 : shaft	12: nut
7 : cylinder	L : lever
10: sensor	S : screw

Fig. 2

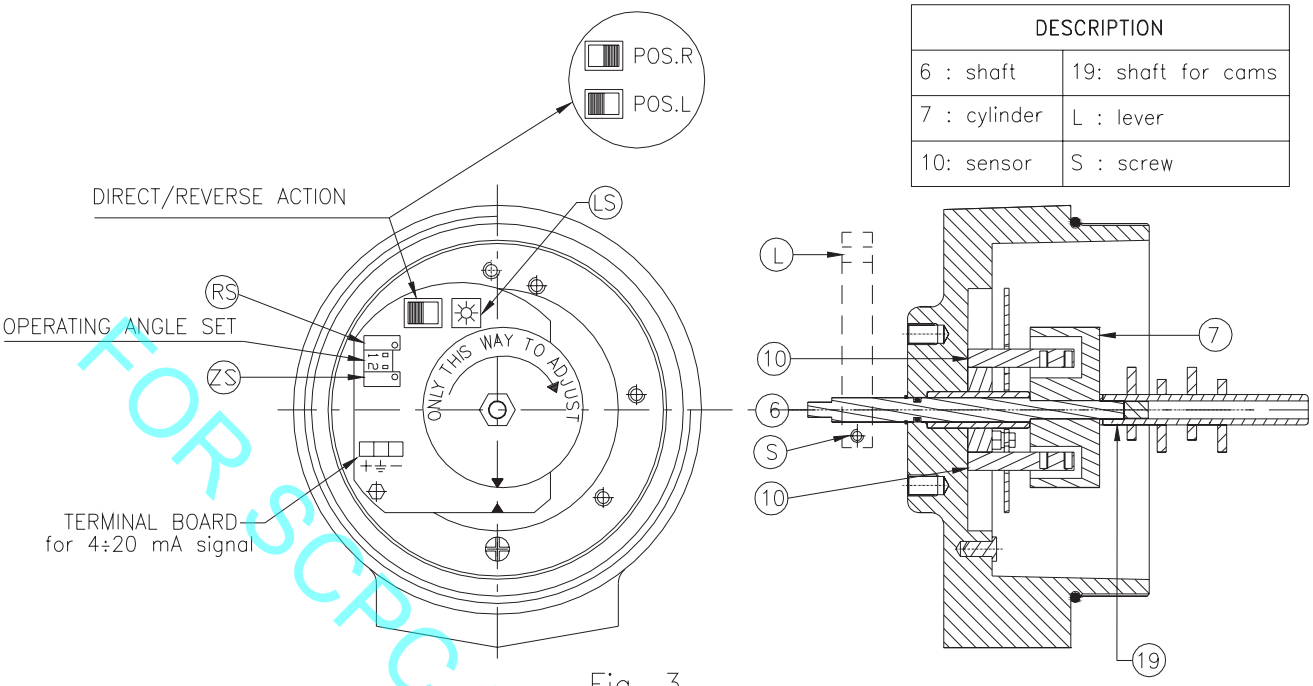
FOR SCPC REFERENCE ONLY

- Togliere il coperchio dal corpo del trasmettitore
- Procedere al collegamento ad un'alimentazione 12÷32 Vcc (Fig.1), utilizzando un cavo schermato.
- Solo per i modelli senza la scritta 'Only this way to adjust' sulla parte superiore del cilindro: se c'è un albero con attacco std., allentare la vite di fissaggio(S) che collega l'albero (6) alla leva (L) in modo tale che l'albero si possa muovere liberamente; se invece vi è un attacco del tipo VDI/VDE 3845, allentare il dado (12) .
- Settare l'attuatore al 50% della corsa
- Ruotare manualmente il tamburo fino a che l'indice su di esso sia allineato con quello presente sulla scheda elettronica ed il segnale in uscita corrisponda a 12 mA

Si definisce azione diretta ed inversa quanto segue:
azione diretta: una rotazione in senso anti-orario del tamburo (7) produce un aumento del segnale in uscita
azione inversa: una rotazione oraria del tamburo (7) produce un aumento del segnale in uscita
Per i modelli dotati di commutatore per il settaggio dell'azione diretta/inversa (si veda fig.2-3):

- Remove the cover from the transmitter housing.
- Connect a 12 to 32 Vdc supply (see Fig.1) using a screened cable.
- Just for the models without the writing 'Only this way to adjust' on the cylinder's topside: in case of standard shaft, loosen the allen screw (S) which connects the shaft (6) to the lever (L). The shaft will become free to move. Otherwise, if a VDI/VDE 3845 connection is used, loosen the nuts (12) .
- Set the actuator at 50% of the stroke.
- Rotate manually the cylinder till the index on the cylinder is aligned with that on the electronic board and the output signal is equal to 12 mA.

Definition of direct or reverse action:
direct action: the counter-clockwise rotation of the cylinder (7) corresponds to the increasing of the output signal
reverse action: the clockwise rotation of the cylinder (7) corresponds to the increasing of the output signal
For the models equipped with a commutator to choose the direct/reverse action (see fig.2-3):

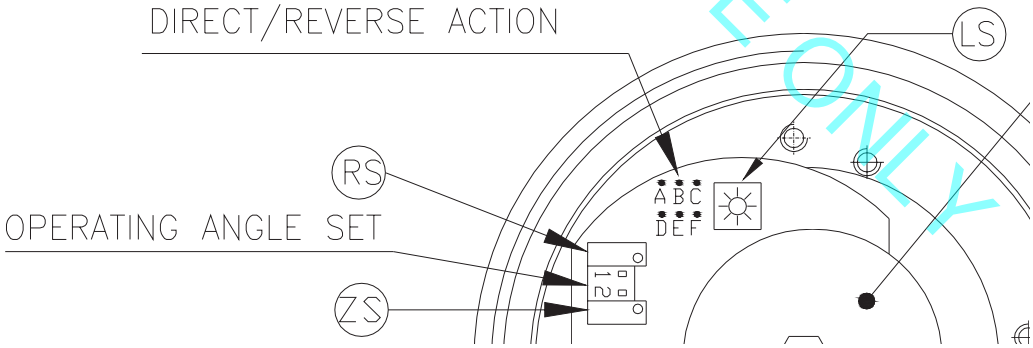


DESCRIPTION	
6 : shaft	19: shaft for cams
7 : cylinder	L : lever
10: sensor	S : screw

Fig. 3

per scegliere l'azione diretta, settare il commutatore 'azione diretta/inversa' nella POS.R; per scegliere l'azione inversa, settare il commutatore 'azione diretta/inversa' nella POS.L
(Per i modelli dotati di 6 pins per il settaggio dell'azione diretta/inversa (si veda Fig.4/allegato 1):per scegliere l'azione diretta, collegare i pins B-C e E-F; per scegliere l'azione inversa, collegare i pins A-B e D-E.)

to choose direct action, switch the commutator 'direct/reverse action' to POS.R; to choose reverse action, switch the commutator 'direct/reverse action' to POS.L
For the models equipped with 6 pins to choose the direct/reverse action (see Fig.4/enclosure 1):
to choose direct action, connect pins B-C and E-F; to choose reverse action, connect pins A-B and D-E



- Impostare l'angolo operativo α posizionando l'interruttore 1 e 2 nel seguente modo:

- To set the operating angle α, position the switch 1 and 2 in the following way:

$65° \leq \alpha \leq 90°$

$45° \leq \alpha < 65°$

$30° \leq \alpha < 45°$

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Solo per i modelli senza la scritta 'Only this way to adjust' sulla parte superiore del cilindro: se è montato un albero con connessione std, avvitare la vite di bloccaggio (S) per bloccare la leva (L) con l'albero (6); se c'è una connessione del tipo VDI/VDE 3845, connettere semplicemente l'albero (6) all'albero dell'attuatore e bloccare il dado (12) (si veda Fig.4/allegato 1).

- Effettuare una corsa completa del servomotore; l'angolo operativo sarà distribuito in modo simmetrico rispetto all'indice presente sulla scheda elettronica.
- Settare l'attuatore con corsa 0% e regolare il trimmer ZS per ottenere un segnale in uscita di 4 mA.
- Settare l'attuatore con corsa 100% e regolare il trimmer RS per ottenere un segnale in uscita di 20 mA.
- Settare l'attuatore con corsa 0% e 100% e verificare che il segnale in uscita sia rispettivamente 4 e 20 mA.
-

NOTA: La linearità è impostata in fabbrica e la posizione del trimmer LS non deve essere in alcun caso modificata.

2.5 SETTAGGIO DEL TRASMETTITORE DI POSIZIONE CON FINE CORSA

Seguire le istruzioni contenute nel par.2.4 e quindi:

- Eseguire i cablaggi elettrici alla morsettiera come indicato sullo schema elettrico allegato. Verificare che l'alimentazione elettrica sia adeguata rispetto alle caratteristiche elettriche dei fine corsa
- Sollevare e ruotare manualmente la camma finchè fa commutare il fine corsa in corrispondenza dell'angolo desiderato (si veda Fig.3/allegato 1)
- Verificare il buon funzionamento dei fine corsa utilizzando un tester.

3. FINE CORSA MOD.2130

3.1 GENERALITA'

La custodia per fine corsa STI mod.2130 è adatto per installazioni su organi rotanti per angoli fino a 90°. E' stato progettato per soddisfare richieste su applicazioni industriali quali ad esempio attuatori pneumatici.
La custodia per fine corsa STI mod.2130 è anche disponibile con certificazione EExd IIC T6 in accordo alla norme EN 50014/50018 per utilizzo in atmosfere potenzialmente esplosive.
Inoltre, può contenere una vasta gamma di fine corsa sia meccanici che induttivi fino ad un numero di 4. I fine corsa, collegati ad una

Just for the models without the writing 'Only this way to adjust' on the cylinder's topside: if a standard shaft is used, tighten the allen screw (S) so that the lever (L) and the shaft (6) are locked together; in case of a VDI/VDE 3845 connection, simply connect the transmitter shaft to the actuator shaft and tighten the nuts (12) (see Fig.4/enclosure1).

- Make a complete actuator stroke: the operating angle shall be symmetric to the index on the electronic board.
- Set the actuator at 0% stroke and adjust the ZS trimmer: the output signal shall be 4 mA.
- Set the actuator at 100% stroke and adjust the RS trimmer: the output signal shall be 20 mA.
- Set the actuator at 0% and 100% stroke and check the output signal is respectively 4 and 20 mA.

NOTE: Linearity is factory set and should not vary the set of the LS trimmer in any case.

2.5 POSITION TRANSMITTER SETTING WITH LIMIT SWITCHES.

Follow the instructions in par.2.4 from and:

- Install all the electrical connections to the terminal strip as per the electrical diagram enclosed. Make sure that the power supply does not exceed the electrical rating of the limit switches.
- Lift and rotate by hand the cam until it makes trigger the limit switches at the required angle (see Fig.3/enclosure1)
- Check the function of the limit switches by a tester.

3 LIMIT SWITCHES MOD.2130.

3.1 GENERAL FEATURES

The limit switches STI mod. 2130 is suitable for installation on rotary motion It has been designed to meet the most various industrial application requirements, i.e. on board of pneumatic actuators.
For use in potentially explosive atmospheres, an explosion proof certification EExd IIC T6 according to EN 50014 and EN50018 is also available.
The enclosure has the capability to house a wide range of either mechanical or inductive limit switches (up to 4). A shaft with manually-adjustable cam is used for the

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morsettiera, possono essere settati facilmente attraverso delle camme regolabili manualmente.

3.2 INSTALLAZIONE

Installare i fine corsa sull'attuatore usando la connessione VDI/VDE 3845 o , se è montato un albero con attacco std, utilizzando un kit di montaggio adeguato (non fornito con i fine corsa), generalmente costituito da una leva di accoppiamento.
Nel caso di albero con attacco std, effettuare una corsa completa del servomotore per assicurarsi che la leva si muova liberamente.

Assialità d'accoppiamento steli è necessaria per evitare usura e surriscaldamento

accurate and easy setting of the limit switches. Limit switches are prewired to a terminal strip.

3.2 INSTALLATION

Install the limit switches on the actuator by using a VDI/VDE 3845 connection. If the actuator is provided with a standard shaft, a suitable mounting kit, not supplied with the enclosure for limit switches and generally made by a coupling lever, shall be used.
In case of standard shaft, it is also necessary to check the lever is free to move by operating the actuator throughout a complete stroke.

Axial stems coupling is necessary to prevent wear and overheating

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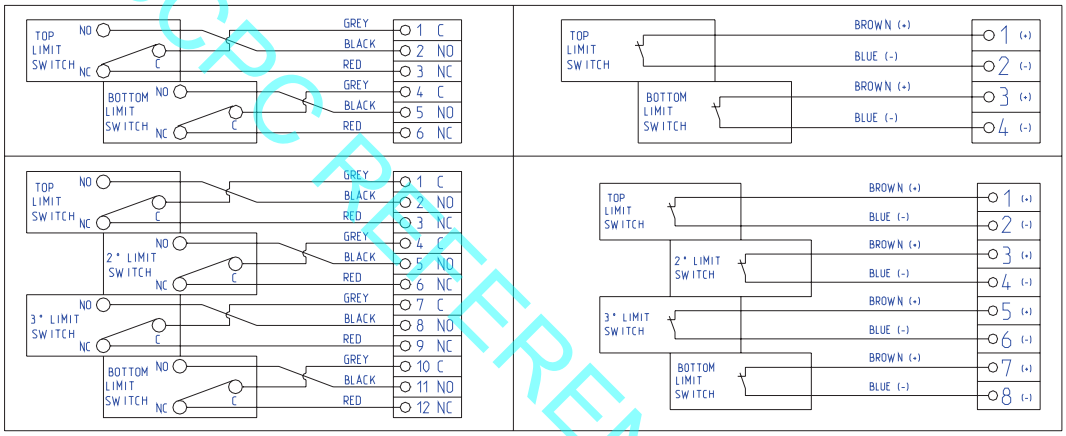


Mechanic LS *Inductive LS*

3.3 SETTAGGIO DEI FINE CORSA

Eseguire i cablaggi elettrici alla morsettiera come indicato sullo schema elettrico allegato. Assicurarsi che l'alimentazione elettrica sia adeguata alle caratteristiche elettriche dei fine corsa.
Sollevare e ruotare manualmente la camma finchè fa commutare il fine corsa in corrispondenza dell'angolo desiderato.
Verificare il buon funzionamento dei fine corsa utilizzando un tester.

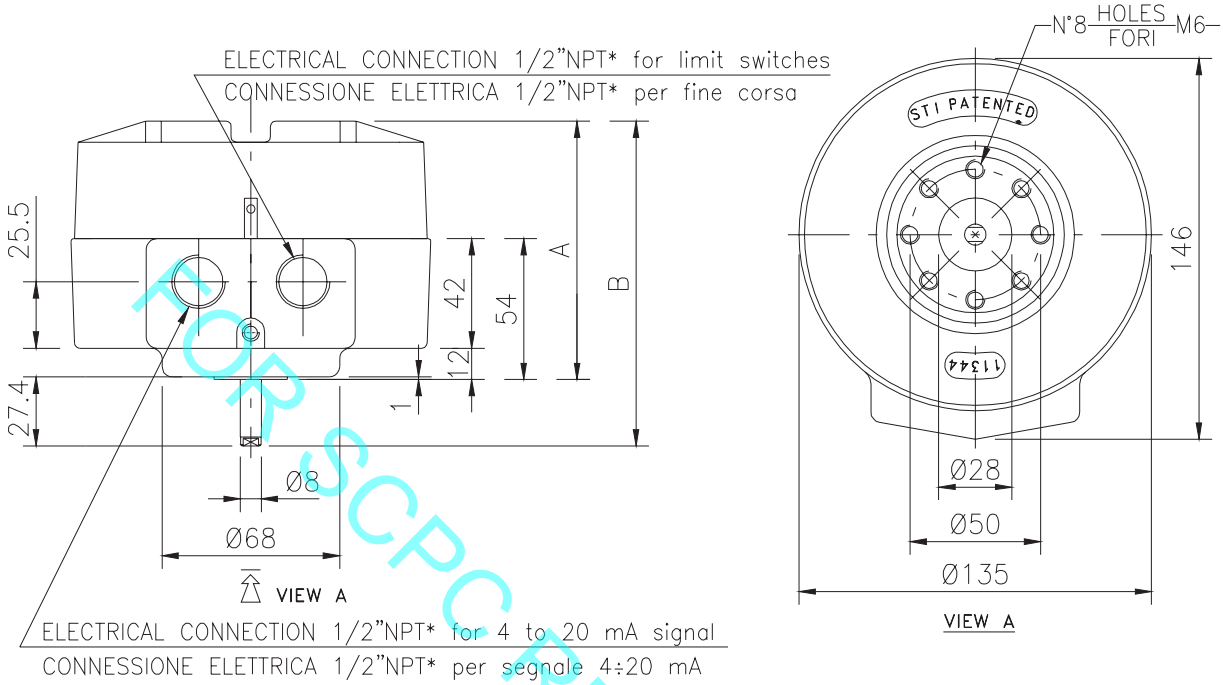
3.3 SETTING OF POSITION SENSORS

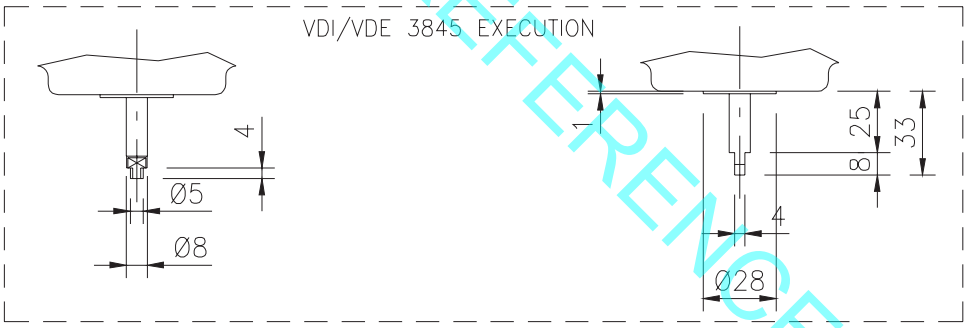
Install all the electrical connections to the terminal strip as per the electrical diagram enclosed. Make sure that the power supply does not exceed the electrical rating of the limit switches.
Lift and rotate by hand the cam until it makes trigger the limit switches at the required angle
Check the function of the limit switches by a tester.

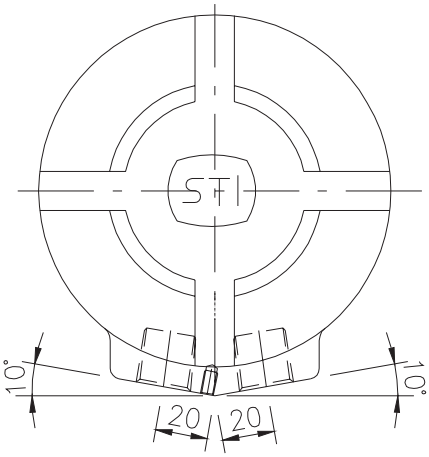
	Honeywell **V5B010CB3D**	Honeywell **V5J012BB1D**	Pepperl+Fuchs **NJ2-V3-N**
Tensione nominale *Nominal voltage*	250Vac 16 A	30Vdc, 125Vac 100 mA	8V
Temperatura d'esercizio *Ambient temperature*	-55°C +85°C	-55°C +85°C	-25°C +100°C
Tipo di contatto *Contact type*	Silver cadmium oxide	Gold alloy	Inductive
Materiali custodia *Housing material*	Thermoplastic	Thermoplastic	PBT

INSTRUCTION MANUAL
Position transmitter – Limit switches

TR







	POSITION TRANSMITTER ONLY 2030 SERIES		POSITION TRANSMITTER AND/OR LIMIT SWITCHES 2030 or 2130 SERIES	
	STD SHAFT	VDI/VDE 3845 EXECUTION	STD SHAFT	VDI/VDE 3845 EXECUTION
A	99	99	150	150
B	125.4	131	176.4	182

* :M20X1.5 execution is also available
:è disponibile anche l'esecuzione M20X1.5

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INSTRUCTION MANUAL
Position transmitter – Limit switches

TR

PRODUCT DESCRIPTION

2		3		-			-		-		

Type
- **0** = pos. transmit.with or w/out L/S
- **1** = limit switches only

Protection degree
- **0** = IP66
- **1** = Eexd IIC T6 and IP66

Limit switches number
- **0** = no limit switches
- *1* = *n° 1*
- **2** = **n° 2**
- *3* = *n° 3*
- **4** = **n° 4**

Limit switches type
- **O** = no limit switches
- **M** = mechanical with SPDT contacts(2 SPDT=1 DPDT) mod.V5B010CB3D
- *I* = *inductive proximity-two wires mod.NJ2-V3-N*
- *G* = *mechanical with gold alloy SPDT contacts mod.V5J012BB1D*

Electrical connection
- *1* = *n° 1 threaded hole 1/2" NPT*
- **2** = **n° 2 threaded holes1/2" NPT (one hole plugged if not needed)**
- *3* = *n° 1 threaded hole M20x1,5*
- *4* = *n° 2 threaded holes M20X1,5(one hole plugged if not needed)*

Operating shaft
- **S** = S.T.I. standard
- **N** = according VDI/VDE 3845

Material housing
- = Aluminum
- **SS** = AISI 316

Notes: **-options in bold are standard;options in italic are available only on application**
-standard shaft is to be used to substitute transmitters 2010 and 2020 and on top-mounted actuator support
-VDI/VDE 3845 shaft is to be used for 90° applications and with long-stroke cam adaptor

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CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

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LMYP-1 Limay,Philippine

Electrical Actuator Data Sheet For Fan

送風機電動作動器資料

本文編號 RF2-0580-DOC-2-0

REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	CERTIFIED / PROJECT NO.	PAGE
						CERTIFIED	
2						PROJ.	
1						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/01/27	PROJECT NO. **LMYP-1**	PAGE 1

ROTORK ACTUATOR SIZING CALCULATION

Customer	CCMC / FHI	Project No.:	RF2—0578/0581
Tag:	PA Fan	Date:	Nov. 15, 2013
Required Torque	540kgm	Q'ty:	4sets

No.	Item	Calculation	Remark
1	Valve Size		
2	Max. Torque	540kgm x 9.8=5292Nm	
3	Angle of travel	90 deg.	
4	Worm Gear Unit	MOW8	
5	Worm Gear Ratio	60:1	
6	Total Gear Ratio	60:1	
7	Actuator Output R.P.M.	21	
8	Worm Gear Output R.P.M	21/60=0.35	
9	Time Open & Close	0.25/0.35x60=43sec	
10	Gear Input Torque	5292/60x0.417=211Nm	
11	Actuator Output Torque	IQM35/24RPM output Torque 271Nm>211Nm	
12	Motor KW	1.43KW	
13	Motor Enclosure	IP68	
14	Motor Power	460VAC / 3ph / 60Hz	
15	Unit Type & Size	IQM35 / MOW8	

FOR SCEC REFERENCE ONLY

ROTORK ACTUATOR SIZING CALCULATION

Customer	CCMC / FHI	Project No.:	RF2—0577/0580
Tag:	SA Fan	Date:	Nov. 15, 2013
Required Torque	320kgm	Q'ty:	4sets

No.	Item	Calculation	Remark
1	Valve Size		
2	Max. Torque	320kgm x 9.8=3136Nm	
3	Angle of travel	90 deg.	
4	Worm Gear Unit	MOW7	
5	Worm Gear Ratio	60:1	
6	Total Gear Ratio	60:1	
7	Actuator Output R.P.M.	21	
8	Worm Gear Output R.P.M	21/60=0.35	
9	Time Open & Close	0.25/0.35x60=43sec	
10	Gear Input Torque	3136/60x0.417=125Nm	
11	Actuator Output Torque	IQM25/24RPM output Torque 152Nm>125Nm	
12	Motor KW	0.93KW	
13	Motor Enclosure	IP68	
14	Motor Power	460VAC / 3ph / 60Hz	
15	Unit Type & Size	IQM25 / MOW7	

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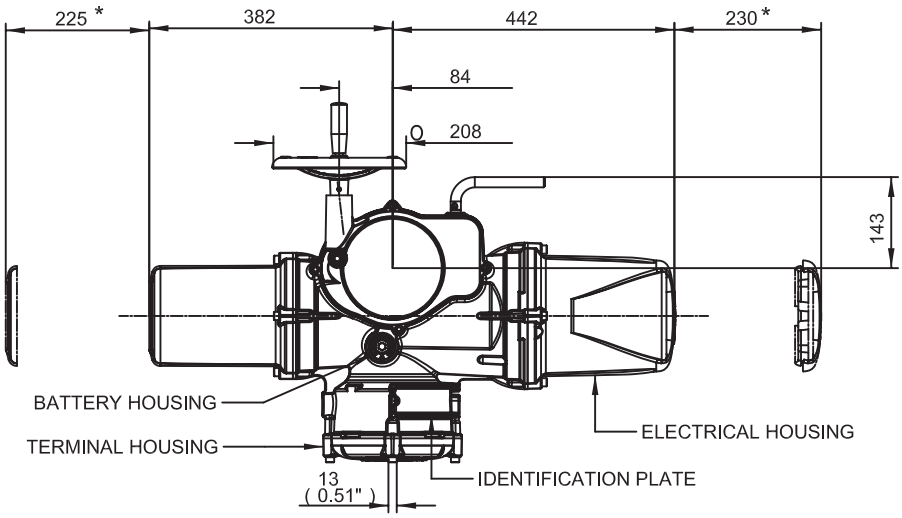

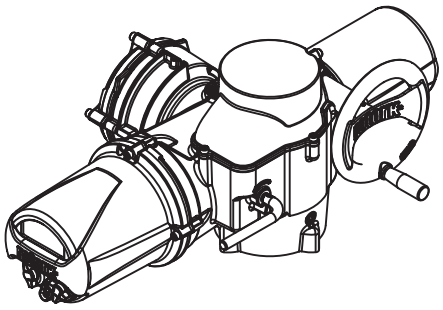

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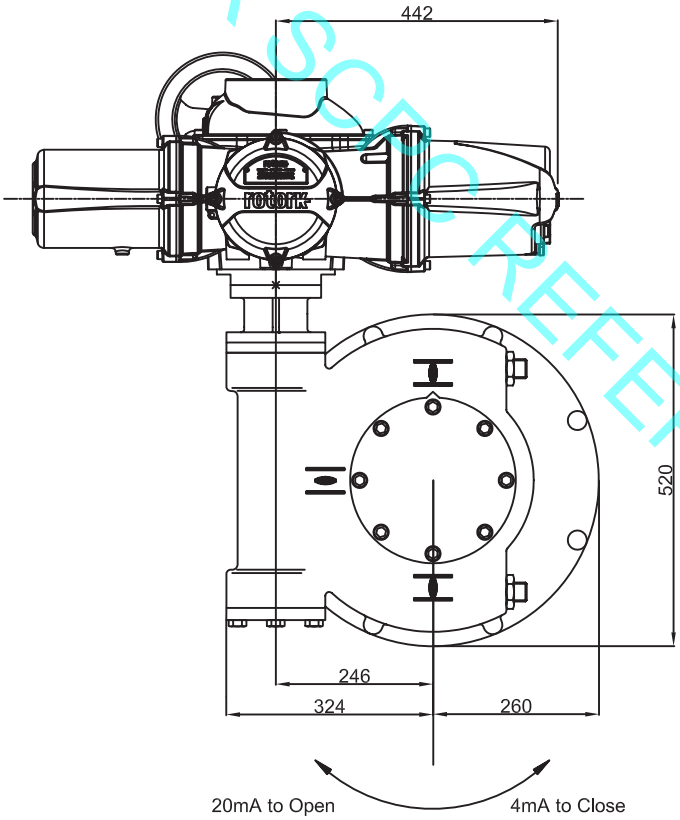
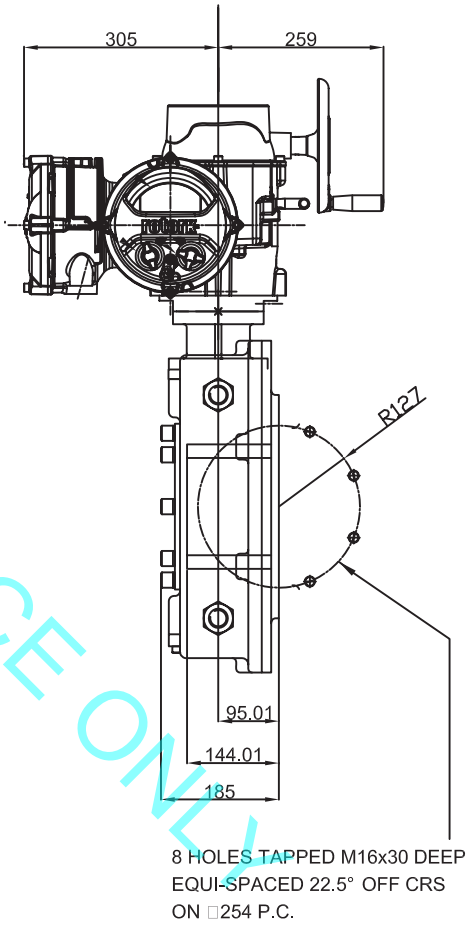
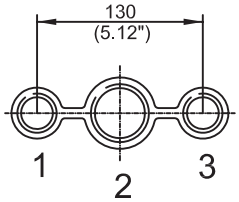




CONDUIT ENTRIES



DEFAULT HOLE SIZES:
HOLES 1 & 3 TAPPED M25
HOLE 2 TAPPED M40

Project	CCMC/FHI DVOP-1 / LMYP-1
Job No.	RF2-0578 /0581
Tag	P.A. Fan
Type	IQM35+MOW8
Q'ty	Q'ty: 4 sets

Title

DIMENSIONAL DATA SHEET FOR IQM35/MOW8 WITH SIDE HANDWHEEL

Date	A4	Doc Number	Ver No	Scale	Sheet No
170512		IQ3-3SP+BC	1	1:12	1 of 1

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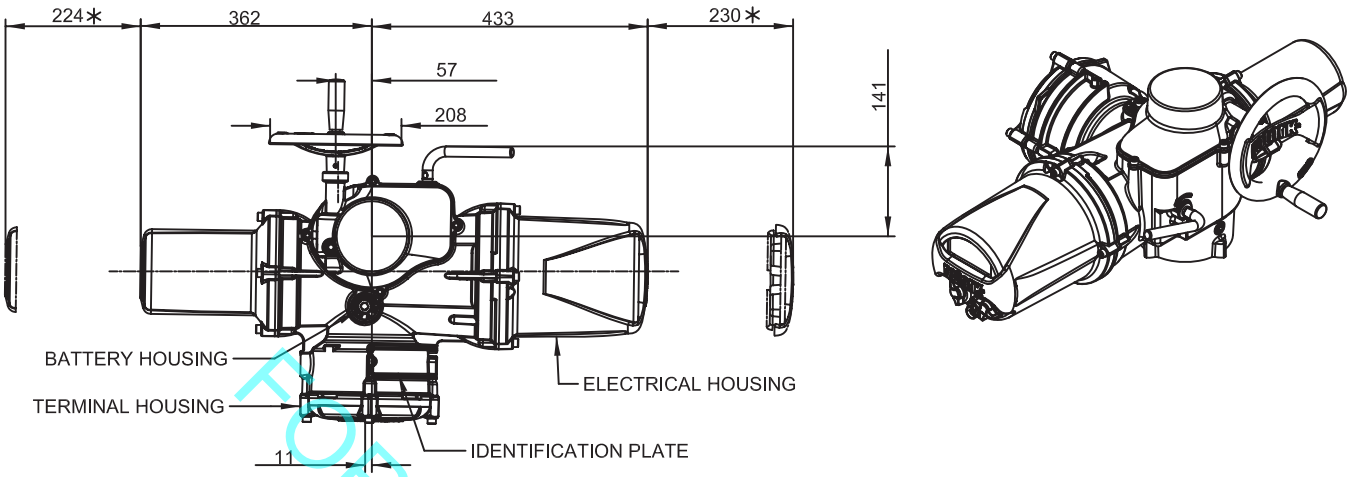
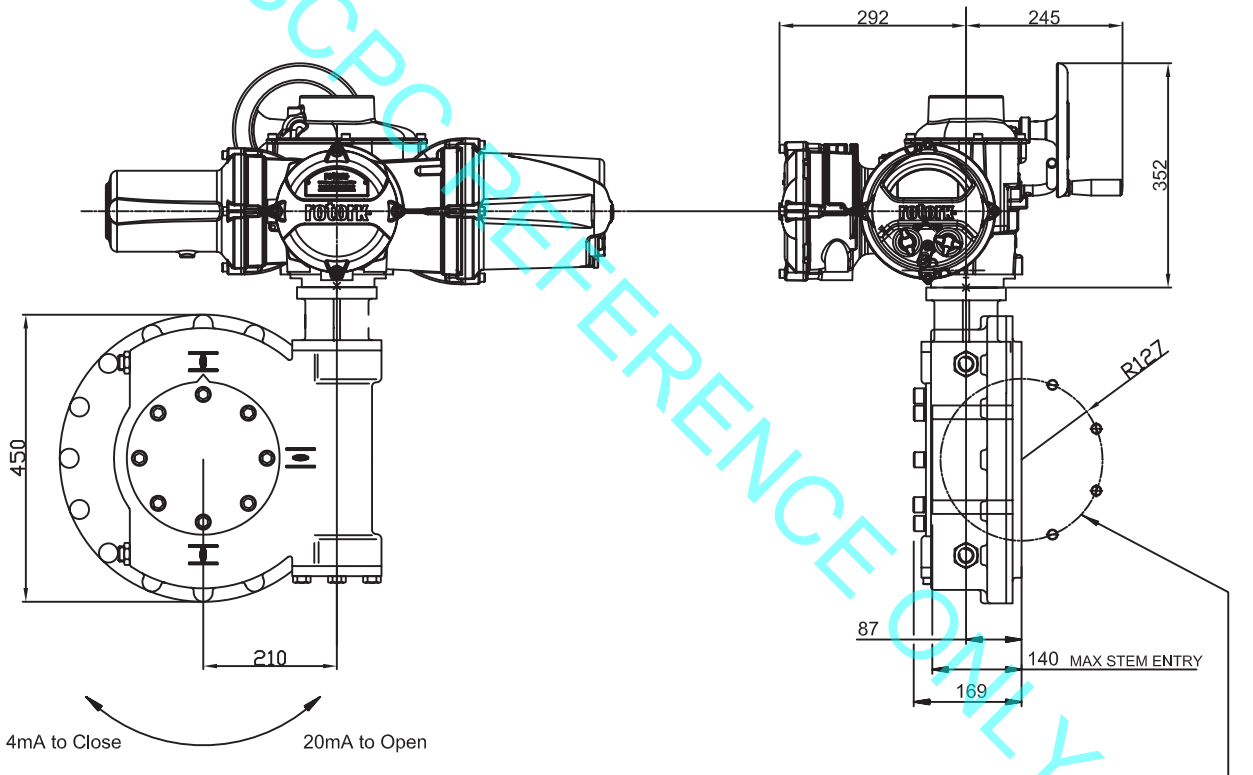
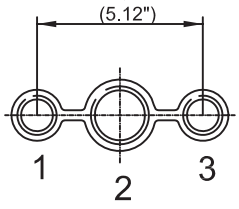


FOR SCPC REFERENCE ONLY



CONDUIT ENTRIES



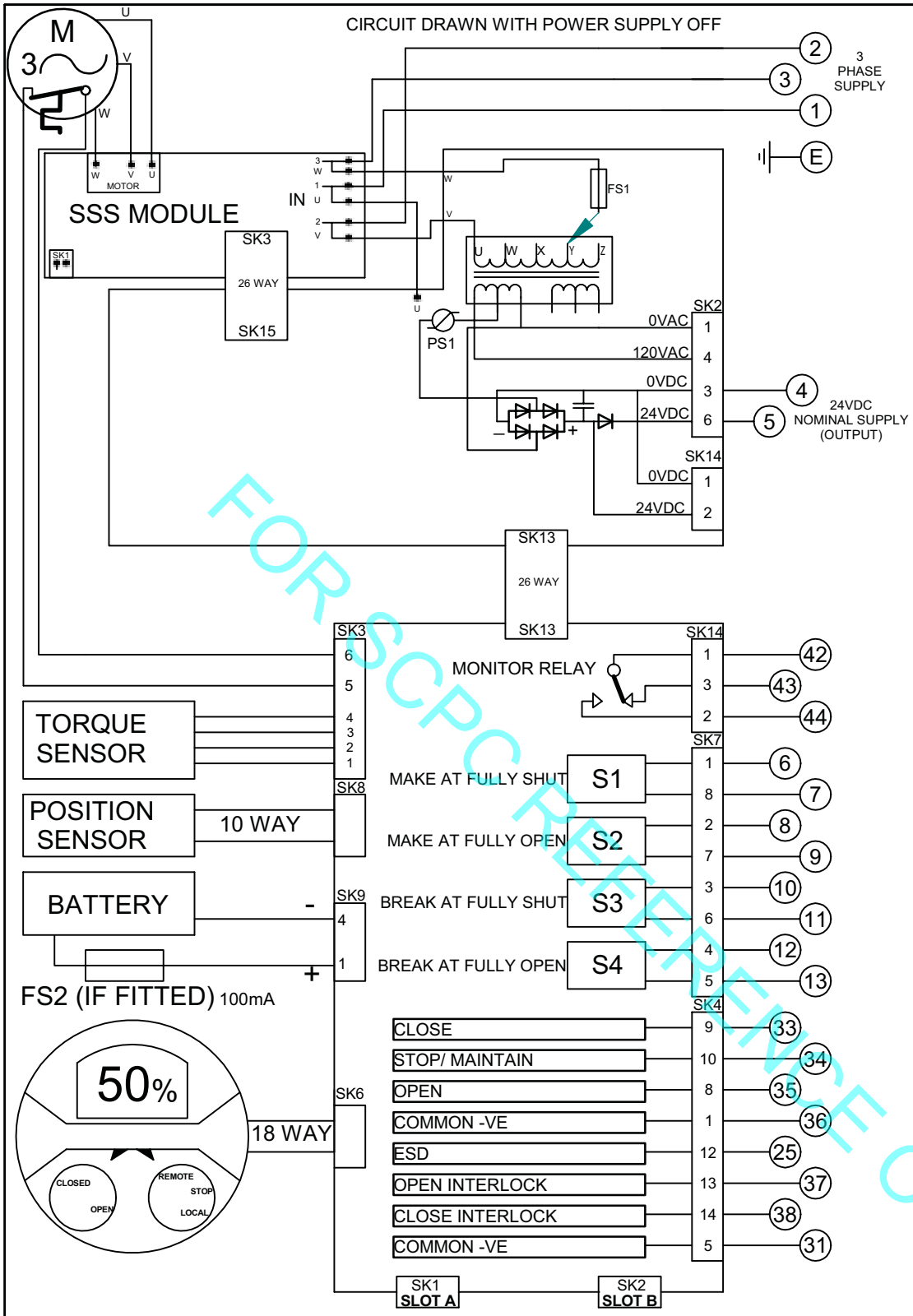
DEFAULT HOLE SIZES:
HOLES 1 & 3 TAPPED M25
HOLE 2 TAPPED M40

Project	CCMC/FHI DVOP-1 / LMYP-1
Job No.	RF2-0577 /0580
Tag	S.A. Fan
Type	IQM25+MOW7
Q'ty	Q'ty: 4 sets

Title

DIMENSIONAL DATA SHEET FOR IQM25/MOW7
WITH SIDE HANDWHEEL

Date	A4	Doc Number	Ver No	Scale	Sheet No
170512		IQ3-2SP+BC	1	1:12	1 of 1



FOR TYPICAL REMOTE CONTROL DETAILS, SEE DOCUMENT
RWS102

TRANSFORMER TAPPING OPTIONS	
Tap	Nominal 50/60Hz
TYPE 1	
W	220/230
X	380/400
Y	415/420
Z	440/460
FUSE FS1=250mA ANTI-SURGE	
TYPE 2	
W	346/380
X	480/500
Y	240/240
Z	550/575
FUSE FS1=250mA ANTI-SURGE	

REFER TO SHEET 2 FOR NOTES & OPTION PCB'S IF FITTED

THE ACTUATOR MUST BE PROTECTED USING SUITABLY RATED HIGH SPEED SEMI-CONDUCTOR FUSES ON THE INCOMING SUPPLY.
SUGGESTED FUESES :-
IQ10-20:10 Amp FERRAZ G330010
IQ25-35:20 Amp FERRAZ K330013
ALTERNATIVES =
IQ10-20:10 Amp SIBA 50-179-06/10A
IQ25-35:20 Amp SIBA 50-179-06/20A
OR EQUIVALENT PROTECTIVE DEVICE.

FOR SCPC REFERENCE ONLY

Iss	Date	Chkd	Revision Details
1	290812	AMW	First Issue.
2	031012	PMJ	NOTE 2 UPDATED.
3	100113	PMJ	Re-track of the 24V DC pion of SK14 to behind the diode.

www.rotork.com

IQM + FOLOMATIC + CPT

ROTORK CONTROLS LTD
BATH, BA1 3JQ
ENGLAND
Tel:01225-733200

ROTORK CONTROLS INC
ROCHESTER
NY 14624, USA
Tel:585-247-2304

Drawn by: PMJ
Date : 290812
Base WD: 203B0000
Job No : - -
MI No : - -

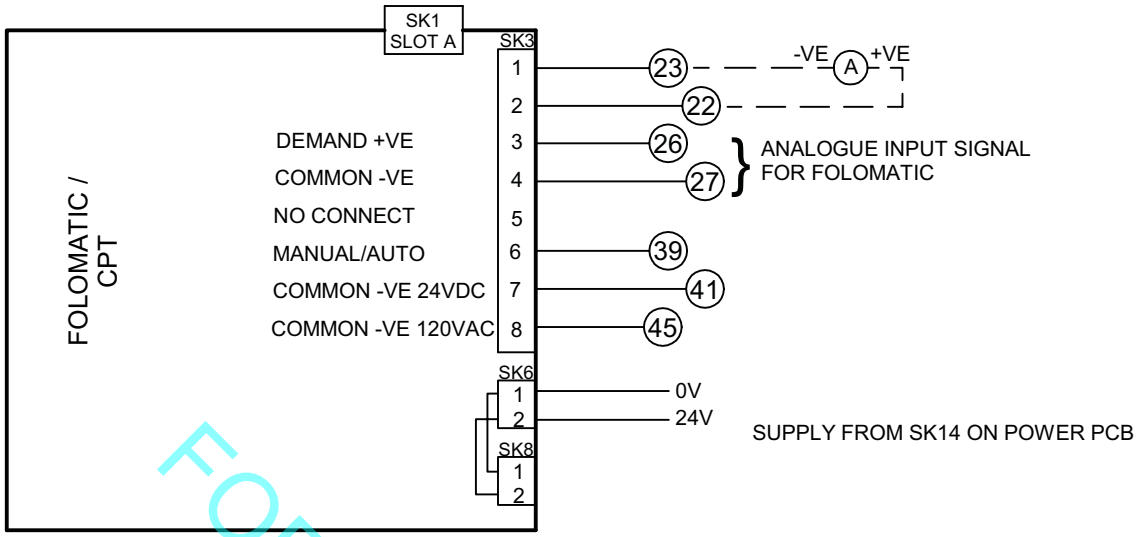
Circuit Diagram Number **203B0000**
B1 C1 B2 C2

Issue No **3**

Sheet **1** of 2

CIRCUIT DRAWN WITH POWER SUPPLY OFF

SLOT A



FOR MANUAL/AUTO CONNECTIONS AND DETAILS OF FOLOMATIC/CPT FUNCTIONS REFER TO ROTORK PUBLICATION PUB002-041

FOR SCPC REFERENCE ONLY

NOTES

1.FUSES:
-PS1 is a self-resetting fuse.
-Refer to publication PUB002-039 for approved fuses FS1 and FS2.

2.REMOTE CONTROL:
-For typical remote control circuits refer to:
  -RWS indicated or PUB002-041.
-For DC and AC control, connect –ve/0V to terminal 36.
-(For negative switch / positive common, refer to RWS indicated).
-Control signal threshold voltages:
  -DC: “on” ≥16Vdc / “off” ≤8Vdc, max 60Vdc.
  -AC: “on” ≥60Vac / “off” ≤40Vac, max 120Vac.
-Control signal duration to be 300ms minimum.
-Maximum current drawn from remote control signals is:
  -8mA at 24Vdc or 12mA at 120Vac.
-Supply provided on terminals 4 & 5:
  -Intended for remote control.
  -Max external load 5W at 24Vdc / 5VA at 120Vac

3.INDICATION:
-For typical position, status and alarm indication see PUB002-041.
-“S” contacts are user configurable and are shown in their default setting.
-Refer to PUB002-040 for functions and configuration instructions.
-Monitor Relay indicates actuator availability for remote control (shown “unavailable”). It can be configured to exclude local/remote selection.
-Refer to PUB002-040 for monitored functions and configuration instructions.
-Voltage applied to indication contacts must not exceed 150Vac
-Individual Switch current must not exceed 3.5A inductive, 5A resistive and no more than 8A in total for all 4 contacts.

4.BATTERY:
-Battery maintains local and remote “S” contact indication only.
-Refer to installation manual for approved replacement battery types.

See Sheet 1 for all Revision details/information

Circuit Diagram Number	Issue No	Sheet
203B0000	3	2 of 2

460 3 Phase 60Hz

ACTUATOR	TORQUE		MOTOR	LOCKED ROTOR		RATED	MODULATING LOAD			
RPM	Seat. lbs.ft.	Mod. lbs.ft.	Poles	Amps	P.F.	Amps	Amps	Kw	P.F.	Effy.%
IQM10										
21	25	12.5	4	1.50	0.85	0.44	0.32	0.10	0.60	63.00
29	25	12.5	4	2.00	0.86	0.55	0.44	0.12	0.55	62.00
43	23	11.5	4	2.00	0.86	0.61	0.46	0.15	0.64	62.00
57	20	10	4	2.00	0.86	0.65	0.47	0.15	0.66	62.00
IQM12										
21	45	25	4	3.10	0.82	0.73	0.57	0.19	0.64	66.00
29	40	25	4	3.10	0.82	0.77	0.59	0.24	0.68	74.00
43	40	22	4	3.10	0.82	0.99	0.65	0.28	0.70	77.00
57	35	20	4	3.10	0.82	1.04	0.70	0.31	0.73	76.00
IQM20										
21	90	60	4	6.10	0.85	1.25	0.97	0.46	0.69	86.00
29	80	60	4	4.60	0.80	1.38	1.07	0.57	0.78	86.00
43	60	50	4	5.30	0.79	1.32	1.14	0.64	0.86	82.00
57	50	40	4	4.60	0.80	1.39	1.14	0.62	0.80	85.00
86	40	35	4	5.30	0.79	1.42	1.24	0.74	0.88	85.00
IQM25										
21	150	113	4	8.80	0.80	2.08	1.68	0.87	0.76	85.00
29	150	113	4	8.80	0.80	2.49	2.01	1.07	0.82	82.00
43	120	95	4	8.80	0.80	2.68	2.27	1.21	0.83	81.00
57	100	75	4	8.80	0.80	2.68	2.16	1.16	0.83	81.00
86	100	75	4				apply			
IQM35										
21	400	200	4	20.00	0.86	4.76	2.89	1.53	0.74	90.00
29	400	200	4	22.00	0.86	6.20	3.32	1.90	0.79	91.00
43	300	187	4	20.00	0.86	6.20	3.96	2.39	0.85	89.00
57	230	150	4	15.50	0.87	5.75	3.79	2.31	0.88	87.00
86	160	150	4	15.50	0.87	6.02	5.48	3.17	0.92	79.00

FOR SCPC REFERENCE ONLY

All data is approximate

5. LUB OIL SUPPLY SYSTON

FOR SCPC REFERENCE ONLY

建成機械股份有限公司
CHENG-CHEN MACHINERY CO., LTD.

DESCRIPTION OF SYSTEM
MAINTENANCE INSTRUCTIONS

FOR SCPC REFERENCE ONLY

LUB OIL – SUPPLY – SYSTEM

Purchaser： FHI/LMYP-1
Vendor： CCMC
Fan Serial NO.： RF2-0581

TABLE OF CONTENTS

	Page
Technical data of the system	3
Technical data of the main components	4
State at delivery of the system	5
Erection of the system	6
References to start of operation	7 - 9
Maintenance instructions	10 - 20
Events of failure and repair of same	21 - 22
Parts list	23

FOR SCPC REFERENCE ONLY

°°) Installed in the system and described are those Components only for which a page-no. is given

Technical data of the main components

P.A.FAN

PUMP

Design：	Gear Pump		
Flow rate：	max.	34	1/min
Operating pressure：		500	Kpa
Power required：		0.75	Kw

Motor

Design：	Squirrel cage Motor	
Speed：	1800	1/min
Voltage：	460	V
Frequency：	60	HZ
Protection：	IP 55	
Construction：	B3/B5	
Power：	0.75	Kw

Oil Filter

Design：	Double-Switch Filter		
Flow rate：	max.	90	1/min
Operating pressure：		500	Kpa
Loss of pressure：		10	Kpa

Design and computation acc.：
（×）AD-Code
（ ）ASME Sect. VIII

Performance
（ ）WITH
（×）WITHOUT

FOR SCPC REFERENCE ONLY

Oil Cooler

P.A.FAN

Design：	Single Tube Heat Exchanger	
Cooling capacity：	10	Kw
Oil flow：	34	1/min
Operating pressure oil-sided：	500	Kpa max.
Oil inlet temperature：	60	℃
Oil outlet temperature：	40	℃
Water inlet temperature：	30	℃
Water outlet temperature：	30	l/min
Operating pressure water-sided：	150	Kpa
Design and computation acc.：	（×）AD-Code （ ）ASME Sect. VIII	
Performance：	（ ）WITH （×）WITHOUT	

FOR SCPC REFERENCE ONLY

State at delivery

The system will be supplied

（×）In seaworthy packing

（ ）In a planking

（ ）Unpacked

The system is coated with a prime paint

（ ）Priming coat with alkyd resin rust prevention

（ ）Covering paint acc. RAL 5002

（×）without covering paint

Existing apertures (pipe connections) are sealed with plastic caps.

The system is completely assembled.

FOR SCPC REFERENCE ONLY

Erection of the system

Transportation

The oil supply system shall be suspended only on the transportation posts provided for this purpose.

The system shall be transported only in an unfilled condition.

Erection, alignment, fastening

When erecting the oil supply system, please observe that the bottom (foundation) should be straight. The system should be erected on a horizontal level.
Following the alignment of the system, it should be fastened with stone bolts (sealed).

Connection of pipes

When connecting the pipes, please observe that the junction between system and pipes should be free from tension.

Electrical connection

( ) Will be performed on the spot

(×) All electrical components are completely installed.
Connections are led onto terminal box.

( ) All electrical components are completely installed.
Connections are led onto switch box.

FOR SORC REFERENCE ONLY

References to start of operation

The following rules shall be observed prior to starting operation：

（×） Fill oil into the tank through inlet aperture

（×） Check function of float switch

（×） Observe arrow of rotation direction when connecting the pump aggregate.

（×） The air inlet screen on the ventilation wing of the motor should not be covered

（×） The safety valve or pressure limiting valve should be adjusted to decreased pressure prior to starting the plant. When adjusting to higher values subsequent to start, observe pressure on the manometer

（×） Observe pollution indicator on the filter

（×） Adjust pressure switch

（ ） Control function of heating

（×） Connect cooling water an cooler

（×） Connect oil return flow to tank

（×） The ventilation screws (or cocks) on the oil cooler shall be opened until the cooler is filled (which becomes evident by oil or water penetration). Sub-sequently re-close screws (cocks)

（ ） Open pressure compensation pipe of the twin filter

FOR SCRC REFERENCE ONLY

Adjustment of pressure relief valve (overflow valve)

If there is no circulatory measuring instrument available you can proceed from following rule of thumb：

If there is-measured on whole pumping quantity of pump-only a low component current to be led off (hence, it will be required extensive the complete quantity at the consumer) then the valve has to be closed widely. At a completely closed valve the hole pumping quantity go to the consumer.

Adjusting required pressure with flow control valve beforer consumer. Reading at manometer in front of consumer, if need overflow valve has to be regulated.

If there is a pump available with mechanical drive, so you have to take care that safety valve of this pump is high adjusted to avoid that it will be opened premature.

FOR SCPC REFERENCE ONLY

Adjustment of press switch

Open completely safety valve. Start equipment. Turn off safety valve slowly as far as a sufficient large quantity is flowing across the flow control valve to the consumer.

Close slowly flow control valve until manometer shows the adjustable value. On adjustable values "increasing" turning back press switch to value under controlled value and then slowly shifting higher until press switch reacts on (use continuity indicator). With controlled values "falling" turn press switch slowly higher over controlled value and than go back slowly until press switch reacts on.

FOR SCPC REFERENCE ONLY

Maintenance

Next pages contents maintenance directions for the following components：

（×） Oil tank
（ ） Heating
（×） Pump aggregate
（ ） Safety valve
（×） Overflow valve
（ ） Pressure control valve
（×） Oil cooler
（×） Oil filter
（×） Thermometer
（×） Manometer
（×） Pressure switch
（×） Level switch
（ ） Differential pressure gauge
（ ） Temperature control device
（ ） Cooling water regulator
（ ） Flow controller
（ ） Flow volumeter

FOR SCPC REFERENCE ONLY

Oil Tank

The tank serves to receive the lubricating medium.
The tank bottom is installed in sloping position to the discharge connection.

On the upper part, the tank is closed with a cover which is screwed all around. An oil-resistant rubber band is placed between cover and tank, sealing the tank splash-proof.

Mounted to the tank are：

（×） Cover for cleaning aperture

（×） Oil drain screw

（ ） Oil drain cock

（×） Oil level gauge

（×） Thermometer

（ ） Radiator

（ ） Temperature regulator

（×） Filling and ventilation filter

（×） Level switch

Maintenance

The tank should be thoroughly cleaned with each oil change.

The oil change intervals depends on local conditions such as type of oil, service stress on the oil, accumulation of impurities. Oil samples should be taken at least half yearly and tested to further usability of the existing oil.

FOR SCDC REFERENCE ONLY

Pump aggregate

Feeding of the oil circulating system is effected by a pump aggregate. Rotary current motors are used exclusively as drives.

The pump bearings are relieved from pressure and lubricated by means of the conveying medium.

Bearings of the electric motors are filled with grease for a duration of approx. 20.000 operating hours.

Maintenance

Roller bearings on the motor shall be dismantled every 20.000 operating hours, cleaned and filled up to 3/4 level with grease saponified with lithium.

Beyond that pump aggregates do not require any special maintenance provided the following points with be observed：

Clamp on the motor three-phase, and observe arrow of the rotation direction.

Measure the motor nominal current and adjust correctly the thermal excess current contactor.

Avoid dust collection on the motor.

Air inlet screen on the ventilation blade shall not be covered.

Suction pipes shall be checked as to impermeability to ensure that no air may be taken in.

FOR SCPC REFERENCE ONLY

Pressure relif valve (overflow valve)

Pressure relief valves serves for fusing of circuits (fluids, gases) at excess pressure.

Maintenance

Maintenance of pressure relief valves is not necessary. In case a pressure relief valve fails it should be replaced completely.

Exchange of the pressure relief valve during operation of the system is not possible.

FOR SCPC REFERENCE ONLY

Oil cooler

The oil cooler serves the purpose of re-cooling heated oil to normal operation temperature. The heat exchange takes place by means of water.

Maintenance

On the water side, pipe pollution occurs mainly by deposits of boiler scale. Depending on the degree of water hardness, cleaning shall be performed more or less frequently (noticeable by gradually reduced cooling capacity).

Cleaning is necessary by gradually reduced cooling capacity.

Having switched off the oil and water supply and discharged the water at the water chamber, the reversing cover shall be removed. The pipes shall be passed through by a steel brush and blown clean with compressed air.

On the oil side, scalding by means of exhaust vapor and rinsing with acetone solution, from time to time, will suffice.

Penetration of oil or water at commissure between cooler jacket and water chamber showes leakage of the gasket.

Going over of water into is not possible. So the system is protected.

The gaskets are to be replaced not later than with every maintenance of the cooler.

FOR SORC REFERENCE ONLY

Filter

The filter serves for purifying the oil. It is installed in the pressure pipe. The filter installed in this system is a

（×） Double-switch filter

（ ） Single filter

Two chambers exist at the double-switch filter. By means of reversing cock, one of the chambers may be put out of operation while simultaneously the other chamber is starting operation. The position of the switching lever indicates the filter chamber presently in operation. Cleaning of the filter during operation is thus possible.

For the single filter, cleaning process during operation is not possible.

Maintenance

When initially starting operation, it is recommended to reverse the filter to the remaining clean chamber after 30 operating hours at the latest (double-switch filter), or to clean the filter respectively (single filter).

Otherwise, the filter should be cleaned after every 200 operation hours. The degree of pollution may be ascertained by means of a built-in pollution indicator.

By loosening the screws on the housing, the strainer insert may be removed. If the filter chamber is provided with a drain screw, this should be opened beforehand and the oil should be allowed to drip into a pan. Strainer insert to be cleaned with petroleum ether or compressed air

Do not use sharp or pointed articles for cleaning.

Prior to assembly of the filter, check gaskets and replace them, if necessary.

FOR SCBC REFERENCE ONLY

Thermometer

A thermometer is/thermometers are installed for controlling the oil temperature. Range 0－100℃.

Maintenance

Damaged thermometers shall be replaced completely.

Exchange of the thermometer during operation is possible due to a protective sleeve. Subsequent to loosening the counter nut, the thermometer can be unscrewed.

FOR SCPC REFERENCE ONLY

Manometer

A manometer is installed to allow optical observation of the operating pressure.

Maintenance

Damaged manometers shall be replaced completely.

Exchange of the manometer during operation is possible due to a shut-off valve being mounted in front.

FOR SCPC REFERENCE ONLY

Pressure Switch

Pressure switches are used for watching and limiting an adjusted pressure automatically. They start and finish regulation and control operations and working processes.

Maintenance

Pressure switches shall, if necessary, replaced completely.

Exchange of pressure switches during operation is possible due to a sleeve valve.

FOR SCPC REFERENCE ONLY

Manometer

A manometer is installed to allow optical observation of the operating pressure.

Maintenance

Damaged manometers shall be replaced completely.

Exchange of the manometer during operation is possible due to a shut-off valve being mounted in front.

FOR SCPC REFERENCE ONLY

Level Switch

Level switches are used in favour in such cases, where a sufficient control of liquid levels is required. They are used as well for regulation of pumps and valves as for controlling a limiting value.

Maintenance

Level switches does not require a special main tenance. Defective level indicators should be exchanged completely.

FOR SCPC REFERENCE ONLY

Events of Failure and Repair of Same

Failure	Possible Cause	Repairs
Pump is not conveying any oil	Clutch between motor and pump is defective	Replacement of clutch
	Safety valve/pressure limiting valve is adjusted to low	Adjustment of valves to higher value
	Wrong rotation direction of pump (please note rotation arrow on pump)	Clamp on DC-motor correctly
	Strainer with return flap does not open up	Check strainer for function, possibly replace strainer
	Oil level too low	Refill oil check level switch
Pump is conveying too little oil	Pump is drawing air	Check flange connections on suction-side of pump and seal same
Electric motor heating up excessively	Power consumption (capacity consumption of pump) is too high	Check pump for damages
	Closing-up of fan blade cover	Lay open cover
Excessive oil discharge temperature on cooler	Water inflow temperature is too higt-thermostat on cooling water regulator adjust too high	Correct adjustment of thermostat
	Defective cooling water regulator	Exchange regulators
	Pollution of cooling pipes on water side	Clean cooling pipes

FOR SCPC REFERENCE ONLY

Failure	Possible Cause	Repairs
Excessive oil discharge temperature on cooler	Excessive oil quantity-overflow valve adjusted incorrectly	Correct adjustment of overflow valve
	Excessive oil inflow temperature excessive frictional heat/loss of heat	Eliminate causes
	Excessive oil inflow temperature-heating does not switch off, defective regulator	Check regulator, possibly replace
Excessive operating pressure	Pressure limiting valve incorrectly adjusted/defective	Correct adjustment of pressure limiting valve/exchange same
	Filter polluted (see differential pressure indicator)	Clean filter, or switch to reverse, respectively
	Shut-off element in pressure pipe not fully opened	Check shut-off elements and open same, if applicable
	Oil-side pollution of cooler	Clean cooler

FOR SCPC REFERENCE ONLY

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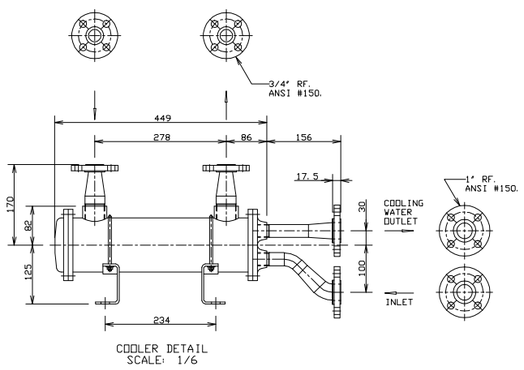
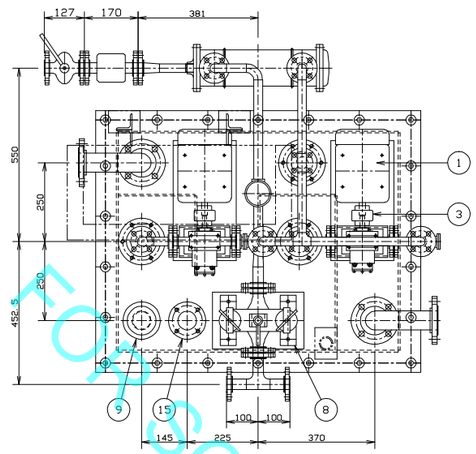


COOLER DETAIL
SCALE: 1/6



ITEM	DESCRIPTION	MATERIAL	Q'TY	REMARKS
1	MOTOR	TECO	2	SEE NOTE
2	OIL PUMP	NDP	2	TDP-220HB
3	RUBBER COUPLING	C. M	2	CM-67
4	SUCTION FILTER	SUS304	2	MF-06 3/4" NPT
5	CHECK VALVE	SCS13	2	ANSI 150# 3/4" RF
6	REGULATOR VALVE	NDP	1	TDP-3VBD 3/4"
7	COOLER	HAWEI	1	SL-408W
8	DOUBLE FILTER	A126B	1	25A 40MESH
9	BREATHER FILTERS	EPE	1	EF4-50-R2"
10	TEMP. GAUGE	BCH	1	5" Dia BTA
11	PRESS. GAUGE	WIKA	2	0 ~ 6Kg/Cm²
12	PRESS. SWITCH	WIKA	1	PCS2TG
13	PRESS. SWITCH	WIKA	1	PCS2MG
14	PRESS DIFF. SWITCH	WIKA	1	DCLH2G
15	OIL LEVEL DETECTOR	FINETEK	1	
16	OIL LEVEL GAUGE	ELESA	1	HCX. 254-M12
17	OIL TANK	SUS304	1	
18	DRAIN & COCK + PLUG	SUS304	2	NPT 1-1/2"
19	BALL VALVE	SUS304	2	1" ANSI #150 FLANGE TYPE
20	SIGHT CALSS	SUS304	1	1" ANSI #150 FLANGE TYPE
21	ELBOW	SUS304	3	3/4" NPT

NOTE:

1). THREE PHASE MOTOR : 1HPX4PX60HZX460V
INSULATION : F CLASS
S.F. : 1.15
IP : 55
FRAME NO. : 80
STANDARD. : IEC

2). GEAR PUMP TYPE : TDP-220HB
DISCHARGE CAPACITY : 34 L/min
AT 6 Kg/Cm² AND 1750 RPM.
WITH SAFETY VALVE

3). REGULATION VALVE TYPE: TDP-3VBD 3/4"

4). HEAT EXCHANGER TYPE : SL-408W
MATERIAL TUBES : C1220T(DCuT)
COOLING SURFACE : 0.8 m²
COOLING WATWR RATE : 2 T/H
MAX. WORKING PRESS. : 10 Kg/cm²
MAX. WORKING TEMP. : 80°C

5). THERMOMETER TYPE : BCH
INDICATING RANGE : 0-100°C
CONNECTION : 1/2" NPT
DIA. : 5"

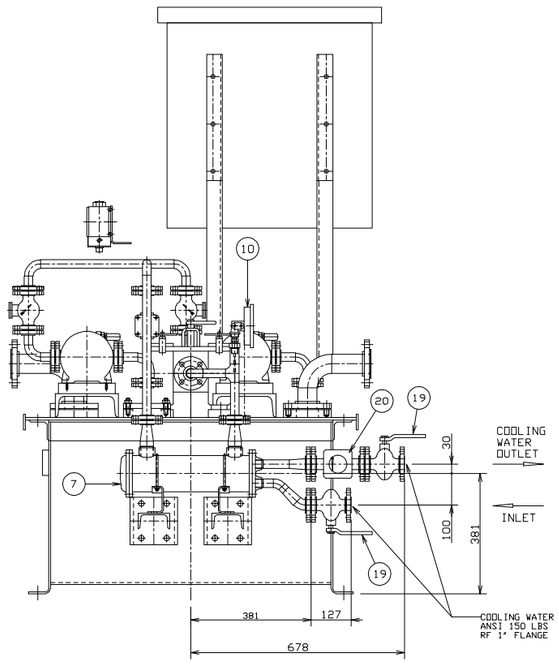
6). MANOMETER + PULSATION DAMPENER : BBL+BCE
INDICATING RANGE : 0-10Kg/cm²
CONNECTION : 1/2" NPT
DIA. : 4-1/2"

7). PRESS. SWITCH TYPE : PCS2TG
SETTING POINT : 0.4Kg/Cm²

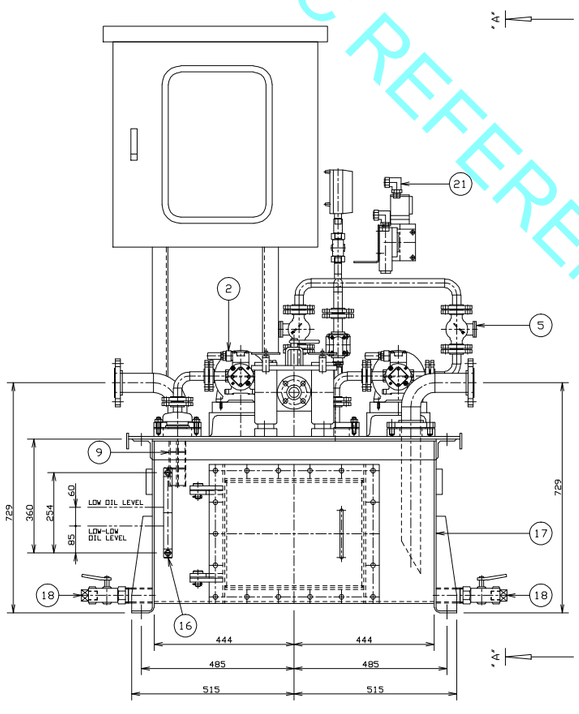
8). PRESS. SWITCH TYPE : PCS2MG
SETTING POINT : 1.3Kg/Cm²

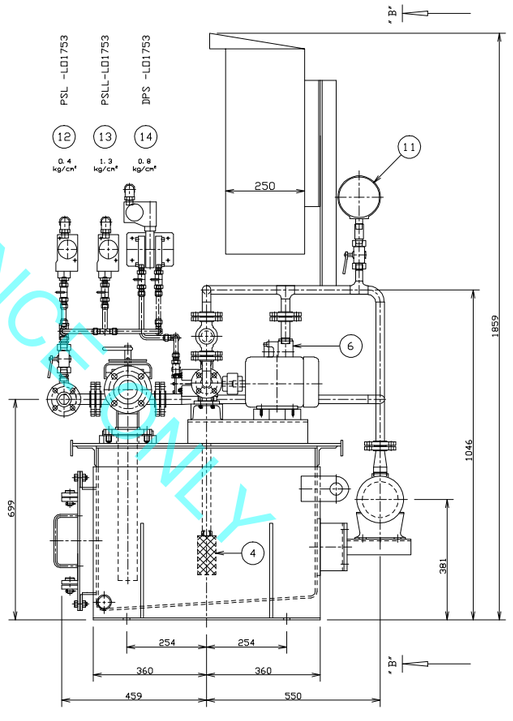
9). PRESS DIFF. SWITCH : DCLH2G
SETTING POINT : 0.8Kg/CM²

10). OIL LEVEL DETECTOR : FDEEM3S220340
PROTECTION : IP65
CONNECTION : 1-1/2" X 5Kg/cm²



VIEW "B"-"B"





VIEW "A"-"A"

REV.	DESCRIPTION	REVISED	CHECKED	APPROVED
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫
F	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
E	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
D	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
C	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫
B	TO COMPLY WITH CUSTOMER'S REQUIRED	陳玉樹	陳玉樹	傅泰錫

SCALE: 1/10

APPD	CKD
傅泰錫	陳玉樹
10/30/'13	10/30/'13
DESN	DWN
陳玉樹	陳玉樹
10/30/'13	10/30/'13
PROJECT NO.	SHEET NO.
RF2-0581	NO. OF

CHENG-CHEN MACHINERY CO., LTD.
TAIPEI, TAIWAN HEADOFFICE TEL:(02)25946171
SHIN FONG FACTORY TEL:(035)981611 FAX:035)981149

NAME: TDP-220HBX4PX60HZ 34L/min
P. A. FAN CIRCULATING OIL SYSTEMS

DWG NO. A-RF2-0581-100 UNIT MM REV 0

COPY

ENGINEER

REG. NO.	:
P. T. R. NO.	:
DATE	:
PLACE	:
T. I. N.	:

SHEET CONTENT

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R. O. C.

PROJECT TITLE / LOCATION
SMC LIMAY POWER PLANT PROJECT
LIMAY, BATAAN, PHILIPPINES

OWNER / ADDRESS
SMC Consolidated Power Corporation
4D SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

FOR REFERENCE ONLY

6. APPENDIX

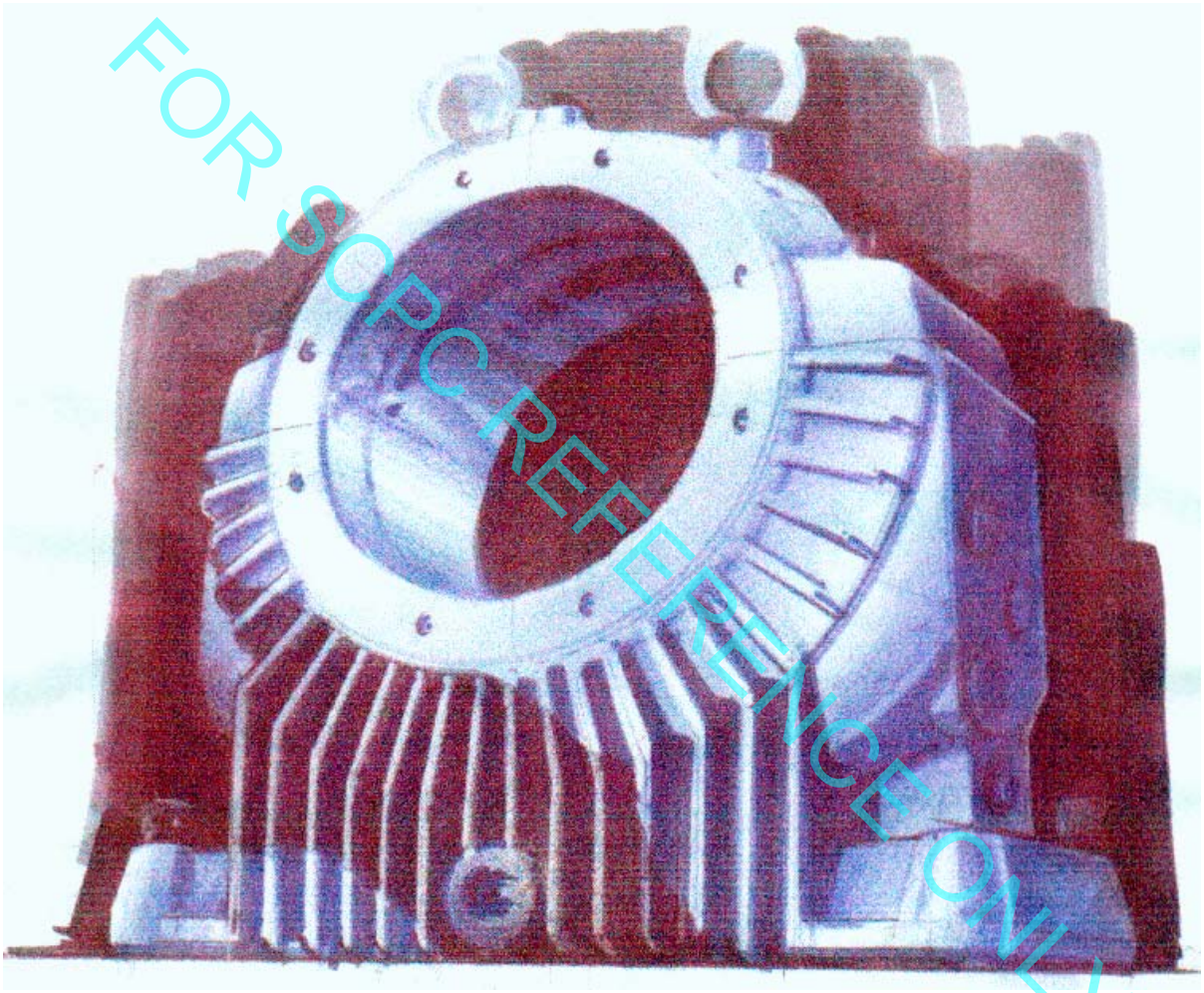
FOR SCPC REFERENCE ONLY

ZOLLERN

Power transmission

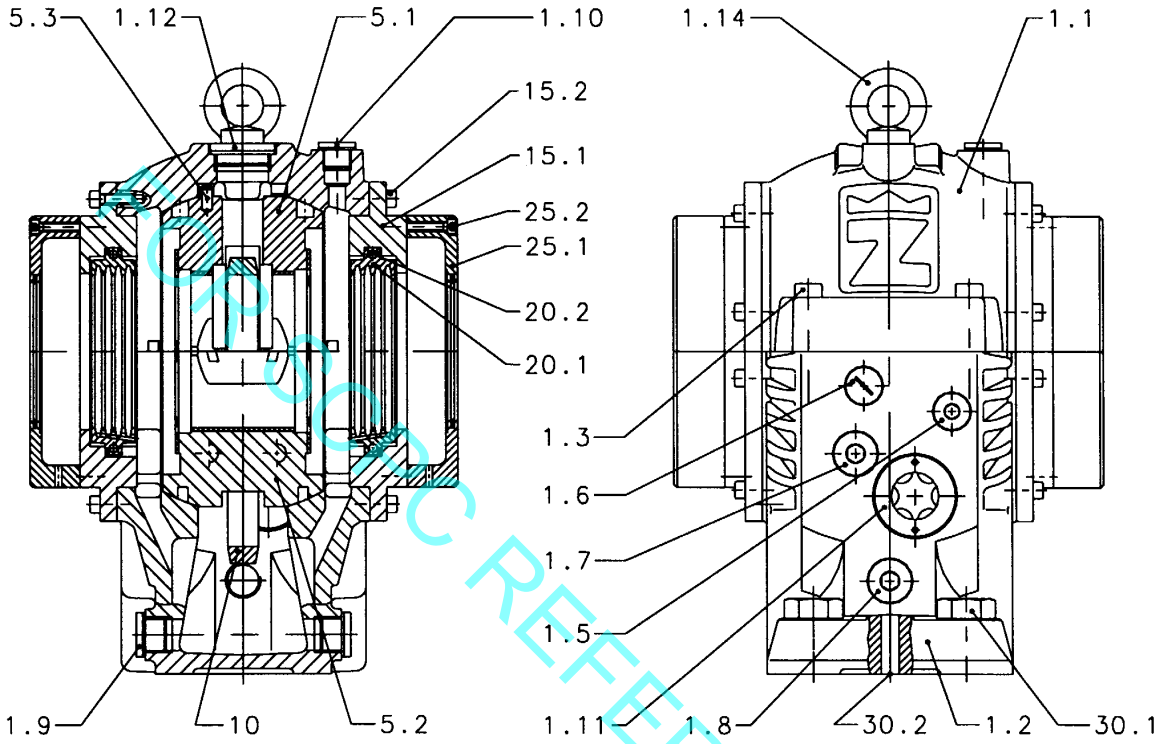
Plain bearings

Installation and Operating Instructions ZR



FOR SOPC REFERENCE ONLY

Parts List for Pedestal Bearing ZR



FOR SCRC REFERENCE ONLY

1.1	Upper bearing housing*	5.1	Bearing shell upper half*
1.2	Lower bearing housing*	5.2	Bearing shell lower half*
1.3	Bearing housing cover screws*	5.3	Anti-rotation pin*
1.5	Sealing plug with seal* (Oil inlet for circulated oil)	10	Loose oil ring
1.6	Cable screw or plug PG7	15.1	Seal carrier
1.7	Sealing plug with seal*(Temperature sensor port)	15.2	Seal carrier retaining screws*
1.8	Sealing plug with seal* (Connection for heater, sump thermometer or cooler)	20.1	Floating labyrinth seal*
		20.2	Spring key for floating labyrinth seal*
1.9	Sealing plug with seal* (drain)	25.1	Bolt on baffle
1.10	Sealing plug with seal* (oil fill)	25.2	Bolt on baffle retaining screws*
1.11	Oil level sight gauge (or oil outlet for circulated oil)	30.1	Bearing pedestal fixing screw
1.12	Oil sight window*	30.2	Pre-drilled dowel location
1.14	Eye Bolt*		

*These parts can only be supplied as sub-assemblies. Spares parts are listed in section 11

990225/h520

1 Safety Recommendations

Zollern pedestal bearings (type ZR) are incorporated into various types of machinery, any safety regulations specific to the machinery must be observed.

During bearing operation, assembly, maintenance and repair, accidents and injuries must be avoided at all costs. The following points require special attention:

- During maintenance and repair ensure that rotating machinery is at a standstill, isolated from electrical supply with interlocks such the supply cannot be resumed unintentionally.
- Loads and speeds must not exceed those specified in Zollern's data sheets, drawings or calculation sheets.
- Rotating parts must be fixed so that accidental movement is not possible.
- BEWARE: after use bearings will be hot, there is a danger from burning during repairs and maintenance if the bearing is not allowed to cool.
- The shaft tolerances set out in our specifications and drawings may not be exceeded.
- The bearing may become detached if screws are allowed to loosen. Tighten the screws to the specified torque.

Alterations to the ZR bearing may only be carried out with the express permission of Zollern, who must also be consulted in the event of load specifications being exceeded. Our qualified team of fitters can carry out repairs and service if required avoiding errors caused by untrained personnel.

2 Bearing Types and Designations

Type	Z	Bearing for horizontal mounted shaft
Housing	R	Finned pedestal bearing
Cooling	N	Natural convection
	Z	Forced oil with external cooling
	W	Water cooler in oil sump
Lubrication	L	Cylindrical bore with loose oil ring
Thrust bearing	B	Plain white metal with oil grooves
	K	Tapered pad for bi-directional rotation
	D	Tapered pad for uni-directional rotation
	A	Round tilted pads, spring supported
	Q	Without thrust bearing
Note	Insulated bearing shells are designated 'insulated'	
Example : ZRZLK 9-90	Finned pedestal bearing with forced oil lubrication externally cooled having plain cylindrical bore with loose oil ring. Tapered thrust pad for bi-directional rotation size 9, shaft diameter 90mm	

3 Introduction

Please pay careful attention to all points covered by this manual to ensure the safe and efficient operation of the bearing. Care needs to be taken to keep all bearing parts and the workplace clean. Provision is made for thermometers (1.7, 1.8), oil level gauges (1.11), oil inlet (1.5) and outlets to be fitted to either side of the bearing. These devices can be fitted to either left or the right as required. Where a bearing is force oil lubricated, the oil outlet pipe is fitted into the tapping for the oil level gauge (1.11). Oil drain plugs (1.9), which can also act as an oil flow connections, are positioned on both sides of the bearing case.

With electrically insulated bearings, there is a non-conductive layer on the spherical seat of the bearing housing. This layer is both wear-resistant and temperature-resistant. The shaft seals (see chapter 6 - "shaft seals") and the anti-rotating pins (5.3) are manufactured from non-conductive material and prevent any spark erosion of either bearing shell or. Any temperature control instrumentation installed by the user should be insulated using appropriate measures, such as isolated protective piping and plastic screws. Zollern can, of course, provide help as and when required.

Eyebolts (1.14) are fastened to the upper part of the bearing to assist handling of the complete and assembled bearing. It is important to remember that they are not suitable for the handling of the assembled machine and that the eyebolts are only to be used as tension elements.

4 Packing

All bearings are delivered fully assembled without oil. The oil ring (10) and other accessories are packed as separate items. The bearings are protected against corrosion and other influences by a preserving film, according to individual application.

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5 Installation

Once the upper housing (1.1), the seal carrier (15.1), the shaft seals (20.1) and the bearing shell (5.1 and 5.2) have been dismantled, carefully clean the interior of the bearing and the bearing shell and check for any damage caused during transport.

Ensure that the shaft is clean and that dimensions and tolerances correspond to our recommendations. We recommend that the appearance of the bearing surface on the shaft is checked and that it is checked that non-moving seals have not become jammed in the shaft (see chapter 6 “shaft seals”).

Proceed with installation as follows:

- Position lower bearing casing (1.2) onto the machinery bed without securing the bearing base fixing screws (30.1) so as to allow alignment of the bearing.
- Place the lower half of the bearing shell (5.2) into the lower part of the bearing casing (1.2) and align with the split line.
- Then lower the shaft slowly (ATTENTION take care that thrust bearing surfaces are not damaged) until there is a clearance of 2mm between shaft and bearing surface. Align the casing lower half (1.2) to 0.3-0.4mm to the shaft (the more precise the alignment the better the performance of the bearing seals).
- Tighten bearing base fixing screws (30.1).
- Lubricate the shaft in the bearing area (use the same oil as is to be used during operation of the bearing) lower the shaft into the bearing shell lower half.
- Now follows assembly of the loose oil ring (10), which must first be disassembled. Position both halves of the oil ring on the shaft and around the lower half of the shell using the notch provided, then press both halves together on the dowel pins. Following this, tighten the fixing screws to the following torque values:

Bearing Size	09 - 14	18 - 28
Torque (Nm)	1.4	2.7

- Coat the surface of the split line in the lower part of the bearing housing with a thin film of non-hardening sealing compound
- Place the upper part of the bearing shell (5.1) onto the lower part of the shell (5.2) (it is impossible to turn both halves of the shell together as they would become misaligned).
- Lower the upper part of the housing (1.1) onto the lower half. Ensure that the upper part is correctly located such that there is no clearance at the parts’ join.
- Tighten the cover screws (1.3) crossways to the following torque values :

Bearing size	9	11	14	18	22	28
Torque (Nm)	41	71	172	336	580	1160

With machines that are prone to vibration, it is essential that measures be taken to prevent screws from loosening. To this end secure with Loctite or similar thread locking compound.

- Assemble the seals on either side of the casing. These are normally floating labyrinth seals although other types are sometimes fitted. The correct assembly of the seal being used is described in chapter 6.

Assembly of the seal carriers (15.1) to casing to be carried out as follows:

- Coat the split lines and flange surfaces of the seal carrier with non-hardening sealing compound.
- Place the seal carrier (15.1) around the floating labyrinth seal, push both into the bearing housing and tighten the seal carrier screws (15.2) to the following torque values :

Bearing Size	09 - 14	18 - 28
Torque (Nm)	10.5	26

5.1 Additional Information for Bearings with External Oil Supply

When using an external oil supply the pipe diameters should be chosen such that the inlet flow does not exceed 1.5 m/s and the outlet flow 0.15m/s.

If the inclination of the oil outlet pipes for structural or design reasons cannot be set to 15°, pipes with correspondingly larger diameters need to be used. Pipes with inclinations or diameters which are too small may create back-flow, which could in turn lead to overflowing or leakage.

Heat-treated (welded or hot-worked) pipes and/or those showing any signs of rust or dirt must be scoured and cleaned prior to assembly. Following assembly of the pipes the entire oil circulating system must be flushed to prevent dirt entering the bearing or fittings.

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Prior to cleaning all measuring and switch fittings (pressure control, flow gauge) and their connections must be removed. The bearings must not be left intact during cleaning, and following cleaning all filters must be rinsed. All connections to the bearing must be free of vibration and tension.

5.2 Additional Information for Bearings with Water Cooler in the Oil Sump

When attaching the water cooler care must be taken to ensure that there are no leaks from the cooler or its connecting pipes. The joint seals must not be damaged. During fitting a stop tap or valve should be provided in the inlet. Any oil should be free to drain from the housing (no pressure should be required). The cooling flow can be in either direction.

6 Shaft Seals

Zollern bearings can be supplied with various seal types or seal systems depending on the required application, with seals conforming to DIN 40050 or IEC 529. Special seals can be supplied for particular applications on request.

6.1 Floating Labyrinth Seals

For standard applications ZR bearings are equipped with floating labyrinth seals (see fig. 2) which achieve IP44. The labyrinth seals float on the shaft at circumferential speeds above 100 m/s. They are manufactured from a rigid, unbreakable and temperature resistant polymer. The seals are seated within a seal carrier or directly seated in the housing such that they are not subject to loading due to radial shaft displacements.

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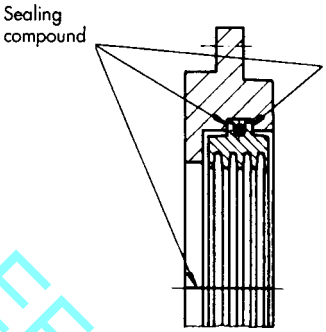

Fig. 2

Floating labyrinth seal
Protection level IP 44

Oil drain holes are provided in the lower part of the floating labyrinth seal. An anti rotation pin on the split line prevents the seal from being assembled incorrectly and damaged.

- Build the seals up prior to bearing assembly so that the hook and eye of the spring key (20.2 & 21.2) are separated.
- Prepare the seal for installation and coat the surface of the split line and the exterior of the spring guide (see fig. 2) with a thin layer of non-hardening sealing compound.
- Place the lower part of the seal onto the shaft and turn it into the correct position (the oil drain holes in the seal must be facing towards the bearing and the anti rotating pin must sit in the groove of the split line).
- Place the upper part of seal onto the lower
- Insert the spring key through the sealing groove and lock into place.
- Check the alignment of the seal and housing split line

6.2 Floating Labyrinth Seal with Bolt on Baffle

The bolt on baffle can be combined with either a floating labyrinth seal (chapter 6.1) or a rigid seal (chapter 6.4) to achieve IP55. The inserted seal strip is temperature resistant withstanding high circumferential speed. The coefficient of friction is minimal as is the ‘stick/slip’ coefficient. The electrical insulation properties are very good and independent of temperature and environmental influences.

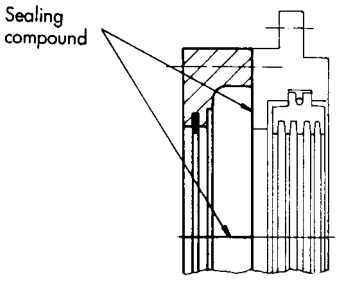

Fig. 3

Floating
Labyrinth Seal with
additional rigid seal
Protection level IP 55

Insertion to be done as follows:

- Spread the sealing compound onto the surface of the split line and coat the movement surface with the seal carrier of the floating labyrinth seal (fig. 3)
- Next gently press the lower part (recognized by the drain hole) from underneath against the shaft and tighten the fixing bolts equally.
- Finally position the upper part, align on the split line and attach with the bolts.

6.3 Floating Labyrinth Seal with Labyrinth Ring

The labyrinth is clamped onto the shaft and inserted into a groove in the seal carrier of the floating labyrinth seal (fig. 4), thus preventing dirt and water from entering the bearing. The groove can be filled with grease where necessary.

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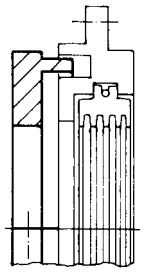
Fig. 4

Floating labyrinth
seal with dust flinger
Protection level IP 54

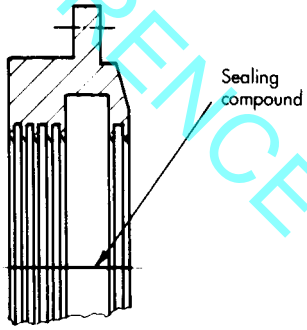


After placing onto the shaft the labyrinth ring must be so placed that it is not possible for it to run into the groove. The axial movement of the shaft is limited.

6.4 The Rigid Seal

Fig. 5 shows a rigid seal with two separate labyrinth systems. Where axial movement is minimal a shaft collar can be provided between the labyrinths.

Fig. 5
Rigid Seal
Protection level IP 44



- A thin layer of sealing compound initially retains the rigid seal. During assembly the lower half of the seal is lightly pressed onto the shaft from below and the screws then tightened (during operation the shaft is raised).

ATTENTION the baffles must be removed during assembly and disassembly of the shaft or damage to the shaft may ensue.

7 Commissioning

Having removed the oil sight glass (1.12) or the plug (1.10) fill to level with the recommended oil grade. The oil is at the correct level when it reaches the middle of the oil level glass (1.11). The minimum oil level is when the oil only reaches the lower part of the sight level glass when the machine is stationary.

WARNING

Insufficient oil level will lead to ineffective lubrication, bearing overheating and subsequent damage leading to failure. Too high a level of oil will not hinder the operation of the bearing, but could lead to leakage at the seals. Prior to commissioning of the bearing it must be checked for leaks. All oil port plugs and securing screws are correctly fitted and tightened.

ATTENTION

Should, at any time during commissioning or operation, the bearing temperature exceed that specified by Zollern the machine should be stopped and the causes investigated.

8 Maintenance

The oil level must be checked at regular intervals with the machine at standstill. The minimum oil level is when the oil only reaches the lower part of the sight level glass (1.11).

The bearing housing must be kept clean as heat transfer can be hindered by dirt and dust leading to bearing overheating and subsequent damage leading to failure.

Should, during operation, there be unusual variations in bearing temperature or oil level the causes must be investigated.

Our technicians are available should any assistance be required.

8.1 Oil change intervals

For self contained and bearings using mineral oils we recommend flushing and oil change after 8000 operating hours, for forced oil lubricated bearings we recommend an interval of 20000 hours. Longer intervals may be possible subject to Zollern approval. Shorter intervals are necessary where there is frequent starting or where oil temperatures are high or where there is an excessively dirty environment. To change the oil, take out the oil drain port which is positioned at the centre beneath the bearing (allow the oil to drain whilst hot). Should unusual oil discoloration or smell be apparent the causes must be investigated. Should any chemical cleaning compounds be used these must be thoroughly flushed away prior to further operation.

The oil is refilled through the oil sight glass (1.12) or plug (1.10). Use only the recommended oil grade as shown by the manufacturer's plate. The oil is at the correct level when it reaches the middle of the oil level glass (1.12). There is always a possibility of overfilling - too high a level of oil will hinder the operation of the bearing.

WARNING

If regular checks are not made there is a chance of insufficient oil leading to ineffective lubrication, bearing overheating and subsequent damage leading to failure.

8.2 Inspection

Should unusual oil colour changes or increases in temperature be apparent an inspection must be made. Remove the upper half of the bearing housing (1.1) leaving the lower half in place. Prior to disassembly thoroughly clean the bearing casing such that no dirt can accidentally enter the bearing itself.

8.3 Disassembly

- Loosen the cover screws (1.3) and the retaining screws (15.2) on the upper half of the seal carrier.
- Raise and remove the upper casing half (1.1).
- Dismantle the floating labyrinth seals (20.1 & 21.1) by raising the upper half and tilting it then open the spring key and dismantle together with the lower half.
- Carefully raise the upper half of the bearing shell (5.1).
- Release the loose oil ring (10) screws, separate and remove both parts.
- Remove any probes or sensors if present.
- Raise the shaft far enough to give sufficient clearance to turn the lower shell (5.2) half through 180° and lift it away.

8.4 Visual Check

During disassembly check that all bearing surfaces remain in good condition. Bearing in mind the length of bearing service the shell should show consistent polishing of the surface. Any light score marks should be smoothed prior to rebuilding of the bearing. The shaft seals should be changed should they show excessive wear or have broken edges.

9 Transport and Storage

During transportation of the machine the bearings should never be subject to shock loading, and must be protected from environmental humidity. Care must be taken to prevent dry sliding within the bearing. Storage facilities must be dry and not subject to severe variations in temperature or humidity. In good conditions the bearings can be stored for up to one year.

10 Troubleshooting

10.1 Increased Bearing Temperature

Cause: lack of lubrication

Solution: Regularly check the oil level, inspect the bearing surface for score marks. Check that the filter does not restrict oil flow.

10.2 Excessive scoring of the bearing shell

Cause: Dirt particles in oil

Solution: Change oil (possibly polish out scores on bearing surface). Change oil filter.

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11 Ordering Spare Parts

Every bearing has a part and order number embossed onto the manufacturer's plate. In order to ensure the delivery of the correct parts please quote these numbers in all correspondence or return the part to Zollern for repair or replacement.

The following parts are supplied as sub-assemblies:

Sub-assembly	Position (fig. 1)
Bearing Shell	5.1 to 5.3
Loose oil ring	10
Seal Carrier	15.1
Floating labyrinth seal	20.1 & 20.2
Bolt on baffle with Teflon bronze strips	25.1
Labyrinth ring	N/A

12 Service

Our qualified team of fitters can carry out repairs and service if required avoiding errors caused by untrained personnel.

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Werk Herbertingen
-Gleitlagertechnik-

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Fax: +49.7586.959-715

INSTRUCTION MANUAL
for
DODGE®
SLEEVOIL® STL Pillow Blocks
and
SLEEVOIL® SSL Pillow Blocks
Sizes $3^{15}/_{16}$" thru 8" PLXC

FOR SCPC REFERENCE ONLY

SLEEVOIL SSL Plain Pillow Blocks
(Short Series Housing and
PLXC Liner with Pipes Removed
and Vented Plugs Installed)

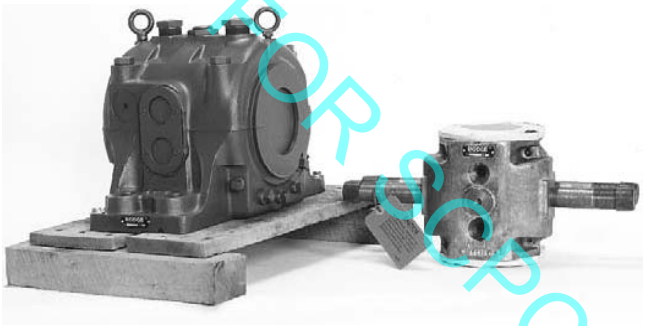
SLEEVOIL STL PLXC Pillow Blocks
(Standard Series Housing
and PLXC Liner)

⚠ WARNING:

Rotating parts can cause injury to persons and damge to property. Because of the possible danger to person(s) or property from accidents which may result from the improper use of products, it is important that correct procedures be followed. Products must be used in accordance with the engineering information specified in the catalog. Proper installation, maintenance and operation procedures must be observed. The instructions in the instruction manuals must be followed. Inspections should be made as necessary to assure safe operation under prevailing conditions. Proper guards and other suitable safety devices or procedures as may be desirable or as may be specified in safety codes should be provided, and are neither provided by Baldor Electric Company nor are the responsibility of Baldor Electric Company. This unit and its associated equipment must be installed, adjusted and maintained by qualified personnel who are familiar with the construction and operation of all equipment in the system and the potential hazards involved. When risk to persons or property may be involved, a holding device must be an integral part of the driven equipment beyond the speed reducer output shaft.

INSTALLATION

The design of this pillow block allows the use of the PLXC liner. This is accomplished by using a "universal" grommet for the application. Complete pillow blocks are shipped in two packages, the housing and its associated parts are in one package and the liner in the other.



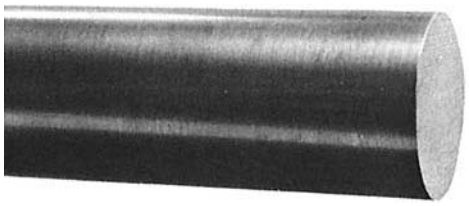
Liners are shipped with pipes installed. If water cooling is not required, remove the pipes and install the vented plugs. Any optional accessories are packaged separately.

Before mounting the pillow block read all instructions in this manual to become familiar with the mounting procedure and pillow block parts nomenclature.

All instructions in this manual pertain to both standard (STL) and short series (SSL) pillow blocks with the exception of step 8 following which does not apply to SSL style

bearings

1. Check **mounting structure** making sure it is rigid, level and well supported. Inspect shaft to ensure it is smooth (32 micro-inch finish or better), free of burrs or rough spots, clean shaft in the bearing area. Standard shaft tolerance should be +.000/-.002 on all sizes unless otherwise specified on shaft detail.



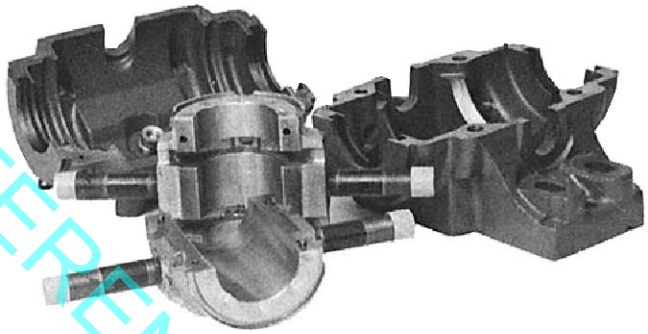
2. Disassemble and thoroughly clean all parts of the pillow block. Housing caps and liner caps are matched to their bases and are not interchangeable. Housings and liners can be interchanged as assemblies only.

⚠CAUTION

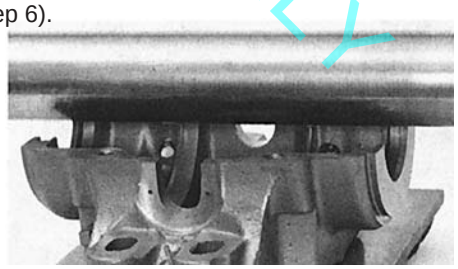
Parts can be easily damaged. Liner assembly has critical machined surfaces which are easily damaged. Use care in handling to protect these surfaces. Liner parts should be placed on a soft, CLEAN surface.

⚠CAUTION

To avoid bearing and liner damage, take care when reinstalling coolant pipes. Over tightening may result in liner damage. Use pipe sealant and tighten securely.



3. Position **housing base** on pedestal so that oil gage is in the position specified on the construction drawing. Also check construction drawings for special modifications on housing and/or liner. **Do not tighten housing base to pedestal.** Apply oil to the spherical seats in the housing base. Because of the closely controlled liner to housing fit, it may be necessary to bolt the housing base down to allow the lower liner to slip into the housing seat in Step 4. Therefore, do a preliminary alignment check at this point (See Step 6).



4. Note location of **thermocouple holes in liner base.** Set liner base in housing base so that thermocouple holes in housing align with holes in liner. Thermocouple holes in housing base are opposite the side of housing marked “X.” Apply oil to liner bearing surface. **Note: Liner has been machined to close tolerances. Scraping of bore is not recommended.** It may be necessary to bolt housing base

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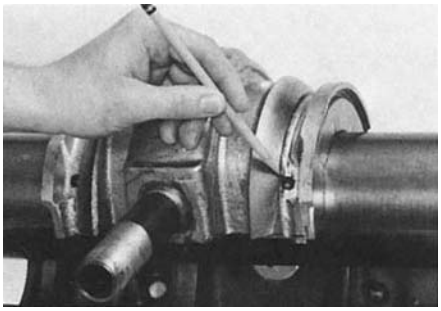
down to seat liner as noted in Step 3.



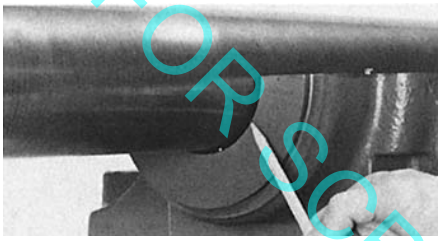
FOR SCPC REFERENCE ONLY

⚠CAUTION

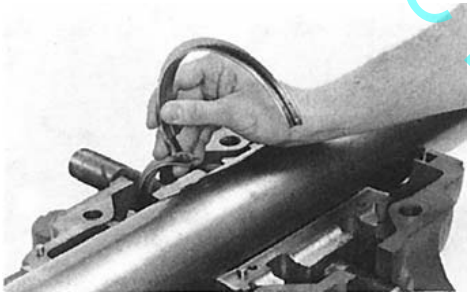
Care should be taken when reinstalling coolant pipes. Use pipe sealant and tighten securely. Overtightening may result in liner damage.

5. Apply oil to **shaft** in the bearing area and set shaft in place.

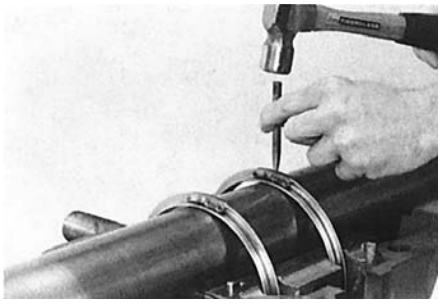
6. Check alignment of pillow block by noting clearance between housing and shaft at each end of the housing — clearance should be uniform within $^{1}/_{32}$". Shim bearing pedestal where possible, otherwise use full length shims under base as required. Alignment of pillow block should be as accurate as possible. The self-alignment feature of the unit is to compensate for normal shaft deflection and possible settling of the supports.



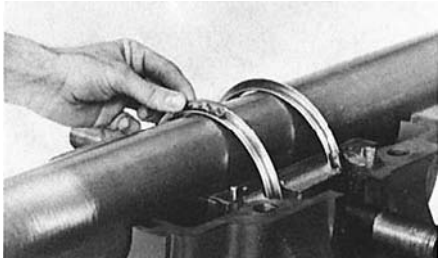
7. Place **oil rings** around outside of lower liner and over shaft.



Peen screws to ensure that they are secure.

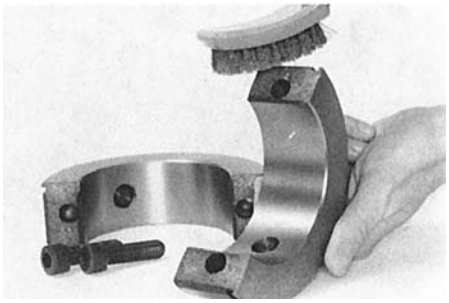


Make sure rings rotate freely on shaft. **Note:** If pillow block is to be arranged for circulating oil, read section headed "Circulating Oil."

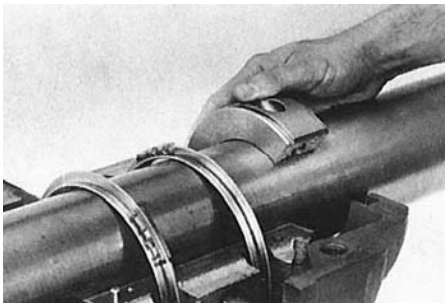


Step 8 does not apply to short series (SSL) style pillow blocks (and where integral collars are used).

8. Thrust Collars in a fixed unit, should now be installed. Remove clamp screws from **thrust collars** and clean cracked joint with wire brush. Back off set screws to clear inside of collar.



Place one collar half on shaft so that flinger groove is next to liner base in the nonexpansion (fixed) bearing.



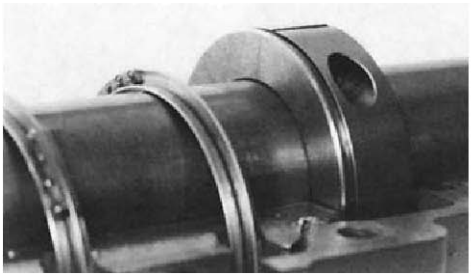
Rotate collar half around shaft and place other half in position. Bring halves together at joint, making sure match at joint is perfect, and insert clamp screws. There should be no offset at collar face. Tap halves together and torque clamp screws to value given in Table 1. Repeat this operation for opposite end of bearing.

Assemble two collars on one bearing only.

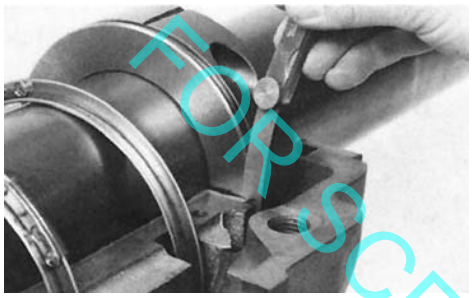
TABLE 1. — Collar and Plunger Screw Torque Values.

STANDARD SLEEVOIL SIZE	COLLAR				PLUNGER SCREW	
	CLAMP SCREW		SET SCREW			
	SCREW SIZE (SOC. HEX)	WRENCH TORQUE (IN.-LBS.)	SCREW SIZE (SOC. HEX)	WRENCH TORQUE (IN.-LBS.)	WRENCH SIZE (IN.-LBS.)	WRENCH TORQUE (IN.-LBS.)
$3^{15}/_{16}$	$^{1}/_{2}$-13NC	1425	$^{3}/_{4}$-10NC	1750	$^{3}/_{8}$	850
$4^{7}/_{16}$	$^{1}/_{2}$-13NC	1425	$^{3}/_{4}$-10NC	1750	$^{1}/_{2}$	1250
$4^{15}/_{16}$	$^{1}/_{2}$-13NC	1425	$^{3}/_{4}$-10NC	1750	$^{1}/_{2}$	1250
$5^{7}/_{16}$	$^{1}/_{2}$-13NC	1425	$^{3}/_{4}$-10NC	1750	$^{1}/_{2}$	1250
6	$^{5}/_{8}$-11NC	2800	$^{7}/_{8}$-9NC	3500	$^{5}/_{8}$	2500
7	$^{5}/_{8}$-11NC	2800	$^{7}/_{8}$-9NC	4100	$^{5}/_{8}$	2500
8	$^{3}/_{4}$-10NC	5000	1-8NC	5000	$^{5}/_{8}$	2500

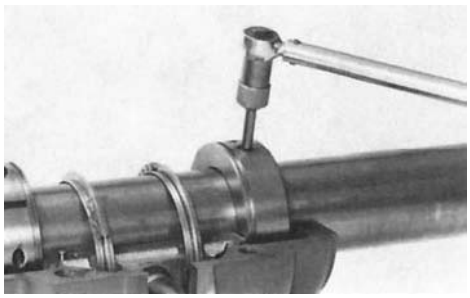
FOR SCRC REFERENCE ONLY



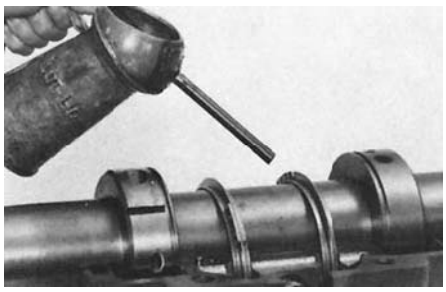
Tap collar up to face of liner allowing a total of .008" to .014" on sizes $3^{15}/16$ and $4^{7}/16$, and .010'' to .016" on sizes $4^{15}/16$" through 8".



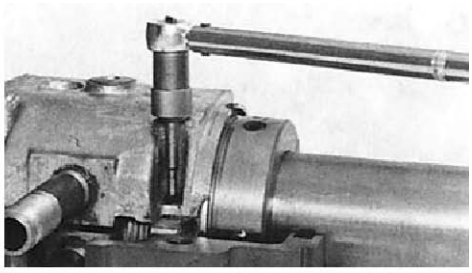
Tighten (Soc. Hex) set screws on both collars to torque specified in Table 1. Collars should run parallel to end faces of liner within .002".



9. Apply oil to bearing surface of **liner cap.** Locate cap in place on lower liner making sure oil rings are in their cavities and are free to rotate.



Install and torque liner cap screws to value given in Table 2. **Note:** If liner cap does not seem to fit properly, ensure match marks are aligned and try again. These SLEEVOIL liners have match marks permanently stamped at the joint near one end starting in June of 1988. These match marks permanently ensure parts stay paired and critical orientation of assemblies is maintained.



10. Tighten housing base to pedestal. Torque bolts to value given in Table 2.

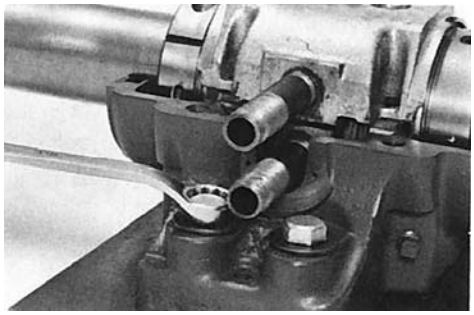
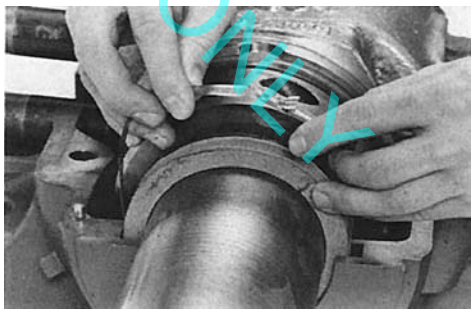


Table 2.

STANDARD SLEEVOIL SIZE	LINER CAP SCREWS		HOUSING MTG. BOLTS	
	SCREW SIZE	TORQUE (IN.-LB.)	BOLT SIZE	TORQUE (IN.-LB.)
$3^{15}/16$	$^{5}/16$-18	132	$^{7}/8$	2040
$4^{7}/16$	$^{5}/16$-18	132	1	3000
$4^{15}/16$	$^{3}/8$-16	240	1	3000
$5^{7}/16$	$^{3}/8$-16	240	$1^{1}/8$	4200
6	$^{1}/2$-13	600	$1^{1}/4$	6000
7	$^{1}/2$-13	600	$1^{1}/2$	10000
8	$^{1}/2$-13	600	$1^{3}/4$	11500

11. Thread **dust seal** and **seal retainer** into groove at end of housing base and around shaft. Slide free end of seal retainer through clasp and pull tightly.

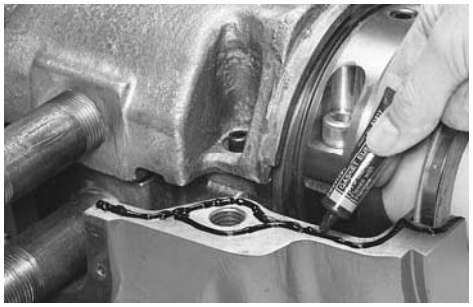


FOR SCPC REFERENCE ONLY

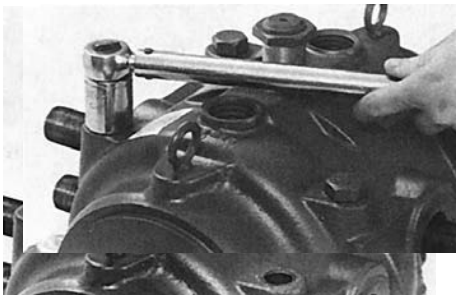
Hold clasp with screwdriver and pull free end of retainer as tightly as possible with pliers. After tightening, it should reduce chances for leakage, a non-hardening sealantbe difficult to move seal from side to side. Cut off excess material.

Aluminum seals, when specified for the application, are packaged separately. When using aluminum seals, discard standard (cork) seals and seal retainers pack-aged with the housing. To install aluminum seals, wrap the O-ring (rubber cord) around the shaft in the seal area and cut it to fit the shaft. For most effective sealing, ends of O-ring must meet. Cementing the ends together is rec-ommended. Disassemble one seal and place one half on shaft over the O-ring installed. Locate O-ring in seal groove and rotate seal half around shaft into housing base groove. Install other half of seal and tighten clamp screws to torque listed in Table 3. Install second seal in same manner. **If using End Closure, install neoprene disc(s) at this time. Consult construction drawing for type of seal recommended.**

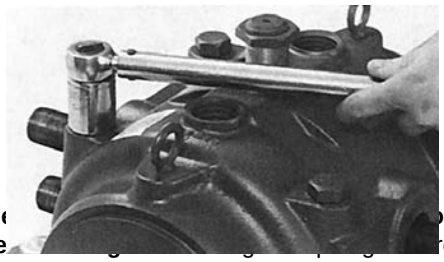
12. We have replaced SLEEVOIL housing gaskets with 515 Gasket Eliminator. Apply Gasket Eliminator to SLEEVOIL housing base along outer contour of joint. **Note:** Special care should be taken at grommet area.



Tighten housing bolts to torque given in Table 3. To reduce Back **off plunger screw** and locate housing cap on base taking care not to damage dust seals or gasket material.

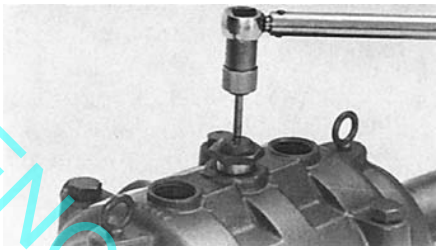


chances for leakage, a non-hardening sealant may be used under cap bolts.

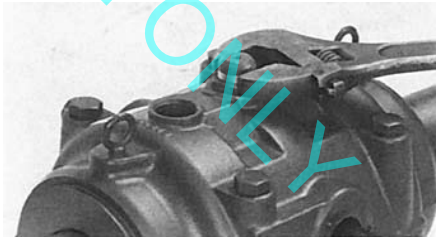


The plunge **ousing bolts have** **rew** to torque given in Table 1. These SLEEVOIL housings have match marks permanently stamped on the water pipe grommet pad starting in June of 1988. These match marks permanently ensure that parts stay paired and critical orientation of assemblies is maintained.

Cap Loaded Bearings: If shaft must be held down to install cap, tighten plunger screws tightly with shaft held down. Mark position of plunger screw. Loosen plunger screw one complete turn and loosen shaft hold down. Then tighten plunger screw while tightening shaft hold-down until plunger screw is tightened to the mark. Do not over-tighten shaft hold-down as this can misalign the bearing. Remove shaft hold-down and tighten plunger screw locknut. **Note: Do not tighten plunger screw on accompanying base loaded bearing until cap loaded bearing has been installed and hold down removed.**



Base Loaded Bearings: Tighten plunger screw locknut.



AOP USA REFERENCE ONLY

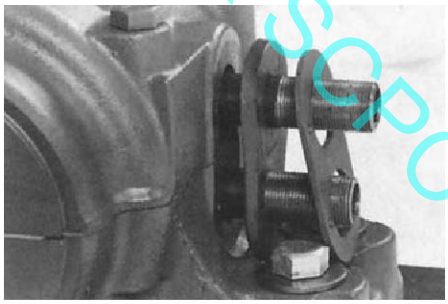
13. Plain Liner Replacement: When using a Plain Liner, remove pipes from PLXC liner assembly and install vent plugs, install grommet and grommet plate over openings in the sides of the housing. To reduce chances for oil leakage, a non-hardening sealant may be used.

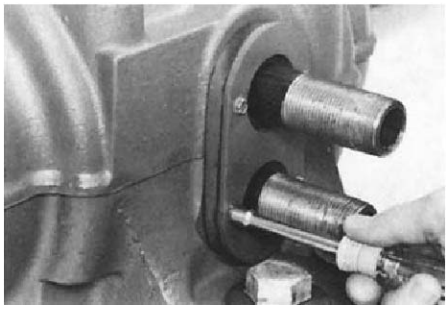
FOR SCPC REFERENCE ONLY

Table 3.

STANDARD SLEEVOIL SIZE	HOUSING CAP BOLTS		SEAL CLAMP SCREWS	
	BOLT SIZE	TORQUE (IN.-LB.)	THREAD SIZE	TORQUE (IN.-LB.)
$3^{15}/_{16}$	$^{3}/_{4}$-10	1900	10-24	12
$4^{7}/_{16}$	$^{7}/_{8}$-9	1700	10-24	12
$4^{15}/_{16}$	$^{7}/_{8}$-9	1700	10-24	12
$5^{7}/_{16}$	1-8	2600	1/4-20	33
6	1-8	2600	1/4-20	33
7	$1^{1}/_{8}$-7	3600	$^{5}/_{16}$-18	65
8	$1^{1}/_{8}$-7	3600	$^{5}/_{16}$-18	65

Externally Cooled Liner: XC liners can be air, water or oil cooled; check drawing for details. When PLXC Liners (with coolant pipes) are to be used, install grommet and grommet plate over pipes and tighten screws. To reduce chances for oil leakage a non-hardening sealant may be used on pipes and housing faces. PLXC liners replace both older plain and XC styles.





14. The **oil level gauge** may be located any distance from the pillow block by the use of a coupling and pipe of the desired length. The extended pipe must be supported so that it remains straight and perfectly level**. Use a spirit level — do not guess.** Use pipe sealer on all connections. **Note:** If bearing is to be arranged for circulating oil, discard oil level gauge.

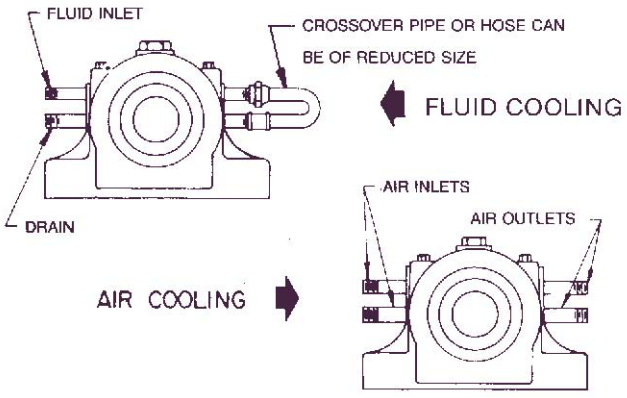
15. Remove all pipe plugs and flush liner bore and housing thoroughly with solvent or cleaner. Reinstall pipe plugs using pipe sealer. Tighten securely.

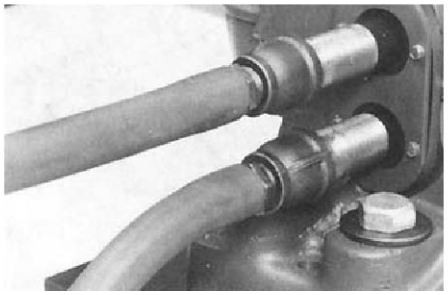
FOR SCPC REFERENCE ONLY

17. Each housing base has predrilled holes for doweling bearing to base plate.

FOR SHOP REFERENCE ONLY

16. Check construction drawing and/or equivalent manu-al to determine if coolant pipes are to be connected. If no connection is required, pipes can remain open. Otherwise, make **pipe connections** required for coolant, making sure that all pipe lengths are correct and unions are well aligned. Careless fitting will result in serious pre-loading of bearing. Lengths of flexible hose between pil-low block and rigid piping are recommended to avoid pre-loading of bearing. If water or glycol is used for coolant, the flexible hose can be of reduced size. A regulating valve should be placed ahead of the inlet and a sight drain at the outlet for liquid coolants. The recommended method of pipe connection for liquid coolants is to con-nect the inlet to one pipe and the outlet to the other pipe on the same side of the bearing. A return is then used to connect the two pipes on the other side of the bearings (See illustration). Adjust coolant flow rate specified on construction drawing or to suit conditions (See Table 4). When using air as the cooling medium, connect an inlet to each pipe on one side of the bearing and an outlet to the pipes on the other side.





LUBRICATION and OPERATION

Since the satisfactory operation of the pillow block depends almost entirely on the oil film being maintained between the shaft and liner bearing surface, it is recommended that a high grade straight mineral oil with rust and oxidation (R & O) inhibitors and anti-foam agents be used. Check equipment specifications for specific recommendation of oil viscosity by equipment manufacturer. If no viscosity is recommended, the following chart can be used as a guide. Information regarding qualities and properties of specific oils should be referred to the lubri-cant manufacturer.

Recommended Oil Viscosity
if not specified by equipment manufacturer.

Ambient Temp. Fahr. During Start Up	Speed	SAE/ISO Oil Required
Below –10°	All	Consult Equipment Manufacturer
–10° to 32°	All	SAE 10/ISO32
32° to 70°	Low High	SAE 20/ISO68 SAE 10/ISO32
Above 70°	Low	SAE 30/ISO100
	High	SAE 10/ISO32 for Light Loads SAE 20/ISO68 for Heavy Loads

Use high grade, high quality, well refined petroleum oils of the straight mineral type with rust and oxidation inhibitor and anti-foam agent only.

Approximate viscosity:
SAE 10 —183 SUS at 100°F; 46 SUS at 210°F
SAE 20 —348 SUS at 100°F; 57 SUS at 210°F
SAE 30 —489 SUS at 100°F; 65 SUS at 210°F
ISO32 —158 SUS at 100°F; 44 SUS at 210°F
ISO68 —335 SUS at 100°F; 55 SUS at 210°F
ISO100 —495 SUS at 100°F; 66 SUS at 210°F

Oil film temperature in liners during normal operation should not exceed 180°F. If in doubt, consult equipment manufacturer.

Low ambient and operating temperatures can be as harmful to the bearing as high temperatures. A heater and thermoswitch would be required for such applications.

Fill the pillow block with oil to the top of the center circle in the oil gauge.

After placing into operation, remove inspection covers and check to make sure oil rings are bringing up oil. Operation should be checked frequently during the first few days. After some running of base loaded bearings only, loosen plunger screw 1/4 turn, then retighten to spec-ified torque. This will allow the liner to align with the shaft. For cap loaded bearings, follow installation procedure. If noise develops, check alignment of housing, collar runout, plunger screw and all operating parts. Check all points and make sure all screws and nuts are tightened after several days operation. **Maintain oil level at center of oil gauge at all times while unit is in operation.**

Oil Maintenance Schedule

Drain, flush, and refill with oil after 2 to 3 weeks of initial break-in operation. Since the satisfactory operation of the bearing depends entirely on an oil film being maintained between the shaft and the bearing liner surface, it is rec-ommended that an oil analysis be performed at these regular intervals.

- Every 3 months for 24 hour/day service
- Every 6 months for 8 hour/day service

Acceptability of oil should be referred to the lubricant manufacture. If oil quality is acceptable then repeat this procedure in 3 month intervals. Visually check oil for con-tamination between oil analysis checks. Oil service life depends upon several factors such as ambient condi-tions, operating temperature and frequency of bearing starts and stops. It is recommended that the oil be changed at least once per year for unfiltered static appli-cations. Removing contaminants through the use of either the OLF (Oil Level and Filtration) Unit or a circulat-ing oil system can extend oil service life. Consult equipment manufacturer for more information.

Check construction drawing for proper coolant flow rate. If not given, use flow rate shown in the following table as a general starting point for proper cooling. Lower coolant inlet temperature and (or) lower ambient temperature require less flow. If the coolant inlet temperature exceeds 90°F additional coolant will be required. Anti-freeze type additives may be used with the water in cold operating environments, otherwise purge all coolant from the liner by blowing out with compressed air or steam anytime bearing coolant is subject to freezing.

Any questions on installation, maintenance or operation should be referred to the equipment manufacturer.

⚠ CAUTION

High air pressure may damage equipment. The fluid or air pressure should never exceed 120 p.s.i.

FOR SPC REFERENCE ONLY

TABLE 4.

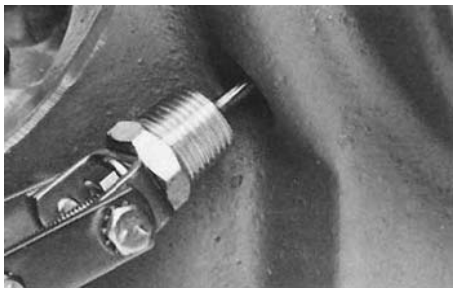
Pillow Block Size	Max. Speed for Ring Oiling (RPM) †	Normal Water Flow Rate (GPM) ▲	Normal Air Flow Rate (CFM) ★	Pressure Drop Req'd. for Air Cooling (PSI) ‡	Oil Flow Rate for Circ. Oil (GPM) ■	Oil Volume*		
						Fl. Oz. ●	Qts. Approx. ●	Liters Approx. ●
$3^{15}/_{16}$	2910	1.75	52	.19	3.0	50	$1^{1}/_{2}$	1.45
$4^{7}/_{16}$	2582	2.25	63	.22	3.5	64	2	1.90
$4^{15}/_{16}$	2321	2.50	77	.11	3.5	90	$2^{3}/_{4}$	2.70
$5^{7}/_{16}$	2107	3.00	90	.12	5.0	100	$3^{1}/_{8}$	2.95
6	1910	4.00	104	.18	5.5	132	$4^{1}/_{8}$	3.90
7	1637	5.00	156	.40	6.5	240	$7^{1}/_{2}$	7.10
8	1432	6.50	173	.50	7.5	320	10	9.50

† Converted from 3000 FPM. Speeds above 3000 require Circulating Oil.
▲ Based on 90°F ambient temperature and 90°F water inlet temperature. Cooler ambient and (or) water inlet temperatures require less flow. Pressure drop across bearing — Approx 1 PSI.
★ Based on 90°F ambient and 90°F air inlet temperatures. Cooler ambient and (or) air inlet temperatures require less flow.
‡ Minimum back pressure required for proper air circulation through the liner coolant chambers.
■ Flow rate required at 120°F oil inlet temperature, 90°F ambient temperature and at 6000 fpm shaft speed under maximum radial load. Slower speeds and lighter loads require less flow.
* Volume of oil required to fill pillow block to top of center circle in oil gage. Applies to both XC and Plain Pillow Blocks.
● 32 fl. oz. = 1 qt = 94636 liters.

FOR SCRD REFERENCE ONLY

OPTIONS

Thermocouple. A thermocouple is used to sense the temperature of the pillow block liner in the shaft area, and if a pre-set thermal limit is exceeded it can trigger a warning or, if desired, send a signal to disconnect electrical power to the motor.



Two $^{1}/_{2}$-14 N.P.S.F. threaded holes on the side opposite the "X" marking have been provided as a standard means of thermocouple mounting. Also furnished with the pillow block is a $^{1}/2$" x $^{1}/8$" adapter bushing to allow adaption of thermocouples from several manufacturers. Spring loaded thermocouples are recommended for positive contact.

Install the thermocouple in the location shown on the construction drawing. Make certain the probe extends into the liner thermocouple hole. For non-expansion bearing, the thermocouple should be located in the hole nearest to the end carrying the thrust load. When the direction of the thrust load is not known, two thermocouples are recommended. For expansion bearings, the thermocouple can be located at either end. Depending on thermocouple and bearing size it may be necessary to employ a system of pipe nipples and couplings to achieve proper probe penetration and housing clearance. Use sealant on all threaded connections including thermocouple mounting thread. Apply sealant to the pipe plug furnished and install it in the other hole.

Thermostat and Immersion Heater. An immersion heater is used to maintain a minimum oil temperature in the pillow block. The thermostat, used in conjunction with the heater, monitors oil temperature in the pillow block and regulates the immersion heater.

If the pillow block has been machined for a heater and thermostat, the threaded mounting holes are usually located in the end face of the housing base. Install heater and thermostat using sealant on all threaded connections to prevent oil leakage.

⚠CAUTION

Heater can be damaged. Turn heater off before filling or drawing oil from pillow block. Heater will overheat when not submerged in oil.

⚠WARNING

Risk of shock. When installing heater and thermoswitch, follow directions and safety procedures recommended by the manufacturer. Install wiring in accordance with the *National Electric Code* and local codes.

Circulating Oil. If the housing has been arranged for circulating oil, the inlets are located in the housing cap. Pipes extend through the housing cap and are directed toward the inspection holes in the liner. The standard drain arrangements are shown at bottom of page. Consult manufacturer's drawing for exact dimensions and location of drain(s).

Install oil supply lines to the $^{1}/_{4}$" N.P.T. external (male) threaded inlets so that each inlet will receive an equal amount of oil. Make sure that the oil flows from the pipes directly into the inspection holes. Drain piping should be vented and of adequate size to drain oil from the bearing at the specified flow rate. The housing drain must be directed straight down into a return drain sloping away at a 15 degree or greater angle from the horizontal. The oil-ing system must have a means of filtering the oil to remove any contaminating particles. A 25 micron filter or better is recommended.

When baffles are installed in oiling grooves of upper liner, oil rings cannot be used and should be discarded. Bend baffle so that free end rests on lower liner joint. Since the housing drain is arranged to maintain the proper oil level, then oil rings may be used as a safety measure or back-up oiling system without oil baffles.

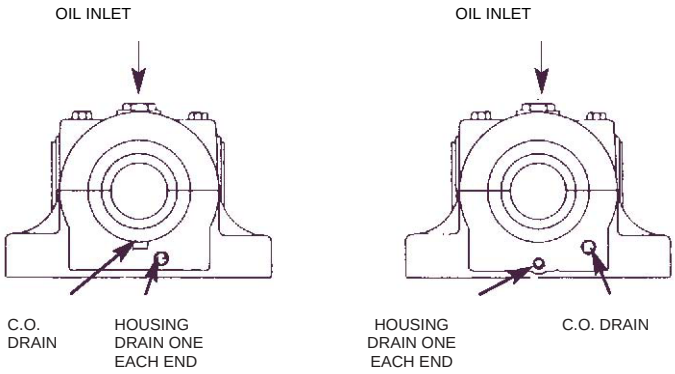
NOTE: Auxillary seals are recommended for outdoor applications, contaminated atmospheres, and high vol-ume of air flowing over the bearing.

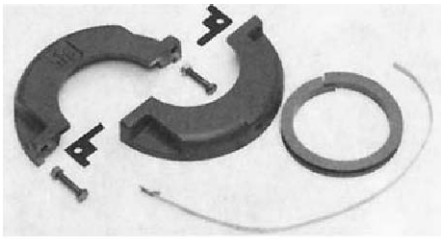
NOTE: Bearings should NOT be stored outdoors before installation. For extended or outdoor storage, contact equipment manufacturer for special precautions against corrosion.

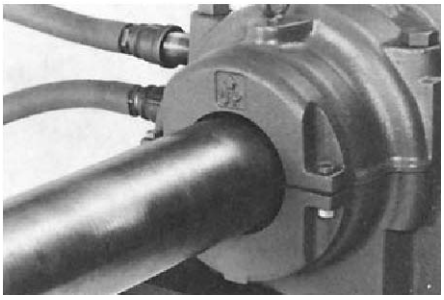
NOTE: Bearings (and shafts) allowed to set idle for extended periods after being run MUST be protected against corrosion. If the unit can not be run for several minutes at least once a week, consult equipment manufacturer for special lubrication instructions.

STL PL PB **SSL PL**

PB





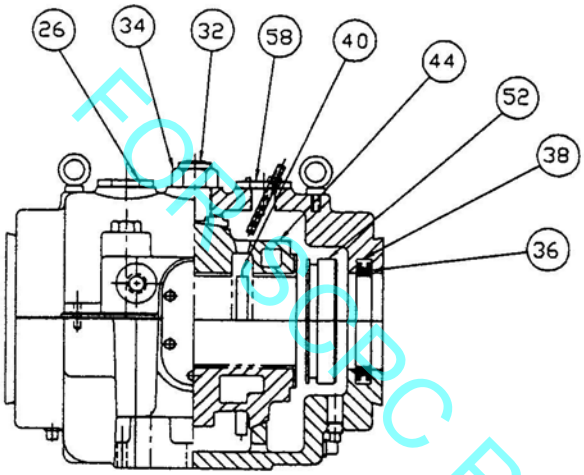


End Closure Kits and Auxiliary Seal Kits are available and may be added at any time without any machining to the pillow block. Refer to the instructions packaged with the kits for installation details.

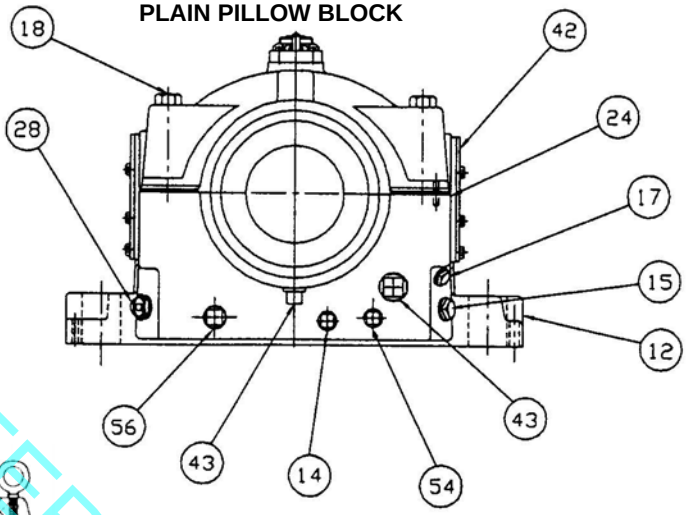
FOR SCPC REFERENCE ONLY

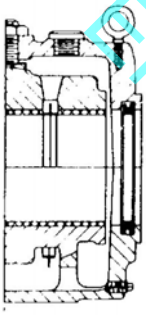
Note: The two-digit numbers are for reference only. Order These SLEEVOIL pillow blocks and liners have name-parts by the six-digit part numbers in the Parts List. Each six-digit number is a complete identification of the part or assembly

These SLEEVOIL pillow blocks and liners have nameplates attached starting in June 1988 identified by a six-digit part number which fully identifies the housing and/or liner. Liner nameplates are pinned to the SLEEVOIL liner cap near an oil ring inspection hole. Housing nameplates are pinned to the housing foot parallel to the shaft. Refer to these part numbers when ordering replacement parts.

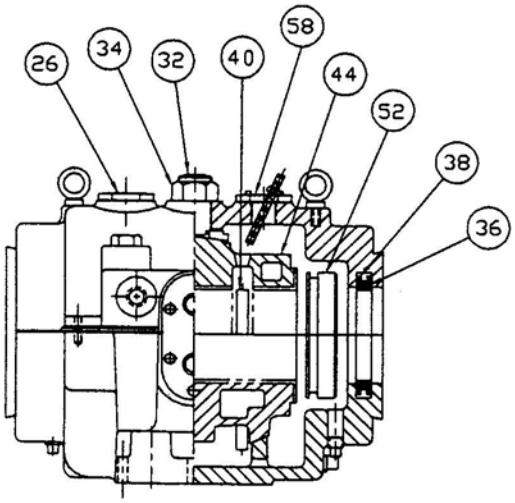


PLAIN PILLOW BLOCK

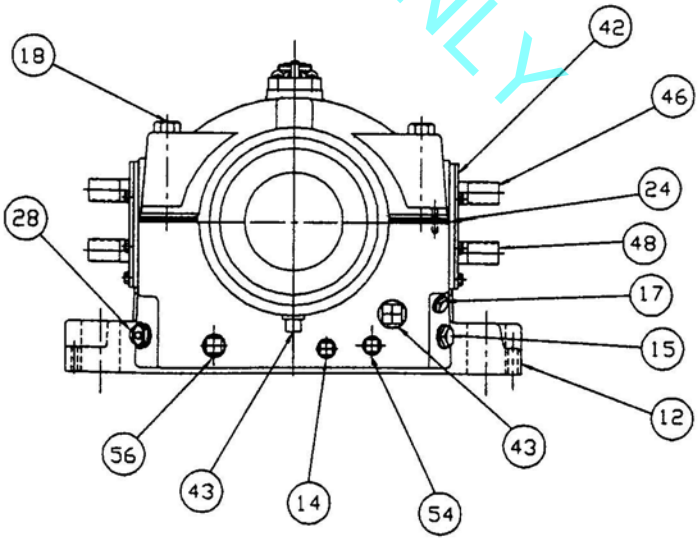




SHORT SERIES
(PLAIN AND PLXC)
END BELL
CONFIGURATION



PLXC PILLOW BLOCK



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SLEEVOIL STL AND SSL PILLOW BLOCKS

Item	Description	No. Req'd.	Replacement Part Numbers						
			3-15/16"	4-7/16"	4-15/16"	5-7/16"	6	7	8
●	STL PLXC Non-Exp. PB (A)	1	134222	134223	134224	134225	134226	134227	134228
	STL PLXC Exp. PB (B)	1	134200	134201	134202	134203	134204	134205	134206
●	SSL PLXC Exp. PB (B)	1	134207	134208	134209	134210	134211	134212	134213
●	STL Modular HSG*	1	132782	132783	132784	132688	132689	132690	132691
●	SSL Modular HSG*	1	132718	132719	132720	132920	132921	132922	132923
12	STL HOUSING MACH	1	132779	132780	132781	132663	132664	132665	132666
●	SSL HOUSING MACH	1	132739	132740	132741	132930	132931	132932	132993
14	▲ Drain Plug	1	430012	430012	430012	430012	430012	430012	430012
15	▲ Oil Level Plug	1	430012	430012	430012	430014	430014	430014	430014
17	▲ Thermocouple Plug	2	430012	430012	430012	430012	430012	430012	430012
●	▲ Thermocouple Adapter	1	430081	430081	430081	430081	430081	430081	430081
18	▲ Housing Bolt	4	411190	411196	411197	411205	411207	411226	411227
24	▲ Gasket Eliminator	1	427359	427359	427359	—	—	—	—
	▲ Gasket Eliminator	2	—	—	—	427359	427359	427359	427359
26	▲ Inspection Cover	2	432197	432199	432199	432199	432199	432199	432199
28	▲ Oil Gage	1	430139	430139	430139	430135	430135	430135	430135
32	▲ Plunger Screw Assembly	1	390522	390523	390524	390525	390571	390572	390572
34	▲ Plunger Screw Locknut	1	133368	133369	133369	133369	133369	133369	133369
36	▲ Dust Seal	2	133567	133568	133549	133614	133616	133617	133618
38	▲ Seal Retainer	2	133579	133579	133579	133579	133579	133579	133579
40	▲ Oil Ring	2	130056	130057	130058	130059	130060	130062	130065
42	▲ Grommet Kit	1	132192	132192	132193	132193	132198	132198	132198
43	▲ Circulating Oil Drain Plug (STL)	2	430014	430014	430016	430016	430016	430017	430017
	▲ Circulating Oil Drain Plug (SSL)	2	430014	430016	430017	430017	430017	430017	430017
●	Auxiliary Seal Kit	2	432181	432184	432187	133932	133933	133937	133938
●	Housing End Cap Kit	1	432190	432193	432196	132546	132547	132548	132549
●	Circulating Oil Grommet Kit	1	432153	430198	430198	430155	430155	430155	430155
44	▲ PLXC LINER ASSEMBLY ■	1	132952	132953	132954	132955	132956	132957	132958
46	▲ Water Pipe—Upper Liner Half	2	430171	430171	430174	430174	430176	430176	430177
48	▲ Water Pipe—Lower Liner Half	2	430171	430171	430174	430174	430176	430176	430177
●	▲ Dowel Pin	2	420053	420053	420064	420064	420080	420080	420080
●	▲ Liner Cap Screw	2	417084	417084	417117	417114	417184	417184	417184
●	▲ Vented Plug Kit	1	462996	462996	462997	462997	462997	462997	462997
52	Thrust Collar*	2	133285	133292	133293	133294	133295	133296	133297
54	▲ Thermostat Plug	1	430012	430012	430012	430012	430012	430012	430012
●	Thermostat	1	133116	133116	133116	133116	133116	133116	133116
56	▲ Heater Plug	1	430014	430014	430014	430014	430016	430016	430017
●	Heater	1							
24	DUST SEAL KIT	1	389828	389829	389830	389831	389832	389833	389834
36	▲ Gasket Eliminator	1	427359	427359	427359	—	—	—	—
38	▲ Dust Seal	2	133567	133568	133549	133614	133616	133617	133618
24	▲ Seal Retainer	2	133579	133579	133579	133579	133579	133579	133579
	▲ Gasket Eliminator	2	—	—	—	427359	427359	427359	427359
OPTIONAL ACCESSORIES									
●	Auxiliary Seal Kit	2	432181	432184	432187	133932	133933	133937	133938
●	Housing End Cap Kit	1	432190	432193	432196	132546	132547	132548	132549
58	Circulating Oil Grommet Kit	1	432153	430198	430198	430155	430155	430155	430155

★Both Housings include all parts immediately following marked "▲" and a two piece housing.
▲ These parts make up the assemblies under which they are listed.
● Not shown on drawing.
† One grommet kit consists of 2 rubber grommets — 2 grommet plates and 8 grommet screws.
■ Liner Assemblies also include a 2-piece liner. PLXC liners replace both older plain and XC styles.
* 2 required for one non-expansion (fixed) pillow block. Includes all hardware required for mounting. Cannot be used with SSL style Pillow Blocks.
**For SLEEVOIL SSL housing only.
x Plunger Screw Assembly consists of 1 plunger screw, 1 washer and 1 button assembled together.
(A) STL PLXC Non-Expansion Pil Blk includes Modular HSG, Liner Assy and (2) Split Thrust Collars.
(B) STL PLXC Expansion or SSL PLXC Expansion Pil Blk includes Modular HSG and Liner Assy.

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This material is not intended to provide operational instructions. Appropriate instruction manuals and precautions should be studied prior to installation, operation or maintenance of equipment.

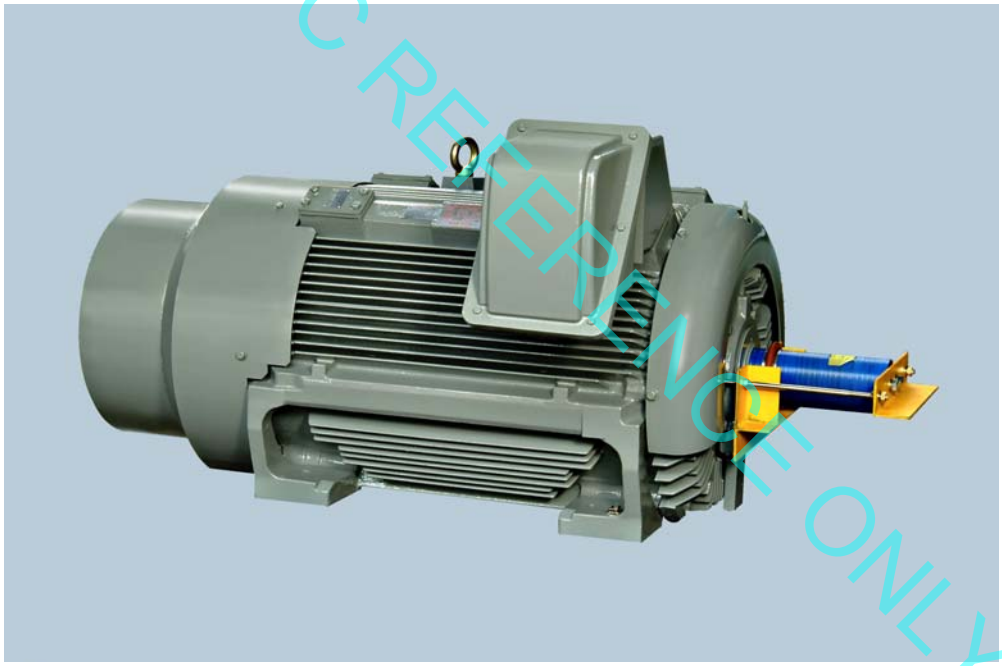
INSTRUCTIONS

FOR

THREE PHASE

INDUCTION MOTORS

FOR SCPC REFERENCE ONLY



TECO

TECO Electric & Machinery Co., Ltd.

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1. INTRODUCTION

This and the following instructions address the more common situations encountered in motor installation, operation and maintenance. For the TECO motor warranty to be and to remain in effect, the motor must be installed and operated in strict accordance with the outline drawing, motor nameplates and these instructions and must not be altered or modified in any unauthorized manner.

During these installations and operation of motors in heavy industrial applications there is a danger of live electrical parts and rotating parts. Therefore to prevent injury and/or damage the basic planning work for installation, transport, assembly, operation, etc.… needs to be done and checked by authorized and competent personnel only.

Since these instructions cannot cover every eventuality of installation, operation and maintenance, the following points should however be considered and checked.

- The technical data and information on permissible use such as assembly, connection, ambient and operating conditions given in the related catalogue, operating instructions, nameplates and other production documentation.
- The general erection and safety regulations.
- The local and plant-specific specifications and requirements.
- The proper use of transport, lifting devices and tools.
- The use of personal protective equipment.

Following indications should be observed when reading these instructions.

Safety instructions are marked as follows：

Warning of electric hazards for personnel.

Warning of dangers for personnel.

ATTENTION！
Warning of damage for the motor or installation.

FOR SCPC REFERENCE ONLY

2. ACCEPTING, INSPECTION, STORAGE, TRANSPORTATION

2.1 Inspection upon receipt

Check the following points upon receipt：

- Are the nameplate ratings identical with what you ordered ?
- Are dimensions and color in compliance with your specifications ?
- Are the nameplate ratings for heater, thermal protector, temperature detector, etc. identical with what you ordered ?
- Is there any damage ?
- Are all accessories and accompanying instruction manuals in good order ?
- Please ensure that the arrowhead indicator really indicates direction of revolution.
- If there are any specific requirements, please ensure they are in conformity with your specification.

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2.2 Storage

When motors are not in operation, the following precautionary measures should be undertaken to assure best performance.

2.2.1 Place

(a) High and dry, well-ventilated without direct sun, dust or corrosive gas.

(b) Not located near to a boiler or freezer.

(c) Entirely free from vibration and easy for movements.

(d) Motors should be put on pallets to prevent moisture.

2.2.2 Well protection

Motors should be well shielded from dust, but under well-ventilated circumstances. For those water-cooling motors or using bearings with water-cooling coils, please make sure the water already dried off to prevent tube corrosion or danger of frost.

2.2.3 Moisture prevention

Since moisture can be very detrimental to electrical components, the motor temperature should be maintained about 3℃ above the dew point temperature by providing either external or internal heat. If the motor is equipped with space heaters, they should be energized at the voltage shown by the space heater nameplate attached to the motor. Incandescent light bulbs can be placed within the motor to provide heat. However, if used, they must not be allowed to come in contact with any parts of the motor because of the concentrated hot spot that could result.

2.2.4 Insulation resistance test

Even during storage, the insulation resistance should be kept above the specified values.

(a) For measurement of insulation resistance and acceptable standard values, please refer to measures stated in 4.1.2 "Measurement of insulation resistance".

(b) Insulation resistance test should be performed once every three months.

2.2.5 Long period storage

If the motor is not in operation for a long period (one week and above) after installation or has been in operation but stopped for a period of time, the following precautions must be taken.

(a) Protect the motor as measures stated in 2.2.3.

(b) Insulation resistance test should be performed as stated in 2.2.4.

(c) Bearing protection per 2.2.6.

(d) Operation test should be performed once every three months.

(e) Storage maintenance is to be documented for warranty data.

2.2.6 Bearing protection

(a) If the motor has been provided with a shaft shipping brace to prevent shaft movement during transit, it must be removed before operating the motor.
It is very important that this brace be reinstalled exactly as it was originally, before the motor is moved from storage or any time when the motor is being transported. This prevents axial rotor movement that might damage the bearings.

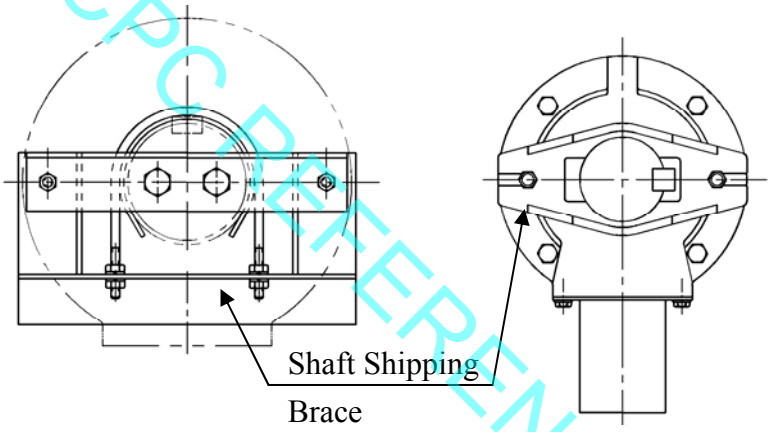

Fig. 1

(b) Motors equipped with sleeve bearings are shipped from the factory with the bearing oil reservoirs drained. In storage, the oil reservoirs should be properly filled to the center of the oil level gauge with a good grade of rust inhibiting oil. To keep the bearing journals well oiled and to prevent rusting, the motor shaft should be rotated several revolutions about every month ensuring the shaft does not come to rest in its original position. While the shaft is rotating, it should be pushed to both extremes of the endplay. If the motor is not in operation over six months, dismount the upper cover of sleeve bearing housing and check the anti-corrosion protection.

(c) Motors with anti-friction bearings are properly lubricated with the correct grade of grease at the factory and no further greasing is required in storage. If the motor is not in operation over three months, add grease to each bearing per lubrication nameplate. The shaft should be rotated several revolutions about every month to maintain proper distribution of the grease within the bearings.

(d) Tilt-pad bearings are a type of sleeve bearing used in special design applications. Due to the nature of this bearing, a loose oil ring for delivering lubricant cannot be provided. Therefore, during the storage interval, oil must be periodically manually introduced into the pads and housing to prevent the occurrence of oxidation of the precision machined components.

FOR SCPC REFERENCE ONLY

(1) Remove the pipe plug from the bearing cap located above the tilt-bearing shell.
(2) Pour in approximately one cup of oil every month and rotate the shaft a few revolutions about every two (2) weeks.

For long periods of storage, the oil that accumulates in the housing should be removed.

(e) The bearing assembly parts of motors with oil mist lubrication are put on with anti-rust oil, so they can be preserved for several months in good conditions. The motor should be stored indoor & well-ventilated environment and prevent to contact with contaminated or corrosive air. The following points should be noted：

(1) During preservation, the Inpro seal can not prevent the moisture to go through into the bearings. Please use the oil mist to lubricate the bearings every two (2) weeks.
(2) If the color of flow out oil is changed, the bearing should be rusted or having contamination in it. Please contact with us.
(3) Avoid using grease as it will plug the vent/drain.
(4) All assembly surfaces are painted with seal bonds, don't disassemble them anytime.
(5) Don't remove the plugs in vent/drain to prevent the moisture.
(6) Don't apply any force on the Inpro seal to prevent damage.
(7) The Inpro seal is a labyrinth type seal. Therefore it can not contain a pressure differential.

2.2.7 Prevent rusting

ATTENTION！

Cares should be taken to keep parts such as fitting surface, key, shaft extension and axial central hole from any collision with foreign matters. Grease should also be generously applied to prevent rusting.

FOR SCDC REFERENCE ONLY

2.3 Transportation

ATTENTION！

To keep the rotating parts of motors from moving, thus preventing damage and scratching during transportation, they should be held securely with a locking device. Remove all transit clamps before operating the motor. It is very important that this device be reinstalled exactly as it was originally, before the motor is moved from storage or any time when the motor is being transported.

The vertical mounting type motors should be transported in the vertical position.

Do not use the hoisting hook/eyebolts to lift more that the motor itself. They are designed to support the motor only.

Make sure the hoisting hook is correctly attached to the eyebolt(s) or lug(s) of the motor and that the eyebolt(s)/lug(s) are fully screwed in before hoisting. Also note such parts as fan cover, ventilation box, bracket, slip-ring, etc. may have their own hoisting lugs which can only carry their own weight. Nothing extra should be attached while hoisting.

Do not twist the steel wires and make sure the eyebolts have been firmly screwed and the sling angle is correct.

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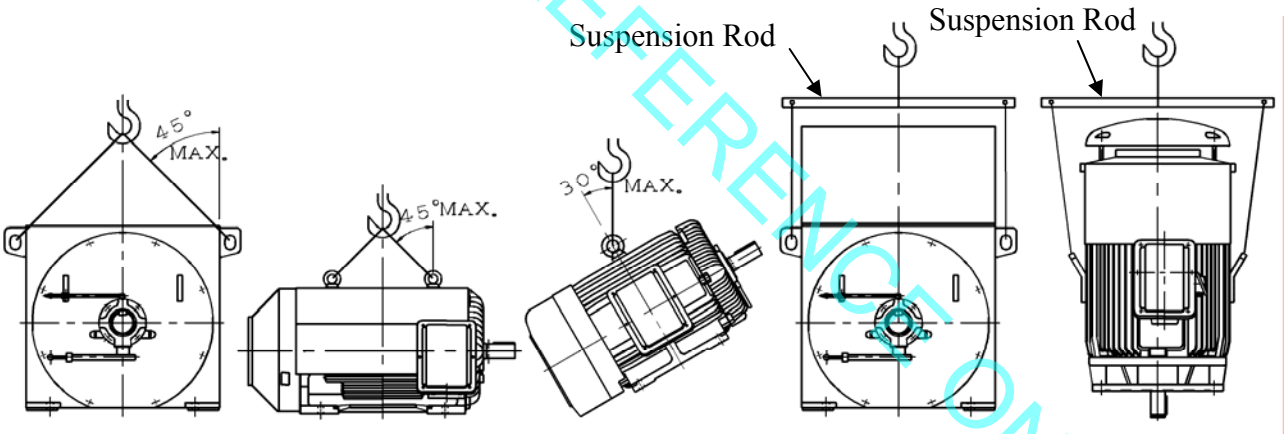

Fig. 2

3. INSTALLATION

3.1 Site and environment for motor installation

3.1.1

Standard environment and site conditions for the installation of motors are usually set as follows：

(a) Ambient temperature：-20 ~ +40 ℃

(b) Humidity：Relative humidity below 95%RH for totally-enclosed types, and below 80%RH for semi-enclosed types.

(c) Elevation：below 1000 meters.

(d) Harmful gases, liquids, dusts, high moisture should be absent.

(e) Foundations should be strong and free of vibration.

For those water-cooling motors or using bearings with water-cooling coils, the ambient temperature shall not below 0℃ to prevent danger of frost. If there are any special environmental conditions, please inform us upon ordering.

3.1.2 Ventilation and space

(a) Installation area should be well-ventilated.

(b) The installation space should be large enough to facilitate heat dissipation and maintenance.

3.2 Foundation

3.2.1 Soleplate & common bed

Use rigid and solid soleplate or common bed as foundation.

ATTENTION !

For best motor performance, it is advisable to use a soleplate or common bed, particularly when using a shaft coupling.

If the soleplate or common bed does't have enough stiffness, the critical speed of motors or equipment will then be changed. This change may cause a large vibration (resonance) and decrease the life of machines.

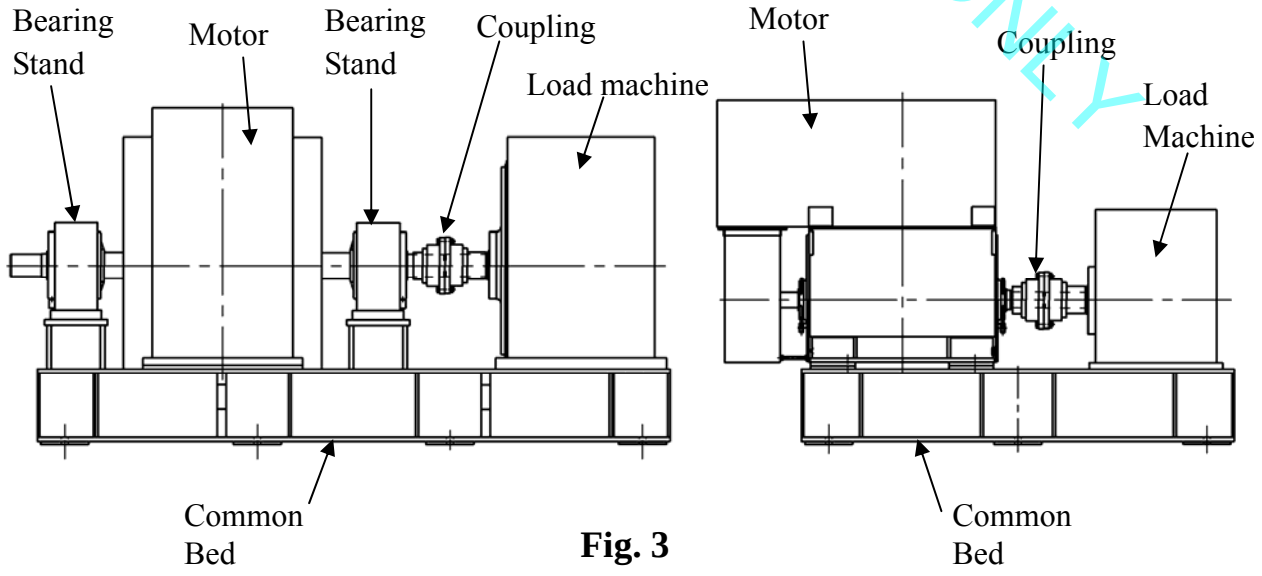

Fig. 3

FOR SCPC REFERENCE ONLY

3.2.2 Installation

(a) Select an appropriate foundation surface for the soleplate or common bed which will be considered the ultimate level.

(b) Align the position of the common bed with reference to that level.

(c) Align the level accuracy at least at four points such as bearing mounting, shaft extension etc. The accuracy should be within 0.04mm (1.5mil).

(d) Soleplate or common bed should be embedded in concrete foundation as illustrated in Fig.4. Stiff pads should also be installed beneath the wedges which are welded together at various spots about 400-500mm (15-20inches) apart etc., to enable foundation to carry evenly the weight of the whole motor.

(e) The base should be sturdy and rigid to keep it flat and level.

(f) Make sure the mortar and concrete are completely dry, and the precision of the level is acceptable, then set the motor on the mounting foundation.

(g) Accurately install shaft couplings, belt sheaves etc., then weld the wedges solid to prevent untoward change in position.

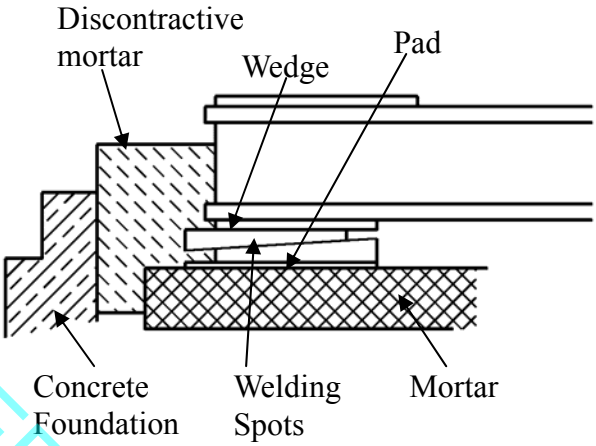

Fig. 4

3.2.3 The foundation of vertical induction motor (Also the foundation of pump)

(a) Foundation of motor/pump must be rigid and secure to provide adequate support. There must be no vibration, twisting, misalignment etc. due to inadequate foundations.

(b) A massive concrete foundation is preferred in order to minimize vibration. Rigidity and stability are enhanced by prop plate and foundation bolt. As shown in Fig.5 and Fig.6.

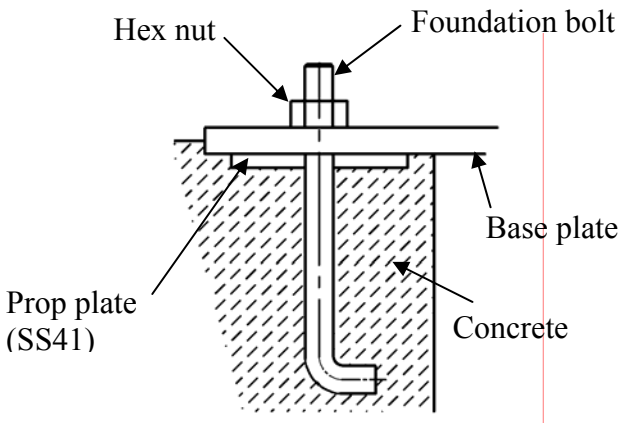

Fig. 5

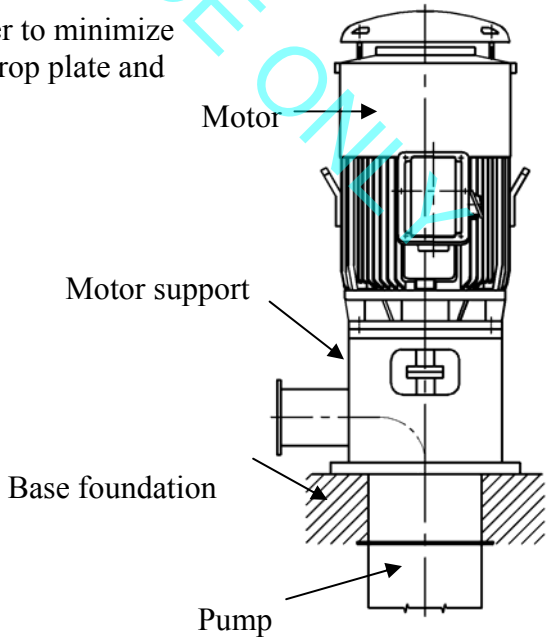

Fig. 6

FOR SCPC REFERENCE ONLY

3.2.4 Installation of vertical motor

(a) All mounting surfaces must be clean and level.

(b) Foundation must be leveled at least at 4 points and guaranteed to be below 0.04mm flat and level.

(c) Make sure the mortar and concrete are completely dry, and the precision of the level is acceptable, then set the motor on the mounting foundation.

(d) Accurately install shaft couplings.

3.3 Installation of shaft coupling

3.3.1 General

ATTENTION！

Motors must always be accurately aligned, and this applies especially where they are directly coupled.

Incorrect alignment can lead to bearing failure, vibration and even shaft fracture. As soon as bearing failure or vibration is detected, the alignment should be checked.

3.3.2 Mounting procedure

Field application of a coupling to the motor shaft should follow the procedures recommended by the coupling manufacturer. The motor shaft extension must not be subjected to either extreme heat or cold during coupling installation.

3.3.3 Safety attention

ATTENTION！

Basically, the coupling should be heated and pushed onto the shaft extension with slight axial force. Do not hammer coupling to prevent bearing damage.

3.3.4 End-play

Although the sleeve bearings are equipped with thrust faces, these are intended only to provide momentary axial restraint of rotor movement either during start-up or when operating the motor disconnected from the driven equipment. They must not be operated under a constant thrust load unless they were originally designed for this condition.

Motors with either sleeve or anti-friction bearings are suitable for connection to the driven load through a flexible coupling. Coupling solidly to the load is not acceptable. With sleeve bearings, the flexible coupling should be of the limited end float type to prevent the possibility of any end thrust from the load being transmitted to the motor bearings, which could cause bearing damage.

The recommended limits of end float for couplings are as follows：

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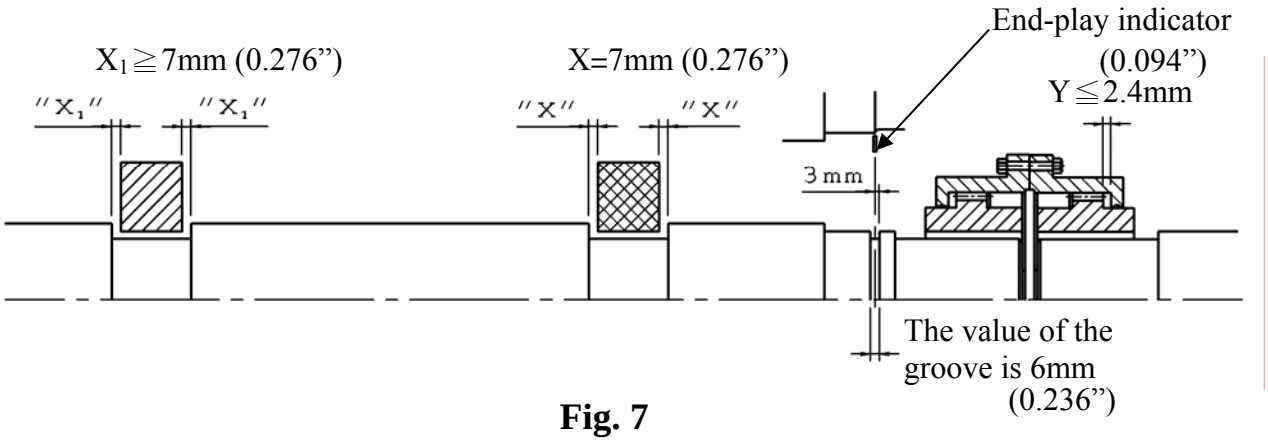

Fig. 7

(a) When the motor is in operation after installation, be sure that the end-play indicator is within the 6mm of the groove on the shaft or aligned to the shaft shoulder immediately outboard of the drive-end bearing to assure there is low friction between shaft and bearing.

(b) Unless otherwise specified, the designed end-play value X of the groove for TECO motors in general is within 7mm (0.276”) as illustrated in Fig. 7. In essence, the end-play indicator is adjusted to point at the center of the groove or the drive-end shaft shoulder; thus X equals to 7± 1mm (0.276”±0.039”) or so, and the end-play value (Y) of the couplings should equal or be smaller than 2.4mm (0.094”).

(c) If the desired value Y is greater than 3mm (0.118”) caused for instance by a thrust load and/or load machine with large end-play, please inform us when ordering.

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3.3.5 Thermal growth

In aligning the motor (and rotor) axially with the driven equipment, consideration should be given not only to the end-play indicator position but also to axial shaft expansion and increase in shaft centerline height due to thermal effects. In general, the axial shaft growth for motors can be disregarded since neither bearing is fixed and any shaft growth due to temperature increase will produce an elongation away from the coupling.

Shaft height growth (change in shaft centerline elevation) for TEFC machines can be calculated as follows：

$$\triangle = (0.0005) \times (\text{motor foot to shaft centerline dimension})$$

For non-TEFC machines, divide this number by 2.

3.3.6 Alignment

It is desirable, in normal operation, that the motor operate on its magnetic center, so that no axial force is exerted on the coupling.

The motor shaft and the driven shaft should be aligned within the following tolerances in both angular and parallel alignment：

Unit：mm

TIR	Range of rotating speed	Solid coupling	Flexible coupling
C	2500rpm and above	0.03	0.03
	Below 2500rpm	0.04	0.05
A	2500rpm and above	0.03	0.03
	Below 2500rpm	0.03	0.04

Angular misalignment is the amount by which the centerlines of driver and driven shaft are skewed. It can be measured using a dial indicator set up as shown in Fig.8. The couplings are rotated together through 360 degrees so that the indicator does not measure run out of the coupling hub face. The shafts should be forced against either the in or out extreme of their end float while being rotated.

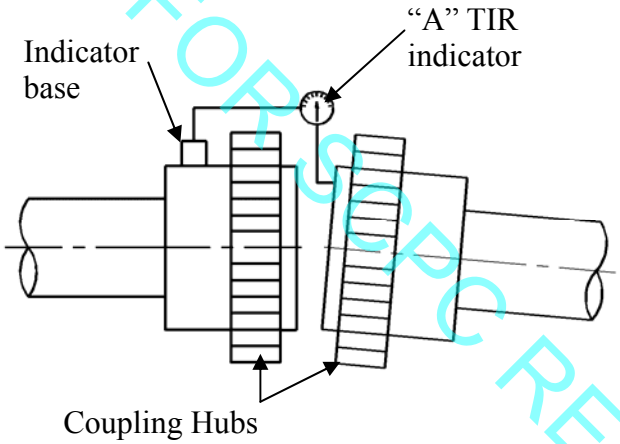

Fig. 8

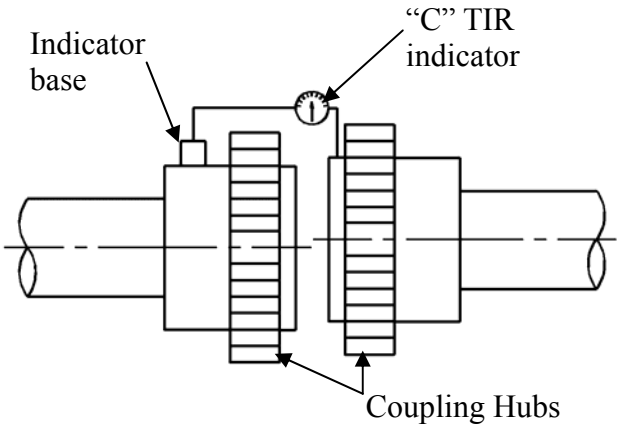

Fig. 9

Parallel misalignment is the amount by which the centerlines of the driver and driven shafts are out of parallel. It can be measured using a dial indicator set up as shown in Fig.9. Again, the couplings are rotated together through 360 degrees so that the indicator does not measure runout of the coupling hub outside diameter.

TIR = Total indicator reading (by dial indicator)

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3.3.7 Dowel

After the motor has been properly aligned with the driven equipment and the hold-down bolts have been installed and tightened, for motors with fabricated frame, at least two dowel pins should be installed in two diagonally opposite motor feet.

3.3.8 Installation of shaft coupling (Vertical hollow shaft motor only)

Bolted coupling as Fig.10

(a) Bearings are provided to absorb some upward shaft thrust when the coupling is fitted.

(b) The coupling is fastened with bolts.

(c) This coupling type is not auto-release type.

Note：Standard high thrust motors can absorb momentary upthrust load up to 30% of the standard downthrust load. If the upthrust is long duration (over 10 seconds) and/or exceeds 30% of the standard high thrust rating, special design arrangements are required and a standard motor is not suitable.

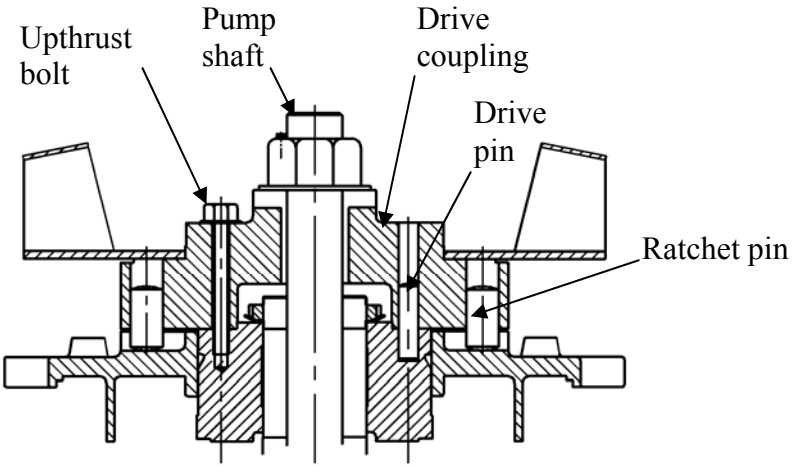

Fig. 10

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3.3.9 Non-reverse ratchet/coupling, as Fig.9 (If necessary)

The non-reverse coupling is also a bolted type and,

(a) It prevents the pump and motor from rotating in the reverse direction.

(b) It also prevents damage from over speeding and damage to pump shaft and bearings.

(c) The ratchet pins are lifted by the ratchet teeth and are held clear by centrifugal force and friction as the motor comes up to speed.

(d) When power is removed, speed decreases, and the pins fall. At the instant of reversal, a pin will catch in a ratchet tooth and prevent backward rotation.

(e) When installing the non-reverse coupling, do not use lubricant. Lubrication will interfere with proper operation. The top half of the coupling should seat solidly on the lower half and the pins should touch the bottom of the pockets between the teeth in the plate.

(f) As with the bolted coupling, the upthrust capabilities are 30% of the standard high thrust rating for downthrust.

ATTENTION !

Do not apply non-reverse ratchets on applications in which the pump reversal time from shutdown (the instant the stop button is pressed) to zero speed is less than one second.

3.3.10 Removal of redundant shaft key

When the length of coupling hub is different from the length of shaft key, the motor may have a high vibration level due to this unbalance condition. The removal of redundant shaft key is necessary, shown as Fig.11.

Method (1)：

After installing the coupling, use a grinding wheel to remove the redundant key (hatch area).

Method (2)：

Before installing the coupling, calculate the different length between coupling hub and shaft key, then cut the half of this different value (hatch area) to achieve approximate-balance condition.

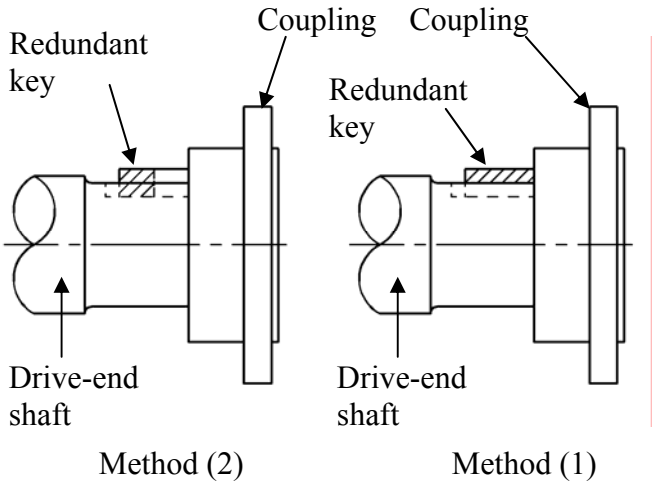

Fig. 11

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3.4 Installation for belt drive

In general, power transmission through direct flexible coupling is appropriate for large motors. Such motors are not suitable for belt, chain or gear connection unless specially designed for such service. However, for small and medium motors of which outputs within the ranges shown on table below, it is acceptable to use belt transmission as indicated. Beyond these ranges, do not apply belt sheaves unless specially designed.

3.4.1 Diameter of sheaves

The diameter ratio between conveyance sheaves should not be greater than 5 to 1 for flat belts, and 8 to 1 for V-belt. It is also advisable to limit the belt velocity to under 35 m/sec to limit belt abrasion and vibration. The smaller the outer diameter of the V-belt sheave, the greater the shaft bending stress will be. If the bending stress is in excess of the shaft fatigue stress, the shaft may break. Therefore, please inform us when you have decided the size of the sheaves and the length of the belts upon ordering.

ATTENTION！

Place the sheave and belt as close as possible to the motor body (it is advisable to make x as shown in Fig.12 equal to 0) to reduce the bending moment and improve shaft life.

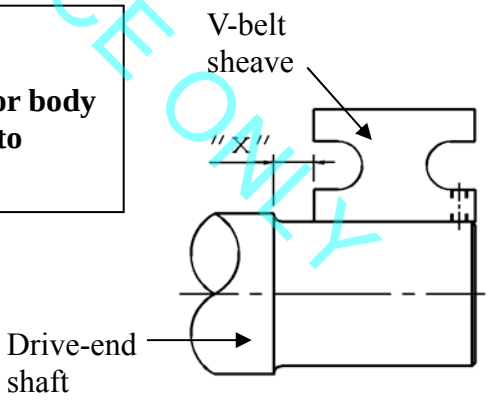

Fig. 12

3.4.2 Table of belt-sheave application for general electric motors

Output (kW)			V-Belt Sheave							
			Conventional V-Belt				Narrow V-Belt			
4P	6P	8P	V-Belt Type	Number of Belts	Min. PCD (mm)	Max. Width (mm)	V-Belt Type	Number of Belts	Min. PCD (mm)	Max. Width (mm)
11	--	--	B	4	160	82	3V	4	125	48
--	11	--	B	5	170	101	3V	5	140	59
--	--	11	B	5	190	101	3V	6	160	69
15	--	--	B	5	170	101	3V	6	125	69
--	15	--	B	5	224	101	3V	6	160	69
--	--	15	C	4	224	111	5V	3	180	60
18.5	--	--	B	5	200	101	3V	6	140	69
--	18.5	--	C	4	224	111	5V	3	180	60
--	--	18.5	C	5	224	136	5V	4	180	78
22	--	--	B	5	224	101	3V	6	160	69
--	22	--	C	5	224	136	5V	4	180	78
--	--	22	C	5	250	136	5V	4	200	78
30	--	--	C	5	224	136	5V	4	180	78
--	30	--	C	5	265	136	5V	4	224	78
--	--	30	C	6	265	162	5V	5	224	95
37	--	--	C	6	224	162	5V	4	200	78
--	37	--	C	6	265	162	5V	4	224	78
--	--	37	C	7	280	187	5V	5	250	95
45	--	--	C	6	265	162	5V	4	224	78
--	45	--	C	7	280	187	5V	5	224	95
--	--	45	C	7	315	187	5V	6	250	113
55	--	--	C	7	265	187	5V	5	224	95
--	55	--	C	8	300	213	5V	6	250	113
--	--	55	D	5	355	196	5V	6	280	113
75	--	--	C	8	315	213	5V	6	250	113
--	75	--	D	6	355	233	5V	6	315	113
--	--	75	D	6	400	233	5V	6	355	113
--	90	--	D	6	400	233	5V	6	355	113
--	--	90	D	6	425	233	8V	4	355	124
--	110	--	D	7	400	270	8V	4	355	124
--	132	110	D	7	450	270	8V	4	400	124
--	160	132	D	9	450	344	8V	4	450	124

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3.5 Conveyance with chain or gear

3.5.1 Loading capacity

Make sure the loading capacity of shaft and bearings is appropriate for the size and installation position (overhung) of chain and gear. If necessary, please contact us to ensure the shaft and bearings will meet your requirements.

3.5.2

Pay close attention to ensure the parallelism of shafts.

3.5.3

The teeth of couplings should be correctly and precisely matched; the force conveyance centers should lie on the same line.

3.5.4

There should be no skip, jumping, vibration or unusual noises.

ATTENTION！

Do not hammer the conveyance devices such as couplings, belt sheaves, chain wheels, gears etc. onto the shaft. Those shaft fitments should be fitted and removed only by means of suitable devices. Heat shrinking may be a better alternative to avoid damaging bearings and other components.

The exposed rotating parts should be covered to prevent accidents.

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3.6 Electrical connections

All interconnecting wiring for controls and grounding should be in strict accordance with local requirements such as the USA National Electrical Code and UK IEE wiring regulations.

Wiring of motor and control, overload protection and grounding should follow the instructions of connection diagrams attached.

3.6.1 Power

The rated conditions of operation for the motor are as shown on the nameplate. Within the limits, given below, of voltage and frequency variation from the nameplate values, the motor will continue to operate but with performance characteristics that may differ from those at the rated conditions：

+/- 10% of rated voltage

+/- 5% of rated frequency

+/- 10% combined voltage and frequency variation so long as frequency variation is no more than +/- 5% of rated

Operating the motor at voltages and frequencies outside of the above limits can result in both unsatisfactory motor performance and damage to or failure of the motor.

3.6.2 Main lead box

The main lead box furnished with the motor has been sized to provide adequate space for the make-up of the connections between the motor lead cables and the incoming power cables.

The bolted joints between the motor lead and the power cables must be made and insulated in a workman-like manner following the best trade practices.

3.6.3 Grounding

Either fabricated motors or fan cooled cast frame motors are all provided with grounding pads or bolts.

The motor must be grounded by a proper connection to the electrical system ground.

3.6.4 Rotation direction

The rotation direction of the motor will be as shown by either a nameplate on the motor or the outline drawing. The required phase rotation of the incoming power for this motor rotation may also be stated. If either is unknown, the correct sequence can be determined in the following manner：While the motor is uncoupled from the load, start the motor and observe the direction of rotation. Allow the motor to achieve full speed before disconnecting it from the power source. Refer to the operation section of these instructions for information concerning initial start-up. If resulting rotation is incorrect, it can be reversed by interchanging any two (2) incoming cables.

3.6.5 Auxiliary devices

Auxiliary devices such as resistance temperature detectors, thermocouples, thermoguards, etc., will generally terminate on terminal blocks located in the auxiliary terminal box on the motor. Other devices may terminate in their own enclosures elsewhere on the motor. Such information can be obtained by referring to the outline drawing. Information regarding terminal designation and the connection of auxiliary devices can be obtained from auxiliary drawings or attached nameplates.

If the motor is provided with internal space heaters, the incoming voltage supplied to them must be exactly as shown by either a nameplate on the motor or the outline drawing for proper heater operation.

Caution must be exercised anytime contact is made with the incoming space heater circuit as space heater voltage is often automatically applied when the motor is shutdown.

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4. OPERATION

4.1 Examination before start

4.1.1 Wiring check

When motors are installed in good manner, ensure the wiring is according to the diagram. Also, the following points should be noted：

(a) Make sure all wiring is correct.

(b) Ensure the sizes of cable wires are appropriate and all connections are well made for the currents they will carry.

(c) Ensure all connections are properly insulated for the voltage and temperature they will experience.

(d) Ensure the capacity of fuse, switches, magnetic switches and thermo relays etc. are appropriate and the contactors are in good condition.

(e) Make sure that frame and terminal box are grounded.

(f) Make sure that the starting method is correct.

(g) Make sure switches and starters are set at their right positions.

(h) Motor heaters must be switched off when the motor is running.

4.1.2 Measurement of insulation resistance

During and immediately after measuring, the terminals must not be touched as they may carry residual dangerous voltages. Furthermore, if power cables are connected, make sure that the power supplies are clearly disconnected and there are no moving parts.

(a) For rated voltage below 1000V, measured with a 500VDC megger.
For rated voltage above 1000V, measured with a 1000VDC megger.

(b) In accordance with IEEE 43-2000, there are three recommendation minimum insulation resistance values. These values corrected to 40℃ are：

(1) kV+1 in Megohms for most windings made before 1970, all field windings and windings not otherwise described.

(2) 100 Megohms for most DC armatures and AC windings built after about 1970 with form wound coils.

(3) 5 Megohms for machines with random wound stator coils and for form wound coils rated below 1kV.

ATTENTION！

After measurement the winding must be grounded for discharging the winding.

(c) On a new winding, where the contaminant causing low insulation resistance is generally moisture, drying the winding through the proper application of heat will normally increase the insulation resistance to an acceptable level. The following are several accepted methods for applying heat to a winding：

(1) If the motor is equipped with space heaters, they can be energized to heat the winding.

(2) Direct current (as from a welder) can be passed through the winding. The total current should not exceed approximately 20% of rated full load current. If the motor has only three leads, two must be connected together to form one circuit through the winding. In this case,

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one phase will carry the full applied current and each of the others, one-half each. If the motor has six leads (3 mains and 3 neutrals), the three phase should be connected into one series circuit.

Ensure there is adequate guarding so live parts cannot be touched.

(3) Heated air can be either blown directly into the motor or into a temporary enclosure surrounding the motor. The source of heated air should preferably be electrical as opposed to fueled (such as kerosene) where a malfunction of the fuel burner could result in carbon entering the motor.

ATTENTION！

Caution must be exercised, when heating the motor with any source of heat other than self contained space heaters, to raise the winding temperature at a gradual rate to allow any entrapped moisture to vaporize and escape without rupturing the insulation. The entire heating cycle should extend over 15-20 hours.

Insulation resistance measurements can be made while the winding is being heated. However, they must be corrected to 40℃ for evaluation since the actual insulation resistance will decrease with increasing temperature. As an approximation for a new winding, the insulation resistance will approximately halve for each 10℃ increase in insulation temperature above the dew point temperature.

(d) Should the resistance fail to attain the specified value even after drying, careful examination should be undertaken to eliminate all other possible causes, if any.

4.1.3 Power source

(a) Ensure the capacity of the power source is sufficient.

(b) Ensure the supply voltage and frequency ratings are identical to those on the nameplate.

(c) Voltage variation should be confined to within ±10% of the rated value and the phase to phase voltages should be balanced.

4.1.4 Bearing lubrication

(a) For sleeve bearing motors, the oil reservoir must be filled with oil to the correct level. On self-lubricated bearings, the standstill oil level will be at the center of the oil gauge. The proper oil is a rust and oxidation inhibited, turbine grade oil. Refer to the lubrication nameplate for the recommended viscosity.

(b) Motors which are supplied with provision for flood lubrication have an inlet orifice to meter the oil flow to the bearing. Refer to the outline drawing for these values. If the supply pressure does not match that stated on the outline, the orifice size must be adjusted to produce the specified flow rate. The drain adapter (also provided) has a weir plate fixed to the inside of the pipe to permit the establishment of the proper oil level. This weir plate must be located at the bottom of the pipe and must be parallel to the plane of the motor feet. To ensure optimum flow, the drain line should be vented to the atmosphere.

Oil inlet temperature：	Normal below	50℃
	Alarm	60℃
	Trip	65℃

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(c) If the motor is in storage for over three (3) months, refilling of some new oil should be undertaken before operation to prevent bearing damage due to dry friction. The oil level should be kept at the center of the oil gauge. If necessary, drain some oil after refilling.

(d) Grease lubricant type

(1) The bearings have been well greased at factory before delivery. However, regreasing is required if a significant period has elapsed between manufacture and use or in storage. Fill new grease until it overflows and the old grease is entirely replaced.

(2) Unless otherwise specified, SHELL Alvania R3 is the standard applied to TECO motors.

4.1.5 Oil-mist lubrication motors

(a) Before operation：

Please check the rotor & stator parts of the Inpro seal, and make sure that they don't contact with each other. The 0.03mm thickness gap gauge is suggested to inspect the gap between rotor and stator as shown in Fig.12. If they are touched in one side, it means that this seal is in the wrong position. You can use a wooden or rubber hammer to adjust this seal gently. If this can't be done, this seal may be failed and have to be checked. Please contact with us.

(b) Lubricant：

Please use ISO VG68 or the same viscosity oil except lubrication nameplate specified others.

(c) About the oil mist lubrication generation system：

The users have to prepare an oil mist lubrication system, and apply suitable inlet oil quantity and pressure. Please follow the instruction of oil mist lubrication system what you used. Avoid high oil pressure going into the inside of the motor to damage the insulation or oil leakage.

(d) Operation：

Before first operation or long time without running, please turn on the oil mist lubrication system at least 5 hours to lubricate the bearings. If you use the central control system, the mist system shall operate continuously for a minimum of 48 hours. After making sure that the pressure is stable and flow out oil is in good conditions, then you could operate this motor.

(e) Appendix：

Fitting or reclassifier of oil-mist lubrication system：

Reclassifiers or fittings are devices which convert small particle-size dry mist to larger particle sizes which can lubricate bearings. Please use correct fittings or reclassifiers.

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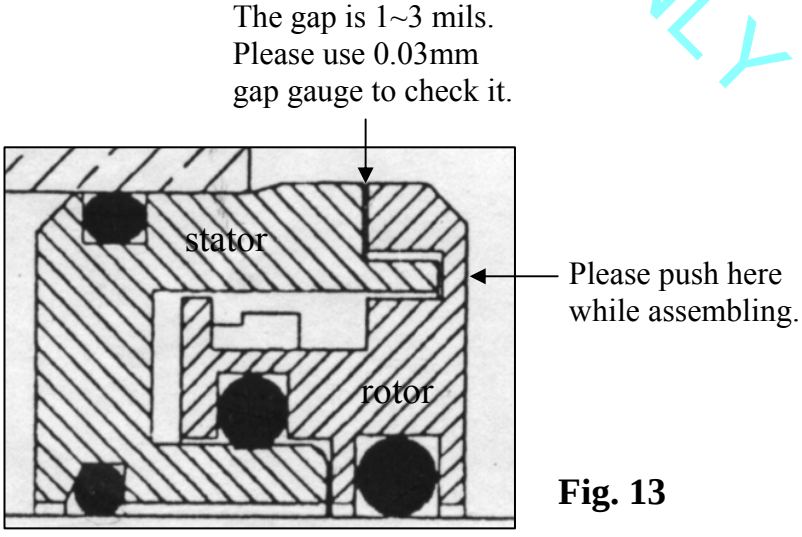

Fig. 13

4.1.6 Cooling water for the cooler on water-cooled motors

Make sure the quality, volume and inlet temperature of cooling water for the motors are normal before the machine is in operation.

Water：General tower water or industrial water, the suspension solid shall below 20 μ m/l
Volume：Please see outline drawing
Inlet temperature：Normal below 30℃ (86℉)；above 5℃ (41℉)
Alarm 35℃ (95℉)
Trip 40℃ (104℉)
Special temperature settings will be noted in outline drawings.

4.1.7 Remove all locks

ATTENTION！

Make sure all locks which fasten the movable parts of the motors during transportation are dismantled and the shaft can rotate freely.

4.1.8 Clean before starting

ATTENTION！

Ensure there are no foreign matters or tools inside the motors before starting motors.

4.1.9 Transmission system check

Make sure the transmission system, including belts, screws, bolts, nuts and set pins are in good condition.

The keys fitted to the shaft extensions are held by plastic tape only to prevent them falling out during transportation or handling. The shaft key shall be removed to avoid flying out, when the motor is operated prior to the couplings etc. being fitted to the shaft extension.

4.1.10 Test run

Make sure the items above are examined. Test the motor running with or without load. Record and check according to "Maintenance" at 15 minutes intervals during the first three hours of operation. Then regular examinations should take place at longer intervals. If all goes well the motor can be classified as "in good order".

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4.2 Starting operation

4.2.1 Starting load

Initially run the motor unloaded prior to coupling to other machines. Unless otherwise specified, a motor usually starts with light load which is then gradually increased proportional to the square of speed and at last reach 100% load at full load speed.

4.2.2 Starting

Too frequent starts can harm the motors. The following restrictions should be observed：

(a) Motor can be restarted should the initial start fail. Two starts are generally permissible when the motor is cold.

(b) Motor can be started only once when it is at normal running temperature.

(c) Should additional starts be necessary beyond the conditions stated above, the following restrictions should be noted：

(1) Let the motor cool down for 60 minutes before restarting, fully loaded.
(2) Let the motor cool down for 30 minutes before restarting, unloaded.
(3) Two inching starts can be regarded as one normal start.

(d)

ATTENTION！

If the motor rotor fails to start turning within one or two seconds, shut off the power supply immediately.
Investigate thoroughly and take corrective action before attempting a restart.

Possible reasons for not starting are：

(1) Too low a voltage at the motor terminals.

(2) The load is too much for the rotor to accelerate.

(3) The load is frozen up mechanically.

(4) All electrical connections have not been made.

(5) Single phase power has been applied.

(6) Any combination of the above.

4.2.3 Rotating direction

(a) Most TECO motors are bi-directional. However, when some special types, such as high speed 2P, certain large capacity motors, those with a non-reverse ratchet etc. should rotate in one direction, please ensure the rotation is in conformity with the directional arrow-mark shown on the attached nameplate.

(b) To reverse a bi-directional motor, cut the power and wait until the motor stops. Then interchange any two of the three phases.

4.2.4 Power source, Voltage, Current

(a) Ensure the voltage and frequency of the power source are identical to the ratings shown on the nameplate.

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(b) Voltage variation should be confined to within ±10% of the rating and the three phase voltages should be in full balanced.

(c) Ensure the motor phase currents, when without load, are within ±5% of the average values.

4.2.5 Frequency

Frequency variation should be confined to within ±5% of the rating. The aggregate variation of voltage and frequency should be confined to within ±10% of the absolute value of the ratings.

4.2.6 Starting time and unusual noises

ATTENTION！

Starting time is longer for the motors with large inertia. However, if starting time is longer than usual or if there is difficulty in starting, or there is abnormal noise, do not run the motor and refer to TECO.

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4.2.7 Sleeve bearing oil rings (sleeve bearing types only)

As the oil ring is used to carry lubricant to sleeve bearings, frequently check to ensure the oil ring is in motion.

4.2.8 Bearing temperature rise

Following the initial start-up, the bearing temperatures should be closely monitored. The rate of rise in bearing temperature is more indicative of impending trouble than is the actual temperature.

ATTENTION！

If the rate of rise in temperature is excessive or if the motor exhibits excessive vibration or noise, it should be shut down immediately and a thorough investigation made as to the cause before it is operated again.

If the bearing temperature rise and motor operation appear to be normal, operation should continue until the bearing temperatures stabilize.

Recommended limits on bearing temperature are as follows：

Sleeve Bearings.	Alarm temperature.	Trip temperature
• By permanently installed detector	90℃(194℉)	95℃(203℉)
Anti-Friction Bearings.		
• By permanently installed detector	95℃(203℉)	100℃(212℉)

ATTENTION！(For sleeve bearing)

(1) It must be noted that when operating flood lubricated sleeve bearings without outside lubrication supplied, the bearing temperature must not be allowed to exceed 85℃ total temperature.

(2) Under normal condition, for the self-lube bearing, the rate of temperature rise should be from 11℃ to 14℃ for the first ten (10) minutes after starting up and approximately 22℃ at thirty (30) minutes. The rate of bearing temperature rise is a function of the natural ventilation and operating conditions.

(3) When the rate of bearing temperature rise is less than 1℃ per half hour, the bearing temperature is considered to be stabilized.

(4) If the total bearing temperature exceeds 95℃, the motor should be shut down immediately.

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4.2.9 Noise and Vibration

ATTENTION！

Any abnormal noise or vibration should be immediately investigated and corrected. Increased vibration can be indicative of a change in balance due to mechanical failure of a rotor part, a stator winding problem or a change in motor alignment.

5. MAINTENANCE

5.1 Major points in regular inspection and maintenance

For safety, maintenance and repairs must only be carried out by properly trained personnel.

Some testing, such as insulation resistance, usually requires the motor to be stopped and isolated from power supplie(s).

Routine inspection and maintenance are usually performed by looking, listening, smelling and simple meters.

High temperature may arise under operating conditions on the motor surfaces, so that touching should be prevented or avoided.
Keep away from moving and live parts.
Unless deemed necessary, do not remove guards whilst assessing the motor.

Timely replacement of worn parts can assure longevity and prevent breakdown.

Routine inspection and regular inspection and maintenance are important in preventing breakdown and lengthening service life.

Owing to the varied time and circumstances motors are used, it is difficult to set the items and periods for regular inspection and maintenance. However, as a guide it is recommended to be performed periodically according to factory maintenance program. Generally, the inspection scope determined by the following factors：

(a) Ambient temperature.
(b) Starting and stopping frequency.
(c) Troublesome parts usually affecting motor functions.
(d) Easily abraded parts.
(e) The important position of motor in the operational system of a factory should be duly recognized. Therefore, its health and wellbeing should be fully protected, especially when it is operating in severe conditions.

FOR SCPC REFERENCE ONLY

5.2 Motor windings

(a) Measurement of insulation resistance and standards to determine quality of insulation resistance, please refer to measures stated in 4.1.2 "Measurement of insulation resistance".

(b) Inspection of coil-ends：

(1) Grease and dust accumulated on coils may cause insulation deterioration and poor cooling effect.

(2) Moisture must not accumulate. Keep coils warm when motor is not in use if moisture can be seen.

(3) Discoloring. This is mainly caused by overheat.

(c) Ensure no untoward change of wedges from original position.

(d) Ensure the binding at the coil end is in its normal position.

FOR SCBC REFERENCE ONLY

5.3 Clean the interior of the motor

(a) After a motor is in operation for some time, accumulation of dust, carbon powder and grease etc., on the inside is unavoidable, and may cause damage. Regular cleaning and examination is necessary to assure top performance.

(b) Points to note during cleaning：

(1) If using compressed air or a blower：

a) Compressed air should be free of moisture.

b) Maintain air pressure at 4 kg/cm^2, since high pressure can cause damage to coils.

(2) Vacuum

Vacuum cleaning can be used, both before and after other methods of cleaning, to remove loose dirt and debris. It is a very effective way to remove loose surface contamination from the winding without scattering. Vacuum cleaning tools should be non-metallic to avoid any damage to the winding insulation.

(3) Wiping

Surface contamination on the winding can be removed by wiping using a soft, lint-free wiping material. If the contamination is oily, the wiping material can be moistened (not dripping wet) with a safety type petroleum solvent.

In hazardous locations, a solvent such as inhibited methyl chloroform may be used, but must be used sparingly and immediately removed. While this solvent is non-flammable under ordinary conditions, it is toxic and proper health and safety precautions should be followed while using it.

ATTENTION！

Solvents of any type should never be used on windings provided with abrasion protection. Abrasion protection is a gray, rubber-like coating applied to the winding end-turns.

Adequate ventilation must always be provided in any area where solvents are being used to avoid the danger of fire, explosion or health hazards. In confined areas (such as pits) each operator should be provided with an air line respirator, a hose mask or a self-contained breathing apparatus. Operators should wear goggles, aprons and suitable gloves. Solvents and their vapors should never be exposed to open flames or sparks and should always be stored in approved safety containers.

(4) Keep core ducts completely clean. The difference in temperature rise could be around 10℃ before and after cleaning.

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5.4 Clean the exterior of the motor

(a) On open ventilated motors, screens and louvers over the inlet air openings should not be allowed to accumulate any build-up of dirt, lint, etc. that could restrict free air movement.

ATTENTION！

Screens and louvers should never be cleaned or disturbed while the motor is in operation because any dislodged dirt or debris can be drawn directly into the motor.

(b) If the motor is equipped with air filters, they should be replaced (disposable type) or cleaned and reconditioned (permanent type) at a frequency that is dictated by conditions. It is better to replace or recondition filters too often than not often enough.

(c) Totally enclosed air-to-air cooled and totally enclosed fan cooled motors require special cleaning considerations. The external fan must be cleaned thoroughly since any dirt build-up not removed can lead to unbalance and vibration. All of the tubes of the air-to-air heat exchanger should be cleaned using a suitable tube brush having synthetic fiber bristles (not wire of any type).

5.5 Maintenance of anti-friction bearing

5.5.1 Frequency of relubrication

The life of grease varies greatly as a result of types of model, revolution speed, temperature, operational conditions etc. It is, therefore, impossible to be precise about replenishment intervals. However, for normal direct coupling transmission, the periods shown as Table 1 may be used as a guide.

Remarks：

(a) The periods shown in Table 1 should be halved where bearings are used for belt drive and/or in dirty, or high ambient temperature or high humidity environments.

(b) Please refer to the lubrication nameplate, if attached to the motor.

(c) For bearing numbers outside the range of Table 1, please contact TECO.

(d) If the periods referred to Table 1 for drive-end bearing and opposite drive-end bearing are different, for the convenience of maintenance operation, please take the shorter one the required grease replenishment period of these bearings.

TABLE 1

Bearing number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM	3000 RPM	3600 RPM
62XX 63XX 72XX 73XX	6210										
	12									2000Hrs	
	13										
	14									1000Hrs	
	15										
	16									720 Hrs	
	17							2000Hrs			
	18			3000Hrs							
	20										
	22										
	24							1500Hrs			
	26										
	28					2000Hrs		1000Hrs			
	30										
	32							500 Hrs			
	34					1500Hrs					
	36										
	38			2000Hrs		1000Hrs					

Bearing number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM
NU2XX NU3XX	NU214								
	15							2000Hrs	
	16								
	17								
	18			3000Hrs				1500Hrs	
	20								
	22							1000Hrs	
	24								
	26					2000Hrs			
	28							500 Hrs	
	30								
	32								
	34			2000Hrs		1000Hrs			
	36								
	38	2000Hrs							
	40								
	44			1000Hrs					
	48	1000Hrs							

Bearing number		600 RPM	720 RPM	750 RPM	900 RPM	1000 RPM	1200 RPM	1500 RPM	1800 RPM
222XX 223XX	22220							300Hrs	
	22								
	24			1000Hrs		500 Hrs			
	26								
	28								
	30					300 Hrs			
	32			500 Hrs					
	34								
	36								
	38	500 Hrs							
	40			300 Hrs					
	44								
	48	300 Hrs							

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5.5.2 Kinds of grease

SHELL Alvania R3 grease is standard for TECO motors except some special models for which special grease will be shown on the lubrication nameplate. Please use identical grease or its equivalents when maintaining.

ATTENTION !

Do not mix different kinds of grease.
Mixing grease with different type of thickeners may destroy its composition and physical properties. Even if the thickeners are of the same type, possible differences in the additive may cause detrimental effects.

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5.5.3 Grease quantity

The amount of grease per replenishment depends on the type, size and construction of the bearings. The maximum amount of one replenishment for each bearing is shown in Table 2.

TABLE 2

Bearing No.		Amount of replenishment
62XX 72XX NU2XX 222XX	6210	30g
	6212	40
	6213	50
	6214	50
	6215	60
	6216	60
	6217	80
	6218	80
	6220	100
	6222	120
	6224	120
	6226	140
	6228	160
	6230	180
	6232	200
	6234	250
	6236	300
	6238	350
	6240	400
	6244	450
	6248	500

Bearing No.		Amount of replenishment
63XX 73XX NU3XX 223XX	6310	40g
	6312	60
	6313	80
	6314	80
	6315	100
	6316	100
	6317	120
	6318	120
	6320	160
	6322	220
	6324	270
	6326	300
	6328	400
	6330	450
	6332	500
	6334	600
	6336	700
	6338	800
	6340	900
	6344	900
	6348	900

* Fill new grease until it overflows and the old grease is entirely replaced.

5.5.4 Re-greasing

If relubrication is to be performed when the motor is running, stay clear of rotating parts.

It is advisable to re-grease when the motor is running to allow the new grease to be evenly distributed inside the bearing.

Before re-greasing, the inlet fitting should be thoroughly cleaned to prevent any accumulated dirt from being carried into the bearing with the new grease. The outlet of grease drainage should be opened to allow the proper venting of old grease.

Use a grease gun to pump grease through grease nipple into bearings. After re-greasing, operate the motor for 10-30 minutes to allow any excess grease to vent out.

5.5.5 Oil relubrication (For oil lubrication types only)

Maintain proper lubrication by checking the oil level periodically and adding oil when necessary. Because of the initial clearing action of the bearing and the expansion of the oil as it comes up operating temperature, the oil level will be higher after the motor has been in operation for a while than it is with the motor at standstill.

Overfilling should be avoided not only because of the possibility that expansion may force the oil over the oil sleeve and on to the rotor, but also because too high an operating oil level prevents the bearing form clearing itself of excess oil. The resultant churning can cause extra loss, high temperatures, and oxidized oil. If, during operation, the oil level goes above the maximum shown on the sight gauge, drain enough oil to bring the level back within the recommended operating range.

Do not permit the operating level to fall below the minimum shown on the gauge.

ATTENTION !

Should it ever become necessary to add excessive amount of make-up oil, investigate immediately for oil leaks.

Change the oil at regular intervals. The time between oil changes depends upon the severity of operating conditions and, hence, must be determined by the motor user. Two or three changes a year is typical, but special conditions, such as high ambient temperature, may require more frequent changes. Avoid operating the motor with oxidized oil.

Use only good grade, oxidation-corrosion-inhibited turbine oils produced by reputable oil companies. The viscosity of the oil to be used depends upon the type and size of the bearing, its load and speed, the ambient temperature, and the amount and temperature of the cooling water (if used). The lubrication nameplate or instructions with each motor specifies the viscosity range of oil suitable for average conditions. The usual oil viscosity recommendations are summarized in Table 3 and Table 4. Operation in ambient temperatures that are near or below freezing may require preheating the oil or the use of special oil. Whenever the motor is disassembled for general cleaning and reconditioning, the bearing housing may be washed out with a suitable cleaning solvent. Be sure that the oil metering hole is clear, and then dry the housing thoroughly before reassembly, and ensure all traces of cleaning solvent have been removed.

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Table 3 oil viscosity for vertical motors**

Bearing Type	Oil viscosity	Range of pole
Angular contact ball (72XX，73XX)	ISO VG32 (150 SSU/100˚F)	2 pole
	ISO VG68 (300 SSU/100˚F)	4 pole and above
Spherical roller (293XX，294XX)	ISO VG68 (300 SSU/100˚F)	4 pole
	ISO VG150 (700 SSU/100˚F)	6 pole and above
RENK Vertical Bearing Inserts Type EV or Kingsbury Type	ISO VG68 (300 SSU/100˚F)	4 pole and above

Table 4 oil viscosity for horizontal motors**

Bearing Type	Oil viscosity	Range of pole
Sleeve bearing	ISO VG32 (150 SSU/100˚F)	2 pole
	ISO VG46 (200 SSU/100˚F)	4 pole
	ISO VG68 (300 SSU/100˚F)	6 pole and above

**Remark: When a lubrication nameplate attached to the motor, use lubrication oil it stipulates.

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5.5.6 Cleaning and installation of bearings

(a) Apply the proper amount of grease to disassembled parts of the bearing after they have been thoroughly cleaned with high quality cleaning oil. Then protect them from contamination before and during assembly.

(b) Bearing installation

ATTENTION！

Before installing the bearings, make sure that the shaft mounted parts inside the bearings are in place before installation.

Since the bearing is a high precision component, it is important to avoid ingression of dust and foreign matter, and hammering during cleaning and installation. Use extreme care and ensure clean conditions during installation and assembly.

ATTENTION！

The best way for bearing installation is heat shrinking. Knocking and hammering during installation should be avoided absolutely.

The bearing should be heated in a bath of clean oil at temperature of approx. 80℃. After warming, slide the bearings in place quickly and nimbly so that it has not shrunk before being fully in position.

Grease the bearing after the temperature returns to normal, and then reassemble the motor.

5.6 Maintenance of sleeve bearing

5.6.1 Daily inspections

(a) Ensure the volume and quality of lubricating oil are in compliance with specifications.

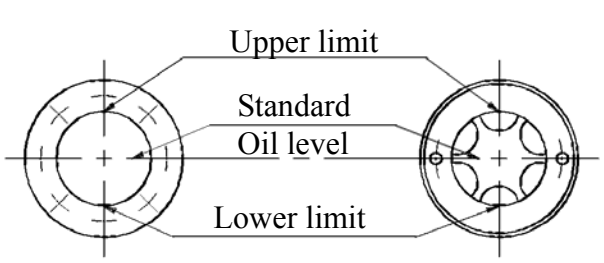

Fig. 14

(b) Ensure there is motion of the oil ring and it is not clamped.

(c) The indicator of the shaft endplay should be restricted within the specified range of the red groove of the shaft or the ±3mm range of the drive-end shaft shoulder, or the bearing may be damaged.

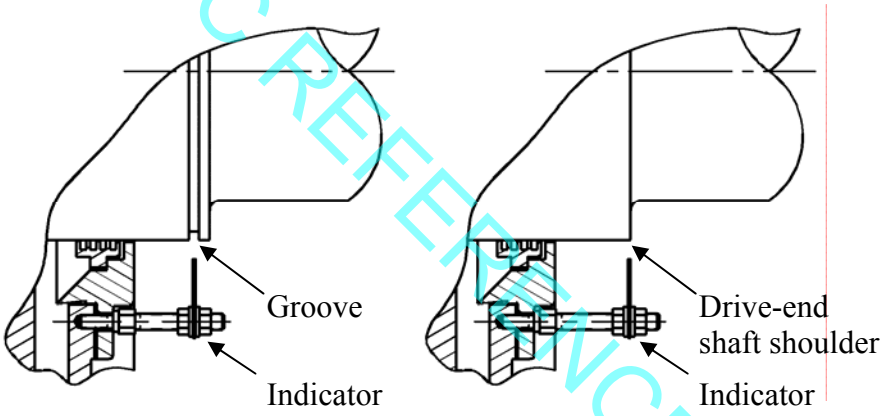

Fig. 15

5.6.2 Regular examination

(a) Periodical change of oil

The oil reservoirs of self (not flood) lubricated bearings should be drained and refilled about every six (6) months. More frequent changes may be needed on high-speed (3600-rpm) motors or if severe oil discoloration or contamination occurs. In conditions where contamination does occur, it may be advisable to flush the reservoir with kerosene to remove any sediment before new oil is added. Proper care must be taken to thoroughly drain the reservoir of the flushing material before refilling with the new oil.

Refill the reservoir to the center of oil sight glass with a rust and oxidation inhibited turbine grade oil. Refer to the outline and lubrication nameplate for the correct viscosity.

(b) Quantity of lubrication oil

Please refer to the lubrication nameplate for oil quantity.

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(c) Oil viscosity

ISO	Equivalents		Viscosity	
	ESSO	MOBIL	cSt，40℃	SUS/100℉
VG32	ESSO TERESSO 32	MOBIL DTE OIL LIGHT	32	150
VG46	ESSO TERESSO 46	MOBIL DTE OIL MEDIUM	46	200
VG68	ESSO TERESSO 68	MOBIL OIL HEAVY MEDIUM	68	300
VG150	ESSO NUTO 150	MOBIL DTE OIL EXTRA HEAVY	150	700

5.6.3 Disassembly

Prior to disassembling, ensure the power supplies are disconnected and there are no moving parts.

The bearing sleeve is of the spherically seated, self-aligning type. The opposite drive end bearing is normally insulated for larger motors (or when specified). On some motors, the insulation is bonded to the spherical seat of the bearing housing.

ATTENTION！

Extreme care must be exercised in removing the bearing sleeve from the insulated support to avoid damaging this insulation.

The following is the recommended procedure for removing the bearing sleeve:

(a) Remove the oil drain plug in the housing bottom and drain the oil sump.

(b) Remove all instrumentation sensors that are in contact with the bearing sleeve. These would include resistance temperature detectors, thermocouples, thermometers, etc.

(c) Remove the socket head bolts holding the bearing cap and the inner air seal. The end cover plate must also be removed if the non-drive end bearing is being disassembled. Remove the bearing cap and top half of the inner air seal. Place them on a clean, dry surface to avoid damage to the parting surfaces.

(d) Remove the top half of the bearing sleeve using suitable eyebolts in the tapped holes provided. Lift the bearing top straight up and avoid any contact with the shoulders of the shaft journals that might damage the thrust faces of the bearing. Place on a clean, dry surface taking care to prevent damage to either the parting surfaces or the locating pins that are captive in the top bearing half.

(e) Remove the screws at the partings in the oil ring and dismantle the ring by gently tapping the dowel pin ends with a soft face mallet. Remove the ring halves and immediately reassemble them to avoid any mix up in parts or damage to the surface at the partings.

(f) Pulls up on the garter spring surrounding the floating labyrinth seal and carefully slip out the top half. Rotate the garter spring until the lock is visible. Twist counter-clockwise to disengage the lock, remove the garter spring, and then rotate the lower half of the seal out of the groove in the bearing housing. Note the condition of these floating labyrinth seals. If they are cracked or chipped, they must be replaced. Do not attempt to reuse a damaged seal.

(g) To remove the bottom bearing half, the shaft must be raised a slight amount to relieve pressure on the bearing. On the drive end, this can be done by jacking or lifting on the shaft

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extension. Protect the shaft. On the non-drive end, jacking or lifting can be done using bolts threaded into the tapped holes provided in the shaft end.

(h) Roll the bottom bearing half to the top of the shaft journal and then lift it using suitable eyebolts threaded into the holes provided. Again avoid any contact with the shaft shoulders that could damage the bearing thrust faces. Place the lower bearing half on a clean, dry surface to protect the parting surfaces.

⚠ **Use extreme care when rolling out the lower bearing half. Keep the hands and fingers well clear of any position where they might be caught by the bearing half if it were accidentally released and rotated back to its bottom position. Serious personal injury could result.**

(i) Protect the shaft journal by wrapping it with clean, heavy paper or cardboard.

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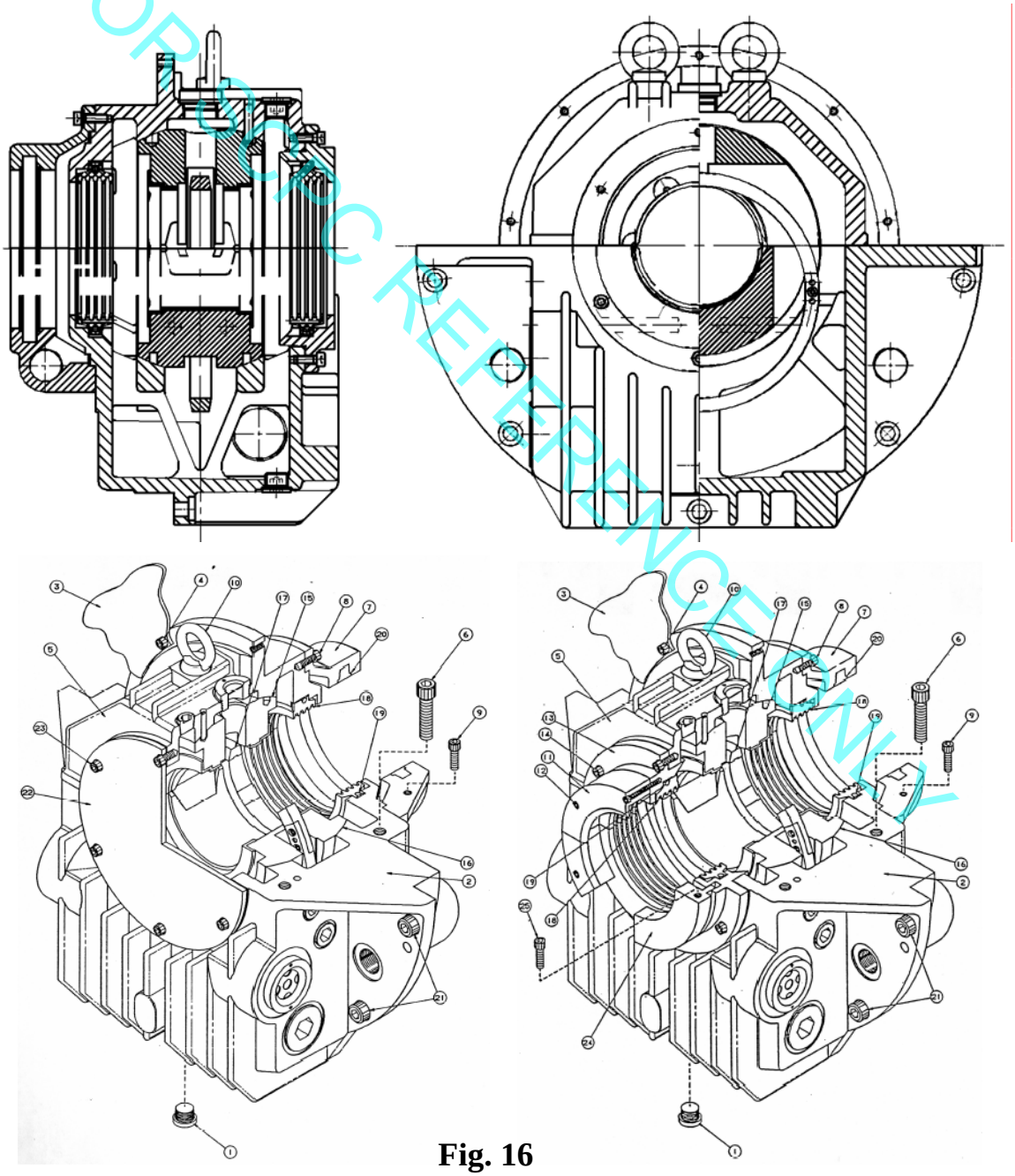

Fig. 16

5.6.4 Reassembly

Bearing reassembly is basically a reverse of the disassembly procedures outlined above, with the following suggestion:

(a) The interior of the bearing housing should be cleaned and then flushed with clean oil or kerosene.

(b) The bearing halves and the shaft journal should be wiped clean using lint-free cloth soaked with clean oil.

(c) All parts should be carefully inspected for nicks, scratches, etc., in any contact surfaces. Such imperfections should be removed by an appropriate method such as stoning, scraping, filing, etc., followed by thorough cleaning.

(d) Before installing the floating labyrinth seal halves, observe their condition. Do not attempt to use a cracked or chipped seal. The bottom half seal has a set of drilled holes in its side face. These must be placed at the bottom toward the inside of the bearing so that accumulating oil may drain back into the housing.

(e) Put a bead of Curil-T around the seal half O.D.'s on both sides adjacent to the garter spring groove. This will prevent oil by-passing the seal around its outside.

(f) Place the bottom seal half on top of the shaft and roll it into position. Install the top half and insert the garter spring pulling up on both ends to permit engaging the lock. Run a bead of Curil-T around the O.D.'s on both sides adjacent to the garter spring groove on this half also.

(g) Carefully reassemble the two oil ring halves. Inspect the dowel pins for burrs and straightness and make any corrections required. Do not force the ring halves together. Excessive force may alter the roundness or flatness of the oil ring which can change its oil delivery performance.

(h) Some of the pipe plugs in the housing are metric thread type. These are identified as those which have a copper, lead, or similar material washer. If these plugs are removed, be careful not to lose the washers. Before reassembly, inspect the washers and replace them as required.

(i) Before installing the bearing cap, observe the position of the floating labyrinth seal. The "tab" must be on top to engage the pocket. Failure to position the seal properly will result in damage when the cap is assembled.

ATTENTION !

(1) Curil-T is the only approved compound for use in the assembly of the bearings on this motor. Other products may harden and impede the operation.

(2) During the reassembly of the bearing parts, a thin layer of Curil-T should be applied to all gaskets and machined interface surface. This suggestion does not apply to the machined surfaces of the bearing liner halves.

(3) When seating the bearing shell, apply a thin layer of lube oil at the spherical surface of the liner. Slowly roll the lower bearing liner into the bearing housing making sure that splitted surface of the liner and the housing are flush. Gradually lower the shaft onto the bearing. The weight of the shaft will help rotate the bearing liner so that the babbitt surface of the liner will match the slope of the journal. Sometimes it is required to use a rubber mallet to tap lightly on the bearing housing while slowly rolling the shaft to help this seating operation.

FOR SCR REFERENCE ONLY

5.7 Maintenance of slip ring (For Wound Rotor only)

Ensure motor is disconnected from power supplies and there are no accessible moving parts before maintenance operation.

5.7.1 Adjustment of carbon brush

(a) Brush pressure for normal operation：

- Electro-graphite brush…200 g/cm^2 ±10%

  When frequent vibrations are evident or the brush is small (area below 0.5 cm^2), the pressure should be greater than as shown.

(b) Adjustment of brush pressure：

The brush pressure should be adjusted to keep normal operation as it wears.

- The brush pressure may be reduced after use, so it is necessary to re-adjust. For adjustment, please turn adjusting screw, pressure adjusting pin or pressure adjusting plate as shown in Fig.17 to obtain the correct tension (=0.23×brush cross sectional area in cm^2) ±10% kg.

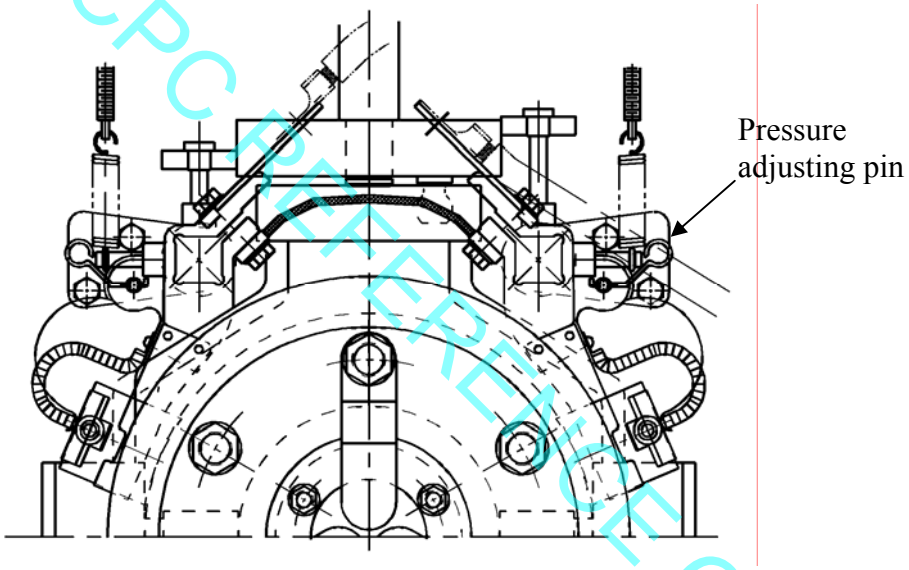

Fig. 17

(c) Brush pressure need not be adjusted if constant force spring is used as shown in Fig.18 and Fig. 19.

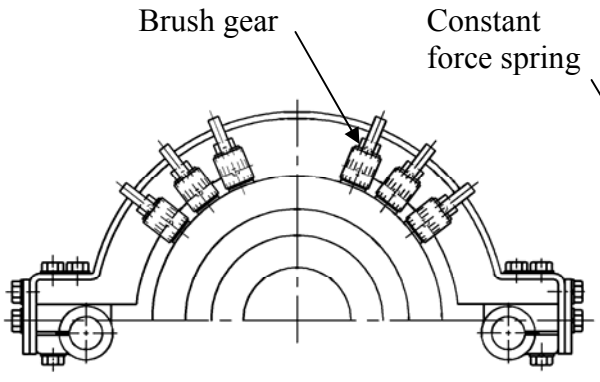

Fig. 18

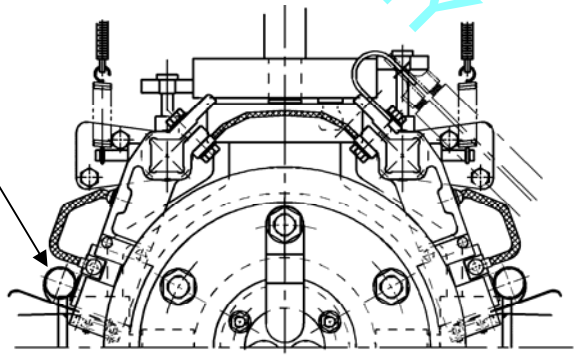

Fig. 19

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5.7.2 Brush replacement

The carbon brush is a part of the equipment which easily worn away, replace it after it is worn to 1/2~3/5 of original size.

(a) Brush material

The brush material is important to the performance of the motor. Only the most appropriate materials are chosen by TECO, and are listed on the nameplate of motor. It is important to know this when you replace the brush, so a recommended type is used.

(b) Dimensions

Brush, holder and gap between them, please refer to CNS 2322 C4051 or JIS C2802.

ATTENTION !

The gap between a brush and its holder is important for good performance and safety of the motor.

(c) Adjustment of new brushes (Shown in Fig.20)

(1) Polish the new brush with a file until it assumes the appropriate contour of the slip ring which it touches.

(2) Place sandpaper (JIS R6252 NO.40~...50) on the slip ring with the abrasive face of the paper against the brush to induce a closer contact by rubbing against each other.

(3) Repeat item 2 with fine sand-paper (JIS R6252 NO. 100 to 200) until the contact surface between brush and slip ring exceeds 80%.

(4) Finally, clean the contaminated slip ring and brush with clean cloth or compressed air.

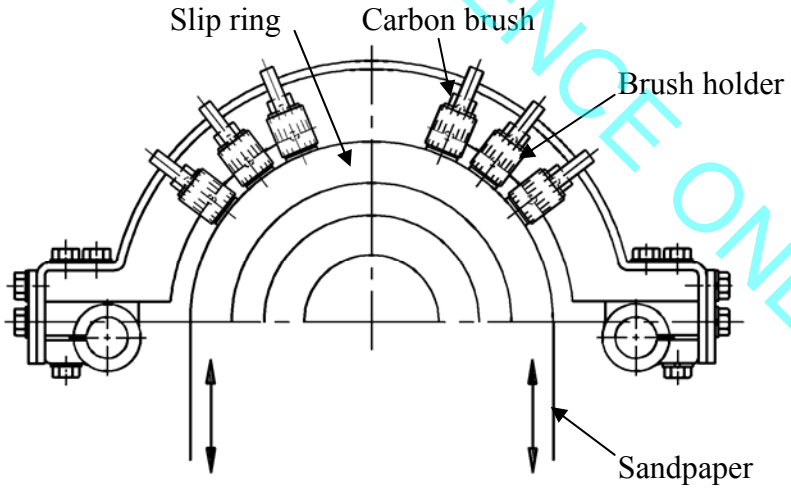

Fig. 20

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5.8 Maintenance of non-reverse ratchet mechanism (For Vertical high Thrust Motor only)

5.8.1 Non-reverse ratchet mechanism

In the pump piping system, a check valve and a stop valve should be installed in the discharge line. The check valve, placed between the pump and the stop valve, is to protect the pump from reverse flow and excessive back pressure. The stop valve is used in priming, starting and when shutting down the pump. It is advisable to close the stop valve before stopping the pump. This is especially important when the pump is operated against a high static head.

TECO vertical high thrust motors are equipped with non-reverse ratchet (N.R.R.) mechanism only when requested by the pump manufacturer. Typical construction of N.R.R. mechanism is shown as Fig.21 below.

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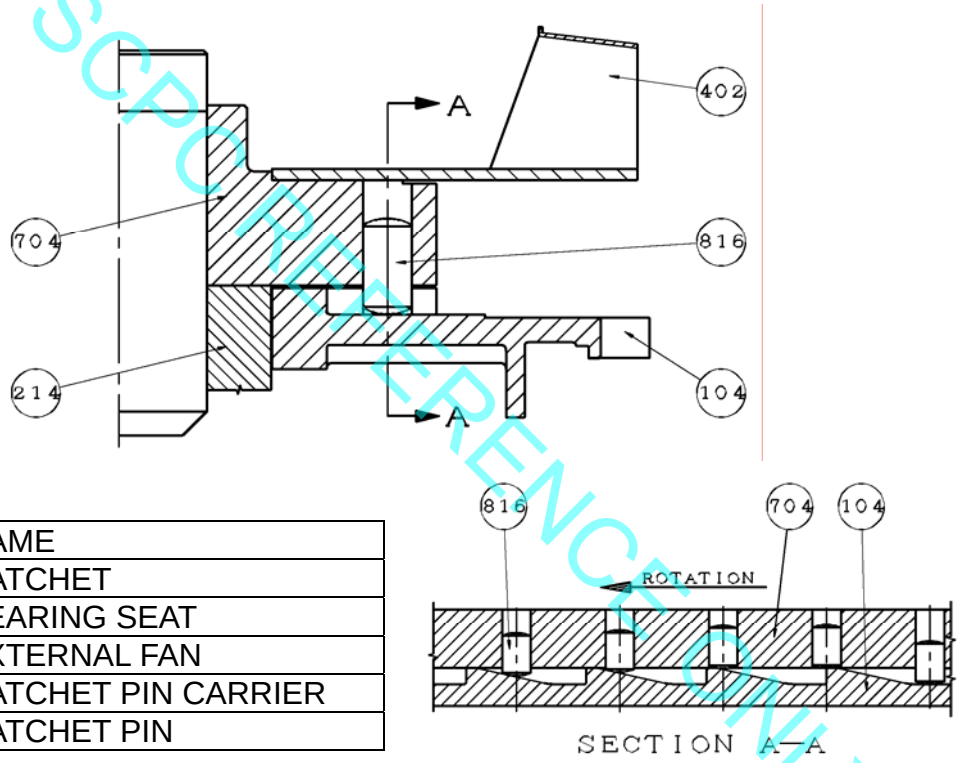

ITEM	NAME
104	RATCHET
214	BEARING SEAT
402	EXTERNAL FAN
704	RATCHET PIN CARRIER
816	RATCHET PIN

Fig. 21

The N.R.R. mechanism keeps the pump and motor from rotating in the reverse direction. Thus prevents damage from over speeding and damage to water-lubricated pump shaft bearings when, on shutdown, the falling water column tends to drive the pump in the reverse direction.

In normal operation, the ratchet pins are lifted by the ratchet teeth and are held clear by centrifugal force and friction as the motor comes up to speed. When power is removed, the speed decreases and the pins fall. At the instant of reversal, a pin will catch in a ratchet tooth and prevent backward rotation.

5.8.2 Service life

The service life of ratchet pins depends not only on the reverse shock load between the pin and ratchet tooth when pump stopped but also the frequency of pump starting and stop in application.

Provided that the pins are deformed due to thus reverse shock load, then the up and down motion of ratchet pins could be sluggish or jammed and that unusual noises shall arise.

The recommended replacement period for these ratchet pins is every three (3) years. If the reverse shock load is greater than 30% of motor rated torque or the starting frequency is more than twice per day, then the replacement period is to be halved.

ATTENTION！

The check valve and stop valve in the discharge line should be regularly inspected and maintained to assure the normal function of these valves. This is important to protect the pump and motor from damage and increase the service life of N.R.R. mechanism.

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6. FAULT FINDING AND RECOGNITION

Kinds of Breakdown	Symptoms	Possible causes	Remedies
Fail to start without load	Motionless and soundless	Power-off	Consult power company
		Switch-off	Switch-on
		No fuse	Install fuse
		Broken wiring	Check wiring and repair
		Broken lead	Check wiring and repair
		Broken windings	Check windings and repair
	Fuse blowing. (Automatic switch trips off, slow start with electromagnetic noise)	Short circuit of circuit switches	Check circuit switches and replace
		Incorrect wiring	Check wiring according to nameplate
		Poor contact at terminals	Lock tightly
		Windings grounded	Factory repair
		Broken windings	Factory repair
		Poor contact of circuit switches	Check and repair
		Broken wiring	Check and repair
		Poor contact of starting switches	Check and repair
		Short circuit of starting switches	Check and repair
		Incorrect connections of starting switches	Connect according to nameplate
Loading after start	Fuse blowing. Fail to restart due to trip-off of automatic switch	Insufficient capacity of fuse	Replace fuse if wiring permits
		Overload	Lighten load
		High load at low voltage	Check circuit capacity and reduce load
	Overheating motor	Overload or intermittent overload	Lighten load
		Under-voltage	Check circuit capacity and power source
		Over-voltage	Check power source
		Ventilation duct clogged	Remove the foreign matter in the duct
		Ambient temperature exceeds 40°C	Correct insulation class to B or F, or lower ambient temperature.
		Friction between rotor and stator	Factory repair
		Fuse blown (Single-phase rotating)	Install the specified fuse
		Poor contact of circuit switches	Check and repair
		Poor contact of circuit starting switches	Check and repair
		Unbalanced three-phase voltage	Check circuit or consult power company

FOR SCHOOL REFERENCE ONLY

Kinds of Breakdown	Symptoms	Possible causes	Remedies
Loading after start	Speed falls sharply	Voltage drop	Check circuit and power source
		Sudden overload	Check machine
		Single-phase rotating	Check circuit and repair
	Switch overheat	Insufficient capacity of switch	Replace switch
		High load	Lighten load
	Bearing overheating	High belt tension	Adjust belt tension
		Slack belt tension	Adjust belt tension
		Misalignment between motor and machine shafts	Re-align
		Over speed of bearing outer-ring	Adjust bracket
		High bearing noise	Replace the damaged bearing
Noise	Electromagnetic noise induced by electricity	Occurrence from its first operation	May be normal
		Sudden sharp noise and smoking	Short circuit of windings Should be repaired at factory
	Bearing noise	Noise of low shishi or Thru-Thru	May be normal
		Kala-Kala as result of poor lubrication	Grease
		Kulo-Kulo as a result of deteriorated grease	Clean bearing and grease
		Sa-Sa or larger noise	Replace the damaged bearing
	Mechanical noise caused by machinery	Loose belt sheave	Adjust key and lock the screw
		Loose coupling or skip	Adjust the position of couplings, lock key and screw
		Loose screw on fan cover	Lock fan cover screw tightly
		Fan rubbing	Adjust fan position
		Rubbing as a result of ingression of foreign matters	Clean motor interior and ventilation ducts
		Wind noise	Noise induced by air flowing through ventilation ducts
		Induced by conveyance machine	Repair machine
Vibration	Electromagnetic vibration	Short circuit of windings	Factory repair
		Open circuit of rotor	Factory repair
	Mechanical vibration	Unbalanced rotor	Factory repair
		Unbalanced fan	Factory repair
		Broken fan blade	Replace fan
		Unsymmetrical centers between belt sheaves	Align central points
		Central points of couplings do not lie on the same level	Adjust the central points of couplings to the same level
		Improper mounting installation	Lock the mounting screws
		Motor mounting bed is not strong enough	Reinforce mounting bed
		Mounting bed vibration caused by near machines	Eliminate the vibration source near motor

Remarks:

(1) Circuit switches: These include knife switches, electromagnetic switches, fuse and other connection switch etc.

(2) Starting switches: These include Delta-Star starters, compensate starters, reactance starters, resistor starters, starting controllers etc.

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CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

LMYP-1 Limay,Philippine

Vibration Sensor Data Sheet For Fan

風機震動感測器資料

本文編號 RF2-0580-DOC-4-0

						CERTIFIED	
						PROJ.	
						MGR. _____ DATE _____	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2013/11/20	PROJECT NO.	PAGE
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	**LMYP-1**	1

MARKER: MARTRIX

ST5491E-022-020-00

1. S.A.FAN : 4Pcs/Set 8 Sets
2. P.A.FAN : 4Pcs/Set 8 Sets
3. I.D.FAN : 4Pcs/Set 8 Sets

FOR SCPC REFERENCE ONLY

MODEL ST5491E
2 WIRE SEISMIC
VIBRATION TRANSMITTER

Installation Manual M9344

FOR SOLD REFERENCE ONLY

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Houston, TX 77064, USA
Tel: 1-281-940-1802
After Hours Technical Assistance: 1-713-702-8805
Fax: 1-713-559-9421
E-mail: info@metrix1.com
www.metrix1.com

Revision: F

- **Provides 4-20 mA signal proportional to velocity vibration**
- **Direct connection to PLC, DCS or other recording monitor**
- **Two wire, loop powered, simplifies field wiring**
- **Different mounting stud types, see table (page 3)**

The Model ST5491E Seismic Vibration Transmitter is a new version of an established series of transmitters from Metrix. The ST5491E is the indicating version of Metrix Model ST5484E and comes with an integral 2-1/2 digit LCD indicator for local vibration readout. It incorporates state-of-the-art electronic components and circuits. It also offers several new options in terms of features and configurations. Be sure to study the configuration of the transmitter before putting it in service.

The transmitter combines a vibration sensor and signal conditioner in a single package. They provide the ideal solution for sensing machinery vibration level and transmitting a proportional 4-20 mA signal directly to a PLC, DCS, monitor, or computer. Typical applications include low and medium speed machines, such as fans, blowers, pumps, motors, and centrifuges. The transmitter can be supplied with optional screw cover elbow fittings for connecting flexible conduit.

The transmitter is all solid state, has no moving parts, and is encapsulated in a stainless steel housing. The LCD indicator is packaged in a sealed module that can be positioned in 90° increments with the elbow fitting to allow proper viewing when the transmitter is installed. Each transmitter is factory calibrated to the sensitivity marked on the label.

Verify Transmitter Configuration

Be sure to determine the transmitter configuration before installing it. See inside for configuration details.

ST5491E-XXX-XXX-XX
1st XXX denotes measurement range
2nd XXX denotes mounting type, area use rating, and top connection (including signal output option)
3rd XX denotes optional filtering

Specifications:

Vibration Range: 4 to 20 mA output proportional to velocity. Refer to “How to Select A” for ranges. Nonstandard ranges available.

Accuracy: 2%

Indicator: 2 ½ digits, ips or mm/s

Frequency Response: Standard: 2 - 1500 Hz, available up to 2000 Hz Refer to “How to Select E/F”. 12 dB / oct high pass and low pass filters are used.

Axis Orientation: Any

Supply Voltage (Vs): 13 to 30 VDC

Isolation: 500Vrms, circuit to case

Electrical Connection:
Flying leads - 457 mm (24 in.) length, 18 AWG

Maximum Load Resistance (RL):
RL = 50 x (Vsupply-13) ohms

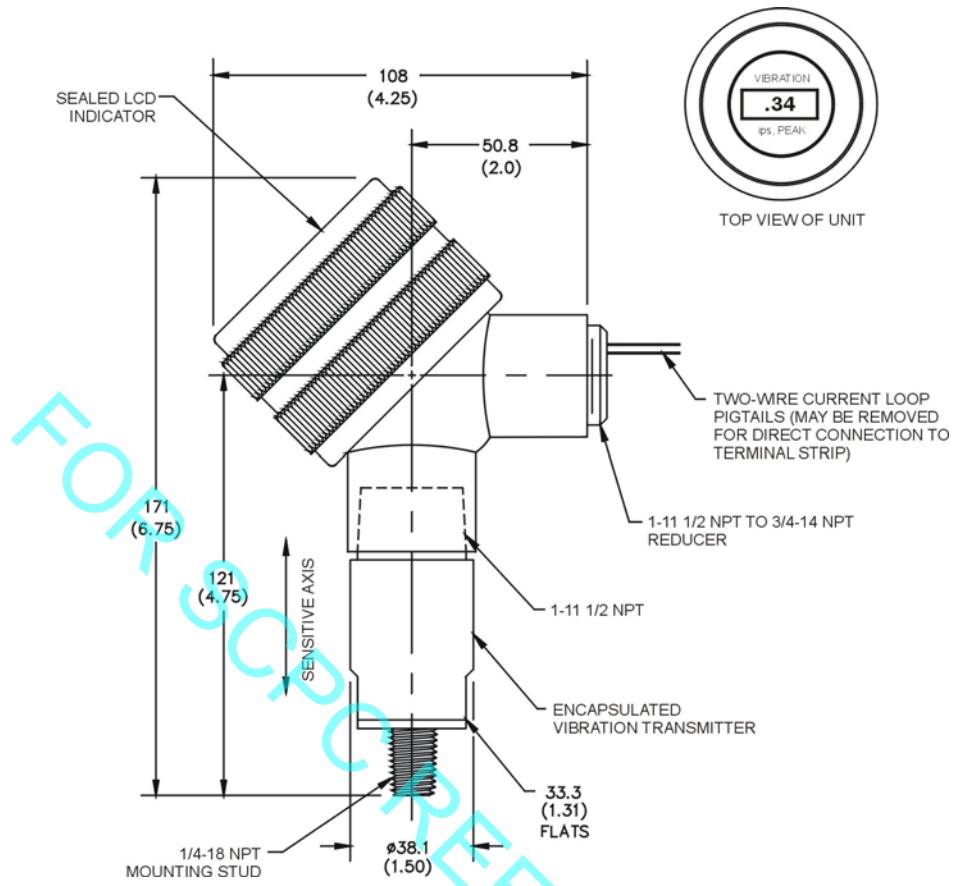
Service Temp. Rating:
-10°C to +70°C (+14°F to +156°F)

Enclosure Materials: 303 SST/316 SST

Enclosure Environmental Rating: NEMA 4X, IP 67

COPY FOR REFERENCE ONLY

OUTLINE AND DIMENSIONS



ST5491E CONFIGURATION

A B C D E F

ST5491E - □□□-□□□-□□

A - Full Scale

0	2	1	= 1.0 ips (25mm/s), pk
0	2	2	= 20.0 mm/s, pk
0	2	3	= 100mm/s, pk
0	2	4	= 2.0 ips (50mm/s), pk

B - Mounting

0	= Integral 1/4" NPT
1	= Integral 1/2" NPT
2	= 3/8-24 UNF X 1/2"
4	= M8 X 1-12
5	= M10 X 1.25-12

For 316 SST Housing Material, add 10 to dash number.
Ex: -0 for 1/4 NPT (303 SST)
-10 for 1/4 NPT (316 SST)

C - Hazardous Area Rating

1	= Non-hazardous & CSA/NRTL/C (for all connections) Class I, Div 2, Grps C & D
2	= CSA/NRTL/C for Class I, DIV 1, Grps C-D & Class II, DIV 1, Grps E-G (available with flying leads only)

D - Connection

0	= 4-20 mA; Flying leads (C = 1 or 2)

E - HP Filter

0	= No Filter (2 Hz)
1	= 5 Hz
2	= 10 Hz
3	= 20 Hz
4	= 50 Hz
5	= 100 Hz
6	= 200 Hz

F - LP Filter

0	= No Filter (1500 Hz)
1	= 500 Hz
2	= 1000 Hz
3	= 2000 Hz

The complete Model Number, showing the configuration and serial number, is etched onto the body of the Transmitter.

FOR SCPC REFERENCE ONLY

INSTALLATION

1. MOUNTING It is important the transmitter body be solidly mounted to the machine surface. See the section on transducer placement. Different machine preparations are required for the two basic transmitter mounting styles; NPT (National Pipe Thread) and machine thread (UNF and Metric). Transmitters with the NPT type mounting stud are secured by the thread engagement and the base of the transmitter does not come in contact with the machine surface. On the other hand, transmitters with the machine thread studs must make contact with the machine surface. The base of the transmitter must make square and direct contact. This requires preparing the surface of the machine with a 1 1/2 inch counter bore (surfacing tool). This tool can be used with a portable drill equipped with a magnetic base but care must be taken so that the tapped and threaded hole is perpendicular to the machined surface. The transmitter must make contact all the way around its base surface. Contact Metrix for more detailed instructions for using a counter bore.

If installing a transmitter with a standard 1/4 inch NPT stud, drill a hole using a 7/16 inch bit, 5/8 inch deep. Then tap using a 1/4 - 18 NPT (tapered pipe tap). Hand tighten the transmitter and then turn an additional 1 to 2 turns using a wrench on the wrench flats. Do not use a pipe wrench. A pipe wrench can apply extreme forces to the body and potentially damage electronic components. A minimum of five (5) threads of engagement should be made. A 1/4 inch to 1/2 inch NPT bushing is available for mounting the transmitter in existing 1/2 inch NPT holes. Also, a Model 7084 Flange Adapter can be used between the transmitter and the machine surface when there is not enough surface thickness to drill and tap a hole. The flange adaptor mounts with three small screws. See accessories.
If installing a transmitter with one of the straight machined thread sizes, follow standard drill and tap procedures. Be sure to not drill a hole larger that the counter bore pilot diameter before using the counter bore to prepare the machine surface. Drill out the hole with the correct tap drill size after preparing the surface.

The sensitive axis of the transmitter is in line with the mounting stud. The transmitter can be oriented in any (0 to 360 degree) position.

2. WIRING The ST5491E is connected like any other loop powered transmitter. The following is a summary based on area designations.

2.A SAFE AREA INSTALLATION Connect the field wiring in accordance with the appropriate diagram on page 4. The ST5491E transmitter requires a minimum of 13 VDC for proper operation. The minimum loop power supply voltage required is 13 VDC plus 1 volt for each 50 ohms of loop resistance.

ST5491E Example:

Signal wire resistance	= 10 ohms
Resistance of receiver	= 250 ohms
Total loop resistance	= 260 ohms

Minimum supply voltage = 260 (1V/50 ohms) + 13V = 18.2 VDC

The maximum loop power supply voltage that may be applied is 30 VDC. The maximum loop resistance (RL) is calculated by the equation:

RL = 50 (VS - 13) ohms
Example: RL = 50 (24 - 13) = 550 ohms for 24 VDC loop supply.

2.B EXPLOSION-PROOF INSTALLATION IN HAZARDOUS LOCATIONS ST5491E transmitters are CSA certified explosion-proof, Class I (C & D); Class II (E, F & G,). Connect the field wiring in accordance with the appropriate diagram on page 4. Refer to paragraph 2.A for loop voltage and resistance requirements. All conduit and junction boxes used must be certified explosion-proof for the class, division and group required by the application. Installation of the transmitter must meet all of the explosion-proof installation requirements of the local governing agency and facility safety procedures.

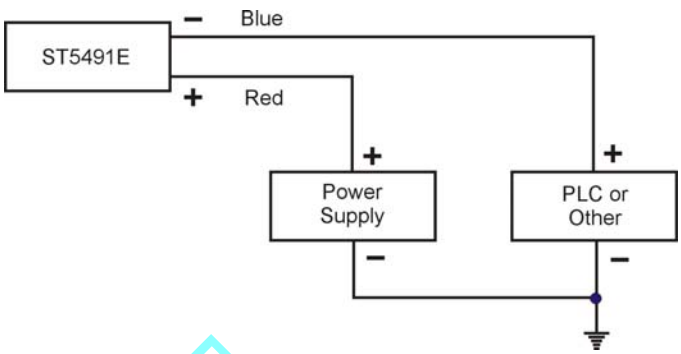
3. ELECTROMAGNETIC COMPATIBILITY In order to meet the requirements of CE electromagnetic compatibility in areas of high electromagnetic interference, the field wiring must be:

1. shielded twisted pair cable enclosed in grounded metallic conduit, or

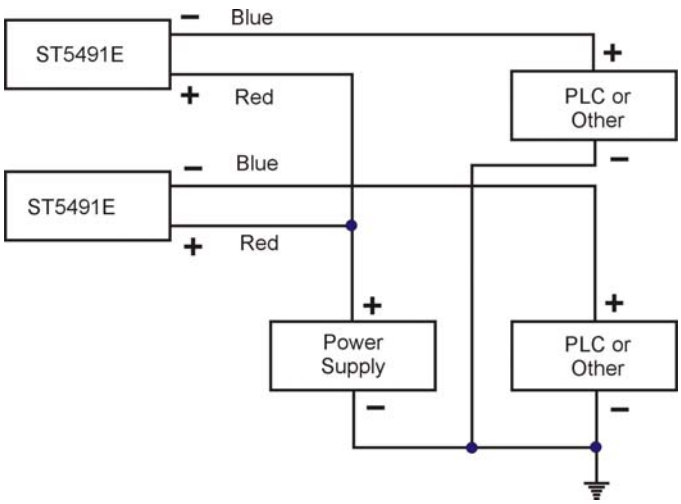
2. double shielded twisted pair cable with a metallic body cable gland fitting and with the outer shield grounded.

FOR SCPC REFERENCE ONLY

WIRING DIAGRAMS



Single Transmitter



Multiple Transmitter

FOR SCD REFERENCE ONLY

Use standard two-conductor, twisted pair, shielded wiring for the long run to the instrumentation enclosure. The transmitter is connected like any other loop powered end device.

CONNECTION TO PLC OR OTHER INDICATING INSTRUMENT

The first step in configuring the PLC, DCS, or other recording instrument is to determine the source of power. The ST5491E requires loop power be supplied to it. Some Analog Input channels, on a PLC or DCS, for example, provide this power from within. If they do not provide power, an external power supply must be provided. Connect the transmitter field wiring using standard instrumentation practices.

Scaling of the display is on the basis of the range of the transmitter. The measurement parameter name is "vibration" and the units are "ips" (inches per second) or "mm/s" (millimeters per second). The example below is based on a standard 1.0 ips transmitter.

Vibration Level	Transmitter Output	PLC (or other) Scale
None	4mA +/-0.1mA	0.00 ips
1.0 ips	20mA +/-0.4mA	1.00 ips

In order to prevent nuisance alarms, due to transient operating conditions, program a time delay into the alarm. Momentary "jolts" that can occur at start-up or during some operating condition changes, do not reflect a machines steady state operating condition. The indicated vibration level must cross the threshold level and stay above it for a preset time before any alarm action is taken. A 2 to 3 second delay is normally applied to most machinery. Consult Metrix if you have a question about your machines operating characteristics.

Some rough starting machinery may also need a start-up time lockout for alarms. A start-up lockout is different than a time delay! Both may be needed.

TYPICAL TRANSMITTER PLACEMENT

The ST5491E measures seismic vibration at the attachment point on the machine. Its sensitive direction is through its cylindrical body. It will not measure side to side motion. It measures velocity vibration in ips (inches per second). Velocity vibration is a common measurement for a large variety of machinery.

Typical transmitter mounting for casing vibration measurements is in the horizontal direction at the bearing housings. Usually the horizontal direction is more sensitive because of the way machinery is typically mounted. Figure 2 shows the flying leads being spliced in the conduit elbow. Please be aware the splice can not be made at this point if in a hazardous area. In this case, make the splice in a second conduit hub (meeting splicing requirements) at the end of the flexible conduit.

TYPICAL TRANSMITTER PLACEMENT (con't)

The standard lead length is 24 inches. We recommend using the "Y" conduit elbow because it prevents the conduit from extending too far away from the transmitter, limiting the likelihood of getting snagged while working near the machinery. If a 1 inch to 3/4 inch reducer is used at the elbow, a smaller diameter flexible conduit can be used. It is always a good idea to limit the mass that "hangs" on the transmitter. The flexible conduit should make a bend in order to be sure any rigid conduit motion can not feed into the transmitter body.

TYPICAL APPLICATION AND MOUNTING ARRANGEMENT

Figure 1

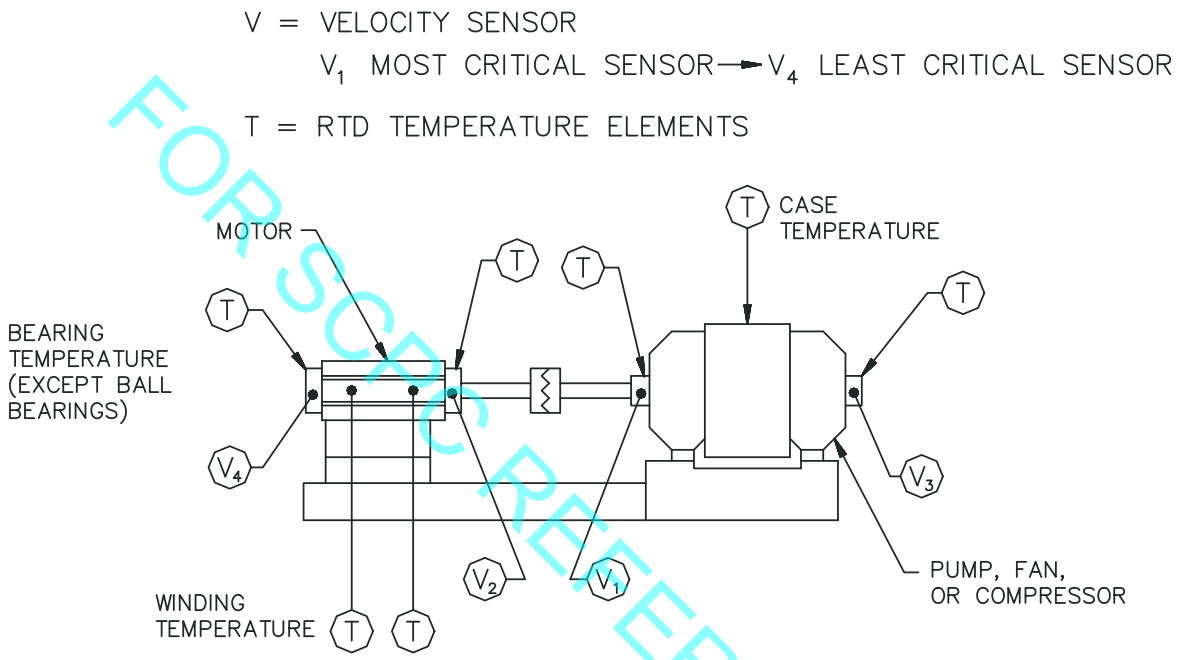
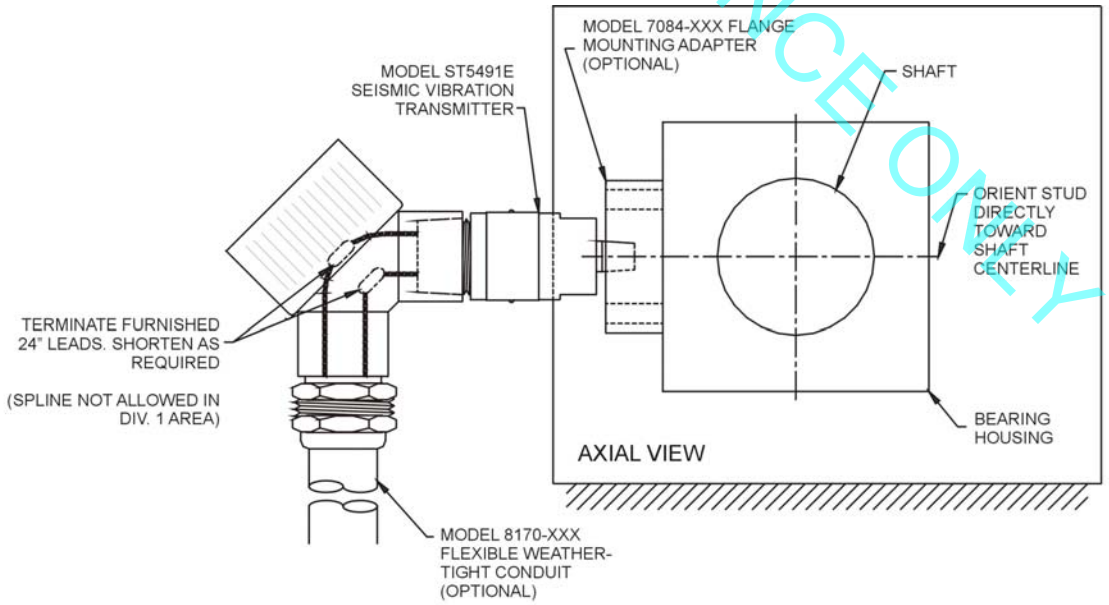


Figure 2



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CALIBRATION

The ST5491E transmitter has been factory calibrated for the full scale vibration level marked on the label. If the calibration is in doubt the unit can be verified in the field by following the procedures outlined below. Note that there are no Zero and Span adjustments on the transmitter.

1. Zero Verification In the absence of vibration the output current should be 4 mA ±0.1 mA. If the ambient vibration is more than 2% of full scale, the transmitter should be removed from the machine and placed on a vibration free surface for this measurement. Often a piece of foam can be used to isolate the transmitter from external motion.

2. Span Verification Subject the transmitter to a known vibration within the full scale range marked on the label. If using a portable vibration shaker where it can be tested at full scale, the output should be 20 mA ±0.4 mA. If using another reference, like a similar machine, calculate the current value on a proportional basis.

The output level can also be compared to a vibration level read on a portable vibration meter. This reading should be taken as close to the transmitter location as possible.

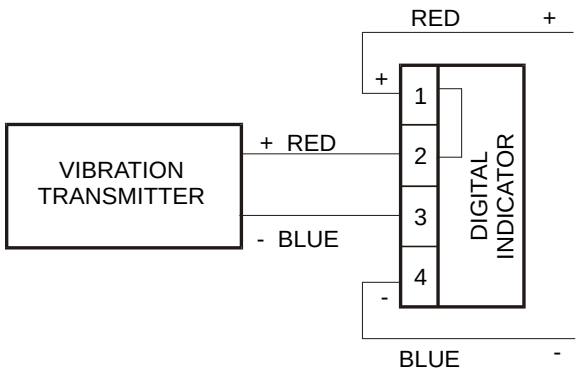
If you shake the transmitter by hand you can produce a high output as long as there are no high pass filters. Check the configuration for filters. There is no way of knowing the expected output when shaking it by hand.

3. Digital Indicator Calibration of the Digital Indicator requires partial disassembly of the ST5491E. Remove the Digital Indicator threaded cover. Carefully remove the electronics module using a small flat head screwdriver. Note how the transmitter wires are connected before disconnecting them. Remove the retaining wire from the bottom of the module and carefully remove the electronics board assembly from the module housing. Connect the electronics module is series with a current loop that can be accurately set from 4.0 mA to 20.0 mA +/- 0.1 mA. NOTE: IF A VOLTAGE SOURCE IS USED TO GENERATE A CURRENT, A LOAD RESISTANCE MUST BE CONNECTED IN SERIES WITH THE ST5491E INDICATOR MODULE. FAILURE TO DO SO MAY DAMAGE THE UNIT.

Set the current loop to the zero current reading of the transmitter and adjust the ZERO (R7) control for a reading of .00. Next set the current loop to the full scale current value and adjust the SPAN (R6) control for the proper indication. Alternately adjust the ZERO and SPAN controls as adjustments interact.

Label Full Scale	Reading
1.0 ips	1.00
19.9 mm/s	19.9*
100 mm/s	100

After calibration, reassemble the module, reconnect the wiring, and reassemble the unit.



FOR SCHOOL REFERENCE ONLY

OPTIONAL MOUNTING & CONNECTION

The ST5491E is available with different mounting options. The performance of the transmitter does not change (except for the dynamic output option, discussed).

OPTIONAL MOUNTING ACCESSORIES

8201-001, Conduit Union
Fits between transmitter and 8200-001 conduit elbow to facilitate installation and wiring where there is not enough room to rotate the elbow. Suitable for Class I, Div. 1 (Grps A, B, C & D) and Class II, Div. 1 (Grps E, F & G), hazardous areas. Material: zinc plated steel.

7084-001, Stainless Steel Flange Mount Adaptor
Provides a means to surface mount transmitters rather than NPT stud (½": NPT center hole). Three equally spaced 6.6 mm (0.26') diameter mounting holes on 38 mm (1.50") diameter circle.
7084-002, Flange Mount Adaptor
Same as 7084-001, except center hole is ¼" NPT. Material: stainless steel

8253-002, Bushing
Bushing for ½" NPT mount when screwed onto standard ¼" NPT base. Material: stainless steel.

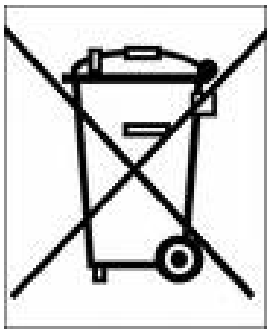
OPTIONAL CONNECTION ACCESSORIES

8169-75-002-XXX, Two-wire, Cable Assembly
2 conductor (20 AWG) twisted, shielded PVC jacketed cable, with plated steel grip for cable strain relief, male ¾" NPT end. Specify -XXX for length in feet. Example: 8169-75-002-010 =10 ft (3.1M). Material: zinc plated steel.

93818-004, Cable Grip Strain Relief Fitting
3/4" NPT male thread to cable grip. Diameter range: .156" to .25". Complete with sealing ring and locknut. Hot dip / mechanically galvanized finish. Suitable for NEMA 4 enclosures.

93818-018, Armored Cable Grip Strain Relief Fitting
3/4" NPT male thread to cable grip. Armor diameter range: .40" to .50". Complete with sealing ring and locknut. Hot dip / mechanically galvanized finish. Suitable for NEMA 4 enclosures.

ENVIROMENTAL INFORMATION



This electronic equipment was manufactured according to high quality standards to ensure safe and reliable operation when used as intended. Due to its nature, this equipment may contain small quantities of substances known to be hazardous to the environment or to human health if released into the environment. For this reason, Waste Electrical and Electronic Equipment (commonly known as WEEE) should never be disposed of in the public waste stream. The "Crossed-Out Waste Bin" label affixed to this product is a reminder to dispose of this product in accordance with local WEEE regulations. If you have questions about the disposal process, please contact Metrix Customer Services.

FOR SCPC REFERENCE ONLY

建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

LMYP-1 Limay,Philippine

Motor Data Sheet For Fan

送引風機馬達資料

本文編號 RF2-0580-DOC-1-0

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0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	CERTIFIED	
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C	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/05/23	MGR. _____ DATE _____	
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A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2013/11/20	LMYP-1	1

I.D.FAN

3300KW * 6P * 6600V * 3Ø * 60HZ

FOR SCPC REFERENCE ONLY

DOCUMENTATION LIST

Order No. FE14A702B1
Type AEZKS2064420Y4001 **Customer**
Total Sets 1 **User**
Issued Date JUL.22.2014 **Customer PO.** P1022461-1

Item	Drawing No.	Drawing Name	Rev	Item No.	Note
1	3A057H186-72068	Specification Table	04	--	Change
2	3A040S706	Outline Dimensions	03	--	--
3	4A040Y033X1	Primary Terminal Box Schematic Drawing	02	01	Change
4	3B040Z580	Heater Terminal Box Schematic Drawing	07	10	--
5	3A040D751	WDG & BTG RTD Common Terminal Box Schematic Drawing	28	03	--
6	4A040Y229	Shaft Schematic Drawing	01	--	--
7	3A061H403	SCHEMATIC STATOR CONNECTION	00	--	--
8	3A061H574	SCHEMATIC SPACE HEATER	00	--	--
~~9~~	~~FE14A702B1-T-I-N~~	~~T-N/I-N Curve~~	~~00~~	~~--~~	Delete
10	FE14A702B1-T-I-L	T-N/I-N VS Load Curve	00	--	Add
11	FE14A702B1-E-P-L	Eff & Pf VS Load Curve	00	--	--
12	FE14A702B1-TIME	Time-Current and Thermal Limit Curve	00	--	--
13	FE14A702B1-DS	Data Sheet	00	--	--

FOR SCPC REFERENCE ONLY

承 認 用

本承認圖，請於文到7個工作日內簽回，以利工程進行，否則視爲承認論。

承認者：
工號：

TECO ELEC. & MACH. CO., LTD.

DOC NO. FE14A702B1
REV.04 1 / 1

SPECIFICATION TABLE OF 3-PHASE SQUIRREL CAGE INDUCTION MOTOR	CUSTOMER		USER	
	INQ. NO.		EQUIPMENT	
	JOB NO.	FE14A702B1	MACHINE	
	TOTAL SETS	1	ITEM NO.	

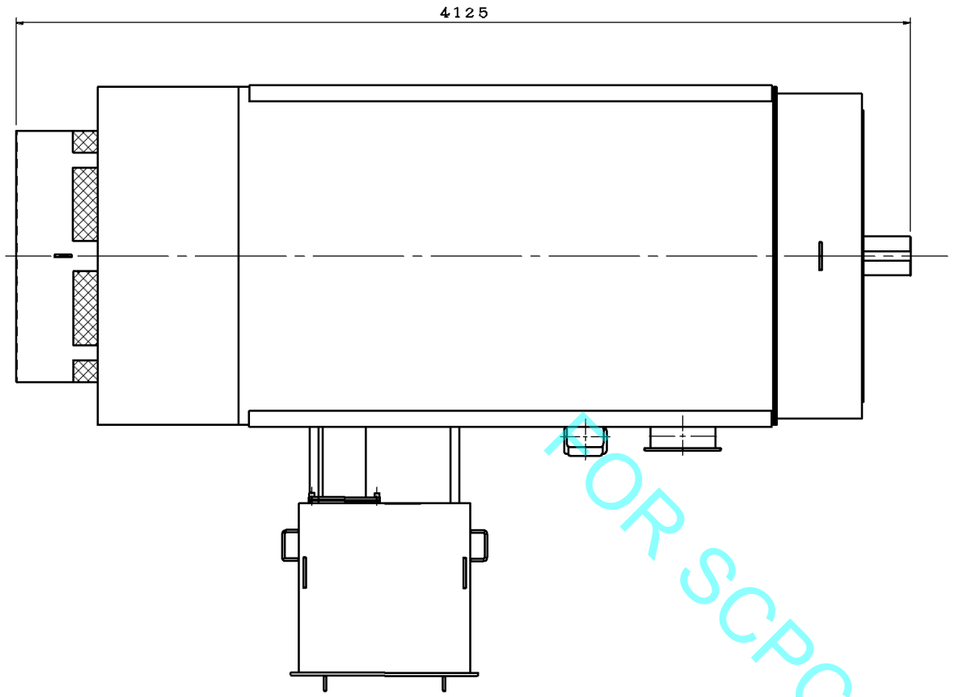
Item	Terms	Description			
1	Model	AEZK-S2			
2	Code or Standard	Dimensions	Frame Assignment	Performance	Test
		IEC	TECO	IEC	IEC
3	Rating	3300 kW 6 Pole 6600 Volt 3 Phase 60 Hz			
4	Service Duty	Continuous Rating , S.F. 1.15			
5	Starting Method	Soft Start			*
6	Rotation	Facing The Drive End : CW			
7	Drive Method	Direct Coupling			
8	Environment	Amb. Temp. : -20 ~ 40 ℃			
		Humidity : Less Than 90 %RH			
		Altitude : Up to 100 M			
9	Enclosure & Protection	IPW55 : Totally Enclosed			Outdoor
10	Cooling	IC611 : Fan Cooled With Built-in Air Cooler			
11	Mounting	IM1001 : HS, Foot			
12	Dimensions	Dr# 3A040S706 (REV.03)		Frame No ： 630D	
13	Frame & Bracket	Frame : Steel Plate		Bracket : Steel Plate	
14	Fan & Fan Cover	Fan :Steel Plate		Fan Cover :Steel Plate	
15	Terminal Box	Steel Plate			
16	Lead Terminals	(TLK95-16)X3			
17	Lubricant	Oil Viscosity : ISO VG46			
18	Painting	Color : MUNSELL 7.5GY 5/4,STANDARD TECO DR#3A057D659			
19	Stator Winding	Ins. Class F			
20	Rotor Conductor	Cu-Alloy		GD^2 ： 1074 Kg-m^2	
21	Starting Performance	LRC≦ 2275 Amp		LRT/FLT 170 %	
22	Operating Performance	Hz/V	60/6600	Break Down Torque	
		%Load	100	270 %FLT	
		Amp.	350		
		Eff.%	96.0	Temp. Rise Limit. (Res.)	
		P.F.%	86.0	Stator 90 ℃	
		R.P.M.	1193	at S.F.1.15	
23	Approximate Weight	Motor ： 15000 kg			
24	Note	1. With Space Heater : 1φ 220V 1050W 2. With Winding RTD : PT 100Ω/0℃ 6pcs 3. With Bearing RTD : PT 100Ω/0℃ 2pcs 4. With Surge Capacitor & Lightning Arrester 5. Corrosion Proof 6. Noise : Below 83dBA at 1 Meter Distance No Load 7. Vibration:Below 2.5mm/s(0-P) On Bearing Housing No Load 8. Tolerances on declared values shall be in accordance with IEC 9. Load GD^2 : 15000 Kg-m^2 Note ： [*] at the end represent revised item ◦			

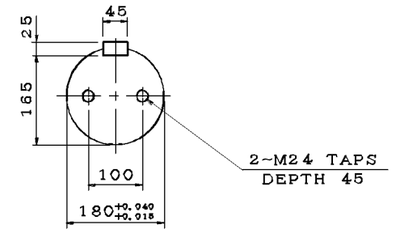
FOR SCR REFERENCE ONLY

APPD.	Kenny.Shyu	JUL. 22 2014	TECO ELEC. & MACH. CO., LTD.	DWG NO. 3A057H186-72068
CHKD.	AIYU	JUL. 22 2014		REV.04 1/1
DWN.	CF.WENG	JUL. 09 2014		

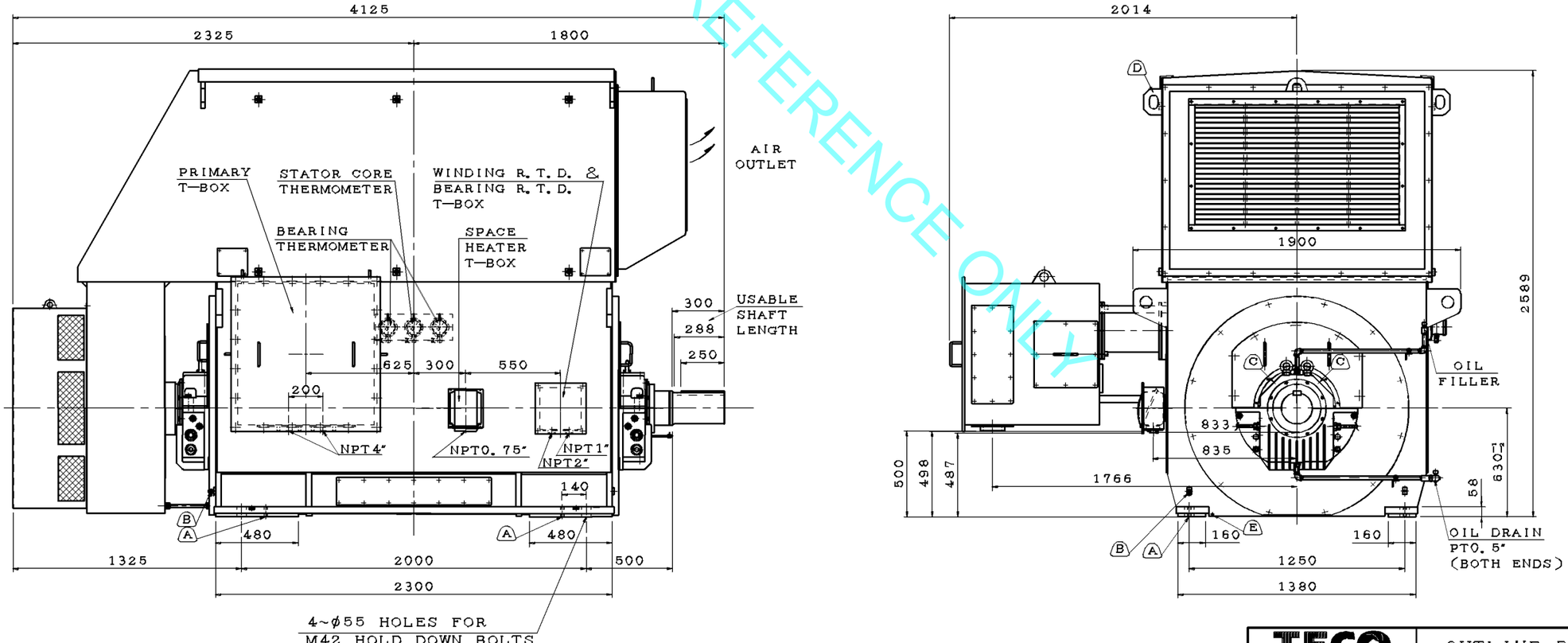
TYPE	OUTPUT		POLE	TIME RATING	VOLTAGE V	Hz	SYN. SPEED R. P. M.
	HP.	kW.					
AEZK-S2		3300	6	CONT	6600	60	1200

TOTALLY ENCLOSED AIR TO AIR COOLED TYPE. SQUIRREL CAGE ROTOR





DRIVE END
(ENLARGED VIEW)



FOR SCPC REFERENCE ONLY

DWN.	P. LIN	OCT 07 2013
CHKD.	S. WANG	OCT 11 2013
APPD.	C. SHIH	OCT 11 2013

TECO
DATE

OUTLINE DIMENSIONS
3-PHASE INDUCTION MOTOR

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040S706
REV:03

第三角法

TYPE	OUTPUT HP.	OUTPUT kW.	POLE	TIME RATING	VOLTAGE V	Hz	SYN. SPEED R. P. M.
AEZK-S2		3300	6	CONT	6600	60	1200

TOTALLY ENCLOSED AIR TO AIR COOLED TYPE. SQUIRREL CAGE ROTOR

NOTE:
1. DIMENSIONS IN mm.
2. FRAME NO. 630D
3. F CLASS INSULATION, S.F.: 1.15.
4. FOR DIRECT FLEXIBLE COUPLING.
5. SLEEVE TYPE BEARING, AXIAL THRUST LOAD NOT ALLOWED.
6. THE MOTOR ENDPLAY IS ±7 mm. A LIMITED END FLOAT TYPE COUPLING IS REQUIRED TO LIMIT ENDPLAY TO ±2.4 mm.
7. BEARING SIZE: DRIVE END: 18—180(UNINSULATED)
   NON—DRIVE END: 18—160(INSULATED)
8. THE NON—DRIVE END BEARING LINER(SHELL) IS INSULATED FROM THE HOUSING. METAL CONNECTIONS MADE TO THE BEARING SHELL MUST BE INSULATED TO PREVENT AN INSULATION SHORT CIRCUIT. METAL CONNECTIONS MADE TO THE HOUSING DO NOT NEED TO BE INSULATED.
9. BEARING LUBRICATION:SELF LUBRICATION.
   A. OIL VISCOSITY: ISO VG46 (180~220 SSU AT 100°F)
   B. OIL CAPACITY: 10.0L FOR DRIVE END.
   10.0L FOR NON—DRIVE END.
10. WITH SPACE HEATER: 1φ, 220V, 1050W.
11. WITH WINDING R.T.D.: PT 100Ω/0°C, 6PCS.
12. WITH BEARING R.T.D.: PT 100Ω/0°C, 2PCS.
13. WITH BEARING THERMOMETER:WITH N.O. CONTACTS, 2PCS.
14. WITH STATOR CORE THERMOMETER:WITH N.O. CONTACTS, 1PCS.
15. WITH SURGE CAPACITOR(6.6 kV, 3φ) 1PCS,
    LIGHTNING ARRESTER(9.0 kV, 1φ) 3PCS,
    PROVIDED IN PRIMARY T—BOX.
16. NOISE:BELOW 85dBA AT 1 METER DISTANCE ON LOAD.
17. VIBRATION:BELOW 2.5mm/s(0—P) ON BEARING HOUSING NO LOAD.
18. ENCLOSURE: IPW55, OUTDOOR SERVICE.
19. CORROSION PROOF.
20. MOTOR APPROX. WEIGHT: 15000 kg.

Ⓐ M20 VERTICAL JACKING HOLE, ONE HOLE PER FOOT.

Ⓑ FOUR M12 TAPPED GROUNDING PADS, WITH TWO SETS GROUNDING STUD &(R100—12) TERMINALS LOCATED DIAGONALLY OPPOSITE.

03 Ⓒ NPT1/4" HOLE, PROVISION FOR SUPPLY SCOPE OF CCMC.

Ⓓ FOUR LIFTING LUGS ON AIR COOLER FOR LIFTING AIR COOLER ONLY. DO NOT USE FOR LIFTING ENTIRE MOTOR.

Ⓔ TWO M10 TAPPED DRAIN HOLES, LOCATED DIAGONALLY OPPOSITE.

FOR SCPC REFERENCE ONLY

DWN.	P. LIN	OCT 07 2013
CHKD.	S. WANG	OCT 11 2013
APPD.	C. SHIH	OCT 11 2013

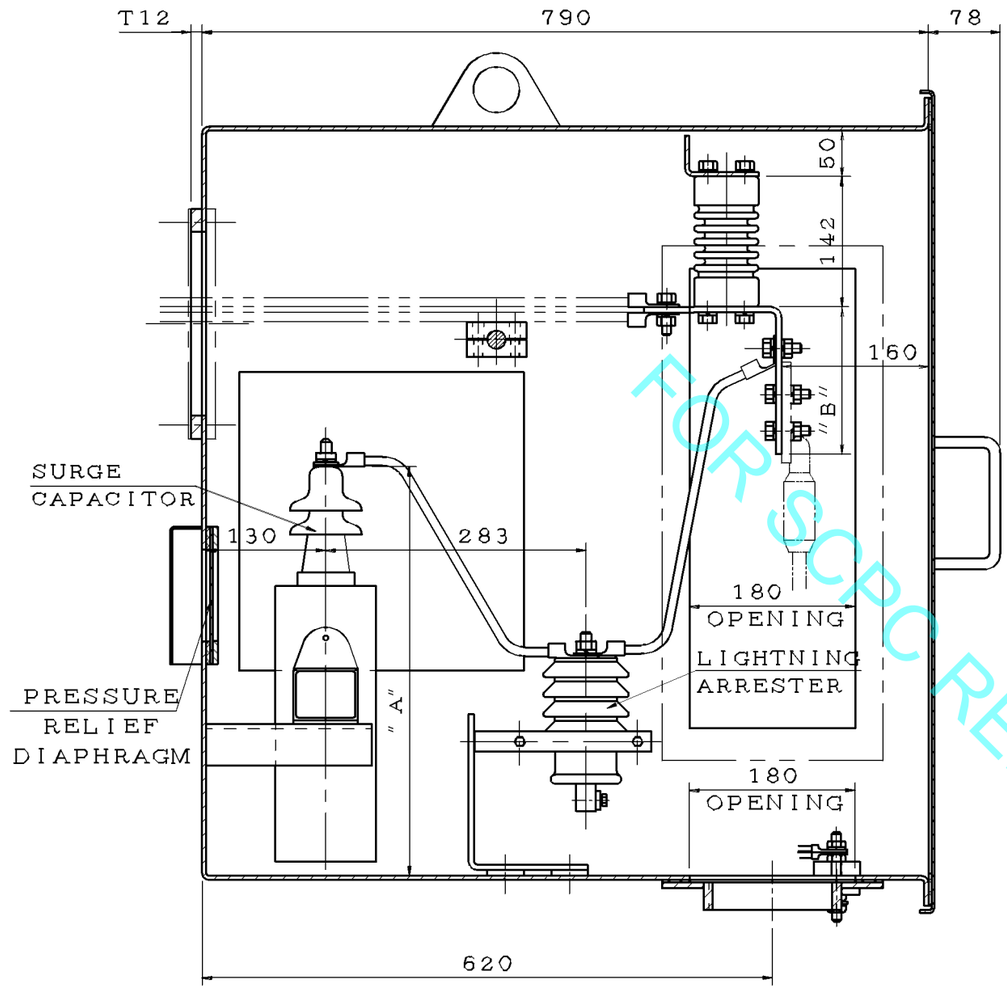
TECO
DATE

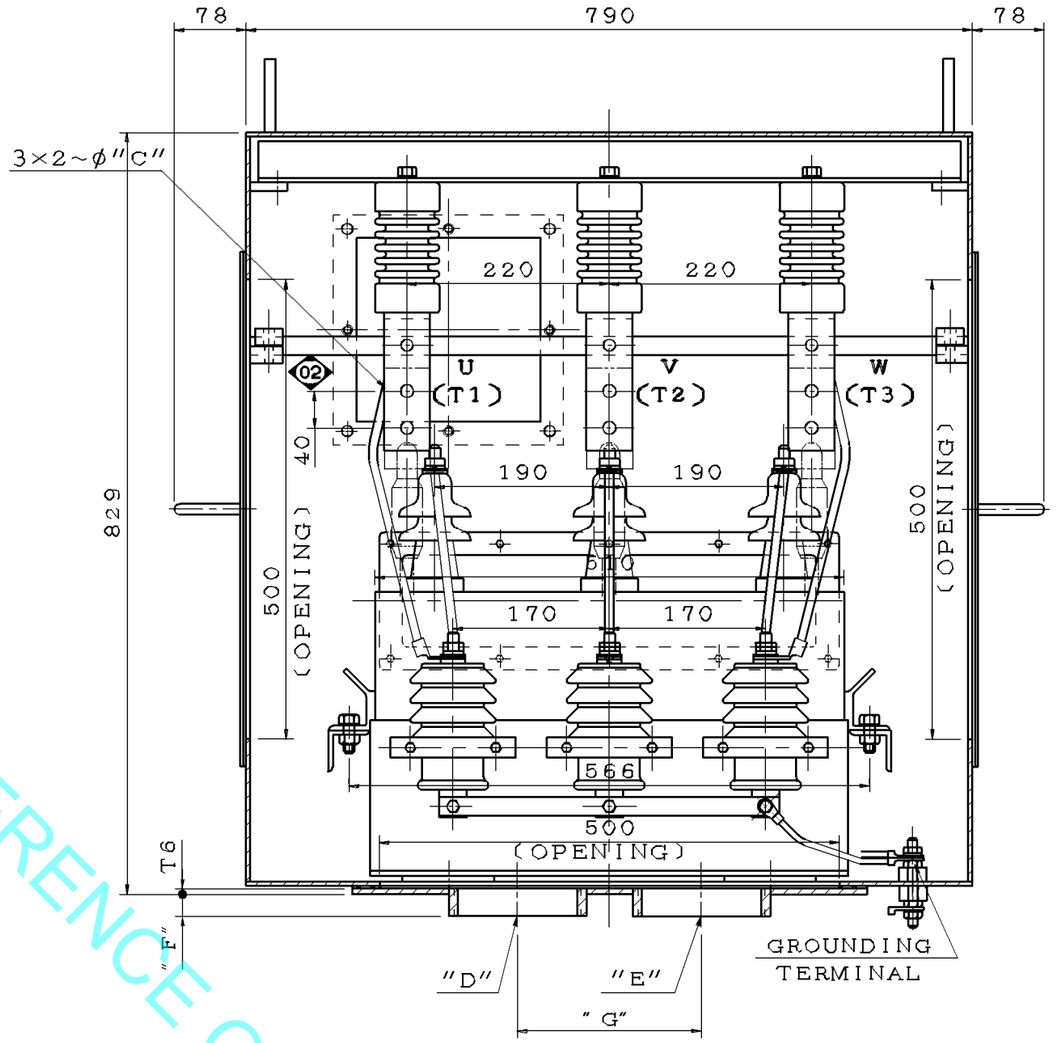
OUTLINE DIMENSIONS
3—PHASE INDUCTION MOTOR

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040S706
REV:03

第三角法





ITEM	A	B	C	D	E	F	G
01	605	150	14	NPT4"	NPT4"	24	200
02							
03							
04							
05							

NOTE:
1. DIMENSIONS IN mm.
2. PRIMARY T—BOX.

FOR SCPC REFERENCE ONLY

DWN.	W. LIU	FEB 19 2014
CHKD.	K. LIN	MAR 03 2014
APPD.	C. SHIH	MAR 03 2014

TECO

DATE

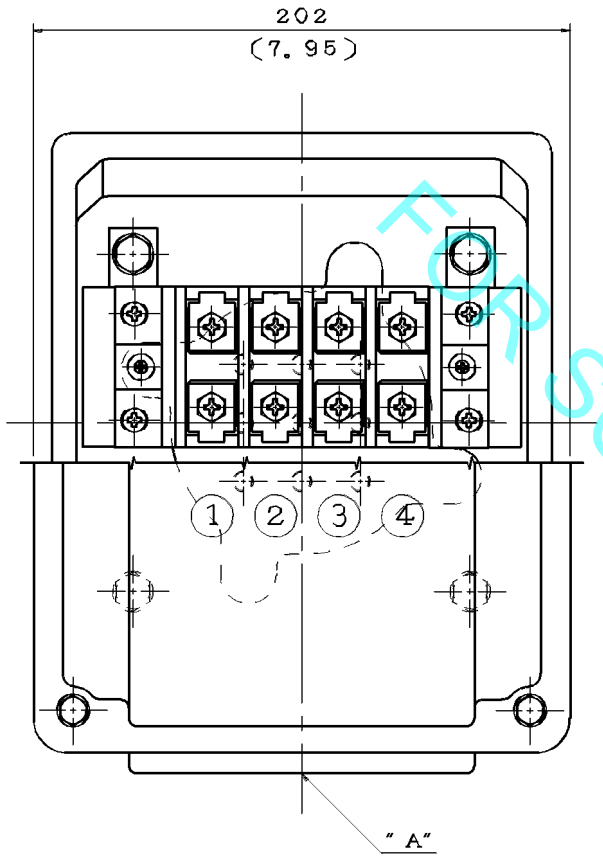
SCHEMATIC DRAWING

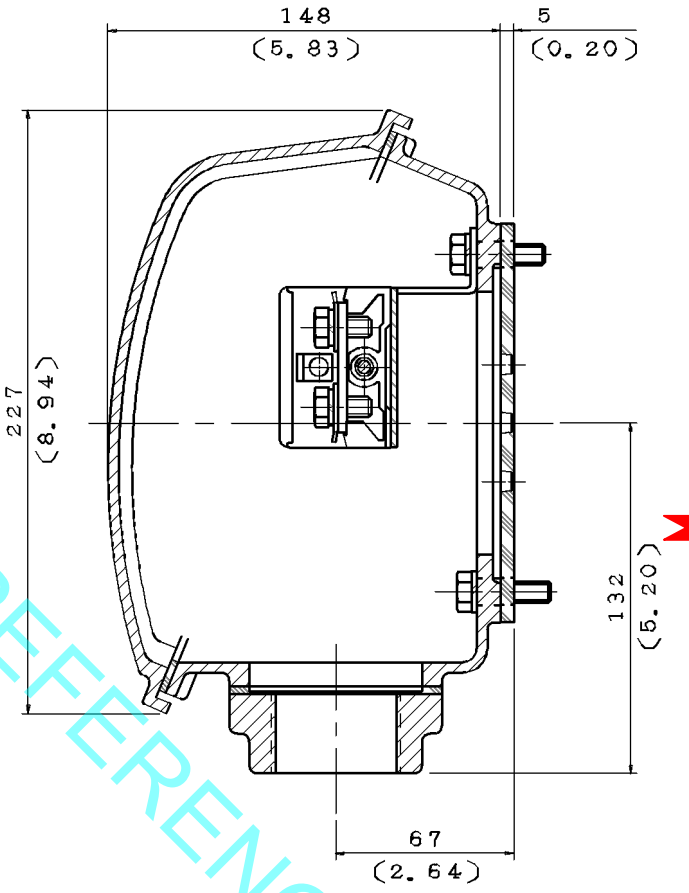
TERMINAL BOX

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

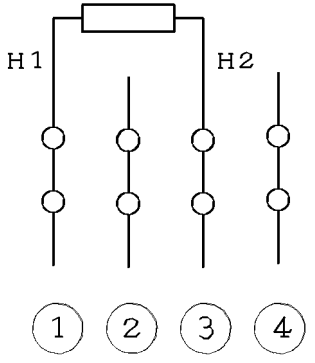
DWG NO. 4A040Y033X1

REV:02





ITEM	A
01	PF-1"
02	PF-1.5"
03	PF-2"
04	NPT-1"
05	NPT-1.5"
06	NPT-2"
07	M25×1.5
08	M32×1.5
09	M50×1.5
10	NPT-0.75"
11	M20×1.5
12	NPT-0.25"
13	NPT-0.5"
14	
15	
16	
17	
18	
19	
20	



FOR SCPC REFERENCE ONLY

NOTE:
1. DIMEMSION IN mm(INCH).
2. TW-36.
3. SPACE HEATER T-BOX.
4. ENCLOSURE: IPW55.

DWN.	S. WANG	NOV 21 2008
CHKD.	S. WANG	NOV 22 2008
APPD.	C. WANG	NOV 25 2008

TECO

DATE

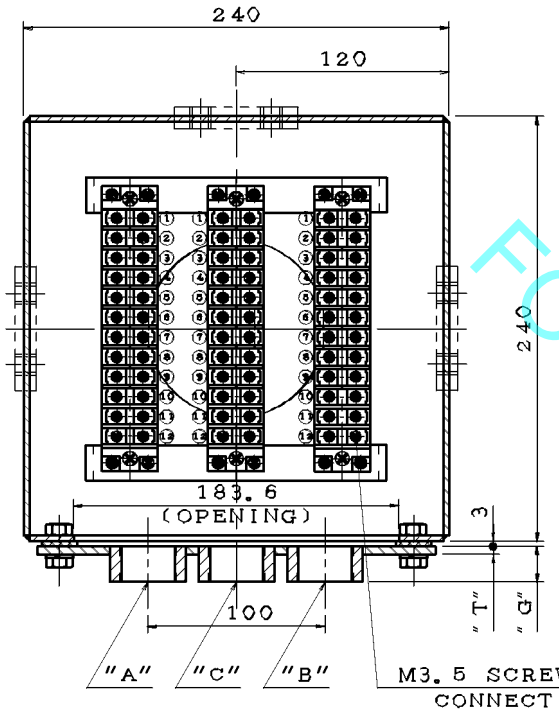
SCHEMATIC DRAWING

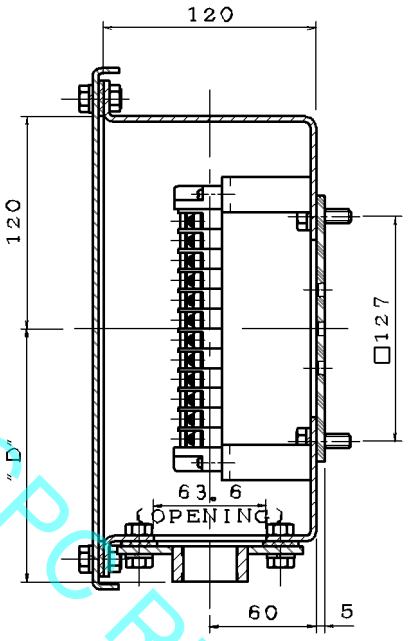
TERMINAL BOX

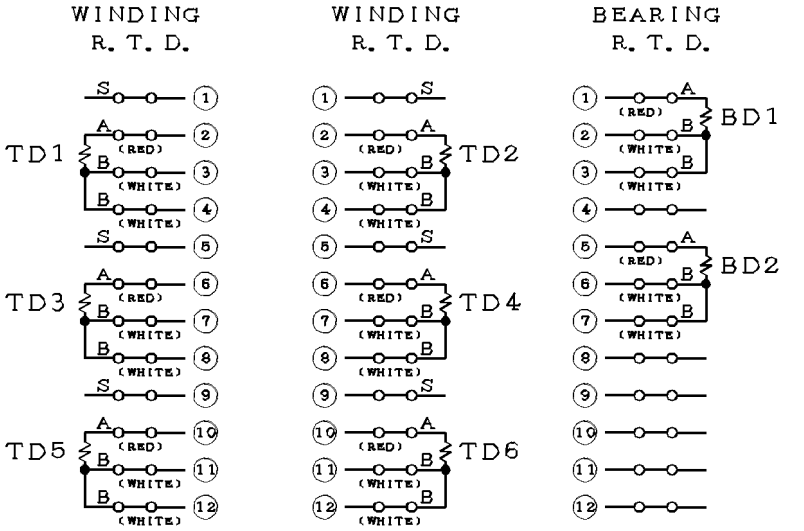
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3B040Z580

REV:07







ITEM	A	B	C	D	G	T
01	0	0	0	127.5	4.5	4.5
02	M50×1.5	M20×1.5	0	143	20	4.5
03	NPT2"	NPT1"	0			
04	PF2"	PF1"	0			
05	PT2"	PT1"	0			
06	0	0	PF1.25"			
07	M20×1.5	M20×1.5	0			
08	0	0	M50×1.5			
09	NPT0.75"	NPT0.75"	NPT0.75"			
10	0	0	PF1"			
11	NPT1.5"	NPT0.75"	0			
12	0	0	M20×1.5			
13	0	0	NPT1.5"			
14	NPT1.5"	NPT1"	0			
15	0	0	NPT2"			
16	PF1.5"	PF1.5"	0			
17	0	0	PG16			
18	PF1"	PF1"	0			

ITEM	A	B	C	D	G	T
19	0	0	PF2"	143	20	4.5
20	G2"	G1"	0			
21	0	0	NPT1"			
22	NPT1"	NPT1"	0			
23	M50×1.5	M50×1.5	0			
24	M32×1.5	M50×1.5	0			
25	M32×1.5	M20×1.5	0			
26	PF0.75"	PF0.75"	0			

NOTE:
1. DIMENSIONS IN mm.
2. WINDING R.T.D. & BEARING R.T.D. T—BOX.
3. TD1 & TD2 FOR U PHASE
   TD3 & TD4 FOR V PHASE
   TD5 & TD6 FOR W PHASE
   BD1 FOR DRIVE END BEARING
   BD2 FOR NON—DRIVE END BEARING.

FOR SCPC REFERENCE ONLY

DWN.	C. LEONG	FEB. 19. 2001
CHKD.	C. WANG	FEB. 19. 2001
APPD.	T. CHEN	FEB. 19. 2001

TECO
DATE

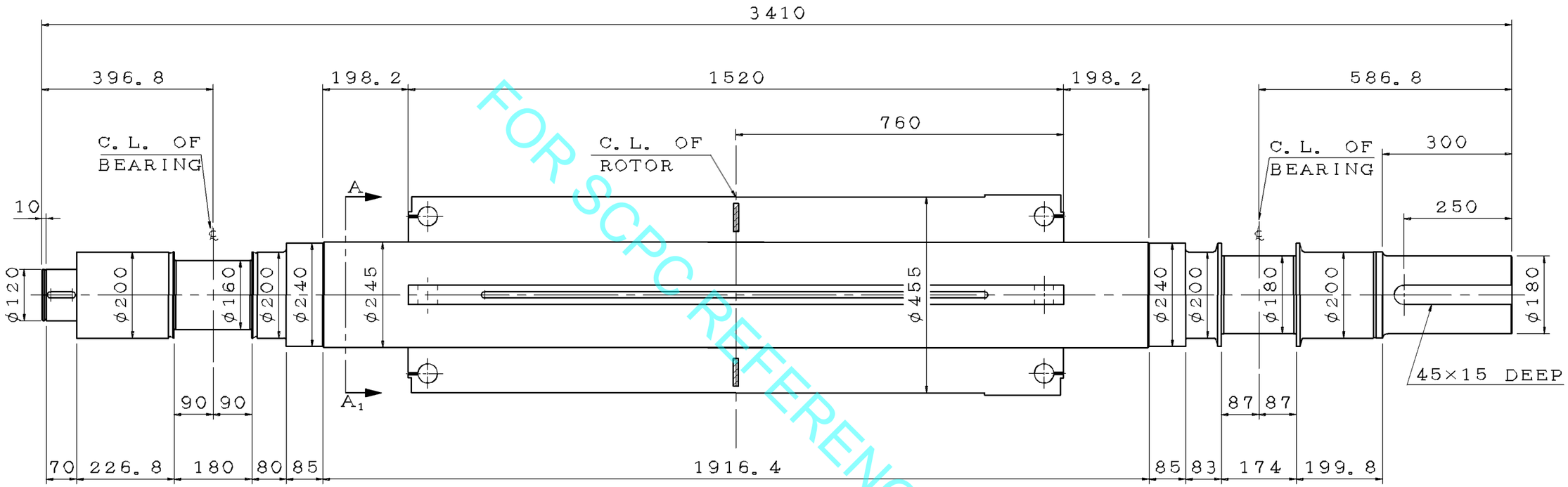
SCHEMATIC DRAWING
TERMINAL BOX

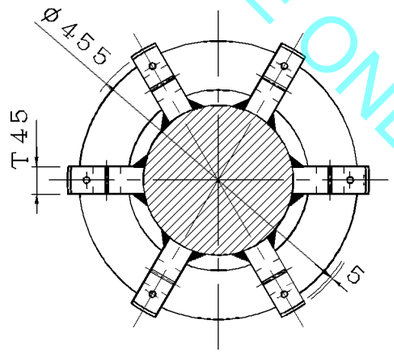
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040D751
REV:28

NON—DRIVE END

DRIVE END





A—A1 VIEW

FOR SCPC REFERENCE ONLY

NOTE:
1. DIMENSIONS IN mm.

TORSIONAL ANALYSIS DATA

MOTOR WR²: 268.5 kg—m²

SHAFT TORSIONAL STIFFNESS: 1.06E09 kgf—mm/rad
(FROM ROTOR CENTERLINE TO COUPLING END OF SHAFT) (FOR REFERENCE ONLY)

SHAFT MATERIAL: JIS G 4051 S45C (AISI 1045 EQUIV.)
SPIDER PLATE: ASTM A36

SHAFT ULTIMATE TENSILE STRENGTH: 58 kgf/mm² MIN

SHAFT YIELD STRENGTH IN TENSION: 35 kgf/mm² MIN

SHAFT SHEAR MODULUS OF RIGIDITY: 8300 kgf/mm²

DWN.	P. LIN	OCT 08 2013
CHKD.	S. WANG	OCT 11 2013
APPD.	C. SHIH	OCT 11 2013

TECO

DATE

SCHEMATIC DRAWING

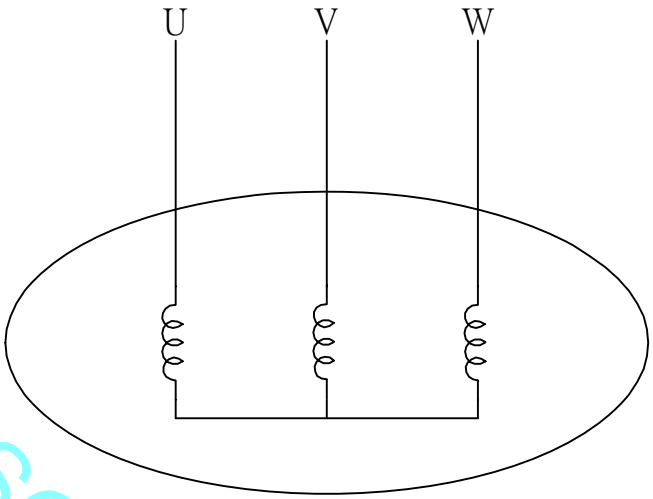
SHAFT

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

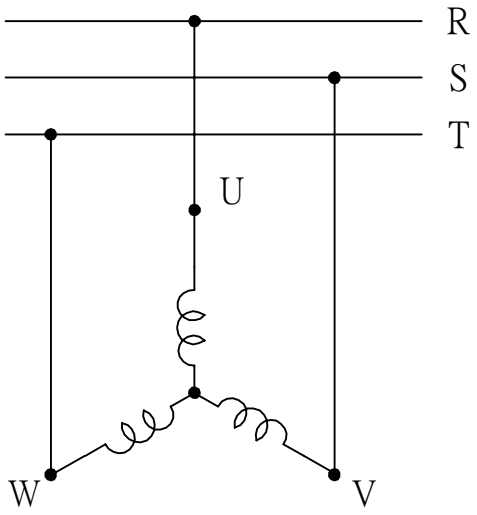
DWG NO. 4A040Y229

REV:00

DATE	SCHEMATIC WYE CONN 3 LEADS	MODEL



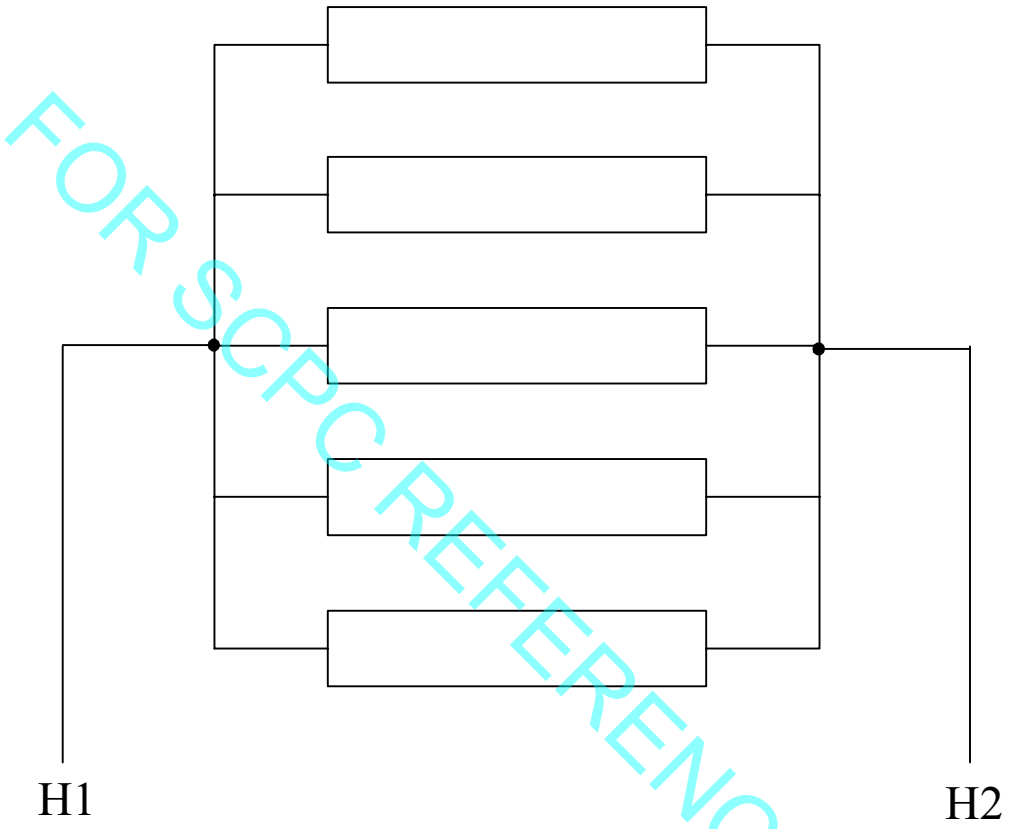
SCHEMATIC - WYE CONN - 3 LEADS

CONNECTION	ROTATION (VIEWED FROM DRIVE END)
	

FOR SCDC REFERENCE ONLY

DWN.	S.HUANG	MAR・03・2003	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	T.HSIAO	MAR・03・2003		3A061H403
APPD.	T.HSIAO	MAR・03・2003		

DATE	SCHEMATIC SPACE HEATER	MODEL



FOR SCPC REFERENCE ONLY

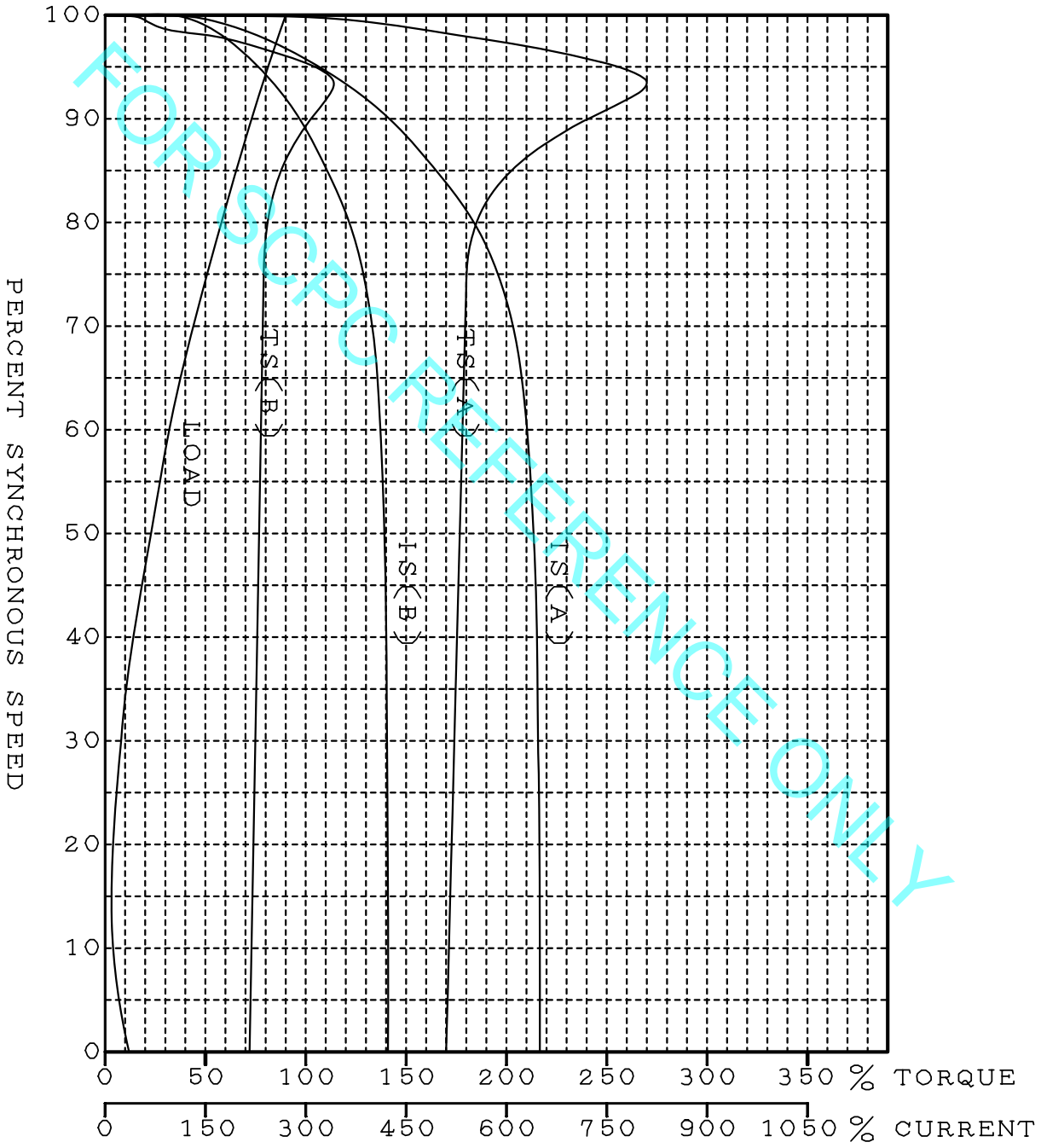
DWN.	ANDREA	JUL • 03 • 2012	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	HB.LIN	JUL • 03 • 2012		3 A 0 6 1 H 5 7 4
APPD.	H.CHIANG	JUL • 03 • 2012		

INDUCTION MOTOR STARTING CHARACTERISTICS

T—N/I—N CURVE

TYPE:AEZK KW:3300 VOLTS:6600 ORDER NO:FE14A702B1
HZ:60 POLES:6 RPM(FLS):1193

VOLTS: (A)100% (B):65%



FOR SCPC REFERENCE ONLY

DWG.	HEDY	20140709
APPD.	T. HSIAO	20140709

TECO Electric & Machinery Co., Ltd.

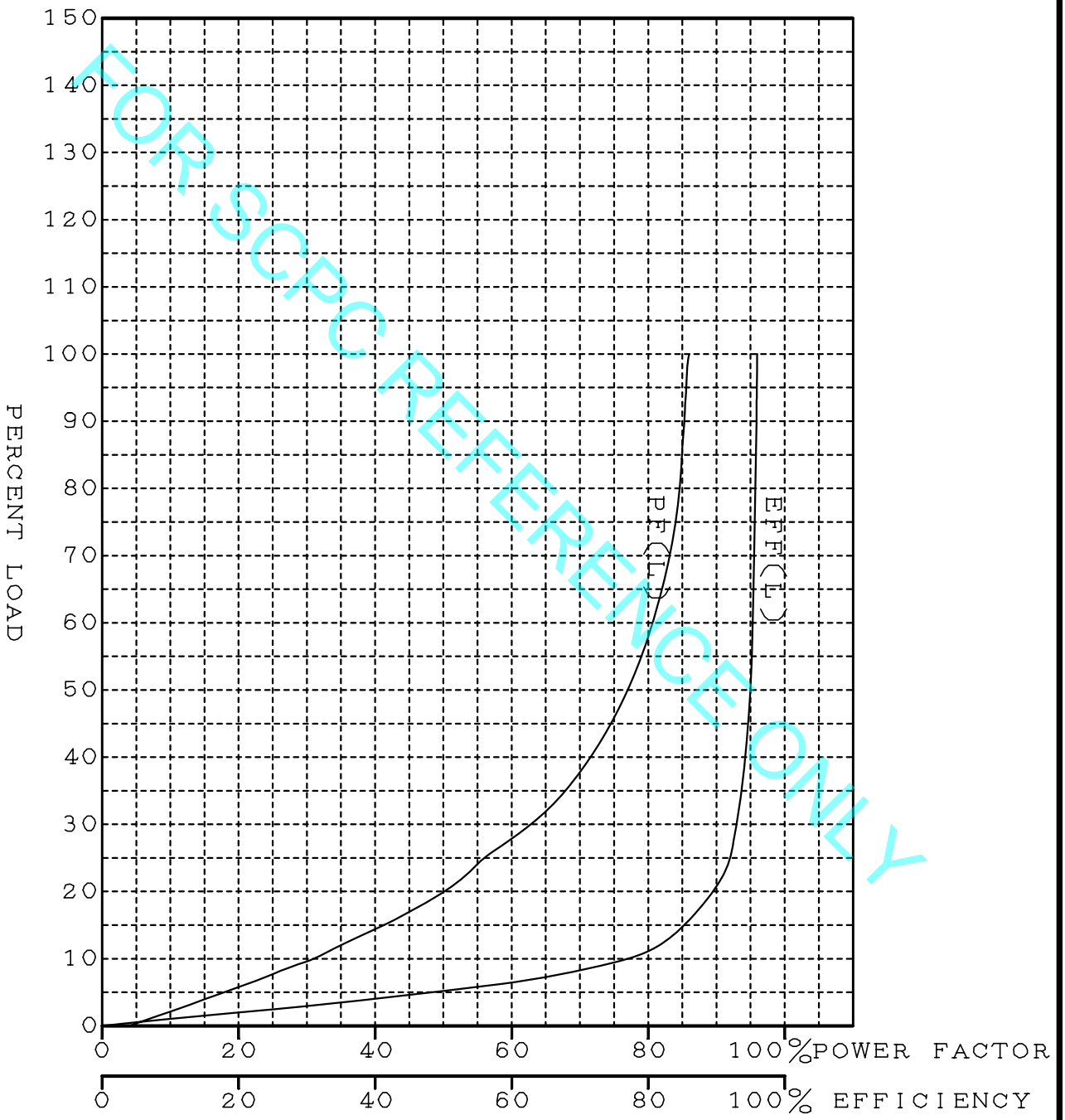
DWG NO. REV:00
FE14A702B1
T—I—L

Input Ver:02

INDUCTION MOTOR STARTING CHARACTERISTICS

Efficiency & Power Factor Vs Load Curve

TYPE:AEZK KW:3300 VOLTS:6600 ORDER NO:FE14A702B1
HZ:60 POLES:6 RPM(FLS):1193



FOR SCPC REFERENCE ONLY

DWG.	CF. WENG	20131002
APPD.	T. HSIAO	20131002

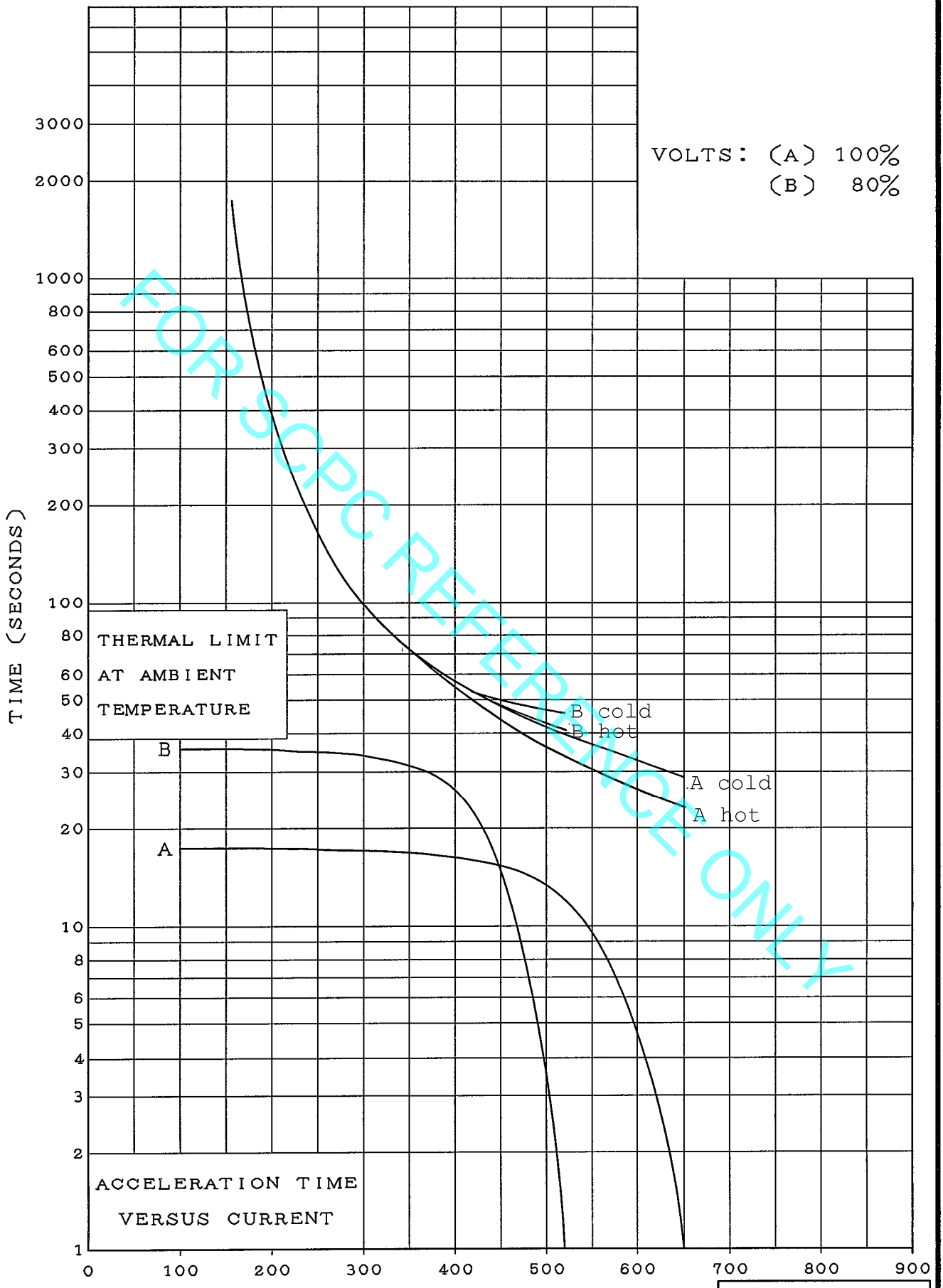
TECO Electric & Machinery Co., Ltd.

DWG NO. REV:00
FE14A702B1
E—P—L

Input Ver:01

TIME — CURRENT AND THERMAL LIMIT CURVES

LOAD GD² (KG—M²):15000 MOTOR GD² (KG—M²):1074
TYPE:AEZK POLE:6 KW:3300 NO. : FE14A702B1



FOR SCPC REFERENCE ONLY

DWG.	CF. WENG	20131002
APPD.	T. HSIAO	20131002

TECO Electric & Machinery Co., Ltd.

DWG NO. REV:00
FE14A702B1
TIME

Input Ver:02

P.A.FAN

3700KW * 4P * 6600V * 3Ø * 60HZ

FOR SCPC REFERENCE ONLY

DOCUMENTATION LIST

Order No. FE14A701B1
Type AEZKS2044960Y4001 **Customer**
Total Sets 1 **User**
Issued Date AUG.9.2014 **Customer PO.** P1022461-1

Item	Drawing No.	Drawing Name	Rev	Item No.	Note
1	3A057H186-72076	Specification Table	04	--	Change
2	3A040S710	Outline Dimensions	03	--	--
3	4A040Y033X1	Primary Terminal Box Schematic Drawing	02	01	Change
4	3B040Z580	Heater Terminal Box Schematic Drawing	07	10	--
5	3A040D751	WDG & BTG RTD Common Terminal Box Schematic Drawing	28	03	--
6	4A040Y260	Shaft Schematic Drawing	00	--	--
7	3A061H403	SCHEMATIC STATOR CONNECTION	00	--	--
8	3A061H574	SCHEMATIC SPACE HEATER	00	--	--
~~9~~	~~FE14A701B1-T-I-N~~	~~T-N/I-N Curve~~	~~00~~	~~--~~	Delete
10	FE14A701B1-T-I-L	T-N/I-N VS Load Curve	00	--	Add
11	FE14A701B1-E-P-L	Eff & Pf VS Load Curve	00	--	--
12	FE14A701B1-TIME	Time-Current and Thermal Limit Curve	00	--	--
13	FE14A701B1-DS	Data Sheet	00	--	--

FOR SCPC REFERENCE ONLY

承 認 用

本承認圖，請於文到7個工作日內簽回，以利工程進行，否則視爲承認論。

承認者：
工號：

TECO ELEC. & MACH. CO., LTD.

DOC NO. FE14A701B1
REV.04 1 / 1

SPECIFICATION TABLE OF 3-PHASE SQUIRREL CAGE INDUCTION MOTOR	CUSTOMER		USER	
	INQ. NO.		EQUIPMENT	
	JOB NO.	FE14A701B1	MACHINE	
	TOTAL SETS	1	ITEM NO.	

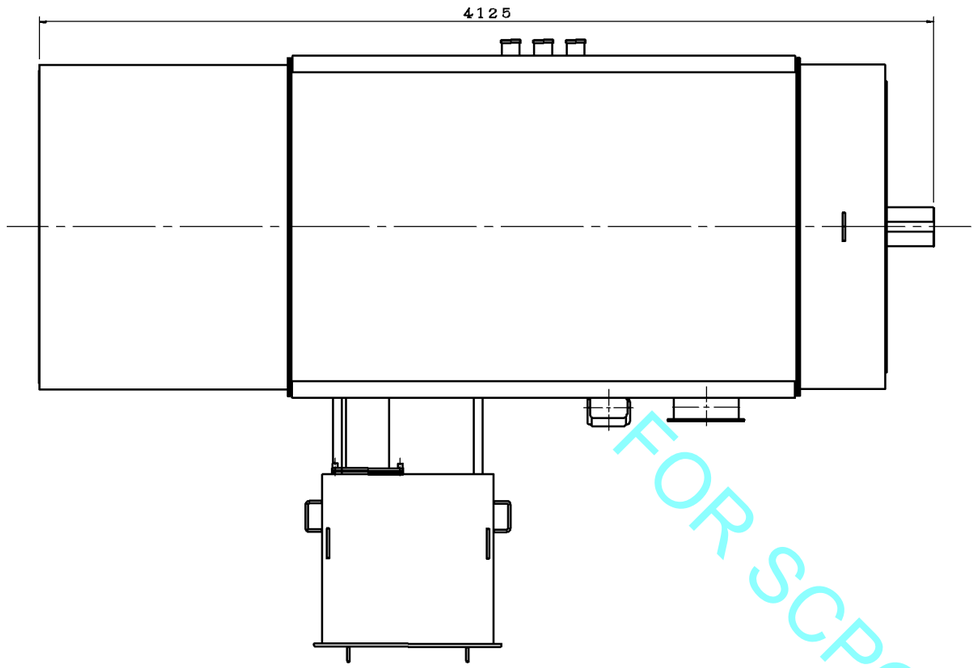
Item	Terms	Description			
1	Model	AEZK-S2			
2	Code or Standard	Dimensions	Frame Assignment	Performance	Test
		IEC	TECO	IEC	IEC
3	Rating	3700 kW 4 Pole 6600 Volt 3 Phase 60 Hz			
4	Service Duty	Continuous Rating , S.F. 1.15			
5	Starting Method	Soft Start			*
6	Rotation	Facing The Drive End : CW			
7	Drive Method	Direct Coupling			
8	Environment	Amb. Temp. : -20 ~ 40 ℃			
		Humidity : Less Than 90 %RH			
		Altitude : Up to 100 M			
9	Enclosure & Protection	IPW55 : Totally Enclosed			Outdoor
10	Cooling	IC611 : Fan Cooled With Built-in Air Cooler			
11	Mounting	IM1001 : HS, Foot			
12	Dimensions	Dr# 3A040S710(REV.03)		Frame No ： 630D	
13	Frame & Bracket	Frame : Steel Plate		Bracket : Steel Plate	
14	Fan & Fan Cover	Fan :Steel Plate		Fan Cover :Steel Plate	
15	Terminal Box	Steel Plate			
16	Lead Terminals	(TLK95-16)X3			
17	Lubricant	Oil Viscosity : ISO VG46			
18	Painting	Color : MUNSELL 7.5GY 5/4,STANDARD TECO DR#3A057D659			
19	Stator Winding	Ins. Class F			
20	Rotor Conductor	Cu-Alloy		GD^2 ： 679 Kg-m^2	
21	Starting Performance	LRC≦ 2424 Amp		LRT/FLT 150 %	
22	Operating Performance	Hz/V	60/6600	Break Down Torque	
		%Load	100	250 %FLT	
		Amp.	373		
		Eff.%	95.8	Temp. Rise Limit. (Res.)	
		P.F.%	90.5	Stator 90 ℃	
		R.P.M.	1788	at S.F.1.15	
23	Approximate Weight	Motor ： 13850 kg			
24	Note	1. With Space Heater : 1φ 220V 1050W 2. With Winding RTD : PT 100Ω/0℃ 6pcs 3. With Stator Core Thermometer : N.O. 1pcs 4. With Bearing Thermometer : N.O. 2pcs 5. With Surge Capacitor & Lightning Arrester 6. Corrosion Proof 7. Tolerances on declared values shall be in accordance with IEC 8. Load GD^2 : 5000 Kg-m^2 Note ： [*] at the end represent revised item ◦			

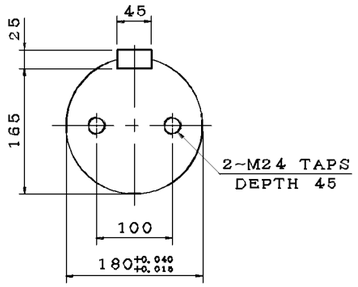
FOR SCR REFERENCE ONLY

APPD.	Kenny.Shyu	AUG. 09 2014	TECO ELEC. & MACH. CO., LTD.	DWG NO. 3A057H186-72076
CHKD.	AIYU	AUG. 09 2014		REV.04 1/1
DWN.	CF.WENG	JUL. 09 2014		

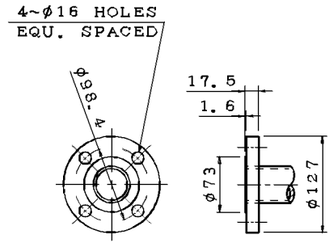
TYPE	OUTPUT HP.	OUTPUT kW.	POLE	TIME RATING	VOLTAGE V	Hz	SYN. SPEED R. P. M.
AEZK-S2		3700	4	CONT	6600	60	1800

TOTALLY ENCLOSED AIR-TO-AIR COOLED TYPE, SQUIRREL CAGE ROTOR

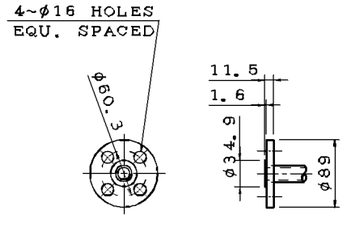




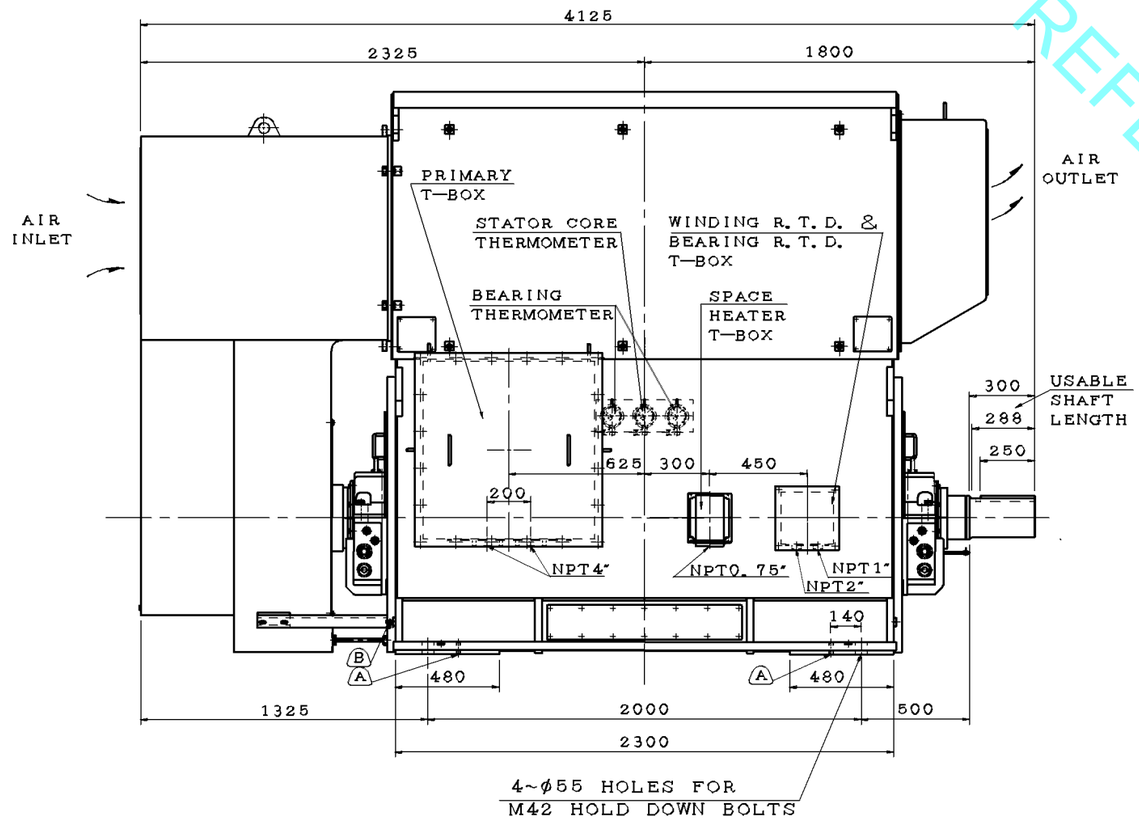
DRIVE END
(ENLARGED VIEW)

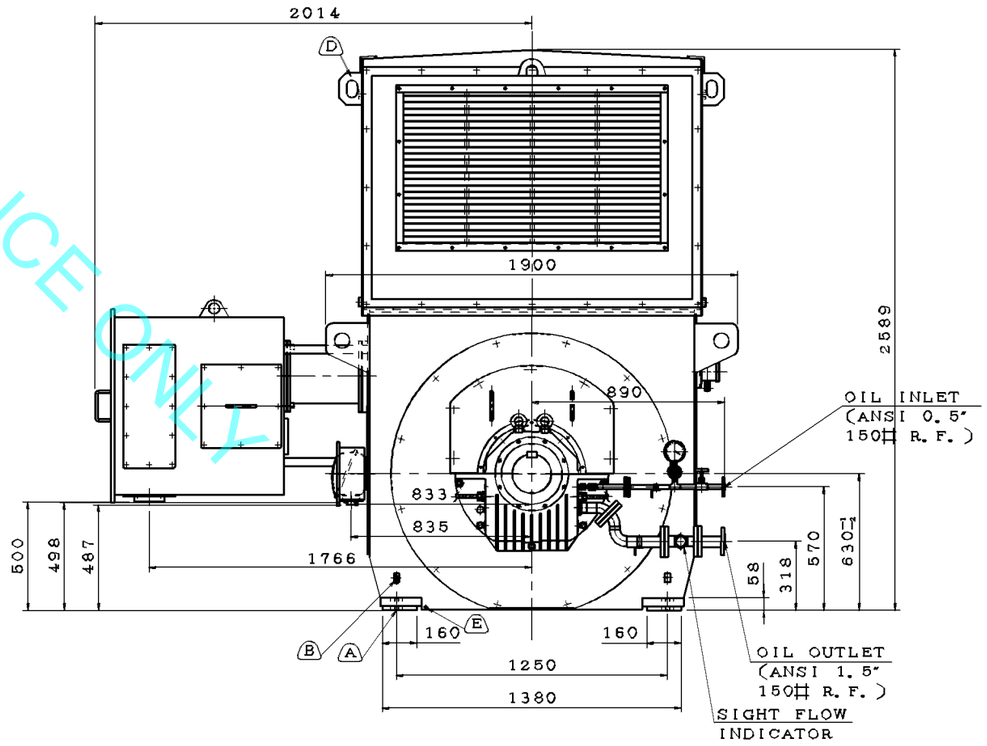


OIL OUTLET-1.5" NOM.
ANSI 150LB R.F. FLANGE
(ENLARGED VIEW)



OIL INLET-0.5" NOM.
ANSI 150LB R.F. FLANGE
(ENLARGED VIEW)





FOR SCPC REFERENCE ONLY

DWN.	P. LIN	OCT 14 2013
CHKD.	S. WANG	OCT 17 2013
APPD.	C. SHIH	OCT 17 2013

TECO
DATE

OUTLINE DIMENSIONS
3-PHASE INDUCTION MOTOR

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040S710
REV:03

第三角法

TYPE	OUTPUT HP.	OUTPUT kW.	POLE	TIME RATING	VOLTAGE V	Hz	SYN. SPEED R. P. M.
AEZK-S2			4	CONT	6600	60	1800

TOTALLY ENCLOSED AIR-TO-AIR COOLED TYPE, SQUIRREL CAGE ROTOR

NOTE:
1. DIMENSIONS IN mm.
2. FRAME NO. 630D
3. F CLASS INSULATION, S.F.: 1.15.
4. FOR DIRECT FLEXIBLE COUPLING.
5. SLEEVE TYPE BEARING, AXIAL THRUST LOAD NOT ALLOWED.
6. THE MOTOR ENDPLAY IS ±7 mm. A LIMITED END FLOAT TYPE COUPLING IS REQUIRED TO LIMIT ENDPLAY TO ±2.4 mm.
7. BEARING SIZE: DRIVE END: 18-200(UNINSULATED)
   NON-DRIVE END: 18-180(INSULATED)
8. THE NON-DRIVE END BEARING LINER(SHELL) IS INSULATED FROM THE HOUSING. METAL CONNECTIONS MADE TO THE BEARING SHELL MUST BE INSULATED TO PREVENT AN INSULATION SHORT CIRCUIT. METAL CONNECTIONS MADE TO THE HOUSING DO NOT NEED TO BE INSULATED.
9. BEARING LUBRICATION:SELF LUBRICATION.
   A. OIL VISCOSITY: ISO VG46 (180~220 SSU AT 100°F)
   B. OIL FLOW RATE: DRIVE END: 6.0~7.5 L/min.
   : NON-DRIVE END: 4.5~5.6 L/min.
   C. OIL INLET PRESSURE: 0.8~1.0 kgf/cm².
   D. OIL INLET ORIFICE DIAMETER: DRIVE END: 3.2mm.
   NON-DRIVE END: 2.8mm.
   E. OIL INLET TEMPERATURE: 25°C~49°C.
   F. EXTERNAL OIL SUPPLY UNIT TO BE PREPARED BY THE CUSTOMER.
10. WITH SPACE HEATER: 1φ, 220V, 1050W.
11. WITH WINDING R.T.D.: PT 100Ω/0°C, 6PCS.
12. WITH BEARING R.T.D.: PT 100Ω/0°C, 2PCS.
13. WITH BEARING THERMOMETER:WITH N.O. CONTACTS, 2PCS.
14. WITH STATOR CORE THERMOMETER:WITH N.O. CONTACTS, 1PCS.
15. WITH SURGE CAPACITOR(6.6 kV, 3φ) 1PCS,
    LIGHTNING ARRESTER(9.0 kV, 1φ) 3PCS,
    PROVIDED IN PRIMARY T-BOX.
16. NOISE:BELOW 85dBA AT 1 METER DISTANCE ON LOAD.
17. VIBRATION:BELOW 2.5mm/s(0-P) ON BEARING HOUSING NO LOAD.
18. ENCLOSURE: IPW55, OUTDOOR SERVICE.
19. CORROSION PROOF.
20. MOTOR APPROX. WEIGHT: 13850 kg.

Ⓐ M20 VERTICAL JACKING HOLE, ONE HOLE PER FOOT.

Ⓑ FOUR M12 TAPPED GROUNDING PADS, WITH TWO SETS GROUNDING STUD &(R100-12) TERMINALS LOCATED DIAGONALLY OPPOSITE.

03 Ⓒ NPT1/4" HOLE, PROVISION FOR SUPPLY SCOPE OF CCMC.

Ⓓ FOUR LIFTING LUGS ON AIR COOLER FOR LIFTING AIR COOLER ONLY. DO NOT USE FOR LIFTING ENTIRE MOTOR.

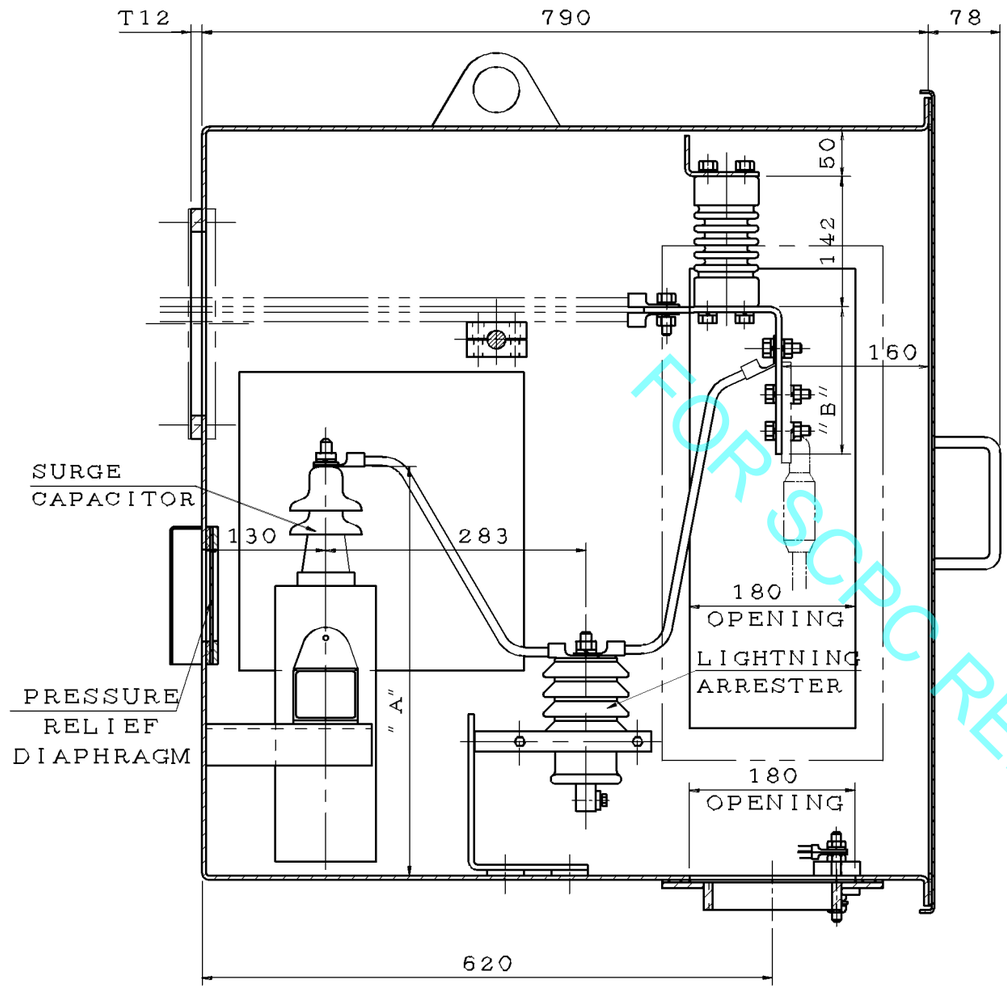
Ⓔ TWO M10 TAPPED DRAIN HOLES, LOCATED DIAGONALLY OPPOSITE.

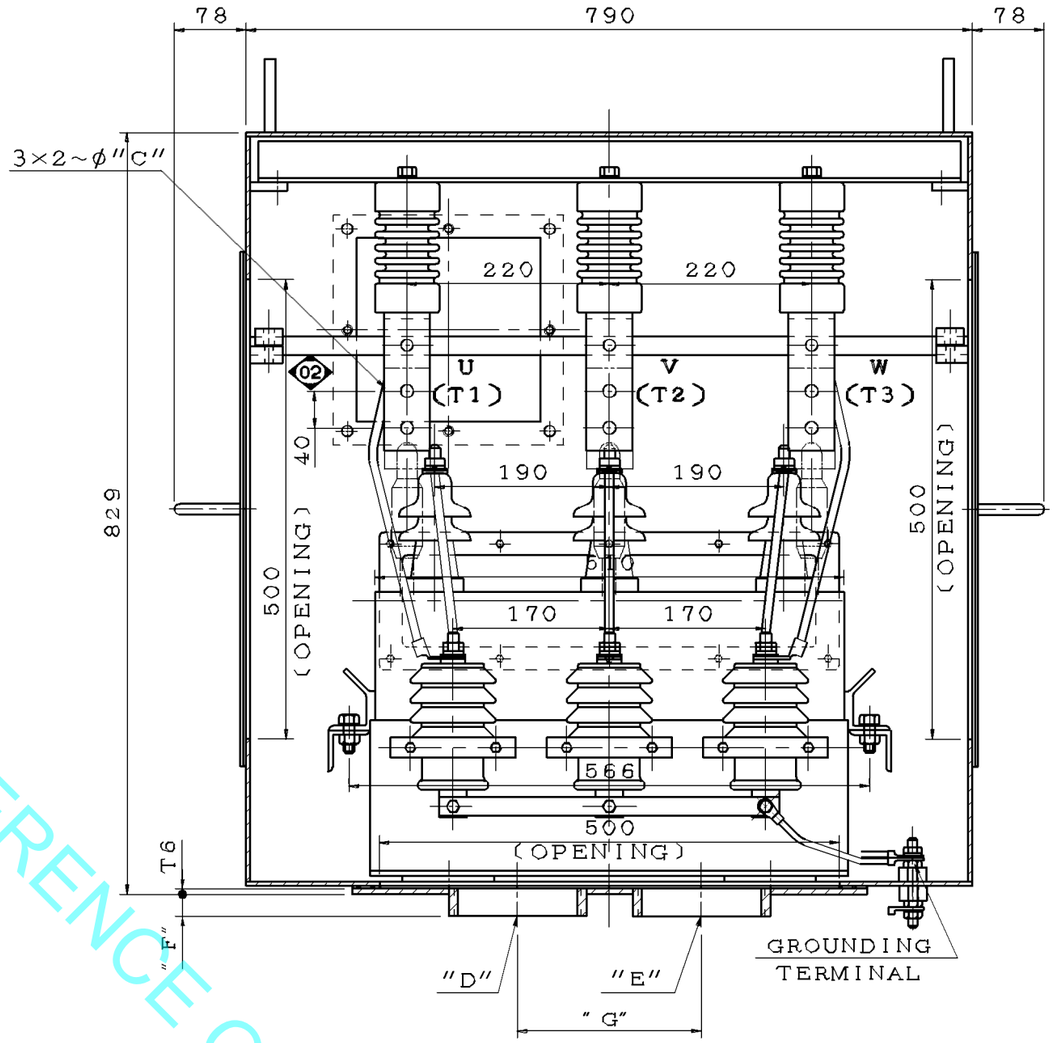
FOR SCPC REFERENCE ONLY

DWN.	P. LIN	OCT 14 2013
CHKD.	S. WANG	OCT 17 2013
APPD.	C. SHIH	OCT 17 2013

TECO
DATE
OUTLINE DIMENSIONS
3-PHASE INDUCTION MOTOR
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.
DWG NO. 3A040S710
REV:03
2/4

第三角法





FOR SCPC REFERENCE ONLY

ITEM	A	B	C	D	E	F	G
01	605	150	14	NPT4"	NPT4"	24	200
02							
03							
04							
05							

NOTE:
1. DIMENSIONS IN mm.
2. PRIMARY T—BOX.

DWN.	W. LIU	FEB 19 2014
CHKD.	K. LIN	MAR 03 2014
APPD.	C. SHIH	MAR 03 2014

TECO

DATE

SCHEMATIC DRAWING

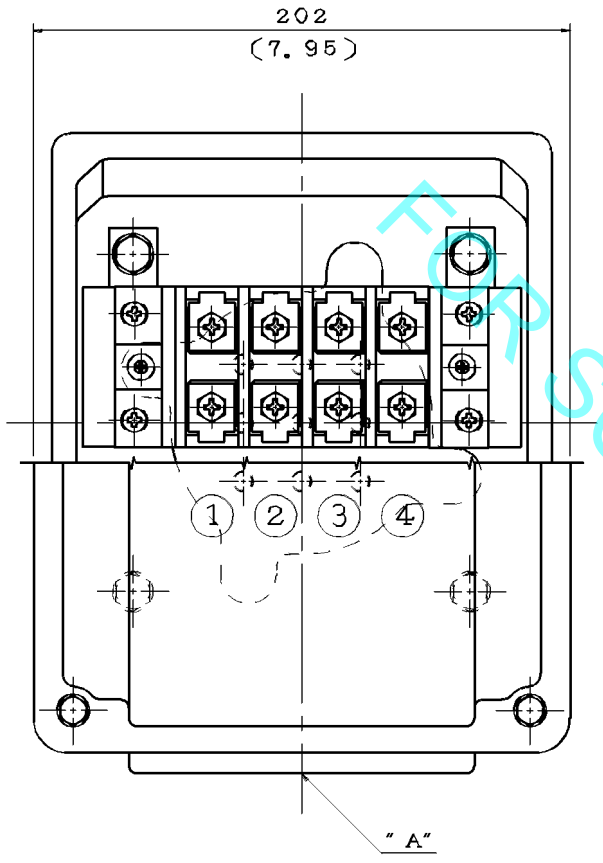
TERMINAL BOX

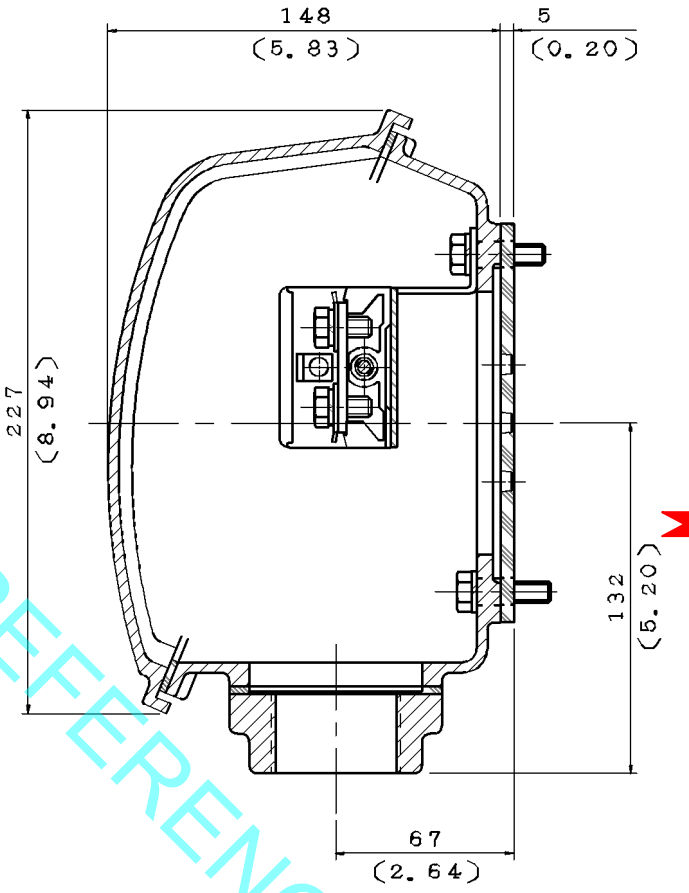
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 4A040Y033X1

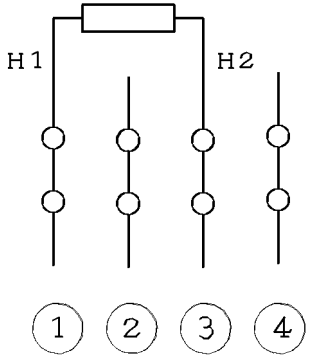
REV:02

1/1





ITEM	A
01	PF-1"
02	PF-1.5"
03	PF-2"
04	NPT-1"
05	NPT-1.5"
06	NPT-2"
07	M25×1.5
08	M32×1.5
09	M50×1.5
10	NPT-0.75"
11	M20×1.5
12	NPT-0.25"
13	NPT-0.5"
14	
15	
16	
17	
18	
19	
20	



FOR SCPC REFERENCE ONLY

NOTE:
1. DIMEMSION IN mm(INCH).
2. TW-36.
3. SPACE HEATER T-BOX.
4. ENCLOSURE: IPW55.

DWN.	S. WANG	NOV 21 2008
CHKD.	S. WANG	NOV 22 2008
APPD.	C. WANG	NOV 25 2008

TECO

DATE

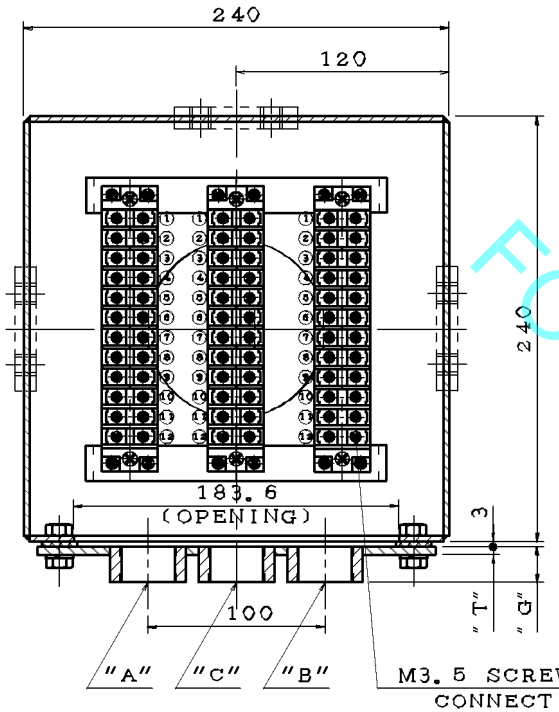
SCHEMATIC DRAWING

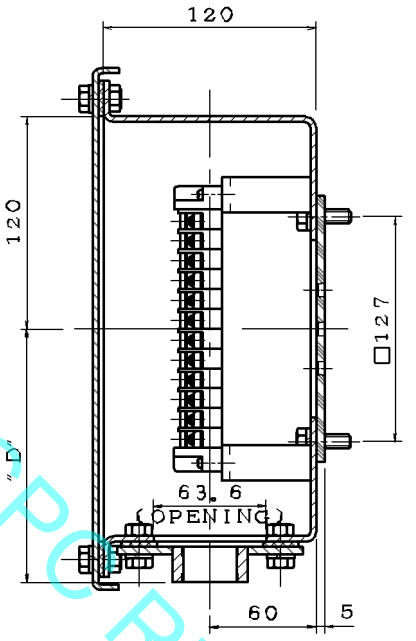
TERMINAL BOX

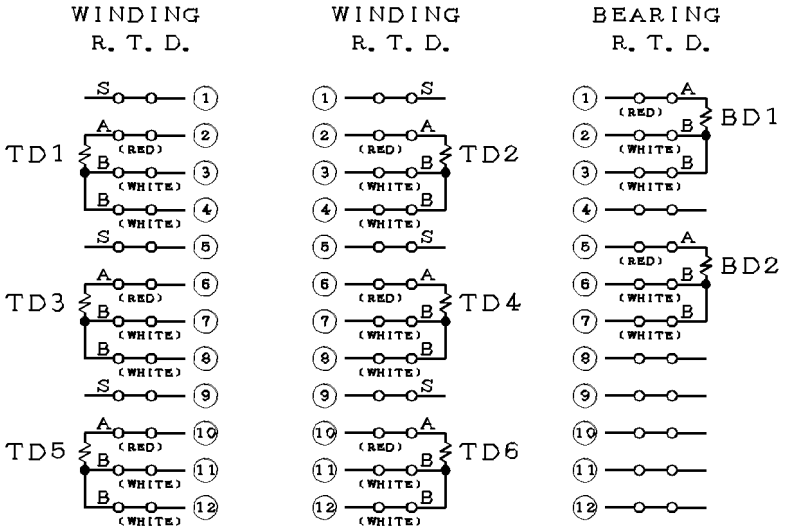
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3B040Z580

REV:07







ITEM	A	B	C	D	G	T
01	0	0	0	127.5	4.5	4.5
02	M50×1.5	M20×1.5	0	143	20	4.5
03	NPT2"	NPT1"	0			
04	PF2"	PF1"	0			
05	PT2"	PT1"	0			
06	0	0	PF1.25"			
07	M20×1.5	M20×1.5	0			
08	0	0	M50×1.5			
09	NPT0.75"	NPT0.75"	NPT0.75"			
10	0	0	PF1"			
11	NPT1.5"	NPT0.75"	0			
12	0	0	M20×1.5			
13	0	0	NPT1.5"			
14	NPT1.5"	NPT1"	0			
15	0	0	NPT2"			
16	PF1.5"	PF1.5"	0			
17	0	0	PG16			
18	PF1"	PF1"	0			

ITEM	A	B	C	D	G	T
19	0	0	PF2"	143	20	4.5
20	G2"	G1"	0			
21	0	0	NPT1"			
22	NPT1"	NPT1"	0			
23	M50×1.5	M50×1.5	0			
24	M32×1.5	M50×1.5	0			
25	M32×1.5	M20×1.5	0			
26	PF0.75"	PF0.75"	0			

NOTE:
1. DIMENSIONS IN mm.
2. WINDING R.T.D. & BEARING R.T.D. T—BOX.
3. TD1 & TD2 FOR U PHASE
   TD3 & TD4 FOR V PHASE
   TD5 & TD6 FOR W PHASE
   BD1 FOR DRIVE END BEARING
   BD2 FOR NON—DRIVE END BEARING.

FOR SCPC REFERENCE ONLY

DWN.	C. LEONG	FEB. 19. 2001
CHKD.	C. WANG	FEB. 19. 2001
APPD.	T. CHEN	FEB. 19. 2001

TECO
DATE

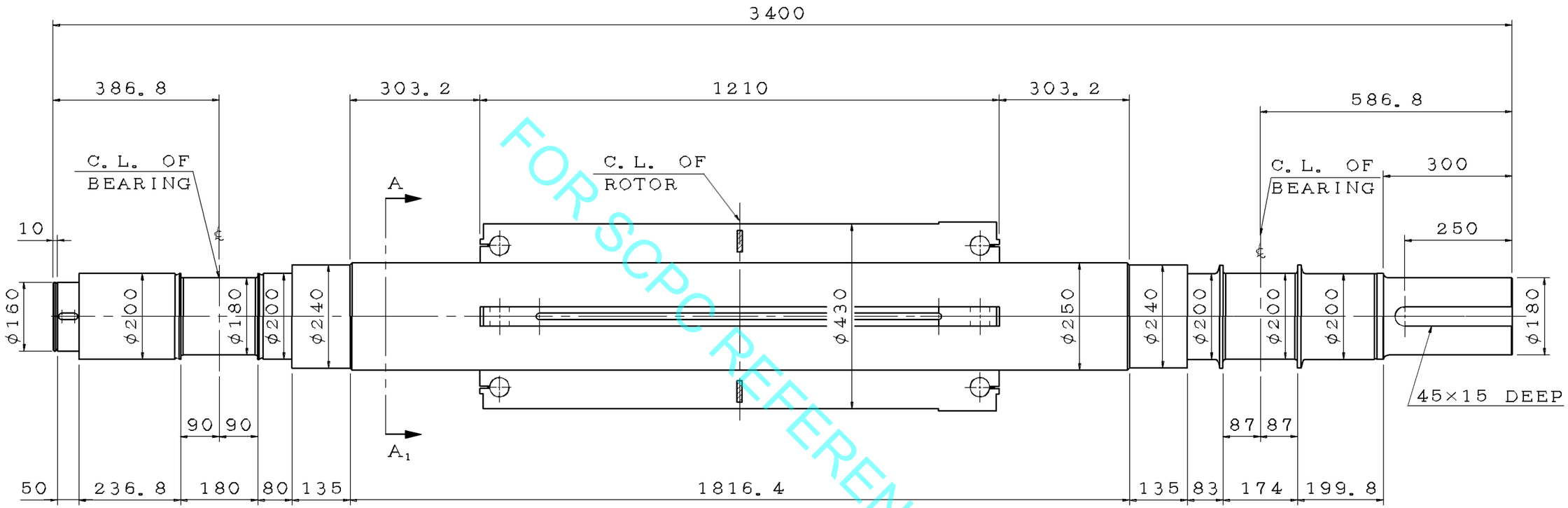
SCHEMATIC DRAWING
TERMINAL BOX

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

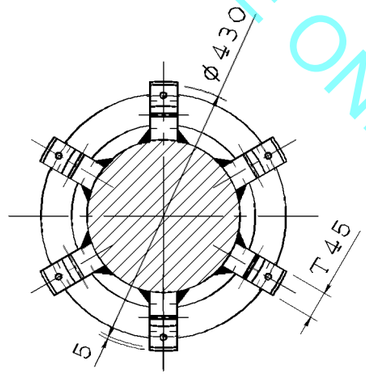
DWG NO. 3A040D751
REV:28

NON—DRIVE END

DRIVE END



FOR SCPC REFERENCE ONLY



A—A1 VIEW

TORSIONAL ANALYSIS DATA

MOTOR WR²: 169.8 kg—m²

SHAFT TORSIONAL STIFFNESS: 1.10E09 kgf—mm/rad
(FROM ROTOR CENTERLINE TO COUPLING END OF SHAFT) (FOR REFERENCE ONLY)

SHAFT MATERIAL: JIS G 4051 S45C (AISI 1045 EQUIV.)
SPIDER PLATE: ASTM A36

SHAFT ULTIMATE TENSILE STRENGTH: 58 kgf/mm² MIN

SHAFT YIELD STRENGTH IN TENSION: 35 kgf/mm² MIN

SHAFT SHEAR MODULUS OF RIGIDITY: 8300 kgf/mm²

NOTE:
1. DIMENSIONS IN mm

DWN.	P. LIN	OCT 15 2013
CHKD.	S. WANG	OCT 17 2013
APPD.	C. SHIH	OCT 17 2013

TECO

DATE

SCHEMATIC DRAWING

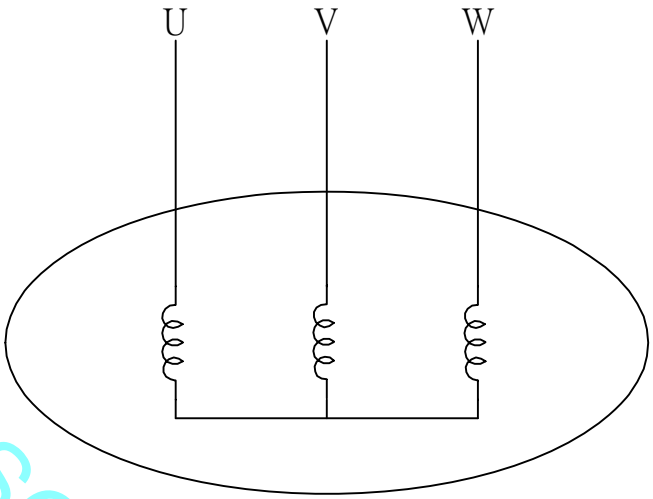
SHAFT

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

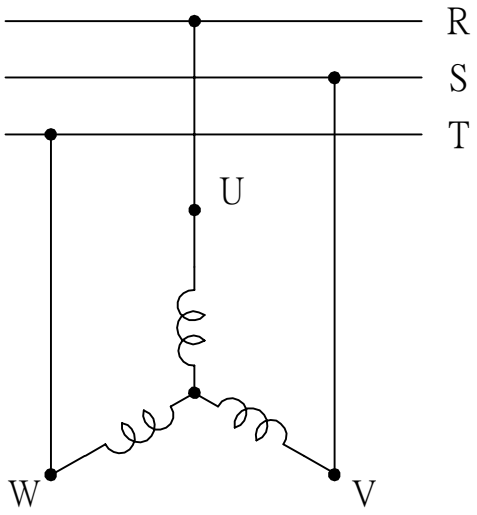
DWG NO. 4A040Y260

REV:00

DATE	SCHEMATIC WYE CONN 3 LEADS	MODEL



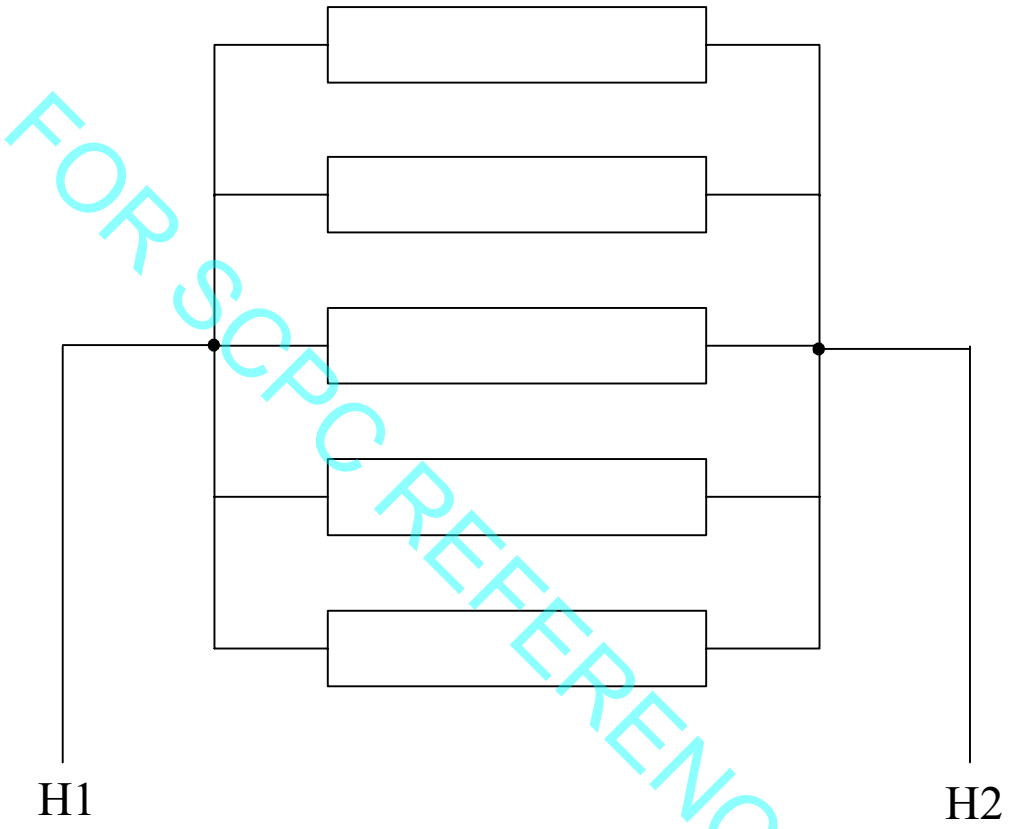
SCHEMATIC - WYE CONN - 3 LEADS

CONNECTION	ROTATION (VIEWED FROM DRIVE END)
	

FOR SCDC REFERENCE ONLY

DWN.	S.HUANG	MAR・03・2003	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	T.HSIAO	MAR・03・2003		3A061H403
APPD.	T.HSIAO	MAR・03・2003		

DATE	SCHEMATIC SPACE HEATER	MODEL



FOR SCPC REFERENCE ONLY

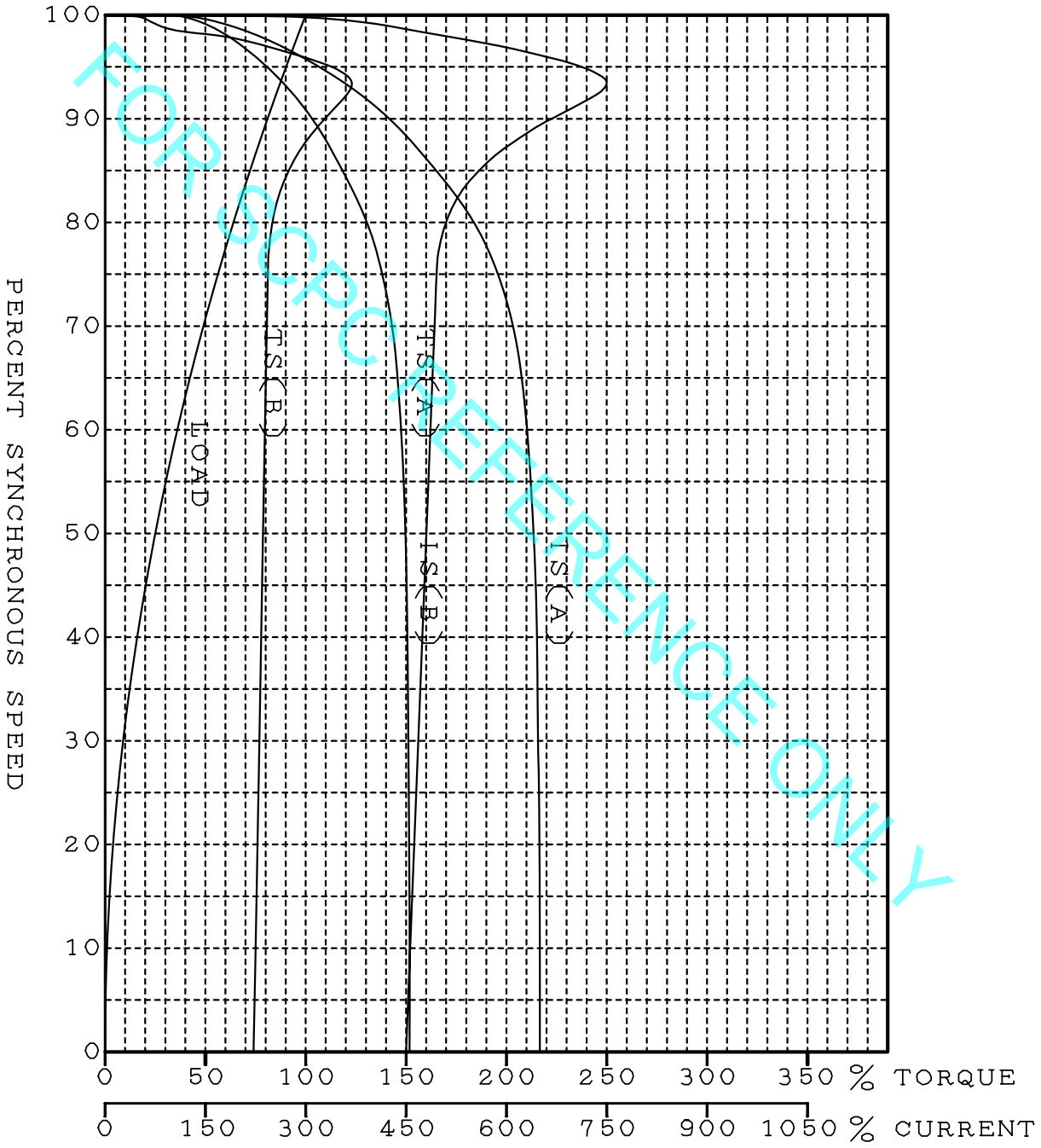
DWN.	ANDREA	JUL • 03 • 2012	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	HB.LIN	JUL • 03 • 2012		3 A 0 6 1 H 5 7 4
APPD.	H.CHIANG	JUL • 03 • 2012		

INDUCTION MOTOR STARTING CHARACTERISTICS

T—N/I—N CURVE

TYPE:AEZK KW:3700 VOLTS:6600 ORDER NO:FE14A701B1
HZ:60 POLES:4 RPM(FLS):1788

VOLTS: (A)100% (B):70%



FOR SCPC REFERENCE ONLY

DWG.	HEDY	20140709
APPD.	T. HSIAO	20140709

TECO Electric & Machinery Co., Ltd.

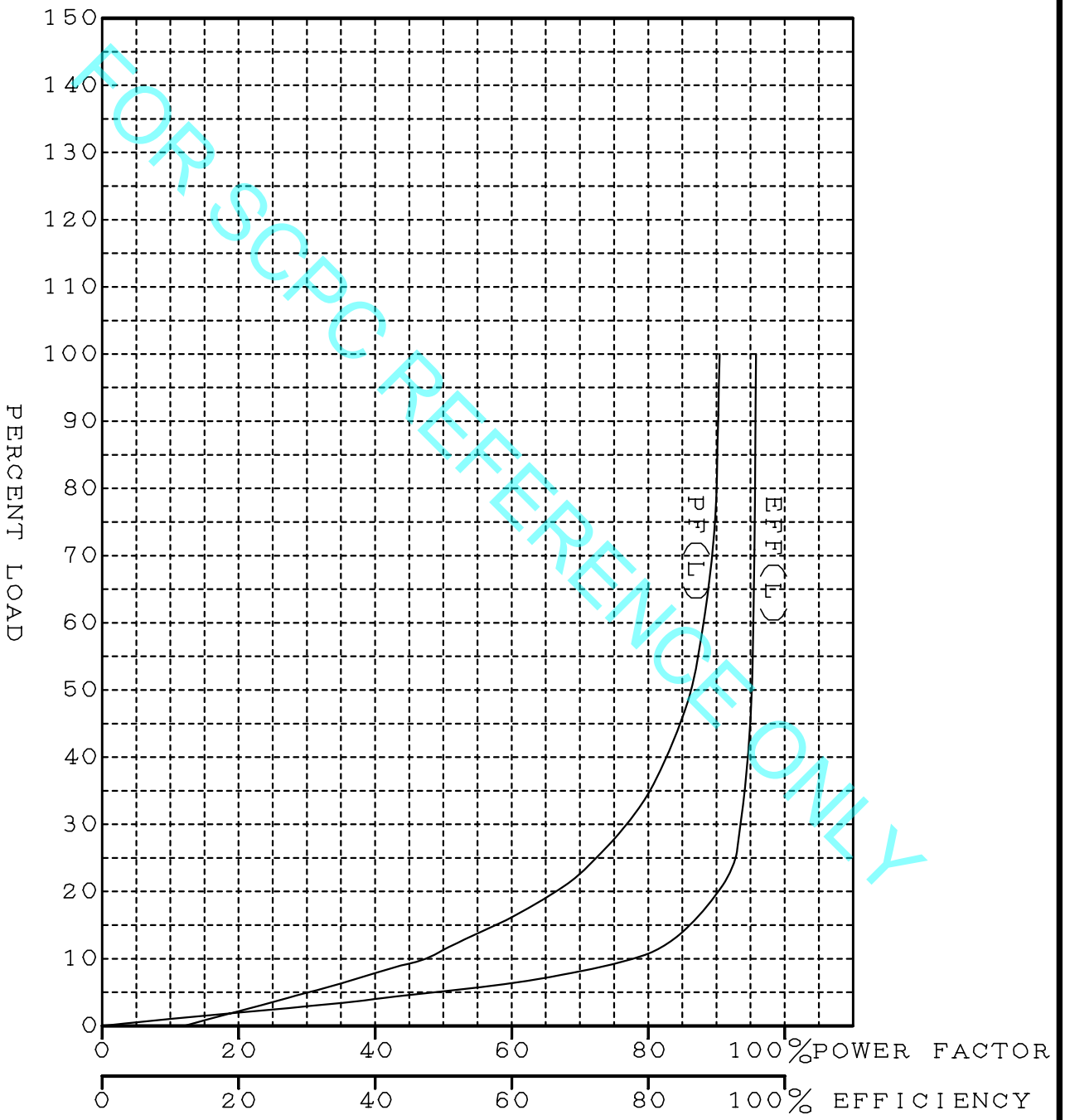
DWG NO. REV:00
FE14A701B1
T—I—L

Input Ver:02

INDUCTION MOTOR STARTING CHARACTERISTICS

Efficiency & Power Factor Vs Load Curve

TYPE:AEZK KW:3700 VOLTS:6600 ORDER NO:FE14A701B1
HZ:60 POLES:4 RPM(FLS):1788



FOR SCPC REFERENCE ONLY

DWG.	HEDY	20131007
APPD.	T.HSIAO	20131007

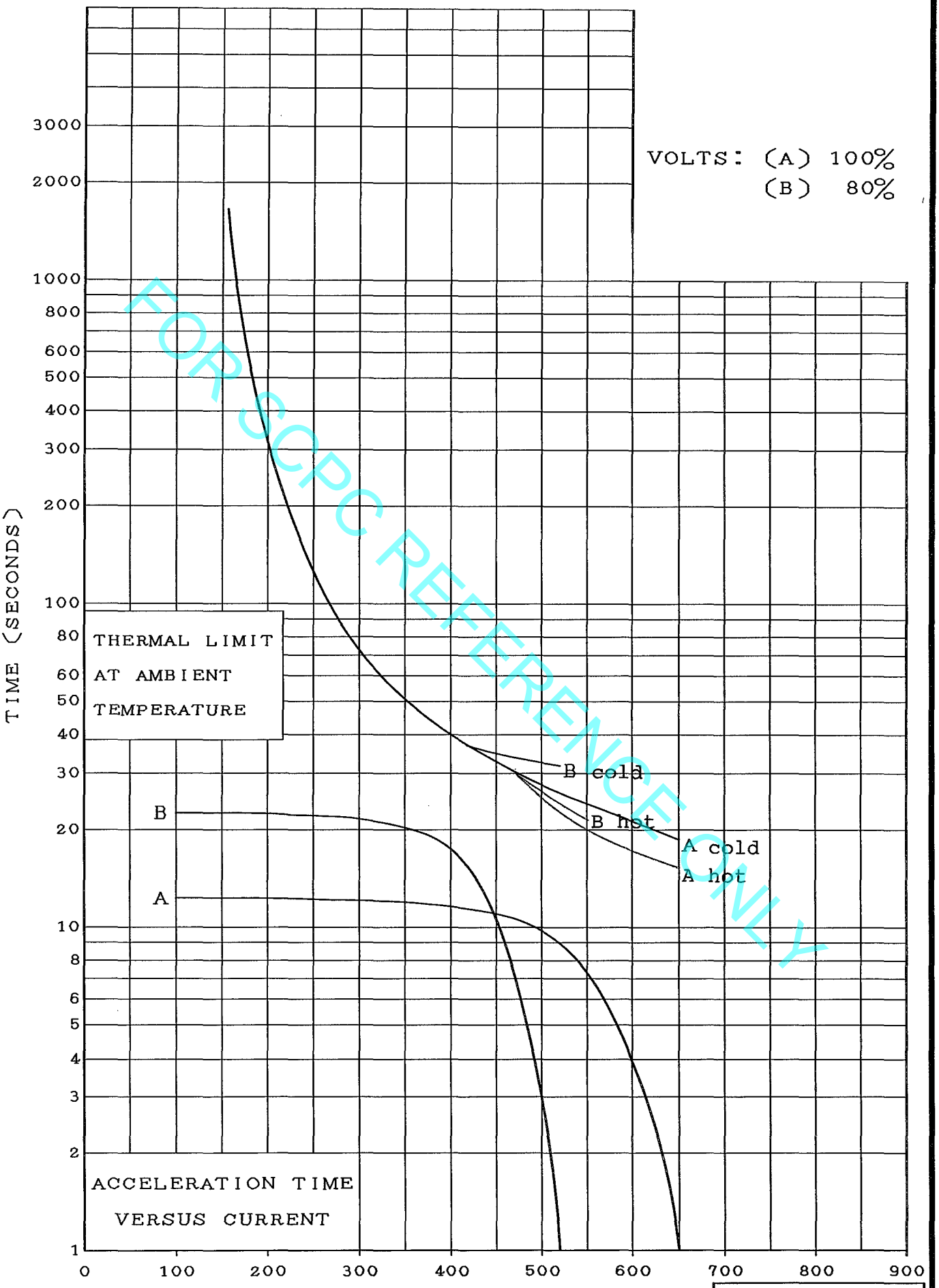
TECO Electric & Machinery Co., Ltd.

DWG NO. REV:00
FE14A701B1
E—P—L

Input Ver:01

TIME — CURRENT AND THERMAL LIMIT CURVES

LOAD GD² (KG—M²):5000 MOTOR GD² (KG—M²):679
TYPE:AEZK POLE:4 KW:3700 NO. :FE14A701B1



FOR SCPC REFERENCE ONLY

DWG.	HEDY	20131007
APPD.	T. HSIAO	20131007

TECO Electric & Machinery Co., Ltd.

DWG NO. REV:00
FE14A701B1
TIME

S.A.FAN

1850KW * 4P * 6600V * 3Ø * 60HZ

FOR SCPC REFERENCE ONLY

DOCUMENTATION LIST

Order No. FD14A202B1
Type AECKS2042480Y4003 **Customer**
Total Sets 1 **User**
Issued Date JUL.22.2014 **Customer PO.** P1022461-1

Item	Drawing No.	Drawing Name	Rev	Item No.	Note
1	3A057H186-72064	Specification Table	04	--	Change
2	3A040S702	Outline Dimensions	04	--	Change
3	4B040D700X1	Primary Terminal Box Schematic Drawing	02	01	Change
4	3B040Z580	Heater Terminal Box Schematic Drawing	07	10	--
5	3A040U362	WDG & BRG RTD Common Terminal Box Schematic Drawing	07	08	--
6	3A061H404	SCHEMATIC STATOR CONNECTION	00	--	--
7	3A061H573	SCHEMATIC SPACE HEATER	00	--	--
~~8~~	~~FD14A202B1-T-I-N~~	~~T-N/I-N Curve~~	~~01~~	~~--~~	Delete
9	FD14A202B1-T-I-L	T-N/I-N VS Load Curve	00	--	Add
10	FD14A202B1-E-P-L	Eff & Pf VS Load Curve	01	--	--
11	FD14A202B1-TIME	Time-Current and Thermal Limit Curve	01	--	--
12	FD14A202B1-DS	Data Sheet	01	--	--

FOR SCPC REFERENCE ONLY

承 認 用

本承認圖，請於文到7個工作日內簽回，以利工程進行，否則視爲承認論。

承認者：
工號：

TECO ELEC. & MACH. CO., LTD.

DOC NO. FD14A202B1
REV.05 1 / 1

SPECIFICATION TABLE OF 3-PHASE SQUIRREL CAGE INDUCTION MOTOR	CUSTOMER		USER	
	INQ. NO.		EQUIPMENT	
	JOB NO.	FD14A202B1	MACHINE	
	TOTAL SETS	1	ITEM NO.	

Item	Terms	Description			
1	Model	AECK-S2			
2	Code or Standard	Dimensions	Frame Assignment	Performance	Test
		IEC	TECO	IEC	IEC
3	Rating	1850 kW 4 Pole 6600 Volt 3 Phase 60 Hz			
4	Service Duty	Continuous Rating , S.F. 1.15			
5	Starting Method	Soft Start			*
6	Rotation	Facing The Drive End : CCW			
7	Drive Method	Direct Coupling			
8	Environment	Amb. Temp. : -20 ~ 40 ℃			
		Humidity : Less Than 90 %RH			
		Altitude : Up to 100 M			
9	Enclosure & Protection	IPW55 : Totally Enclosed		Outdoor	
10	Cooling	IC611 : Fan Cooled With Built-in Air Cooler			
11	Mounting	IM1001 : HS, Foot			
12	Dimensions	Dr# 3A040S702 (REV.04)	Frame No ： 560C		*
13	Frame & Bracket	Frame : Steel Plate	Bracket : Steel Plate		
14	Fan & Fan Cover	Fan :Steel Plate	Fan Cover :Steel Plate		
15	Terminal Box	Steel Plate			
16	Lead Terminals	(TLK50-10)X3			
17	Lubricant	Oil Viscosity : ISO VG32 (Sleeve Bearings)			
18	Painting	Color : MUNSELL 7.5GY 5/4,STANDARD TECO DR#3A057D659			
19	Stator Winding	Ins. Class F			
20	Rotor Conductor	Cu-Alloy	GD^2 ： 260 Kg-m^2		
21	Starting Performance	LRC≦ 1248 Amp	LRT/FLT 150 %		
22	Operating Performance	Hz/V	60/6600	Break Down Torque	
		%Load	100	250 %FLT	
		Amp.	192		
		Eff.%	95.6	Temp. Rise Limit. (Res.)	
		P.F.%	88.0	Stator 90 ℃	
		R.P.M.	1782	at S.F.1.15	
23	Approximate Weight	Motor ： 9200 kg			
24	Note	1. With Space Heater : 1φ 220V 550W 2. With Winding RTD : PT 100Ω/0℃ 6pcs 3. With Bearing RTD : PT 100Ω/0℃ 2pcs 4. With Stator Core Thermometer : N.O. 1pcs 5. With Surge Capacitor & Lightning Arrester 6. With Bearing Thermometer : With N.O. Contacts 2pcs 7. Corrosion Proof 8. Tolerances on declared values shall be in accordance with IEC 9.Load GD^2 : 3000 Kg-m^2 10. Noise : Below 85dBA at 1 Meter Distance No Load Note ： [*] at the end represent revised item ◦			

FOR SCR REFERENCE ONLY

APPD.	Kenny.Shyu	JUL. 22 2014	TECO ELEC. & MACH. CO., LTD.	DWG NO. 3A057H186-72064
CHKD.	AIYU	JUL. 22 2014		REV.04
DWN.	CF.WENG	JUL. 09 2014		1/2

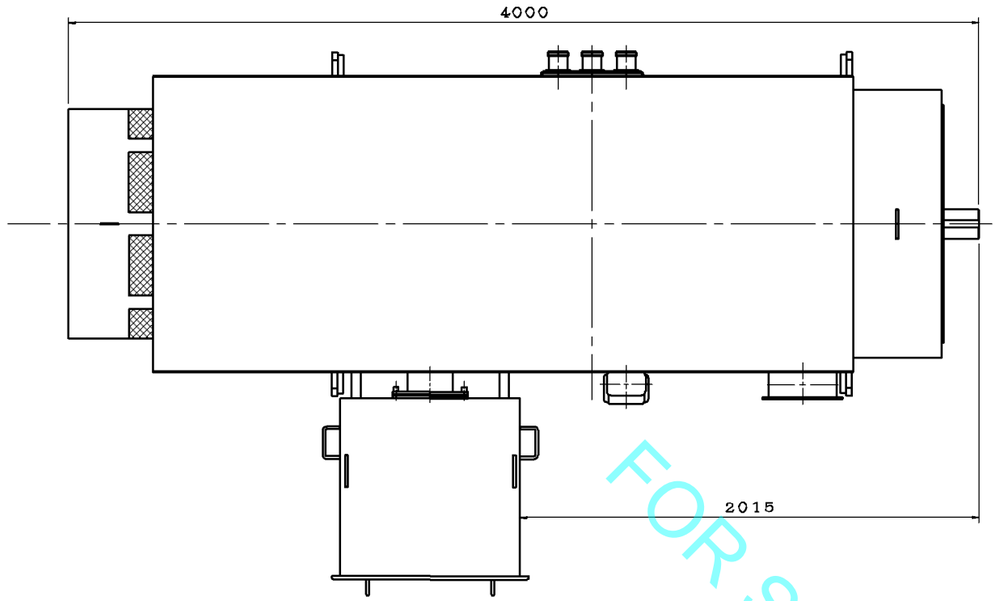
Item	Terms	Description
24	Note	11. Vibration : Below 2.5mm/s(0-P) on Bearing Housing No Load.

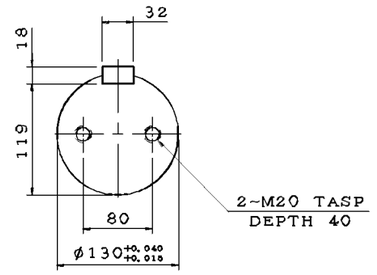
FOR SCPC REFERENCE ONLY

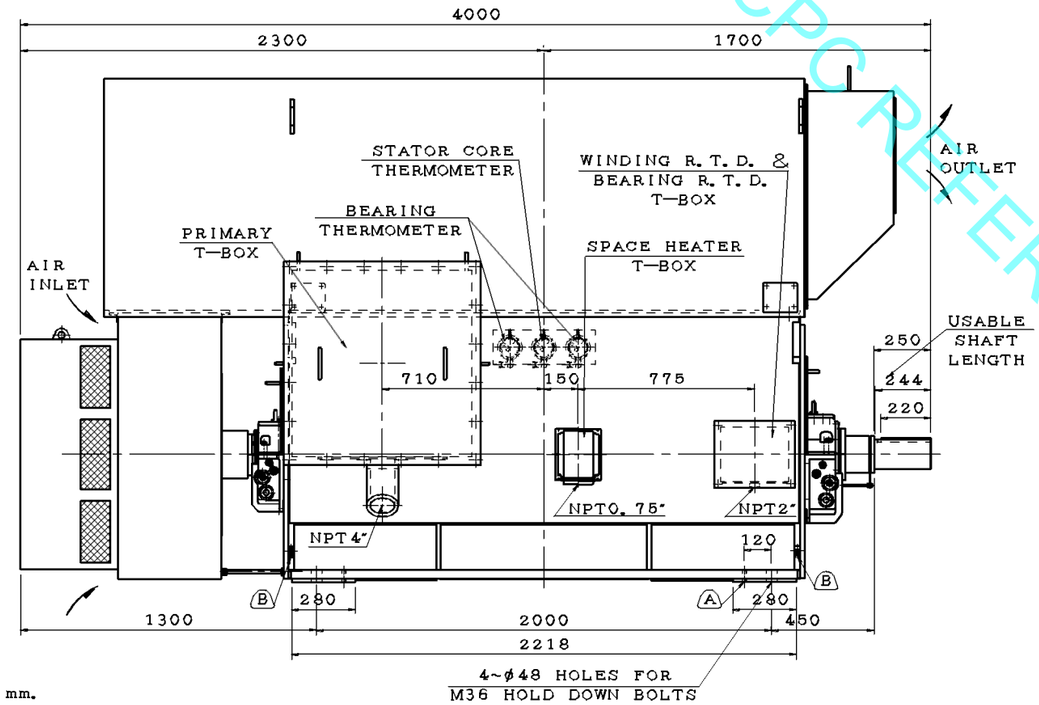
Note：[*] at the end represent revised item。

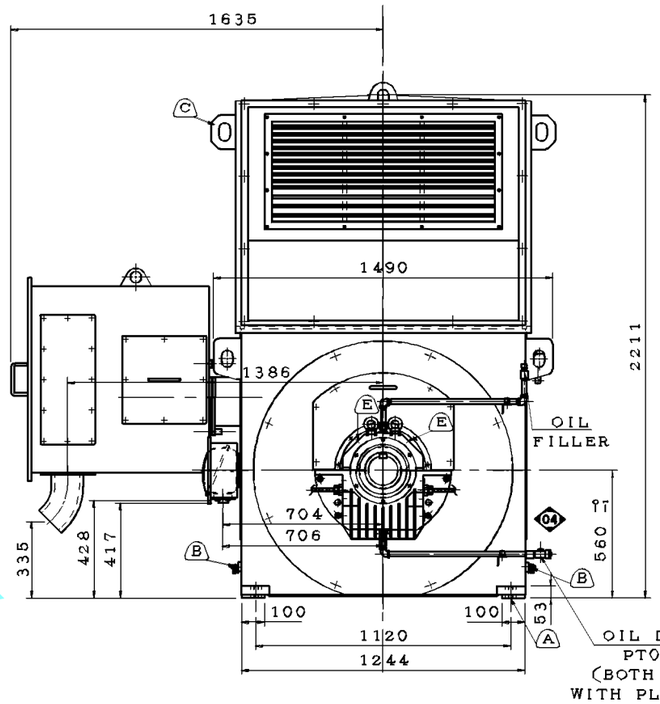
TYPE	OUTPUT HP.	OUTPUT kW.	POLE	TIME RATING	VOLTAGE V	Hz	SYN. SPEED R. P. M.
AECK-S2		1850	4	CONT	6600	60	1800

TOTALLY ENCLOSED AIR-TO-AIR COOLED TYPE. SQUIRREL CAGE ROTOR









NOTE:
1. DIMENSIONS IN mm.
2. FRAME NO. 560.
3. F CLASS INSULATION, S.F.: 1.15.
4. FOR DIRECT FLEXIBLE COUPLING.
5. SLEEVE TYPE BEARING, AXIAL THRUST LOAD NOT ALLOWED.
6. THE MOTOR ENDPLAY IS ±7 mm. A LIMITED END FLOAT TYPE COUPLING IS REQUIRED TO LIMIT ENDPLAY TO ±2.4 mm.
7. BEARING SIZE: DRIVE END: 14-125(UNINSULATED)
   NON-DRIVE END: 11-100(INSULATED)
8. THE NON-DRIVE END BEARING LINER(SHELL) IS INSULATED FROM THE HOUSING. METAL CONNECTIONS MADE TO THE BEARING SHELL MUST BE INSULATED TO PREVENT AN INSULATION SHORT CIRCUIT. METAL CONNECTIONS MADE TO THE HOUSING DO NOT NEED TO BE INSULATED.
9. BEARING LUBRICATION:SELF LUBRICATION.
   A. OIL VISCOSITY: ISO VG32 (140~160 SSU AT 100°F)
   B. OIL CAPACITY: 6.3L FOR DRIVE END.
   4.2L FOR NON-DRIVE END.
10. WITH SPACE HEATER: 1ϕ, 220V, 550W.
11. WITH WINDING R.T.D.: PT 100Ω/0°C, 6PCS. & TRANSMITTERS:ROSEMOUNT 248H A I6 N O NS F6, 6PCS.
12. WITH BEARING R.T.D.: PT 100Ω/0°C, 2PCS. TRANSMITTERS:ROSEMOUNT 248H A I6 N O NS F6, 2PCS.
13. WITH BEARING THERMOMETER:WITH N.O. CONTACTS, 2PCS.
14. WITH STATOR CORE THERMOMETER:WITH N.O. CONTACTS, 1PCS.
15. WITH SURGE CAPACITOR(6.6 kV, 3ϕ) 1PCS. LIGHTNING ARRESTER(9.0 kV, 1ϕ) 3PCS. PROVIDED IN PRIMARY T-BOX.
16. NOISE:BELOW 85dBA AT 1 METER DISTANCE ON LOAD.
17. VIBRATION:BELOW 2.5mm/s(0-P) ON BEARING HOUSING NO LOAD.
18. ENCLOSURE: IPW55, OUTDOOR SERVICE.
19. CORROSION PROOF.
20. MOTOR APPROX. WEIGHT: 9200 kg.

(A) M20 VERTICAL JACKING HOLE, ONE HOLE PER FOOT.

(B) FOUR M12 TAPPED GROUNDING PADS, WITH TWO SETS GROUNDING STUD &(R38-12) TERMINALS LOCATED DIAGONALLY OPPOSITE.

(C) FOUR LIFTING LUGS ON COOLER FOR LIFTING COOLER ONLY. DO NOT USE FOR LIFTING ENTIRE MOTOR.

(E) NPT1/4" HOLE, PROVISION FOR SUPPLY SCOPE OF CCMC.

FOR SCPC REFERENCE ONLY

DWN.	S. WANG	OCT 01 2013
CHKD.	S. WANG	OCT 11 2013
APPD.	C. SHIH	OCT 11 2013

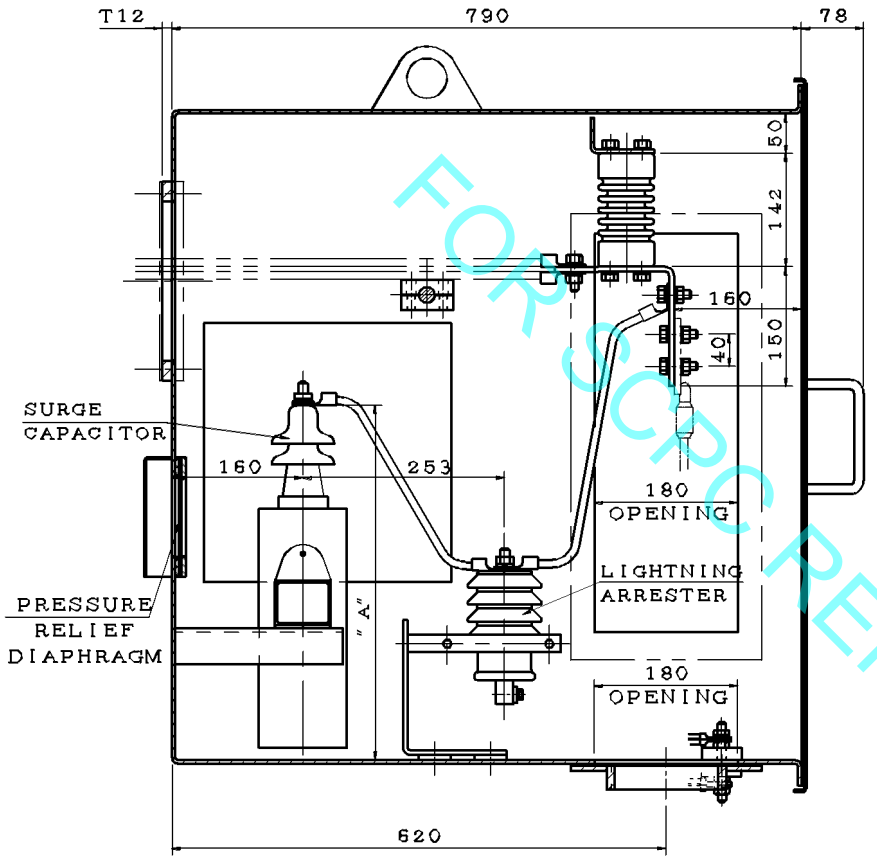
TECO
DATE

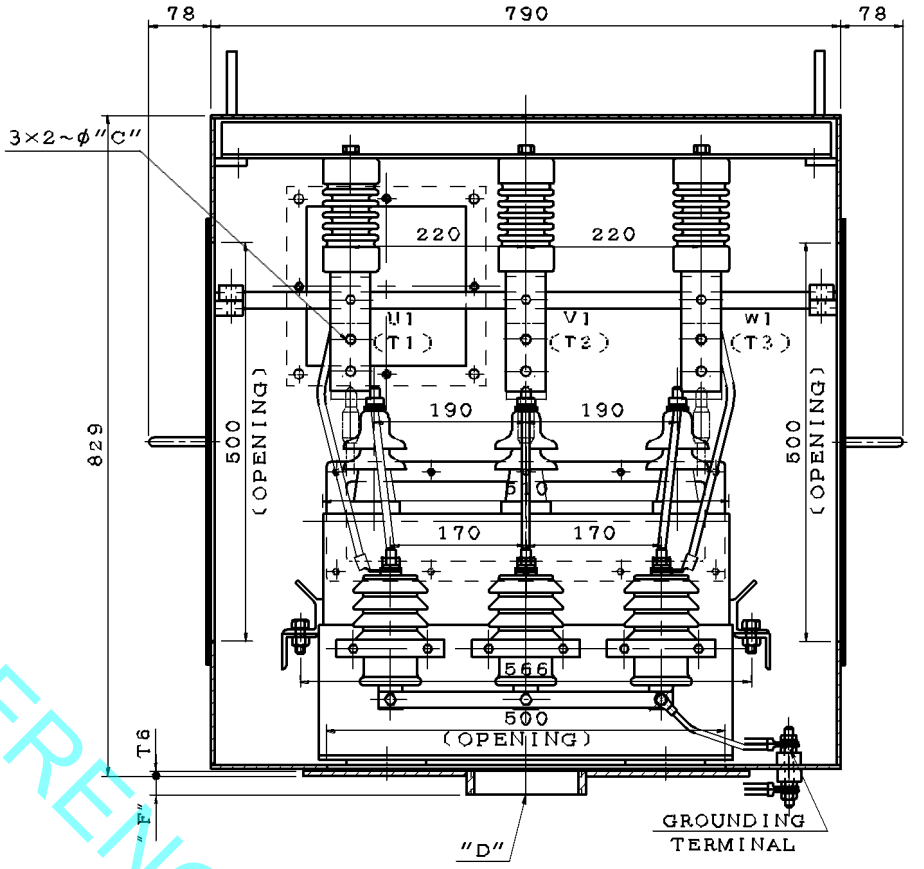
OUTLINE DIMENSIONS
3-PHASE INDUCTION MOTOR

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040S702
REV:04

第三角法





ITEM	A	B	C	D	F
01	605	150	14	NPT4"	24
02	605	150	17	NPT4"	24
03					
04					
05					

NOTE:
1. DIMENSIONS IN mm.
2. PRIMARY T-BOX.

FOR SCPC REFERENCE ONLY

DWN.	S. SHIH	JAN 27 2014
CHKD.	Y. CHIOU	JAN 27 2014
APPD.	C. SHIH	JAN 27 2014

TECO

DATE

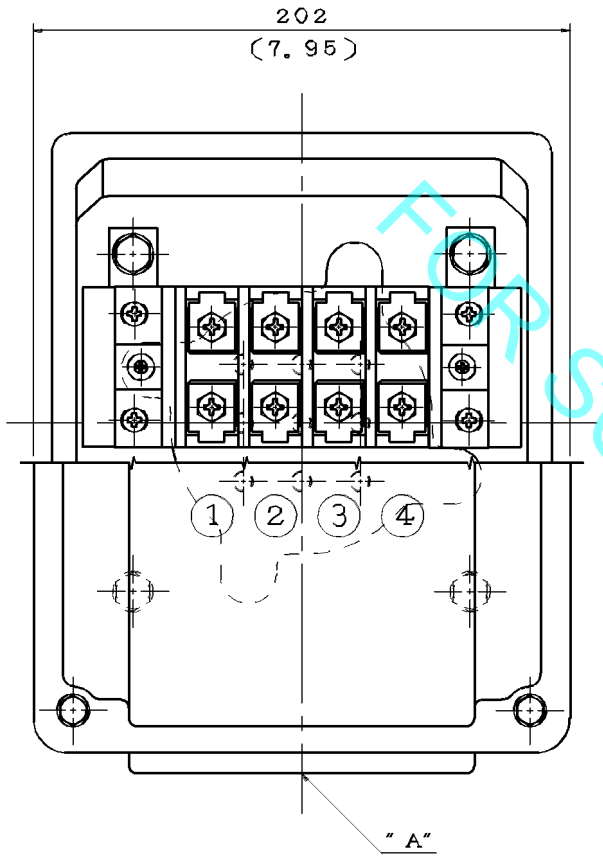
SCHEMATIC DRAWING

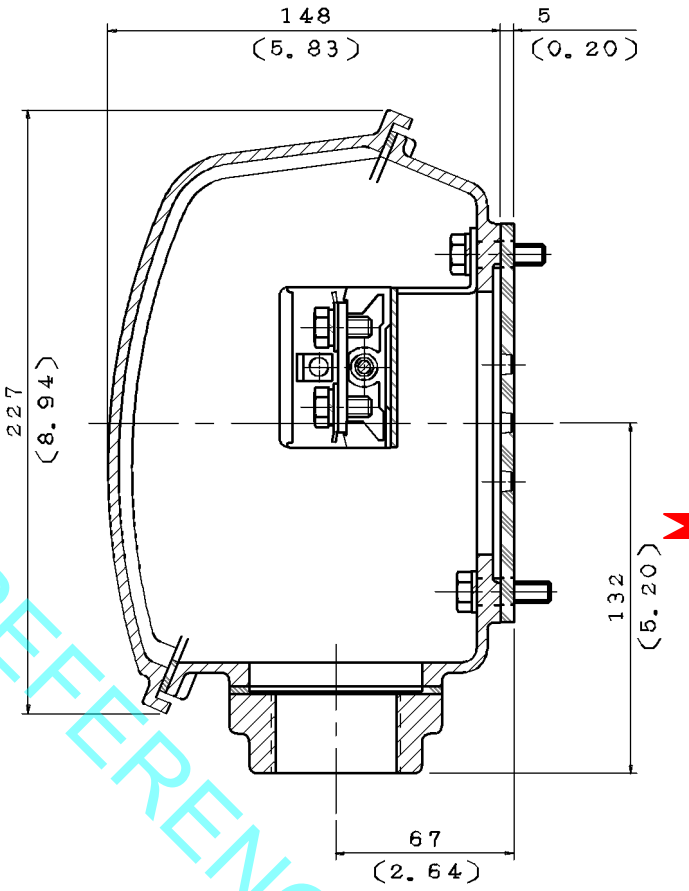
TERMINAL BOX

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

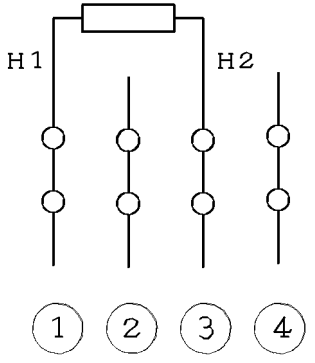
DWG NO. 4B040D700X1

REV:02





ITEM	A
01	PF-1"
02	PF-1.5"
03	PF-2"
04	NPT-1"
05	NPT-1.5"
06	NPT-2"
07	M25×1.5
08	M32×1.5
09	M50×1.5
10	NPT-0.75"
11	M20×1.5
12	NPT-0.25"
13	NPT-0.5"
14	
15	
16	
17	
18	
19	
20	



FOR SCPC REFERENCE ONLY

NOTE:
1. DIMEMSION IN mm(INCH).
2. TW-36.
3. SPACE HEATER T-BOX.
4. ENCLOSURE: IPW55.

DWN.	S. WANG	NOV 21 2008
CHKD.	S. WANG	NOV 22 2008
APPD.	C. WANG	NOV 25 2008

TECO

DATE

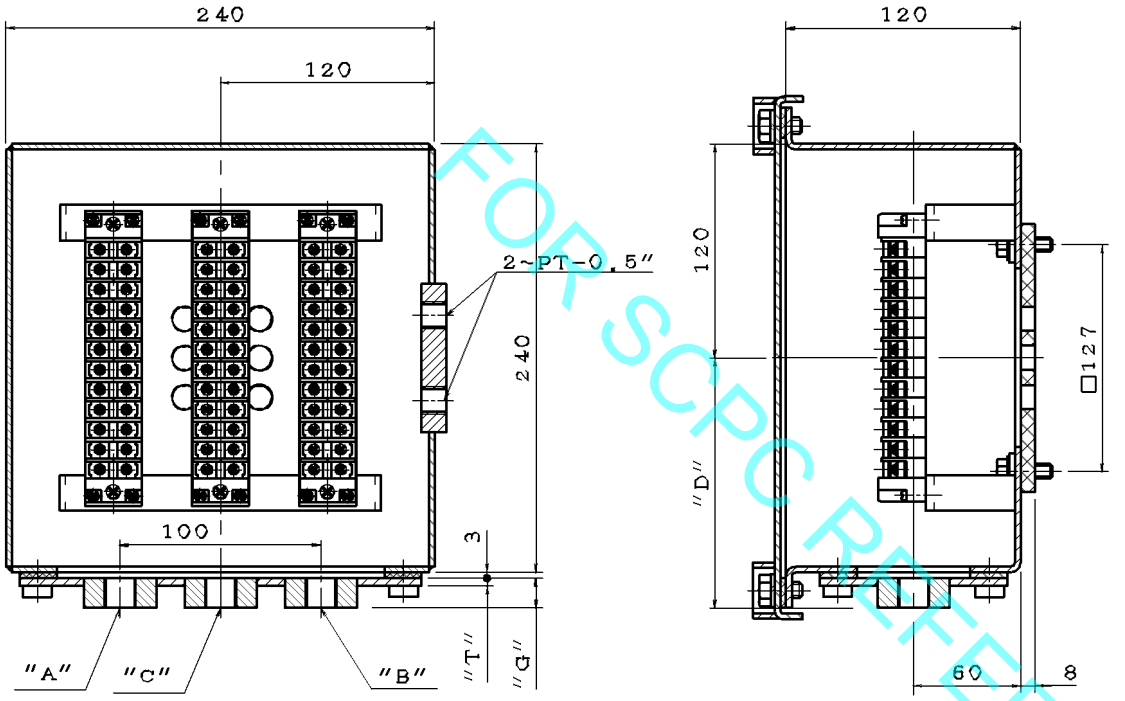
SCHEMATIC DRAWING

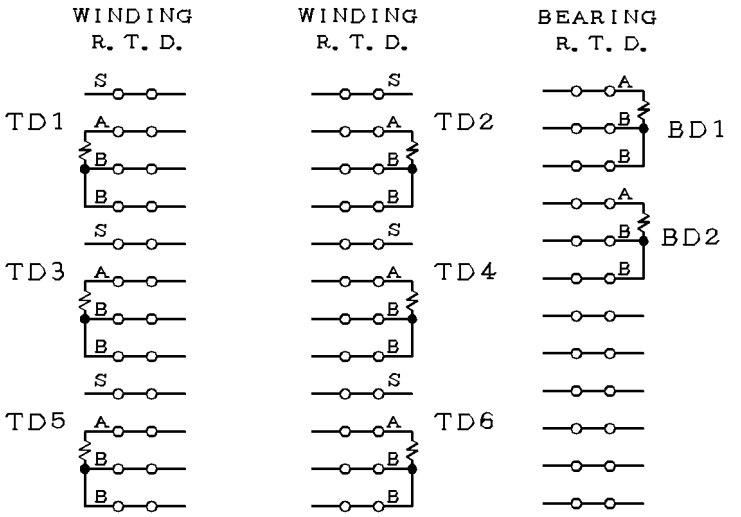
TERMINAL BOX

東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3B040Z580

REV:07





ITEM	A	B	C	D	G	T
01	NPT1"	NPT1"	0	143	20	4.5
02	0	0	NPT1.5"	143	20	4.5
03	NPT2"	NPT2"	0	143	20	4.5
04	0	0	0	127.5	4.5	4.5
05	PF2"	PF1"	0	143	20	4.5
06	0	0	PF1.5"	143	20	4.5
07	NPT1"	NPT2"	0	143	20	4.5
08	NPT2"	NPT1"	0	143	20	4.5
09						
10						
11						
12						

NOTE:
1. DIMENSIONS IN mm.
2. WINDING R.T.D. & BEARING R.T.D. T-BOX.
3. TD1 & TD2 FOR U PHASE
   TD3 & TD4 FOR V PHASE
   TD5 & TD6 FOR W PHASE
   BD1 FOR DRIVE END BEARING
   BD2 FOR NON-DRIVE END BEARING.

FOR SCPC REFERENCE ONLY

DWN.	S. WU	AUG. 05. 2003
CHKD.	H. LIU	AUG. 05. 2003
APPD.	C. WANG	AUG. 05. 2003

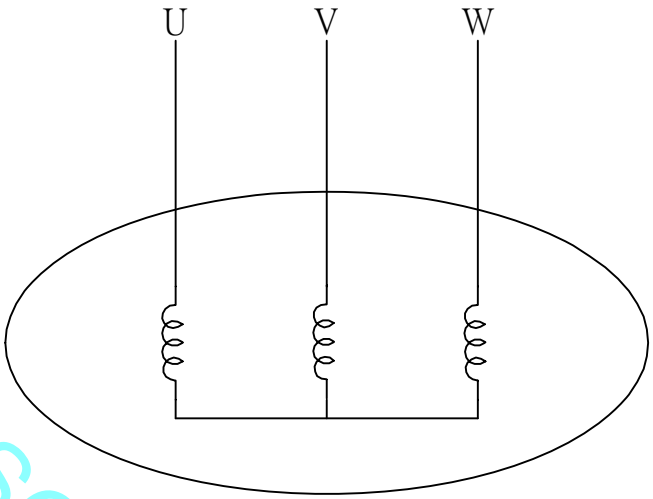
TECO
DATE

SCHEMATIC DRAWING
TERMINAL BOX

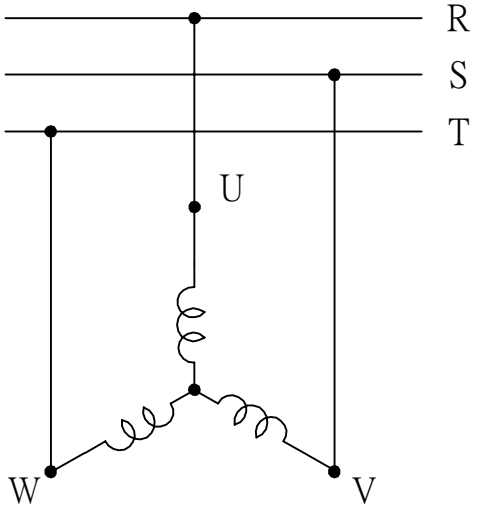
東元電機股份有限公司
TECO Elec. & Mach. Co., Ltd.

DWG NO. 3A040U362
REV:07

DATE	SCHEMATIC WYE CONN 3 LEADS	MODEL



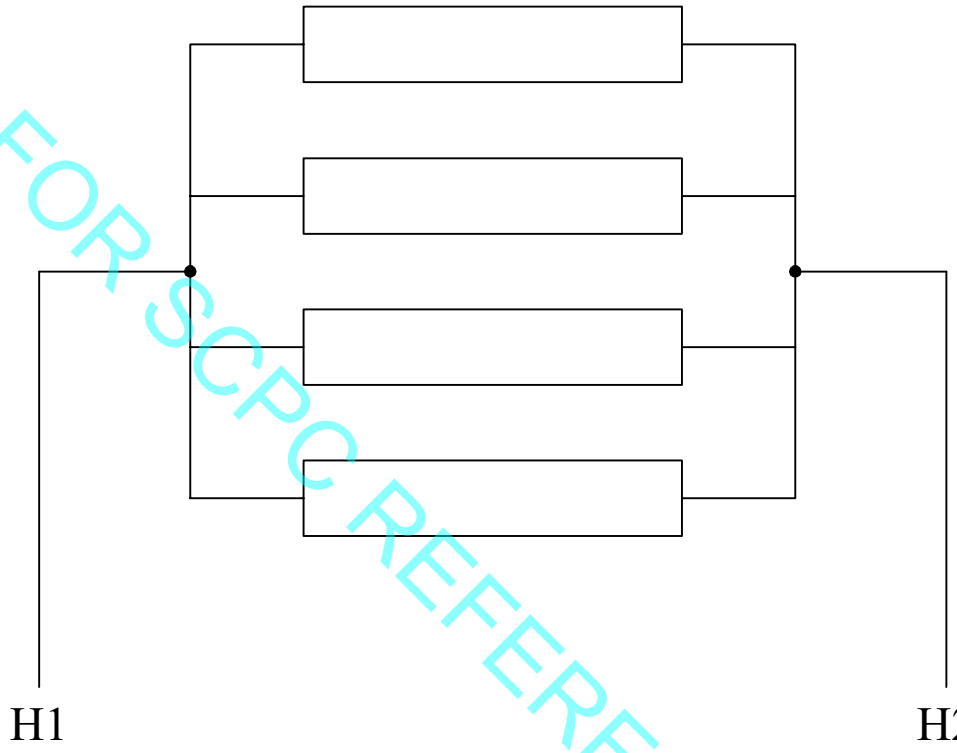
SCHEMATIC - WYE CONN - 3 LEADS

CONNECTION	ROTATION (VIEWED FROM DRIVE END)
	

FOR SCDC REFERENCE ONLY

DWN.	S.HUANG	MAR・03・2003	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	T.HSIAO	MAR・03・2003		3A061H404
APPD.	T.HSIAO	MAR・03・2003		

DATE	SCHEMATIC SPACE HEATER	MODEL



FOR SCPC REFERENCE ONLY

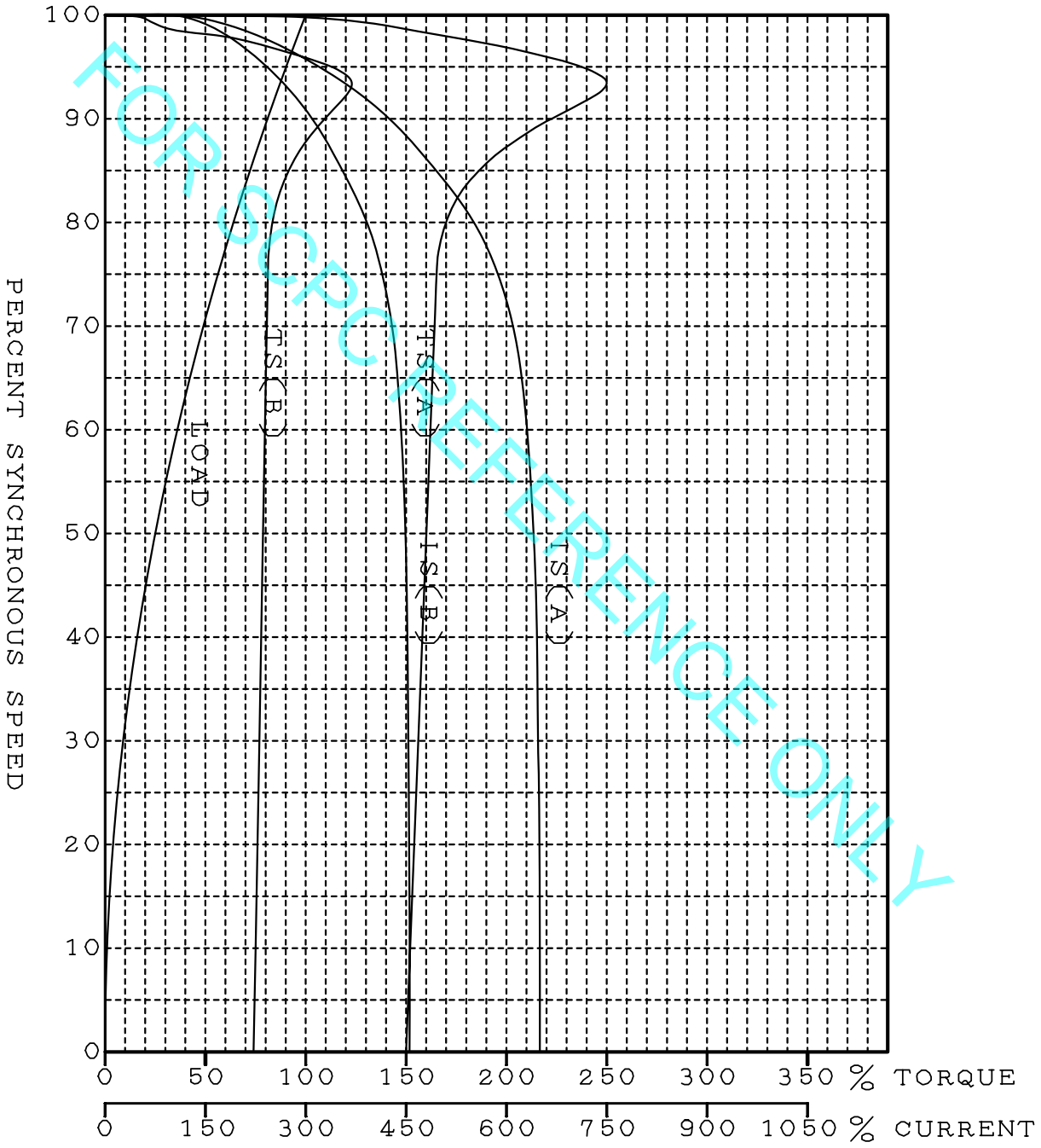
DWN.	ANDREA	JUL・03・2012	TECO ELEC. & MACH. CO., LTD.	DWG NO. REV: 00
CHKD.	HB.LIN	JUL・03・2012		3A061H573
APPD.	H.CHIANG	JUL・03・2012		

INDUCTION MOTOR STARTING CHARACTERISTICS

T—N/I—N CURVE

TYPE:AECK KW:1850 VOLTS:6600 ORDER NO:FD14A202B1
HZ:60 POLES:4 RPM(FLS):1782

VOLTS: (A)100% (B):70%



FOR SCPC REFERENCE ONLY

DWG.	HEDY	20140709
APPD.	T. HSIAO	20140709

TECO Electric & Machinery Co., Ltd.

DWG NO. REV:00
FD14A202B1
T—I—L

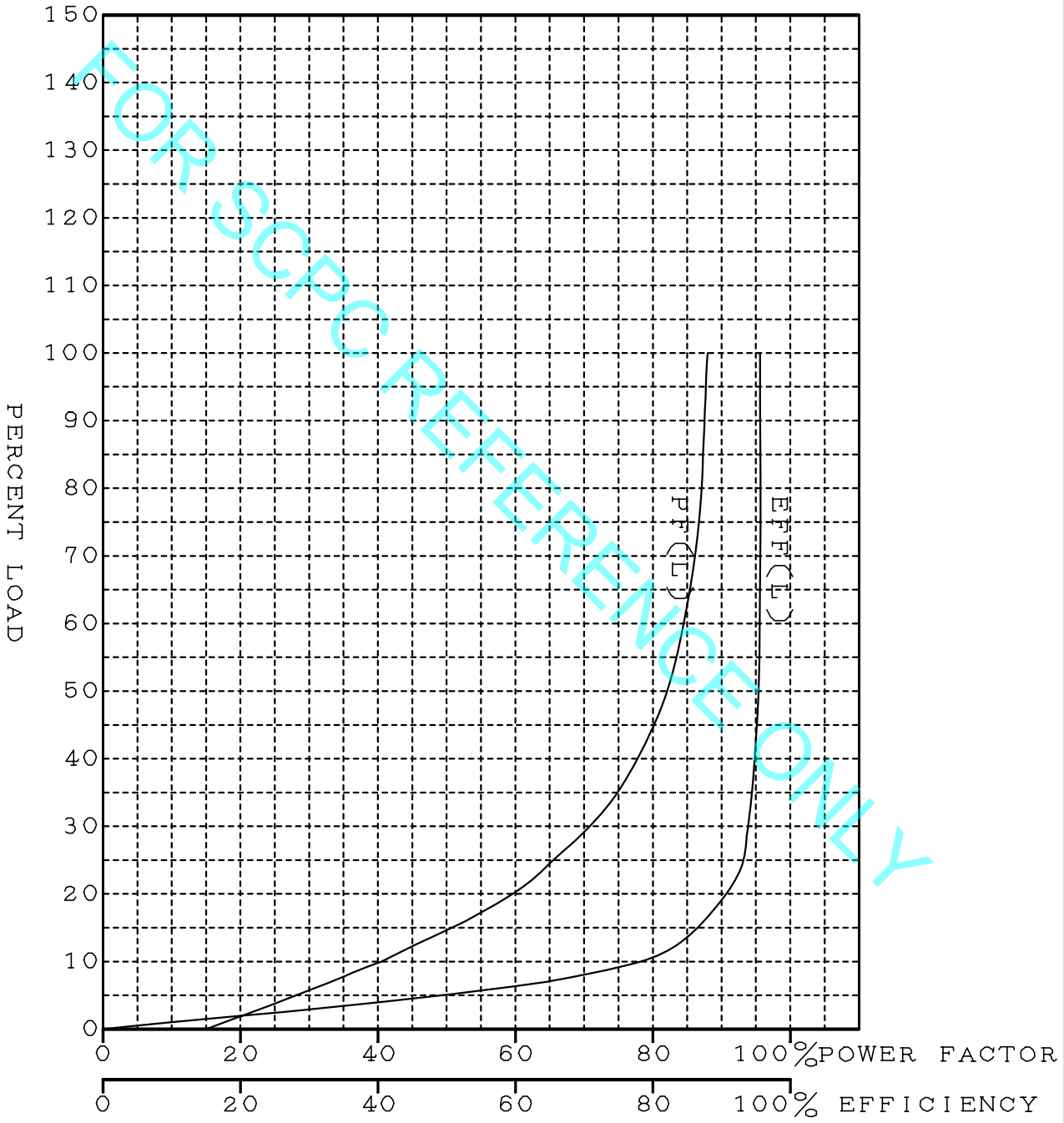
Input Ver:06

INDUCTION MOTOR STARTING CHARACTERISTICS

Efficiency & Power Factor Vs Load Curve

TYPE:AECK KW:1850 VOLTS:6600 ORDER NO:FD14A202B1

HZ:60 POLES:4 RPM(FLS):1782



FOR SCPC REFERENCE ONLY

DWG.	HB. LIN	20131028
APPD.	T. HSIAO	20131028

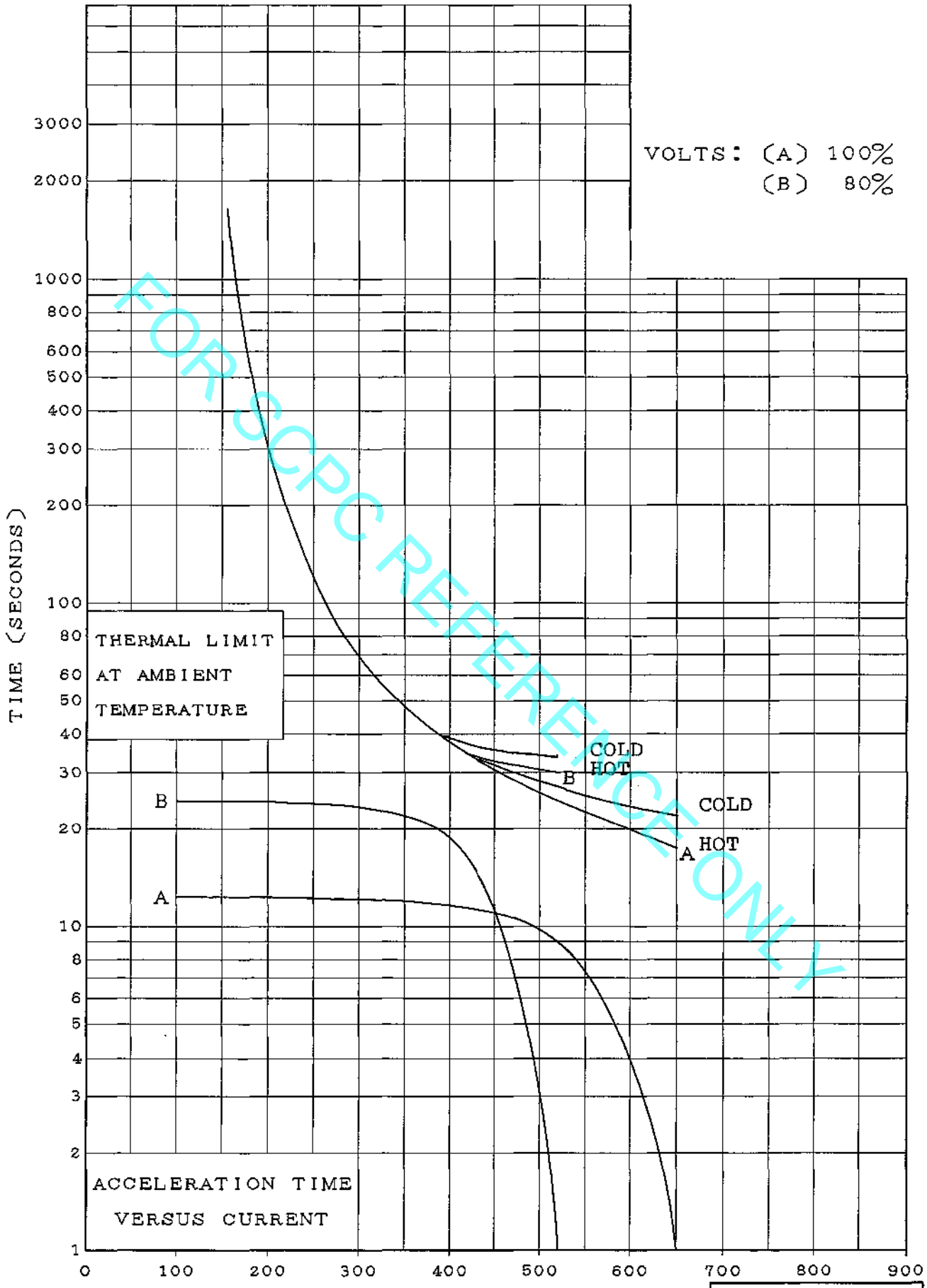
TECO Electric & Machinery Co., Ltd.

DWG NO. REV:01
FD14A202B1
E—P—L

Input Ver:01

TIME — CURRENT AND THERMAL LIMIT CURVES

LOAD GD² (KG—M²):3000 MOTOR GD² (KG—M²):260
TYPE:AECK POLE:4 KW:1850 NO. :FD14A202B1



FOR SCPC REFERENCE ONLY

DWG.	HB. LIN	20131018
APPD.	T. HSIAO	20131018

TECO Electric & Machinery Co., Ltd.

DWG NO. REV:01
FD14A202B1
TIME

Input Ver:03

建成機械股份有限公司
CHENG-CHEN MACHINERY CO,LTD.

台朔重工股份有限公司

FOR SCPC REFERENCE ONLY

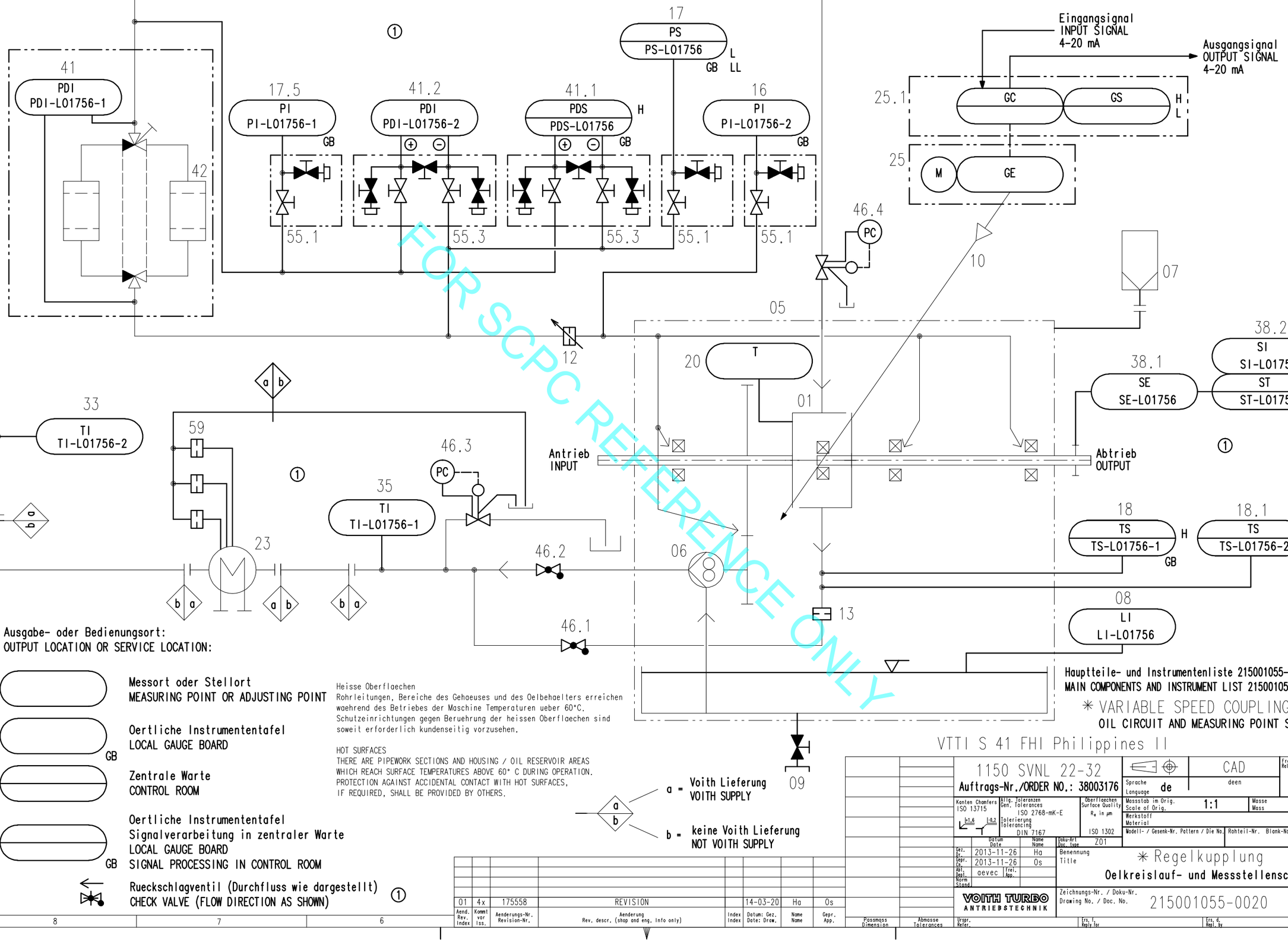
LMYP-1 Limay,Philippine

HYDRAULIC COUPLING FOR I.D.FAN

引風機液壓聯軸器資料

本文編號 RF2-0582-DOC-7-0

						CERTIFIED	
						PROJ.	
0	FOR CONSTRUCTION	陳玉樹	陳玉樹	傅泰錫	2015/01/14	MGR. _____ DATE _____	
B	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/04/17	CLINT _____ DATE _____	
A	ISSUED FOR APPROVE	陳玉樹	陳玉樹	傅泰錫	2014/01/27	PROJECT NO.	PAGE
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	**LMYP-1**	1



FOR SCPC REFERENCE ONLY

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Ausgabe- oder Bedienungsort:
OUTPUT LOCATION OR SERVICE LOCATION:

Messort oder Stellort
MEASURING POINT OR ADJUSTING POINT

Oertliche Instrumententafel
LOCAL GAUGE BOARD
GB

Zentrale Warte
CONTROL ROOM

Oertliche Instrumententafel
Signalverarbeitung in zentraler Warte
LOCAL GAUGE BOARD
SIGNAL PROCESSING IN CONTROL ROOM
GB

Rueckschlagventil (Durchfluss wie dargestellt)
CHECK VALVE (FLOW DIRECTION AS SHOWN) ①

Heisse Oberflaechen
Rohrleitungen, Bereiche des Gehaeuses und des Oelbehaelters erreichen waehrend des Betriebes der Maschine Temperaturen ueber 60°C. Schutzeinrichtungen gegen Beruehrung der heissen Oberflaechen sind soweit erforderlich kundenseitig vorzusehen.

HOT SURFACES
THERE ARE PIPEWORK SECTIONS AND HOUSING / OIL RESERVOIR AREAS WHICH REACH SURFACE TEMPERATURES ABOVE 60° C DURING OPERATION. PROTECTION AGAINST ACCIDENTAL CONTACT WITH HOT SURFACES, IF REQUIRED, SHALL BE PROVIDED BY OTHERS.

a = Voith Lieferung
VOITH SUPPLY

b = keine Voith Lieferung
NOT VOITH SUPPLY

Hauptteile- und Instrumentenliste 215001055-0030
MAIN COMPONENTS AND INSTRUMENT LIST 215001055-0030

* VARIABLE SPEED COUPLING
OIL CIRCUIT AND MEASURING POINT SCHEME

VTTI S 41 FHI Philippines II

Aend. Rev. Index	Kommt vor Iss.	Aenderungs-Nr. Revision-Nr.	Aenderung Rev. descr. (shop and eng. info only)	Index Index	Datum: Gez. Date: Draw.	Name Name	Gepr. App.
01	4x	175558	REVISION		14-03-20	Ha	Os

	Datum Date	Name Name
Gez. Dr.	2013-11-26	Ha
Gepr. Ch.	2013-11-26	Os
Abt. Dept.	aevec	Freig. App.
Norm Stand.		

1150 SVNL 22-32
Auftrags-Nr./ORDER NO.: 38003176

CAD
deen
Sprache Language: de
Freigabevermerk Released for microf.

Kanten Chamfers ISO 13715
Allg. Toleranzen Gen. Tolerances ISO 2768-mK-E
Oberflaechen Surface Quality R_a in µm
Tolerierung Tolerancing DIN 7167
ISO 1302

Massstab im Orig. Scale of Orig. 1:1
Masse Mass
Werkstoff Material
Modell- / Gesenk-Nr. Pattern / Die No.
Rohteil-Nr. Blank-No.

Doku-Art Doc. type Z01

Benennung Title
* Regelkupplung
Oelkreislauf- und Messstellenschema

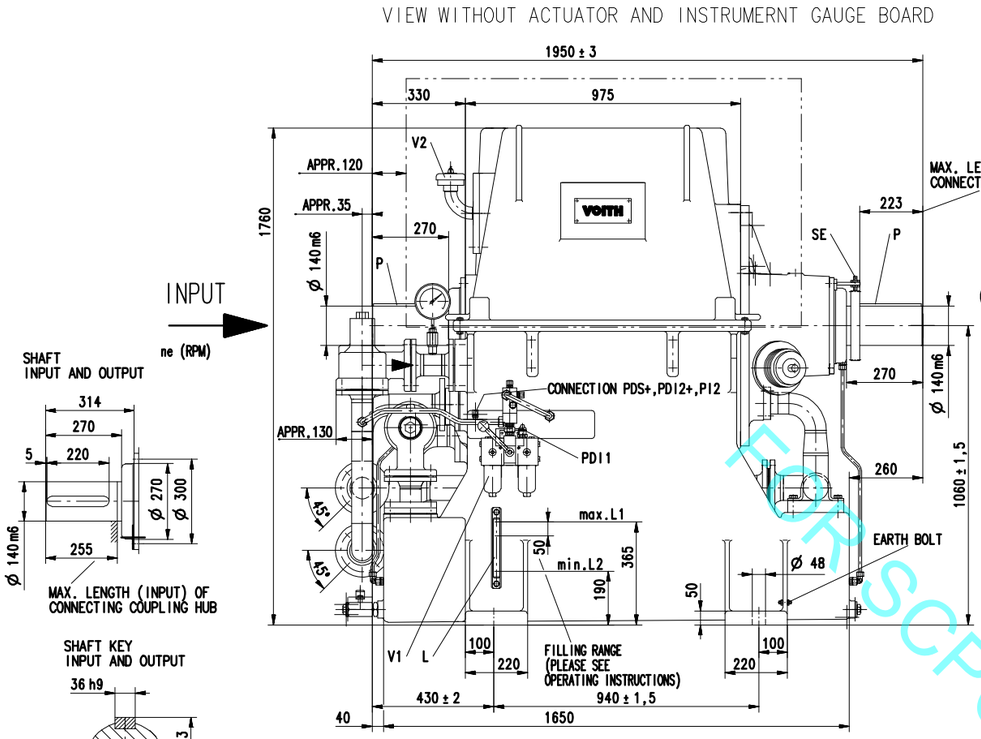
VOITH TURBO ANTRIEBSTECHNIK

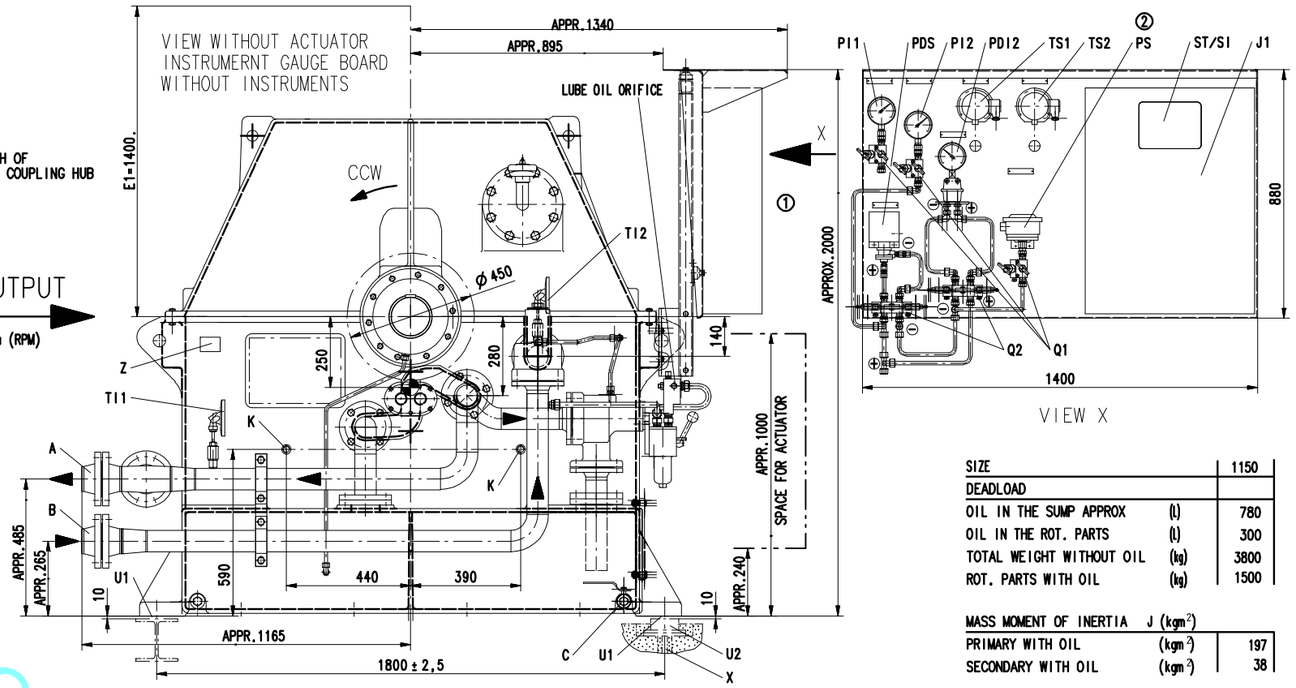
Zeichnungs-Nr. / Doku-Nr. Drawing No. / Doc. No. 215001055-0020

Blatt Sheet 1
v. 1 Bl.

Passmass Dimension
Abmasse Tolerances
Ursprr. Refer.
Ers. f. Reply for
Ers. d. Repl. by

DIN A2





SIZE		1150
DEADLOAD		
OIL IN THE SUMP APPROX	(l)	780
OIL IN THE ROT. PARTS	(l)	300
TOTAL WEIGHT WITHOUT OIL	(kg)	3800
ROT. PARTS WITH OIL	(kg)	1500

MASS MOMENT OF INERTIA J (kgm²)		
PRIMARY WITH OIL	(kgm²)	197
SECONDARY WITH OIL	(kgm²)	38

DYNAMIC FOUNDATION LOADING f dyn (N)		
f dyn ≤ ne/60	(Hz)	
F dyn = y (ne/1000) ²	(N)	y=1000

VIBRATIONS VDI 2056 GROUP T

SCOOP TUBE STROKE		
DIRECTION OF ROTATION "CCW"	(mm)	305

NO. FROM OIL CIRCUIT AND MEASURING POINT SCHEME 215001055-0020		
	A	WELD-NECK FLANGE TO HEAT EXCHANGER 3" ANSI B16.5 150 LBS - MATERIAL: ASTM A182 F316L
	B	WELD-NECK FLANGE FROM HEAT EXCHANGER 3" ANSI B16.5 150 LBS - MATERIAL: ASTM A182 F316L
09	C	OIL DISCHARGE SET G 1 1/4
	D	THREAD FOR ADJUSTING SCREW M16
	E1	CLEARANCE FOR ROT. PARTS REMOVAL
25	G	ACTUATOR CONTRAC RHD 500-10
25.1	G1	ELECTRONICS FOR ACTUATOR EBN 853
	J1	TERMINAL BOX
	K	THREAD CONNECTION G 3/4 FOR HEAT EXCHANGER VENTING
08	L	OIL LEVEL SIGHT GLASS
	L1	OIL LEVEL AT SCOOP TUBE POSITION 0 % DURING OPERATION
	L2	OIL LEVEL AT SCOOP TUBE POSITION 100 % DURING OPERATION
	O	CLEARANCE FOR SCOOP TUBE REMOVAL
	P	KEY AS 36x20x220 DIN 6885
46.3	PC1	OVERFLOW VALVE
46.4	PC2	PRESSURE RELIEF VALVE
41	PDI1	VISUAL MAINTENANCE INDICATOR
41.2	PDI2	DIFFERENTIAL PRESSURE INDICATOR
41.1	PDS	DIFFERENTIAL PRESSURE SWITCH
17	PS	PRESSURE SWITCH
16	PI1	PRESSURE INDICATOR (LUBE OIL PRESSURE)
17.5	PI2	PRESSURE INDICATOR (WORKING OIL PRESSURE FLOW UPSTREAM OF COUPLING)
55.1	Q1	2-VALVE MANIFOLD
55.3	Q2	5-VALVE MANIFOLD
46.1	R1	NON RETURN VALVE
46.2	R2	NON RETURN VALVE
38.1	SE	SPEED MEASURING PICKUP
38.2	ST/SI	SPEED MEASURING TRANSDUCER AND SPEED INDICATOR (INSTALLED IN TERMINAL BOX "J1")
18	TS1	CAPILLARY DIAL THERMOMETER WITH CONTACTS (WORKING OIL TEMPERATURE AT SCOOP TUBE OUTLET - ALARM)
18.1	TS2	CAPILLARY DIAL THERMOMETER WITH CONTACTS (WORKING OIL TEMPERATURE AT SCOOP TUBE OUTLET - TRIP)
35	TI1	BIMETALLIC THERMOMETER (WORKING OIL TEMPERATURE FLOW UPSTREAM OF HEAT EXCHANGER)
33	TI2	BIMETALLIC THERMOMETER (WORKING OIL TEMPERATURE FLOW DOWNSTREAM OF HEAT EXCHANGER)
	U1	SHIMS FOR FOUNDATION 10 mm
	U2	FOUNDATION RAIL / FOUNDATION PLATES - NOT VOITH SUPPLY
42	V1	DOUBLE FILTER
07	V2	VENT FILTER
	W	CENTRE OF GRAVITY ⊕ WITHOUT OIL
	X	STONE BOLT - NOT VOITH SUPPLY
	Z	NAMEPLATE

THE OIL VOLUME OF HEAT EXCHANHER AND PIPINGS IS NOT INCLUDED IN THE FILLING CAPACITY.

PIPE CONNECTION TOLERANCE TO EN ISO 13920-D

MOUNT THE COVERS FOR ROTATING PARTS ACCORDING TO THE LEGAL REGULATIONS!

THE TURBO COUPLING IS A MACHINE WITH PIPINGS, HEAT EXCHANGERS AND ONE-WALL SUMP FOR MINERAL OIL. ADDITIONAL REGULATIONS CONCERNING THE ENVIRONMENT, IF ANY, HAVE TO BE DETERMIND AT SITE!

HOT SURFACES
THERE ARE PIPEWORK SECTIONS AND HOUSING / OIL RESERVOIR AREAS WHICH REACH SURFACE TEMPERATURES ABOVE 60° C DURING OPERATION. PROTECTION AGAINST ACCIDENTAL CONTACT WITH HOT SURFACES, IF REQUIRED, SHALL BE PROVIDED BY OTHERS.

VTTI S 41 FHI Philippines II

Rev.	Iss.	Revision-Nr.	Rev. descr. (shop and eng. info only)	Index	Date: draw.	Name	App.
02	1x	176509	REVISION		14-04-09	Ha	Os
01	1x	175558	REVISION		14-03-20	Ha	Os

140 H6	+0,025 / 0	Dr.	2013-11-26	Ha
140 m6	+0,040 / +0,015	Ch.	2013-11-26	Os
36 h9	0 / -0,062	Dept.	aevec	App.
36 JS9	+0,031 / -0,031	Stand.		
Dimension	Tolerances	Refer.		

1150 SVNL 22-32
ORDER NO.: 38003176

CAD | Released for manuf.
Language: en | de, en
Chamfers ISO 13715 | Gen. tolerances ISO 2768-mK | Surface quality Ra in µm | Scale of orig. 1:10 | Mass 0 kg
Material
Tolerancing ISO 1101 DIN 7167 | ISO 1302 | Pattern / Die No. | Blank-No.
Doc. type Z01

Title: VARIABLE SPEED COUPLING
ASSEMBLY PLAN

VOITH TURBO ANTRIEBSTECHNIK
Drawing No. / Docu-No. 215001055-0010
Sheet 1 of 1 Sh.
Repl. for | Repl. by

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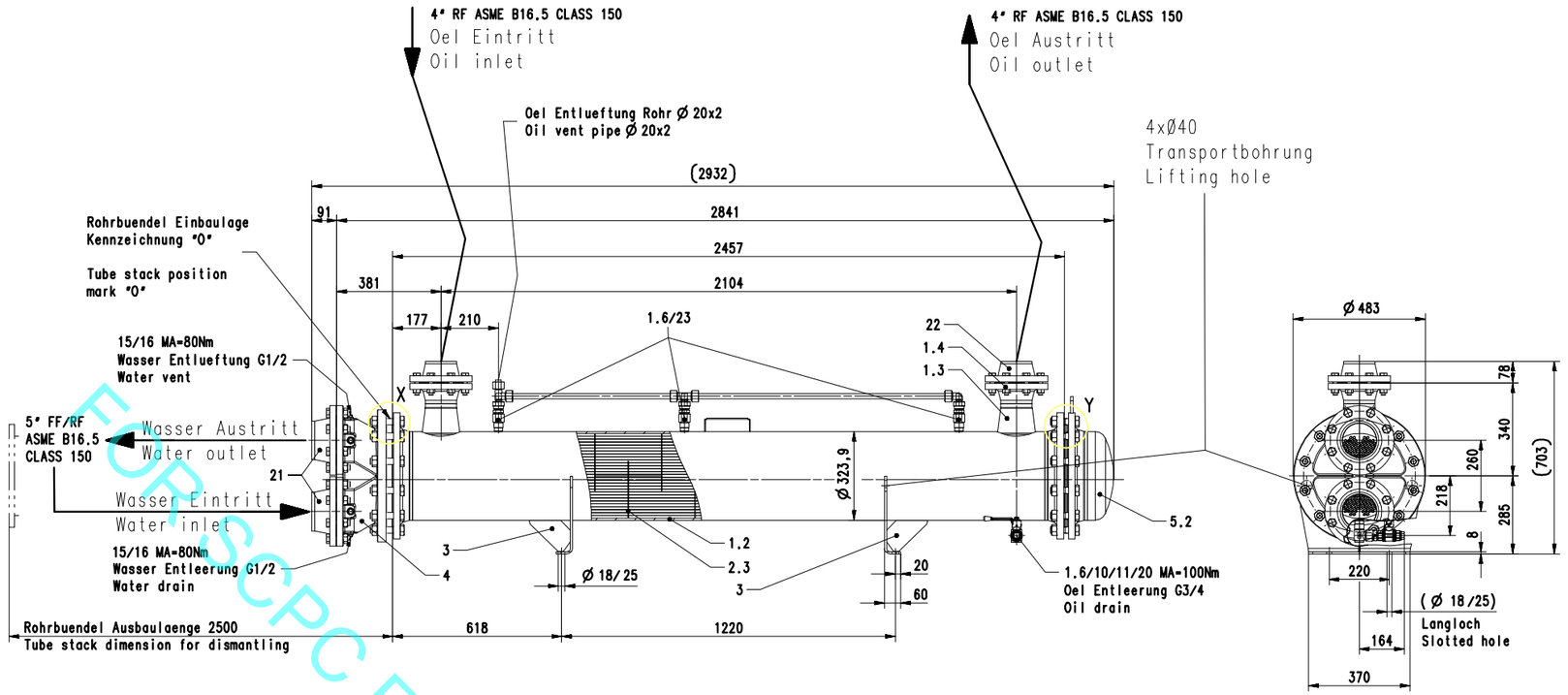
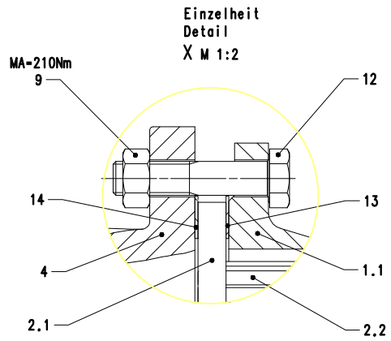
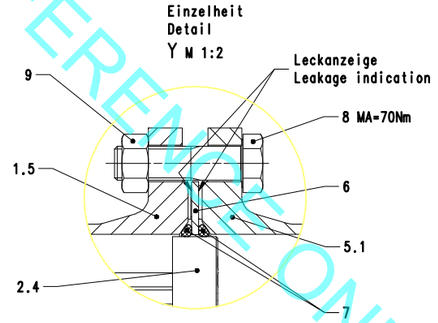
Pos	Benennung Designation	Groesse Size	DIN/EN/ISO	Werkstoff Material
	Waermeaustauscher Heat exchanger			
	Mantel Baugruppe Shell			
1.1	Vorschweissflansch Welding neck flange	DN300PN16	EN 1092-1	P245GH DIN EN 10222-2
1.2	Mantelrohr Shell pipe	Ø324x7	DIN EN 10220	P235 DIN EN 10217
1.3	Reduzierstueck Reduction	Ø139.7x114.3	DIN 2616-2	P235TR2 EN 10216-1
1.4	Vorschweissflansch Welding neck flange	4" RF CLASS 150	ASME B16.5	C 21
1.5	Vorschweissflansch Welding neck flange	DN300PN16	EN 1092-1	P245GH DIN EN 10222-2
1.6	Anschweissauge Socket	G 3/4 Ø33.7x6.3x32	DIN EN 10220	P235GH EN 10216-2
	Rohrbuendel Baugruppe Tube stack			
2.1	Rohrboden starr Stationary tube sheet		EN 1982	CuAl10Ni CC333G
2.2	Rohr Tube	Ø10	DIN EN 10217-7	X5CrNi1810 (1.4301)
2.3	Umlenkblech Baffle		DIN 1541	DC 01 DIN EN 10130
2.4	Rohrboden beweglich Floating tube sheet		EN 1982	CuAl10Ni CC333G
3	Befestigung angeschweisst Support welded			S235JR DIN EN 10025
4	Anschlussflansch Channel			EN-GJL-250 DIN EN 1561
5	Umlenkkammer Bonnet			Stahl Steel
5.1	Vorschweissflansch Welding neck flange	DN300PN16	EN 1092-1	P245GH DIN EN 10222-2
5.2	Kloepperboden Torispherical	Ø324x8	DIN 28011	P265GH EN 10028-2
6	Blockring Pressure ring		DIN 1543	S235JR DIN EN 10025
7	RDR O-Ring	Ø300 x 8		NBR 80
8	Skt. Schraube Hex. bolt	M24 x 90	ISO 4014	8.8-A
9	Skt. Mutter Hex. nut	M24	ISO 4032	8-A
10	Verschlussschraube Screw plug	G 3/4	DIN 910	Ms
11	Dichtring Sealing ring	A27x32	DIN 7603	Cu
12	Skt. Schraube Hex. bolt	M24 x 110	ISO 4014	8.8-A
13	Flachdichtung Flat gasket	DN300PN16	DIN 2690	Klinger - SIL C 4400
14	Flachdichtung mit Steg Flat gasket with web	DN300PN16		Klinger - SIL C 4400
15	Verschlussschraube Screw plug	G 1/2	DIN 910	Ms
16	Dichtring Sealing ring	A21x26	DIN 7603	Cu
17	Schildtraegerblech Nameplate bracket		DIN 1542	S235JR DIN EN 10025
18	Leistungsschild Nameplate			X2CrNiMo17122 (1.4404)
19	Zylinderkopfniet Socket head rivet			Edelstahl Stainless steel
20	Kugelhahn Ball valve	G 3/4		Ms
21	Vorschweissflansch Welding neck flange	5" RF CLASS 150	ASME B16.5	Edelstahl Stainless steel
22	Vorschweissflansch Welding neck flange	4" RF CLASS 150	ASME B16.5	Edelstahl Stainless steel
23	Blende Orifice plate	G 3/4 Ø 3,0		9SMn28K







Heisse Oberflaechen
Teile des Waermeaustauschers und evt. vorhandener Anbauteile koennen waehrend des Betriebes Temperaturen ueber 60°C erreichen. Schutzeinrichtungen gegen Beruehrung der heissen Oberflaechen sind nicht im Voith Lieferumfang enthalten.

Hot surfaces
There are sections of the heat exchanger and possibly mounted accessories which may reach temperatures above 60°C during operation.
Protection devices, if required, shall be provided by others.

Assembly plan
* Heat exchanger

* Der Chloridgehalt des Kuehlwassers darf 50 mg/l (ppm) nicht ueberschreiten.

* The chloride content of the cooling water shall not exceed 50 mg/l (ppm).

VTTI S 41 Philippines II

Herstell-Nr.: Serial NO.:	Betriebsdaten Operating data	Mantelraum Shell side	Rohrraum Tube side
8210864- 8210867	Medium Fluid	Oel Oil	* Wasser * Water
	Betriebsvolumenstrom V Operating flow rate V	38.1 m³/h	53.9 m³/h
	Mindestvolumenstrom gegen Korrosion Min. Flow rate against of corrosion		35 m³/h
	Ein-/Austrittstemperatur Inlet/Outlet temperature	92 °C / 65 °C	50 °C / 58 °C
Stueckzahl: Quantity: 4	Zul. Betriebsueberdruck Max. Admissible working pressure	10 bar	10 bar
	Pruefueberdruck Test pressure	15 bar	15 bar
	Zul. Betriebstemperatur Max. Admissible working temperature	120 °C	70 °C
Gewicht leer / gefuellt Weight empty / filled 603 kg / 776 kg	Inhalt Contents	95 L Litres	78 L Litres
	Einbau- und Betriebsanweisung Installation and operating instructions	3.626-007101	

Typ:L1 N4 11 17 13M
ORDER-NO.: 38003176
CAD
Sprache deen de en
Freigabevermerk Released for manf.
Kanten Chamfers ISO 13715
Allg. Toleranzen Gen. Tolerances ISO 2768-mK-E
Oberflächen Surface Quality R_z in µm
Toleranzfeld Tolerance DIN 7167
ISO 1302
Maßstab im Orig. Scale of Orig. 1:10
Masse Mass
Werkstoff Material
Datum Date 2013-12-03 zzphaf
Name Name sga
Gez. Dr. Dwn.
Gepr. Chkd.
Norm Appd.
Benennung Title
Einbauplan
* Waermeaustauscher
Zeichnungs-Nr. / Doku-Nr. Drawing No. / Doc. No. 215001055-0780
Blatt Sheet 1
VOITH TURBO ANTRIEBSTECHNIK

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VOITH TURBO

POWER TRANSMISSION
Variable Speed Drives

Voithstrasse 1
D-74564 Crailsheim
Tel.: 07951 / 32 - 0
Fax : 07951 / 32 - 650

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Code	:	VTTI S 41 FHI Philippines II
VC product	:	1150 SVNL 22-32
Order number	:	38003176
Drawing number	:	215001055-0050en
Designation	:	Variable speed turbo coupling - Logic Flow Chart
Symbolic files	:	
File	:	J:\Turbo\ai-VTCR\air\BA-Doku\215\215001001-1500\1055\215001055-0050en.pdf
Department	:	aevec

Issued on : 13-12-02 by : MMar Number of pages : 17

				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Cover sheet	Order No.: 38003176	
				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 1
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7	Description of symbols part 3
8	Description of symbols part 4
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10	Start procedure part 1
11	Start procedure part 2
12	Monitoring of lube oil pressure
13	Monitoring of pressure drop across lube oil duplex filter
14	Monitoring of oil temperature
15	Monitoring of actuator
16	Shutdown procedure part 1
17	Shutdown procedure part 2

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Revision list

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SPACE FOR INPUT SIGNALS

SPACE FOR LOGIC SYMBOLS AND GENERAL COMMENTS

SPACE FOR OUTPUT SIGNALS

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GENERAL COMMENTS

Identification of symbols for Logic Flow Chart refer to the following 4 pages of this document.

For item numbers, see Oil Circuit and Measuring Point Scheme 215001055-0020.

Before test run and after each overhaul all switch points and connections have to be checked or adjusted.

For switch points, see Main Components and Instrument List 215001055-0030.

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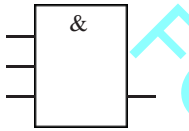
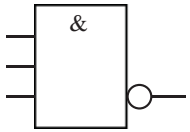
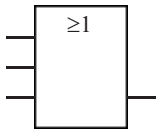
This is the input side



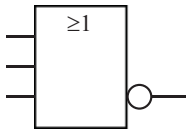
AND
The output will be in the 1-state if all the connected inputs are in their defined 1-state.



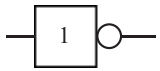
NAND
The output will be in the 0-state if all the connected inputs are in their defined 1-state.



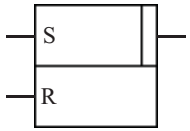
OR
The output will be in the 1-state if any of the connected inputs is in its defined 1-state.



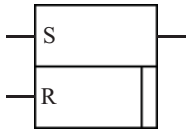
NOR
The output will be in the 0-state if any of the connected inputs is in its defined 1-state.



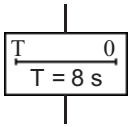
NOT (Negator/Inverter)
The output will be in the 0-state if the input is in the defined 1-state.



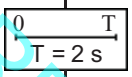
Bistablelogic function symbol with prevailing S input



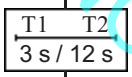
Bistable logic function symbol with prevailing R input



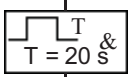
ON - delay (8 sec)



OFF - delay (2 sec)



ON - delay (3 sec)
and
OFF - delay (12 sec)



The output pulse time is always 20 sec irrespective of the input signal duration.

This is the output side.

The letter indicates the signal, the number indicates the page where the signal is going to.

K6

Example of a connection to another page

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				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Description of symbols part 1	Order No.: 38003176	
				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 5
Index	Rev. No.	Date	Name	Checked						17 P.

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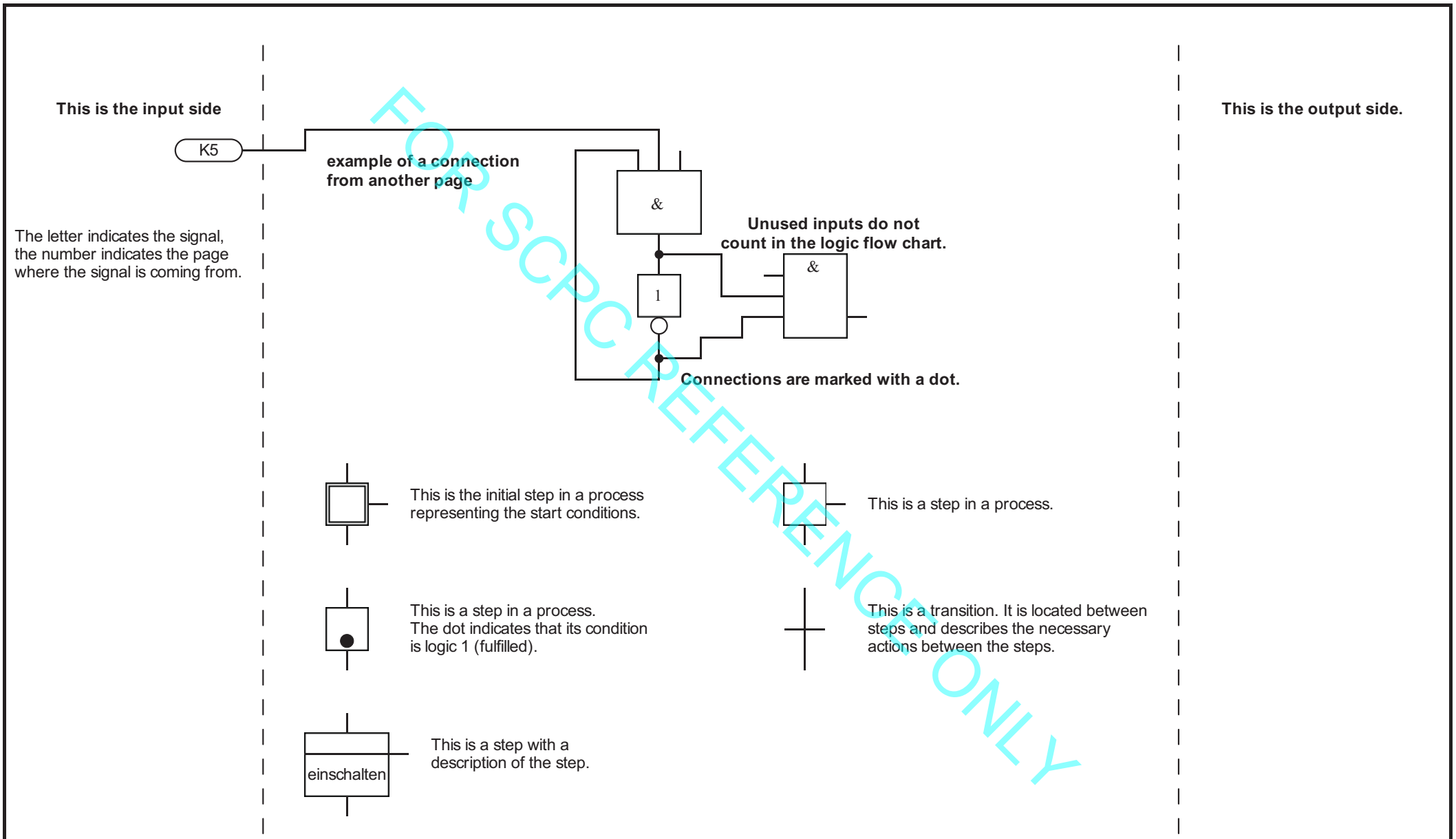

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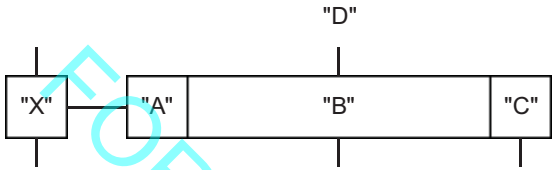
This is the input side

This is the output side



The symbol on the right-hand side to the step symbol is an action symbol.
The action symbol could exist of 3 fields (field "A", "B" and "C").

Commands above and from left of the action symbol are input signals
and are to be treated like an AND connection.
Commands beneath the action symbol are output signals.

Field "A" or "C" are to be used only if required.

Field "B" describes the action if step "X" is fulfilled.

Field "A" descibes the correlation between the step "X" and the action "B",
the following letters are to be used:

S	stored
D	delayed
L	time limited
P	pulse shape
C	conditional

Letter combinations can be used, therefore the order the letters are written is important.

For example: "A" = DSL The input signal "X" will be delayed, stored and the action "B" limited to a certain time.

Field "C" describes the feedback of the command "B".
The following symbols can be used:

A	Command output
R	Command effect is achieved (response control)
X	Fault signal, command effect is not achieved
1,2...	No. of feedback

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For example:

"A" = DSL — The input signal "X" will be delayed, stored and the action "B" limited to a certain time.

If letter C (conditional) is used the action is only enabled if the condition described under "B" is fulfilled.

For example:

"A" = SLC
"B" = lube oil pump ON
"D" = oil level ok

The input signal "X" will be stored and the lube oil pump will be switched ON for a limited time if the oil level is ok.

Field "C" indicates the feedback of the action to be carried out.
In the above example "C" would indicate logic 1 if the lube oil pump is ON.

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Index	Rev. No.	Date	Name	Checked						17 P.

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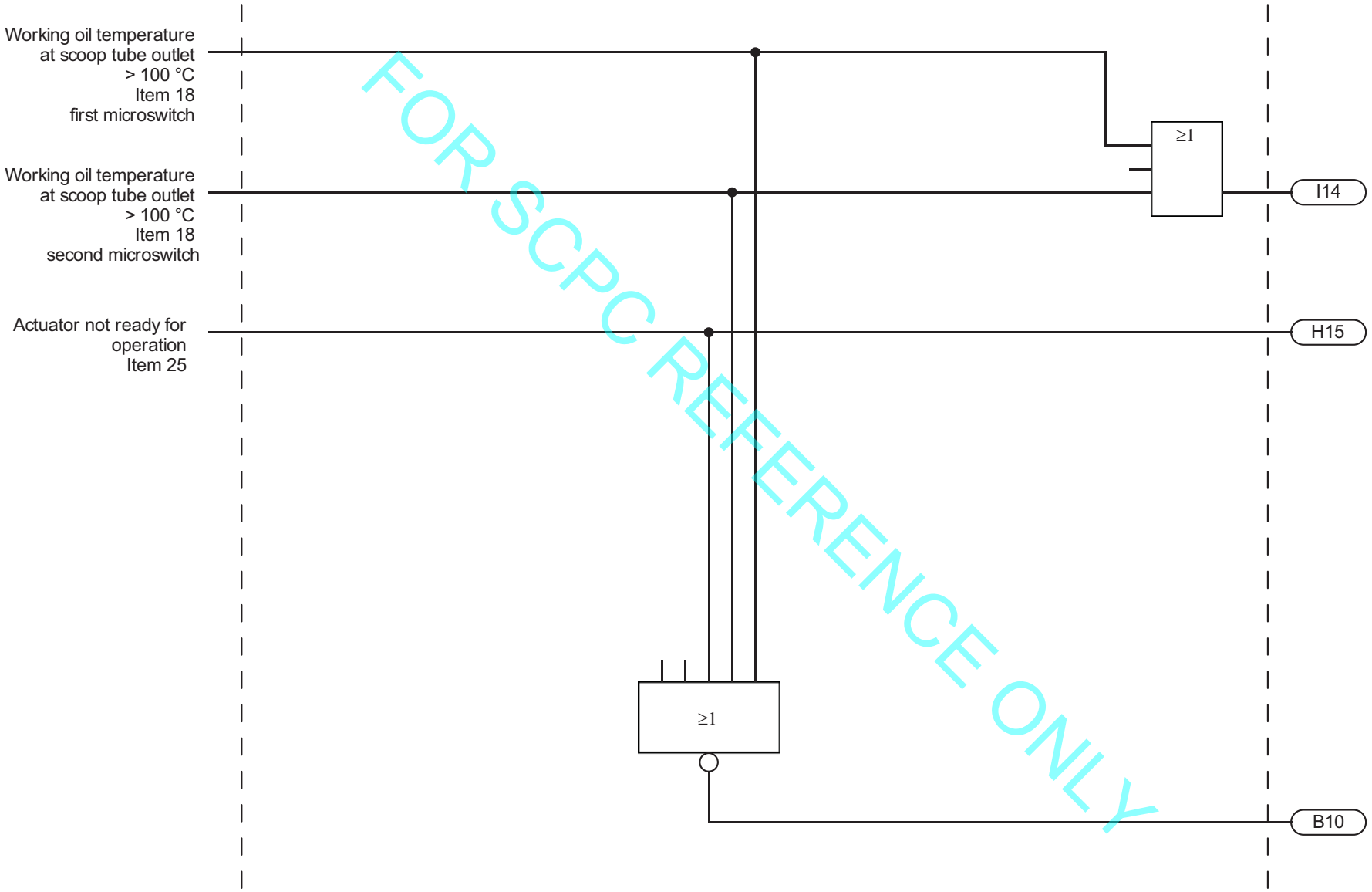

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				Editor	MMar					
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Index	Rev. No.	Date	Name	Checked						17 P.

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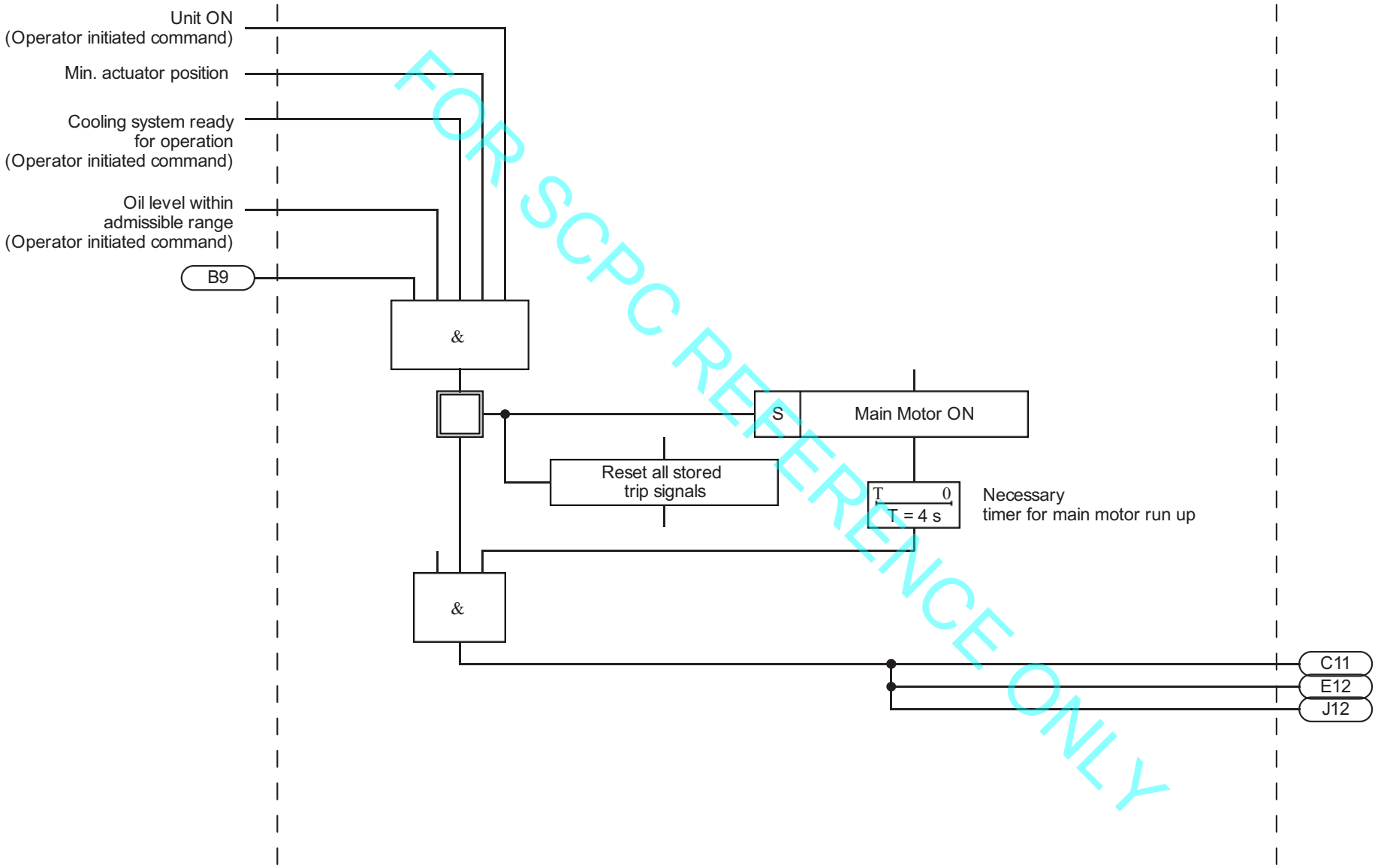

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				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 10
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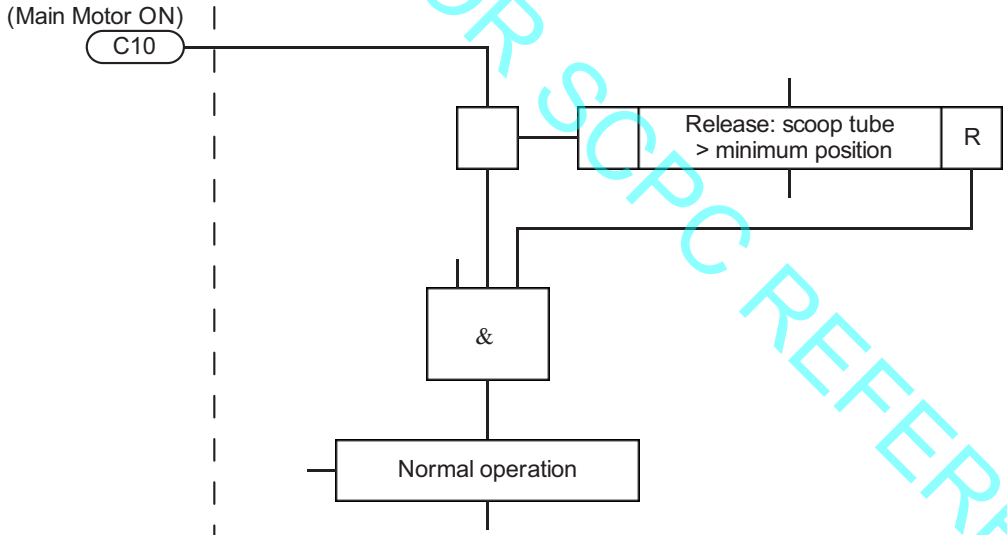

FOR SCPC REFERENCE ONLY

				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Start procedure part 2	Order No.: 38003176	
				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 11
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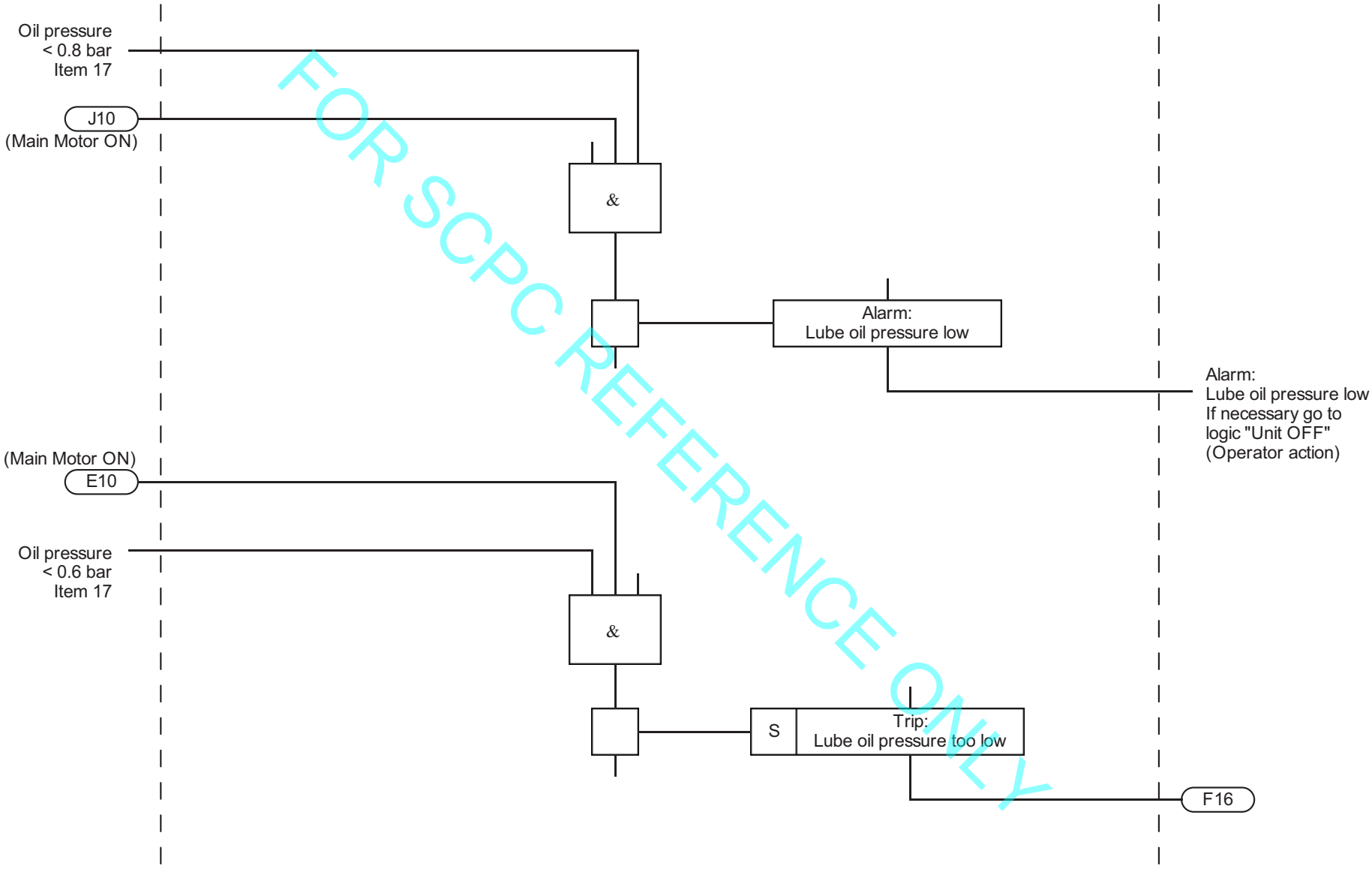

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				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Monitoring of lube oil pressure	Order No.: 38003176	
				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 12
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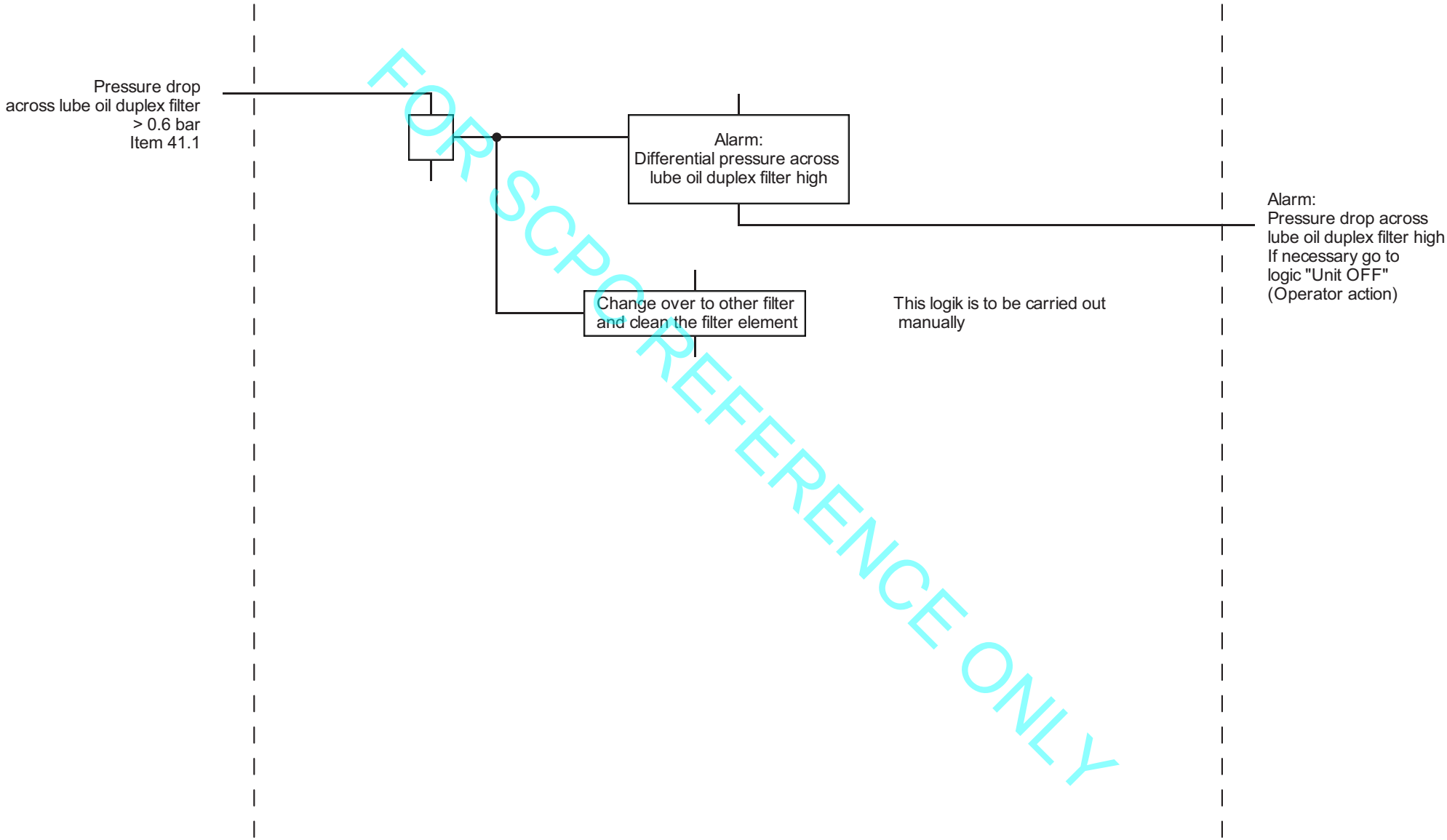

FOR SCP REFERENCE ONLY

				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Monitoring of pressure drop across lube oil duplex filter	Order No.: 38003176	
				Editor	MMar					
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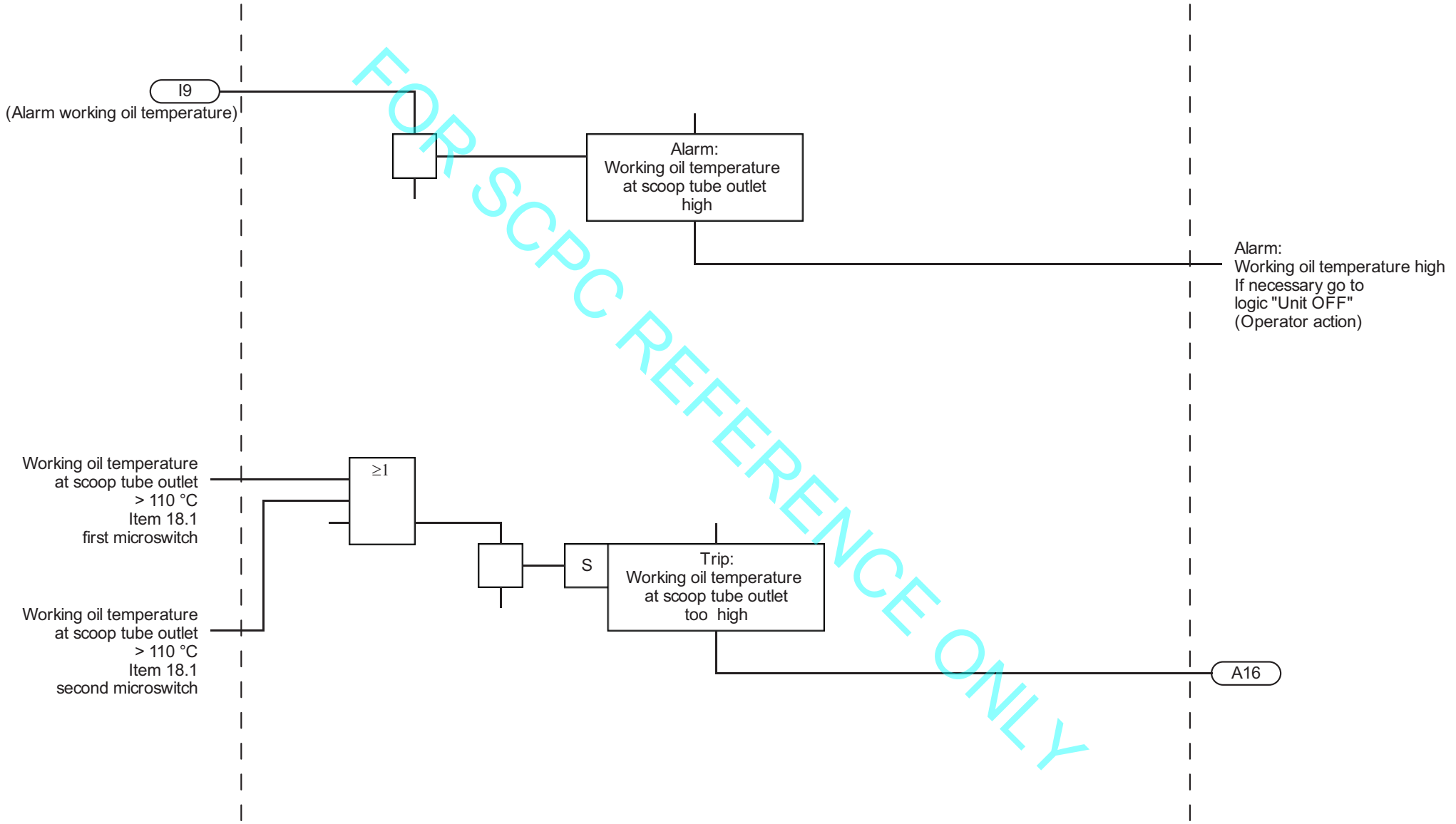

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				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Monitoring of oil temperature	Order No.: 38003176	
				Editor	MMar					
				Date		1150 SVNL 22-32	Dept. aevec		Drawing No.: 215001055-0050en	Page 14
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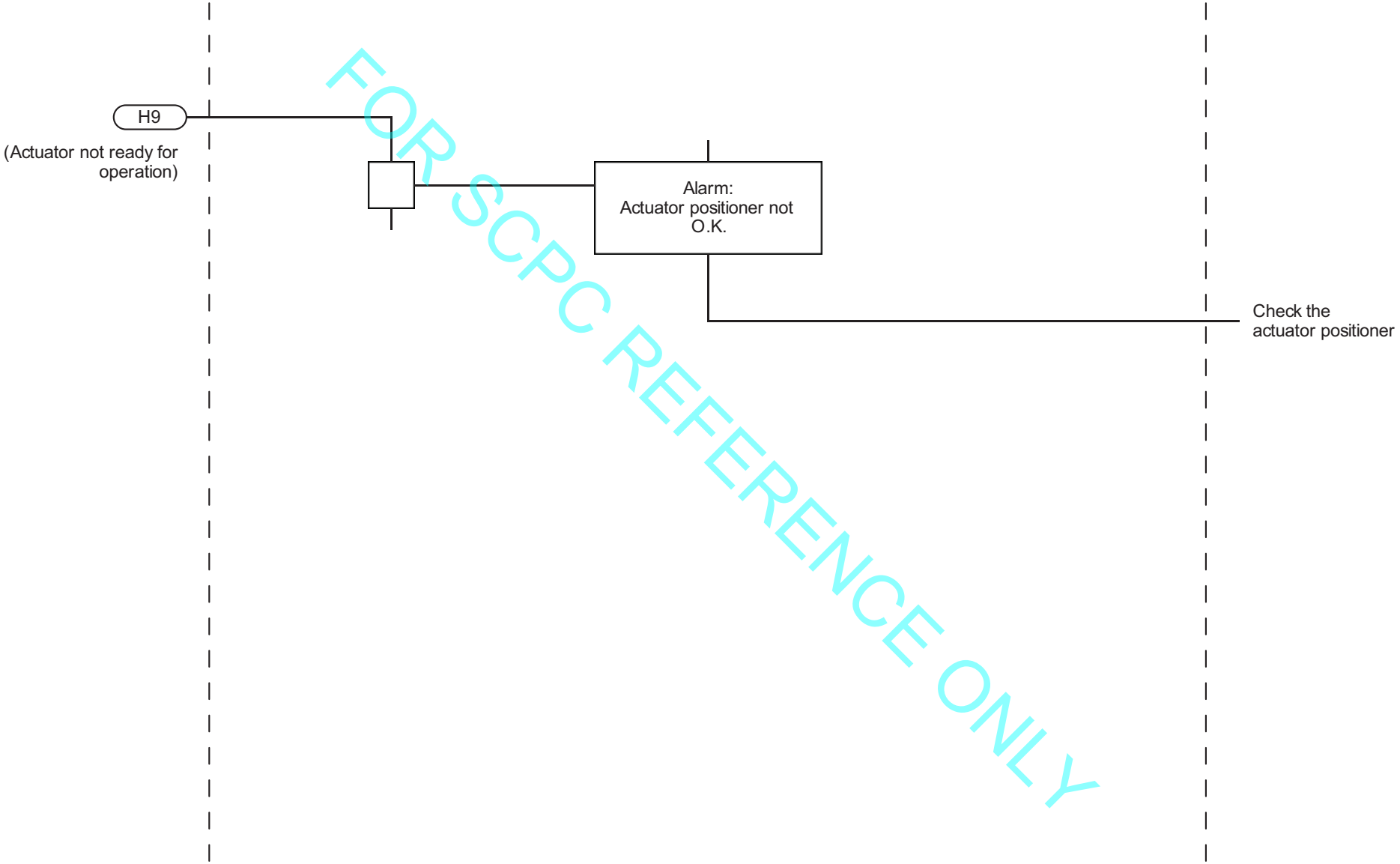

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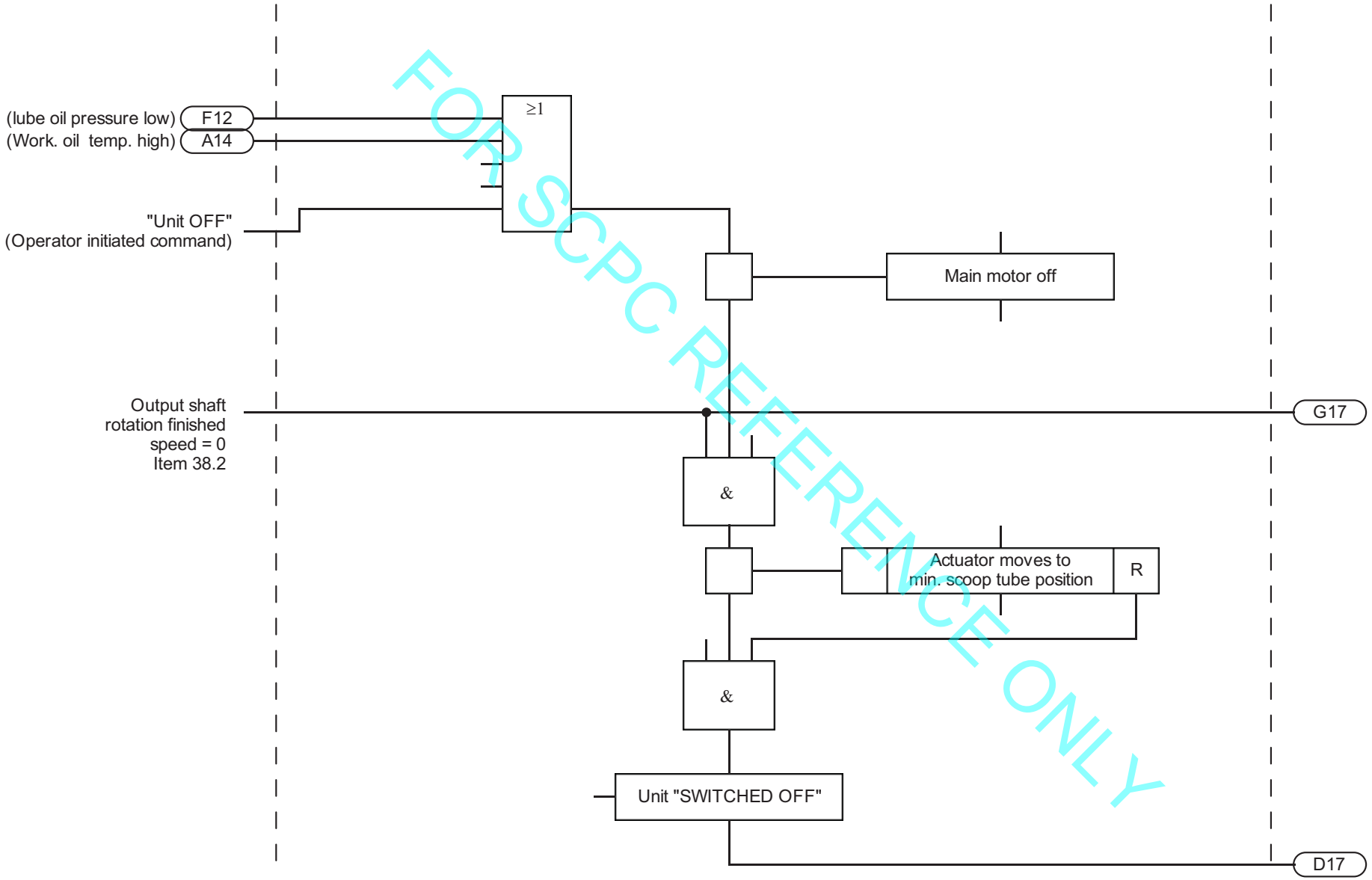

FOR SCPC REFERENCE ONLY

				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Shutdown procedure part 1	Order No.: 38003176	
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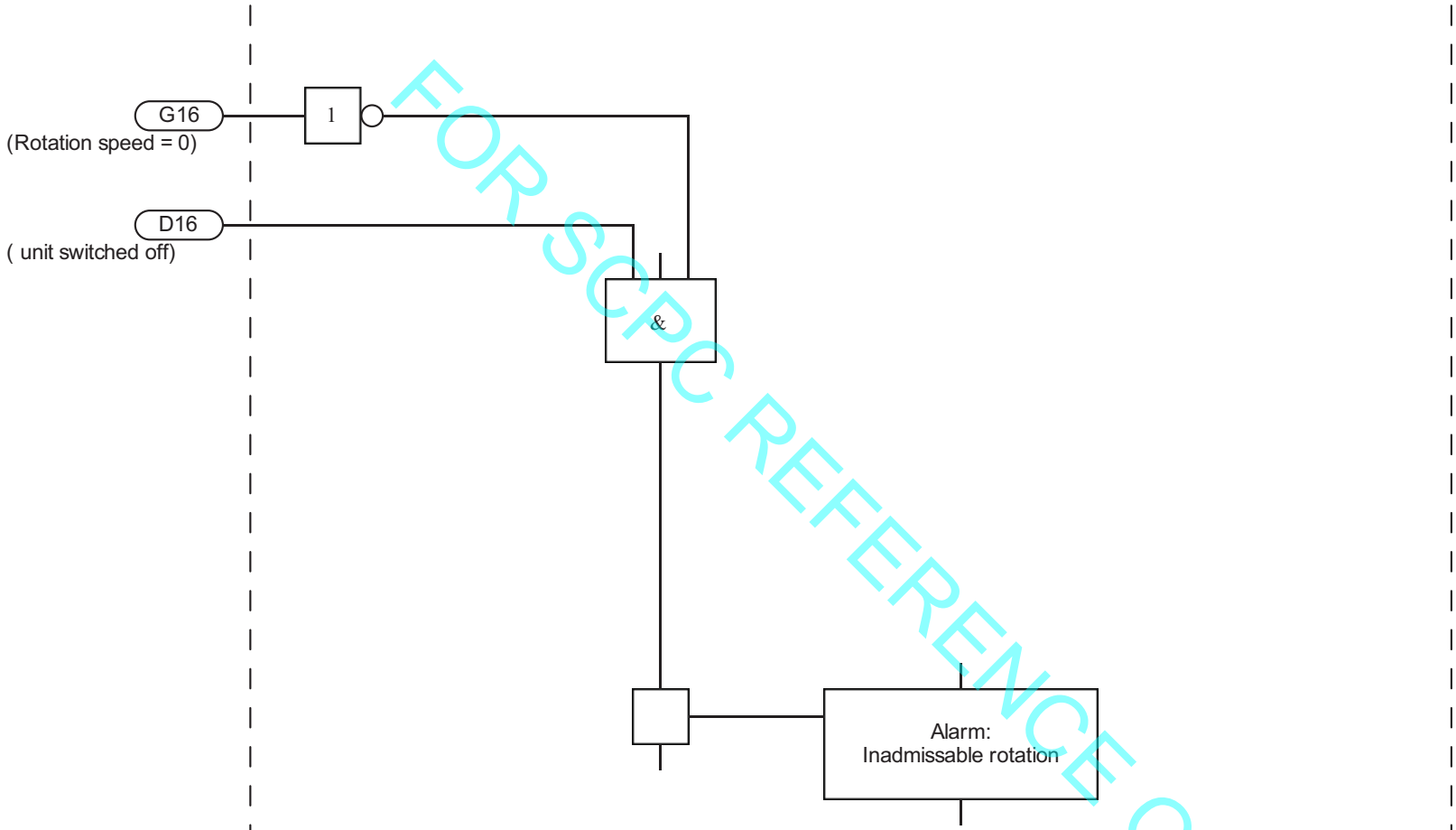

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				Date	13-12-02	VTTI S 41 FHI Philippines II	VOITH TURBO	Shutdown procedure part 2	Order No.: 38003176	
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Main Components and Instrument List

Revision	Item / TAG No. Voith	Item / TAG No. Customer	Component or instrument Type Make	Measuring range --- Adjusted range	Process connection --- Electrical connection	Ingress protection	Measurement / measuring point, location	Set value	Nominal value during opera-tion	Remarks
	01		Runner parts							
	05		Coupling housing							
	06		Filling pump							
	07		Vent filter Type: TLF I 2-32 G 25 Make: Eppensteiner							
1)	08	LI-LO1756	Oil level sight glass Type: FSA 254.2.0/12 Make: Flutec							
	09		Oil discharge set Type: 406-4 Make: Schwabe		G 1 1/4					
	10		Scoop tube							
	12		Lube oil orifice				Lube oil downstream of lube oil filter			
	13		Orifice							
1) 2)	16	PI-LO1756-2	Pressure indicator Type: 233.50.100 Make: Wika	0 - 1 bar 0 – 1.1 kg/cm²	1/2 NPT	IP 65	Lube oil pressure downstream of lube oil orifice		0.3 bar 0 – 0.31 kg/cm²	Dial size 100 mm Housing material: stainless steel filled with glycerine

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Main Components and Instrument List

Revision	Item / TAG No. Voith	Item / TAG No. Customer	Component or instrument Type Make	Measuring range --- Adjusted range	Process connection --- Electrical connection	Ingress protection	Measurement / measuring point, location	Set value	Nominal value during operation	Remarks
1) 2)	17	PS-LO1756	Pressure switch Type: MW1-XX Make: Wika	0 - 2,5 bar bar 0 – 2.55 kg/cm²	1/4 NPT --- 1/2 NPT	IP 65	Lube oil pressure downstream lube oil filter	Alarm: < 0.8 bar < 0.82 kg/cm² Trip: < 0.6 bar < 0.61 kg/cm²	> 1.2 bar > 1.22 kg/cm²	2xSPDT contacts 125 VAC / 1 A Housing material : stainless steel
1) 2)	17.5	PI-LO1756-1	Pressure indicator Type: 233.50.100 Make: Wika	0 - 4 bar 0 – 4.08 kg/cm²	1/2 NPT	IP 65	Working oil pressure upstream of coupling		> 1.8 bar > 1.84 kg/cm²	Dial size 100 mm Housing material: stainless steel filled with glycerine
1)	18	TS-LO1756-1	Capillary dial thermometer with contacts Type: 8523-22-10 / 034 SA 19 Make: Juchheim	0 - 160 ℃	G 1/2 A --- M20x1,5	IP 55	Working oil temperature at scoop tube outlet	Alarm: > 100 ℃	< 100 ℃	2 microswitches as sequence-controlled contact contact distance 2℃ Dial size 100 mm Housing material : stainless steel
1)	18.1	TS-LO1756-2	Capillary dial thermometer with contacts Type: 8523-22-10 / 034 SA 19 Make: Juchheim	0 - 160 ℃	G 1/2 A --- M20x1,5	IP 55	Working oil temperature at scoop tube outlet	Trip: > 110 ℃	< 100 ℃	2 microswitches as sequence-controlled contact contact distance 2℃ Dial size 100 mm Housing material : stainless steel
	20		Fusible plug					Melting point: 160 ℃		
	23		Heat exchanger Type: L1N4 1117 13M Make: Voith							
	25		Actuator Type: Contrac RHD 500-10 Make: ABB				Scoop tube position			

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Main Components and Instrument List

Revision	Item / TAG No. Voith	Item / TAG No. Customer	Component or instrument Type Make	Measuring range --- Adjusted range	Process connection --- Electrical connection	Ingress protection	Measurement / measuring point, location	Set value	Nominal value during operation	Remarks
	25.1		Electronic Type: Contrac EBN 853 Make: ABB		--- 2x M20x1,5 1x M25x1,5	IP 66	Scoop tube position			Input signal : 0/4 – 20 mA Output signal : 0/4 – 20 mA 2 integrated limit switches for 0 and 100 % scoop tube position 2 x NO contacts Contact rating: max. 60 V / 150 mA Power supply: 190-260 V AC; 60 Hz; 1Ph
1)	33	TI-LO1756-2	Bimetallic thermometer Type: 50 EE=070-E -108-0/150℃ Make: Ashcroft	0 – 150 ℃	1/2 NPT A	IP 65	Working oil temperature downstream of heat exchanger		< 90 ℃	Dial size: 5” Stainless steel housing with stainless steel thermowell Process connection: G 1/2 A
1)	35	TI-LO1756-1	Bimetallic thermometer Type: 50 EE=070-E -108-0/150℃ Make: Ashcroft	0 – 150 ℃	1/2 NPT A	IP 65	Working oil temperature upstream of heat exchanger		< 100 ℃	Dial size: 5” Stainless steel housing with stainless steel thermowell Process connection: G 1/2 A
1)	38.1	SE-LO1756	Pick up for speed measuring Type: MRS-300-M12-10-S Make: Rechner Sensor		--- M12x1	IP 67	Output shaft			
1)	38.2	SI-LO1756 ST-LO1756	Measuring transducer with speed indication Type: D1553.120 U2 Make: Braun			IP 65				Installed in a local junction box with sight glass 85-265 V AC/DC Input signal: 0-1200 Hz Output signal: 0/4 - 20 mA Indication: 0-1200 rpm

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Main Components and Instrument List

Revision	Item / TAG No. Voith	Item / TAG No. Customer	Component or instrument Type Make	Measuring range --- Adjusted range	Process connection --- Electrical connection	Ingress protection	Measurement / measuring point, location	Set value	Nominal value during operation	Remarks
1) 2)	41	PDI-LO1756-1	Visual maintenance indicator Type: AOC.31717.0,6.P Make: Internormen	0.6 bar 0.61 kg/cm²	G 1/2 A		Pressure drop on duplex filter	Red pin visible > 0.6 bar > 0.61 kg/cm² Change over duplex filter	< 0.3 bar < 0 – 0.31 kg/cm²	
1) 2)	41.1	PDS-LO1756	Differential pressure switch Type: 101RN-KK3-M4-C1A Make: SOR	0.2 – 2.0 bar 0.20 – 2.04 kg/cm²	1/4 NPT --- 3/4 NPTF	NEMA 4X IP 65	Pressure drop on duplex filter	Alarm: > 0.6 bar > 0.61 kg/cm² Change over duplex filter	< 0.3 bar < 0 – 0.31 kg/cm	1 x DPDT Kontakt 250 VAC / 15 A 125 VDC / 0,4 A Housing material : Aluminum
1) 2)	41.2	PDI-LO1756-2	Diff. pressure indicator Type: 733.51.100 Make: Wika	0 – 1.0 bar 0 – 1.1 kg/cm²	G 1/4	IP 65	Pressure drop on duplex filter		< 0.3 bar < 0 – 0.31 kg/cm	Dial size 100 mm Housing material: 1.4571 filled with glycerine
	42		Duplex filter Type: DU 40.31044.25 G. Make: Internormen		G 1/2		Lube oil			Filter elements: stainless steel grade of filtration: 25 µm
	46.1		Check valve Type: RK 44 Make: Gestra		DN 80 PN 16					Opening pressure 0.07 bar Material: bronze
	46.2		Check valve Type: RK 44 Make: Gestra		DN 65 PN 16					Opening pressure 0.07 bar Material: bronze
	46.3		Overflow valve Make: Voith		Flange DN 65					Housing material: EN-GJL-200
	46.4		Pressure relief valve Make: Voith		Flange DN 65					Housing material: EN-GJL-200

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Main Components and Instrument List

Revision	Item / TAG No. Voith	Item / TAG No. Customer	Component or instrument Type Make	Measuring range --- Adjusted range	Process connection --- Electrical connection	Ingress protection	Measurement / measuring point, location	Set value	Nominal value during opera-tion	Remarks
	55.1		2-valve manifold Type: N 342.44.483.21 Make: Schneider		Instrument: 1/2 NPT Process: 1/2 NPT Test: 1/4 NPT					Housing material: 1.4404
	55.3		5-valve manifold Type: N 342.42.482.01 Make: Schneider		Instrument: 1/2 NPT Process: 1/2 NPT Test: 1/4 NPT					Housing material: 1.4404
	59		Orifice				Heat exchanger venting			

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VOI1203E/16.Jul.1996

VOITH

VOITH TURBO GmbH & Co. KG

Voithstrasse 1 74564 Crailsheim
Tel: 07951 / 32-0
Fax: 07951 / 32-650

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Client : Voith Turbo Co., Ltd.
Unit designation : 1150 SVNL 22-32
Order number : 38 003 176
Drawing number : 215 001055-0040 en
Code word : VTTI S 41 FHI Philippines II

Regulating and Control : ABB
Equipment - Type : CONTRAC RHD 500-10 / EBN 853

File name without \EPLAN\P: \VOI\GA_2013\215 001 055-0040 EN

CAD - Symbolic files : DIN_WUP/VC_SYMBL

Designation : WIRING DIAGRAM

Department : airea-Ga

Created at : 03.12.2013
Revised edition : 09.04.2014 by : Gaugler

according design specification c127

1	176549	09.04.14	Ga	Date	03.12.2013	VTTI S 41 FHI Philippines II	VOITH TURBO	cover sheet	Order no.: 38 003 176
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+J1/2	table of contents		09. 04. 2014	Gaugler
+J1/3	junction box J1 / view / layout		09. 04. 2014	Gaugler
+J1/4	pressure switch		09. 04. 2014	Gaugler
+J1/5	temperature switch		09. 04. 2014	Gaugler
+J1/6	differential pressure switch		09. 04. 2014	Gaugler
+J1/7	power supply 220VAC / enclosure heating		09. 04. 2014	Gaugler
+J1/8	power electronics EBN 853		09. 04. 2014	Gaugler
+J1/9	power electronics EBN 853 / actuator Contrac RHD 500-10		09. 04. 2014	Gaugler
+J1/10	speed measuring transducer		09. 04. 2014	Gaugler
+J1/11	parameterlist		09. 04. 2014	Gaugler

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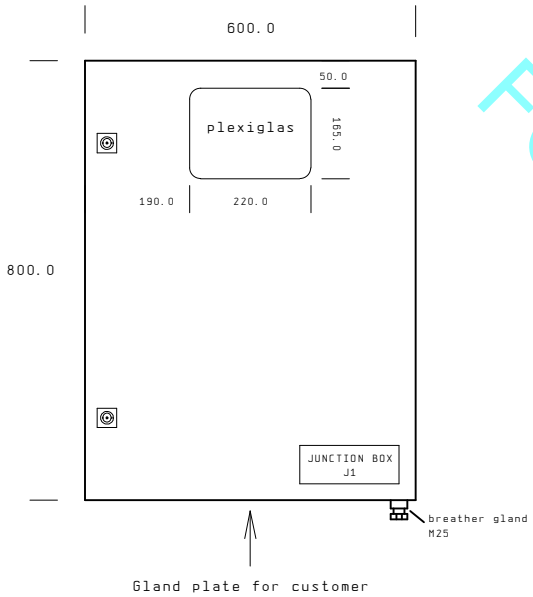
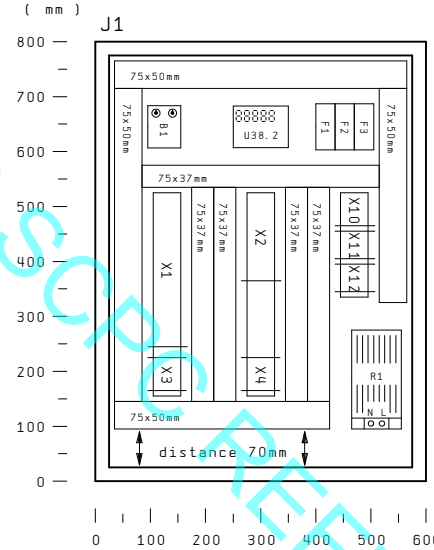
junction box J1

front view



layout
mounting plate

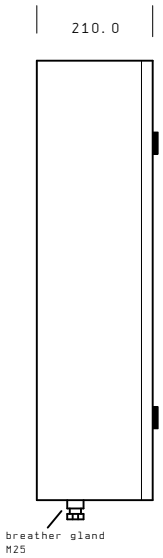


Notes:

Terminal strip: X1/X2/X3/X4/X11/X12
Supplier: PHOENIX
Type: UK 2,5 N / USLKG 2,5 N
(0,2-4 mm²)

Terminal strip: X10
Supplier: PHOENIX
Type: UK 5 N / USLKG 5
(0,2-6 mm²)

side view



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Supplier: Eldon

Type: ASR 0806021

junction box dimensions: 800x600x210 mm

degree of protection: IP 65 / NEMA 4

material: stainless steel 304

weight: approx. 50kg

scale: 1:10

ITEM NO.	Symbol designation	Type designation	VTCR - Art. number
1	J1	ASR 0806021	4 180 062 0
2	J1-B1	ETF300	204.008 984 10
3	J1-U38.2	D 1553.120 U2 / 85-265VAC/DC	4 201 074 009
4	J1-F1	FAZ-S6/2	4 222 947 0
5	J1-F2	FAZ-S1/2	4 178 991 0
6	J1-F3	FAZ-S4/2	4 221 910 0
7	J1-R1	SK 3105.370 / 130 - 150 W	204.011 267 10

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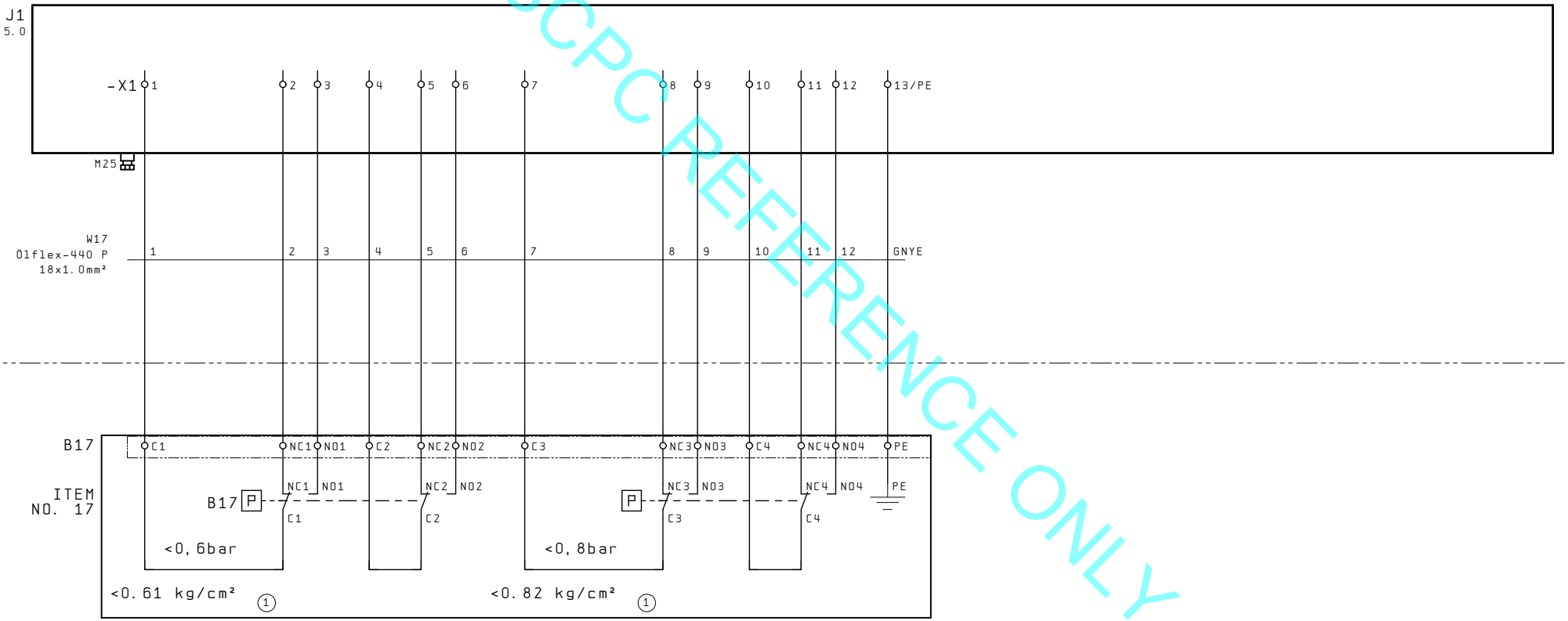

pressure switch
lube oil

PS-L01756 (1)

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3 5

1 176549 3x	09.04.14	Ga	Date	03.12.2013	VTTI S 41 FHI Philippines II		VOITH TURBO	pressure switch	Order no.: 38 003 176	= +J1
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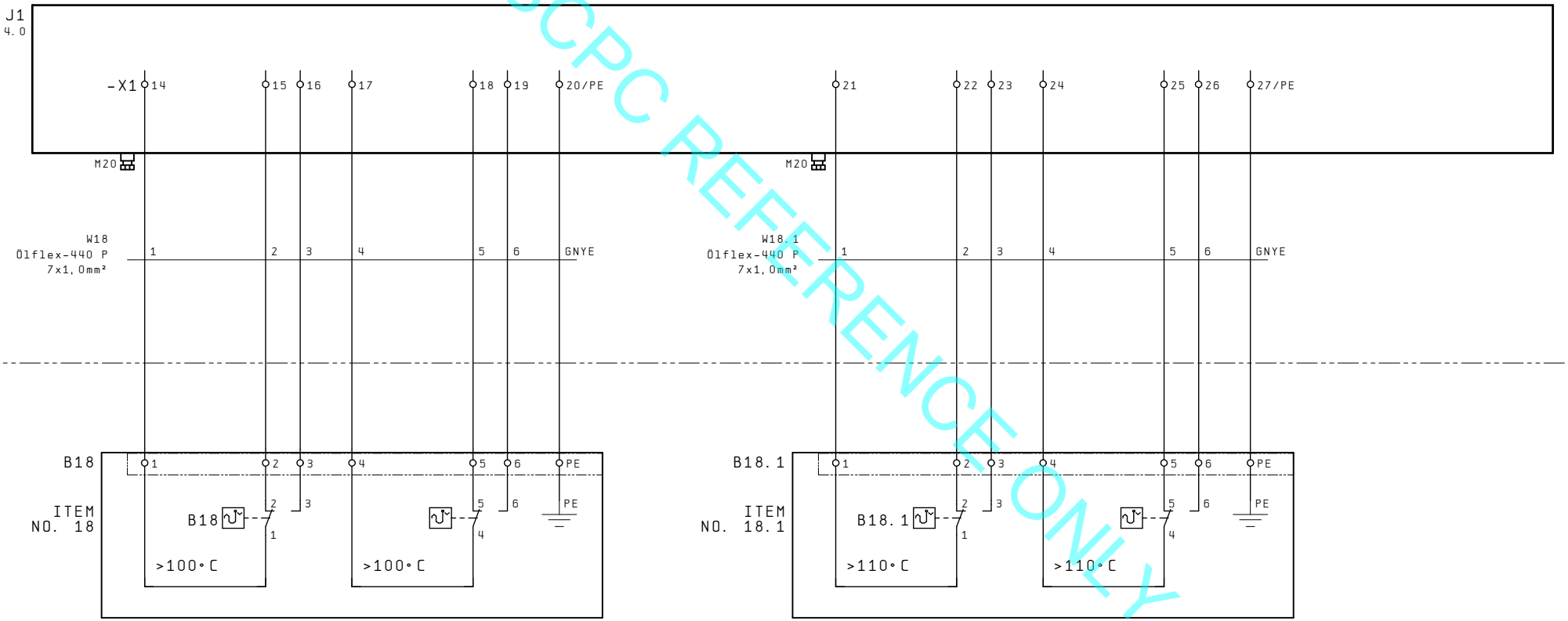

working oil temperature
at scoop tube outlet

TS-L01756-1 ①

working oil temperature
at scoop tube outlet

TS-L01756-2 ①

4

6

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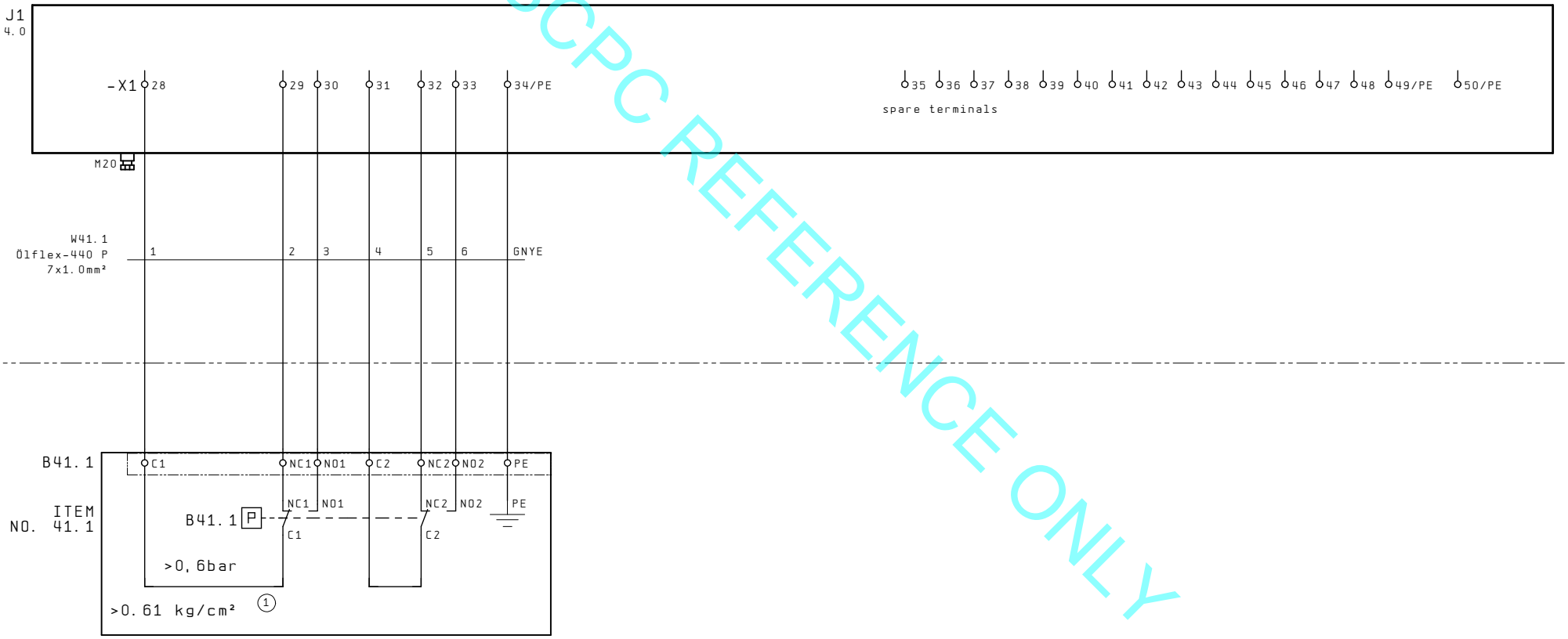

differential pressure switch
lube oil

PDS-L01756 (1)

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5 7

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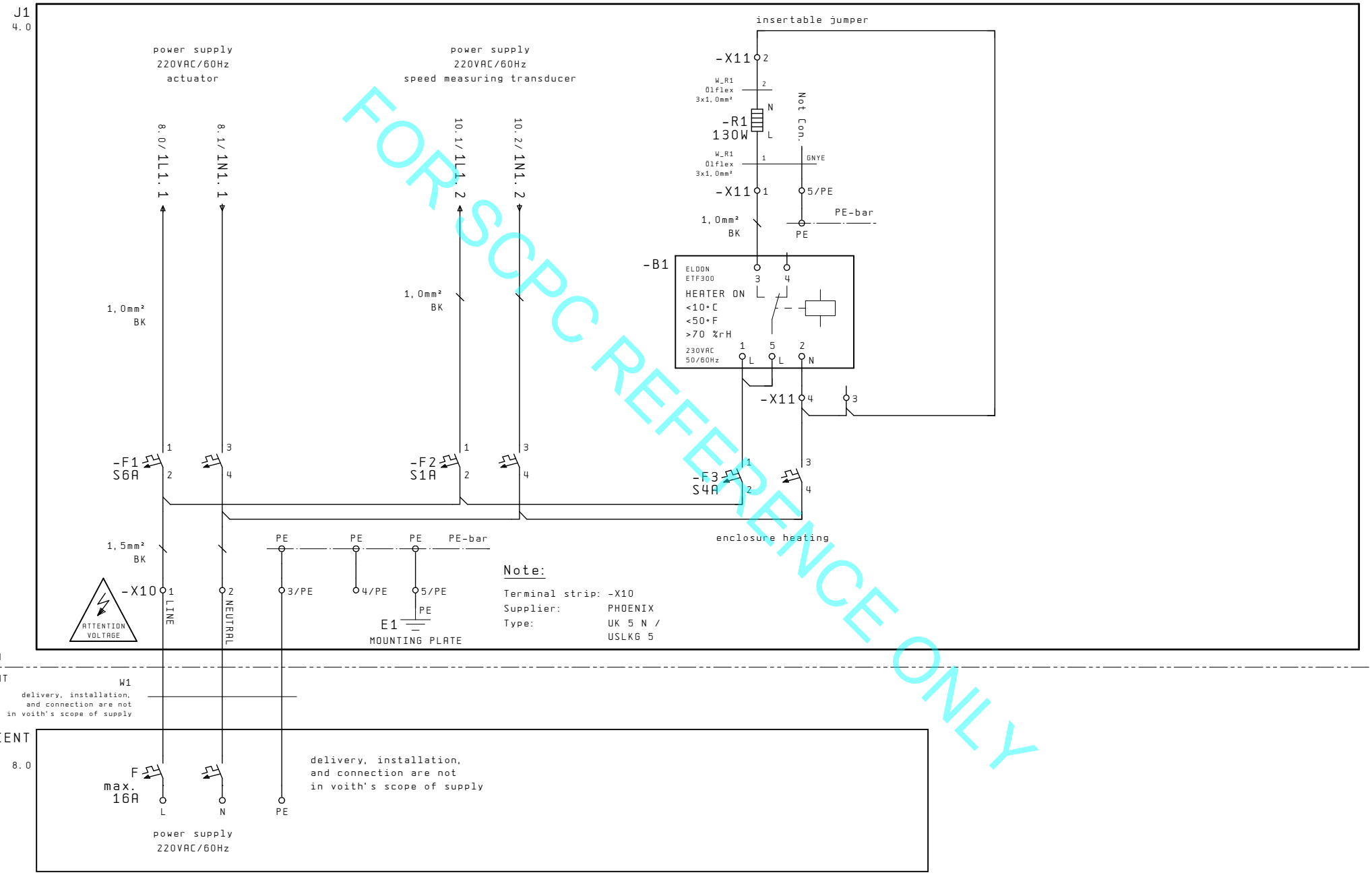

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CLIENT

7.0

delivery, installation, and connection are not in Voith's scope of supply

option manual / automatic

S+E y+

S-E y-

manual S1 automatic

X 1 2 3 4 5 6 7 8

control system

setvalue 4...20mA

actual value 4...20mA

PE

W2

delivery, installation, and connection are not in Voith's scope of supply

W3

delivery, installation, and connection are not in Voith's scope of supply

CLIENT

VOITH

J1

4.0

7.1/1L1.1

7.1/1N1.1

connection must be removed at option manual/automatic function

1,0mm² BU

-X12 1 2 3/PE

-X2 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18/PE 19/PE

20 21 22 23 24/PE 25/PE

M16 M32 M20

W25.1/A Ölflex-440 P 3x1,0mm² 1 2 GNYE

W25.1/B Ölflex-440 CP 18x0,5mm² 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 GNYE

W25.1/C Ölflex-440 CP 5x0,5mm² 1 2 3 4 GNYE

Y25.1

9.0

ITEM NO. 25.1

Y25.1 L N PE

POWER SUPPLY

190 ... 260VAC

47,5 ... 63 Hz

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 SPARE SPARE SPARE PE

BE1 BE1 BE1 BA1 BA2 BA3 Uv 24V

potential-free contact max. 60V/150mA

manual automatic

scoop tube position increase

scoop tube position decrease

ready for operation

end position 0%

end position 100%

24VDC

9-8: ok

9-7: fault

26 27 30 31 PE PE

RB

I U

set value 4...20mA

actual value 4...20mA

Contrac

Power electronics

Type: EBN 853

7

9

FOR SPEC REFERENCE ONLY

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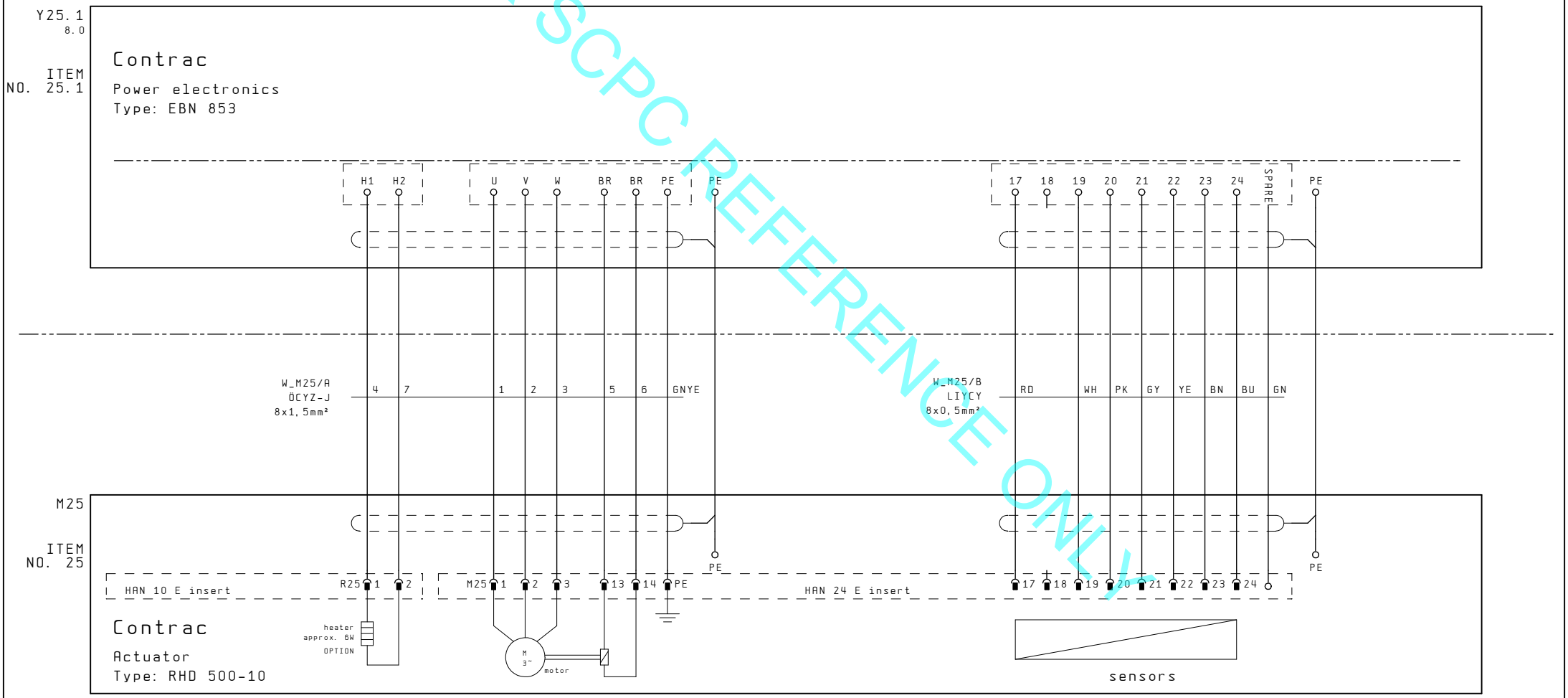

8 actuator 10

1	176549	09.04.14	Ga	Date	03.12.2013	VTTI S 41 FHI Philippines II	VOITH TURBO	power electronics EBN 853 / actuator Contrac RHD 500-10
				Editor	Gaugler	1150 SVNL 22-32	Dep.: airea-Ga	
				Date	09.04.2014			
Revisions		Date	Name	Checked		Original	Replacing	Replaced by

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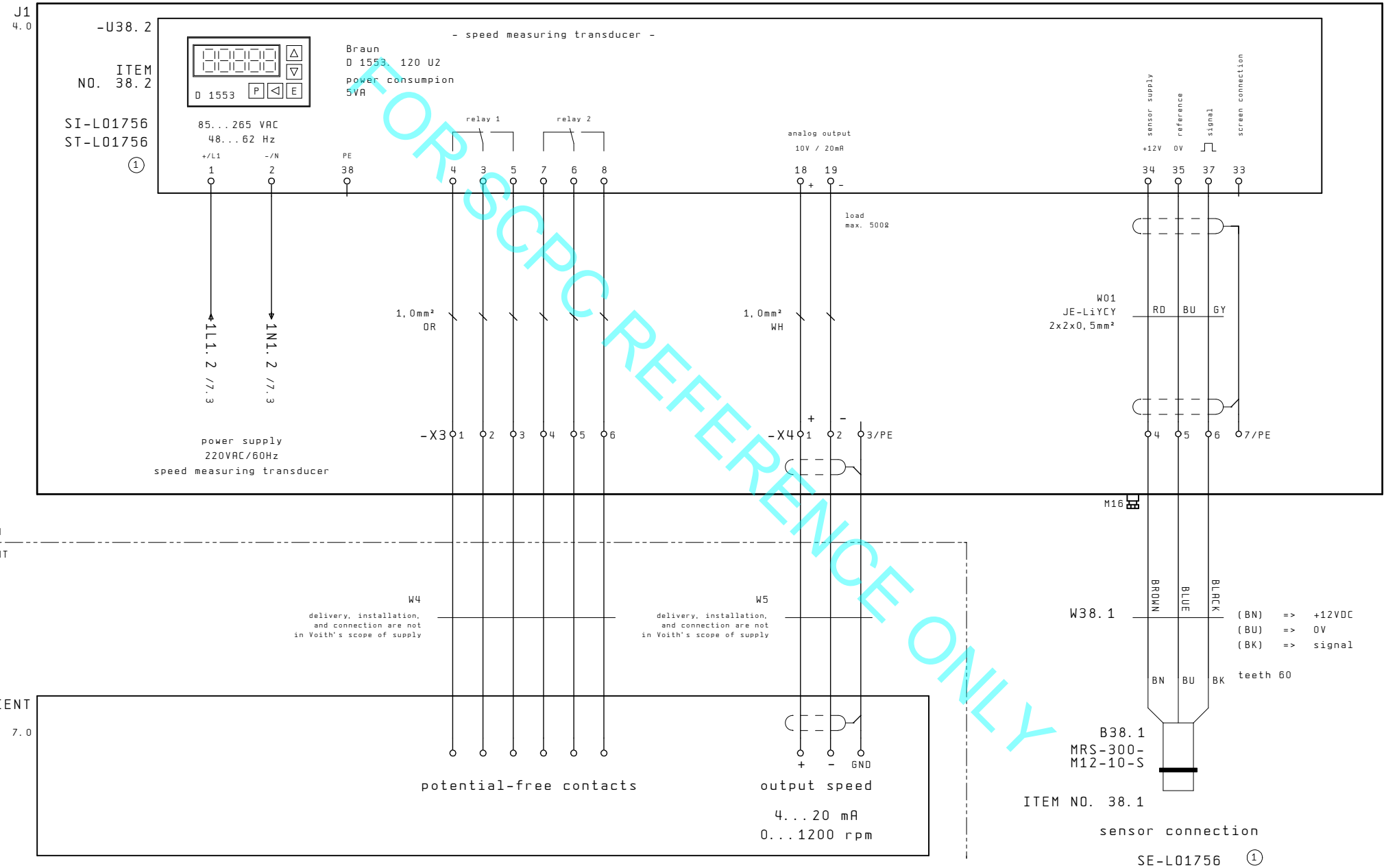
item no. see oil circuit and measuring point scheme 215 001055-0020



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1	176549 2x	09.04.14	Ga	Date	03.12.2013	VTTI S 41 FHI Philippines II	VOITH TURBO	speed measuring transducer	Order no.: 38 003 176
				Editor	Gaugler	1150 SVNL 22-32	Dep.: airea-Ga		= +J1
				Date	09.04.2014				Page 10
Revisions		Date	Name	Checked		Original	Replacing	Replaced by	Drawing number: 215 001055-0040 en 11 P.

Summary of programming steps and initial parameters
equipment: U38.2

step No.	comments page	parameter	data set on delivery	actual parameter
P00.00	7	access code figure	0000	0000
.01	7	new code figure		0000
.02	7	lock function(0 = key locked, 1 = unlocked)	1	1
P01.00	8	input decimals of input frequency	0 = none	0
.01	8	value of input frequency (Hz)	10000	01200
.02	8	decimals of variable	0 = none	0
.03	8	value decimals of variable in appl. terms	10000	01200
.04	8	low end for shut - off "	00001	00001
.05	8	adjustable predivider (01..99)	01	01
P02.00	9	zeroed LEDs in display	0 = none	0
.01	9	display updating sequence (X.X sec)	0.3 (sec)	0.3
.02	9	starter time period (XXX sec)	000 (sec)	000
.03	9	active level of force inputs B/D (l=low, h=high)	3 = h/h	3
.04	9	enable/disable of sensor failure detection	0=disabl.	0
P03.00	10	setpoint SP1 (in selected terms)	10000	01300
.01	10	hysteresis bandwidth for SP1 (xx % of SP)	05 (%)	1
.02	10	hyst. position for SP1 (0=above, 1=below, 2=symm.)	1=below	1
.03	11	relay condition if n > SP1 (0=energized, 1=deener.)	1=deener	1
.04	11	starter function for SP1 (0=disable, 1=enable)	0=disabl.	0
.05	11	starter state of SP1 (0 = < SP, 1 = > SP)	1 = > SP	1
.06	11	force state of SP1 (0 = < SP, 1 = > SP)	1 = > SP	1
.07	11	fault state of SP1 (0 = < SP, 1 = > SP)	1 = > SP	1
.08	11	LED 1 on/off in n > SP1		1
P04.00	12	setpoint SP2 (in selected terms)	09000	01300
.01	12	hysteresis bandwidth for SP2 (xx % of SP)	05 (%)	1
.02	12	hyst. position for SP2 (0=above, 1=below, 2=symm.)	1=below	1
.03	12	relay condition if n > SP2 (0=energized, 1=deener.)	1=deener	1
.04	12	starter function for SP2 (0=disable, 1=enable)	0=disabl.	0
.05	12	starter state of SP2 (0 = < SP, 1 = > SP)	1 = > SP	1
.06	12	force state of SP2 (0 = < SP, 1 = > SP)	1 = > SP	1
.07	12	fault state of SP2 (0 = < SP, 1 = > SP)	1 = > SP	1
.08	12	LED 2 on/off in n > SP2		1
P05.00	12	setpoint SP3 (in selected terms)	08000	programm-selection 05 not available
.01	12	hysteresis bandwidth for SP3 (xx % of SP)	05 (%)	
.02	12	hyst. position for SP3 (0=above, 1=below, 2=symm.)	1=below	
.03	12	relay condition if n > SP3 (0=energized, 1=deener.)	1=deener	
.04	12	starter function for SP3 (0=disable, 1=enable)	0=disabl.	
.05	12	starter state of SP3 (0 = < SP, 1 = > SP)	1 = > SP	
.06	12	force state of SP3 (0 = < SP, 1 = > SP)	1 = > SP	
.07	12	fault state of SP3 (0 = < SP, 1 = > SP)	1 = > SP	
.08	12	LED 3 on/off in n > SP3		
P06.00	12	setpoint SP4 (in selected terms)	07000	programm-selection 06 not available
.01	12	hysteresis bandwidth for SP4 (xx % of SP)	05 (%)	
.02	12	hyst. position for SP4 (0=above, 1=below, 2=symm.)	1=below	
.03	12	relay condition if n > SP4 (0=energized, 1=deener.)	1=deener	
.04	12	starter function for SP4 (0=disable, 1=enable)	0=disabl.	
.05	12	starter state of SP4 (0 = < SP, 1 = > SP)	1 = > SP	
.06	12	force state of SP4 (0 = < SP, 1 = > SP)	1 = > SP	
.07	12	fault state of SP4 (0 = < SP, 1 = > SP)	1 = > SP	
.08	12	LED 4 on/off in n > SP4		
P07.00	13	analog output high end	10000	01200
.01	13	low end	00000	00000
.02	13	zero level (0=no, 1=with live zero)	0	1
.03	13	mode: 0=voltage, 1=current	1	1
.04	13	analog out. at sensor fault (0=n.change, 1=min, 2=max)	1	0
P08.00	13	date interface: baudrate (see table)	1 = 9600	1
.01	13	: device address No.	001	001

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parameterlist
VOITH TURBO
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Revisions

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INSPECTION & TEST PLAN (ITP)

GLOSSARY OF ABBREVIATIONS :

H HOLD POINT. WORK SHALL NOT PROCEED UNTIL RELEVANT INSPECTION COMPLETED.

W WITNESS POINT. WORK WILL PROCEED WITHOUT FURTHER RESTRICTION.

R REVIEW OF ACTIVE DOCUMENTS OR CERTIFICATES.

PT LIQUID PENETRANT TESTING.

RT RADIOGRAPHIC TESTING.

VT VISUAL INSPECTION.

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REV.	DATE	CONTENT	CUSTOMER	APPROVED	REVIEWED	PREPARED
0	08-29-'13	INITIAL ISSUE				

INSPECTION STAGE	DESCRIPTION OF INSPECTION	REFERENCE DOCUMENT	REPORTING EVIDENCE	INSPECTION		
				CCMC	FHI	OWNER
1. REVIEW OF DOCUMENT	(a) WELDING PROCEDURE SPECFFICATIONS AND PROCEDURE QUALIFICATION RECORDS	ASME SEC.IX	APPROVED WPS & PQR	R		
	(b) WELDERS / WELDING OPERATOR QUALIFICATION	ASME SEC.IX	APPROVED WQR	R		
2. FABRICATION OF FAN CASING	(a) MATERIAL VERIFICATION	DRAWING	MILL CERTIFICATE	H		
	(b) VT OF WELD SEAMS	DRAWING		H		
3. IMPELLER	(a) MATERIAL VERIFICATION	DRAWING	MILL CERTIFICATE	H		
	(b) PT OF WELD SEAMS	DRAWING	REPORT	H		
	(c) RT OF WELD SEAMS	DRAWING	REPORT	R		
4. ROTOR ASSEMBLY	DYNAMIC BALANCE TEST	ISO 1940 / G2.5	REPORT	H		
5. FAN ASSEMBLY	(a) FINAL VT CHECK	DRAWING		H		
	(b) PAINTING INSPECTION	DRAWING	REPORT	H		
6. MOTOR	PERFORMANCE TEST	CNS1373 / C 4040	REPORT	H		
REMARK:						

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